

Enterprise COBOL for z/OS
6.4

Language Reference



Note

Before using this information and the product it supports, be sure to read the general information under [“Notices” on page 811](#).

Seventh edition (28 April 2026 update)

This edition applies to Version 6.4 of IBM® Enterprise COBOL for z/OS® (program number 5655-EC6) and to all subsequent releases and modifications until otherwise indicated in new editions. Make sure that you are using the correct edition for the level of the product.

You can view or download softcopy publications free of charge in the [Enterprise COBOL for z/OS library](#). Because Enterprise COBOL for z/OS supports the continuous delivery (CD) model and publications are updated to document the features delivered under the CD model, it is a good idea to check for updates once every two months.

It is our intention to update the product documentation for this release periodically, without updating the order number. If you need to uniquely refer to the version of your product documentation, refer to the order number with the date of update.

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Preface

About this information

This information describes the COBOL language supported by IBM Enterprise COBOL for z/OS, referred to in this information as Enterprise COBOL.

See the *IBM Enterprise COBOL for z/OS Programming Guide* for information and examples that will help you write, compile, and debug programs and classes.

How to read the syntax diagrams

Throughout the document, diagrams illustrate Enterprise COBOL syntax.

Use the following description to read the syntax diagrams in this document:

- Read the syntax diagrams from left to right, from top to bottom, following the path of the line.

The **▶▶** symbol indicates the beginning of a syntax diagram.

The **—▶** symbol indicates that the syntax diagram is continued on the next line.

The **▶—** symbol indicates that the syntax diagram is continued from the previous line.

The **—▶◀** symbol indicates the end of a syntax diagram.

Diagrams of syntactical units other than complete statements start with the **▶—** symbol and end with the **—▶** symbol.

- Required items appear on the horizontal line (the main path).



- Optional items appear below the main path.



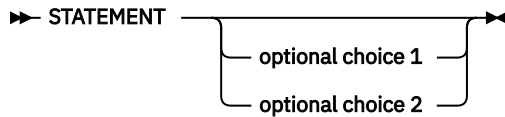
- When you can choose from two or more items, they appear vertically, in a stack.

If you *must* choose one of the items, one item of the stack appears on the main path.



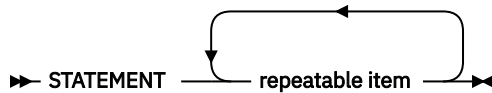
If choosing one of the items is optional, the entire stack appears below the main path.

Format



- An arrow returning to the left above the main line indicates an item that can be repeated.

Format

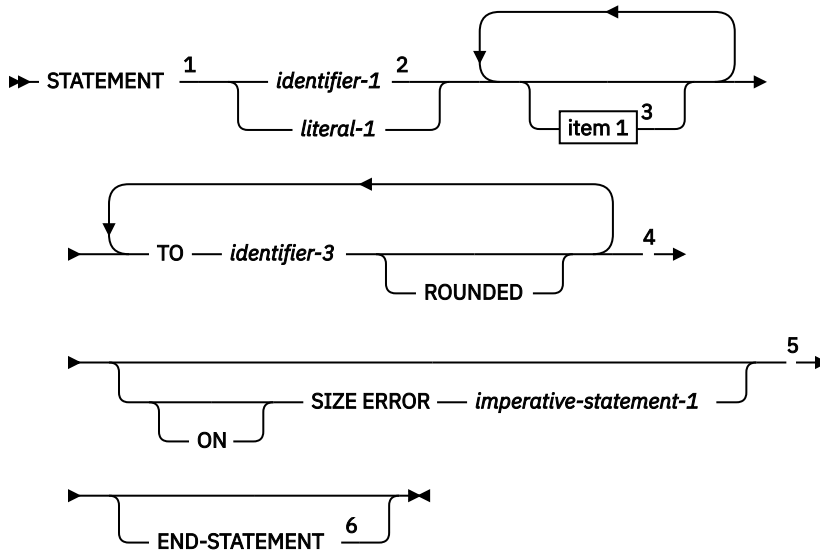


A repeat arrow above a stack indicates that you can make more than one choice from the stacked items, or repeat a single choice.

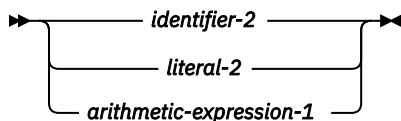
- Variables appear in *italic lowercase letters* (for example, *parm*). They represent user-supplied names or values.
- If punctuation marks, parentheses, arithmetic operators, or other such symbols are shown, they must be entered as part of the syntax.

The following example shows how the syntax is used.

Format



item 1



Notes:

- ¹ The **STATEMENT** keyword must be specified and coded as shown.
- ² This operand is required. Either *identifier-1* or *literal-1* must be coded.

³ The item 1 fragment is optional; it can be coded or not, as required by the application. If item 1 is coded, it can be repeated with each entry separated by one or more COBOL separators. Entry selections allowed for this fragment are described at the bottom of the diagram.

⁴ The operand *identifier-3* and associated TO keyword are required and can be repeated with one or more COBOL separators separating each entry. Each entry can be assigned the keyword ROUNDED.

⁵ The ON SIZE ERROR phrase with associated *imperative-statement-1* is optional. If the ON SIZE ERROR phrase is coded, the keyword ON is optional.

⁶ The END-STATEMENT keyword can be coded to end the statement. It is not a required delimiter.

How to use examples

This information shows numerous examples of sample COBOL statements, program fragments, and small programs to illustrate the coding techniques being described. The examples of program code are written in lowercase, uppercase, or mixed case to demonstrate that you can write your programs in any of these ways.

To more clearly separate some examples from the explanatory text, they are presented in a monospace font.

COBOL keywords and compiler options that appear in text are generally shown in SMALL UPPERCASE. Other terms such as program variable names are sometimes shown in *an italic font* for clarity.

If you copy and paste examples from the PDF format documentation, make sure that the spaces in the examples (if any) are in place; you might need to manually add some missing spaces to ensure that COBOL source text aligns to the required columns per the [COBOL reference format](#). Alternatively, you can copy and paste examples from the HTML format documentation and the spaces should be already in place.

IBM extensions

IBM extensions generally add features, syntax, or rules that are not specified in the ANSI and ISO COBOL standards that are listed in [Appendix J, “Industry specifications,” on page 797](#). In this document, the term 85 COBOL Standard refers to those standards.

Extensions range from minor relaxation of rules to major capabilities, such as XML support, Unicode support, object-oriented COBOL for Java™ interoperability, and DBCS character handling.

The rest of this document describes the complete language without identifying extensions. You will need to review [Appendix A, “IBM extensions and COBOL standards,” on page 737](#) and the *Compiler options* in the *Enterprise COBOL Programming Guide* if you want to use only standard language elements.

Obsolete language elements

Obsolete language elements are elements that are categorized as *obsolete* in the 85 COBOL Standard. Those elements are not part of the 2002 COBOL Standard.

This does *not* imply that IBM will remove the 85 COBOL Standard obsolete elements from a future release of Enterprise COBOL.

The following language elements are categorized as obsolete by the 85 COBOL Standard:

- ALTER statement
- AUTHOR paragraph
- Comment entry
- DATA RECORDS clause
- DATE-COMPILED paragraph

- DATE-WRITTEN paragraph
- DEBUG-ITEM special register
- Debugging sections
- ENTER statement
- GO TO without a specified procedure-name
- INSTALLATION paragraph
- LABEL RECORDS clause
- MEMORY SIZE clause
- MULTIPLE FILE TAPE clause
- RERUN clause
- REVERSED phrase
- SECURITY paragraph
- Segmentation module
- STOP *literal* format of the STOP statement
- USE FOR DEBUGGING declarative
- VALUE OF clause
- The figurative constant ALL *literal* with a length greater than one, when the figurative constant is associated with a numeric or numeric-edited item

DBCS notation

Double-Byte Character Set (DBCS) strings in literals, comments, and user-defined words are delimited by shift-out and shift-in characters.

In this document, the shift-out delimiter is represented pictorially by the < character, and the shift-in character is represented pictorially by the > character. The single-byte EBCDIC codes for the shift-out and shift-in delimiters are X'0E' and X'0F', respectively.

The <> symbol denotes contiguous shift-out and shift-in characters. The >< symbol denotes contiguous shift-in and shift-out characters.

DBCS characters are shown in this form: D1D2D3. Latin alphabet characters in DBCS representation are shown in this form: .A.B.C. The dots that precede the letters represent the hexadecimal value X'42'.

Notes:

- In EBCDIC DBCS data containing mixed single-byte and double-byte characters, double-byte character strings are delimited by shift-out and shift-in characters.
- In ASCII DBCS data containing mixed single-byte and double-byte characters, double-byte character strings are not delimited by shift-out and shift-in characters.

Acknowledgment

The following extract from Government Printing Office Form Number 1965-0795689 is presented for the information and guidance of the user:

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Note: The Conference on Data Systems Languages (CODASYL), mentioned above, is no longer in existence.

Additional documentation and support

Enterprise COBOL provides Portable Document Format (PDF) versions of the entire library for this version and for previous versions on the library page at <http://www.ibm.com/support/docview.wss?uid=swg27036733>. These documents are also available in Japanese.

Support information is also available at https://www.ibm.com/support/home/product/B984385H82239E03/Enterprise_COBOL_for_z/OS.

Summary of changes

This section lists the key changes that have been made to this document since Enterprise COBOL for z/OS 6.4. The changes that are described in this information have an associated cross-reference for your convenience.

For a complete list of new and improved features in Enterprise COBOL for z/OS 6.4 and COBOL 6.4 with PTFs installed, see What is new in Enterprise COBOL for z/OS 6.4 and COBOL 6.4 with PTFs installed in the *Enterprise COBOL for z/OS What's New*.

Enterprise COBOL for z/OS 6.4 with PTFs installed

- PH48667: A problem is fixed for using figurative constant HIGH-VALUES with fixed byte-length UTF-8 data items of a length not a multiple of 4 bytes. (“[Figurative constants](#)” on page 15)
- PH53631: Enhanced the ON EXCEPTION phrase support to deal with exceptions in the non-OO COBOL/Java interoperability framework. (“[CALL statement](#)” on page 319)
- PH56036 and PH56037: An optional alternate logic path is introduced for VSAM files that use the ACCESS IS DYNAMIC mode. The alternate logic path uses a direct read-by-key request instead of a point to a record by key. (“[Access modes](#)” on page 149)
- PH57297: You can use UTF-8 (PIC U) data items as the arguments to the STRING and UNSTRING statements. (“[STRING statement](#)” on page 462 and “[UNSTRING statement](#)” on page 470)

Note: COBOL Runtime LE APAR PH57264 (for AMODE 31) or APAR PH57265 (for AMODE 64) must also be applied on all systems where programs that make use of this new feature are linked or run.

- PH57397: With a function prototype, you can define the function name, parameters, and returning value of a user-defined function or other non-COBOL external functions such as C functions and invoke these functions. This is part of the 2014 COBOL Standard.

- [Part 3, “IDENTIFICATION DIVISION,” on page 99](#)
 - [Chapter 20, “FUNCTION-ID paragraph,” on page 113](#)
 - [Chapter 27, “Procedure division structure,” on page 257](#)
 - [“The PROCEDURE DIVISION header” on page 258](#)
 - [“USING phrase” on page 260](#)
 - [“RETURNING phrase” on page 263](#)
 - PH57398: You can use the ENCODING phrase of the JSON GENERATE and JSON PARSE statements to specify the encoding of the JSON document. ([“JSON GENERATE statement” on page 369](#) and [“JSON PARSE statement” on page 382](#))
- Note:** COBOL Runtime LE APAR PH57152 must also be applied on all systems where programs that make use of this new feature are linked or run.
- PH57400: You can use dynamic-length and UTF-8 (PIC U) data items as the arguments to the JSON GENERATE and JSON PARSE statements. ([“JSON GENERATE statement” on page 369](#) and [“JSON PARSE statement” on page 382](#))
- Note:** COBOL Runtime LE APAR PH57152 must also be applied on all systems where programs that make use of this new feature are linked or run.
- PH58384: You can use the NAME IS OMITTED phrase to parse an anonymous JSON array in addition to an anonymous JSON object. ([“JSON GENERATE statement” on page 369](#) and [“JSON PARSE statement” on page 382](#))
 - PH59733: You can generate and parse JSON null values by using the JSON GENERATE and JSON PARSE statements. ([“JSON GENERATE statement” on page 369](#) and [“JSON PARSE statement” on page 382](#))
 - PH62425: When compiling and running programs containing COBOL code with the ALLOCATE statement, where *data-name-2* references a USAGE POINTER-32 data item, and the LP(64) compiler option in effect, the compiler will issue the diagnostic message IGYPA3441-S when LOC 64 is specified, or allocate storage below the 2 GB bar when LOC 64 is not specified. ([“ALLOCATE statement” on page 313](#))

Enterprise COBOL for z/OS 6.4

Improved COBOL/Java interoperability

You can use JAVA-CALLABLE and JAVA-SHAREABLE directives to make your COBOL applications read/write accessible for Java applications. The CALL statement is enhanced to enable the compiler to call a static Java method.

- [JAVA-CALLABLE](#)
- [JAVA-SHAREABLE](#)
- [“CALL statement” on page 319](#)

User-defined functions support

You can define your own functions by specifying a FUNCTION-ID paragraph in the IDENTIFICATION DIVISION and invoke them by using a reference to a function identifier. This is part of the 2002 COBOL Standard.

- [Part 3, “IDENTIFICATION DIVISION,” on page 99](#)
- [Chapter 20, “FUNCTION-ID paragraph,” on page 113](#)
- [“REPOSITORY paragraph” on page 132](#)
- [Chapter 27, “Procedure division structure,” on page 257](#)
 - [“The PROCEDURE DIVISION header” on page 258](#)

- [“USING phrase” on page 260](#)
- [“RETURNING phrase” on page 263](#)
- [CONTENT-OF](#)

PERFORM ... UNTIL EXIT support

You can specify EXIT in place of a condition in a PERFORM statement. If the UNTIL phrase with the EXIT reserved word is specified, execution proceeds exactly as if the same PERFORM statement were coded with *condition-1* specified, except that *condition-1* never evaluates as true. ([“PERFORM with UNTIL phrase” on page 422](#))

How to send your comments

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Part 1. COBOL language structure

Chapter 1. Characters

The most basic and indivisible unit of the COBOL language is the *character*. The basic character set includes the letters of the Latin alphabet, digits, and special characters.

In the COBOL language, individual characters are joined to form *character-strings* and *separators*. Character-strings and separators, then, are used to form the words, literals, phrases, clauses, statements, and sentences that form the language.

The basic characters used in forming character-strings and separators in source code are shown in [Table 1 on page 3](#).

For certain language elements, the basic character set is extended with the EBCDIC Double-Byte Character Set (DBCS).

DBCS characters can be used in forming user-defined words.

The content of alphanumeric literals, comment lines, and comment entries can include any of the characters in the computer's compile-time character set, and can include both single-byte and DBCS characters.

Runtime data can include any characters from the runtime character set of the computer. The runtime character set of the computer can include alphanumeric characters, DBCS characters, national characters, and UTF-8 characters. National characters are represented in UTF-16, a 16-bit encoding form of Unicode. UTF-8 characters are represented in UTF-8, a variable length encoding form of Unicode (1 to 4 bytes for each character).

When the NSYMBOL (NATIONAL) compiler option is in effect, literals identified by the opening delimiter N" or N' are national literals and can contain any single-byte or double-byte characters, or both, that are valid for the compile-time code page in effect (either the default code page or the code page specified for the CODEPAGE compiler option). Characters contained in national literals are represented as national characters at run time.

For details, see [“User-defined words with DBCS characters” on page 12](#), [“DBCS literals” on page 41](#), and [“National literals” on page 46](#).

Literals identified by the opening delimiter U" or U' are UTF-8 literals. They can contain any single-byte or double-byte characters, or both, that are valid for the compile-time code page in effect (either the default code page or the code page specified for the CODEPAGE compiler option). Characters contained in UTF-8 literals are represented as UTF-8 characters at run time.

Table 1. Basic COBOL character set. This table lists basic COBOL character set.			
Character	Meaning	Use	Example
	Space	Punctuation character	01 WS-A PIC X(10).
+	Plus sign	Arithmetic operator	COMPUTE WS-A = WS-B + WS-C.
		Editing character	01 WS-A PIC +9(3).

Table 1. Basic COBOL character set. This table lists basic COBOL character set. (continued)

Character	Meaning	Use	Example
-	Minus sign or hyphen	Arithmetic operator	COMPUTE WS-A = WS-B - WS-C.
		Editing character	01 WS-A PIC -9(3).
		Continuation character	01 WS- VAR PIC X(27) VALUE - 'THIS MULTI-LINE TEXT'.
		COBOL word element	01 WS-A PIC 9(3).
*	Asterisk	Arithmetic operator	COMPUTE WS-A = WS-B * WS-C.
		Editing character	01 WS-A PIC **9.
		Comment character	* THIS IS COMMENT LINE.
/	Forward slash or solidus	Arithmetic operator	COMPUTE WS-A = WS-B / WS-C.
		Editing character	01 WS-DATE PIC 99/99/99.
		Continuation character	/01 WS- VAR PIC X(27) VALUE / 'THIS MULTI-LINE TEXT'.
=	Equal sign	Assignment character	COMPUTE WS-A = WS-B / WS-C.
		Relation character	IF WS-A = 10

Table 1. Basic COBOL character set. This table lists basic COBOL character set. (continued)			
Character	Meaning	Use	Example
\$	Currency sign ¹	Editing character	01 WS-DATE PIC \$\$\$99.
,	Comma	Editing character	01 WS-DATE PIC 99,999.
		Punctuation character	MOVE 10 TO WS-A, WS-B.
;	Semicolon	Punctuation character	MOVE 10 TO WS-A; WS-B.
.	Decimal point or period	Editing character	01 WS-DATE PIC 99.999.
		Punctuation character	MOVE 10 TO WS-A, WS-B.
"	Quotation mark ²	Punctuation character	01 WS-VAR PIC X(5) VALUE "HELLO".
'	Apostrophe	Punctuation character	01 WS-VAR PIC X(5) VALUE 'HELLO'.
(	Left parenthesis	Punctuation character	IF (WS-A = 10) AND (WS-B = 5)
)	Right parenthesis	Punctuation character	IF (WS-A = 10) AND (WS-B = 5)
>	Greater than	Relation character	IF WS-A > 10
<	Less than	Relation character	IF WS-A < 10
:	Colon	Relation character	MOVE WS- VAR(1:10) TO WS-VAR1.
_	Underscore	User-defined word element	01 WS_VAR PIC X(10).
A - Z	Alphabet (uppercase)	Alphabetic characters	/
a - z	Alphabet (lowercase)	Alphabetic characters	/

Table 1. Basic COBOL character set. This table lists basic COBOL character set. <i>(continued)</i>			
Character	Meaning	Use	Example
0 - 9	Numeric characters	Numeric characters	/
1. The currency sign is the character with the value X'5B', regardless of the code page in effect. The assigned graphic character can be the dollar sign or a local currency sign. 2. The quotation mark is the character with the value X'7F'.			

Chapter 2. Character sets and code pages

A *character set* is a set of letters, numbers, special characters, and other elements used to represent information. The term *code page* refers to a coded character set.

A character set is independent of a coded representation. A *coded character set* is the coded representation of a set of characters, where each character is assigned a numerical position, called a *code point*, in the encoding scheme. ASCII and EBCDIC are examples of types of coded character sets. Each variation of ASCII or EBCDIC is a specific coded character set.

Each code page that IBM defines is identified by a *code page name*, for example IBM-1252, and a *coded character set identifier* (CCSID), for example 1252.

Enterprise COBOL provides the CODEPAGE compiler option for specifying a coded character set for use at compile time and run time for code-page-sensitive elements, such as:

- The encoding of literals in the source program
- The default encoding for data items described with USAGE DISPLAY or DISPLAY-1
- The default encoding for XML parsing and XML generation

Some COBOL operations can override the encoding established by the CODEPAGE compiler option, for example:

- The DISPLAY-OF and NATIONAL-OF intrinsic functions can specify a CCSID as *argument-2*.
- The XML PARSE and XML GENERATE statements can specify a code page in the ENCODING phrase.

For further details about the CODEPAGE compiler option, see *CODEPAGE* in the *Enterprise COBOL Programming Guide*.

If you do not specify a code page, the default is code page IBM-1140, CCSID 1140.

The encoding of national and UTF-8 data is not affected by the CODEPAGE compiler option. The encoding for national literals and data items described with usage NATIONAL is UTF-16BE (big endian), CCSID 1200. A reference to *UTF-16* in this document is a reference to UTF-16BE. The encoding for UTF-8 literals and data items described with usage UTF-8 is UTF-8, CCSID 1208.

Character encoding units

A *character encoding unit* (or *encoding unit*) is the unit of data that COBOL treats as a single character at run time. In this information, the terms *character* and *character position* refer to a single encoding unit.

The size of an encoding unit for data items and literals depends on the USAGE clause of the data item or the category of the literal as follows:

- For data items described with USAGE DISPLAY and for alphanumeric literals, an encoding unit is 1 byte, regardless of the code page used and regardless of the number of bytes used to represent a given *graphic character*.
- For data items described with USAGE DISPLAY-1 (DBCS data items) and for DBCS literals, an encoding unit is 2 bytes.
- For data items described with USAGE NATIONAL and for national literals, an encoding unit is 2 bytes.
- For data items described with USAGE UTF-8 and for UTF-8 literals, an encoding unit is 1 byte.

The relationship between a graphic character and an encoding unit depends on the type of code page used for the data item or literal. See the following types of runtime code pages:

- Single-byte EBCDIC
- EBCDIC DBCS
- Unicode UTF-16
- Unicode UTF-8

See the following sections for the details of each type of code page.

Also see the section *Specifying the encoding* in the *Enterprise COBOL Programming Guide*.

Single-byte code pages

You can use a single-byte EBCDIC code page in data items described with USAGE DISPLAY and in literals of category alphanumeric. An encoding unit is 1 byte and each graphic character is represented in 1 byte. For these data items and literals, you need not be concerned with encoding units.

EBCDIC DBCS code pages

USAGE DISPLAY

You can use a mixture of single-byte and double-byte EBCDIC characters in data items described with USAGE DISPLAY and in literals of category alphanumeric. Double-byte characters must be delimited by shift-out and shift-in characters. An encoding unit is 1 byte and the size of a graphic character is 1 byte or 2 bytes.

When alphanumeric data items or literals contain DBCS data, programmers are responsible for ensuring that operations do not unintentionally separate the multiple encoding units that form a graphic character. Care should be taken with reference modification, and truncation during moves should be avoided. The COBOL runtime system does not check for a split between the encoding units that form a graphic character or for the loss of shift-out or shift-in codes.

To avoid problems, you can convert alphanumeric literals and data items described with usage DISPLAY to national data (UTF-16) by moving the data items or literals to data items described with usage NATIONAL or by using the NATIONAL-OF intrinsic function. You can then perform operations on the national data with less concern for splitting graphic characters. You can convert the data back to USAGE DISPLAY by using the DISPLAY-OF intrinsic function.

USAGE DISPLAY-1

You can use double-byte characters of an EBCDIC DBCS code page in data items described with USAGE DISPLAY-1 and in literals of category DBCS. An encoding unit is 2 bytes and each graphic character is represented in a single 2-byte encoding unit. For these data items and literals, you need not be concerned with encoding units.

Unicode UTF-16

You can use UTF-16 in data items described with USAGE NATIONAL. National literals are stored as UTF-16 characters regardless of the code page used for the source program. An encoding unit for data items of usage NATIONAL and national literals is 2 bytes.

For most of the characters in UTF-16, a graphic character is one encoding unit. Characters converted to UTF-16 from an EBCDIC, ASCII, or EUC code page are represented in one UTF-16 encoding unit. Some of the other graphic characters in UTF-16 are represented by a *surrogate pair* or a *combining character sequence*. A surrogate pair consists of two encoding units (4 bytes). A combining character sequence consists of a base character and one or more *combining marks* or a sequence of one or more combining marks (4 bytes or more, in 2-byte increments). In data items of usage NATIONAL, each 2-byte encoding unit is treated as a character.

When national data contains surrogate pairs or combining character sequences, programmers are responsible for ensuring that operations on national characters do not unintentionally separate the multiple encoding units that form a graphic character. Care should be taken with reference modification, and truncation during moves should be avoided. The COBOL runtime system does not check for a split between the encoding units that form a graphic character.

Unicode UTF-8

You can use UTF-8 in data items described with the “U” symbol in their picture clause. These items are usage UTF-8. UTF-8 literals are stored as UTF-8 characters regardless of the code page used for the

source program. An encoding unit for data items of usage UTF-8 and UTF-8 literals is 1 byte. In data items of usage UTF-8, each Unicode code point represented in the data item is encoded using between 1 and 4 encoding units.

For most of the characters in UTF-8, a graphic character is 1 to 4 encoding units (i.e., a character typically corresponds to a single Unicode code point). Characters converted to UTF-8 from an EBCDIC, ASCII, or EUC code page are represented in 1 to 4 UTF-8 encoding units. However, some of the other graphic characters in UTF-8 are represented by a *combining character sequence*, which corresponds to a sequence of Unicode code points. In particular, a combining character sequence consists of a base character and one or more *combining marks* or a sequence of one or more combining marks, with each combining mark being represented by 1 to 4 encoding units.

When UTF-8 data contains combining character sequences, programmers are responsible for ensure that operations on UTF-8 characters do not unintentionally separate the multiple groupings of encoding units that form graphic characters. You must pay attention to the reference modification and avoid truncations during moves. The COBOL runtime system does not check for a split between the groups of encoding units that form a graphic character.

Chapter 3. Character-strings: COBOL words and literals

A *character-string* is a character or a sequence of contiguous characters that forms a COBOL word, a literal, a PICTURE character-string, or a comment-entry. A character-string is delimited by separators.

A *separator* is a string of contiguous characters used to delimit character strings. Separators are described in detail under [Chapter 4, “Separators,” on page 49](#).

Character strings and certain separators form *text words*. A text word is a character or a sequence of contiguous characters (possibly continued across lines) between character positions 8 and 72 inclusive in source text, library text, or pseudo-text. For more information about pseudo-text, see [“Pseudo-text” on page 61](#).

Source text, library text, and pseudo-text can be written in single-byte EBCDIC and, for some character-strings, DBCS. (The compiler cannot process source code written in ASCII or Unicode.)

You can use single-byte and double-byte character-strings to form the following items:

- COBOL words
- Literals
- Comment text

You can use only single-byte characters to form PICTURE character-strings.

COBOL words with single-byte characters

A COBOL *word* is a character-string that forms a user-defined word, a system-name, or a reserved word. The maximum size of a COBOL user-defined word is 30 bytes. The number of characters that can be specified depends on the code page indicated by the compile-time locale.

Except for arithmetic operators and relation characters, each character of a COBOL word is selected from the following set:

- Latin uppercase letters A through Z
- Latin lowercase letters a through z
- digits 0 through 9
- - (hyphen)
- _ (underscore)

The hyphen cannot appear as the first or last character in such words. The underscore cannot appear as the first character in such words. Most user-defined words (all except section-names, paragraph-names, priority-numbers, and level-numbers) must contain at least one alphabetic character. Priority numbers and level numbers need not be unique; a given specification of a priority-number or level-number can be identical to any other priority-number or level-number.

In COBOL words (but not in the content of alphanumeric, DBCS, national, and UTF-8 literals), each lowercase single-byte alphabetic letter is considered to be equivalent to its corresponding single-byte uppercase alphabetic letter.

The following rules apply for all COBOL words:

- A reserved word cannot be used as a user-defined word or as a system-name.
- The same COBOL word, however, can be used as both a user-defined word and as a system-name. The classification of a specific occurrence of a COBOL word is determined by the context of the clause or phrase in which it occurs.

User-defined words with DBCS characters

There are the rules for forming user-defined words with DBCS characters.

The rules are:

Contained characters

DBCS user-defined words can contain only double-byte characters, and must contain at least one DBCS character that is not in the set A through Z, a through z, 0 through 9, hyphen, and underscore (DBCS representation of these characters has X'42' in the first byte).

DBCS user-defined words can contain characters that correspond to single-byte EBCDIC characters and those that do not correspond to single-byte EBCDIC characters. DBCS characters that correspond to single-byte EBCDIC characters follow the normal rules for COBOL user-defined words; that is, the characters A - Z, a - z, 0 - 9, the hyphen (-), and the underscore (_) are allowed. The hyphen cannot appear as the first or last character. The underscore cannot appear as the first character. Any of the DBCS characters that have no corresponding single-byte EBCDIC character can be used in DBCS user-defined words.

Uppercase and lowercase letters

In COBOL words, each lowercase single-byte encoded character "a" through "z" is considered to be equivalent to its corresponding single-byte encoded uppercase character. DBCS-encoded uppercase and lowercase letters are not equivalent.

Value range

DBCS user-defined words can contain characters whose values range from X'41' to X'FE' for both bytes.

Maximum length

14 characters

Continuation

Words formed with DBCS characters cannot be continued across lines.

Use of shift-out and shift-in characters

DBCS user-defined words begin with a shift-out character and end with a shift-in character.

User-defined words

A user-defined word is a COBOL word that must be supplied by the user to satisfy the format of a clause or statement.

The following sets of user-defined words are supported. The second column indicates whether DBCS characters are allowed in words of a given set.

User-defined word	DBCS characters allowed?
Alphabet-name	Yes
Class-name (of data)	Yes
Condition-name	Yes
Data-name	Yes
File-name	Yes
Index-name	Yes
Level-numbers: 01–49, 66, 77, 88	No
Library-name	No

User-defined word	DBCS characters allowed?
Mnemonic-name	Yes
Object-oriented class-name	No
Paragraph-name	Yes
Priority-numbers: 00–99	No
Program-name	No
Record-name	Yes
Section-name	Yes
Symbolic-character	Yes
Text-name	No
XML-schema-name	Yes

The maximum length of a user-defined word is 30 bytes, except for level-numbers and priority-numbers. Level-numbers and priority numbers must each be a one-digit or two-digit integer.

A given user-defined word can belong to only one of these sets, except that a given number can be both a priority-number and a level-number. Each user-defined word within a set must be unique, except for priority-numbers and level-numbers and except as specified in [Chapter 8, “Referencing data names, copy libraries, and PROCEDURE DIVISION names,” on page 67](#).

The following types of user-defined words can be referenced by statements and entries in the program in which the user-defined word is declared:

- Paragraph-name
- Section-name

The following types of user-defined words can be referenced by any COBOL program, provided that the compiling system supports the associated library or other system and that the entities referenced are known to that system:

- Library-name
- Text-name

The following types of names, when they are declared within a configuration section, can be referenced by statements and entries in the program that contains the configuration section or in any program contained within that program:

- Alphabet-name
- Class-name
- Condition-name
- Mnemonic-name
- Symbolic-character
- XML-schema-name

The function of each user-defined word is described in the clause or statement in which it appears.

Related references

[Appendix H, “Context-sensitive words,” on page 791](#)

System-names

A *system-name* is a character string that has a specific meaning to the system.

There are three types of system-names:

- Computer-name
- Language-name
- Implementor-name

There are four types of implementer-names:

- Environment-name
- External-class-name
- External-fileid
- Assignment-name

The meaning of each system-name is described with the format in which it appears.

Computer-name can be written in DBCS characters, but the other system-names cannot.

Function-names

A *function-name* specifies the mechanism provided to determine the value of an intrinsic function or a user-defined function.

The same word, in a different context, can appear in a program as a user-defined word or a system name. For a list of intrinsic function names and their definitions, see [Table 63 on page 515](#).

Intrinsic function names may be used as user-defined function-names, except for LENGTH, RANDOM, SIGN, SUM, and WHEN-COMPILED.

Reserved words

A *reserved word* is a character-string with a predefined meaning in a COBOL source unit.

Reserved words are listed in [Appendix G, “Reserved words,” on page 773](#). There are six types of reserved words:

- Keywords
- Optional words
- Figurative constants
- Special character words
- Special object identifiers
- Special registers

Keywords

Keywords are reserved words that are required within a given clause, entry, or statement. Within each format, such words appear in uppercase on the main path.

Optional words

Optional words are reserved words that can be included in the format of a clause, entry, or statement in order to improve readability. They have no effect on the execution of the program.

Figurative constants

See [“Figurative constants” on page 15](#).

Special character words

There are five types of *special character words*, which are recognized as special characters only when represented in single-byte characters:

- **Arithmetic operators:** + - / * **

See [“Arithmetic expressions” on page 266](#).

- **Relational operators:** < > = <= >=

See [“Conditional expressions” on page 268](#).

- **Floating comment indicators:** *>

See [“Floating comment indicators \(*>\)” on page 60](#).

- **Pseudo-text delimiters in COPY and REPLACE statements:** ==

See [“COPY statement” on page 696](#) and [“REPLACE statement” on page 708](#).

- **Compiler directive indicators:** >>

See [Chapter 115, “Compiler directives,” on page 717](#).

Special object identifiers

COBOL provides two special object identifiers, SELF and SUPER:

SELF

A special object identifier that you can use in the PROCEDURE DIVISION of a method. SELF refers to the object instance used to invoke the currently executing method. You can specify SELF only in places that are explicitly listed in the syntax diagrams.

SUPER

A special object identifier that you can use in the PROCEDURE DIVISION of a method only as the object identifier in an INVOKE statement. When used in this way, SUPER refers to the object instance used to invoke the currently executing method. The resolution of the method to be invoked ignores any methods declared in the class definition of the currently executing method and methods defined in any class derived from that class. Thus, the method invoked is inherited from an ancestor class.

Special registers

See [“Special registers” on page 17](#).

Figurative constants

Figurative constants are reserved words that name and refer to specific constant values. The reserved words for figurative constants and their meanings are listed in this section.

ZERO, ZEROS, ZEROES

Represents the numeric value zero (0) or one or more occurrences of the character zero, depending on context.

When the figurative constant ZERO, ZEROS, or ZEROES is used in a context that requires an alphanumeric character, an alphanumeric character zero is used. When the context requires a national character zero, a national character zero is used (value NX'0030'). When the context cannot be determined, an alphanumeric character zero is used.

SPACE, SPACES

Represents one or more blanks or spaces. SPACE is treated as an alphanumeric literal when used in a context that requires an alphanumeric character, as a DBCS literal when used in a context that requires a DBCS character, as a national literal when used in a context that requires a national character.

The EBCDIC DBCS space character has the value X'4040', the national space character has the value NX'0020', and the UTF-8 space character has the value UX'20'.

HIGH-VALUE, HIGH-VALUES

Represents one or more occurrences of the character that has the highest ordinal position in the collating sequence used.

HIGH-VALUE is treated as an alphanumeric literal in a context that requires an alphanumeric character. For alphanumeric data with the EBCDIC collating sequence, the value is X'FF'. For other alphanumeric data, the value depends on the collating sequence in effect.

HIGH-VALUE is treated as a national literal when used in a context that requires a national literal. The value is national character NX'FFFF'.

HIGH-VALUE is treated as a UTF-8 literal when used in a context that requires a UTF-8 literal. The value is UTF-8 character UX'F48FBFBF' corresponding to Unicode code point U+10FFFF, except when HIGH-VALUE is used in a move or compare operation with a fixed byte-length UTF-8 data item that has a length that is not a multiple of 4 bytes. In that case, when HIGH-VALUE is moved into or compared against the final 3, 2 or 1 byte(s) of the UTF-8 data item, the value of HIGH-VALUE is UX'EFBFBF' (U+FFFF), UX'DFBF' (U+07FF), and UX'7F' (U+007F), respectively.

When the context cannot be determined, an alphanumeric context is assumed and the value X'FF' is used.

Usage note: You should not use HIGH-VALUE (or a value assigned from HIGH-VALUE) in a way that results in conversion between one data representation and another. X'FF' does not represent a valid EBCDIC character, and NX'FFFF' does not represent a valid national character. Conversion of either the alphanumeric or the national HIGH-VALUE representation to another representation results in a substitution character. For example, conversion of X'FF' to UTF-16 would give a substitution character, not NX'FFFF'.

LOW-VALUE, LOW-VALUES

Represents one or more occurrences of the character that has the lowest ordinal position in the collating sequence used.

LOW-VALUE is treated as an alphanumeric literal in a context that requires an alphanumeric character. For alphanumeric data with the EBCDIC collating sequence, the value is X'00'. For other alphanumeric data, the value depends on the collating sequence in effect.

LOW-VALUE is treated as a national literal when used in a context that requires a national literal. The value is national character NX'0000'.

LOW-VALUE is treated as a UTF-8 literal when used in a context that requires a UTF-8 literal. The value is UTF-8 character UX'00' corresponding to Unicode code point U+0000.

When the context cannot be determined, an alphanumeric context is assumed and the value X'00' is used.

QUOTE, QUOTES

Represents one or more occurrences of:

- The quotation mark character ("), if the QUOTE compiler option is in effect
- The apostrophe character ('), if the APOST compiler option is in effect

QUOTE or QUOTES represents an alphanumeric character when used in a context that requires an alphanumeric character, represents a national character when used in a context that requires a national character, and represents a UTF-8 character when used in a context that requires a UTF-8 character. The national character value of quotation mark is NX'0022'. The national character value of apostrophe is NX'0027'. The UTF-8 character value of quotation mark is UX'22'. The UTF-8 character value of apostrophe is UX'27'.

QUOTE and QUOTES cannot be used in place of a quotation mark or an apostrophe to enclose an alphanumeric literal, a DBCS literal, a national literal, or a UTF-8 literal.

ALL literal

literal can be an alphanumeric literal, a DBCS literal, a national literal, a UTF-8 literal, or a figurative constant other than the ALL literal.

When *literal* is not a figurative constant, ALL *literal* represents one or more occurrences of the string of characters that compose the literal.

When *literal* is a figurative constant, the word ALL has no meaning and is used only for readability.

The figurative constant ALL *literal* must not be used with the CALL, INSPECT, INVOKE, STOP, or STRING statements.

symbolic-character

Represents one or more of the characters specified as a value of the *symbolic-character* in the SYMBOLIC CHARACTERS clause of the SPECIAL-NAMES paragraph.

symbolic-character always represents an alphanumeric character; it can be used in a context that requires a national or UTF-8 character only when implicit conversion of alphanumeric to national or UTF-8 characters is defined. (It can be used, for example, in a MOVE statement where the receiving item is of class national or UTF-8 because implicit conversion is defined when the sending item is alphanumeric and the receiving item is national or UTF-8.)

NULL, NULLS

Represents a value used to indicate that data items defined with USAGE POINTER, USAGE PROCEDURE-POINTER, USAGE FUNCTION-POINTER, USAGE OBJECT REFERENCE, or the ADDRESS OF special register do not contain a valid address. NULL can be used only where explicitly allowed in the syntax formats. NULL has the value zero.

The singular and plural forms of NULL, ZERO, SPACE, HIGH-VALUE, LOW-VALUE, and QUOTE can be used interchangeably. For example, if DATA-NAME-1 is a five-character data item, each of the following statements moves five spaces to DATA-NAME-1:

```
MOVE SPACE      TO DATA-NAME-1
MOVE SPACES     TO DATA-NAME-1
MOVE ALL SPACES TO DATA-NAME-1
```

When the rules of COBOL permit any one spelling of a figurative constant name, any alternative spelling of that figurative constant name can be specified.

You can use a figurative constant wherever *literal* appears in a syntax diagram, except where explicitly prohibited. When a numeric literal appears in a syntax diagram, only the figurative constant ZERO (or ZEROS or ZEROES) can be used. Figurative constants are not allowed as function arguments except in an arithmetic expression, where the expression is an argument to a function.

The length of a figurative constant depends on the context of its use. The following rules apply:

- When a figurative constant is specified in a VALUE clause or associated with a data item (for example, when it is moved to or compared with another item), the length of the figurative constant character-string is equal to 1 or the number of character positions in the associated data item, whichever is greater.
- When a figurative constant, other than the ALL literal, is not associated with another data item (for example, in a CALL, INVOKE, STOP, STRING, or UNSTRING statement), the length of the character-string is one character.

Special registers

Special registers are reserved words that name storage areas generated by the compiler. Their primary use is to store information produced through specific COBOL features. Each such storage area has a fixed name, and must not be defined within the program.

For programs with the RECURSIVE attribute, for programs compiled with the THREAD option, and for methods, storage for the following special registers is allocated on a per-invocation basis:

- ADDRESS OF
- JSON-CODE
- JSON-STATUS
- RETURN-CODE
- SORT-CONTROL
- SORT-CORE-SIZE
- SORT-FILE-SIZE
- SORT-MESSAGE
- SORT-MODE-SIZE
- SORT-RETURN

- TALLY
- XML-CODE
- XML-EVENT
- XML-INFORMATION
- XML-NAMESPACE
- XML-NAMESPACE-PREFIX
- XML-NNAMESPACE
- XML-NNAMESPACE-PREFIX
- XML-NTEXT
- XML-TEXT

For the first call to a program after a cancel of that program, or for a method invocation, the compiler initializes the special register fields to their initial values.

For the following four cases:

- Programs that have the INITIAL clause specified
- Programs that have the RECURSIVE clause specified
- Programs compiled with the THREAD option
- Methods

the following special registers are reset to their initial value on each program or method entry:

- IGY-JAVAIOP-CALL-EXCEPTION
- JSON-CODE
- JSON-STATUS
- RETURN-CODE
- SORT-CONTROL
- SORT-CORE-SIZE
- SORT-FILE-SIZE
- SORT-MESSAGE
- SORT-MODE-SIZE
- SORT-RETURN
- TALLY
- XML-CODE
- XML-EVENT

Further, in the above four cases, values set in ADDRESS OF special registers persist only for the span of the particular program or method invocation.

In all other cases, the special registers will not be reset; they will be unchanged from the value contained on the previous CALL or INVOKE.

Unless otherwise explicitly restricted, a special register can be used wherever a data-name or identifier that has the same definition as the implicit definition of the special register can be used. Implicit definitions, if applicable, are given in the specification of each special register.

You can specify an alphanumeric special register in a function wherever an alphanumeric argument to a function is allowed, unless specifically prohibited.

If qualification is allowed, special registers can be qualified as necessary to provide uniqueness. (For more information, see [“Qualification”](#) on page 67.)

ADDRESS OF

The ADDRESS OF special register references the address of a data item in the LINKAGE SECTION, the LOCAL-STORAGE SECTION, or the WORKING-STORAGE SECTION.

For 01 and 77 level items in the LINKAGE SECTION, the ADDRESS OF special register can be used as either a sending item or a receiving item. For all other operands, the ADDRESS OF special register can be used only as a sending item.

The ADDRESS OF special register is implicitly defined as USAGE POINTER.

If LP(32) is in effect, 4 bytes are allocated for the special register; if LP(64) is in effect, 8 bytes are allocated for the special register.

A function-identifier is not allowed as the operand of the ADDRESS OF special register.

DEBUG-ITEM

The DEBUG-ITEM special register provides information for a debugging declarative procedure about the conditions that cause debugging section execution.

DEBUG-ITEM has the following implicit description:

```
01  DEBUG-ITEM.  
   02  DEBUG-LINE      PICTURE IS X(6).  
   02  FILLER          PICTURE IS X VALUE SPACE.  
   02  DEBUG-NAME      PICTURE IS X(30).  
   02  FILLER          PICTURE IS X VALUE SPACE.  
   02  DEBUG-SUB-1     PICTURE IS S9999 SIGN IS LEADING SEPARATE CHARACTER.  
   02  FILLER          PICTURE IS X VALUE SPACE.  
   02  DEBUG-SUB-2     PICTURE IS S9999 SIGN IS LEADING SEPARATE CHARACTER.  
   02  FILLER          PICTURE IS X VALUE SPACE.  
   02  DEBUG-SUB-3     PICTURE IS S9999 SIGN IS LEADING SEPARATE CHARACTER.  
   02  FILLER          PICTURE IS X VALUE SPACE.  
   02  DEBUG-CONTENTS PICTURE IS X(n).
```

Before each debugging section is executed, DEBUG-ITEM is filled with spaces. The contents of the DEBUG-ITEM subfields are updated according to the rules for the MOVE statement, with one exception: DEBUG-CONTENTS is updated as if the move were an alphanumeric-to-alphanumeric elementary move without conversion of data from one form of internal representation to another.

After updating, the contents of the DEBUG-ITEM subfields are:

DEBUG-LINE

The source-statement sequence number (or the compiler-generated sequence number, depending on the compiler option chosen) that caused execution of the debugging section.

DEBUG-NAME

The first 30 characters of the name that caused execution of the debugging section. Any qualifiers are separated by the word 'OF'.

DEBUG-SUB-1, DEBUG-SUB-2, DEBUG-SUB-3

Always set to spaces. These subfields are documented for compatibility with previous COBOL products.

DEBUG-CONTENTS

Data is moved into DEBUG-CONTENTS, as shown in the following table.

<i>Table 2. DEBUG-ITEM subfield contents</i>			
Cause of debugging section execution	Statement referred to in DEBUG-LINE	Contents of DEBUG-NAME	Contents of DEBUG-CONTENTS
<i>procedure-name-1</i> ALTER reference	ALTER statement	<i>procedure-name-1</i>	<i>procedure-name-n</i> in TO PROCEED TO phrase
GO TO <i>procedure-name-n</i>	GO TO statement	<i>procedure-name-n</i>	Spaces
<i>procedure-name-n</i> in SORT or MERGE input/output procedure	SORT or MERGE statement	<i>procedure-name-n</i>	"SORT INPUT", "SORT OUTPUT", or "MERGE OUTPUT" (as applicable)
PERFORM statement transfer of control	This PERFORM statement	<i>procedure-name-n</i>	"PERFORM LOOP"
<i>procedure-name-n</i> in a USE procedure	Statement causing USE procedure execution	<i>procedure-name-n</i>	"USE PROCEDURE"
Implicit transfer from a previous sequential procedure	Previous statement executed in previous sequential procedure ¹	<i>procedure-name-n</i>	"FALL THROUGH"
First execution of first nondeclarative procedure	Line number of first nondeclarative procedure-name	Name of first nondeclarative procedure	"START PROGRAM"
1. If this procedure is preceded by a section header, and control is passed through the section header, the statement number refers to the section header.			

IGY-JAVAIOP-CALL-EXCEPTION

The IGY-JAVAIOP-CALL-EXCEPTION special register provides access to the Java exception object produced when an exception is encountered during a call from COBOL to a static Java method using the CALL statement.

When a call to a static Java method is issued from a COBOL program using a CALL statement with a literal target of the form 'Java.java-class-name.java-static-method-name' and a Java exception occurs during the call, the IGY-JAVAIOP-CALL-EXCEPTION special register can be accessed from COBOL statements specified in the ON EXCEPTION phrase of the CALL statement to refer to the Java exception object of the associated Java exception. Having access to the object reference for this Java exception object allows user exception handling code in COBOL to use JNI functions to perform various tasks during exception handling, such as re-throwing the exception. However, it is not necessary to reference this special register in such exception handling code. Knowing that an exception occurred and performing the appropriate COBOL error handling is often sufficient for an interoperable application.

Notes:

- When an exception occurs in a call to a static Java method using the CALL statement, the exception is cleared before the code specified in the ON EXCEPTION phrase is run. To propagate the exception up the chain of callers so that other callers can provide caller-specific handling for the exception, it is possible to re-throw the exception using the JNI Throw() function.
- Whenever a COBOL program makes calls to JNI functions, the program should either be compiled with the DLL option in effect, or the JNI calls must be under the scope of a CALLINTERFACE DLL directive. It is not necessary to link the program as a DLL, however. Furthermore, at runtime it is necessary to include the directory containing the DLL file libjvm.dll in your LIBPATH environment variable. When the JAVAIOP(...,JAVA64,...) option is in effect, the directory containing the libjvm31.dll must be added to the front of your LIBPATH.
- Finally, you will need to link your application with side deck file igzcnj2.x (for pure AMODE 31 applications), igzxjni2.x (for mixed AMODE 31 COBOL with AMODE 64 Java applications), or igzqjni2.x

(for pure AMODE 64 applications). These side deck files reside in the lib subdirectory of the COBOL install directory in the z/OS UNIX file system. The side deck files can also be found as members in the relevant SCEELIB run time data set.

See the following example:

```
DATA DIVISION.
LINKAGE SECTION.
COPY JNI SUPPRESS.
:
:
PROCEDURE DIVISION.
:
:   set address of JNIEnv to JNIEnvPtr
:   set address of JNINativeInterface to JNIEnv
:
:   CALL 'Java.com.acme.MyClass.myMethod' USING ...
:   ON EXCEPTION
:       <user-error-handling-code>
:       *> re-throw exception
:       >>CALLINTERFACE DLL
:       CALL THROW USING BY VALUE JNIEnvPtr
:                               IGY-JAVAIOP-CALL-EXCEPTION
:   GOBACK
:   END-CALL
```

When the LP(32) compiler option is in effect, IGY-JAVAIOP-CALL-EXCEPTION has the implicit definition:

```
01 IGY-JAVAIOP-CALL-EXCEPTION PIC 9(9) USAGE COMP-5 VALUE 0.
```

When the LP(64) compiler option is in effect or the JAVAIOP(...,JAVA64,...) compiler option is in effect, IGY-JAVAIOP-CALL-EXCEPTION has the implicit definition:

```
01 IGY-JAVAIOP-CALL-EXCEPTION PIC 9(18) USAGE COMP-5 VALUE 0.
```

The value of IGY-JAVAIOP-CALL-EXCEPTION is reset to 0 at the beginning of every program invocation.

JNIENVPTR

The JNIENVPTR special register references the Java Native Interface (JNI) environment pointer. The JNI environment pointer is used in calling Java callable services.

JNIENVPTR is implicitly defined as USAGE POINTER, and cannot be specified as a receiving data item.

The JNIENVPTR special register cannot be used with AMODE 64 Java.

For information about using JNIENVPTR and JNI callable services, see *Accessing JNI services* in the *Enterprise COBOL Programming Guide*.

JSON-CODE

The JSON-CODE special register is used to indicate either that a JSON GENERATE or JSON PARSE statement executed successfully or that an exception occurred during JSON generation or parsing.

The JSON-CODE special register has the implicit definition:

```
01 JSON-CODE PICTURE S9(9) USAGE BINARY VALUE 0.
```

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

At termination of a JSON GENERATE or JSON PARSE statement, JSON-CODE contains either zero, indicating successful completion of JSON generation or parsing, or a nonzero error code, indicating that an exception occurred during JSON generation or parsing. JSON GENERATE and JSON PARSE exception codes are detailed in *JSON GENERATE exceptions* and *JSON PARSE conditions and associated codes and runtime messages* in the *Enterprise COBOL Programming Guide*.

Related references

[“JSON-STATUS” on page 22](#)

JSON-STATUS

The JSON-STATUS special register is used to indicate either that a JSON PARSE statement executed successfully or that a nonexception condition occurred during the JSON parse operation.

The JSON-STATUS special register has the implicit definition:

```
01 JSON-STATUS PICTURE S9(9) USAGE BINARY VALUE 0.
```

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

During execution of a JSON PARSE statement, nonexception conditions result in reason codes in the JSON-STATUS special register, but do not terminate execution of the statement. At termination of a JSON PARSE statement, JSON-STATUS contains either zero, indicating successful completion of the JSON parse operation, or a nonzero status value, representing one or more nonexception conditions that occurred prior to the exception condition. JSON-STATUS reason codes are detailed in *Nonexception conditions and corresponding values of JSON-STATUS* in the *Enterprise COBOL Programming Guide*.

Related references

[“JSON-CODE” on page 21](#)

LENGTH OF

The LENGTH OF special register contains the number of bytes used by a data item.

LENGTH OF creates an implicit special register that contains the current byte length of the data item referenced by the identifier.

For example:

```
IDENTIFICATION DIVISION.  
PROGRAM-ID. LengthOfExample.  
  
DATA DIVISION.  
WORKING-STORAGE SECTION.  
01 MyString PIC X(20) VALUE 'Hello, COBOL!'.  
01 StringLength PIC 9(3).  
  
PROCEDURE DIVISION.  
    MOVE LENGTH OF MyString TO StringLength.  
    DISPLAY 'Length of MyString is ' StringLength.  
    STOP RUN.
```

The output is:

```
Length of MyString is 020
```

For data items described with usage DISPLAY-1 (DBCS data items) and data items described with usage NATIONAL, each character occupies 2 bytes of storage. For data items described with usage UTF-8, each character occupies between 1 and 4 bytes.

LENGTH OF can be used in the PROCEDURE DIVISION anywhere a numeric data item that has the same definition as the implied definition of the LENGTH OF special register can be used.

If LP(32) is in effect, the LENGTH OF special register has the implicit definition:

```
PICTURE 9(9) USAGE IS BINARY.
```

If LP(64) is in effect, the LENGTH OF special register has the implicit definition:

```
PICTURE 9(18) USAGE IS BINARY.
```

If the data item referenced by the identifier contains the GLOBAL clause, the LENGTH OF special register is a global data item.

The LENGTH OF special register can appear within either the starting character position or the length expressions of a reference-modification specification. However, the LENGTH OF special register cannot be applied to any operand that is reference-modified.

The LENGTH OF operand cannot be a function, but the LENGTH OF special register is allowed in a function where an integer argument is allowed.

If the LENGTH OF special register is used as the argument to the LENGTH function, the result is 4 when LP(32) is in effect, and the result is 8 when LP(64) is in effect.

If the ADDRESS OF special register is used as the argument to the LENGTH function, the result is 4 when LP(32) is in effect, and the result is 8 when LP(64) is in effect.

LENGTH OF cannot be either of the following items:

- A receiving data item
- A subscript

When the LENGTH OF special register is used as a parameter on a CALL statement, it must be passed BY CONTENT or BY VALUE.

When a table element is specified, the LENGTH OF special register contains the length in bytes of one occurrence. When referring to a table element, the element name need not be subscripted.

A value is returned for any identifier whose length can be determined, even if the area referenced by the identifier is currently not available to the program.

A separate LENGTH OF special register exists for each identifier referenced with the LENGTH OF phrase. For example:

```
MOVE LENGTH OF A TO B
DISPLAY LENGTH OF A, A
ADD LENGTH OF A TO B
CALL "PROGX" USING BY REFERENCE A BY CONTENT LENGTH OF A
```

The intrinsic function LENGTH can also be used to obtain the length of a data item. For data items of usage NATIONAL, the length returned by the LENGTH function is the number of national character positions. For data items of usage UTF-8, the length returned by the LENGTH function is the number of UTF-8 character positions. Thus, the LENGTH OF special register and the LENGTH intrinsic function have different results for data items of usage NATIONAL and UTF-8. Also, for table elements, the intrinsic function LENGTH requires a subscript, while the LENGTH OF special register does not. For all other data items, the result is the same.

The LENGTH intrinsic function, when applied to a null-terminated alphanumeric literal, returns the number of bytes in the literal prior to but not including the terminating null. (The LENGTH special register does not support literal operands.) For details about null-terminated alphanumeric literals, see [“Null-terminated alphanumeric literals” on page 41](#).

LINAGE-COUNTER

A separate LINAGE-COUNTER special register is generated for each FD entry that contains a LINAGE clause. When more than one is generated, you must qualify each reference to a LINAGE-COUNTER with its related file-name.

The implicit description of the LINAGE-COUNTER special register is in one of the following cases:

- If the LINAGE clause specifies a data-name, LINAGE-COUNTER has the same PICTURE and USAGE as that data-name.
- If the LINAGE clause specifies an integer, LINAGE-COUNTER is a binary item with the same number of digits as that integer.

For more information, see [“LINAGE clause” on page 189](#).

The value in LINAGE-COUNTER at any given time is the line number at which the device is positioned within the current page. LINAGE-COUNTER can be referred to in PROCEDURE DIVISION statements; it must not be modified by them.

LINAGE-COUNTER is initialized to 1 when an OPEN statement for its associated file is executed.

LINAGE-COUNTER is automatically modified by any WRITE statement for this file. (See [“WRITE statement” on page 476](#).)

If the file description entry for a sequential file contains the LINAGE clause and the EXTERNAL clause, the LINAGE-COUNTER data item is an external data item. If the file description entry for a sequential file contains the LINAGE clause and the GLOBAL clause, the LINAGE-COUNTER data item is a global data item.

You can specify the LINAGE-COUNTER special register wherever an integer argument to a function is allowed.

RETURN-CODE

The RETURN-CODE special register can be used to pass a return code to the calling program or operating system when the current COBOL program ends.

When a COBOL program ends:

- If control returns to the operating system, the value of the RETURN-CODE special register is passed to the operating system as a user return code. The supported user return code values are determined by the operating system, and might not include the full range of RETURN-CODE special register values.
- If control returns to a calling program, the value of the RETURN-CODE special register is passed to the calling program. If the calling program is a COBOL program, the RETURN-CODE special register in the calling program is set to the value of the RETURN-CODE special register in the called program.

The RETURN-CODE special register has the implicit definition:

```
01 RETURN-CODE GLOBAL PICTURE S9(4) USAGE BINARY VALUE ZERO.
```

When used in nested programs, this special register is implicitly defined with the GLOBAL clause in the outermost program.

The following examples show how to set the RETURN-CODE special register:

- `COMPUTE RETURN-CODE = 8.`
- `MOVE 8 to RETURN-CODE.`

The RETURN-CODE special register does not return a value from an invoked method or from a program that uses CALL ... RETURNING. For more information, see [“INVOKE statement” on page 362](#) or [“CALL statement” on page 319](#).

You can specify the RETURN-CODE special register in a function wherever an integer argument is allowed.

The RETURN-CODE special register does not return information from a service call for a Language Environment® callable service. For more information, see *Using Language Environment callable services* in the *Enterprise COBOL Programming Guide* and the *Language Environment Programming Guide*.

SHIFT-OUT and SHIFT-IN

You can specify the SHIFT-OUT and SHIFT-IN special registers in a function wherever an alphanumeric argument is allowed.

The SHIFT-OUT and SHIFT-IN special registers are implicitly defined as alphanumeric data items of the format:

```
01 SHIFT-OUT GLOBAL PICTURE X(1) USAGE DISPLAY VALUE X"0E".
01 SHIFT-IN  GLOBAL PICTURE X(1) USAGE DISPLAY VALUE X"0F".
```

When used in nested programs, these special registers are implicitly defined with the global attribute in the outermost program.

These special registers represent EBCDIC shift-out and shift-in control characters, which are unprintable characters.

These special registers cannot be receiving items. SHIFT-OUT and SHIFT-IN cannot be used in place of the keyboard control characters when you are defining DBCS user-defined words or specifying EBCDIC DBCS literals.

The following example shows how SHIFT-OUT and SHIFT-IN might be used:

```
DATA DIVISION.
WORKING-STORAGE.
01 DBCSGRP.
   05 SO      PIC X.
   05 DBCSITEM PIC G(3) USAGE DISPLAY-1.
   05 SI      PIC X.

PROCEDURE DIVISION.
  MOVE SHIFT-OUT TO SO
  MOVE G"<D1D2D3>" TO DBCSITEM
  MOVE SHIFT-IN TO SI
  DISPLAY DBCSGRP
```

SORT-CONTROL

The SORT-CONTROL special register is the name of an alphanumeric data item.

Restriction: The SORT-CONTROL special register is not applicable to sorting a table with the format 2 SORT statement.

The SORT-CONTROL special register has the implicit definition:

```
01 SORT-CONTROL GLOBAL PICTURE X(8) USAGE DISPLAY VALUE "IGZSRTCD".
```

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

This register contains the ddname of the data set that holds the control statements used to improve the performance of a sorting or merging operation.

You can provide a DD statement for the data set identified by the SORT-CONTROL special register. Enterprise COBOL will attempt to open the data set at execution time. Any error will be diagnosed with an informational message.

You can specify the SORT-CONTROL special register in a function wherever an alphanumeric argument is allowed.

The SORT-CONTROL special register is not necessary for a successful sorting or merging operation.

The sort control file takes precedence over the SORT special registers.

SORT-CORE-SIZE

The SORT-CORE-SIZE special register is the name of a binary data item that you can use to specify the number of bytes of storage available to the sort utility.

Restriction: The SORT-CORE-SIZE special register is not applicable to sorting a table with the format 2 SORT statement.

The SORT-CORE-SIZE special register has the implicit definition:

```
01  SORT-CORE-SIZE GLOBAL PICTURE S9(8) USAGE BINARY VALUE ZERO.
```

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

SORT-CORE-SIZE can be used in place of the MAINSIZE or RESINV control statements in the sort control file:

- The 'MAINSIZE=' option control statement keyword is equivalent to SORT-CORE-SIZE with a positive value.
- The 'RESINV=' option control statement keyword is equivalent to SORT-CORE-SIZE with a negative value.
- The 'MAINSIZE=MAX' option control statement keyword is equivalent to SORT-CORE-SIZE with a value of +999999 or +99999999.

You can specify the SORT-CORE-SIZE special register in a function wherever an integer argument is allowed.

SORT-FILE-SIZE

The SORT-FILE-SIZE special register is the name of a binary data item that you can use to specify the estimated number of records in the sort input file, *file-name-1*.

Restriction: The SORT-FILE-SIZE special register is not applicable to sorting a table with the format 2 SORT statement.

The SORT-FILE-SIZE special register has the implicit definition:

```
01  SORT-FILE-SIZE GLOBAL PICTURE S9(8) USAGE BINARY VALUE ZERO.
```

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

SORT-FILE-SIZE is equivalent to the 'FILSZ=Ennn' control statement in the sort control file.

You can specify the SORT-FILE-SIZE special register in a function wherever an integer argument is allowed.

SORT-MESSAGE

The SORT-MESSAGE special register is the name of an alphanumeric data item that is available to both sort and merge programs.

Restriction: The SORT-MESSAGE special register is not applicable to sorting a table with the format 2 SORT statement.

The SORT-MESSAGE special register has the implicit definition:

```
01  SORT-MESSAGE GLOBAL PICTURE X(8) USAGE DISPLAY VALUE "SYSOUT".
```

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

You can use the SORT-MESSAGE special register to specify the ddname of a data set that the sort utility should use in place of the SYSOUT data set.

The ddname specified in SORT-MESSAGE is equivalent to the name specified on the 'MSGDDN=' control statement in the sort control file.

You can specify the SORT-MESSAGE special register in a function wherever an alphanumeric argument is allowed.

When the ddname that the sort utility uses is not allocated, the COBOL program using the sort can be terminated at execution time with a critical message, ICE158A SYSOUT DD STATEMENT MISSING, and a SOC2 abend.

SORT-MODE-SIZE

The SORT-MODE-SIZE special register is the name of a binary data item that you can use to specify the length of variable-length records that occur most frequently.

Restriction: The SORT-MODE-SIZE special register is not applicable to sorting a table with the format 2 SORT statement.

The SORT-MODE-SIZE special register has the implicit definition:

```
01  SORT-MODE-SIZE GLOBAL PICTURE S9(5) USAGE BINARY VALUE ZERO.
```

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

SORT-MODE-SIZE is equivalent to the 'SMS=' control statement in the sort control file.

You can specify the SORT-MODE-SIZE special register in a function wherever an integer argument is allowed.

SORT-RETURN

The SORT-RETURN special register is the name of a binary data item and is available to both sort and merge programs.

Restriction: The SORT-RETURN special register is not applicable to sorting a table with the format 2 SORT statement.

The SORT-RETURN special register has the implicit definition:

```
01  SORT-RETURN GLOBAL PICTURE S9(4) USAGE BINARY VALUE ZERO.
```

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

The SORT-RETURN special register contains a return code of 0 (successful) or 16 (unsuccessful) at the completion of a sort or merge operation. If the sort or merge is unsuccessful and there is no reference to this special register anywhere in the program, a message is displayed on the terminal.

You can set the SORT-RETURN special register to 16 in an error declarative or input/output procedure to terminate a sort or merge operation before all records are processed. The operation is terminated on the next input or output function for the sort or merge operation.

You can specify the SORT-RETURN special register in a function wherever an integer argument is allowed.

TALLY

The TALLY special register is the name of a binary data item.

See the following definition of a binary data item:

```
01 TALLY GLOBAL PICTURE 9(5) USAGE BINARY VALUE ZERO.
```

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

You can refer to or modify the contents of TALLY.

You can specify the TALLY special register in a function wherever an integer argument is allowed.

WHEN-COMPILED

The WHEN-COMPILED special register contains the date at the start of the compilation.

WHEN-COMPILED is an alphanumeric data item that has the implicit definition:

```
01 WHEN-COMPILED GLOBAL PICTURE X(16) USAGE DISPLAY.
```

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

The WHEN-COMPILED special register has the format:

```
MM/DD/YYhh.mm.ss (MONTH/DAY/YEARhour.minute.second)
```

For example, if compilation began at 2:04 PM on 15 October 2007, WHEN-COMPILED would contain the value 10/15/0714.04.00.

WHEN-COMPILED can be used only as the sending field in a MOVE statement.

WHEN-COMPILED special register data cannot be reference-modified.

The compilation date and time can also be accessed with the intrinsic function WHEN-COMPILED (see [Chapter 112, “WHEN-COMPILED,”](#) on page 687). That function supports four-digit year values and provides additional information.

XML-CODE

The XML-CODE special register is used to communicate status between the XML parser and the processing procedure that was identified in an XML PARSE statement, and to indicate either that an XML GENERATE statement executed successfully or that an exception occurred during XML generation.

The XML-CODE special register has the implicit definition:

```
01 XML-CODE PICTURE S9(9) USAGE BINARY VALUE 0.
```

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

When the XML parser encounters an XML event, it sets XML-CODE and then passes control to the processing procedure. For all events except an EXCEPTION event, XML-CODE contains zero when the processing procedure receives control.

For an EXCEPTION event, the parser sets XML-CODE to an exception code that indicates the nature of the exception. XML PARSE exception codes are discussed in *Handling XML PARSE exceptions* in the *Enterprise COBOL Programming Guide*.

For some XML events, you can set XML-CODE before returning to the parser to control subsequent processing of the document. For details, see *XML-CODE* in the *Enterprise COBOL Programming Guide*.

When the parser returns control to the XML PARSE statement, XML-CODE contains the most recent value set by the processing procedure or the parser. In some cases, the parser overrides the value set by the processing procedure.

At termination of an XML GENERATE statement, XML-CODE contains either zero, indicating successful completion of XML generation, or a nonzero error code, indicating that an exception occurred during XML generation. XML GENERATE exception codes are detailed in *XML GENERATE exceptions* in the *Enterprise COBOL Programming Guide*.

Related concepts

XML-CODE (*Enterprise COBOL Programming Guide*)

Related tasks

Handling XML PARSE exceptions (*Enterprise COBOL Programming Guide*)

Related references

XML GENERATE exceptions (*Enterprise COBOL Programming Guide*)

XML-EVENT

The XML-EVENT special register communicates event information from the XML parser to the processing procedure identified in the XML PARSE statement.

Before passing control to the processing procedure, the XML parser sets the XML-EVENT special register to the name of the XML event. The specific events and the associated special registers that are set depend on the setting of the XMLPARSE compiler option, XMLPARSE(XMLSS) or XMLPARSE(COMPAT).

The parser uses the following special registers when XMLPARSE(XMLSS) is in effect:

- XML-CODE
- XML-EVENT
- XML-TEXT or XML-NTEXT
- XML-NAMESPACE or XML-NNAMESPACE
- XML-NAMESPACE-PREFIX or XML-NNAMESPACE-PREFIX

The parser uses the following special registers when XMLPARSE(COMPAT) is in effect:

- XML-CODE
- XML-EVENT
- XML-TEXT or XML-NTEXT

The parser sets XML-NTEXT to associated XML text when the XML document is in a national data item, and sets XML-TEXT when the XML document is in an alphanumeric data item. When the XMLPARSE(COMPAT) compiler option is in effect, the parser sets XML-NTEXT to the text of any numeric character reference (for events ATTRIBUTE-NATIONAL-CHARACTER and CONTENT-NATIONAL-CHARACTER) regardless of the type of the XML document data item.

When the XMLPARSE(XMLSS) compiler option is in effect, the parser sets XML-NNAMESPACE and XML-NNAMESPACE-PREFIX when the XML document is in a national data item and when the RETURNING NATIONAL phrase is specified in the XML PARSE statement; otherwise, the parser sets XML-NAMESPACE and XML-NAMESPACE-PREFIX.

Table 3 on page 30 shows XML events and special register contents for parsing with the XMLPARSE(XMLSS) and XMLPARSE(COMPAT) options.

XML-EVENT has the implicit definition:

```
01 XML-EVENT USAGE DISPLAY PICTURE X(30) VALUE SPACE.
```

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

XML-EVENT cannot be used as a receiving data item.

Table 3. <i>XML events and associated special register contents</i>		
XML-EVENT	XMLPARSE(XMLSS) ¹	XMLPARSE(COMPAT) ¹
ATTRIBUTE-CHARACTER	n/a ⁵	XML-TEXT or XML-NTEXT contains the single character that corresponds with the predefined entity reference in the attribute value.
ATTRIBUTE-CHARACTERS	XML-TEXT or XML-NTEXT contains the value within quotation marks or apostrophes. This can be a substring of the attribute value.	XML-TEXT or XML-NTEXT contains the value within quotation marks or apostrophes. This can be a substring of the attribute value if the value includes a character reference or an entity reference.
ATTRIBUTE-NAME	For attribute names that are not in a namespace, XML-TEXT or XML-NTEXT contains the attribute name. For attributes with names in a nondefault namespace, attribute names are always prefixed and have the form: <i>prefix:local-part</i> = " <i>AttValue</i> ". XML-TEXT or XML-NTEXT contains the <i>local-part</i> , XML-NAMESPACE or XML-NNAMESPACE contains the namespace identifier, and XML-NAMESPACE-PREFIX or XML-NNAMESPACE-PREFIX contains the <i>prefix</i> .	XML-TEXT or XML-TEXT contains the attribute name (the string to the left of the equal sign).
ATTRIBUTE-NATIONAL-CHARACTER	Regardless of the type of the XML document, XML-TEXT is empty with length zero and XML-NTEXT contains the single national character that corresponds with the numeric character reference. ²	XML-TEXT or XML-NTEXT content is the same as for XMLPARSE(XMLSS).
COMMENT	XML-TEXT or XML-NTEXT contains the text of the comment between the opening character sequence "<!--" and the closing character sequence "-->". This can be a substring of the text.	XML-TEXT or XML-NTEXT always contains the complete text of the comment.
CONTENT-CHARACTER	n/a ⁵	XML-TEXT or XML-NTEXT contains the single character that corresponds with the predefined entity reference in the element content.
CONTENT-CHARACTERS	XML-TEXT or XML-NTEXT contains the character content of the element between start and end tags. This can be a substring of the content.	XML-TEXT or XML-NTEXT contains the character content of the element between start and end tags. This can be a substring of the character content if the content includes a character reference or an entity reference.

Table 3. XML events and associated special register contents (continued)		
XML-EVENT	XMLPARSE(XMLSS) ¹	XMLPARSE(COMPAT) ¹
CONTENT-NATIONAL-CHARACTER	Regardless of the type of the XML document, XML-TEXT is empty with length zero and XML-NTEXT contains the single national character that corresponds with the numeric character reference. ²	XML-TEXT or XML-NTEXT content is the same as for XMLPARSE(XMLSS).
DOCUMENT-TYPE-DECLARATION	XML-TEXT or XML-NTEXT contains the name of the root element, as specified in the document type declaration.	XML-TEXT or XML-NTEXT contains the entire document type declaration, including the opening and closing character sequences "<!DOCTYPE" and ">".
ENCODING-DECLARATION	XML-TEXT or XML-NTEXT contains the value, between quotation marks or apostrophes, of the encoding declaration in the XML declaration.	XML-TEXT or XML-NTEXT content is the same as for XMLPARSE(XMLSS).
END-OF-CDATA-SECTION	All XML special registers except XML-CODE and XML-EVENT are empty with length zero.	XML-TEXT or XML-NTEXT contains the string "]]>".
END-OF-DOCUMENT	All XML special registers except XML-CODE and XML-EVENT are empty with length zero.	XML-TEXT or XML-NTEXT content is the same as for XMLPARSE(XMLSS).
END-OF-ELEMENT	XML-TEXT or XML-NTEXT contains the local part of the end element tag or empty element tag name. If the element name is in a nondefault namespace, XML-NAMESPACE or XML-NNAMESPACE contains the namespace identifier. If the element name is in a namespace and is prefixed (of the form <i>prefix:local-part</i>), XML-NAMESPACE-PREFIX or XML-NNAMESPACE-PREFIX contains the prefix.	XML-TEXT or XML-NTEXT contains the name of the end element tag or empty element tag.
END-OF-INPUT	All XML special registers except XML-CODE and XML-EVENT are empty with length zero. To parse an additional segment of an XML document, move the next segment to <i>identifier-1</i> and set XML-CODE to 1.	n/a ⁶
EXCEPTION	XML-CODE contains the unique return code and reason code that identifies the exception. XML-TEXT or XML-NTEXT contains the document fragment up to the point of the error or anomaly that caused the exception. ⁴ All other XML special registers are empty with length zero.	XML-CODE contains the unique error code that identifies the exception. ³ XML-TEXT or XML-NTEXT contains the part of the document that was successfully scanned, up to and including the point at which the exception was detected.

Table 3. *XML events and associated special register contents* (continued)

XML-EVENT	XMLPARSE(XMLSS) ¹	XMLPARSE(COMPAT) ¹
NAMESPACE-DECLARATION	XML-TEXT and XML-NTEXT are both empty with length zero. XML-NAMESPACE or XML-NNAMESPACE contains the declared namespace identifier. If the namespace is "undeclared" by specifying the empty string, XML-NAMESPACE and XML-NNAMESPACE are empty with length zero. XML-NAMESPACE-PREFIX or XML-NNAMESPACE-PREFIX contains the prefix if the namespace declaration is of the form <i>xmlns:prefix</i> = "<i>namespace-identifier</i>"; otherwise, if the declaration is for the default namespace and thus the attribute name is <i>xmlns</i>, XML-NAMESPACE-PREFIX and XML-NNAMESPACE-PREFIX are both empty with length zero.	n/a⁶ (ATTRIBUTE-NAME and ATTRIBUTE-CHARACTERS events are signaled instead.)
PROCESSING-INSTRUCTION-DATA	XML-TEXT or XML-NTEXT contains the rest of the processing instruction (after the target name), not including the closing sequence ">", but including trailing, and not leading, white space characters. This can be a substring of the processing instruction data.	XML-TEXT or XML-NTEXT always contains the complete processing instruction data.
PROCESSING-INSTRUCTION-TARGET	XML-TEXT or XML-NTEXT contains the processing instruction target name, which occurs immediately after the processing instruction opening sequence, "<?". This event can occur multiple times for a given processing instruction: one occurrence preceding each substring of the data.	XML-TEXT or XML-NTEXT content is the same as for XMLPARSE(XMLSS). This event occurs only once for a given processing instruction.
STANDALONE-DECLARATION	XML-TEXT or XML-NTEXT contains the value, between quotation marks or apostrophes ("yes" or "no"), of the stand-alone declaration in the XML declaration.	XML-TEXT or XML-NTEXT content is the same as for XMLPARSE(XMLSS).
START-OF-CDATA-SECTION	All XML special registers except XML-CODE and XML-EVENT are empty with length zero.	XML-TEXT or XML-NTEXT contains the string "<![CDATA[".
START-OF-DOCUMENT	All XML special registers except XML-CODE and XML-EVENT are empty with length zero.	XML-TEXT or XML-NTEXT contains the entire document.

Table 3. *XML events and associated special register contents* (continued)

XML-EVENT	XMLPARSE(XMLSS) ¹	XMLPARSE(COMPAT) ¹
START-OF-ELEMENT	XML-TEXT or XML-NTEXT contains the local part of the start element tag name or the local part of the empty element tag name. If the element name is in a namespace, XML-NAMESPACE or XML-NNAMESPACE contains the namespace identifier. If the element name is in a namespace and is prefixed (of the form <i>prefix:local-part</i>, XML-NAMESPACE-PREFIX or XML-NNAMESPACE-PREFIX contains the prefix.	XML-TEXT or XML-NTEXT contains the name of the start element tag or empty element tag, also known as the element type.
UNKNOWN-REFERENCE-IN-ATTRIBUTE	n/a⁵ For XMLPARSE(XMLSS), the parser always signals EXCEPTION.	XML-TEXT or XML-NTEXT contains the entity reference name, not including the "&" and ";" delimiters.
UNKNOWN-REFERENCE-IN-CONTENT	n/a⁵ For XMLPARSE(XMLSS), the parser signals UNRESOLVED-REFERENCE or EXCEPTION instead. See "Unresolved references" below for additional details.	XML-TEXT or XML-NTEXT contains the entity reference name, not including the "&" and ";" delimiters.
UNRESOLVED-REFERENCE	XML-TEXT or XML-NTEXT contains the entity name from XML content, not including the "&" and ";" delimiters. See "Unresolved references" below for additional details.	n/a⁶ (The parser signals UNKNOWN-REFERENCE-IN-CONTENT instead.)
VERSION-INFORMATION	XML-TEXT or XML-NTEXT contains the value, between quotation marks or apostrophes, of the version information in the XML declaration.	XML-TEXT or XML-NTEXT content is the same as for XMLPARSE(XMLSS).
1. For all events except EXCEPTION, XML-CODE contains zero. Unless stated otherwise, the namespace XML registers (XML-NAMESPACE, XML-NNAMESPACE, XML-NAMESPACE-PREFIX, and XML-NNAMESPACE-PREFIX) are empty and have length zero. 2. National characters with scalar values greater than 65,535 (X"FFFF") are represented using two encoding units (a "surrogate pair"). Programmers are responsible for ensuring that operations on the content of XML-NTEXT do not split the pair of encoding units that together form a graphic character, thereby forming invalid data. 3. For XMLPARSE(COMPAT), exceptions for encoding conflicts are signaled before parsing begins. For these exceptions, XML-TEXT or XML-NTEXT is either zero length or contains only the encoding declaration value from the document. See <i>XML PARSE exceptions with XMLPARSE(COMPAT) in effect</i> in the <i>Enterprise COBOL Programming Guide</i> for information about XML exception codes. 4. If an END-OF-INPUT XML event previously occurred and the processing procedure provided a new document segment, XML-TEXT or XML-NTEXT contains only the new segment. If the anomaly occurs before parsing begins (for example, the encoding specification is invalid), XML-TEXT or XML-NTEXT are empty with length zero. The fragment might or might not include the anomaly. For a duplicate attribute name, for example, the fragment includes the incorrect attribute. For an invalid character, the fragment includes document text up to, but not including, the invalid character. 5. n/a. Not applicable; occurs only with XMLPARSE(COMPAT). 6. n/a. Not applicable; occurs only with XMLPARSE(XMLSS).		

Unresolved References:

An unresolved entity reference is a reference to the name of an entity that has no declaration in the document type definition (DTD).

The parser signals an UNRESOLVED-REFERENCE event only if all of the following conditions are true:

- The unresolved reference is within element content, not an attribute value.
- The XML document starts with an XML declaration that specifies standalone="no".
- The XML document contains a document type declaration, for example,

```
<!DOCTYPE rootElementName>
```

- If the VALIDATING phrase is specified on the XML PARSE statement, the document type declaration must also specify an external DTD subset, for example:

```
<!DOCTYPE rootElementName SYSTEM "someOther.dtd">
```

Otherwise the parser signals an EXCEPTION event instead of UNRESOLVED-REFERENCE.

XML-INFORMATION

The XML-INFORMATION special register is used to provide additional information to an XML PARSE processing procedure about the status of the parse.

To use XML-INFORMATION, you must compile with the XMLPARSE(XMLSS) compiler option.

The XML-INFORMATION special register has the implicit definition:

```
01 XML-INFORMATION PICTURE S9(9) USAGE BINARY VALUE 0.
```

This register provides a mechanism to easily determine whether an XML EVENT is complete. Sometimes XML content might be split across multiple events and the application must concatenate the pieces of content together. The XML-INFORMATION register is used to indicate whether or not content of the XML event is complete.

The value of the XML-INFORMATION register is set as follows for the various XML events:

- ATTRIBUTE-CHARACTERS
 - 1 indicates that the attribute value in XML-TEXT or XML-NTEXT special register is complete
 - 2 indicates that the attribute value in XML-TEXT or XML-NTEXT special register is not complete
 - 4, 8, 16, ... are reserved for future use
- CONTENT-CHARACTERS
 - 1 indicates that the content value in XML-TEXT or XML-NTEXT special register is complete
 - 2 indicates that the content value in XML-TEXT or XML-NTEXT special register is not complete
 - 4, 8, 16, ... are reserved for future use
- All other events
 - 0 indicates that no additional information is currently available
 - 2, 4, 8, 16, ... are reserved for future use

XML-NAMESPACE

The XML-NAMESPACE special register is defined during XML parsing to contain the identifier of the namespace, if any, associated with the name in XML-TEXT for XML events START-OF-ELEMENT, END-

OF-ELEMENT, and ATTRIBUTE-NAME, and to contain the declared namespace identifier for XML event NAMESPACE-DECLARATION.

The parser sets XML-NAMESPACE to the identifier of the namespace associated with a name before transferring control to the processing procedure when the operand of the XML PARSE statement is an alphanumeric data item and the RETURNING NATIONAL phrase is not specified in the XML PARSE statement.

To use XML-NAMESPACE, you must compile with the XMLPARSE(XMLSS) compiler option.

XML-NAMESPACE is an elementary data item of category alphanumeric. The length of XML-NAMESPACE can vary from 0 through 32,768 bytes. The length at run time is the length of the contained namespace identifier.

There is no equivalent COBOL data description entry.

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

XML-NAMESPACE has a length of zero for:

- The START-OF-ELEMENT, END-OF-ELEMENT, and ATTRIBUTE-NAME XML events if there is no namespace associated with a name
- The NAMESPACE-DECLARATION XML event if the namespace is *undeclared* by specifying the empty string
- All other XML events

When XML-NAMESPACE is set, the XML-NNAMESPACE special register has a length of zero. At any given time, only one of the two special registers XML-NAMESPACE and XML-NNAMESPACE has a nonzero length.

Use the LENGTH function or the LENGTH OF special register to determine the number of bytes that XML-NAMESPACE contains.

XML-NAMESPACE cannot be used as a receiving item.

XML-NNAMESPACE

The XML-NNAMESPACE special register is defined during XML parsing to contain the identifier of the namespace, if any, associated with the name in XML-NTEXT for XML events START-OF-ELEMENT, END-OF-ELEMENT, and ATTRIBUTE-NAME, and to contain the declared namespace identifier for XML event NAMESPACE-DECLARATION.

The parser sets XML-NNAMESPACE to the identifier of the namespace associated with a name before transferring control to the processing procedure when the RETURNING NATIONAL phrase is specified in the XML PARSE statement or the operand of the XML PARSE statement is a national data item.

To use XML-NNAMESPACE, you must compile with the XMLPARSE(XMLSS) compiler option.

XML-NNAMESPACE is an elementary data item of category national. The length of XML-NNAMESPACE can vary from 0 through 16,384 national characters (0 through 32,768 bytes). The length at run time is the length of the contained namespace identifier.

There is no equivalent COBOL data description entry.

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

XML-NNAMESPACE has a length of zero for:

- The START-OF-ELEMENT, END-OF-ELEMENT, and ATTRIBUTE-NAME XML events, if there is no namespace associated with a name
- The NAMESPACE-DECLARATION XML event if the namespace is *undeclared* by specifying the empty string
- All other XML events

When XML-NNAMESPACE is set, the XML-NAMESPACE special register has a length of zero. At any given time, only one of the two special registers XML-NNAMESPACE and XML-NAMESPACE has a nonzero length.

Use the LENGTH function to determine the number of national character positions that XML-NNAMESPACE contains; use the LENGTH OF special register to determine the number of bytes.

XML-NNAMESPACE cannot be used as a receiving item.

XML-NAMESPACE-PREFIX

The XML-NAMESPACE-PREFIX special register is defined during XML parsing to contain the prefix, if any, of the name in XML-TEXT for XML events START-OF-ELEMENT, END-OF-ELEMENT, and ATTRIBUTE-NAME, and to contain the local attribute name for XML event NAMESPACE-DECLARATION.

The namespace prefix is used as an alias for the complete namespace identifier.

The parser sets XML-NAMESPACE-PREFIX before transferring control to the processing procedure when the operand of the XML PARSE statement is an alphanumeric data item and the RETURNING NATIONAL phrase is not specified.

To use XML-NAMESPACE-PREFIX, you must compile with the XMLPARSE(XMLSS) compiler option.

XML-NAMESPACE-PREFIX is an elementary data item of category national. The length of XML-NAMESPACE-PREFIX can vary from 0 through 4,096 bytes. The length at run time is the length of the contained namespace prefix.

There is no equivalent COBOL data description entry.

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

XML-NAMESPACE-PREFIX has a length of zero for:

- The START-OF-ELEMENT, END-OF-ELEMENT, and ATTRIBUTE-NAME XML events if the name does not have a prefix
- The NAMESPACE-DECLARATION XML event if the declaration is for the default namespace, in which case the namespace declaration attribute name is not prefixed.
- All other XML events

When XML-NAMESPACE-PREFIX is set, the XML-NNAMESPACE-PREFIX special register has a length of zero. At any given time, only one of the two special registers XML-NAMESPACE-PREFIX and XML-NNAMESPACE-PREFIX has a nonzero length.

Use the LENGTH function or the LENGTH OF special register to determine the number of bytes that XML-NAMESPACE-PREFIX contains.

XML-NAMESPACE-PREFIX cannot be used as a receiving item.

XML-NNAMESPACE-PREFIX

The XML-NNAMESPACE-PREFIX special register is defined during XML parsing to contain the prefix, if any, of the name in XML-NTXT for XML events START-OF-ELEMENT, END-OF-ELEMENT, and ATTRIBUTE-NAME, and to contain the local attribute name for XML event NAMESPACE-DECLARATION.

The namespace prefix is used as an alias for the complete namespace identifier.

The parser sets XML-NNAMESPACE-PREFIX before transferring control to the processing procedure when the operand of the XML PARSE statement is a national data item or the RETURNING NATIONAL phrase is specified in the XML PARSE statement.

To use XML-NNAMESPACE-PREFIX, you must compile with the XMLPARSE(XMLSS) compiler option.

XML-NNAMESPACE-PREFIX is an elementary data item of category national. The length of XML-NNAMESPACE-PREFIX can vary from 0 through 2048 national character positions (0 through 4096 bytes). The length at run time is the length of the contained namespace prefix.

There is no equivalent COBOL data description entry.

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

XML-NNAMESPACE-PREFIX has a length of zero for:

- The START-OF-ELEMENT, END-OF-ELEMENT, and ATTRIBUTE-NAME XML events if the name does not have a prefix
- NAMESPACE-DECLARATION XML event if the declaration is for the default namespace, in which case the namespace declaration attribute name is not prefixed.
- All other XML events

When XML-NNAMESPACE-PREFIX is set, the XML-NAMESPACE-PREFIX special register has a length of zero. At any given time, only one of the two special registers XML-NNAMESPACE-PREFIX and XML-NAMESPACE-PREFIX has a nonzero length.

Use the LENGTH function to determine the number of national character positions that XML-NNAMESPACE contains; use the LENGTH OF special register to determine the number of bytes.

XML-NNAMESPACE-PREFIX cannot be used as a receiving item.

XML-NTEXT

The XML-NTEXT special register is defined during XML parsing to contain document fragments that are represented in usage NATIONAL.

XML-NTEXT is an elementary data item of category national of the length of the contained XML document fragment. The length of XML-NTEXT can vary from 0 through 67,090,431 *national character positions*. The maximum byte length is 134,180,862.

There is no equivalent COBOL data description entry.

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

The parser sets XML-NTEXT to the document fragment associated with an event before transferring control to the processing procedure in these cases:

- When the operand of the XML PARSE statement is a data item of category national or the RETURNING NATIONAL phrase is specified in the XML PARSE statement
- For the ATTRIBUTE-NATIONAL-CHARACTER event
- For the CONTENT-NATIONAL-CHARACTER event

When XML-NTEXT is set, the XML-TEXT special register has a length of zero. At any given time, only one of the two special registers XML-NTEXT and XML-TEXT has a nonzero length.

Use the LENGTH function to determine the number of national characters that XML-NTEXT contains. Use the LENGTH OF special register to determine the number of bytes, rather than the number of national characters, that XML-NTEXT contains.

XML-NTEXT cannot be used as a receiving item.

XML-TEXT

The XML-TEXT special register is defined during XML parsing to contain document fragments that are represented in usage DISPLAY.

XML-TEXT is an elementary data item of category alphanumeric of the length of the contained XML document fragment. The length of XML-TEXT can vary from 0 through 134,180,862 bytes.

There is no equivalent COBOL data description entry.

When used in nested programs, this special register is implicitly defined with the global attribute in the outermost program.

The parser sets XML-TEXT to the document fragment associated with an event before transferring control to the processing procedure when the operand of the XML PARSE statement is an alphanumeric data item and the RETURNING NATIONAL phrase is not specified in the XML PARSE statement, except for the ATTRIBUTE-NATIONAL-CHARACTER event and the CONTENT-NATIONAL-CHARACTER event.

When XML-TEXT is set, the XML-NTEXT special register has a length of zero. At any given time, only one of the two special registers XML-NTEXT and XML-TEXT has a nonzero length.

Use the LENGTH function or the LENGTH OF special register for XML-TEXT to determine the number of bytes that XML-TEXT contains.

XML-TEXT cannot be used as a receiving item.

Literals

A *literal* is a character-string whose value is specified either by the characters of which it is composed or by the use of a figurative constant.

For more information about figurative constants, see [“Figurative constants” on page 15](#).

For descriptions of the different types of literals, see the following topics:

- [“Alphanumeric literals” on page 38](#)
- [“DBCS literals” on page 41](#)
- [“National literals” on page 46](#)
- [“Numeric literals” on page 45](#)

Alphanumeric literals

Enterprise COBOL provides several formats of alphanumeric literals.

The formats of alphanumeric literals are:

- Format 1: [“Basic alphanumeric literals” on page 38](#)
- Format 2: [“Alphanumeric literals with DBCS characters” on page 39](#)
- Format 3: [“Hexadecimal notation for alphanumeric literals” on page 40](#)
- Format 4: [“Null-terminated alphanumeric literals” on page 41](#)

Basic alphanumeric literals

Basic alphanumeric literals can contain any character in a single-byte EBCDIC character set.

The following format is for a basic alphanumeric literal:

Format 1: Basic alphanumeric literals
<pre>"single-byte-characters" 'single-byte-characters'</pre>

The enclosing quotation marks or apostrophes are excluded from the literal when the program is compiled.

An embedded quotation mark or apostrophe must be represented by a pair of quotation marks (") or a pair of apostrophes ('), respectively, when it is the character used as the opening delimiter. For example:

- "THIS ISN" "T WRONG" returns THIS ISN" T WRONG.
- 'THIS ISN' ' T WRONG' returns THIS ISN' T WRONG.

This is a similar concept to an escape character or escape sequence that is used to represent certain special characters within literals in other programming languages. Other than the quotation mark

and apostrophe, the only other consideration you need to give to special characters in literals is to Double-Byte Character Set (DBCS) literals, which are needed for national languages that contain unique characters or symbols. For details, See [“DBCS literals”](#) on page 41.

The delimiter character used as the opening delimiter for a literal must be used as the closing delimiter for that literal. For example:

```
'THIS IS RIGHT'  
"THIS IS RIGHT"  
'THIS IS WRONG'
```

You can use apostrophes or quotation marks as the literal delimiters independent of the APOST/QUOTE compiler option.

Any punctuation characters included within an alphanumeric literal are part of the value of the literal.

The maximum length of an alphanumeric literal is 160 bytes. The minimum length is 1 byte.

Alphanumeric literals are in the alphanumeric data class and category. (Data classes and categories are described in [“Classes and categories of data”](#) on page 170.)

Alphanumeric literals with DBCS characters

When the DBCS compiler option is in effect, the characters X'0E' and X'0F' in an alphanumeric literal will be recognized as shift codes for DBCS characters. That is, the characters between paired shift codes will be recognized as DBCS characters. Unlike an alphanumeric literal compiled under the NODBCS option, additional syntax rules apply to DBCS characters in an alphanumeric literal.

Alphanumeric literals with DBCS characters have the following format:

Format 2: Alphanumeric literals with DBCS characters
<pre>"mixed-SBCS-and-DBCS-characters" 'mixed-SBCS-and-DBCS-characters'</pre>

" or '

The opening and closing delimiter. The closing delimiter must match the opening delimiter.

mixed-SBCS-and-DBCS-characters

Any mix of single-byte and DBCS characters.

Shift-out and shift-in control characters are part of the literal and must be paired. They must contain zero or an even number of intervening bytes.

Nested shift codes are not allowed in the DBCS portion of the literal.

The syntax rules for single-byte characters in the literal follow the rules for basic alphanumeric literals. The syntax rules for DBCS characters in the literal follow the rules for DBCS literals.

The move and comparison rules for alphanumeric literals with DBCS characters are the same as those for any alphanumeric literal.

The length of an alphanumeric literal with DBCS characters is its byte length, including the shift control characters. The maximum length is limited by the available space on one line in Area B. An alphanumeric literal with DBCS characters cannot be continued.

An alphanumeric literal with DBCS characters is of the alphanumeric category.

Alphanumeric literals with DBCS characters cannot be used:

- As a literal in the following cases:
 - ALPHABET clause

- ASSIGN clause
- CALL statement *program-ID*
- CANCEL statement
- CLASS clause
- CURRENCY SIGN clause
- END PROGRAM marker
- ENTRY statement
- PADDING CHARACTER clause
- PROGRAM-ID paragraph
- RERUN clause
- STOP statement
- XML-SCHEMA clause
- As the external class-name for an object-oriented class
- As the basis-name in a BASIS statement
- As the text-name in a COPY statement
- As the library-name in a COPY statement

Enterprise COBOL statements process alphanumeric literals with DBCS characters without sensitivity to the shift codes and character codes. The use of statements that operate on a byte-to-byte basis (for example, STRING and UNSTRING) can result in strings that are not valid mixtures of single-byte EBCDIC and DBCS characters. See *Processing alphanumeric data items that contain DBCS data* in the *Enterprise COBOL Programming Guide* for more information about using alphanumeric literals and data items with DBCS characters in statements that operate on a byte-by-byte basis.

Hexadecimal notation for alphanumeric literals

Hexadecimal notation can be used for alphanumeric literals.

Hexadecimal notation has the following format:

Format 3: Hexadecimal notation for alphanumeric literals
<code>X"hexadecimal-digits"</code> <code>X'hexadecimal-digits'</code>

X" or X'

The opening delimiter for the hexadecimal notation of an alphanumeric literal.

" or '

The closing delimiter for the hexadecimal notation of an alphanumeric literal. If a quotation mark is used in the opening delimiter, a quotation mark must be used as the closing delimiter. Similarly, if an apostrophe is used in the opening delimiter, an apostrophe must be used as the closing delimiter.

Hexadecimal digits are characters in the range '0' to '9', 'a' to 'f', and 'A' to 'F', inclusive. Two hexadecimal digits represent one character in a single-byte character set (EBCDIC or ASCII). Four hexadecimal digits represent one character in a DBCS character set. A string of EBCDIC DBCS characters represented in hexadecimal notation must be preceded by the hexadecimal representation of a shift-out control character (X'0E') and followed by the hexadecimal representation of a shift-in control character (X'0F'). An even number of hexadecimal digits must be specified. The maximum length of a hexadecimal literal is 320 hexadecimal digits.

The continuation rules are the same as those for any alphanumeric literal. The opening delimiter (X" or X') cannot be split across lines.

The DBCS compiler option has no effect on the processing of hexadecimal notation of alphanumeric literals.

An alphanumeric literal in hexadecimal notation has data class and category alphanumeric. Hexadecimal notation for alphanumeric literals can be used anywhere alphanumeric literals can be used.

See also [“Hexadecimal notation for national literals”](#) on page 47.

Null-terminated alphanumeric literals

Alphanumeric literals can be null-terminated.

The format for null-terminated alphanumeric literals is:

Format 4: Null-terminated alphanumeric literals
<pre>Z"mixed-characters" Z'mixed-characters'</pre>

Z" or Z'

The opening delimiter for a null-terminated alphanumeric literal. Both characters of the opening delimiter (Z" or Z') must be on the same source line.

" or '

The closing delimiter for a null-terminated alphanumeric literal.

If a quotation mark is used in the opening delimiter, a quotation mark must be used as the closing delimiter. Similarly, if an apostrophe is used in the opening delimiter, an apostrophe must be used as the closing delimiter.

mixed-characters

Can be any of the following characters:

- Solely single-byte characters
- Mixed single-byte and DBCS characters
- Solely DBCS characters

However, you cannot specify the single-byte character with the value X'00'. X'00' is the null character automatically appended to the end of the literal. The content of the literal is otherwise subject to the same rules and restrictions as an alphanumeric literal with DBCS characters (format 2).

The length of the string of characters in the literal content can be 0 to 159 bytes. The actual length of the literal includes the terminating null character, and is a maximum of 160 bytes.

A null-terminated alphanumeric literal has data class and category alphanumeric. It can be used anywhere an alphanumeric literal can be used except that null-terminated literals are not supported in ALL *literal* figurative constants.

The LENGTH intrinsic function, when applied to a null-terminated alphanumeric literal, returns the number of bytes in the literal prior to but not including the terminating null. (The LENGTH special register does not support literal operands.)

DBCS literals

The formats and rules for DBCS literals are listed in this section.

Format for DBCS literals

```
G"<DBCS-characters>"
G'<DBCS-characters>'
N"<DBCS-characters>"
N'<DBCS-characters>'
```

G", G', N", or N'

Opening delimiters.

N" and N' identify a DBCS literal when the NSYMBOL(DBCS) compiler option is in effect. They identify a national literal when the NSYMBOL(NATIONAL) compiler option is in effect, and the rules specified in [“National literals” on page 46](#) apply.

The opening delimiter must be followed immediately by a shift-out control character.

For literals with opening delimiter N" or N', when embedded quotes or apostrophes are specified as part of DBCS characters in a DBCS literal, a single embedded DBCS quote or apostrophe is represented by two DBCS quotes or apostrophes. If a single embedded DBCS quote or apostrophe is found, an E-level compiler message will be issued and a second embedded DBCS quote or apostrophe will be assumed.

<

Represents the shift-out control character (X'0E')

>

Represents the shift-in control character (X'0F')

" or '

The closing delimiter. If a quotation mark is used in the opening delimiter, a quotation mark must be used as the closing delimiter. Similarly, if an apostrophe is used in the opening delimiter, an apostrophe must be used as the closing delimiter.

The closing delimiter must appear immediately after the shift-in control character.

DBCS-characters

DBCS-characters can be one or more characters in the range of X'00' through X'FF' for either byte.

Any value will be accepted in the content of the literal, although whether it is a valid value at run time depends on the CCSID in effect for the CODEPAGE compiler option.

Maximum length

28 characters

Continuation rules

Cannot be continued across lines

Where DBCS literals can be used

DBCS literals can be used in the following places:

- DATA DIVISION
 - In the VALUE clause of data description entries that define a data item of class DBCS.
 - In the VALUE OF clause of file description entries.
- PROCEDURE DIVISION
 - In a relation condition when the comparand is a DBCS data item, an elementary data item of class national, a national group item, or an alphanumeric group item
 - As an argument passed BY CONTENT in a CALL statement
 - In the DISPLAY and EVALUATE statements
 - In the following statements:
 - INITIALIZE; for details, see [“INITIALIZE statement” on page 350](#).

- INSPECT; for details, see “INSPECT statement” on page 353.
- MOVE; for details, see “MOVE statement” on page 405.
- STRING; for details, see “STRING statement” on page 462.
- UNSTRING, for details, see “UNSTRING statement” on page 470.
- In figurative constant ALL
- As an argument to the NATIONAL-OF intrinsic function
- Compiler-directing statements COPY, REPLACE, and TITLE

UTF-8 literals

The UTF-8 literal formats that Enterprise COBOL provides are basic UTF-8 literals and hexadecimal notation for UTF-8 literals.

Basic UTF-8 literals

The format and rules for basic UTF-8 literals are listed in this section.

Format 1: Basic UTF-8 literals
U"character-data"
U'character-data'

U" or U'

Opening delimiters. The opening delimiter must be coded as single-byte characters. It cannot be split across lines.

" or '

The closing delimiter. The closing delimiter must be coded as a single-byte character. If a quotation mark is used in the opening delimiter, it must be used as the closing delimiter. Similarly, if an apostrophe is used in the opening delimiter, it must be used as the closing delimiter.

To include the quotation mark or apostrophe used in the opening delimiter in the content of the literal, specify a pair of quotation marks or apostrophes, respectively. For example:

```
U'This literal ' 's content includes an apostrophe ' ;
U'This literal includes " , which is not used in the opening delimiter ' ;
U"This literal includes " " , which is used in the opening delimiter " .
```

character-data

The source text representation of the content of the UTF-8 literal. *character-data* can include any combination of EBCDIC single-byte characters and double-byte characters encoded in the Coded Character Set ID (CCSID) specified by the CODEPAGE compiler option.

DBCS characters in the content of the literal must be delimited by shift-out and shift-in control characters.

character-data can contain the following Unicode escape sequences:

- \uhhhh, where each *h* represents a hexadecimal digit in the range '0' to '9', 'a' to 'f', and 'A' to 'F', inclusive. This Unicode escape sequence represents a Unicode code point from the *Basic Multilingual Plane* (i.e., Unicode code points in the range U+0000 through U+FFFF).
- \U00hhhhh, where each *h* represents a hexadecimal digit in the range '0' to '9', 'a' to 'f', and 'A' to 'F'. This Unicode escape sequence can represent any legal Unicode code point, including code points from the *Supplementary Planes*, specifically, Unicode code points in the range U+10000 through U+10FFFF (e.g., an emoji symbol).

Note:

1. Code points U+D800 through U+DFFF are reserved for the high and low halves of surrogate pairs used by UTF-16. There is no legal encoding of these Unicode code points in UTF-8 and hence `\uD800` through `\uDFFF` and `\U0000D800` through `\U0000DFFF` cannot be specified as Unicode escape sequences in UTF-8 literals.
2. To avoid having a string of characters of the form `\uhhhh` or `\U00hhhhhh` in a UTF-8 literal be interpreted as a Unicode escape sequence, the escape character `'\'` can itself be escaped with `'\'` to cause it to be interpreted literally. Thus, the sequence `\\u00E9` will not be treated as a Unicode escape sequence.

Wherever a Unicode escape sequence appears in a basic UTF-8 literal, it is replaced by the compiler with the corresponding UTF-8 encoding of the Unicode code point, which makes it convenient to represent general Unicode code points in the literal using only EBCDIC characters. For example, `u'caf\u00E9'` represents the string 'café'.

Maximum length

The maximum number of UTF-8 characters that can be represented in a basic UTF-8 literal varies depending on the size (1 to 4 bytes) of each UTF-8 character being represented. However, a maximum of 160 bytes after conversion of the literal characters from the EBCDIC codepage to UTF-8 is allowed before truncation occurs. Truncation will be performed on a character boundary.

If the source content of the literal contains one or more DBCS characters, the maximum length is limited by the available space in Area B of a single source line.

The literal must contain at least one character. Each single-byte character in the literal counts as one character position and each DBCS character in the literal counts as one character position. Shift-in and shift-out delimiters for DBCS characters are not counted.

Continuation rules

When the content of the literal includes DBCS characters, the literal cannot be continued.

When the content of the literal does not include DBCS characters, normal continuation rules apply.

The source text representation of *character-data* is automatically converted to UTF-8 for use at run time. For example, when the literal is moved to or compared with a data item of category UTF-8, it is automatically converted to UTF-8.

Hexadecimal notation for UTF-8 literals

The format and rules for the hexadecimal notation format of UTF-8 literals are listed in this section.

Format 2: Hexadecimal notation for UTF-8 literals
<code>UX"hexadecimal-digits"</code> <code>UX'hexadecimal-digits'</code>

UX" or UX'

Opening delimiters. The opening delimiter must be represented in single-byte characters. It must not be split across lines.

" or '

The closing delimiter. The closing delimiter must be coded as a single-byte character.

If a quotation mark is used in the opening delimiter, it must be used as the closing delimiter. Similarly, if an apostrophe is used in the opening delimiter, it must be used as the closing delimiter.

hexadecimal-digits

Hexadecimal digits in the range '0' to '9', 'a' - 'f', and 'A' to 'F', inclusive. Each group of two hexadecimal digits represents a single encoding unit of a UTF-8 character.

Maximum length

The length of a UTF-8 literal in hexadecimal notation must be from two to 320 hexadecimal digits, excluding the opening and closing delimiters.

Continuation rules

Normal continuation rules apply.

The sequence of bytes represented by *hexadecimal-digits* is validated to ensure that it contains a legal sequence of UTF-8 bytes.

The content of a UTF-8 literal in hexadecimal notation is stored as UTF-8 characters. The resulting content has the same meaning as a basic UTF-8 literal that specifies the same UTF-8 characters.

A UTF-8 literal in hexadecimal notation has data class and category UTF-8 and can be used anywhere that a basic UTF-8 literal can be used.

Numeric literals

A *numeric literal* is a character-string whose characters are selected from the digits 0 through 9, a sign character (+ or -), and the decimal point.

If the literal contains no decimal point, it is an integer. (In this documentation, the word *integer* appearing in a format represents a numeric literal of nonzero value that contains no sign and no decimal point, except when other rules are included with the description of the format.) The following rules apply:

- If the ARITH(COMPAT) compiler option is in effect, one through 18 digits are allowed. If the ARITH(EXTEND) compiler option is in effect, one through 31 digits are allowed.
- Only one sign character is allowed. If included, it must be the leftmost character of the literal. If the literal is unsigned, it is a positive value.
- Only one decimal point is allowed. If a decimal point is included, it is treated as an assumed decimal point (that is, as not taking up a character position in the literal). The decimal point can appear anywhere within the literal except as the rightmost character.

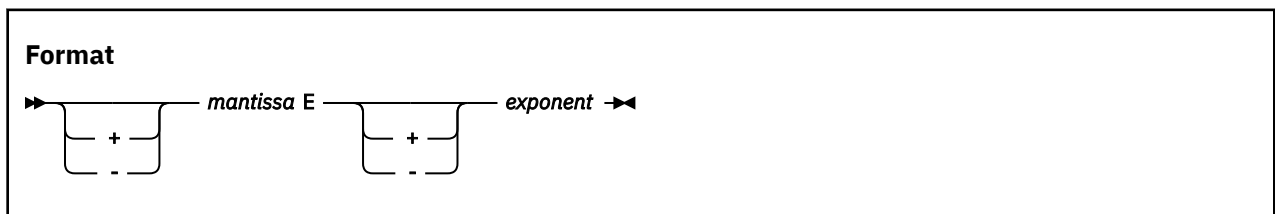
The value of a numeric literal is the algebraic quantity expressed by the characters in the literal. The size of a numeric literal is equal to the number of digits specified by the user.

Numeric literals can be fixed-point or floating-point numbers.

Numeric literals are in the numeric data class and category. (Data classes and categories are described under [“Classes and categories of data”](#) on page 170.)

Rules for floating-point literal values

The format and rules for floating-point literals are listed below.



- The sign is optional before the mantissa and the exponent; if you omit the sign, the compiler assumes a positive number.
- The mantissa can contain between one and 16 digits. A decimal point must be included in the mantissa.
- The exponent is represented by an E followed by an optional sign and one or two digits.
- The magnitude of a floating-point literal value must fall between 0.54E-78 and 0.72E+76. For values outside of this range, an E-level diagnostic message is produced and the value is replaced by either 0 or 0.72E+76, respectively.

National literals

The national literal formats that Enterprise COBOL provides are basic national literals and hexadecimal notation for national literals.

For more information about the formats, see [“Basic national literals” on page 46](#) and [“Hexadecimal notation for national literals” on page 47](#).

Basic national literals

The format and rules for basic national literals are listed in this section.

Format 1: Basic national literals
<pre>N"character-data" N'character-data'</pre>

When the NSYMBOL(NATIONAL) compiler option is in effect, the opening delimiter N" or N' identifies a national literal. A national literal is of the class and category national.

When the NSYMBOL(DBCS) compiler option is in effect, the opening delimiter N" or N' identifies a DBCS literal, and the rules specified in [“DBCS literals” on page 41](#) apply.

N" or N'

Opening delimiters. The opening delimiter must be coded as single-byte characters. It cannot be split across lines.

" or '

The closing delimiter. The closing delimiter must be coded as a single-byte character. If a quotation mark is used in the opening delimiter, it must be used as the closing delimiter. Similarly, if an apostrophe is used in the opening delimiter, it must be used as the closing delimiter.

To include the quotation mark or apostrophe used in the opening delimiter in the content of the literal, specify a pair of quotation marks or apostrophes, respectively. Examples:

```
N'This literal's content includes an apostrophe'  
N'This literal includes ", which is not used in the opening delimiter'  
N"This literal includes "", which is used in the opening delimiter"
```

character-data

The source text representation of the content of the national literal. *character-data* can include any combination of EBCDIC single-byte characters and double-byte characters encoded in the Coded Character Set ID (CCSID) specified by the CODEPAGE compiler option.

DBCS characters in the content of the literal must be delimited by shift-out and shift-in control characters.

Maximum length

The maximum length of a national literal is 80 character positions, excluding the opening and closing delimiters. If the source content of the literal contains one or more DBCS characters, the maximum length is limited by the available space in Area B of a single source line.

The literal must contain at least one character. Each single-byte character in the literal counts as one character position and each DBCS character in the literal counts as one character position. Shift-in and shift-out delimiters for DBCS characters are not counted.

Continuation rules

When the content of the literal includes DBCS characters, the literal cannot be continued. When the content of the literal does not include DBCS characters, normal continuation rules apply.

The source text representation of *character-data* is automatically converted to UTF-16 for use at run time (for example, when the literal is moved to or compared with a data item of category national).

Hexadecimal notation for national literals

The format and rules for the hexadecimal notation format of national literals are listed in this section.

Format 2: Hexadecimal notation for national literals
<pre>NX"hexadecimal-digits" NX'hexadecimal-digits'</pre>

The hexadecimal notation format of national literals is not affected by the NSYMBOL compiler option.

NX" or NX'

Opening delimiters. The opening delimiter must be represented in single-byte characters. It must not be split across lines.

" or '

The closing delimiter. The closing delimiter must be represented as a single-byte character.

If a quotation mark is used in the opening delimiter, a quotation mark must be used as the closing delimiter. Similarly, if an apostrophe is used in the opening delimiter, an apostrophe must be used as the closing delimiter.

hexadecimal-digits

Hexadecimal digits in the range '0' to '9', 'a' - 'f', and 'A' to 'F', inclusive. Each group of four hexadecimal digits represents a single national character and must represent a valid code point in UTF-16. The number of hexadecimal digits must be a multiple of four.

Maximum length

The length of a national literal in hexadecimal notation must be from four to 320 hexadecimal digits, excluding the opening and closing delimiters. The length must be a multiple of four.

Continuation rules

Normal continuation rules apply.

The content of a national literal in hexadecimal notation is stored as national characters. The resulting content has the same meaning as a basic national literal that specifies the same national characters.

A national literal in hexadecimal notation has data class and category national and can be used anywhere that a basic national literal can be used.

Where national literals can be used

National literals can be used in multiple ways.

National literals can be used:

- In a VALUE clause associated with a data item of class national or a VALUE clause associated with a condition-name for a conditional variable that is defined with usage NATIONAL
- In figurative constant ALL
- In a relation condition
- In the WHEN phrase of a format-2 SEARCH statement (binary search)
- In the ALL, LEADING, or FIRST phrase of an INSPECT statement
- In the BEFORE or AFTER phrase of an INSPECT statement
- In the DELIMITED BY phrase of a STRING statement
- In the DELIMITED BY phrase of an UNSTRING statement
- As the method-name in a METHOD-ID paragraph, an END METHOD marker, and an INVOKE statement
- As an argument passed BY CONTENT in the CALL statement
- As an argument passed BY VALUE in an INVOKE or CALL statement

- In the DISPLAY and EVALUATE statements
- As a sending item in the following procedural statements:
 - INITIALIZE
 - INSPECT
 - MOVE
 - STRING
 - UNSTRING
- In the argument list to the following intrinsic functions:
 DISPLAY-OF, LENGTH, LOWER-CASE, MAX, MIN, ORD-MAX, ORD-MIN, REVERSE, UPPER-CASE, USUPPLEMENTARY and UVALID

Note: DBCS literals can't be used in the USUPPLEMENTARY and UVALID functions.

- In the compiler-directing statements COPY, REPLACE, and TITLE

A national literal can be used only as specified in the detailed rules in this document.

PICTURE character-strings

A *PICTURE character-string* is composed of the currency symbol and certain combinations of characters in the COBOL character set. PICTURE character-strings are delimited only by the separator space, separator comma, separator semicolon, or separator period.

A chart of PICTURE clause symbols appears in [Table 12 on page 208](#).

Comments

A *comment* is a character-string that can contain any combination of characters from the character set of the computer.

It has no effect on the execution of the program. There are three forms of comments:

Comment entry (IDENTIFICATION DIVISION)

This form is described under [Chapter 21, “Optional paragraphs,” on page 117](#).

Comment line (any division)

This form is described under [“Comment lines” on page 60](#).

Inline comments (any division)

An inline comment is identified by a floating comment indicator (*>) preceded by one or more character-strings in the program-text area, and can be written on any line of a compilation group.

All characters that follow the floating comment indicator up to the end of area B are comment text.

Character-strings that form comments can contain DBCS characters or a combination of DBCS and single-byte EBCDIC characters.

Multiple comment lines that contain DBCS strings are allowed. The embedding of DBCS characters in a comment line must be done on a line-by-line basis. Words containing those characters cannot be continued to a following line. No syntax checking for valid strings is provided in comment lines.

Chapter 4. Separators

A *separator* is a character or a string of two or more contiguous characters that delimits character-strings. The separators are shown in the following table.

Table 4. <i>Separators</i>	
Separator	Meaning
<i>b</i> ¹	Space
, <i>b</i> ¹	Comma
. <i>b</i> ¹	Period
; <i>b</i> ¹	Semicolon
(	Left parenthesis
)	Right parenthesis
:	Colon
" <i>b</i> ¹	Quotation mark
' <i>b</i> ¹	Apostrophe
U"	Opening delimiter for UTF-8 literal
U'	Opening delimiter for UTF-8 literal
UX"	Opening delimiter for a hexadecimal format UTF-8 literal
UX'	Opening delimiter for a hexadecimal format UTF-8 literal
X"	Opening delimiter for a hexadecimal format alphanumeric literal
X'	Opening delimiter for a hexadecimal format alphanumeric literal
Z"	Opening delimiter for a null-terminated alphanumeric literal
Z'	Opening delimiter for a null-terminated alphanumeric literal
N"	Opening delimiter for a national literal ²
N'	Opening delimiter for a national literal ²
NX"	Opening delimiter for a hexadecimal format national literal
NX'	Opening delimiter for a hexadecimal format national literal
G"	Opening delimiter for a DBCS literal
G'	Opening delimiter for a DBCS literal
==	Pseudo-text delimiter
1. <i>b</i> represents a blank. 2. N" and N' are the opening delimiter for a DBCS literal when the NSYMBOL(DBCS) compiler option is in effect.	

Rules for separators

A separator is a string of one or more punctuation characters.

In the following description, {} (curly braces) enclose each separator, and *b* represents a space. Anywhere a space is used as a separator or as part of a separator, more than one space can be used.

Space {*b*}

A space can immediately precede or follow any separator except:

- The opening pseudo-text delimiter, where the preceding space is required.
- Within quotation marks. Spaces between quotation marks are considered part of the alphanumeric literal; they are not considered separators.

Period {*.b*}, Comma {*,b*}, Semicolon {*;b*}

A separator comma is composed of a comma followed by a space. A separator period is composed of a period followed by a space. A separator semicolon is composed of a semicolon followed by a space.

The separator period must be used only to indicate the end of a sentence, or as shown in formats. The separator comma and separator semicolon can be used anywhere the separator space is used.

- In the IDENTIFICATION DIVISION, each paragraph must end with a separator period.
- In the ENVIRONMENT DIVISION, the SOURCE-COMPUTER, OBJECT-COMPUTER, SPECIAL-NAMES, and I-O-CONTROL paragraphs must each end with a separator period. In the FILE-CONTROL paragraph, each file-control entry must end with a separator period.
- In the DATA DIVISION, file (FD), sort/merge file (SD), and data description entries must each end with a separator period.
- In the PROCEDURE DIVISION, separator commas or separator semicolons can separate statements within a sentence and operands within a statement. Each sentence and each procedure must end with a separator period.

Parentheses { () ... { } }

Except in pseudo-text, parentheses can appear only in balanced pairs of left and right parentheses. They delimit subscripts, a list of function arguments, reference-modifiers, arithmetic expressions, or conditions.

Colon { : }

The colon is a separator and is required when shown in general formats.

Quotation marks { " } ... { ' } }

An opening quotation mark must be immediately preceded by a space or a left parenthesis. A closing quotation mark must be immediately followed by a separator space, comma, semicolon, period, right parenthesis, or pseudo-text delimiter. Quotation marks must appear as balanced pairs. They delimit alphanumeric literals, except when the literal is continued (see [“Continuation lines”](#) on page 58).

Apostrophes { ' } ... { ' }

An opening apostrophe must be immediately preceded by a space or a left parenthesis. A closing apostrophe must be immediately followed by a separator space, comma, semicolon, period, right parenthesis, or pseudo-text delimiter. Apostrophes must appear as balanced pairs. They delimit alphanumeric literals, except when the literal is continued (see [“Continuation lines”](#) on page 58).

Null-terminated literal delimiters { Z" } ... { " }, { Z' } ... { ' }

The opening delimiter must be immediately preceded by a space or a left parenthesis. The closing delimiter must be immediately followed by a separator space, comma, semicolon, period, right parenthesis, or pseudo-text delimiter.

DBCS literal delimiters { G" } ... { " }, { G' } ... { ' }, { N" } ... { " }, { N' } ... { ' }

The opening delimiter must be immediately preceded by a space or a left parenthesis. The closing delimiter must be immediately followed by a separator space, comma, semicolon, period, right parenthesis, or pseudo-text delimiter. N" and N' are DBCS literal delimiters when the NSYMBOL(DBCS) compiler option is in effect.

UTF-8 literal delimiters {U"} ... {"}, {U'} ... {'}, {UX"} ... {"}, {UX'} ... {'}

The opening delimiter must be immediately preceded by a space or a left parenthesis. The closing delimiter must be immediately followed by a separator space, comma, semicolon, period, right parenthesis, or pseudo-text delimiter.

National literal delimiters {N"} ... {"}, {N'} ... {'}, {NX"} ... {"}, {NX'} ... {'}

The opening delimiter must be immediately preceded by a space or a left parenthesis. The closing delimiter must be immediately followed by a separator space, comma, semicolon, period, right parenthesis, or pseudo-text delimiter. N" and N' are DBCS literal delimiters when the NSYMBOL(DBCS) compiler option is in effect.

Pseudo-text delimiters {b==} ... {==b}

An opening pseudo-text delimiter must be immediately preceded by a space. A closing pseudo-text delimiter must be immediately followed by a separator space, comma, semicolon, or period. Pseudo-text delimiters must appear as balanced pairs. They delimit pseudo-text. (See [“COPY statement” on page 696.](#))

Any punctuation character included in a PICTURE character-string, a comment character-string, or an alphanumeric literal is not considered a punctuation character, but is part of the character-string or literal.

Chapter 5. Sections and paragraphs

Sections and paragraphs define a program. Sections and paragraphs are subdivided into sentences, statements, and entries.

Sentences are subdivided into statements, and statements are subdivided into phrases. Entries are subdivided into clauses.

For details, see:

- [“Sentences, statements, and entries” on page 53](#)
- [“Statements” on page 54](#)
- [“Phrases” on page 54](#)
- [“Clauses” on page 54](#)

For more information about sections, paragraphs, and statements, see [“Procedures” on page 265](#).

Sentences, statements, and entries

Unless the associated rules explicitly state otherwise, each required clause or statement must be written in the sequence shown in its format. If optional clauses or statements are used, they must be written in the sequence shown in their formats. These rules are true even for clauses and statements treated as comments.

The syntactical hierarchy follows this form:

- IDENTIFICATION DIVISION
 - Paragraphs
 - Entries
 - Clauses
- ENVIRONMENT DIVISION
 - Sections
 - Paragraphs
 - Entries
 - Clauses
 - Phrases
- DATA DIVISION
 - Sections
 - Entries
 - Clauses
 - Phrases
- PROCEDURE DIVISION
 - Sections
 - Paragraphs
 - Sentences
 - Statements
 - Phrases

Entries

An *entry* is a series of clauses that ends with a separator period. Entries are constructed in the identification, environment, and data divisions.

Clauses

A *clause* is an ordered set of consecutive COBOL character-strings that specifies an attribute of an entry. Clauses are constructed in the identification, environment, and data divisions.

Sentences

A *sentence* is a sequence of one or more statements that ends with a separator period. Sentences are constructed in the PROCEDURE DIVISION.

Statements

A *statement* specifies an action to be taken by the program. Statements are constructed in the PROCEDURE DIVISION.

For descriptions of the different types of statements, see:

- [“Imperative statements” on page 290](#)
- [“Conditional statements” on page 292](#)
- [Chapter 7, “Scope of names,” on page 63](#)
- [Chapter 114, “Compiler-directing statements,” on page 693](#)

Phrases

Each clause or statement in a program can be subdivided into smaller units called *phrases*.

Chapter 6. Reference format

COBOL source text must be written in COBOL *reference format*.

Reference format consists of the following areas in a 72-character line.

Sequence number area

Columns 1 through 6

Indicator area

Column 7

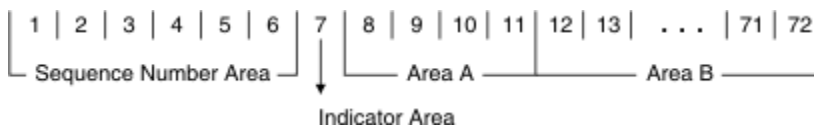
Area A

Columns 8 through 11

Area B

Columns 12 through 72

This figure illustrates reference format for a COBOL source line.



If you want to continue a literal, code a hyphen (-) in the indicator area (column 7) of each continuation line. For details about line continuation rules, see [“Continuation lines”](#) on page 58.

The following topics provide details about these areas:

- [“Sequence number area”](#) on page 55
- [“Indicator area”](#) on page 55
- [“Area A”](#) on page 55
- [“Area B”](#) on page 57
- [“Area A or Area B”](#) on page 59

Sequence number area

The sequence number area can be used to label a source statement line. The content of this area can consist of any character in the character set of the computer.

Indicator area

Use the indicator area to specify the continuation of words or alphanumeric literals from the previous line onto the current line, the treatment of text as documentation, and debugging lines.

See [“Continuation lines”](#) on page 58, [“Comment lines”](#) on page 60, and [“Debugging lines”](#) on page 61.

The indicator area can be used for source listing formatting. A slash (/) placed in the indicator column causes the compiler to start a new page for the source listing, and the corresponding source record to be treated as a comment. The effect can be dependent on the LINECOUNT compiler option. For information about the LINECOUNT compiler option, see *LINECOUNT* in the *Enterprise COBOL Programming Guide*.

Area A

Certain items must begin in Area A.

These items are:

- [Division headers](#)

- [“Section headers” on page 56](#)
- [Paragraph headers or paragraph names](#)
- [Level indicators or level-numbers \(01 and 77\)](#)
- [DECLARATIVES and END DECLARATIVES](#)
- [End program, end class, and end method markers](#)

Division headers

A division header is a combination of words, followed by a separator period to indicate the beginning of a division.

See the following division headers:

- IDENTIFICATION DIVISION.
- ENVIRONMENT DIVISION.
- DATA DIVISION.
- PROCEDURE DIVISION.

A division header (except when a USING phrase is specified with a PROCEDURE DIVISION header) must be immediately followed by a separator period. Except for the USING phrase, no text can appear on the same line.

Section headers

In the environment and procedure divisions, a section header indicates the beginning of a series of paragraphs.

For example:

```
INPUT-OUTPUT SECTION.
```

In the DATA DIVISION, a section header indicates the beginning of an entry; for example:

```
FILE SECTION.
LINKAGE SECTION.
LOCAL-STORAGE SECTION.
WORKING-STORAGE SECTION.
```

A section header must be immediately followed by a separator period.

Paragraph headers or paragraph names

A paragraph header or paragraph name indicates the beginning of a paragraph.

In the ENVIRONMENT DIVISION, a paragraph consists of a paragraph header followed by one or more entries. For example:

```
OBJECT-COMPUTER. computer-name.
```

In the PROCEDURE DIVISION, a paragraph consists of a paragraph-name followed by one or more sentences.

Level indicators (FD and SD) or level-numbers (01 and 77)

A level indicator can be either FD or SD.

A level indicator must begin in Area A and be followed by a space. (See [“FILE SECTION” on page 184.](#)) A level-number that must begin in Area A is a one- or two-digit integer with a value of 01 or 77. It must be followed by a space or separator period.

DECLARATIVES and END DECLARATIVES

DECLARATIVES and END DECLARATIVES are keywords that begin and end the declaratives part of the source unit.

In the PROCEDURE DIVISION, each of the keywords DECLARATIVES and END DECLARATIVES must begin in Area A and be followed immediately by a separator period; no other text can appear on the same line. After the keywords END DECLARATIVES, no text can appear before the following section header. (See [“Declaratives” on page 264.](#))

End program, end class, and end method markers

The end markers are a combination of words followed by a separator period that indicates the end of a COBOL program, method, class, factory, or object definition.

For example:

```
END PROGRAM program-name.  
END CLASS class-name.  
END METHOD "method-name".  
END OBJECT.  
END FACTORY.
```

For programs

program-name must be identical to the *program-name* of the corresponding PROGRAM-ID paragraph. Every COBOL program, except an outermost program that contains no nested programs and is not followed by another batch program, must end with an END PROGRAM marker.

For classes

class-name must be identical to the *class-name* in the corresponding CLASS-ID paragraph.

For methods

method-name must be identical to the *method-name* in the corresponding METHOD-ID paragraph.

For object paragraphs

There is no name in an object paragraph header or in its end marker. The syntax is simply END OBJECT.

For factory paragraphs

There is no name in a factory paragraph header or in its end marker. The syntax is simply END FACTORY.

Area B

Certain items must begin in Area B.

These items are:

- [Entries, sentences, statements, and clauses](#)
- [Continuation lines](#)

Entries, sentences, statements, clauses

The first entry, sentence, statement, or clause begins on either the same line as the header or paragraph-name that it follows, or in Area B of the next nonblank line that is not a comment line. Successive

sentences or entries either begin in Area B of the same line as the preceding sentence or entry, or in Area B of the next nonblank line that is not a comment line.

Within an entry or sentence, successive lines in Area B can have the same format or can be indented to clarify program logic. The output listing is indented only if the input statements are indented. Indentation does not affect the meaning of the program. The programmer can choose the amount of indentation, subject only to the restrictions on the width of Area B. See also [Chapter 5, “Sections and paragraphs,”](#) on [page 53](#).

Continuation lines

Any sentence, entry, clause, or phrase that requires more than one line can be continued in Area B of the next line that is neither a comment line nor a blank line.

The line being continued is a *continued line*; the succeeding lines are *continuation lines*. Area A of a continuation line must be blank.

If there is no hyphen (-) in the indicator area (column 7) of a line, the last character of the preceding line is assumed to be followed by a space.

The following items cannot be continued:

- DBCS user-defined words
- DBCS literals
- Alphanumeric literals containing DBCS characters
- National literals containing DBCS characters

However, alphanumeric literals and national literals in hexadecimal notation can be continued regardless of the kind of characters expressed in hexadecimal notation.

All characters that make up an opening literal delimiter must be on the same line. For example, Z", G", N", NX", or X".

Both characters that make up the pseudo-text delimiter separator, ==, the floating comment indicator, *>, or the compiler directive indicator, >>, must be on the same line.

A compiler directive or compiler directive phrase, which begins with >>, must be specified on the same line.

If there is a hyphen in the indicator area of a line, the first nonblank character of the continuation line immediately follows the last nonblank character of the continued line without an intervening space.

Continuation of alphanumeric and national literals

Alphanumeric and national literals can be continued only when there are no DBCS characters in the content of the literal.

The following rules apply to alphanumeric and national literals that do not contain DBCS characters:

- If the continued line contains an alphanumeric or national literal without a closing quotation mark, all spaces at the end of the continued line (through column 72) are considered to be part of the literal. The continuation line must contain a hyphen in the indicator area, and the first nonblank character must be a quotation mark. The continuation of the literal begins with the character immediately following the quotation mark.
- If an alphanumeric or national literal that is to be continued on the next line has as its last character a quotation mark in column 72, the continuation line must start with two consecutive quotation marks. This will result in a quotation mark as part of the value of the literal.

If the last character on the continued line of an alphanumeric or national literal is a quotation mark in Area B, the continuation line can start with a quotation mark. This will result in two consecutive literals instead of one continued literal.

The rules are the same when an apostrophe is used instead of a quotation mark in delimiters.

If you want to continue a literal such that the continued lines and the continuation lines are part of one literal:

- Code a hyphen in the indicator area of each continuation line.
- Code the literal value using all columns of each continued line, up to and including column 72. (Do not terminate the continued lines with a quotation mark followed by a space.)
- Code a quotation mark before the first character of the literal on each continuation line.
- Terminate the last continuation line with a quotation mark followed by a space.

In the following examples, the number and size of literals created are indicated in the following example:

```
|...+.*..1....+....2....+....3....+....4....+....5....+....6....+....7..
000001      "AAAAAAAAAABBBBBBBBBBCCCCCCCCDDDDDDDDDEEEEEEEEEEE
-          "GGGGGGGGGHHHHHHHHHHIIIIIIIIJJJJJJJJJKKKKKKKKKK
-          "LLLLLLLLLLMMMMMMMMMM"
```

- Literal 000001 is interpreted as one alphanumeric literal that is 120 bytes long. Each character between the starting quotation mark and up to and including column 72 of continued lines is counted as part of the literal.

```
|...+.*..1....+....2....+....3....+....4....+....5....+....6....+....7..
000003      N"AAAAAAAAAABBBBBBBBBBCCCCCCCCDDDDDDDDDEEEEEEEEEEE
-          "GGGGGGGGG"
```

- Literal 000003 is interpreted as one national literal that is 60 national character positions in length (120 bytes). Each character between the starting quotation mark and the ending quotation mark on the continued line is counted as part of the literal. Although single-byte characters are entered, the value of the literals is stored as national characters.

```
|...+.*..1....+....2....+....3....+....4....+....5....+....6....+....7..
000005      "AAAAAAAAAABBBBBBBBBBCCCCCCCCDDDDDDDDDEEEEEEEEEEE
-          "GGGGGGGGGHHHHHHHHHHIIIIIIIIJJJJJJJJJKKKKKKKKKK
-          "LLLLLLLLLLMMMMMMMMMM"
```

- Literal 000005 is interpreted as one literal that is 140 bytes long. The blanks at the end of each continued line are counted as part of the literal because the continued lines do not end with a quotation mark.

```
|...+.*..1....+....2....+....3....+....4....+....5....+....6....+....7..
000010      "AAAAAAAAAABBBBBBBBBBCCCCCCCCDDDDDDDDDEEEEEEEEEEE"
-          "GGGGGGGGGHHHHHHHHHHIIIIIIIIJJJJJJJJJKKKKKKKKKK"
-          "LLLLLLLLLLMMMMMMMMMM"
```

- Literal 000010 is interpreted as three separate literals that have lengths of 50, 50, and 20, respectively. The quotation mark with the following space terminates the continued line. Only the characters within the quotation marks are counted as part of the literals. Literal 000010 is not valid as a VALUE clause literal for non-level-88 data items.

To code a continued literal where the length of each continued part of the literal is less than the length of Area B, adjust the starting column such that the last character of the continued part is in column 72.

Area A or Area B

Certain items can begin in either Area A or Area B.

These items are:

- Level-numbers
- Comment lines
- Floating comment indicators (*>)

- [Compiler-directing statements](#)
- [Debugging lines](#)
- [Pseudo-text](#)
- [Blank lines](#)

Level-numbers

A level-number that can begin in Area A or B is a one- or two-digit integer with a value of 02 through 49, 66, or 88.

A level-number that must begin in Area A is a one- or two-digit integer with a value of 01 or 77. A level-number must be followed by a space or a separator period. For more information, see [“Level-numbers” on page 194](#).

Comment lines

A *comment line* is any line with an asterisk (*) or slash (/) in the indicator area (column 7) of the line, or with a floating comment indicator (*>) as the first character-string in the program text area (Area A plus Area B).

The comment can be written anywhere in the program text area of that line, and can consist of any combination of characters from the character set of the computer.

Comment lines can be placed anywhere in a program, method, or class definition. Comment lines placed before the IDENTIFICATION DIVISION header must follow any control cards (for example, PROCESS or CBL).

Important: Comments intermixed with control cards could nullify some of the control cards and cause them to be diagnosed as errors.

Multiple comment lines are allowed. Each must begin with an asterisk (*) or a slash (/) in the indicator area, or with a floating comment indicator (*>).

For more information about floating comment indicators, see [“Floating comment indicators \(*>\)” on page 60](#).

An asterisk (*) comment line is printed on the next available line in the output listing. The effect can be dependent on the LINECOUNT compiler option. For information about the LINECOUNT compiler option, see *LINECOUNT* in the *Enterprise COBOL Programming Guide*. A slash (/) comment line is printed on the first line of the next page, and the current page of the output listing is ejected.

The compiler treats a comment line as documentation, and does not check it syntactically.

Floating comment indicators (*>)

In addition to the fixed indicators that can only be specified in the indicator area of the source reference format, a floating comment indicator (*>) can be specified anywhere in the program-text area to indicate a comment line or an inline comment.

A floating comment indicator indicates a comment line if it is the first character string in the program-text area (Area A plus Area B), or indicates an inline comment if it is after one or more character strings in the program-text area.

These are the rules for floating comment indicators:

- Both characters (* and >) that form the multiple-character floating indicator must be contiguous and on the same line.
- The floating comment indicator for an inline comment must be preceded by a separator space, and can be specified wherever a separator space can be specified.
- All characters following the floating comment indicator up to the end of Area B are comment text.

Compiler-directing statements

Most compiler-directing statements, including COPY and REPLACE, can start in either Area A or Area B.

BASIS, CBL (PROCESS), *CBL (*CONTROL), DELETE, EJECT, INSERT, SKIP1, SKIP2, SKIP3, and TITLE statements can also start in Area A or Area B.

Compiler directives

Compiler directives must start in Area B.

For more information, see [Chapter 115, “Compiler directives,” on page 717](#).

Debugging lines

A *debugging line* is any line with a D (or d) in the indicator area of the line.

Debugging lines can be written in the ENVIRONMENT DIVISION (after the OBJECT-COMPUTER paragraph), the DATA DIVISION, and the PROCEDURE DIVISION. If a debugging line contains only spaces in Area A and Area B, it is considered a blank line.

See WITH DEBUGGING MODE in [“SOURCE-COMPUTER paragraph” on page 122](#).

Pseudo-text

The character-strings and separators that comprise *pseudo-text* can start in either Area A or Area B.

If, however, there is a hyphen in the indicator area (column 7) of a line that follows the opening pseudo-text delimiter, Area A of the line must be blank, and the rules for continuation lines apply to the formation of text words. See [“Continuation lines” on page 58](#) for details.

Blank lines

A *blank line* contains nothing but spaces in column 7 through column 72. A blank line can be anywhere in a program.

Chapter 7. Scope of names

A user-defined word names a data resource or a COBOL programming element. Examples of named data resources are a file, a data item, or a record. Examples of named programming elements are a program, a paragraph, a method, or a class definition.

The following sections define the types of names in COBOL and explain where the names can be referenced:

- [“Types of names” on page 63](#)
- [“External and internal resources” on page 65](#)
- [“Resolution of names” on page 66](#)

Types of names

In addition to identifying a resource, a name can have global or local attributes. Some names are always global, some names are always local, and some names are either local or global depending on specifications in the program in which the names are defined.

For programs

A *global name* can be used to refer to the resource with which it is associated both:

- From within the program in which the global name is defined
- From within any other program that is contained in the program that defines the global name

Use the GLOBAL clause in the data description entry to indicate that a name is global. For more information about using the GLOBAL clause, see [“GLOBAL clause” on page 185](#).

A *local name* can be used only to refer to the resource with which it is associated from within the program in which the local name is defined.

By default, if a data-name, a file-name, a record-name, or a condition-name definition in a data description entry does not include the GLOBAL clause, the name is local.

For methods

All names defined in methods are implicitly local.

For classes

Names defined in a class definition are global to all the methods contained in that class definition.

For object paragraphs

Names defined in the DATA DIVISION of an object paragraph are global to the methods contained in that object paragraph.

For factory paragraphs

Names defined in the DATA DIVISION of a factory paragraph are global to the methods contained in that factory paragraph.

For user-defined functions

All names defined in user-defined functions are local.

For function prototypes

All names defined in function prototypes are local.

Restriction: Specific rules sometimes prohibit specifying the GLOBAL clause for certain data description, file description, or record description entries.

The following list indicates the names that you can use and whether the name can be local or global:

data-name

data-name assigns a name to a data item.

A data-name is global if the GLOBAL clause is specified either in the data description entry that defines the data-name or in another entry to which that data description entry is subordinate.

file-name

file-name assigns a name to a file connector.

A file-name is global if the GLOBAL clause is specified in the file description entry for that file-name.

record-name

record-name assigns a name to a record.

A record-name is global if the GLOBAL clause is specified in the record description that defines the record-name, or in the case of record description entries in the FILE SECTION, if the GLOBAL clause is specified in the file description entry for the file name associated with the record description entry.

condition-name

condition-name associates a value with a conditional variable.

A condition-name that is defined in a data description entry is global if that entry is subordinate to another entry that specifies the GLOBAL clause.

A condition-name that is defined within the configuration section is always global.

program-name

program-name assigns a name to an external or internal (nested) program. For more information, see [“Conventions for program-names” on page 86](#).

A program-name is neither local nor global. For more information, see [“Conventions for program-names” on page 86](#).

method-name

method-name assigns a name to a method. *method-name* must be specified as the content of an alphanumeric literal or a national literal.

section-name

section-name assigns a name to a section in the PROCEDURE DIVISION.

A section-name is always local.

paragraph-name

paragraph-name assigns a name to a paragraph in the PROCEDURE DIVISION.

A paragraph-name is always local.

basis-name

basis-name specifies the name of source text that is to be included by the compiler into the source unit. For details, see [“BASIS statement” on page 693](#).

library-name

library-name specifies the COBOL library that the compiler uses for including COPY text. For details, see [“COPY statement” on page 696](#).

text-name

text-name specifies the name of COPY text to be included by the compiler into the source unit. For details, see [“COPY statement” on page 696](#).

alphabet-name

alphabet-name assigns a name to a specific character set or collating sequence, or both, in the SPECIAL-NAMES paragraph of the ENVIRONMENT DIVISION.

An alphabet-name is always global.

class-name (of data)

class-name assigns a name to the proposition in the SPECIAL-NAMES paragraph of the ENVIRONMENT DIVISION for which a truth value can be defined.

A class-name is always global.

class-name (object-oriented)

class-name assigns a name to an object-oriented class or subclass.

mnemonic-name

mnemonic-name assigns a user-defined word to an implementer-name.

A mnemonic-name is always global.

symbolic-character

symbolic-character specifies a user-defined figurative constant.

A symbolic-character is always global.

index-name

index-name assigns a name to an index associated with a specific table.

If a data item that possesses the global attribute includes a table accessed with an index, that index also possesses the global attribute. In addition, the scope of that index-name is identical to the scope of the data-name that includes the table.

xml-schema-name

xml-schema-name assigns a name to the system identifier of a file containing an XML schema.

An xml-schema-name is always global.

External and internal resources

The storage associated with a data item or a file connector can be *external* or *internal* to the program or method in which the resource is declared.

A data item or file connector is external if the storage associated with that resource is associated with the run unit rather than with any particular program or method within the run unit. An external resource can be referenced by any program or method in the run unit that describes the resource. References to an external resource from different programs or methods using separate descriptions of the resource are always to the same resource. In a run unit, there is only one representation of an external resource.

A resource is internal if the storage associated with that resource is associated only with the program or method that describes the resource.

External and internal resources can have either global or local names.

A data record described in the WORKING-STORAGE SECTION is given the external attribute by the presence of the EXTERNAL clause in its data description entry. Any data item described by a data description entry subordinate to an entry that describes an external record also attains the external attribute. If a record or data item does not have the external attribute, it is part of the internal data of the program or method in which it is described.

Two programs or methods in a run unit can reference the same file connector in the following circumstances:

- An external file connector can be referenced from any program or method that describes that file connector.
- If a program is contained within another program, both programs can refer to a global file connector by referring to an associated global file-name either in the containing program or in any program that directly or indirectly contains the containing program.

Two programs or methods in a run unit can reference common data in the following circumstances:

- The data content of an external data record can be referenced from any program or method provided that program or method has described that data record.
- If a program is contained within another program, both programs can refer to data that possesses the global attribute either in the program or in any program that directly or indirectly contains the containing program.

The data records described as subordinate to a file description entry that does not contain the EXTERNAL clause or to a sort-merge file description entry, as well as any data items described subordinate to the data description entries for such records, are always internal to the program or method that describes the file-name. If the EXTERNAL clause is included in the file description entry, the data records and the data items attain the external attribute.

Resolution of names

The rules for resolution of names depend on whether the names are specified in a program or in a class definition.

Names within programs

When a program, program B, is directly contained within another program, program A, both programs can define a condition-name, a data-name, a file-name, or a record-name using the same user-defined word. When such a duplicated name is referenced in program B, the following steps determine the referenced resource (these rules also apply to classes and contained methods):

1. The referenced resource is identified from the set of all names that are defined in program B and all global names defined in program A and in any programs that directly or indirectly contain program A. The normal rules for qualification and any other rules for uniqueness of reference are applied to this set of names until one or more resources is identified.
2. If only one resource is identified, it is the referenced resource.
3. If more than one resource is identified, no more than one resource can have a name local to program B. If zero or one of the resources has a name local to program B, the following rules apply:
 - If the name is declared in program B, the resource in program B is the referenced resource.
 - If the name is not declared in program B, the referenced resource is:
 - The resource in program A if the name is declared in program A
 - The resource in the containing program if the name is declared in the program that contains program A

This rule is applied to further containing programs until a valid resource is found.

Names within a class definition

Within a class definition, resources can be defined within the following units:

- The factory data division
- The object data division
- A method data division

If a resource is defined with a given name in the DATA DIVISION of an object definition, and there is no resource defined with the same name in an instance method of that object definition, a reference to that name from an instance method is a reference to the resource in the object DATA DIVISION.

If a resource is defined with a given name in the DATA DIVISION of a factory definition, and there is no resource defined with the same name in a factory method of that factory definition, a reference to that name from a factory method is a reference to the resource in the factory data division.

If a resource is defined within a method, any reference within the method to that resource name is always a reference to the resource in the method.

The normal rules for qualification and uniqueness of reference apply when the same name is associated with more than one resource within a given method data division, object data division, or factory data division.

Chapter 8. Referencing data names, copy libraries, and PROCEDURE DIVISION names

References can be made to external and internal resources. References to data and procedures can be either explicit or implicit.

For more information about rules for qualification, and for explicit and implicit data references, see the following topics:

- [“Uniqueness of reference” on page 67](#)
- [“Data attribute specification” on page 78](#)

Uniqueness of reference

Every user-defined name in a COBOL program is assigned by the user to name a resource for solving a data processing problem. To use a resource, a statement in a COBOL program must contain a reference that uniquely identifies that resource.

To ensure uniqueness of reference, a user-defined name can be qualified. A subscript is required for unique reference to a table element, except as specified in [“Subscripting” on page 72](#). A data-name or function-name, any subscripts, and the specified reference-modifier uniquely reference a data item defined by reference modification.

When the same name has been assigned in separate programs to two or more occurrences of a resource of a given type, and when qualification by itself does not allow the references in one of those programs to differentiate between the identically named resources, then certain conventions that limit the scope of names apply. The conventions ensure that the resource identified is that described in the program containing the reference. For more information about resolving program-names, see [“Resolution of names” on page 66](#).

Unless otherwise specified by the rules for a statement, any subscripts and reference modification are evaluated only once as the first step in executing that statement.

Qualification

A name that exists within a hierarchy of names can be made unique by specifying one or more higher-level names in the hierarchy. The higher-level names are called *qualifiers*, and the process by which such names are made unique is called *qualification*.

Qualification is specified by placing one or more phrases after a user-specified name, with each phrase made up of the word IN or OF followed by a qualifier. (IN and OF are logically equivalent.)

If there is only one 01 level with a given name, that name can be referenced even if it is not unique when the QUALIFY(EXTEND) option is in effect.

You must specify enough qualification to make the name unique; however, it is not always necessary to specify all the levels of the hierarchy. For example, if there is more than one file whose records contain the field EMPLOYEE-NO, but only one of the files has a record named MAIN-RECORD:

- EMPLOYEE-NO OF MAIN-RECORD sufficiently qualifies EMPLOYEE-NO.
- EMPLOYEE-NO OF MAIN-RECORD OF MAIN-FILE is valid but unnecessary.

Qualification rules

The rules for qualifying a name are:

- A name can be qualified even though it does not need qualification except in a REDEFINES clause, in which case it must not be qualified.

- Each qualifier must be of a higher level than the name it qualifies and must be within the same hierarchy.
- If there is more than one combination of qualifiers that ensures uniqueness, any of those combinations can be used.
- If compiler option QUALIFY (EXTEND) is in effect, and if there is only one fully qualified name that matches your combination of qualifiers, that reference will be considered unique, even if the set of qualifiers also matches a partial qualification for a different data item. Fully qualified means every qualifier is specified.

Related references

QUALIFY (*Enterprise COBOL Programming Guide*)

Identical names

When programs are directly or indirectly contained within other programs, each program can use identical user-defined words to name resources.

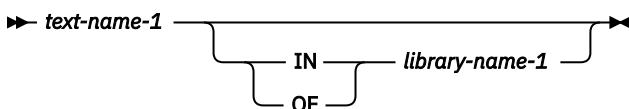
A program references the resources that program describes rather than the same-named resources described in another program, even if the names are different types of user-defined words.

These same rules apply to classes and their contained methods.

References to COPY libraries

If *library-name-1* is not specified, SYSLIB is assumed as the library name.

Format

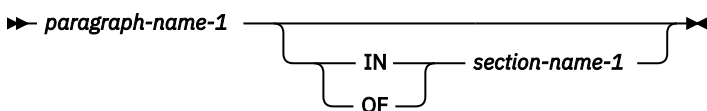


For rules on referencing COPY libraries, see [“COPY statement” on page 696](#).

References to PROCEDURE DIVISION names

PROCEDURE DIVISION names that are explicitly referenced in a program must be unique within a section.

Format 1



Format 2

➡ section-name-1 ➡

A section-name is the highest and only qualifier available for a paragraph-name and must be unique if referenced. (Section-names are described under [“Procedures” on page 265](#).)

If explicitly referenced, a paragraph-name must not be duplicated within a section. When a paragraph-name is qualified by a section-name, the word SECTION must not appear. A paragraph-name need not be qualified when referred to within the section in which it appears. A paragraph-name or section-name that appears in a program cannot be referenced from any other program.

References to DATA DIVISION names

This section discusses the following types of references.

- [“Simple data reference” on page 69](#)
- [“Identifiers” on page 69](#)

Simple data reference

The most basic method of referencing data items in a COBOL program is *simple data reference*, which is *data-name-1* without qualification, subscripting, or reference modification. Simple data reference is used to reference a single elementary or group item.

Format

➡ *data-name-1* ➡

data-name-1

Can be any data description entry.

data-name-1 must be unique in a program.

Identifiers

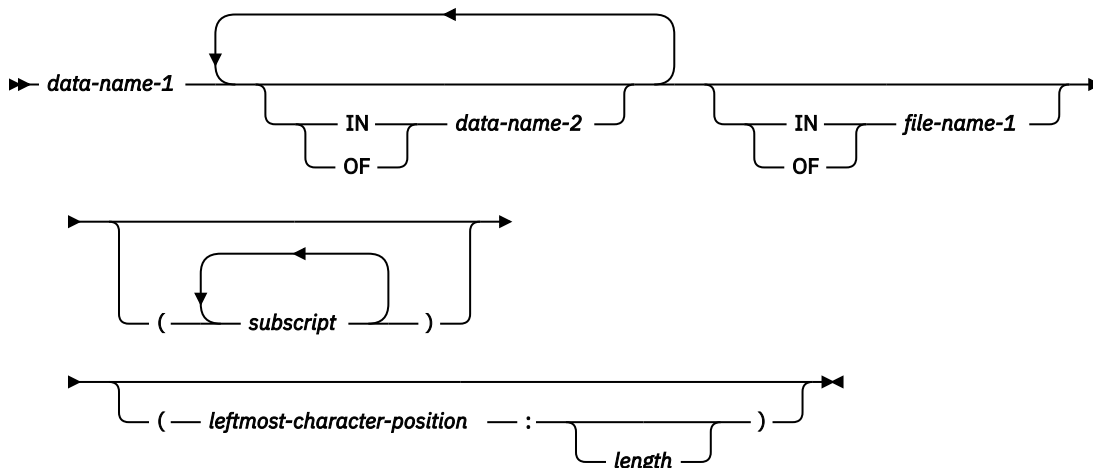
When used in a syntax diagram in this information, the term *identifier* refers to a valid combination of a data-name or function-identifier with its qualifiers, subscripts, and reference-modifiers as required for uniqueness of reference.

Rules for identifiers associated with a format can however specifically prohibit qualification, subscripting, or reference modification.

The term *data-name* refers to a name that must not be qualified, subscripted, or reference modified unless specifically permitted by the rules for the format.

- For a description of qualification, see [“Qualification” on page 67](#).
- For a description of subscripting, see [“Subscripting” on page 72](#).
- For a description of reference modification, see [“Reference modification” on page 75](#).

Format 1



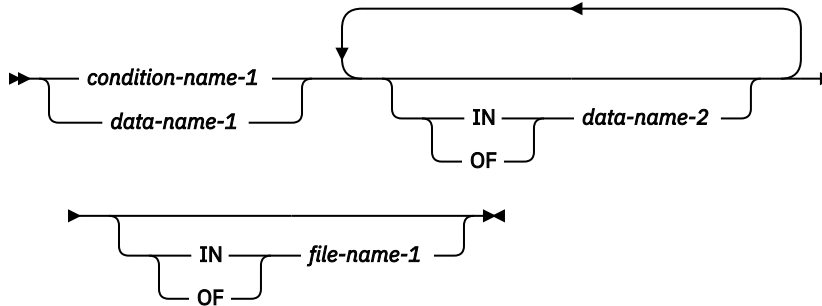
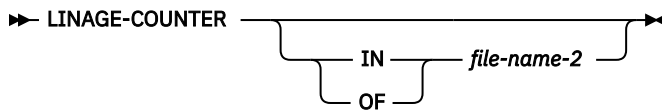
data-name-1 , *data-name-2*

Can be a record-name.

file-name-1

Must be identified by an FD or SD entry in the DATA DIVISION.

file-name-1 must be unique within this program.

Format 2**Format 3*****data-name-1* , *data-name-2***

Can be a record-name.

condition-name-1

Can be referenced by statements and entries either in the program that contains the configuration section or in a program contained within that program.

file-name-1

Must be identified by an FD or SD entry in the DATA DIVISION.

Must be unique within this program.

LINAGE-COUNTER

Must be qualified each time it is referenced if more than one file description entry that contains a LINAGE clause has been specified in the source unit.

file-name-2

Must be identified by the FD or SD entry in the DATA DIVISION. *file-name-2* must be unique within this program.

Duplication of data-names must not occur in those places where the data-names cannot be made unique by qualification.

In the same program, the data-name specified as the subject of the entry whose level-number is 01 that includes the EXTERNAL clause must not be the same data-name specified for any other data description entry that includes the EXTERNAL clause.

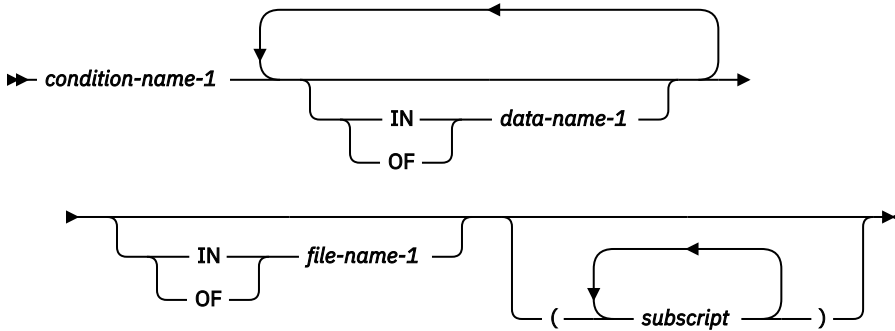
In the same DATA DIVISION, the data description entries for any two data items for which the same data-name is specified must not include the GLOBAL clause.

DATA DIVISION names that are explicitly referenced must either be uniquely defined or made unique through qualification. Unreferenced data items need not be uniquely defined. The highest level in a data hierarchy (a data item associated with a level indicator (FD or SD in the FILE SECTION) or with level-number 01) must be uniquely named if referenced. Data items associated with level-numbers 02 through 49 are successively lower levels of the hierarchy.

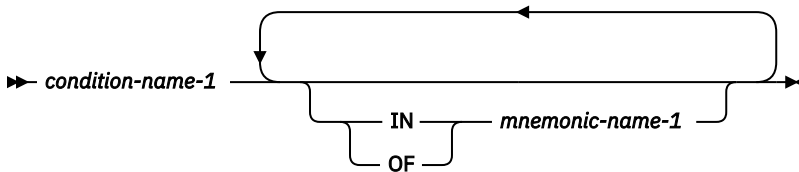
Condition-name

See the syntax and description for details.

Format 1: condition-name in data division



Format 2: condition-name in SPECIAL-NAMES paragraph



condition-name-1

Can be referenced by statements and entries either in the program that contains the definition of *condition-name-1*, or in a program contained within that program.

If explicitly referenced, a condition-name must be unique or be made unique through qualification or subscripting (or both) except when the scope of names by itself ensures uniqueness of reference.

If qualification is used to make a condition-name unique, the associated conditional variable can be used as the first qualifier. If qualification is used, the hierarchy of names associated with the conditional variable itself must be used to make the condition-name unique.

If references to a conditional variable require subscripting, reference to any of its condition-names also requires the same combination of subscripting.

In this information, *condition-name* refers to a condition-name qualified or subscripted, as necessary.

data-name-1

Can be a record-name.

file-name-1

Must be identified by an FD or SD entry in the DATA DIVISION.

file-name-1 must be unique within this program.

mnemonic-name-1

For information about acceptable values for *mnemonic-name-1*, see [“SPECIAL-NAMES paragraph” on page 124](#).

Index-name

An index-name identifies an index. An index can be regarded as a private special register that the compiler generates for working with a table. You name an index by specifying the INDEXED BY phrase in the OCCURS clause that defines a table.

You can use an index-name in only the following language elements:

- SET statements
- PERFORM statements
- SEARCH statements
- Subscripts
- Relation conditions

An index-name is not the same as the name of an index data item, and an index-name cannot be used like a data-name.

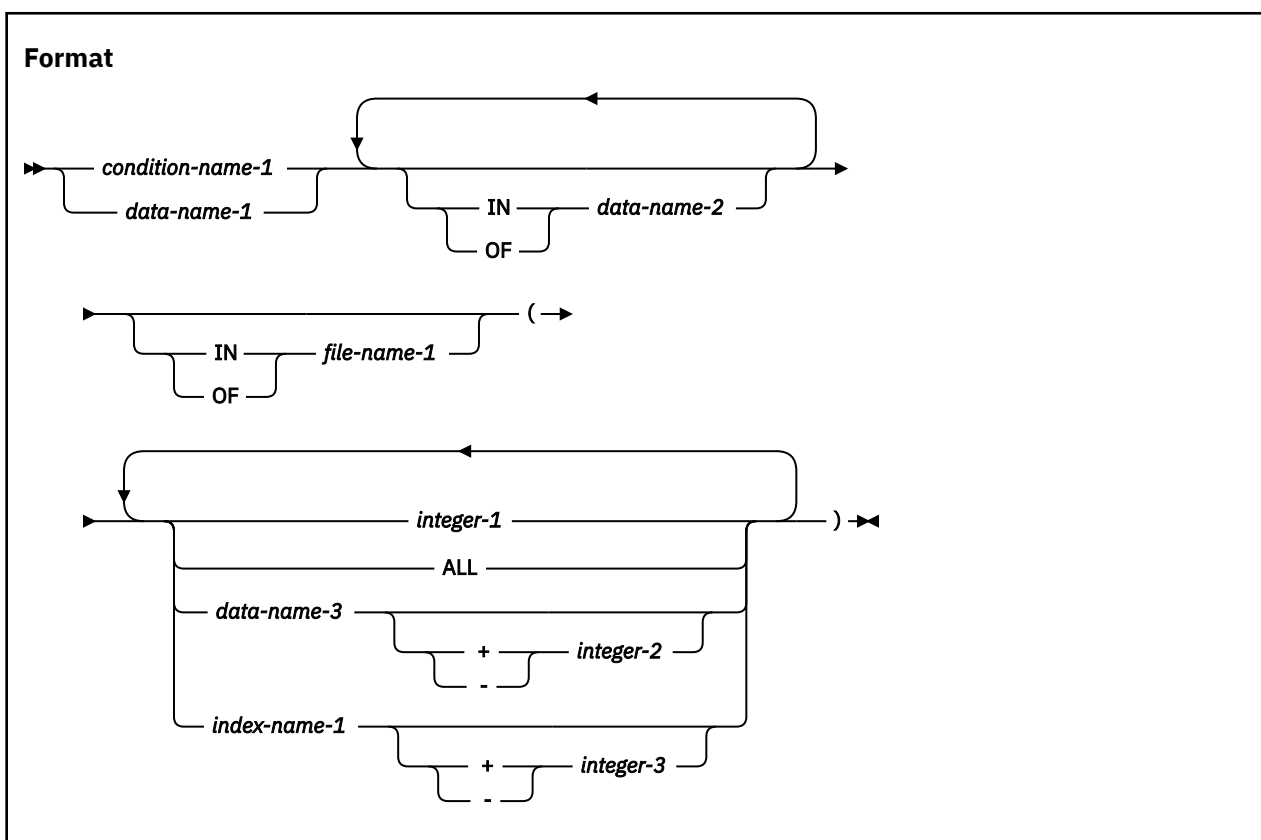
Index data item

An index data item is a data item that can hold the value of an index.

You define an index data item by specifying the USAGE IS INDEX clause in a data description entry. The name of an index data item is a data-name. An index data item can be used anywhere a data-name or identifier can be used, unless stated otherwise in the rules of a particular statement. You can use the SET statement to save the value of an index (referenced by index-name) in an index data item.

Subscripting

Subscripting is a method of providing table references through the use of subscripts. A *subscript* is a positive integer whose value specifies the occurrence number of a table element.



condition-name-1

The conditional variable for *condition-name-1* must contain an OCCURS clause or must be subordinate to a data description entry that contains an OCCURS clause.

data-name-1

Must contain an OCCURS clause or must be subordinate to a data description entry that contains an OCCURS clause.

data-name-2 , file-name-1

Must name data items or records that contain *data-name-1*.

integer-1

Can be signed. If signed, it must be positive.

ALL

Shall not be specified if *condition-name-1* is specified.

Must be used only when the subscripted identifier is used as an intrinsic function argument or to identify a table in a format 2 SORT statement (table SORT statement).

If ALL is specified, the subscript is all of the possible values of a subscript for the associated table as specified in the rules for the functions for which the subscript ALL is allowed.

data-name-3

Must be a numeric elementary item representing an integer.

data-name-3 can be qualified.

index-name-1

Corresponds to a data description entry in the hierarchy of the table being referenced that contains an INDEXED BY phrase that specifies that name.

integer-2 , integer-3

Cannot be signed.

The subscripts, enclosed in parentheses, are written immediately following any qualification for the name of the table element. The number of subscripts in such a reference must equal the number of dimensions in the table whose element is being referenced. That is, there must be a subscript for each OCCURS clause in the hierarchy that contains the data-name including the data-name itself.

When more than one subscript is required, they are written in the order of successively less inclusive dimensions of the data organization. If a multidimensional table is thought of as a series of nested tables and the most inclusive or outermost table in the nest is considered to be the major table with the innermost or least inclusive table being the minor table, the subscripts are written from left to right in the order major, intermediate, and minor.

For example, if TABLE-THREE is defined as:

```
01  TABLE-THREE.
    05  ELEMENT-ONE OCCURS 3 TIMES.
        10  ELEMENT-TWO OCCURS 3 TIMES.
            15  ELEMENT-THREE OCCURS 2 TIMES      PIC X(8).
```

a valid subscripted reference to TABLE-THREE is:

```
ELEMENT-THREE (2 2 1)
```

Subscripted references can also be reference modified. See the third example under “Reference modification examples” on page 76. A reference to an item must not be subscripted unless the item is a table element or an item or condition-name associated with a table element.

Each table element reference must be subscripted except when such reference appears:

- In a USE FOR DEBUGGING statement
- As the subject of a SEARCH statement
- In a REDEFINES clause
- In the KEY IS phrase of an OCCURS clause
- In a format 2 SORT statement (table SORT statement)

In a format 2 SORT statement, subscripting may be specified with the rightmost subscript being the word ALL.

The lowest permissible occurrence number represented by a subscript is 1. The highest permissible occurrence number in any particular case is the maximum number of occurrences of the item as specified in the OCCURS clause.

Subscripting using data-names

When a data-name is used to represent a subscript, it can be used to reference items within different tables. These tables need not have elements of the same size. The same data-name can appear as the only subscript with one item and as one of two or more subscripts with another item. A data-name subscript can be qualified; it cannot be subscripted or indexed. For example, valid subscripted references to TABLE-THREE, assuming that SUB1, SUB2, and SUB3 are all items subordinate to SUBSCRIPT-ITEM, include:

```
ELEMENT-THREE (SUB1 SUB2 SUB3)
```

```
ELEMENT-THREE IN TABLE-THREE (SUB1 OF SUBSCRIPT-ITEM,  
SUB2 OF SUBSCRIPT-ITEM, SUB3 OF SUBSCRIPT-ITEM)
```

Subscripting using index-names (indexing)

Indexing allows such operations as table searching and manipulating specific items. To use indexing, you associate one or more index-names with an item whose data description entry contains an OCCURS clause.

An index associated with an index-name acts as a subscript, and its value corresponds to an occurrence number for the item to which the index-name is associated.

The INDEXED BY phrase, by which the index-name is identified and associated with its table, is an optional part of the OCCURS clause. There is no separate entry to describe the index associated with index-name. At run time, the contents of the index corresponds to an occurrence number for that specific dimension of the table with which the index is associated.

The initial value of an index at run time is undefined, and the index must be initialized before it is used as a subscript. An initial value is assigned to an index with one of the following statements:

- The PERFORM statement with the VARYING phrase
- The SEARCH statement with the ALL phrase
- The SET statement

The use of an integer or data-name as a subscript that references a table element or an item within a table element does not cause the alteration of any index associated with that table.

An index-name can be used to reference any table. However, the element length of the table being referenced and of the table that the index-name is associated with should match. Otherwise, the reference will not be to the same table element in each table, and you might get runtime errors.

Data that is arranged in the form of a table is often searched. The SEARCH statement provides facilities for producing serial and nonserial searches. It is used to search for a table element that satisfies a specific condition and to adjust the value of the associated index to indicate that table element.

To be valid during execution, an index value must correspond to a table element occurrence of neither less than one, nor greater than the highest permissible occurrence number.

For more information about index-names, see [“Index-name” on page 71](#) and [“INDEXED BY phrase” on page 202](#).

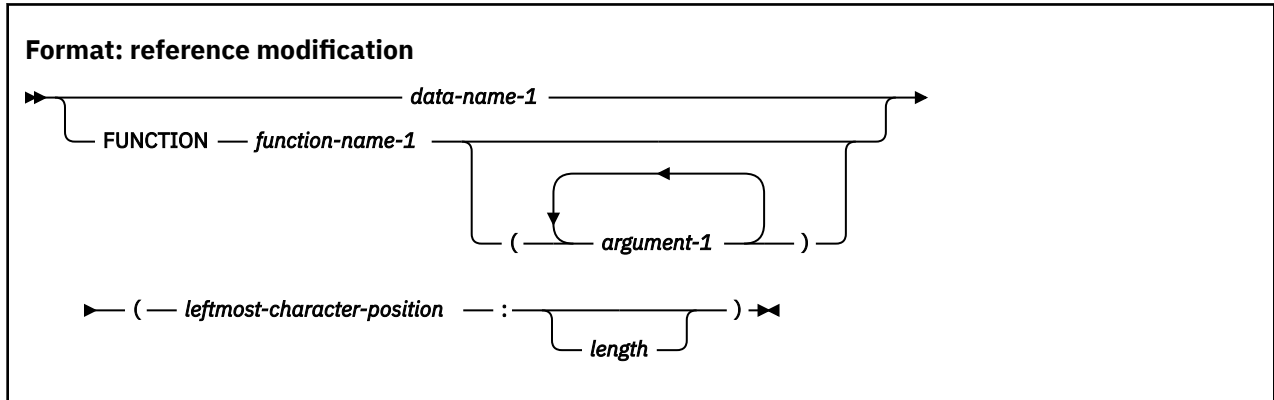
Relative subscripting

In *relative subscripting*, the name of a table element is followed by a subscript of the form data-name or index-name followed by the operator + or -, and a positive or unsigned integer literal.

The operators + and - must be preceded and followed by a space. The value of the subscript used is the same as if the index-name or data-name had been set up or down by the value of the integer. The use of relative indexing does not cause the program to alter the value of the index.

Reference modification

Reference modification defines a data item by specifying a leftmost character and optional length for the data item.



data-name-1

Must reference a data item described explicitly or implicitly with usage DISPLAY, DISPLAY-1, NATIONAL, or UTF-8. A national group item is processed as an elementary data item of category national.

data-name-1 can be qualified or subscripted.

data-name-1 cannot be a dynamic-length group item. If *data-name-1* is a dynamic-length elementary item, it is treated as though it were a fixed-length item whose length is the same as the current length of the dynamic-length elementary item. When used as a reference modified receiver in a PROCEDURE DIVISION statement, the current length of the dynamic-length elementary item is not modified.

leftmost-character-position

Must be an arithmetic expression. The evaluation of *leftmost-character-position* must result in a positive nonzero integer that is less than or equal to the number of characters in the data item referenced by *data-name-1*.

length

Must be an arithmetic expression.

The evaluation of *length* must result in a positive nonzero integer.

The sum of *leftmost-character-position* and *length* minus the value 1 must be less than or equal to the number of character positions in *data-name-1*. If *length* is omitted, the length used will be equal to the number of character positions in *data-name-1* plus 1, minus *leftmost-character-position*.

function-name-1

Must reference an alphanumeric or national function.

For usages DISPLAY-1 and NATIONAL, each character position occupies 2 bytes. Reference modification operates on whole character positions and not on the individual bytes of the characters in usages DISPLAY-1 and NATIONAL. For usage DISPLAY, reference modification operates as though each character were a single-byte character.

Unless otherwise specified, reference modification is allowed anywhere an identifier or function-identifier that references a data item or function with the same usage as the reference-modified data item is permitted.

Each character position referenced by *data-name-1* or *function-name-1* is assigned an ordinal number incrementing by one from the leftmost position to the rightmost position. The leftmost position is assigned the ordinal number one. If the data description entry for *data-name-1* contains a SIGN IS SEPARATE clause, the sign position is assigned an ordinal number within that data item.

If *data-name-1* is described with usage DISPLAY and category numeric, numeric-edited, alphabetic, alphanumeric-edited, or external floating-point, *data-name-1* is operated upon for purposes of reference

modification as if it were redefined as a data item of category alphanumeric with the same size as the data item referenced by *data-name-1*.

If *data-name-1* is described with usage NATIONAL and category numeric, numeric-edited, national-edited, or external floating-point, *data-name-1* is operated upon for purposes of reference modification as if it were redefined as a data item of category national with the same size as the data item referenced by *data-name-1*.

If *data-name-1* is a national group item, *data-name-1* is processed as an elementary data item of category national.

Reference modification creates a unique data item that is a subset of *data-name-1* or a subset of the value referenced by *function-name-1* and its arguments, if any. This unique data item is considered an elementary data item without the JUSTIFIED clause.

When a function is reference-modified, the unique data item has class, category, and usage national if the type of the function is national; otherwise, it has class and category alphanumeric and usage display.

When *data-name-1* is reference-modified, the unique data item has the same class, category, and usage as that defined for the data item referenced by *data-name-1* except that:

- If *data-name-1* has category national-edited, the unique data item has category national.
- If *data-name-1* has usage NATIONAL and category numeric-edited, numeric, or external floating-point, the unique data item has category national.
- If *data-name-1* has usage DISPLAY, and category numeric-edited, alphanumeric-edited, numeric, or external floating-point, the unique data item has category alphanumeric.
- If *data-name-1* references an alphanumeric group item, the unique data item is considered to have usage DISPLAY and category alphanumeric.
- If *data-name-1* references a national group item, the unique data item has usage NATIONAL and category national.

If *length* is not specified, the unique data item created extends from and includes the character position identified by *leftmost-character-position* up to and including the rightmost character position of the data item referenced by *data-name-1*.

Evaluation of operands

Reference modification for an operand is evaluated as follows:

- If subscripting is specified for the operand, the reference modification is evaluated immediately after evaluation of the subscript.
- If subscripting is not specified for the operand, the reference modification is evaluated at the time subscripting would be evaluated if subscripts had been specified.

Reference modification examples

The statements in the examples transfer the first 10 characters of the data-item referenced by WHOLE-NAME to the data-item referenced by FIRST-NAME.

```
77  WHOLE-NAME  PIC X(25).
77  FIRST-NAME  PIC X(10).

77  START-P     PIC 9(4) BINARY VALUE 1.
77  STR-LENGTH  PIC 9(4) BINARY VALUE 10.

...
    MOVE WHOLE-NAME(1:10) TO FIRST-NAME.
    MOVE WHOLE-NAME(START-P:STR-LENGTH) TO FIRST-NAME.
```


The following statement transfers the last 15 characters of the data-item referenced by WHOLE-NAME to the data-item referenced by LAST-NAME.

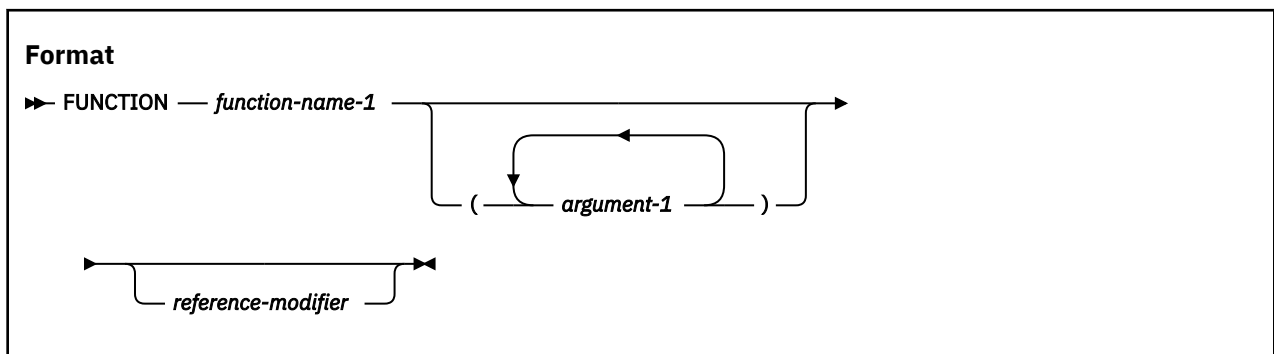
```
77 WHOLE-NAME PIC X(25).
77 LAST-NAME  PIC X(15).
...
MOVE WHOLE-NAME(11:) TO LAST-NAME.
```

The following statement transfers the fourth and fifth characters of the third occurrence of TAB to the variable SUFFIX.

```
01 TABLE-1.
02 TAB OCCURS 10 TIMES PICTURE X(5).
77 SUFFIX PICTURE X(2).
...
MOVE TAB OF TABLE-1 (3) (4:2) TO SUFFIX.
```

Function-identifier

A *function-identifier* is a sequence of character strings and separators that uniquely references the data item that results from the evaluation of a function.



argument-1

Must be an identifier, literal (other than a figurative constant), or arithmetic expression.

For more information about arguments to intrinsic functions, see [Part 7, “Intrinsic functions,” on page 501](#).

For more information about arguments to user-defined functions, see [Chapter 27, “Procedure division structure,” on page 257](#).

function-name-1

function-name-1 must be one of the intrinsic function names or a user-defined function name.

reference-modifier

Can be specified only for functions of the type alphanumeric or national.

A function-identifier that makes reference to an alphanumeric or national function can be specified anywhere that a data item of category alphanumeric or category national, respectively, can be referenced and where references to functions are not specifically prohibited, except as follows:

- As a receiving operand of any statement
- Where a data item is required to have particular characteristics (such as class and category, size, sign, and permissible values) and the evaluation of the function according to its definition and the particular arguments specified would not have these characteristics

A function-identifier that makes reference to an integer or numeric function can be used wherever an arithmetic expression can be used.

Data attribute specification

Explicit data attributes are data attributes that you specify in COBOL coding. *Implicit data attributes* are default values. If you do not explicitly code a data attribute, the compiler assumes a default value.

For example, you need not specify the USAGE of a data item. If USAGE is omitted and the symbol N or symbol U is not specified in the PICTURE clause, the default is USAGE DISPLAY, which is the implicit data attribute. When PICTURE symbol N is used, USAGE DISPLAY-1 is the default when the NSYMBOL(DBCS) compiler option is in effect; USAGE NATIONAL is the default when the NSYMBOL(NATIONAL) compiler option is in effect. Similarly, when PICTURE symbol U is used, USAGE UTF-8 is assumed. These are implicit data attributes.

Chapter 9. Transfer of control

In the PROCEDURE DIVISION, unless there is an *explicit* control transfer or there is no next executable statement, program flow transfers control from statement to statement in the order in which the statements are written. This normal program flow is an *implicit* transfer of control.

In addition to the implicit transfers of control between consecutive statements, implicit transfer of control also occurs when the normal flow is altered without the execution of a procedure branching statement. The following examples show *implicit* transfers of control, overriding statement-to-statement transfer of control:

- After execution of the last statement of a procedure that is executed under control of another COBOL statement, control implicitly transfers. (COBOL statements that control procedure execution are, for example, MERGE, PERFORM, SORT, and USE.) Further, if a paragraph is being executed under the control of a PERFORM statement that causes iterative execution, and that paragraph is the first paragraph in the range of that PERFORM statement, an implicit transfer of control occurs between the control mechanism associated with that PERFORM statement and the first statement in that paragraph for each iterative execution of the paragraph.
- During SORT or MERGE statement execution, control is implicitly transferred to an input or output procedure.
- During XML PARSE statement execution, control is implicitly transferred to a processing procedure.
- During execution of any COBOL statement that causes execution of a declarative procedure, control is implicitly transferred to that procedure.
- At the end of execution of any declarative procedure, control is implicitly transferred back to the control mechanism associated with the statement that caused its execution.

COBOL also provides *explicit* control transfers through the execution of any procedure branching, program call, or conditional statement. (Lists of procedure branching and conditional statements are contained in [“Statement categories” on page 290.](#))

Definition: The term *next executable statement* refers to the next COBOL statement to which control is transferred, according to the rules mentioned previously. There is no next executable statement under the following circumstances:

- When the program contains no PROCEDURE DIVISION
- Following the last statement in a declarative section when the paragraph in which it appears is not being executed under the control of some other COBOL statement
- Following the last statement in a program or method when the paragraph in which it appears is not being executed under the control of some other COBOL statement in that program
- Following the last statement in a declarative section when the statement is in the range of an active PERFORM statement executed in a different section and this last statement of the declarative section is not also the last statement of the procedure that is the exit of the active PERFORM statement
- Following a STOP RUN statement or EXIT PROGRAM statement that transfers control outside the COBOL program
- Following a GOBACK statement that transfers control outside the COBOL program
- Following an EXIT METHOD statement that transfers control outside the COBOL method
- The end program or end method marker

When there is no next executable statement and control is not transferred outside the COBOL program, the program flow of control is undefined unless the program execution is in the nondeclarative procedures portion of a program under control of a CALL statement, in which case an implicit EXIT PROGRAM statement is executed.

Similarly, if control reaches the end of the PROCEDURE DIVISION of a method and there is no next executable statement, an implicit EXIT METHOD statement is executed.

Part 2. COBOL source unit structure

Chapter 10. COBOL program structure

A COBOL source program is a syntactically correct set of COBOL statements.

Nested programs

A *nested program* is a program that is contained in another program. Contained programs can reference some of the resources of the programs that contain them. If program B is contained in program A, it is *directly* contained if there is no program contained in program A that also contains program B. Program B is *indirectly* contained in program A if there exists a program contained in program A that also contains program B. For more information about nested programs, see *Nested programs* in the *Enterprise COBOL Programming Guide*.

Object program

An *object program* is a set or group of executable machine language instructions and other material designed to interact with data to provide problem solutions. An object program is generally the machine language result of the operation of a COBOL compiler on a source program. The term object program also refers to the methods that result from compiling a class definition.

Run unit

A *run unit* is one or more object programs that interact with one another and that function at run time as an entity to provide problem solutions.

Sibling program

Sibling programs are programs that are directly contained in the same program.

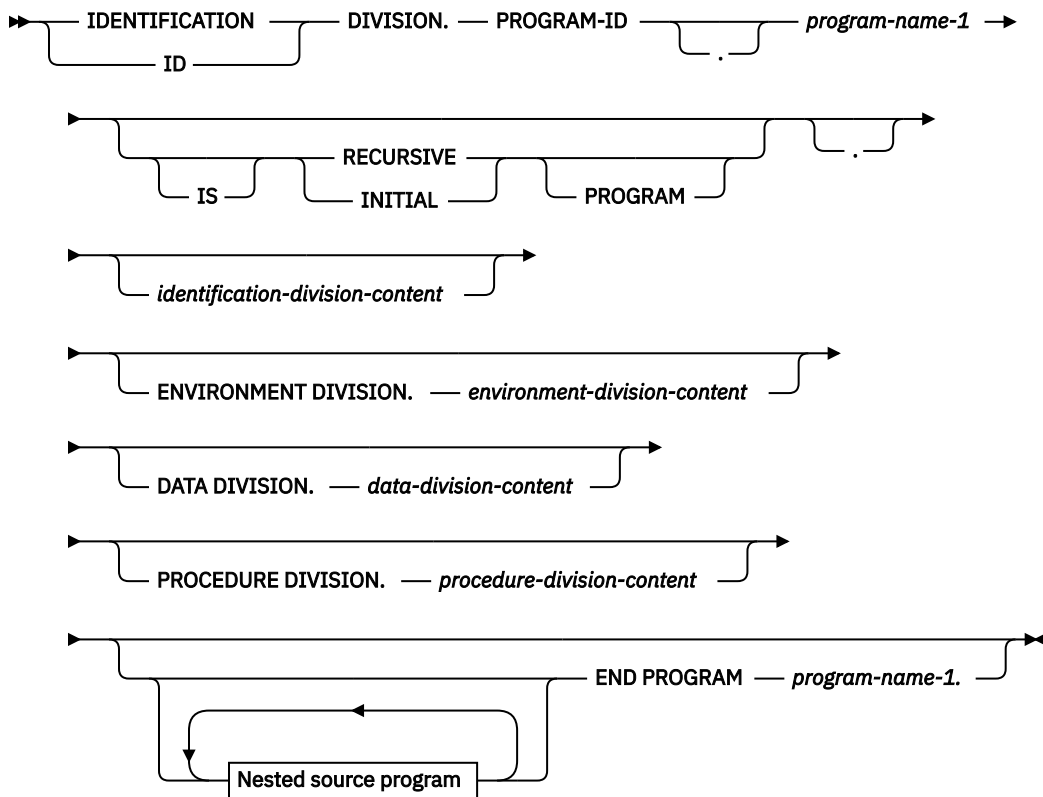
With the exception of the COPY and REPLACE statements and the end program marker, the statements, entries, paragraphs, and sections of a COBOL source program are grouped into the following four divisions:

- IDENTIFICATION DIVISION
- ENVIRONMENT DIVISION
- DATA DIVISION
- PROCEDURE DIVISION

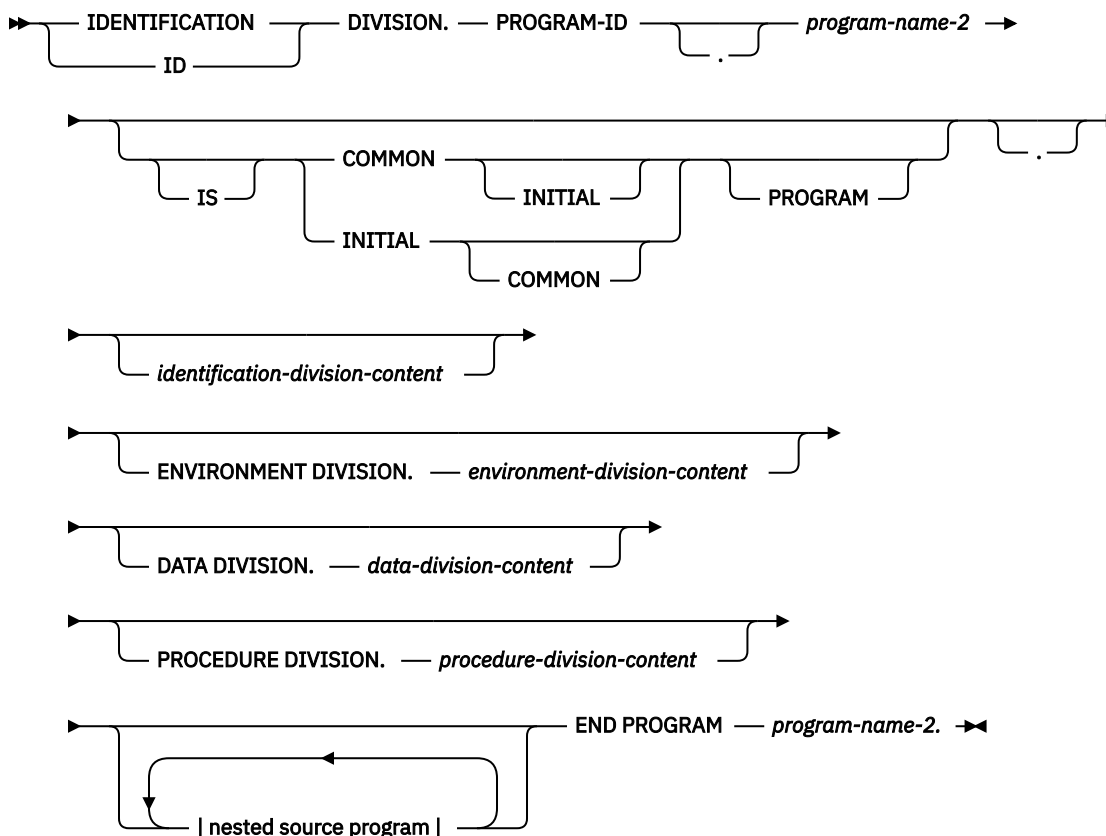
The end of a COBOL source program is indicated by the END PROGRAM marker. If there are no nested programs, the absence of additional source program lines also indicates the end of a COBOL program.

The following format is for the entries and statements that constitute a separately compiled COBOL source program.

Format: COBOL source program



nested source program



A sequence of separate COBOL programs can also be input to the compiler. The following format is for the entries and statements that constitute a sequence of source programs (batch compile).

Format: sequence of COBOL source programs



END PROGRAM *program-name*

An end program marker separates each program in the sequence of programs. *program-name* must be identical to a program-name declared in a preceding program-ID paragraph.

program-name can be specified either as a user-defined word or in an alphanumeric literal. Either way, *program-name* must follow the rules for forming program-names. *program-name* cannot be a figurative constant. Any lowercase letters in the literal are folded to uppercase.

An end program marker is optional for the last program in the sequence only if that program does not contain any nested source programs.

Nested programs

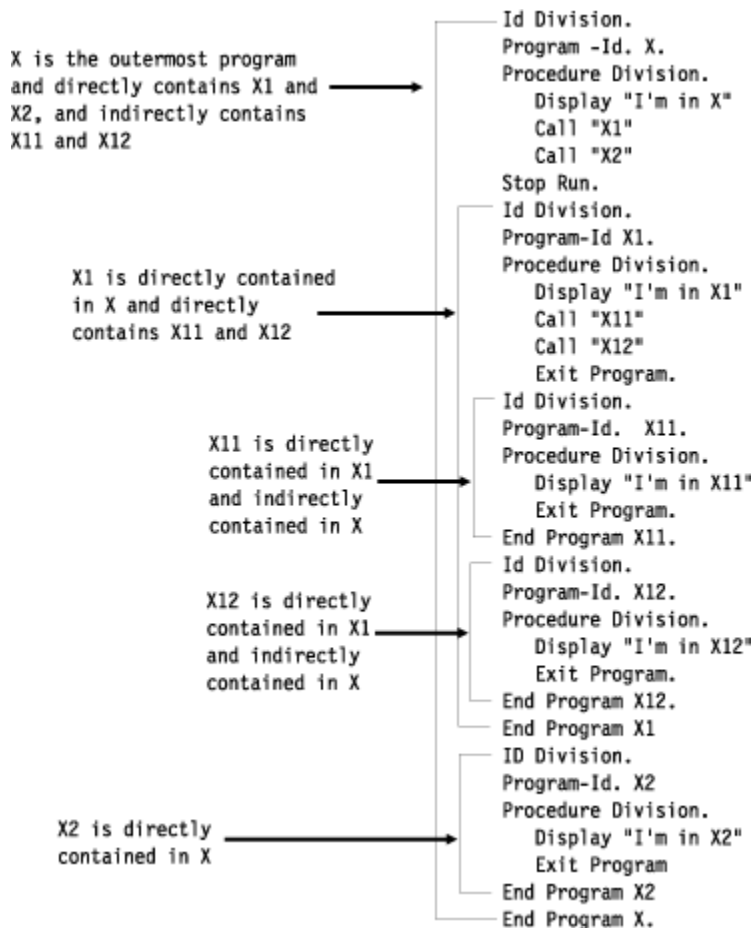
A COBOL program can contain other COBOL programs, which in turn can contain still other COBOL programs. These contained programs are called *nested programs*. Nested programs can be *directly* or *indirectly* contained in the containing program.

Nested programs are not supported for programs compiled with the THREAD option.

In the following code fragment, program Outer-program *directly* contains program Inner-1. Program Inner-1 *directly* contains program Inner-1a, and Outer-program *indirectly* contains Inner-1a:

```
Id division.
Program-id. Outer-program.
  Procedure division.
    Call "Inner-1".
    Stop run.
Id division.
Program-id. Inner-1
  ...
  Call Inner-1a.
  Stop run.
Id division.
Program-id. Inner-1a.
  ...
End Inner-1a.
End Inner-1.
End Outer-program.
```

The following figure describes a more complex nested program structure with directly and indirectly contained programs.



Conventions for program-names

The program-name of a program is specified in the PROGRAM-ID paragraph of the program's IDENTIFICATION DIVISION. A program-name can be referenced only by the CALL statement, the CANCEL statement, the SET statement, or the END PROGRAM marker.

Names of programs that constitute a run unit are not necessarily unique, but when two programs in a run unit are identically named, at least one of the programs must be directly or indirectly contained within another separately compiled program that does not contain the other of those two programs.

A separately compiled program and all of its directly and indirectly contained programs must have unique program-names within that separately compiled program.

Rules for program-names

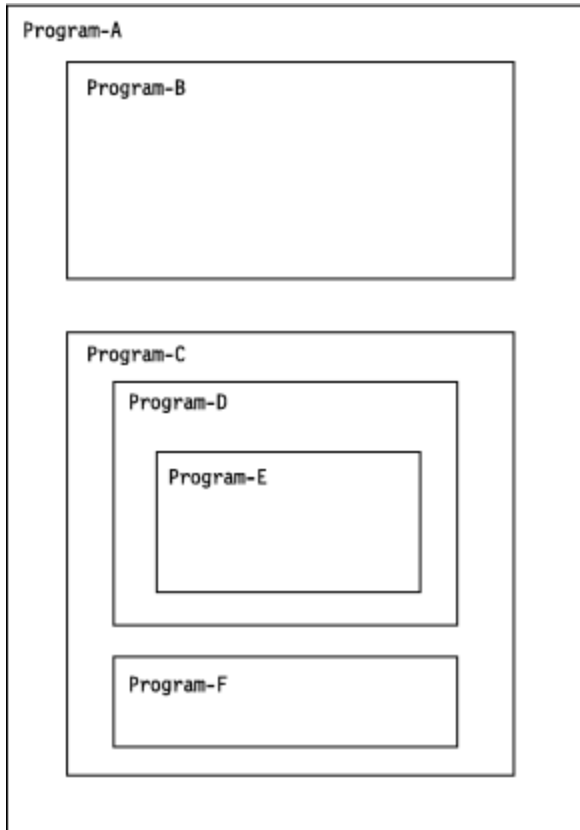
The following rules define the scope of a program-name:

- If the program-name is that of a program that does not possess the COMMON attribute and that program is directly contained within another program, that program-name can be referenced only by statements included in that containing program.
- If the program-name is that of a program that does possess the COMMON attribute and that program is directly contained within another program, that program-name can be referenced only by statements included in the containing program and any programs directly or indirectly contained within that containing program, except that program possessing the COMMON attribute and any programs contained within it.
- If the program-name is that of a program that is separately compiled, that program-name can be referenced by statements included in any other program in the run unit, except programs it directly or indirectly contains.

The mechanism used to determine which program to call is as follows:

- If one of two programs that have the same name as that specified in the CALL statement is directly contained within the program that includes the CALL statement, that program is called.
- If one of two programs that have the same name as that specified in the CALL statement possesses the COMMON attribute and is directly contained within another program that directly or indirectly contains the program that includes the CALL statement, that common program is called unless the calling program is contained within that common program.
- Otherwise, the separately compiled program is called.

The following rules apply to referencing a program-name of a program that is contained within another program. For this discussion, Program-A contains Program-B and Program-C; Program-C contains Program-D and Program-F; and Program-D contains Program-E.



If Program-D does not possess the COMMON attribute, then Program-D can be referenced only by the program that directly contains Program-D, that is, Program-C.

If Program-D does possess the COMMON attribute, then Program-D can be referenced by Program-C (because Program-C contains Program-D) and by any programs contained in Program-C except for programs contained in Program-D. In other words, if Program-D possesses the COMMON attribute, Program-D can be referenced in Program-C and Program-F but not by statements in Program-E, Program-A, or Program-B.

Chapter 11. COBOL class definition structure

Enterprise COBOL provides object-oriented syntax to facilitate interoperation of COBOL and Java programs.

You can use object-oriented syntax to:

- Define classes, with methods and data implemented in COBOL
- Create instances of Java or COBOL classes
- Invoke methods on Java or COBOL objects
- Write classes that inherit from Java classes or from other COBOL classes
- Define and invoke overloaded methods

Basic Java-oriented object capabilities are accessed directly through COBOL language features. Additional capabilities are available to the COBOL programmer by calling services through the Java Native Interface (JNI), as described in *Accessing JNI services* in the *Enterprise COBOL Programming Guide*.

Java programs can be multithreaded, and Java interoperation requires toleration of asynchronous signals. Therefore, to mix COBOL with these Java programs, you must use the thread enablement provided by the `THREAD` compiler option, as described in *THREAD* in the *Enterprise COBOL Programming Guide*.

Java String data is represented at run time in Unicode. The Unicode support provided in Enterprise COBOL with the national data type enables COBOL programs to exchange String data with Java.

The following entities and concepts are used in object-oriented COBOL for Java interoperability:

Class

The entity that defines operations and state for zero, one, or more object instances and defines operations and state for a common object (a factory object) that is shared by multiple object instances.

You create object instances using the `NEW` operand of the COBOL `INVOKE` statement or using a Java class instance creation expression.

Object instances are automatically freed by the Java runtime system's garbage collection when they are no longer in use. You cannot explicitly free individual objects.

Instance method

Procedural code that defines one of the operations supported for the object instances of a class. Instance methods introduced by a COBOL class are defined within the object paragraph of the class definition.

COBOL instance methods are equivalent to public nonstatic methods in Java.

You execute instance methods on a particular object instance by using a COBOL `INVOKE` statement or a Java method invocation expression.

Instance data

Data that defines the state of an individual object instance. Instance data in a COBOL class is defined in the `WORKING-STORAGE SECTION` of the `DATA DIVISION` within the object paragraph of a class definition.

COBOL instance data is equivalent to private nonstatic member data in a Java class.

The state of an object also includes the state of the instance data introduced by inherited classes. Each instance object has its own copy of the instance data defined within its class definition and its own copy of the instance data defined in inherited classes.

You can access COBOL object instance data only from within COBOL instance methods defined in the class definition that defines the data.

You can initialize object instance data with VALUE clauses or you can write an instance method to perform custom initialization.

Factory method, static method

Procedural code that defines one of the operations supported for the common factory object of the class. COBOL factory methods are defined within the factory paragraph of a class definition. Factory methods are associated with a class, not with any individual instance object of the class.

COBOL factory methods are equivalent to public static methods in Java.

You execute COBOL factory methods from COBOL using an INVOKE statement that specifies the class-name as the first operand. You execute them from a Java program using a static method invocation expression.

A factory method cannot operate directly on instance data of its class, even though the data is described in the same class definition; a factory method must invoke instance methods to act on instance data.

COBOL factory methods are typically used to define customized methods that create object instances. For example, you can code a customized factory method that accepts initial values as parameters, creates an instance object using the NEW operand of the INVOKE statement, and then invokes a customized instance method passing those initial values as arguments for use in initializing the instance object.

Factory data, static data

Data associated with a class, rather than with an individual object instance. COBOL factory data is defined in the WORKING-STORAGE SECTION of the DATA DIVISION within the factory paragraph of a class definition.

COBOL factory data is equivalent to private static data in Java.

There is a single copy of factory data for a class. Factory data is associated only with the class and is shared by all object instances of the class. It is not associated with any particular instance object. A factory data item might be used, for example, to keep a count of the number of instance objects that have been created.

You can access COBOL factory data only within COBOL factory methods defined in the same class definition.

Inheritance

Inheritance is a mechanism whereby a class definition (the inheriting class) acquires the methods, data descriptions, and file descriptions written in another class definition (the inherited class). When two classes in an inheritance relationship are considered together, the inheriting class is the *subclass* (or derived class or child class); the inherited class is the *superclass* (or parent class). The inheriting class also indirectly acquires the methods, data descriptions, and file descriptions that the parent class inherited from its parent class.

A COBOL class must inherit from exactly one parent class, which can be implemented in COBOL or Java.

Every COBOL class must inherit directly or indirectly from the java.lang.Object class.

Instance variable

An individual data item defined in the DATA DIVISION of an object paragraph.

Java Native Interface (JNI)

A facility of Java designed for interoperability with non-Java programs.

Java Native Interface (JNI) environment pointer

A pointer used to obtain the address of the JNI environment structure used for calling JNI services. The COBOL special register JNIENVPTR is provided for referencing the JNI environment pointer.

Object reference

A data item that contains information used to identify and reference an individual object. An object reference can refer to an object that is an instance of a Java or COBOL class.

Subclass

A class that inherits from another class; also called a *derived class* or *child class* of the inherited class.

Superclass

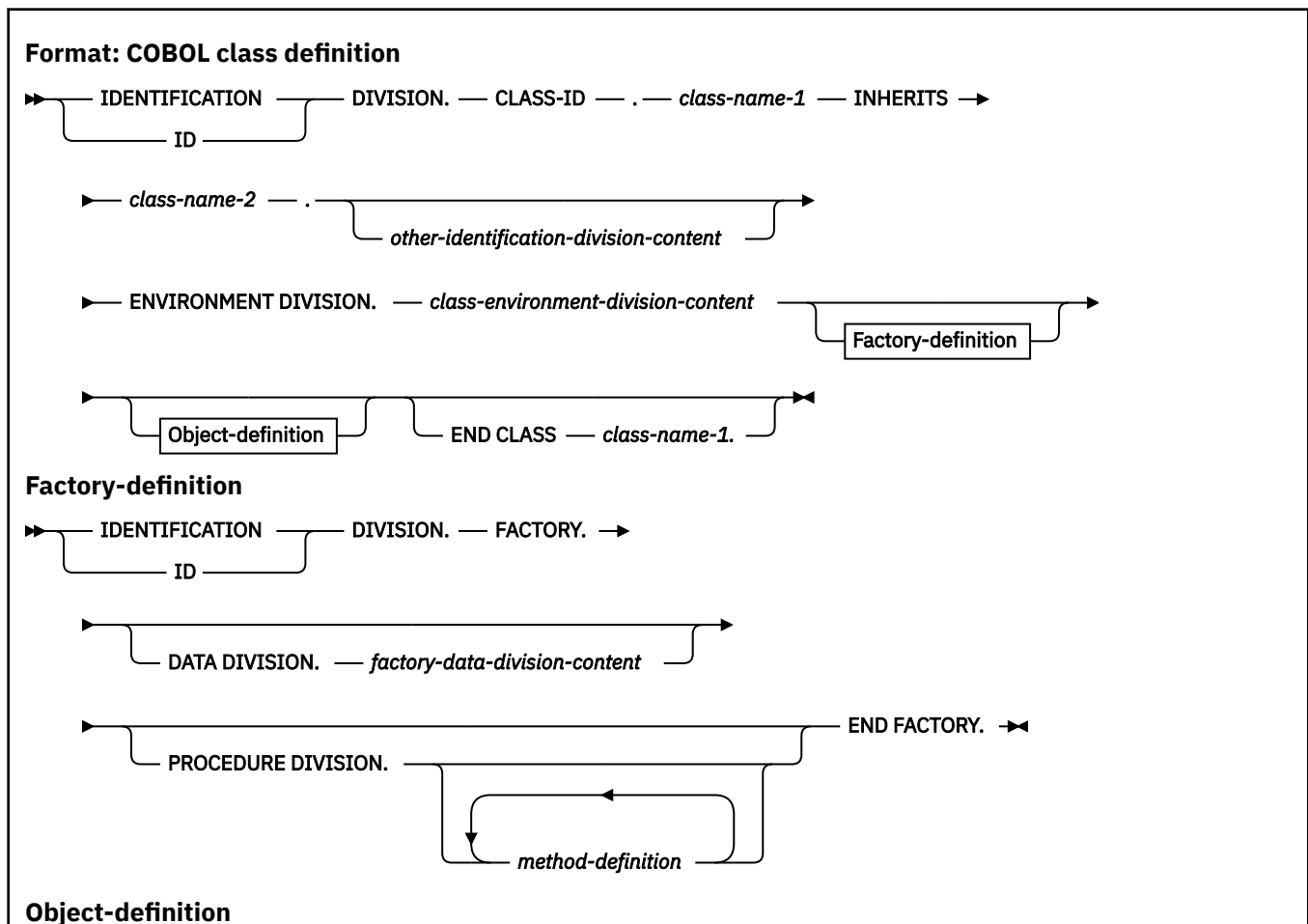
A class that is inherited by another class; also called a *parent class* of the inheriting class.

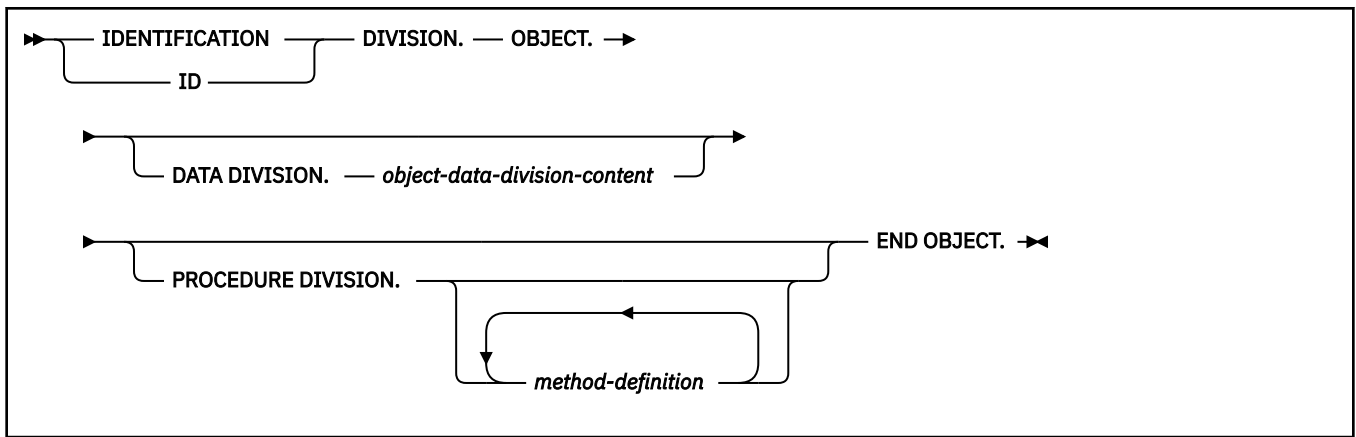
With the exception of the COPY and REPLACE statements and the END CLASS marker, the statements, entries, paragraphs, and sections of a COBOL class definition are grouped into the following structure:

- IDENTIFICATION DIVISION
- ENVIRONMENT DIVISION (configuration section only)
- Factory definition
 - IDENTIFICATION DIVISION
 - DATA DIVISION
 - PROCEDURE DIVISION (containing one or more method definitions)
- Object definition
 - IDENTIFICATION DIVISION
 - DATA DIVISION
 - PROCEDURE DIVISION (containing one or more method definitions)

The end of a COBOL class definition is indicated by the END CLASS marker.

The following format is for a COBOL class definition.





END CLASS

Specifies the end of a class definition.

END FACTORY

Specifies the end of a factory definition.

END OBJECT

Specifies the end of an object definition.

Chapter 12. COBOL method definition structure

A COBOL method definition describes a method. You can specify method definitions only within the factory paragraph and the object paragraph of a class definition.

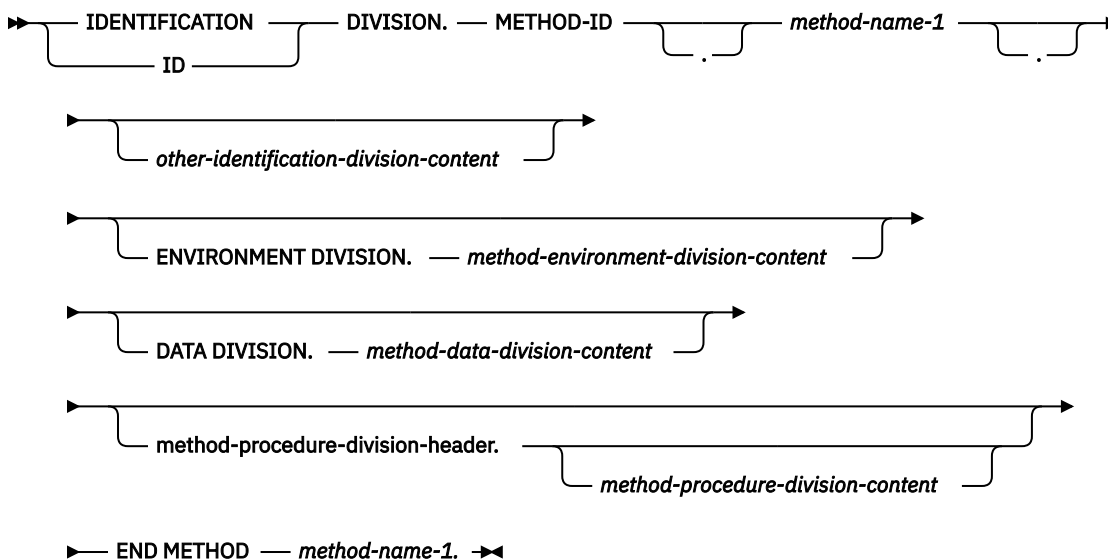
With the exception of COPY and REPLACE statements and the END METHOD marker, the statements, entries, paragraphs, and sections of a COBOL method definition are grouped into the following four divisions:

- IDENTIFICATION DIVISION
- ENVIRONMENT DIVISION (input-output section only)
- DATA DIVISION
- PROCEDURE DIVISION

The end of a COBOL method definition is indicated by the END METHOD marker.

The following format is for a COBOL method definition.

Format: method definition



METHOD-ID

Identifies a method definition. See [Chapter 19, “METHOD-ID paragraph,” on page 111](#) for details.

method-procedure-division-header

Indicates the start of the PROCEDURE DIVISION and identifies method parameters and the returning item, if any. See [“The PROCEDURE DIVISION header” on page 258](#) for details.

END METHOD

Specifies the end of a method definition.

Methods defined in an object definition are *instance methods*. An instance method in a given class can access:

- Data defined in the DATA DIVISION of the object paragraph of that class (*instance data*)
- Data defined in the DATA DIVISION of that instance method (*method data*)

An instance method cannot directly access instance data defined in a parent class, factory data defined in its own class, or method data defined in another method of its class. It must invoke a method to access such data.

Methods defined in a factory definition are *factory methods*. A factory method in a given class can access:

- Data defined in the DATA DIVISION of the factory paragraph of that class (*factory data*)
- Data defined in the DATA DIVISION of that factory method (*method data*)

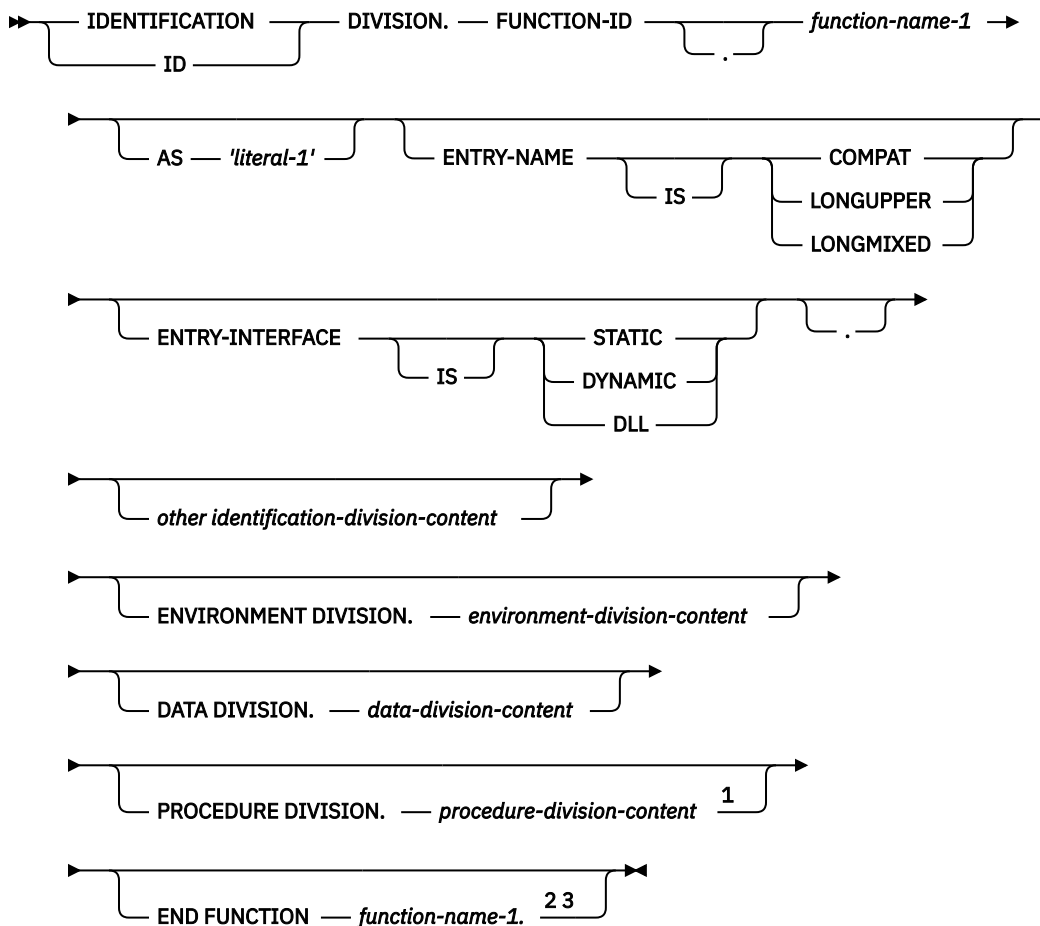
A factory method cannot directly access factory data defined in a parent class, instance data defined in its own class, or method data defined in another method of its class. It must invoke a method to access such data.

Methods can be invoked from COBOL programs and methods, and they can be invoked from Java programs. A method can execute an INVOKE statement that directly or indirectly invokes itself. Therefore, COBOL methods are implicitly recursive (unlike COBOL programs, which support recursion only if the RECURSIVE attribute is specified in the program-ID paragraph.)

Chapter 13. COBOL user-defined function definition structure

A COBOL user-defined function definition describes a user-defined function.

Format: COBOL user-defined function definition



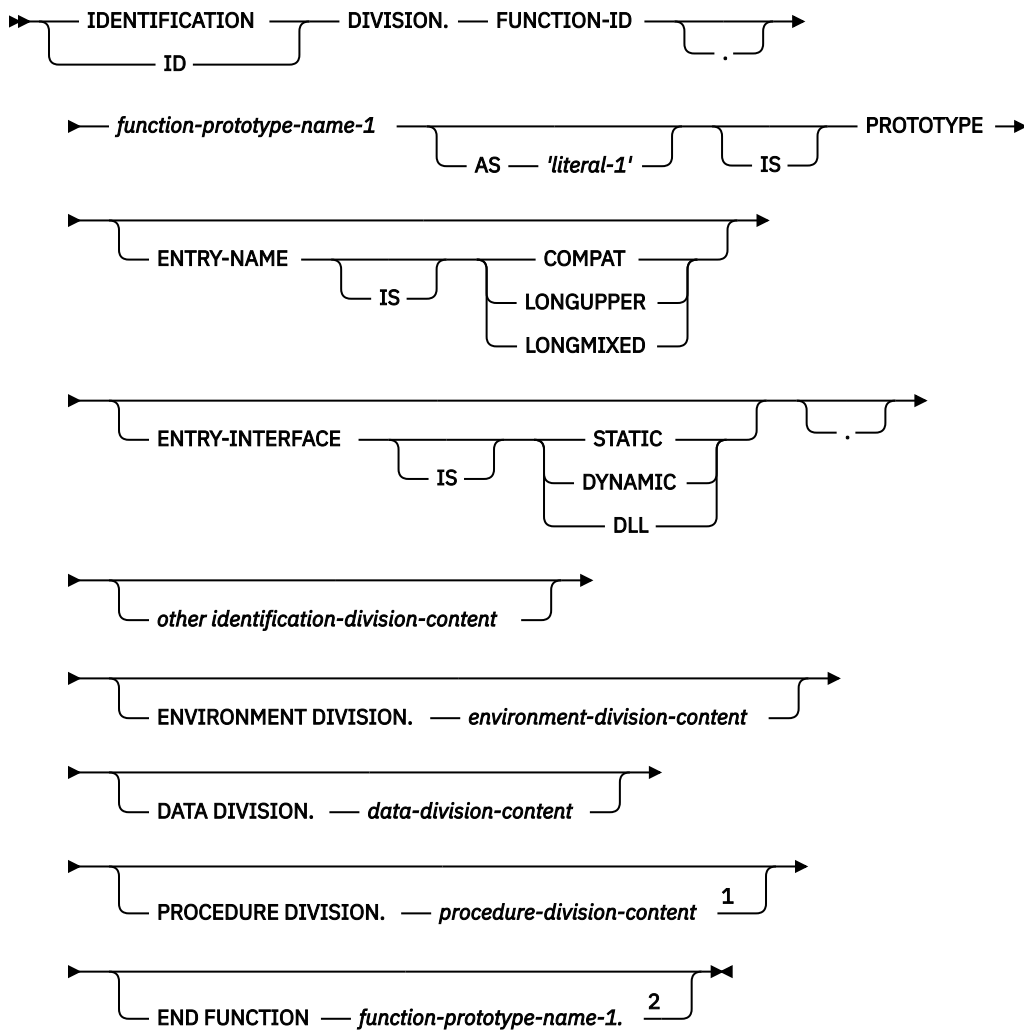
Notes:

- ¹ The RETURNING phrase of the PROCEDURE DIVISION header is required.
- ² User-defined function definitions cannot be nested within any programs, user-defined function definitions, methods, or classes.
- ³ Rules and behaviors applying to programs and program definitions generally also apply to user-defined functions and their definitions, except where explicitly specified.

Chapter 14. COBOL function prototype definition structure

A COBOL function prototype definition describes a function prototype.

Format: COBOL function prototype definition



Notes:

¹ The RETURNING phrase of the PROCEDURE DIVISION header is required.

² Function prototype definitions cannot be nested within any programs, user-defined function definitions, methods, classes, or function prototype definitions.

Part 3. IDENTIFICATION DIVISION

The IDENTIFICATION DIVISION must be the first division in each COBOL source program, factory definition, object definition, method definition, and function definition. The identification division names the program, class, or method and identifies the factory definition, object definition, or function definition. The IDENTIFICATION DIVISION can include the date a program, class, method, or function was written, the date of compilation, and other such documentary information.

Program IDENTIFICATION DIVISION

For a program, the first paragraph of the IDENTIFICATION DIVISION must be the PROGRAM-ID paragraph. The other paragraphs are optional and can appear in any order.

For details, see [Chapter 15, “PROGRAM-ID paragraph,” on page 101.](#)

Class IDENTIFICATION DIVISION

For a class, the first paragraph of the IDENTIFICATION DIVISION must be the CLASS-ID paragraph. The other paragraphs are optional and can appear in any order.

For details, see [Chapter 16, “CLASS-ID paragraph,” on page 105.](#)

Factory IDENTIFICATION DIVISION

A factory IDENTIFICATION DIVISION contains only a factory paragraph header.

For details, see [Chapter 17, “FACTORY paragraph,” on page 107.](#)

Object IDENTIFICATION DIVISION

An object IDENTIFICATION DIVISION contains only an object paragraph header.

For details, see [Chapter 18, “OBJECT paragraph,” on page 109.](#)

Method IDENTIFICATION DIVISION

For a method, the first paragraph of the IDENTIFICATION DIVISION must be the METHOD-ID paragraph. The other paragraphs are optional and can appear in any order.

For details, see [Chapter 19, “METHOD-ID paragraph,” on page 111.](#)

Function IDENTIFICATION DIVISION

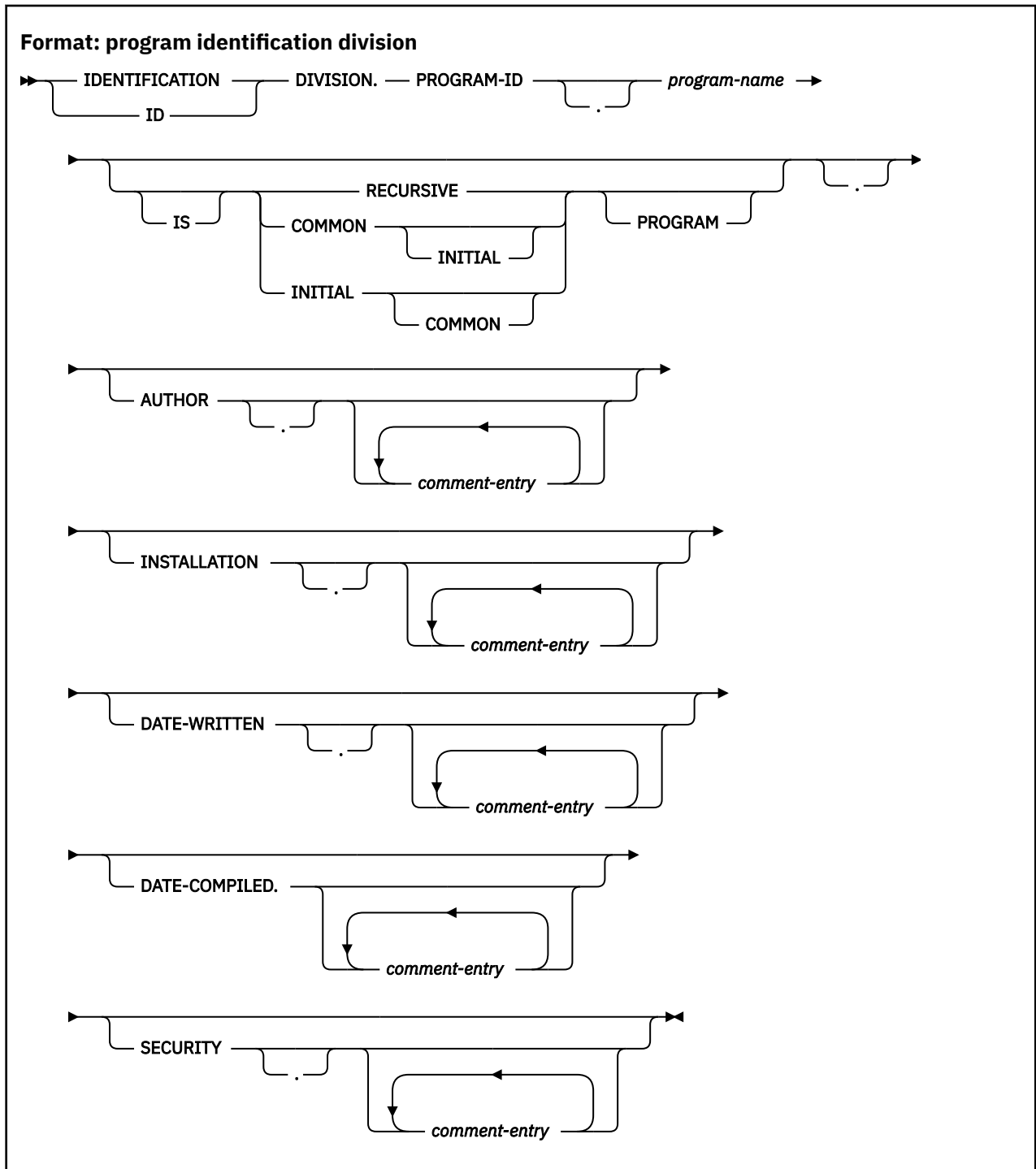
For a user-defined function or function prototype, the first paragraph of the IDENTIFICATION DIVISION must be the FUNCTION-ID paragraph. The other paragraphs are optional and can appear in any order.

For details, see [Chapter 20, “FUNCTION-ID paragraph,” on page 113.](#)

Chapter 15. PROGRAM-ID paragraph

The PROGRAM-ID paragraph specifies the name by which the program is known and assigns selected program attributes to that program. It is required and must be the first paragraph in the IDENTIFICATION DIVISION.

The following format is for a program IDENTIFICATION DIVISION.



program-name

A user-defined word or alphanumeric literal, but not a figurative constant, that identifies your program. It must follow the following rules of formation, depending on the setting of the PGMNAME compiler option:

PGMNAME(COMPAT)

The name can be up to 30 characters in length.

Only the hyphen, underscore, digits 0-9, and alphabetic characters are allowed in the name when it is specified as a user-defined word.

At least one character must be alphabetic.

The hyphen cannot be the first or last character.

If *program-name* is an alphanumeric literal, the rules for the name are the same except that the extension characters \$, #, and @ can be included in the name of the outermost program and the underscore can be the first character.

PGMNAME (LONGUPPER)

If *program-name* is a user-defined word, it can be up to 30 characters in length.

If *program-name* is an alphanumeric literal, the literal can be up to 160 characters in length. The literal cannot be a figurative constant.

Only the hyphen, underscore, digits 0-9, and alphabetic characters are allowed in the name when the name is specified as a user-defined word.

At least one character must be alphabetic.

The hyphen cannot be the first or last character.

If *program-name* is an alphanumeric literal, the underscore character can be the first character.

External program-names are processed with alphabetic characters folded to uppercase.

PGMNAME (LONGMIXED)

program-name must be specified as an alphanumeric literal, which can be up to 160 characters in length. The literal cannot be a figurative constant.

program-name can consist of any character in the range X'41' to X'FE'.

For information about the PGMNAME compiler option and how the compiler processes the names, see *PGMNAME* in the *Enterprise COBOL Programming Guide*.

RECURSIVE

An optional clause that allows COBOL programs to be recursively reentered.

You can specify the RECURSIVE clause only on the outermost program of a compilation unit.

Recursive programs cannot contain nested subprograms.

If the RECURSIVE clause is specified, *program-name* can be recursively reentered while a previous invocation is still active. If the RECURSIVE clause is not specified, an active program cannot be recursively reentered.

The WORKING-STORAGE SECTION of a recursive program defines storage that is statically allocated and initialized on the first entry to a program and is available in a last-used state to any of the recursive invocations.

The LOCAL-STORAGE SECTION of a recursive program (as well as a nonrecursive program) defines storage that is automatically allocated, initialized, and deallocated on a per-invocation basis.

Internal file connectors that correspond to an FD in the FILE SECTION of a recursive program are statically allocated. The status of internal file connectors is part of the last-used state of a program that persists across invocations.

The following language elements are not supported in a recursive program:

- ALTER
- GO TO without a specified procedure-name
- RERUN
- SEGMENT-LIMIT
- USE FOR DEBUGGING

The RECURSIVE clause is required for programs compiled with the THREAD option.

COMMON

Specifies that the program named by *program-name* is contained (that is, nested) within another program and can be called from siblings of the common program and programs contained within them. The COMMON clause can be used only in nested programs. For more information about conventions for program names, see [“Conventions for program-names” on page 86](#).

INITIAL

Specifies that when *program-name* is called, *program-name* and any programs contained (nested) within it are placed in their initial state. The initial attribute is not supported for programs compiled with the THREAD option.

A program is in the initial state:

- The first time the program is called in a run unit
- Every time the program is called, if it possesses the initial attribute
- The first time the program is called after the execution of a CANCEL statement that references the program or a CANCEL statement that references a program that directly or indirectly contains the program
- The first time the program is called after the execution of a CALL statement that references a program that possesses the initial attribute and that directly or indirectly contains the program

When a program is in the initial state:

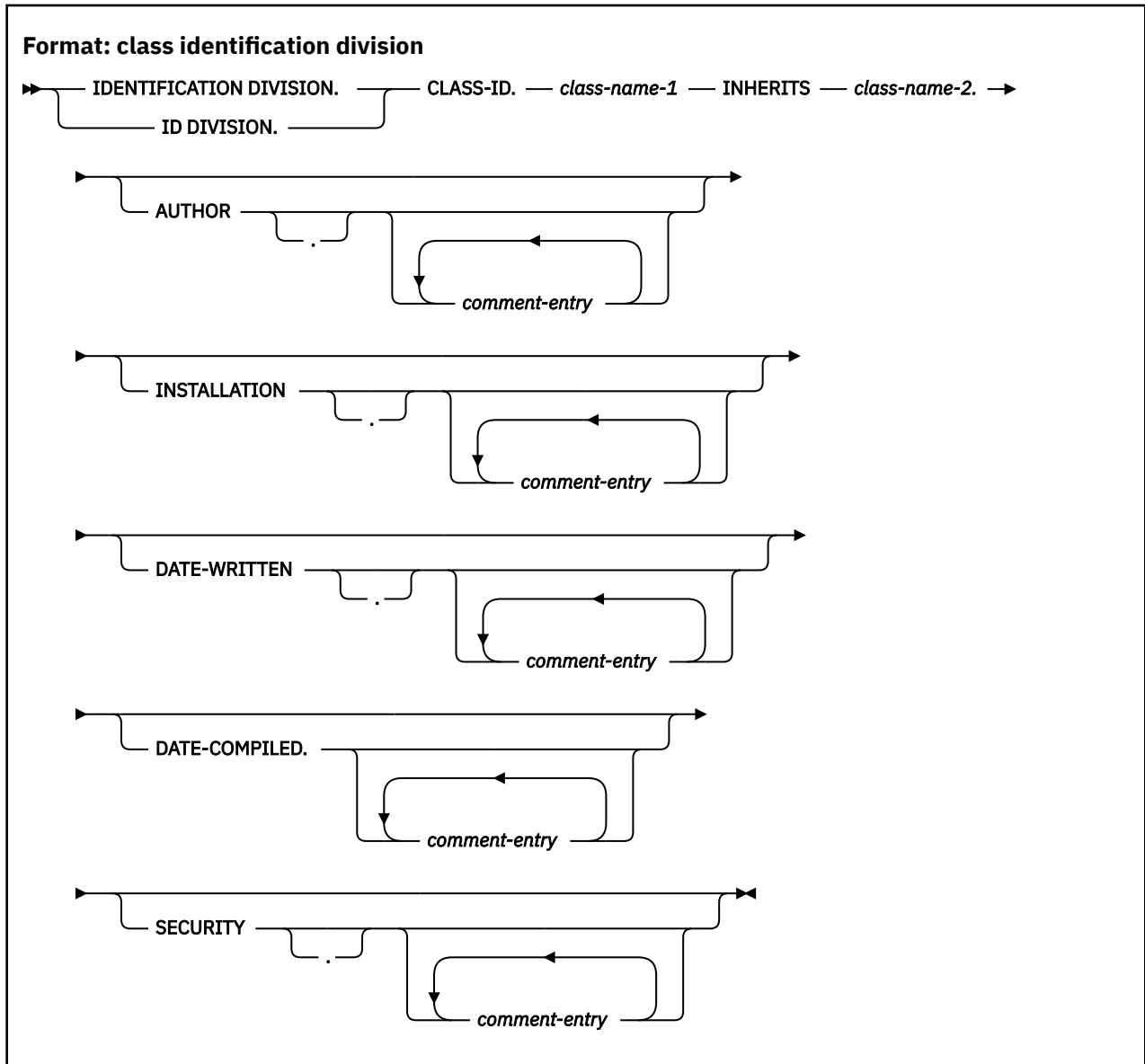
- The program's internal data contained in the WORKING-STORAGE SECTION is initialized. If a VALUE clause is used in the description of the data item, the data item is initialized to the defined value. If a VALUE clause is not associated with a data item, the initial value of the data item is undefined.
- Files with internal file connectors associated with the program are not in the open mode.
- The control mechanisms for all PERFORM statements contained in the program are set to their initial states.
- An altered GO TO statement contained in the program is set to its initial state.

For the rules governing nonunique program names, see [“Rules for program-names” on page 86](#).

Chapter 16. CLASS-ID paragraph

The CLASS-ID paragraph specifies the name by which the class is known and assigns selected attributes to that class. The CLASS-ID paragraph is required and must be the first paragraph in a class IDENTIFICATION DIVISION.

The following format is for a class IDENTIFICATION DIVISION.



class-name-1

A user-defined word that identifies the class. *class-name-1* can optionally have an entry in the REPOSITORY paragraph of the configuration section of the class definition.

INHERITS

A clause that defines *class-name-1* to be a subclass (or derived class) of *class-name-2* (the parent class). *class-name-1* cannot directly or indirectly inherit from *class-name-1*.

class-name-2

The name of a class inherited by *class-name-1*. You must specify *class-name-2* in the REPOSITORY paragraph of the configuration section of the class definition.

General rules

class-name-1 and *class-name-2* must conform to the normal rules of formation for a COBOL user-defined word, using single-byte characters.

See “REPOSITORY paragraph” on page 132 for details on specifying a class-name that is part of a Java package or for using non-COBOL naming conventions for class-names.

You cannot include a class definition in a sequence of programs or other class definitions in a single compilation group. Each class must be specified as a separate source file; that is, a class definition cannot be included in a batch compile.

Inheritance

Every method available on instances of a class is also available on instances of any subclass directly or indirectly derived from that class.

A subclass can introduce new methods that do not exist in the parent or ancestor class and can override a method from the parent or ancestor class. When a subclass overrides an existing method, it defines a new implementation for that method, which replaces the inherited implementation.

The instance data of *class-name-1* is the instance data declared in *class-name-2* together with the data declared in the WORKING-STORAGE SECTION of *class-name-1*. Note, however, that instance data is always private to the class that introduces it.

The semantics of inheritance are as defined by Java. All classes must be derived directly or indirectly from the `java.lang.Object` class.

Java supports single inheritance; that is, no class can inherit *directly* from more than one parent. Only one class-name can be specified in the INHERITS phrase of a class definition.

Chapter 17. FACTORY paragraph

The factory IDENTIFICATION DIVISION introduces the factory definition, which is the portion of a class definition that defines the factory object of the class.

A *factory object* is the single common object that is shared by all object instances of the class. The factory definition contains factory data and factory methods.

The following format is for a factory IDENTIFICATION DIVISION.

Format: factory identification division

► IDENTIFICATION DIVISION. — FACTORY. ◄
 ID

Chapter 18. OBJECT paragraph

The object IDENTIFICATION DIVISION introduces the object definition, which is the portion of a class definition that defines the instance objects of the class.

The object definition contains object data and object methods.

The following format is for an object IDENTIFICATION DIVISION.

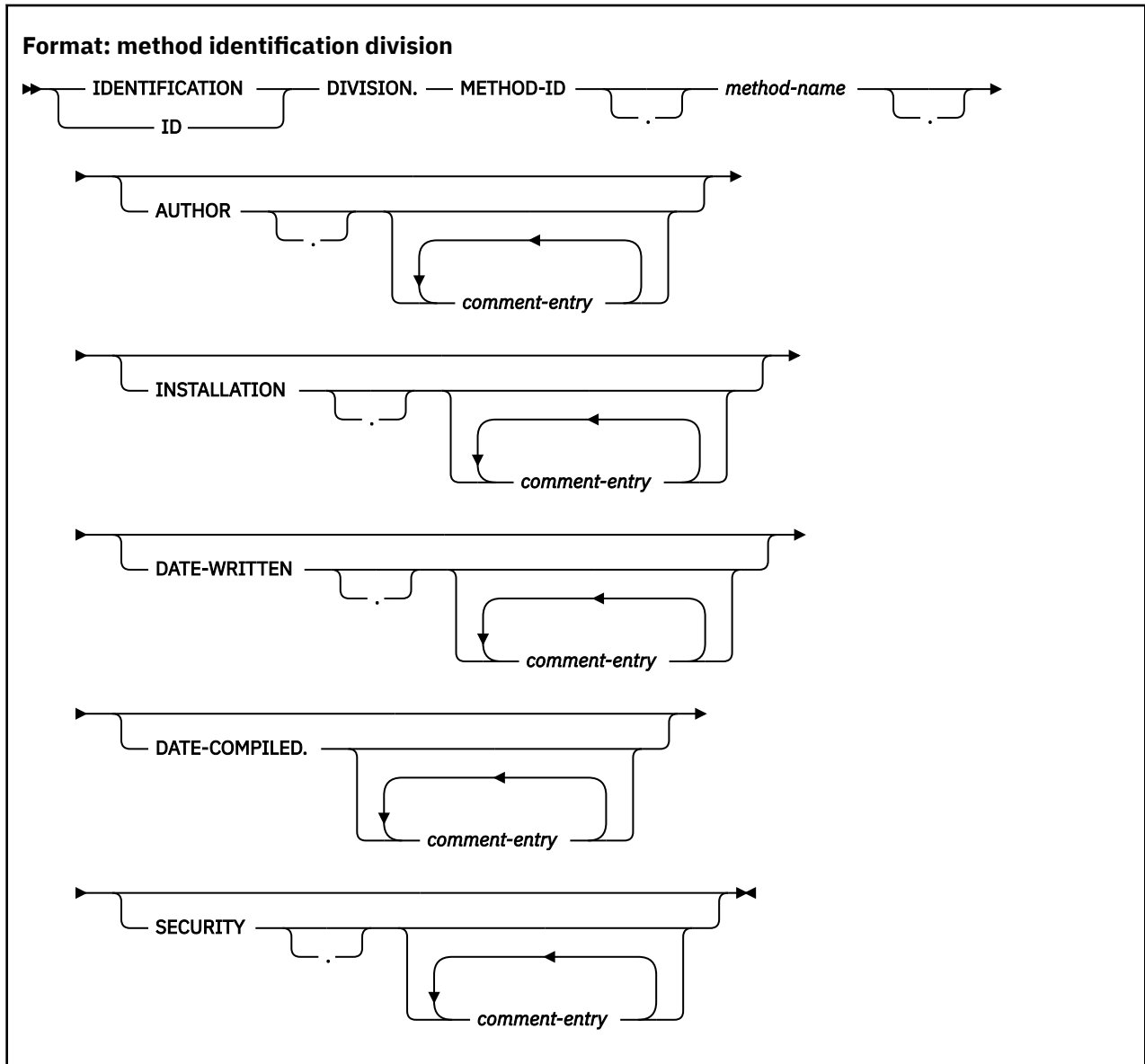
Format: object identification division

► IDENTIFICATION DIVISION. — OBJECT. ◄
 ID

Chapter 19. METHOD-ID paragraph

The METHOD-ID paragraph specifies the name by which a method is known and assigns selected attributes to that method. The METHOD-ID paragraph is required and must be the first paragraph in a method IDENTIFICATION DIVISION.

The following format is for a method IDENTIFICATION DIVISION.



method-name

An alphanumeric literal or national literal that contains the name of the method. The name must conform to the rules of formation for a Java method name. Method names are used directly, without translation. The method name is processed in a case-sensitive manner.

Method signature

The *signature* of a method consists of the name of the method and the number and types of the formal parameters to the method as specified in the PROCEDURE DIVISION USING phrase.

Method overloading, overriding, and hiding

COBOL methods can be *overloaded*, *overridden*, or *hidden*, based on the rules of the Java language.

Method overloading

Method names that are defined for a class are not required to be unique. (The set of methods defined for a class includes the methods introduced by the class definition and the methods inherited from parent classes.)

Method names defined for a class must have unique signatures. Two methods defined for a class and that have the same name but different signatures are said to be *overloaded*.

The type of the method return value, if any, is not included in the method signature.

A class must not define two methods with the same signature but different return value types, or with the same signature but where one method specifies a return value and the other does not.

The rules for overloaded method definitions and resolution of overloaded method invocations are based on the corresponding rules for Java.

Method overriding (for instance methods)

An instance method in a subclass *overrides* an instance method with the same name that is inherited from a parent class if the two methods have the same signature.

When a method overrides an instance method defined in a parent class, the presence or absence of a method return value (the PROCEDURE DIVISION RETURNING data-name) must be consistent in the two methods. Further, when method return values are specified, the return values in the overridden method and the overriding method must have identical data types.

An instance method must not override a factory method in a COBOL parent class, or a static method in a Java parent class.

Method hiding (for factory methods)

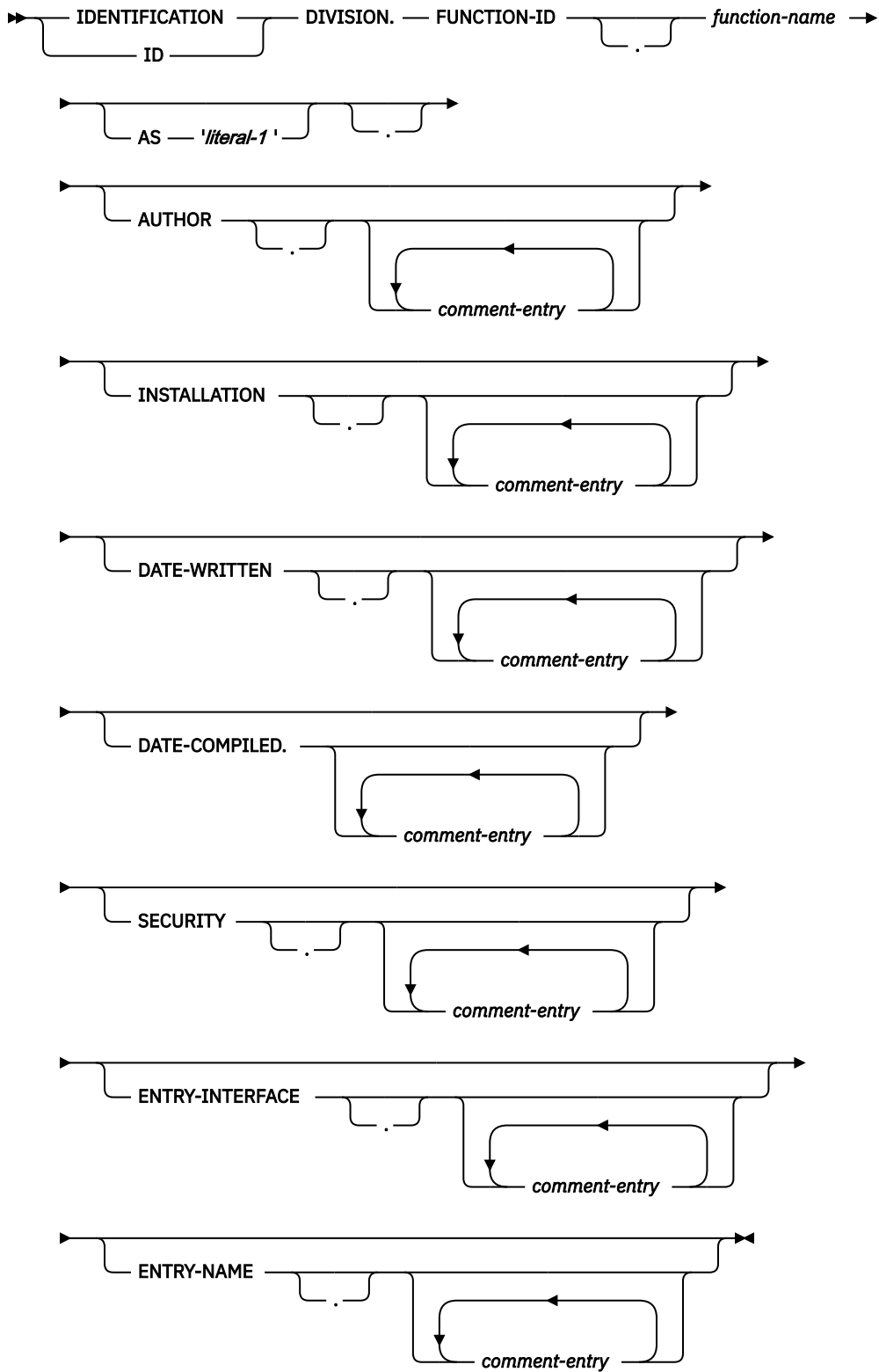
A factory method is said to *hide* any and all methods with the same signature in the superclasses of the method definition that would otherwise be accessible. A factory method must not hide an instance method.

Chapter 20. FUNCTION-ID paragraph

The FUNCTION-ID paragraph specifies the name by which a user-defined function or function prototype is known and assigns selected attributes to that function or function prototype. The FUNCTION-ID paragraph is required and must be the first paragraph in the IDENTIFICATION DIVISION.

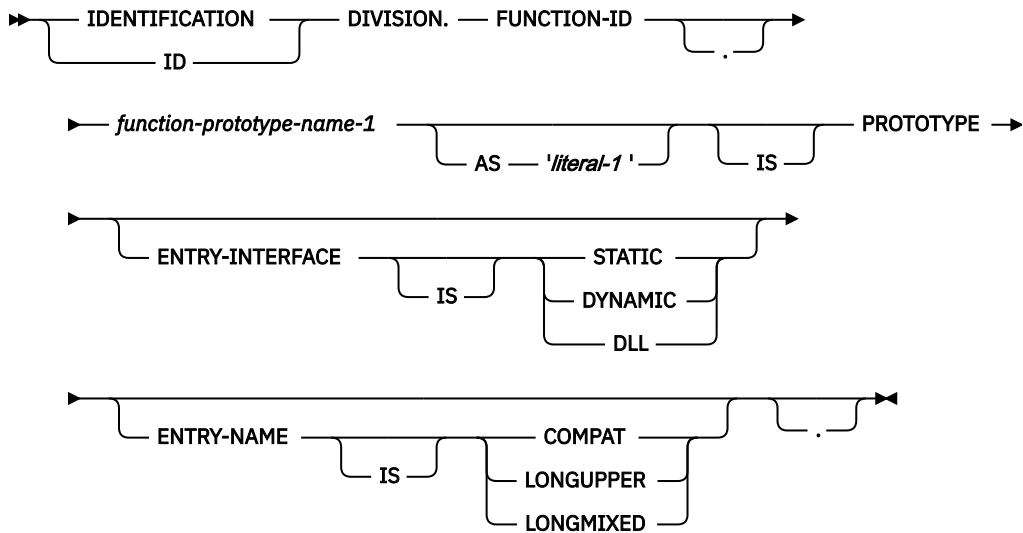
The following format is for a function IDENTIFICATION DIVISION.

Format: function identification division



The following format is for a function prototype definition.

Format: function prototype identification division



function-name-1 or function-prototype-name-1

A user-defined word, but not a figurative constant, that identifies your function or function prototype. It must follow the following rules of formation:

- The name can be up to 30 characters in length.
- The name cannot be a figurative constant.
- Only the hyphen, underscore, digits 0-9, and alphabetic characters are allowed in the name.
- At least one character must be alphabetic.
- A hyphen cannot be the first or last character.

literal-1

- If *literal-1* is specified, *literal-1* is an alphanumeric literal providing the externalized name of the function or function prototype to the operating environment. It may not be a figurative constant.

The externalized name depends on *literal-1* and the PGMNAME option. If *literal-1* is specified, all suboptions of the PGMNAME option are allowed and apply to *literal-1* and do not apply to *function-name-1* and *function-prototype-name-1*.

- If *literal-1* is not specified, the PGMNAME(COMPAT) and PGMNAME(LONGUPPER) compiler options are allowed and apply to *function-name-1* for function definitions, or *function-prototype-name-1* for prototype definitions. The PGMNAME(LONGMIXED) compiler option, in this case, is not allowed.

For more information about the PGMNAME option, see PGMNAME in the *Enterprise COBOL Programming Guide*.

ENTRY-INTERFACE

ENTRY-INTERFACE controls the type of generated code for an invocation to this user-defined function or function prototype. The default is STATIC.

ENTRY-NAME

ENTRY-NAME provides an override of the PGMNAME compiler option for this user-defined function or function prototype.

Chapter 21. Optional paragraphs

Some optional paragraphs in the IDENTIFICATION DIVISION can be omitted.

The optional paragraphs are:

AUTHOR

Name of the author of the program.

INSTALLATION

Name of the company or location.

DATE-WRITTEN

Date the program was written.

DATE-COMPILED

The DATE-COMPILED paragraph provides the compilation date in the source listing. If a comment-entry is specified, the entire entry is replaced with the current date, even if the entry spans lines. If the comment entry is omitted, the compiler adds the current date to the line on which DATE-COMPILED is printed. For example:

```
DATE-COMPILED. 06/30/10.
```

SECURITY

Level of confidentiality of the program.

The *comment-entry* in any of the optional paragraphs can be any combination of characters from the character set of the computer. The comment-entry is written in Area B on one or more lines. The comment-entry must not be written in Area A.

Comment-entries serve only as documentation; they do not affect the meaning of the program. A hyphen in the indicator area (column 7) is not permitted in comment-entries.

You can include DBCS character strings as comment-entries in the IDENTIFICATION DIVISION of your program. Multiple lines are allowed in a comment-entry that contains DBCS character strings.

A DBCS character string must be preceded by a shift-out control character and followed by a shift-in control character. For example:

```
AUTHOR.      <.A.U.T.H.O.R. - .N.A.M.E>, XYZ CORPORATION  
DATE-WRITTEN. <.D.A.T.E>
```

When a comment-entry that is contained on multiple lines uses DBCS characters, shift-out and shift-in characters must be paired on a line.

Part 4. ENVIRONMENT DIVISION

Chapter 22. Configuration section

The configuration section is an optional section for programs, functions, prototypes, and classes, and can describe the computer environment on which the program or class is compiled and executed.

Program configuration section

The configuration section can be specified only in the ENVIRONMENT DIVISION of the outermost program of a COBOL source program.

You should not specify the configuration section in a program that is contained within another program. The entries specified in the configuration section of a program apply to any program contained within that program.

Class configuration section

Specify the configuration section in the ENVIRONMENT DIVISION of a class definition. The repository paragraph can be specified in the ENVIRONMENT DIVISION of a class definition.

Entries in a class configuration section apply to the entire class definition, including all methods introduced by that class.

Method configuration section

The input-output section can be specified in a method configuration section. The entries apply only to the method in which the configuration section is specified.

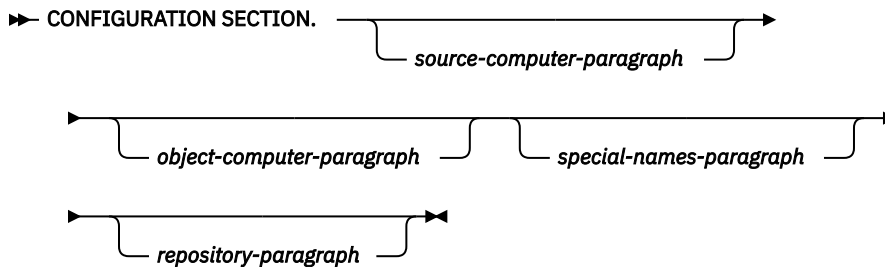
User-defined function configuration section

Specify the configuration section in the ENVIRONMENT DIVISION of a user-defined function definition.

Function prototype configuration section

Specify the configuration section in the ENVIRONMENT DIVISION of a function prototype definition.

Format:

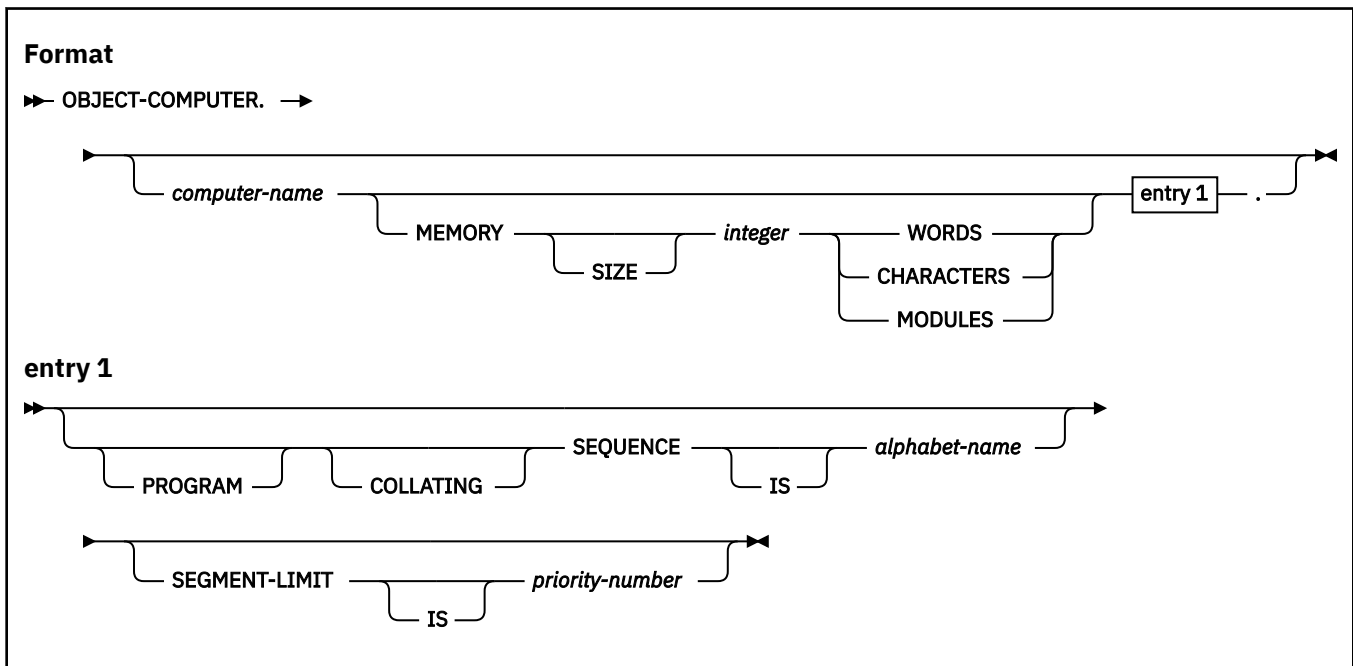


The configuration section can:

- Relate IBM-defined environment-names to user-defined mnemonic names
- Specify the collating sequence
- Specify a currency sign value, and the currency symbol used in the PICTURE clause to represent the currency sign value
- Exchange the functions of the comma and the period in PICTURE clauses and numeric literals
- Relate alphabet-names to character sets or collating sequences
- Specify symbolic characters
- Relate class-names to sets of characters
- Relate object-oriented class names to external class-names and identify class-names that can be used in a class definition or program

OBJECT-COMPUTER paragraph

The OBJECT-COMPUTER paragraph specifies the system for which the object program is designated.



computer-name

A system-name. For example:

```
IBM-system
```

MEMORY SIZE *integer*

integer specifies the amount of main storage needed to run the object program, in words, characters or modules. The MEMORY SIZE clause is syntax checked but has no effect on the execution of the program.

PROGRAM COLLATING SEQUENCE IS *alphabet-name*

The collating sequence used in this program is the collating sequence associated with the specified *alphabet-name*.

The collating sequence pertains to this program and to any programs that this program might contain.

PROGRAM COLLATING SEQUENCE determines the truth value of the following alphanumeric comparisons:

- Those explicitly specified in relation conditions
- Those explicitly specified in condition-name conditions

The PROGRAM COLLATING SEQUENCE clause also applies to any merge or sort keys described with usage DISPLAY, unless the COLLATING SEQUENCE phrase is specified in the MERGE or SORT statement.

The PROGRAM COLLATING SEQUENCE clause does not apply to DBCS data items or data items of usage NATIONAL.

If the PROGRAM COLLATING SEQUENCE clause is omitted, the EBCDIC collating sequence is used. (See [Appendix E, “EBCDIC and ASCII collating sequences,”](#) on page 763.)

SEGMENT-LIMIT IS

The SEGMENT-LIMIT clause is syntax checked but has no effect on the execution of the program.

priority-number

An integer ranging from 1 through 49. All sections with priority-numbers 0 through 49 are fixed permanent segments. See [“Procedures” on page 265](#) for a description of priority-numbers and segmentation support.

Segmentation is not supported for programs compiled with the THREAD option.

All of the OBJECT-COMPUTER paragraph is syntax checked, but only the PROGRAM COLLATING SEQUENCE clause has an effect on the execution of the program.

The OBJECT-COMPUTER paragraph cannot be specified in a function prototype definition.

SPECIAL-NAMES paragraph

The SPECIAL-NAMES paragraph is the name of an ENVIRONMENT DIVISION paragraph in which environment-names are related to user-specified mnemonic-names.

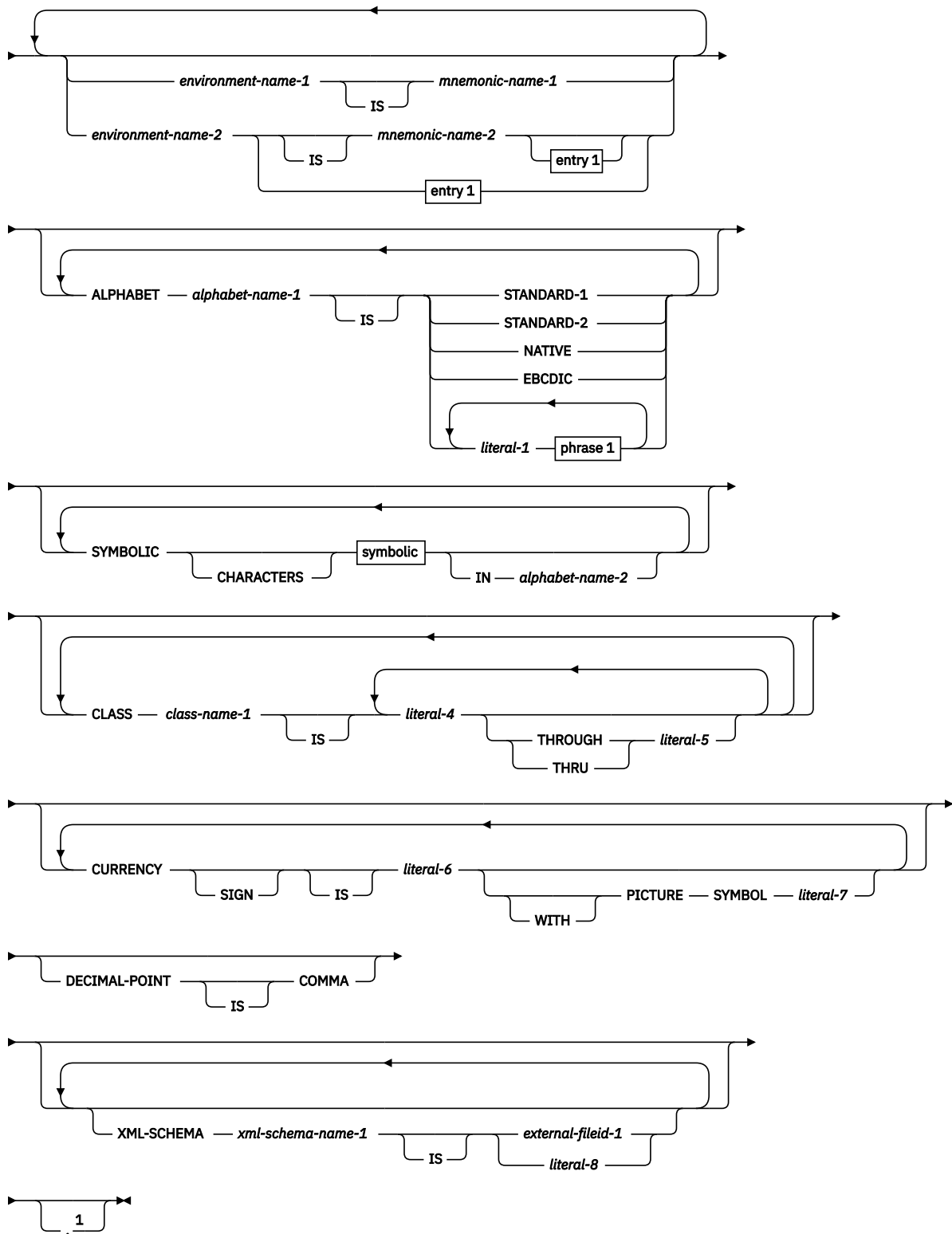
The SPECIAL-NAMES paragraph:

- Relates IBM-specified environment-names to user-defined mnemonic-names
- Relates alphabet-names to character sets or collating sequences
- Specifies symbolic characters
- Relates class names to sets of characters
- Specifies one or more currency sign values and defines a picture symbol to represent each currency sign value in PICTURE clauses
- Specifies that the functions of the comma and decimal point are to be interchanged in PICTURE clauses and numeric literals
- Relates xml-schema-names to ddnames or environment variable names identifying files containing XML schemas

The clauses in the SPECIAL-NAMES paragraph can appear in any order.

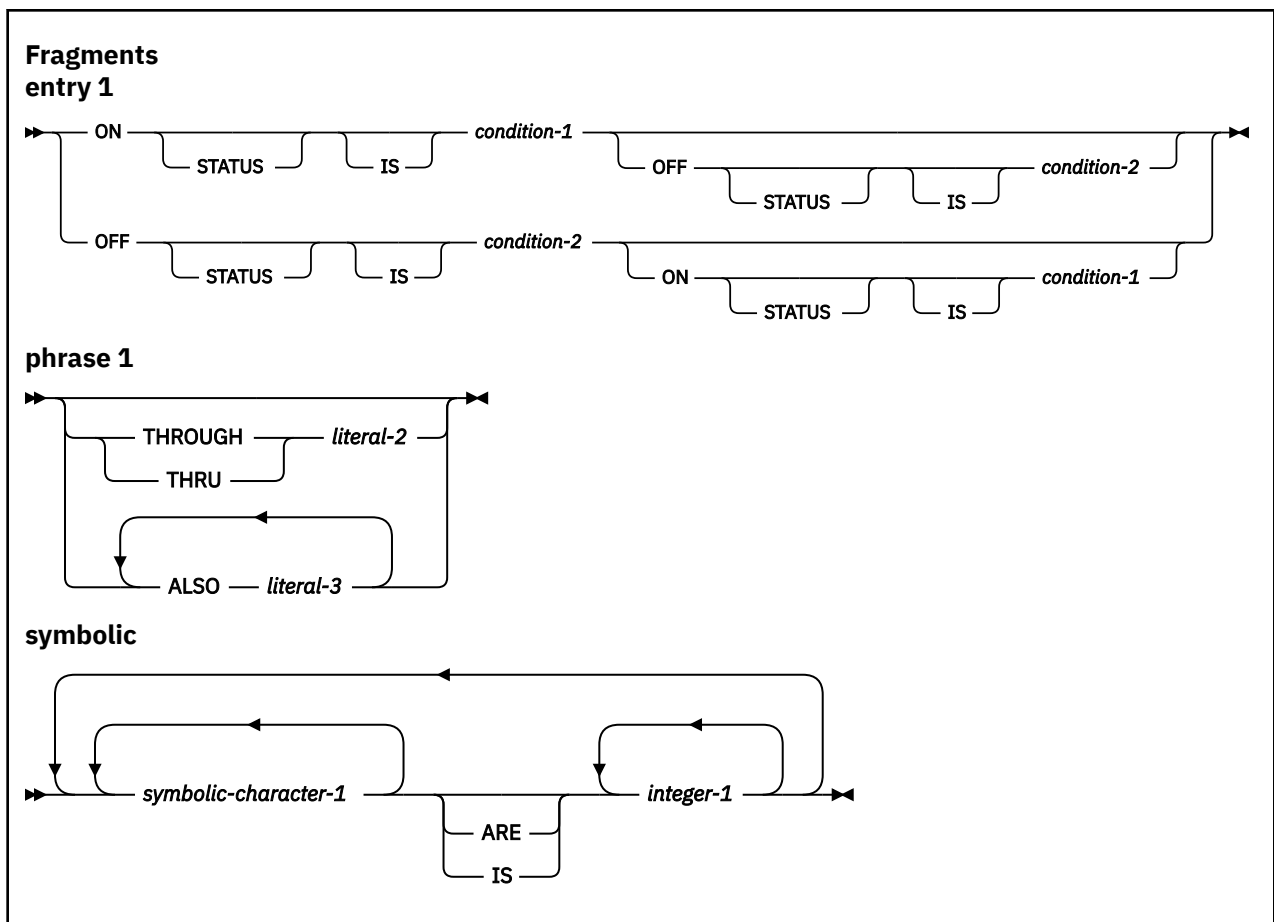
Format: SPECIAL-NAMES paragraph

►► SPECIAL-NAMES. ►



Notes:

¹ This separator period is optional when no clauses are selected. If you use any clauses, you must code the period after the last clause.



environment-name-1

System devices or standard system actions taken by the compiler.

Valid specifications for *environment-name-1* are shown in the following table.

<i>Table 5. Meanings of environment names</i>		
<i>environment-name-1</i>	Meaning	Allowed in
SYSIN SYSIPT	System logical input unit	ACCEPT
SYSOUT SYSLIST SYSLST	System logical output unit	DISPLAY
SYSPUNCH SYSPCH	System punch device	DISPLAY
CONSOLE	Console	ACCEPT and DISPLAY
C01 through C12	Skip to channel 1 through channel 12, respectively	WRITE ADVANCING
CSP	Suppress spacing	WRITE ADVANCING

Table 5. <i>Meanings of environment names</i> (continued)		
environment-name-1	Meaning	Allowed in
S01 through S05	Pocket select 1 through 5 on punch devices	WRITE ADVANCING
AFP-5A	Advanced Function Printing	WRITE ADVANCING

environment-name-2

A 1-byte user-programmable status indicator (UPSI) switch. Valid specifications for *environment-name-2* are UPSI-0 through UPSI-7.

mnemonic-name-1 , mnemonic-name-2

mnemonic-name-1 and *mnemonic-name-2* follow the rules of formation for user-defined names. *mnemonic-name-1* can be used in ACCEPT, DISPLAY, and WRITE statements. *mnemonic-name-2* can be referenced only in the SET statement. *mnemonic-name-2* can qualify *condition-1* or *condition-2* names.

Mnemonic-names and environment-names need not be unique. If you choose a mnemonic-name that is also an environment-name, its definition as a mnemonic-name will take precedence over its definition as an environment-name.

ON STATUS IS, OFF STATUS IS

UPSI switches process special conditions within a program, such as year-beginning or year-ending processing. For example, at the beginning of the PROCEDURE DIVISION, an UPSI switch can be tested; if it is ON, the special branch is taken. (See [“Switch-status condition”](#) on page 283.)

condition-1, condition-2

Condition-names follow the rules for user-defined names. At least one character must be alphabetic. The value associated with the condition-name is considered to be alphanumeric. A condition-name can be associated with the on status or off status of each UPSI switch specified.

In the PROCEDURE DIVISION, the UPSI switch status is tested through the associated condition-name. Each condition-name is the equivalent of a level-88 item; the associated mnemonic-name, if specified, is considered the conditional variable and can be used for qualification.

Condition-names specified in the SPECIAL-NAMES paragraph of a containing program can be referenced in any contained program.

ALPHABET clause

The ALPHABET clause provides a means of relating an alphabet-name to a specified character code set or collating sequence.

The related character code set or collating sequence can be used for alphanumeric data, but not for DBCS or national data.

The ALPHABET clause cannot be specified in a function prototype definition.

ALPHABET alphabet-name-1 IS

alphabet-name-1 specifies a *collating sequence* when used in:

- The PROGRAM COLLATING SEQUENCE clause of the object-computer paragraph
- The COLLATING SEQUENCE phrase of the SORT or MERGE statement

alphabet-name-1 specifies a character code set when used in:

- The FD entry CODE-SET clause
- The SYMBOLIC CHARACTERS clause

STANDARD-1

Specifies the ASCII character set.

STANDARD-2

Specifies the International Reference Version of *ISO/IEC 646, 7-bit coded character set for information interchange*.

NATIVE

Specifies the native character code set. If the ALPHABET clause is omitted, EBCDIC is assumed.

EBCDIC

Specifies the EBCDIC character set.

literal-1 , *literal-2* , *literal-3*

Specifies that the collating sequence for alphanumeric data is determined by the program, according to the following rules:

- The order in which literals appear specifies the ordinal number, in ascending sequence, of the characters in this collating sequence.
- Each numeric literal specified must be an unsigned integer.
- Each numeric literal must have a value that corresponds to a valid ordinal position within the collating sequence in effect.

See Appendix E, “EBCDIC and ASCII collating sequences,” on page 763 for the ordinal numbers for characters in the single-byte EBCDIC and ASCII collating sequences.

- Each character in an alphanumeric literal represents that actual character in the character set. (If the alphanumeric literal contains more than one character, each character, starting with the leftmost, is assigned a successively ascending position within this collating sequence.)
- Any characters that are not explicitly specified assume positions in this collating sequence higher than any of the explicitly specified characters.
- Within one alphabet-name clause, a given character must not be specified more than once.
- Each alphanumeric literal associated with a THROUGH or ALSO phrase must be one character in length.
- When the THROUGH phrase is specified, the contiguous characters in the native character set beginning with the character specified by *literal-1* and ending with the character specified by *literal-2* are assigned successively ascending positions in this collating sequence.

This sequence can be either ascending or descending within the original native character set. That is, if "Z" THROUGH "A" is specified, the ascending values, left-to-right, for the uppercase letters are:

```
ZYXWVUTSRQPONMLKJIHGFEDCBA
```

- When the ALSO phrase is specified, the characters specified as *literal-1*, *literal-3*, ... are assigned to the same position in this collating sequence. For example, if you specify:

```
"D" ALSO "N" ALSO "%"
```

the characters D, N, and % are all considered to be in the same position in the collating sequence.

- When the ALSO phrase is specified and *alphabet-name-1* is referenced in a SYMBOLIC CHARACTERS clause, only *literal-1* is used to represent the character in the character set.
- The character that has the highest ordinal position in this collating sequence is associated with the figurative constant HIGH-VALUE. If more than one character has the highest position because of specification of the ALSO phrase, the last character specified (or defaulted to when any characters are not explicitly specified) is considered to be the HIGH-VALUE character for procedural statements such as DISPLAY and as the sending field in a MOVE statement. (If the

ALSO phrase example given above were specified as the high-order characters of this collating sequence, the HIGH-VALUE character would be %.)

- The character that has the lowest ordinal position in this collating sequence is associated with the figurative constant LOW-VALUE. If more than one character has the lowest position because of specification of the ALSO phrase, the first character specified is the LOW-VALUE character. (If the ALSO phrase example given above were specified as the low-order characters of the collating sequence, the LOW-VALUE character would be D.)

When *literal-1*, *literal-2*, or *literal-3* is specified, the alphabet-name must not be referred to in a CODE-SET clause (see “CODE-SET clause” on page 192).

literal-1, *literal-2*, and *literal-3* must be alphanumeric or numeric literals. All must have the same category. A floating-point literal, a national literal, a DBCS literal, or a symbolic-character figurative constant must not be specified.

CLASS clause

The CLASS clause provides a means for relating a name to the specified set of characters listed in that clause.

The CLASS clause cannot be specified in a function prototype definition.

CLASS *class-name-1* IS

Provides a means for relating a name to the specified set of characters listed in that clause. *class-name-1* can be referenced only in a class condition. The characters specified by the values of the literals in this clause define the exclusive set of characters of which this class consists.

The class-name in the CLASS clause can be a DBCS user-defined word.

literal-4*, *literal-5

Must be category numeric or alphanumeric, and both must be of the same category.

If numeric, *literal-4* and *literal-5* must be unsigned integers and must have a value that is greater than or equal to 1 and less than or equal to the number of characters in the alphabet specified. Each number corresponds to the ordinal position of each character in the single-byte EBCDIC or ASCII collating sequence.

If alphanumeric, *literal-4* and *literal-5* are an actual single-byte EBCDIC character.

literal-4 and *literal-5* must not specify a symbolic-character figurative constant. If the value of the alphanumeric literal contains multiple characters, each character in the literal is included in the set of characters identified by class-name.

Floating-point literals cannot be used in the CLASS clause.

If the alphanumeric literal is associated with a THROUGH phrase, the literal must be one character in length.

THROUGH, THRU

THROUGH and THRU are equivalent. If THROUGH is specified, class-name includes those characters that begin with the value of *literal-4* and that end with the value of *literal-5*. In addition, the characters specified by a THROUGH phrase can be in either ascending or descending order.

CURRENCY SIGN clause

The CURRENCY SIGN clause affects numeric-edited data items whose PICTURE character-strings contain a *currency symbol*.

A currency symbol represents a *currency sign value* that is:

- Inserted in such data items when they are used as receiving items
- Removed from such data items when they are used as sending items for a numeric or numeric-edited receiver

Typically, currency sign values identify the monetary units stored in a data item. For example: '\$', 'EUR', 'CHF', 'JPY', 'HK\$', 'HKD', or X'9F' (hexadecimal code point in some EBCDIC code pages for €, the Euro currency sign). For details on programming techniques for handling the Euro, see *Using currency signs* in the *Enterprise COBOL Programming Guide*.

The CURRENCY SIGN clause specifies a currency sign value and the currency symbol used to represent that currency sign value in a PICTURE clause.

The SPECIAL-NAMES paragraph can contain multiple CURRENCY SIGN clauses. Each CURRENCY SIGN clause must specify a different currency symbol. Unlike all other PICTURE clause symbols, currency symbols are case sensitive. For example, 'D' and 'd' specify different currency symbols.

CURRENCY SIGN IS *literal-6*

literal-6 must be an alphanumeric literal. *literal-6* must not be a figurative constant or a null-terminated literal. *literal-6* must not contain a DBCS character.

If the PICTURE SYMBOL phrase is not specified, *literal-6*:

- Specifies both a currency sign value and the currency symbol for this currency sign value
- Must be a single character
- Must not contain any of the following digits or characters:
 - Digits 0 through 9
 - Alphabetic characters A, B, C, D, E, G, N, P, R, S, U, V, X, Z, their lowercase equivalents, or the space
 - Special characters + - , . * / ; () " = ' (plus sign, minus sign, comma, period, asterisk, slash, semicolon, left parenthesis, right parenthesis, quotation mark, equal sign, apostrophe)
- Can be one of the following lowercase alphabetic characters: f, h, i, j, k, l, m, o, q, t, w, y

If the PICTURE SYMBOL phrase is specified, *literal-6*:

- Specifies a currency sign value. *literal-7* in the PICTURE SYMBOL phrase specifies the currency symbol for this currency sign value.
- Can consist of one or more characters.
- Must not contain any of the following digits or characters:
 - Digits 0 through 9
 - Special characters + - , . ,

PICTURE SYMBOL *literal-7*

Specifies a currency symbol that can be used in a PICTURE clause to represent the currency sign value specified by *literal-6*.

literal-7 must be an alphanumeric literal consisting of one single-byte character. *literal-7* must not contain any of the following digits or characters:

- A figurative constant
- Digits 0 through 9
- Alphabetic characters A, B, C, D, E, G, N, P, R, S, U, V, X, Z, their lowercase equivalents, or the space
- Special characters + - , . * / ; () " = ' ,

If the CURRENCY SIGN clause is specified, the CURRENCY and NOCURRENCY compiler options are ignored. If the CURRENCY SIGN clause is not specified and the NOCURRENCY compiler option is in effect, the dollar sign (\$) is used as the default currency sign value and currency symbol. For more information about the CURRENCY and NOCURRENCY compiler options, see *CURRENCY* in the *Enterprise COBOL Programming Guide*.

DECIMAL-POINT IS COMMA clause

The DECIMAL-POINT IS COMMA clause exchanges the functions of the period and the comma in PICTURE character-strings and in numeric literals.

SYMBOLIC CHARACTERS clause

The SYMBOLIC CHARACTERS clause is applicable only to single-byte character sets. Each character represented is an alphanumeric character.

The SYMBOLIC CHARACTERS clause cannot be specified in a function prototype definition.

SYMBOLIC CHARACTERS *symbolic-character-1*

Provides a means of specifying one or more symbolic characters. *symbolic-character-1* is a user-defined word and must contain at least one alphabetic character. The same symbolic-character can appear only once in a SYMBOLIC CHARACTERS clause. The symbolic character can be a DBCS user-defined word.

The internal representation of *symbolic-character-1* is the internal representation of the character that is represented in the specified character set. The following rules apply:

- The relationship between each *symbolic-character-1* and the corresponding *integer-1* is by their position in the SYMBOLIC CHARACTERS clause. The first *symbolic-character-1* is paired with the first *integer-1*; the second *symbolic-character-1* is paired with the second *integer-1*; and so forth.
- There must be a one-to-one correspondence between occurrences of *symbolic-character-1* and occurrences of *integer-1* in a SYMBOLIC CHARACTERS clause.
- If the IN phrase is specified, *integer-1* specifies the ordinal position of the character that is represented in the character set named by *alphabet-name-2*. This ordinal position must exist.
- If the IN phrase is not specified, *symbolic-character-1* represents the character whose ordinal position in the native character set is specified by *integer-1*.

Ordinal positions are numbered starting from 1.

XML-SCHEMA clause

The XML-SCHEMA clause provides the means of relating *xml-schema-name-1* to an external file identifier: a ddname or environment variable that identifies the actual external file that contains the optimized XML schema.

The external file identifier can be specified as a user-defined word *external-fileid-1* or as an alphanumeric literal *literal-8*, and identifies an existing external z/OS UNIX file or MVS data set that contains the optimized XML schema.

The external file identifier must be either the name specified in the DD statement for the file or the name of an environment variable that contains the file identification information.

The XML-SCHEMA clause cannot be specified in a function prototype definition.

For details on specifying an environment variable, see [“Environment variable contents for an XML schema file”](#) on page 132.

XML-SCHEMA *xml-schema-name-1* IS

xml-schema-name-1 can be referenced only in an XML PARSE statement.

The xml-schema-name in the XML SCHEMA clause can be a DBCS user-defined word.

external-fileid-1

Specifies a user-defined word that must conform to the following rules:

- The user-defined word can contain one to eight characters.
- The user-defined word can contain the characters, A-Z, a-z, 0-9.

- The leading character must be alphabetic.

literal-8

Specifies an alphanumeric literal that must conform to the following rules:

- The literal can contain one to eight characters.
- The literal can contain the characters, A-Z, a-z, 0-9, @, #, and \$.
- The leading character must be alphabetic, @, #, and \$.

The compiler folds *external-fileid-1* or *literal-8* to uppercase to form the ddname or environment variable name for the file.

Environment variable contents for an XML schema file

The environment variable name must be defined using only uppercase because the COBOL compiler automatically folds the external file identifier to uppercase.

For an XML schema in an MVS data set, the environment variable must contain a DSN option in the format shown in the following syntax diagram.

Format: environment variable for XML schema in an MVS data set, DSN option

➤ DSN(*data-set-name* ) ➤

data-set-name must be fully qualified. You must not code blanks within the parentheses.

For an XML schema in a z/OS UNIX file, the environment variable must contain a PATH option in the format shown in the following syntax diagram.

Format: environment variable for XML schema in a z/OS UNIX file, PATH option

➤ PATH(*path-name*) ➤

path-name must be an absolute path name; that is, it must begin with a slash. Special characters in the path name must be "escaped" by preceding them with a backslash. For example, to include a backslash in the path name, code two backslashes in sequence.

For more information about specifying *path-name*, see the description of the PATH parameter in the *z/OS MVS JCL Reference*.

For both formats, blanks at the beginning and end of the environment variable contents are ignored. You must not code blanks between a keyword and the left parenthesis that immediately follows the keyword.

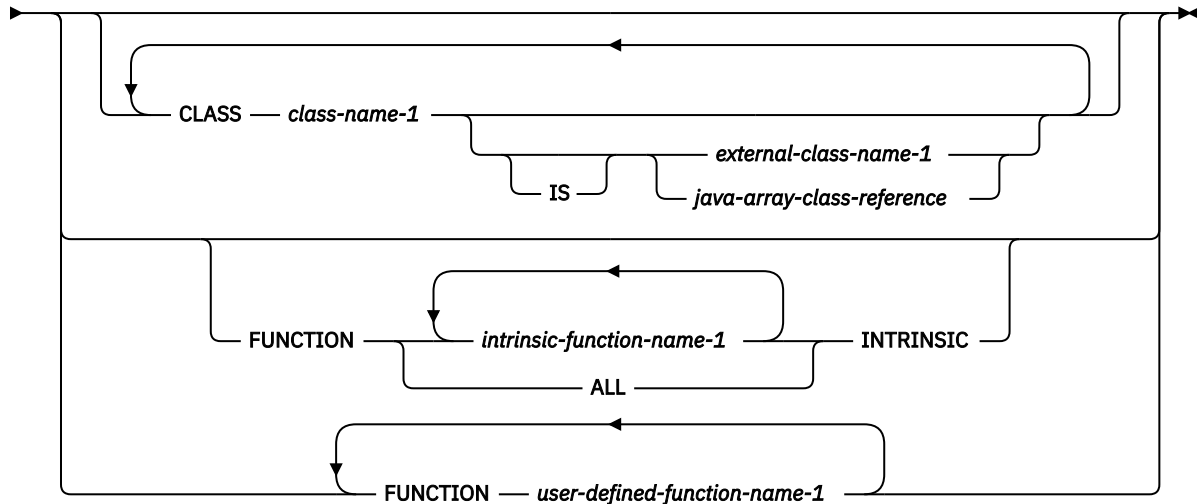
REPOSITORY paragraph

The REPOSITORY paragraph allows specification of class-names that may be used within the scope of the environment division. It also allows declaration of intrinsic function names or user-defined function names that may be used without specifying the keyword FUNCTION.

The REPOSITORY paragraph cannot be specified in a function prototype definition.

Format: REPOSITORY paragraph

➤➤ REPOSITORY. ➤➤



class-name-1

A user-defined word that identifies the class.

external-class-name-1

An alphanumeric literal containing a name that enables a COBOL program to define or access classes with class-names that are defined using Java rules of formation.

The name must conform to the rules of formation for a fully qualified Java class-name. If the class is part of a Java package, *external-class-name-1* must specify the fully qualified name of the package, followed by a period, followed by the simple name of the Java class.

See *Java Language Specification, Third Edition*, by Gosling et al., for Java class-name formation rules.

java-array-class-reference

A reference that enables a COBOL program to access a class that represents an array object, where the elements of the array are themselves objects. *java-array-class-reference* must be an alphanumeric literal with content in the following format:

Format

➤➤ jobjectArray — : — *external-class-name-2* ➤➤

jobjectArray

Specifies a Java object array class.

:

A required separator when *external-class-name-2* is specified. The colon must not be preceded or followed by space characters.

external-class-name-2

The external class-name of the type of the elements of the array. *external-class-name-2* must follow the same rules of formation as *external-class-name-1*.

When the repository entry specifies jobjectArray without the colon separator and *external-class-name-2*, the elements of the object array are of type java.lang.Object.

ALL

If ALL is specified, it is as if all supported Enterprise COBOL intrinsic function names listed in the [Table 63 on page 515](#) were specified.

If ALL is specified, you shall not specify any intrinsic function name as a user-defined word, within the scope of this REPOSITORY paragraph.

intrinsic-function-name-1

The name of a supported Enterprise COBOL intrinsic function.

If any *intrinsic-function-name-1* is specified more than once in the REPOSITORY paragraph, all specifications for that name shall be identical.

The *intrinsic-function-name-1* shall not be specified as a user-defined word within the scope of this REPOSITORY paragraph.

Within the scope of the containing ENVIRONMENT DIVISION, the *intrinsic-function-name-1* may be specified as a function-identifier without being preceded by the keyword FUNCTION, unless specific rules require the use of the keyword FUNCTION.

Note: Since WHEN-COMPILED is both a special register and an intrinsic function name, it may not be specified in the FUNCTION clause of the REPOSITORY paragraph.

user-defined-function-name-1

The name of a user-defined function.

The *user-defined-function-name-1* shall not be specified as a user-defined word within the scope of this REPOSITORY paragraph.

Within the scope of the containing ENVIRONMENT DIVISION, the *user-defined-function-name-1* may be specified as a function-identifier without being preceded by the keyword FUNCTION, unless specific rules require the use of the keyword FUNCTION.

Note: *user-defined-function-name-1* may not be LENGTH, RANDOM, SIGN, SUM, or WHEN-COMPILED.

General rules

This topic lists the general rules of the REPOSITORY paragraph.

1. All referenced class-names must have an entry in the repository paragraph of the COBOL program or class definition that contains the reference. You can specify a given class-name only once in a given repository paragraph.
2. In program definitions, the repository paragraph can be specified only in the outermost program.
3. The repository paragraph of a COBOL class definition can optionally contain an entry for the name of the class itself, but this entry is not required. Such an entry can be used to specify an external class-name that uses non-COBOL characters or that specifies a fully package-qualified class-name when a COBOL class is to be part of a Java package.
4. Entries in a class repository paragraph apply to the entire class definition, including all methods introduced by that class. Entries in a program repository paragraph apply to the entire program, including its contained programs.

Identifying and referencing a class

An external-class-name is used to identify and reference a given class from outside the class definition that defines the class.

The external class-name is determined by using the contents of *external-class-name-1*, *external-class-name-2*, or *class-name-1* (as specified in the repository paragraph of a class), as described in the following list:

1. *external-class-name-1* and *external-class-name-2* are used directly, without translation. They are processed in a case-sensitive manner.

2. *class-name-1* is used if *external-class-name-1* or *java-array-class-reference* is not specified. To create an external name that identifies the class and conforms to Java rules of formation, *class-name-1* is processed as follows:

- The name is converted to uppercase.
- Hyphens are translated to zero.
- Underscores are not translated.
- If the first character of the name is a digit, it is converted as follows:
 - Digits 1 through 9 are changed to A through I.
 - 0 is changed to J.

The class can be implemented in Java or COBOL.

When referencing a class that is part of a Java package, *external-class-name-1* must be specified and must give the fully qualified Java class-name.

For example, the repository entry

```
Repository.  
  Class JavaException is "java.lang.Exception"
```

defines local class-name `JavaException` for referring to the fully qualified external-class-name `"java.lang.Exception."`

When defining a COBOL class that is to be part of a Java package, specify an entry in the repository paragraph of that class itself, giving the full Java package-qualified name as the external class-name.

Chapter 23. Input-Output section

The input-output section of the ENVIRONMENT DIVISION contains FILE-CONTROL paragraph and I-O-CONTROL paragraph.

The exact contents of the input-output section depend on the file organization and access methods used. See [“ORGANIZATION clause” on page 146](#) and [“ACCESS MODE clause” on page 148](#).

The Input-Output section cannot be specified in a function prototype definition.

Program input-output section

The same rules apply to program, method, and user-defined function I-O sections.

Class input-output section

The input-output section is not valid for class definitions.

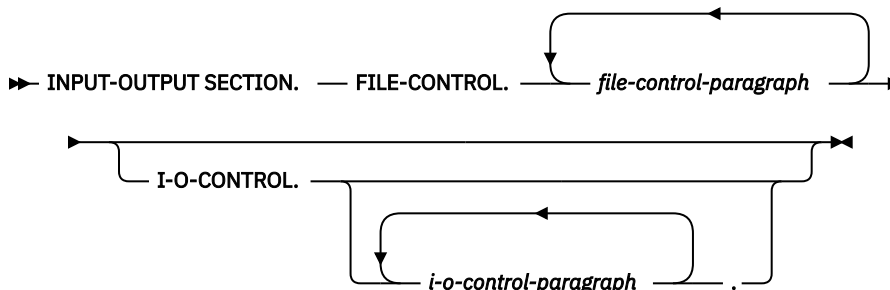
Method input-output section

The same rules apply to program, method, and user defined function I-O sections.

User-defined function input-output section

The same rules apply to program, method, and user-defined function I-O sections.

Format: input-output section



FILE-CONTROL

The keyword FILE-CONTROL identifies the file-control paragraph. This keyword can appear only once, at the beginning of the FILE-CONTROL paragraph. It must begin in Area A and be followed by a separator period.

The keyword FILE-CONTROL and the period can be omitted if no *file-control-paragraph* is specified and there are no files defined in the program.

file-control-paragraph

Names the files and associates them with the external data sets.

Must begin in Area B with a SELECT clause. It must end with a separator period. See [“FILE-CONTROL paragraph” on page 138](#).

file-control-paragraph can be omitted if there are no files defined in the program, even if the FILE-CONTROL keyword is specified.

I-O-CONTROL

The keyword I-O-CONTROL identifies the I-O-CONTROL paragraph.

i-o-control-paragraph

Specifies information needed for efficient transmission of data between the external data set and the COBOL program. The series of entries must end with a separator period. See [“I-O-CONTROL paragraph” on page 154](#).

FILE-CONTROL paragraph

The FILE-CONTROL paragraph associates each file in the COBOL program with an external data set, and specifies file organization, access mode, and other information.

The following formats are for the FILE-CONTROL paragraph:

- Sequential file entries
- Indexed file entries
- Relative file entries
- Line-sequential file entries

The following table lists the different type of files available to programs and methods.

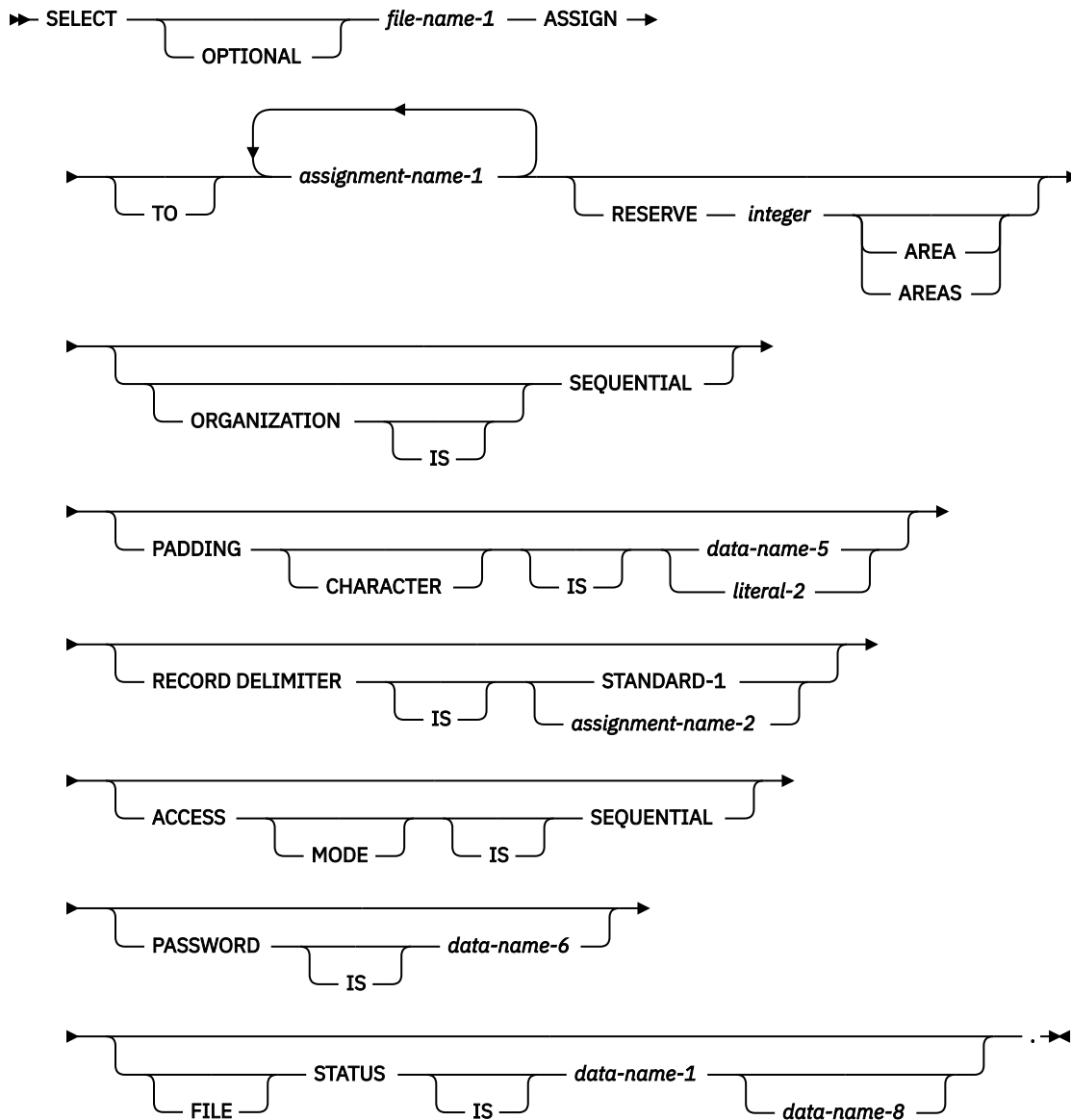
<i>Table 6. Types of files</i>	
File organization	Access method
Sequential	QSAM, VSAM ¹
Relative	VSAM ¹
Indexed	VSAM ¹
Line sequential ²	Text stream I-O
1. VSAM does not support z/OS UNIX files. 2. Line-sequential support is limited to z/OS UNIX files.	

The FILE-CONTROL paragraph begins with the word FILE-CONTROL followed by a separator period. It must contain one and only one entry for each file described in an FD or SD entry in the DATA DIVISION.

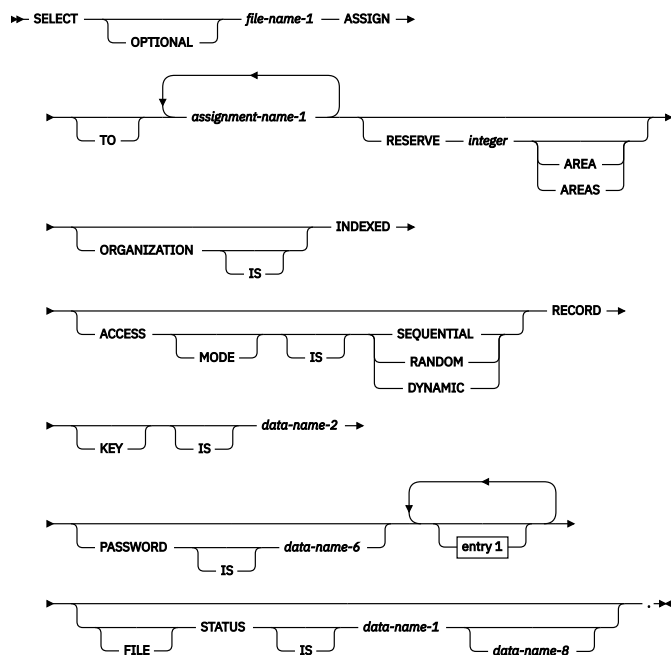
Within each entry, the SELECT clause must appear first. The other clauses can appear in any order, except that the PASSWORD clause for indexed files, if specified, must immediately follow the RECORD KEY or ALTERNATE RECORD KEY data-name with which it is associated.

The *name* component of *assignment-name-1* cannot contain an underscore.

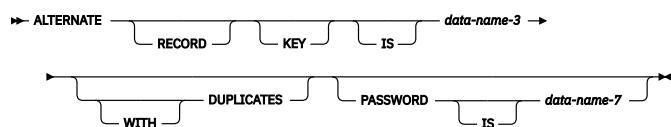
Format 1: sequential-file-control-entry



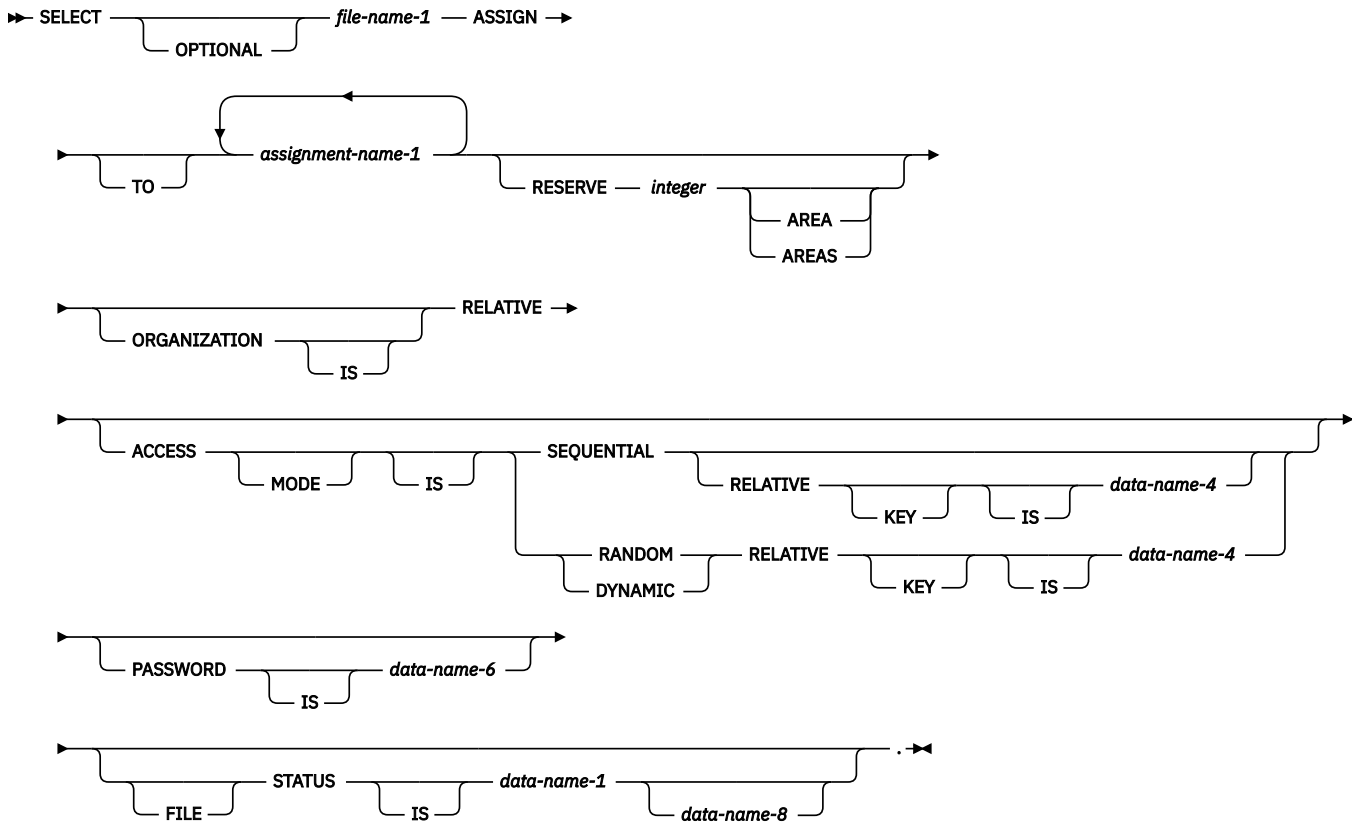
Format 2: indexed-file-control-entry



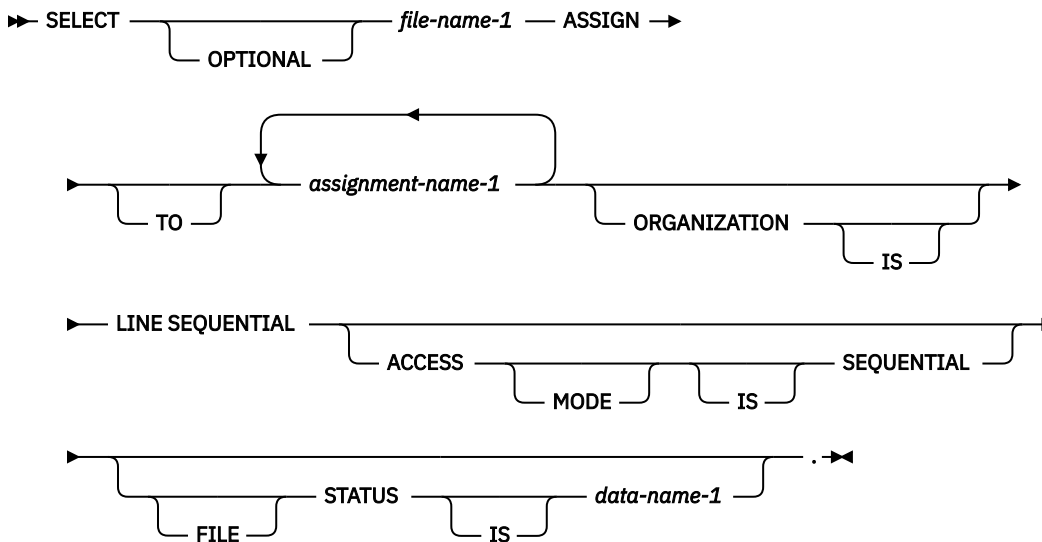
entry 1



Format 3: relative-file-control-entry



Format 4: line-sequential-file-control-entry



SELECT clause

The SELECT clause identifies a file in the COBOL program to be associated with an external data set.

SELECT OPTIONAL

Can be specified only for files opened in the input, I-O, or extend mode. You must specify SELECT OPTIONAL for those input files that are not necessarily available each time the object program is executed. For more information, see [“OPEN statement notes” on page 415](#).

file-name-1

Must be identified by an FD or SD entry in the DATA DIVISION. A file-name must conform to the rules for a COBOL user-defined name, must contain at least one alphabetic character, and must be unique within this program.

When *file-name-1* specifies a sort or a merge file, only the ASSIGN clause can follow the SELECT clause.

If the file connector referenced by *file-name-1* is an external file connector, all file-control entries in the run unit that reference this file connector must have the same specification for the OPTIONAL phrase.

ASSIGN clause

The ASSIGN clause associates the name of a file in a program with the actual external name of the data file.

assignment-name-1

Identifies the external data file. It can be specified as a name or as an alphanumeric literal.

assignment-name-1 is not the name of a data item, and *assignment-name-1* cannot be contained in a data item. It is just a character string. It cannot contain an underscore character.

Any assignment-name after the first is syntax checked, but has no effect on the execution of the program.

assignment-name-1 has the following formats:

Format: assignment-name for QSAM files

► label- S- *name* ►

Format: assignment-name for VSAM sequential file

► label- AS- *name* ►

Format: assignment-name for line-sequential, VSAM indexed, or VSAM relative file

► label- *name* ►

label-

Documents (for the programmer) the device and device class to which a file is assigned. It must end in a hyphen; the specified value is not otherwise checked. It has no effect on the execution of the program. If specified, it must end with a hyphen.

S-

For QSAM files, the S- (organization) field can be omitted.

AS-

For VSAM sequential files, the AS- (organization) field must be specified.

For VSAM indexed and relative files, the organization field must be omitted.

name

A required field that specifies the external name for this file.

It must be either the name specified in the DD statement for this file or the name of an environment variable that contains file allocation information. For details on specifying an environment variable, see [“Assignment name for environment variable” on page 143](#).

name must conform to the following rules of formation:

- If *assignment-name-1* is a user-defined word:
 - The name can contain from one to eight characters.
 - The name can contain the characters A-Z, a-z, and 0-9.
 - The leading character must be alphabetic.
 - The name cannot contain an underscore.
- If *assignment-name-1* is a literal:
 - The name can contain from one to eight characters.
 - The name can contain the characters A-Z, a-z, 0-9, @, #, and \$.
 - The leading character must be alphabetic.
 - The name cannot contain an underscore.

For both user-defined words and literals, the compiler folds *name* to uppercase to form the ddname for the file.

In a sort or merge file, *name* is treated as a comment.

If the file connector referenced by *file-name-1* in the SELECT clause is an external file connector, all file-control entries in the run unit that reference this file connector must have a consistent specification for *assignment-name-1* in the ASSIGN clause. For QSAM files and VSAM indexed and relative files, the name specified on the first *assignment-name-1* must be identical. For VSAM sequential files, it must be specified as AS-*name*.

Assignment name for environment variable

The *name* component of *assignment-name-1* is initially treated as a ddname. If no file has been allocated using this ddname, then *name* is treated as an environment variable.

The environment variable name must be defined using only uppercase because the COBOL compiler automatically folds the external file-name to uppercase.

If this environment variable exists and contains a valid PATH or DSN option (described below), then the file is dynamically allocated using the information supplied by that option.

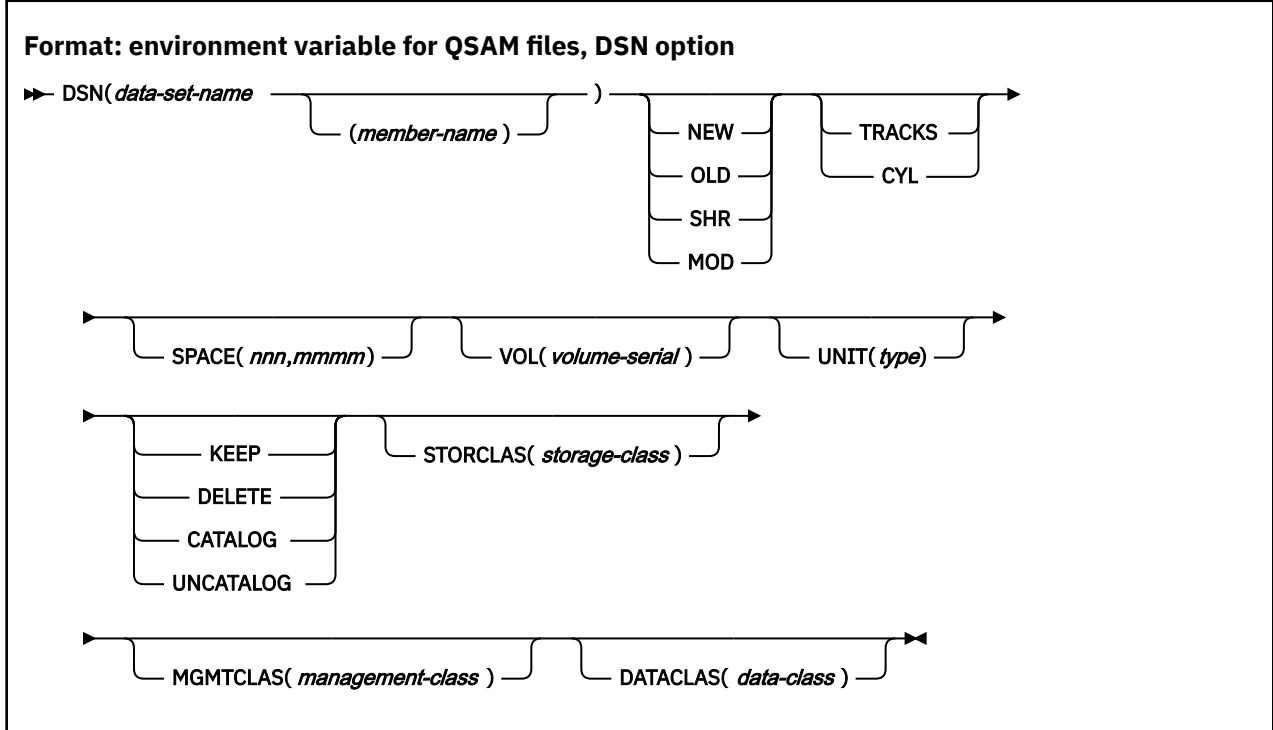
If the environment variable does not contain a valid PATH or DSN option or if the dynamic allocation fails, then attempting to open the file results in file status 98.

The contents of the environment variable are checked at each OPEN statement. If a file was dynamically allocated by a previous OPEN statement and the contents of the environment variable have changed since the previous OPEN, then the previous allocation is dynamically deallocated prior to dynamically reallocating the file using the options currently set in the environment variable.

When the run unit terminates, the COBOL runtime system automatically deallocates all automatically generated dynamic allocations.

Environment variable contents for a QSAM file

For a QSAM file, the environment variable must contain either a DSN or a PATH option in the format shown below.



data-set-name must be fully qualified. The data set must not be a temporary data set; that is, it must not start with an ampersand.

After *data-set-name* or *member-name*, the data set attributes can follow in any order.

The options that follow DSN (such as NEW or TRACKS) must be separated by a comma or by one or more blanks.

Blanks at the beginning and end of the environment variable contents are ignored. You must not code blanks within the parentheses or between a keyword and the left parenthesis that immediately follows the keyword.

COBOL does not provide a default for data set disposition (NEW, OLD, SHR, or MOD); however, your operating system might provide one. To avoid unexpected results when opening the file, you should always specify NEW, OLD, SHR, or MOD with the DSN option when you use environment variables for dynamic allocation of QSAM files.

For information about specifying the values of the data set attributes, see the description of the DD statement in the *z/OS MVS JCL Reference*.

Format: environment variable for QSAM files, PATH option

➡ PATH(*path-name*) ➡

path-name must be an absolute path name; that is, it must begin with a slash. For more information about specifying *path-name*, see the description of the PATH parameter in *z/OS MVS JCL Reference*.

Blanks at the beginning and end of the environment variable contents are ignored. You must not code blanks within the parentheses or between a keyword and the left parenthesis that immediately follows the keyword.

Environment variable contents for a line-sequential file

For a line-sequential file, the environment variable must contain a PATH option in the following format:

Format: environment variable for line-sequential files

➤ PATH(*path-name*) ➤

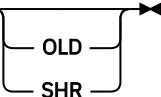
path-name must be an absolute path name; that is, it must begin with a slash. For more information about specifying *path-name*, see the description of the PATH parameter in *z/OS MVS JCL Reference*.

Blanks at the beginning and end of the environment variable contents are ignored. You must not code blanks within the parentheses or between a keyword and the left parenthesis that immediately follows the keyword.

Environment variable contents for a VSAM file

For an indexed, relative, or sequential VSAM file, the environment variable must contain a DSN option in the following format:

Format: environment variable for VSAM files, DSN option

➤ DSN(*data-set-name*)  ➤

data-set-name specifies the data set name for the base cluster. *data-set-name* must be fully qualified and must reference an existing predefined and cataloged VSAM data set.

If an indexed file has alternate indexes, then additional environment variables must be defined that contain DSN options (as above) for each of the alternate index paths. The names of these environment variables must follow the same naming convention as used for alternate index ddnames. That is:

- The environment variable name for each alternate index path is formed by concatenating the base cluster environment variable name with an integer, beginning with 1 for the path associated with the first alternate index and incrementing by 1 for the path associated with each successive alternate index. (For example, if the environment variable name for the base cluster is CUST, then the environment variable names for the alternate indexes would be CUST1, CUST2, ..., .)
- If the length of the base cluster environment variable name is already eight characters, then the environment variable names for the alternate indexes are formed by truncating the base cluster portion of the environment variable name on the right to reduce the concatenated result to eight characters. (For example, if the environment variable name for the base cluster is DATAFILE, then the environment variable names for the alternate clusters would be DATAFIL1, DATAFIL2, ..., .)

The options that follow DSN (such as SHR) must be separated by a comma or by one or more blanks.

Blanks at the beginning and end of the environment variable contents are ignored. You must not code blanks within the parentheses or between a keyword and the left parenthesis that immediately follows the keyword.

COBOL does not provide a default for data set disposition (OLD or SHR); however, your operating system might provide one. To avoid unexpected results when opening the file, you should always specify OLD or SHR with the DSN option when you use environment variables for dynamic allocation of VSAM files.

RESERVE clause

The RESERVE clause allows the user to specify the number of input/output buffers to be allocated at run time for the files.

The RESERVE clause is not supported for line-sequential files.

If the RESERVE clause is omitted, the number of buffers at run time is taken from the DD statement. If none is specified, the system default is taken.

If the file connector referenced by *file-name-1* in the SELECT clause is an external file connector, all file-control entries in the run unit that reference this file connector must have the same value for the integer specified in the RESERVE clause.

ORGANIZATION clause

The ORGANIZATION clause identifies the logical structure of the file. The logical structure is established at the time the file is created and cannot subsequently be changed.

You can find a discussion of the different ways in which data can be organized and of the different access methods that you can use to retrieve the data under [“File organization and access modes” on page 149](#).

ORGANIZATION IS SEQUENTIAL (format 1)

A predecessor-successor relationship among the records in the file is established by the order in which records are placed in the file when it is created or extended.

ORGANIZATION IS INDEXED (format 2)

The position of each logical record in the file is determined by indexes created with the file and maintained by the system. The indexes are based on embedded keys within the file's records.

ORGANIZATION IS RELATIVE (format 3)

The position of each logical record in the file is determined by its relative record number.

ORGANIZATION IS LINE SEQUENTIAL (format 4)

A predecessor-successor relationship among the records in the file is established by the order in which records are placed in the file when it is created or extended. A record in a LINE SEQUENTIAL file can consist only of printable characters.

If you omit the ORGANIZATION clause, the compiler assumes ORGANIZATION IS SEQUENTIAL.

If the file connector referenced by *file-name-1* in the SELECT clause is an external file connector, the same organization must be specified for all file-control entries in the run unit that reference this file connector.

File organization

You establish the organization of the data when you create a file. Once the file has been created, you can expand the file, but you cannot change the organization.

Sequential organization

The physical order in which the records are placed in the file determines the sequence of records. The relationships among records in the file do not change, except that the file can be extended. Records can be fixed length or variable length; there are no keys.

Each record in the file except the first has a unique predecessor record; and each record except the last has a unique successor record.

Indexed organization

Each record in the file has one or more embedded keys (referred to as *key data items*); each key is associated with an index. An index provides a logical path to the data records according to the contents

of the associated embedded record key data items. Indexed files must be direct-access storage files. Records can be fixed length or variable length.

Each record in an indexed file must have an embedded prime key data item. When records are inserted, updated, or deleted, they are identified solely by the values of their prime keys. Thus, the value in each prime key data item must be unique and must not be changed when the record is updated. You tell COBOL the name of the prime key data item in the RECORD KEY clause of the file-control paragraph.

In addition, each record in an indexed file can contain one or more embedded alternate key data items. Each alternate key provides another means of identifying which record to retrieve. You tell COBOL the name of any alternate key data items on the ALTERNATE RECORD KEY clause of the file-control paragraph.

The key used for any specific input-output request is known as the *key of reference*.

Relative organization

Think of the file as a string of record areas, each of which contains a single record. Each record area is identified by a relative record number; the access method stores and retrieves a record based on its relative record number. For example, the first record area is addressed by relative record number 1 and the 10th is addressed by relative record number 10. The physical sequence in which the records were placed in the file has no bearing on the record area in which they are stored, and thus no effect on each record's relative record number. Relative files must be direct-access files. Records can be fixed length or variable length.

Line-sequential organization

In a line-sequential file, each record contains a sequence of characters that ends with a record delimiter. The delimiter is not counted in the length of the record.

When a record is written, any trailing blanks are removed prior to adding the record delimiter. The characters in the record area from the first character up to and including the added record delimiter constitute one record and are written to the file.

When a record is read, characters are read one at a time into the record area until:

- The first record delimiter is encountered. The record delimiter is discarded and the remainder of the record is filled with spaces.
- The entire record area is filled with characters. If the first unread character is the record delimiter, it is discarded. Otherwise, the first unread character becomes the first character read by the next READ statement.
- End-of-file is encountered. The remainder of the record area is filled with spaces.

Records written to line-sequential files must consist of data items described as USAGE DISPLAY or DISPLAY-1 or a combination of DISPLAY and DISPLAY-1 items. A zoned decimal data item either must be unsigned or, if signed, must be declared with the SEPARATE CHARACTER phrase.

A line-sequential file can contain printable characters and control characters. Be aware though that if your file contains a newline character (X'15'), the newline character will function as a record delimiter.

The following clauses are not supported for line-sequential files:

- APPLY WRITE-ONLY clause
- CODE-SET clause
- DATA RECORDS clause
- LABEL RECORDS clause
- LINAGE clause
- I-O phrase of the OPEN statement
- PADDING CHARACTER clause
- RECORD CONTAINS 0 clause

- RECORD CONTAINS clause format 2 (for example: RECORD CONTAINS 100 to 200 CHARACTERS)
- RECORD DELIMITER clause
- RECORDING MODE clause
- RERUN clause
- RESERVE clause
- REVERSED phrase of the OPEN statement
- REWRITE statement
- VALUE OF clause of file description entry
- WRITE ... AFTER ADVANCING *mnemonic-name*
- WRITE ... AT END-OF-PAGE
- WRITE ... BEFORE ADVANCING

PADDING CHARACTER clause

The PADDING CHARACTER clause specifies a character to be used for block padding on sequential files.

data-name-5

Must be defined in the DATA DIVISION as a one-character data item of category alphabetic, alphanumeric, or national, and must not be defined in the FILE SECTION. *data-name-5* can be qualified.

literal-2

Must be a one-character alphanumeric literal or national literal.

For external files, *data-name-5*, if specified, must reference an external data item.

The PADDING CHARACTER clause is syntax checked, but has no effect on the execution of the program.

RECORD DELIMITER clause

The RECORD DELIMITER clause indicates the method of determining the length of a variable-length record on an external medium. It can be specified only for variable-length records.

STANDARD-1

If STANDARD-1 is specified, the external medium must be a magnetic tape file.

assignment-name-2

Can be any COBOL word.

The RECORD DELIMITER clause is syntax checked, but has no effect on the execution of the program.

ACCESS MODE clause

The ACCESS MODE clause defines the manner in which the records of the file are made available for processing. If the ACCESS MODE clause is not specified, sequential access is assumed.

For sequentially accessed relative files, the ACCESS MODE clause does not have to precede the RELATIVE KEY clause.

ACCESS MODE IS SEQUENTIAL

Can be specified in all formats.

Format 1: sequential

Records in the file are accessed in the sequence established when the file is created or extended.
Format 1 supports only sequential access.

Format 2: indexed

Records in the file are accessed in the sequence of ascending record key values according to the collating sequence of the file.

Format 3: relative

Records in the file are accessed in the ascending sequence of relative record numbers of existing records in the file.

Format 4: line-sequential

Records in the file are accessed in the sequence established when the file is created or extended. Format 4 supports only sequential access.

ACCESS MODE IS RANDOM

Can be specified in formats 2 and 3 only.

Format 2: indexed

The value placed in a record key data item specifies the record to be accessed.

Format 3: relative

The value placed in a relative key data item specifies the record to be accessed.

ACCESS MODE IS DYNAMIC

Can be specified in formats 2 and 3 only.

Format 2: indexed

Records in the file can be accessed sequentially or randomly, depending on the form of the specific input-output statement used.

Format 3: relative

Records in the file can be accessed sequentially or randomly, depending on the form of the specific input-output request.

File organization and access modes

File organization is the permanent logical structure of the file. You tell the computer how to retrieve records from the file by specifying the *access mode* (sequential, random, or dynamic).

For details on the access methods and data organization, see [Table 6 on page 138](#).

Sequentially organized data can be accessed only sequentially; however, data that has indexed or relative organization can be accessed in any of the three access modes.

Access modes

See the descriptions of the following types of access modes.

Sequential-access mode

Allows reading and writing records of a file in a serial manner; the order of reference is implicitly determined by the position of a record in the file.

Random-access mode

Allows reading and writing records in a programmer-specified manner; the control of successive references to the file is expressed by specifically defined keys supplied by the user.

Dynamic-access mode

Allows the specific input-output statement to determine the access mode. Therefore, records can be processed sequentially or randomly or both. By default, dynamic access is tuned for positioning to a record and then performing sequential reads. This access mode is suitable for applications that mainly use READ by key, then a few READ NEXT operations. If you access records only by key, use the random-access mode instead. To learn other dynamic-access options to tune and optimize your applications, see "VSAM dynamic access optional logic path" in the *Performance Tuning Guide*.

For external files, every file-control entry in the run unit that is associated with that external file must specify the same access mode. In addition, for relative file entries, *data-name-4* must reference an external data item, and the RELATIVE KEY phrase in each associated file-control entry must reference that same external data item.

Relationship between data organizations and access modes

This section discusses which access modes are valid for each type of data organization.

Sequential files

Files with sequential organization can be accessed only sequentially. The sequence in which records are accessed is the order in which the records were originally written.

Line-sequential files

Same as for sequential files (described above).

Indexed files

All three access modes are allowed.

In the sequential access mode, the sequence in which records are accessed is the ascending order of the record key value. The order of retrieval within a set of records that have duplicate alternate record key values is the order in which records were written into the set.

In the random access mode, you control the sequence in which records are accessed. A specific record is accessed by placing the value of its key or keys in the RECORD KEY data item (and the ALTERNATE RECORD KEY data item). If a set of records has duplicate alternate record key values, only the first record written is available.

In the dynamic access mode, you can change as needed from sequential access to random access by using appropriate forms of input-output statements.

Relative files

All three access modes are allowed.

In the sequential access mode, the sequence in which records are accessed is the ascending order of the relative record numbers of all records that exist within the file.

In the random access mode, you control the sequence in which records are accessed. A specific record is accessed by placing its relative record number in the RELATIVE KEY data item; the RELATIVE KEY must not be defined within the record description entry for the file.

In the dynamic access mode, you can change as needed from sequential access to random access by using appropriate forms of input-output statements.

RECORD KEY clause

The RECORD KEY clause (format 2) specifies the data item within the record that is the prime RECORD KEY for an indexed file. The values contained in the prime RECORD KEY data item must be unique among records in the file.

data-name-2

The prime RECORD KEY data item.

data-name-2 must be described within a record description entry associated with the file. The key can have any of the following data categories:

- Alphanumeric
- Numeric
- Numeric-edited (with usage DISPLAY or NATIONAL)
- Alphanumeric-edited
- Alphabetic
- External floating-point (with usage DISPLAY or NATIONAL)
- Internal floating-point
- DBCS
- National
- National-edited
- UTF-8

Regardless of the category of the key data item, the key is treated as an alphanumeric item. The collation order of the key is determined by the item's binary value order when the key is used for locating a record or for setting the file position indicator associated with the file.

data-name-2 must not reference a variable-length data item. *data-name-2* can be qualified.

If the indexed file contains variable-length records, *data-name-2* need not be contained within the minimum record size specified for the file. That is, *data-name-2* can exceed the minimum record size, but this is not recommended.

The data description of *data-name-2* and its relative location within the record must be the same as those used when the file was defined.

If the file has more than one record description entry, *data-name-2* need be described in only one of those record description entries. The identical character positions referenced by *data-name-2* in any one record description entry are implicitly referenced as keys for all other record description entries for that file.

For files defined with the EXTERNAL clause, all file description entries in the run unit that are associated with the file must have data description entries for *data-name-2* that specify the same relative location in the record and the same length.

ALTERNATE RECORD KEY clause

The ALTERNATE RECORD KEY clause (format 2) specifies a data item within the record that provides an alternative path to the data in an indexed file.

data-name-3

An ALTERNATE RECORD KEY data item.

data-name-3 must be described within a record description entry associated with the file. The key can have any of the following data categories:

- Alphanumeric
- Numeric
- Numeric-edited (with usage DISPLAY or NATIONAL)
- Alphanumeric-edited
- Alphabetic
- External floating-point (with usage DISPLAY or NATIONAL)
- Internal floating-point
- DBCS
- National
- National-edited
- UTF-8

Regardless of the category of the key data item, the key is treated as an alphanumeric item. The collation order of the key is determined by the item's binary value order when the key is used for locating a record or for setting the file position indicator associated with the file.

data-name-3 must not reference a group item that contains a variable-occurrence data item. *data-name-3* can be qualified.

If the indexed file contains variable-length records, *data-name-3* need not be contained within the minimum record size specified for the file. That is, *data-name-3* can exceed the minimum record size, but this is not recommended.

If the indexed file contains variable-length records, *data-name-3* need not be contained within the minimum record size specified for the file. That is, *data-name-3* can exceed the minimum record size, but this is not recommended.

The data description of *data-name-3* and its relative location within the record must be the same as those used when the file was defined. The number of alternate record keys for the file must also be the same as that used when the file was created.

The leftmost character position of *data-name-3* must not be the same as the leftmost character position of the prime record key, or of another alternate record key.

If the DUPLICATES phrase is not specified, the values contained in the ALTERNATE RECORD KEY data item must be unique among records in the file.

If the DUPLICATES phrase is specified, the values contained in the ALTERNATE RECORD KEY data item can be duplicated within any records in the file. In sequential access, the records with duplicate keys are retrieved in the order in which they were placed in the file. In random access, only the first record written in a series of records with duplicate keys can be retrieved.

For files defined with the EXTERNAL clause, all file description entries in the run unit that are associated with the file must have data description entries for *data-name-3* that specify the same relative location in the record and the same length. The file description entries must specify the same number of alternate record keys and the same DUPLICATES phrase.

RELATIVE KEY clause

The RELATIVE KEY clause (format 3) identifies a data-name that specifies the relative record number for a specific logical record within a relative file.

data-name-4

Must be defined as an unsigned integer data item whose description does not contain the PICTURE symbol P. *data-name-4* must not be defined in a record description entry associated with this relative file. That is, the RELATIVE KEY is not part of the record. *data-name-4* can be qualified.

data-name-4 is required for ACCESS IS SEQUENTIAL only when the START statement is to be used. It is always required for ACCESS IS RANDOM and ACCESS IS DYNAMIC. When the START statement is executed, the system uses the contents of the RELATIVE KEY data item to determine the record at which sequential processing is to begin.

If a value is placed in *data-name-4*, and a START statement is not executed, the value is ignored and processing begins with the first record in the file.

If a relative file is to be referenced by a START statement, you must specify the RELATIVE KEY clause for that file.

For external files, *data-name-4* must reference an external data item, and the RELATIVE KEY phrase in each associated file-control entry must reference that same external data item in each case.

The ACCESS MODE IS RANDOM clause must not be specified for file-names specified in the USING or GIVING phrase of a SORT or MERGE statement.

PASSWORD clause

The PASSWORD clause controls access to files.

data-name-6 , data-name-7

Password data items. Each must be defined in the WORKING-STORAGE SECTION of the DATA DIVISION as a data item of category alphabetic, alphanumeric, or alphanumeric-edited. The first eight characters are used as the password; a shorter field is padded with blanks to eight characters. Each password data item must be equivalent to one that is externally defined.

When the PASSWORD clause is specified, at object time the PASSWORD data item must contain a valid password for this file before the file can be successfully opened.

Format 1 considerations:

The PASSWORD clause is not valid for QSAM sequential files.

Format 2 and 3 considerations:

The PASSWORD clause, if specified, must immediately follow the RECORD KEY or ALTERNATE RECORD KEY data-name with which it is associated.

For indexed files that have been completely predefined to VSAM, only the PASSWORD data item for the RECORD KEY need contain the valid password before the file can be successfully opened at file creation time.

For any other type of file processing (including the processing of dynamic calls at file creation time through a COBOL runtime subroutine), every PASSWORD data item for the file must contain a valid password before the file can be successfully opened, regardless of whether all paths to the data are used in this object program.

For external files, *data-name-6* and *data-name-7* must reference external data items. The PASSWORD clauses in each associated file-control entry must reference the same external data items.

FILE STATUS clause

The FILE STATUS clause monitors the execution of each input-output operation for the file.

When the FILE STATUS clause is specified, the system moves a value into the file status key data item after each input-output operation that explicitly or implicitly refers to this file. The value indicates the status of execution of the statement. (See the file status key description under [“Common processing facilities”](#) on page 299.)

data-name-1

The file status key data item can be defined in the WORKING-STORAGE, LOCAL-STORAGE, or LINKAGE SECTION as one of the following items:

- A two-character data item of category alphanumeric
- A two-character data item of category national
- A two-digit data item of category numeric with usage DISPLAY or NATIONAL (an external decimal data item)

data-name-1 must not contain the PICTURE symbol 'P'.

data-name-1 can be qualified.

The file status key data item must not be variably located; that is, the data item cannot follow a data item that contains an OCCURS DEPENDING ON clause.

data-name-8

Must be defined as an alphanumeric group item of 6 bytes in the WORKING-STORAGE SECTION or LINKAGE SECTION of the DATA DIVISION.

Specify *data-name-8* only if the file is a VSAM file (that is, ESDS, KSDS, RRDS).

data-name-8 holds the 6-byte VSAM return code, which is composed as follows:

- The first 2 bytes of *data-name-8* contain the VSAM *return code* in binary format. The value for this code is defined (by VSAM) as 0, 8, or 12.
- The next 2 bytes of *data-name-8* contain the VSAM *function code* in binary format. The value for this code is defined (by VSAM) as 0, 1, 2, 3, 4, or 5.
- The last 2 bytes of *data-name-8* contain the VSAM *feedback code* in binary format. The code value is 0 through 255.

If VSAM returns a nonzero return code, *data-name-8* is set.

If FILE STATUS is returned without having called VSAM, *data-name-8* is zero.

If *data-name-1* is set to zero, the content of *data-name-8* is undefined. VSAM status return code information is available without transformation in the currently defined COBOL FILE STATUS code. User identification and handling of exception conditions are allowed at the same level as that defined by VSAM.

Function code and *feedback code* are set if and only if the *return code* is set to a nonzero value. If they are referenced when the return code is set to zero, the contents of the fields are not dependable.

Values in the *return code*, *function code*, and *feedback code* fields are defined by VSAM. There are no COBOL additions, deletions, or modifications to the VSAM definitions.

For more information, see *DFSMS Macro Instructions for Data Sets*.

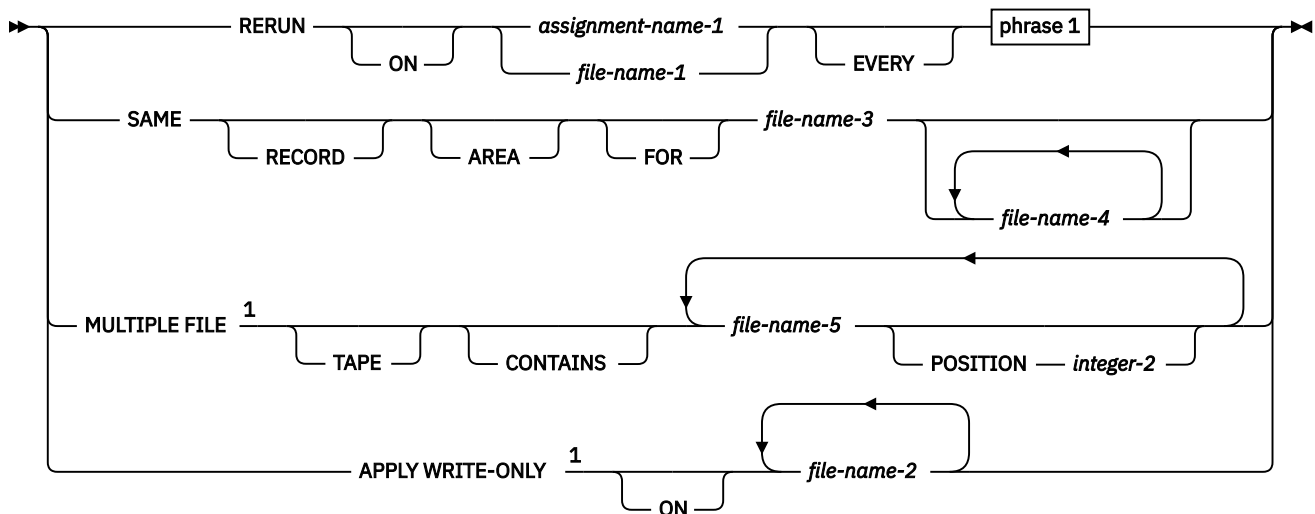
I-O-CONTROL paragraph

The I-O-CONTROL paragraph of the input-output section specifies when checkpoints are to be taken and the storage areas to be shared by different files. This paragraph is optional in a COBOL program.

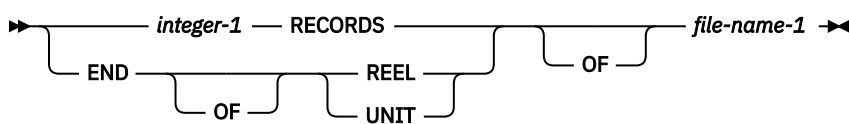
The keyword I-O-CONTROL can appear only once, at the beginning of the paragraph. The word I-O-CONTROL must begin in Area A and must be followed by a separator period.

The order in which I-O-CONTROL paragraph clauses are written is not significant. The I-O-CONTROL paragraph ends with a separator period.

Format: QSAM-i-o-control-entry



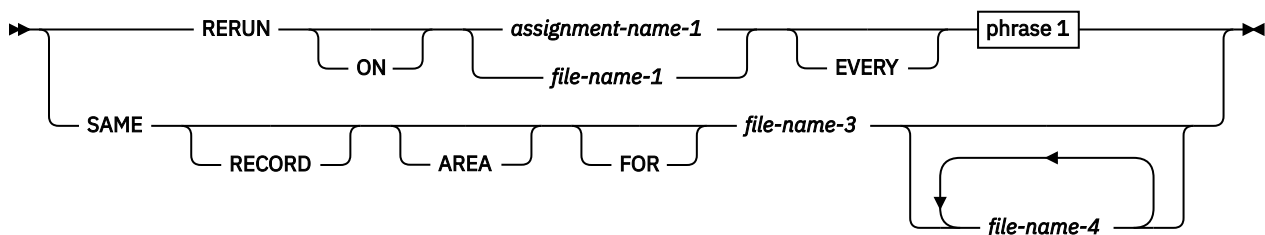
phrase 1



Notes:

¹ The MULTIPLE FILE clause and APPLY WRITE-ONLY clause are not supported for VSAM files.

Format: VSAM-i-o-control-entry



phrase 1

►► *integer-1* — RECORDS — *file-name-1* —►

OF

Format: line-sequential-i-o-control-entry

►► SAME — RECORD — AREA — FOR — *file-name-3* — *file-name-4* —►

↖ ↗

Format: sort/merge-i-o-control-entry

►► RERUN — ON — *assignment-name-1* —►

↖ ↗

►► SAME — RECORD — AREA — FOR — phrase 1 —►

↖ ↗

SORT

SORT-MERGE

phrase 1

►► *file-name-3* — *file-name-4* —►

↖ ↗

RERUN clause

The RERUN clause specifies that checkpoint records are to be taken. Subject to the restrictions given with each phrase, more than one RERUN clause can be specified.

For information regarding the checkpoint data set definition and the checkpoint method required for complete compliance to the 85 COBOL Standard, see *DD statements for defining checkpoint data sets* in the *Enterprise COBOL Programming Guide*.

Do not use the RERUN clause:

- For files described with the EXTERNAL clause
- In programs with the RECURSIVE clause specified
- In programs compiled with the THREAD option
- In methods

file-name-1

Must be a sequentially organized file.

VSAM and QSAM considerations:

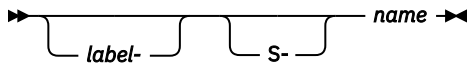
The file named in the RERUN clause must be a file defined in the same program as the I-O-CONTROL paragraph, even if the file is defined as GLOBAL.

assignment-name-1

The external data set for the checkpoint file. It must not be the same assignment-name as that specified in any ASSIGN clause throughout the entire program, including contained and containing programs.

For QSAM files, *assignment-name-1* has the format:

Format: assignment-name for QSAM files



The QSAM file must reside on a tape or direct access device. See also [Appendix I, “ASCII considerations,”](#) on page 793.

SORT/MERGE considerations:

When the RERUN clause is specified in the I-O-CONTROL paragraph, checkpoint records are written at logical intervals determined by the sort/merge program during execution of each SORT or MERGE statement in the program. When the RERUN clause is omitted, checkpoint records are not written.

There can be only one SORT/MERGE I-O-CONTROL paragraph in a program, and it cannot be specified in contained programs. It will have a global effect on all SORT and MERGE statements in the program unit.

EVERY *integer-1* RECORDS

A checkpoint record is to be written for every *integer-1* records in *file-name-1* that are processed.

When multiple *integer-1* RECORDS phrases are specified, no two of them can specify the same value for *file-name-1*.

If you specify the *integer-1* RECORDS phrase, you must specify *assignment-name-1*.

EVERY END OF REEL/UNIT

A checkpoint record is to be written whenever end-of-volume for *file-name-1* occurs. The terms REEL and UNIT are interchangeable.

When multiple END OF REEL/UNIT phrases are specified, no two of them can specify the same value for *file-name-1*.

The END OF REEL/UNIT phrase can be specified only if *file-name-1* is a sequentially organized file.

SAME AREA clause

The SAME AREA clause is syntax checked, but has no effect on the execution of the program. The SAME AREA clause specifies that two or more files that do not represent sort or merge files are to use the same main storage area during processing.

The files named in a SAME AREA clause need not have the same organization or access.

file-name-3* , *file-name-4

Must be specified in the file-control paragraph of the same program. *file-name-3* and *file-name-4* must not reference a file that is defined with the EXTERNAL clause.

- For QSAM files, the SAME clause is treated as documentation.
- For VSAM files, the SAME clause is treated as if equivalent to the SAME RECORD AREA clause.

More than one SAME AREA clause can be included in a program. However:

- A specific file-name must not appear in more than one SAME AREA clause.
- If one or more file-names of a SAME AREA clause appear in a SAME RECORD AREA clause, all the file-names in that SAME AREA clause must appear in that SAME RECORD AREA clause. However, the SAME RECORD AREA clause can contain additional file-names that do not appear in the SAME AREA clause.

- The rule that in the SAME AREA clause only one file can be open at one time takes precedence over the SAME RECORD AREA rule that all the files can be open at the same time.

SAME RECORD AREA clause

The SAME RECORD AREA clause specifies that two or more files are to use the same main storage area for processing the current logical record.

The files named in a SAME RECORD AREA clause need not have the same organization or access.

file-name-3 , file-name-4

Must be specified in the file-control paragraph of the same program. *file-name-3* and *file-name-4* must not reference a file that is defined with the EXTERNAL clause.

All of the files can be opened at the same time. A logical record in the shared storage area is considered to be both of the following ones:

- A logical record of each opened output file in the SAME RECORD AREA clause
- A logical record of the most recently read input file in the SAME RECORD AREA clause

More than one SAME RECORD AREA clause can be included in a program. However:

- A specific file-name must not appear in more than one SAME RECORD AREA clause.
- If one or more file-names of a SAME AREA clause appear in a SAME RECORD AREA clause, all the file-names in that SAME AREA clause must appear in that SAME RECORD AREA clause. However, the SAME RECORD AREA clause can contain additional file-names that do not appear in the SAME AREA clause.
- The rule that in the SAME AREA clause only one file can be open at one time takes precedence over the SAME RECORD AREA rule that all the files can be open at the same time.
- If the SAME RECORD AREA clause is specified for several files, the record description entries or the file description entries for these files must not include the GLOBAL clause.
- The SAME RECORD AREA clause must not be specified when the RECORD CONTAINS 0 CHARACTERS clause is specified.

The files named in the SAME RECORD AREA clause need not have the same organization or access.

SAME SORT AREA clause

The SAME SORT AREA clause is syntax checked but has no effect on the execution of the program.

file-name-3 , file-name-4

Must be specified in the file-control paragraph of the same program. *file-name-3* and *file-name-4* must not reference a file that is defined with the EXTERNAL clause.

When the SAME SORT AREA clause is specified, at least one file-name specified must name a sort file. Files that are not sort files can also be specified. The following rules apply:

- More than one SAME SORT AREA clause can be specified. However, a given sort file must not be named in more than one such clause.
- If a file that is not a sort file is named in both a SAME AREA clause and in one or more SAME SORT AREA clauses, all the files in the SAME AREA clause must also appear in that SAME SORT AREA clause.
- Files named in a SAME SORT AREA clause need not have the same organization or access.
- Files named in a SAME SORT AREA clause that are not sort files do not share storage with each other unless they are named in a SAME AREA or SAME RECORD AREA clause.
- During the execution of a SORT or MERGE statement that refers to a sort or merge file named in this clause, any nonsort or nonmerge files associated with file-names named in this clause must not be in the open mode.

SAME SORT-MERGE AREA clause

The SAME SORT-MERGE AREA clause is equivalent to the SAME SORT AREA clause.

For more details, see [“SAME SORT AREA clause” on page 157](#).

MULTIPLE FILE TAPE clause

The MULTIPLE FILE TAPE clause (format 1) specifies that two or more files share the same physical reel of tape.

This clause is syntax checked, but has no effect on the execution of the program. The function is performed by the system through the LABEL parameter of the DD statement.

APPLY WRITE-ONLY clause

The APPLY WRITE-ONLY clause optimizes buffer and device space allocation for files that have standard sequential organization, have variable-length records, and are blocked.

If you specify this phrase, the buffer is truncated only when the space available in the buffer is smaller than the size of the next record. Otherwise, the buffer is truncated when the space remaining in the buffer is smaller than the maximum record size for the file.

APPLY WRITE-ONLY is effective only for QSAM files.

file-name-2

Each file must have standard sequential organization.

APPLY WRITE-ONLY clauses must agree among corresponding external file description entries. For an alternate method of achieving the APPLY WRITE-ONLY results, see the description of the compiler option, *AWO* in the *Enterprise COBOL Programming Guide*.

Part 5. DATA DIVISION

Chapter 24. DATA DIVISION overview

This overview describes the structure of the DATA DIVISION for programs, object definitions, factory definitions, and methods.

Each section in the DATA DIVISION has a specific logical function within a COBOL program, object definition, factory definition, or method and can be omitted when that logical function is not needed. If included, the sections must be written in the order shown. The DATA DIVISION is optional.

Program data division

The DATA DIVISION of a COBOL source program describes, in a structured manner, all the data to be processed by the program.

Object data division

The object data division contains data description entries for instance object data (instance data). Instance data is defined in the WORKING-STORAGE SECTION of the object paragraph of a class definition.

Factory data division

The factory data division contains data description entries for factory object data (factory data). Factory data is defined in the WORKING-STORAGE SECTION of the factory paragraph of a class definition.

Method data division

A method data division contains data description entries for data accessible within the method. A method data division can contain a LOCAL-STORAGE SECTION or a WORKING-STORAGE SECTION, or both. The term *method data* applies to both. Method data in LOCAL-STORAGE is dynamically allocated and initialized on each invocation of the method; method data in WORKING-STORAGE is static and persists across invocations of the method.

User-defined function data division

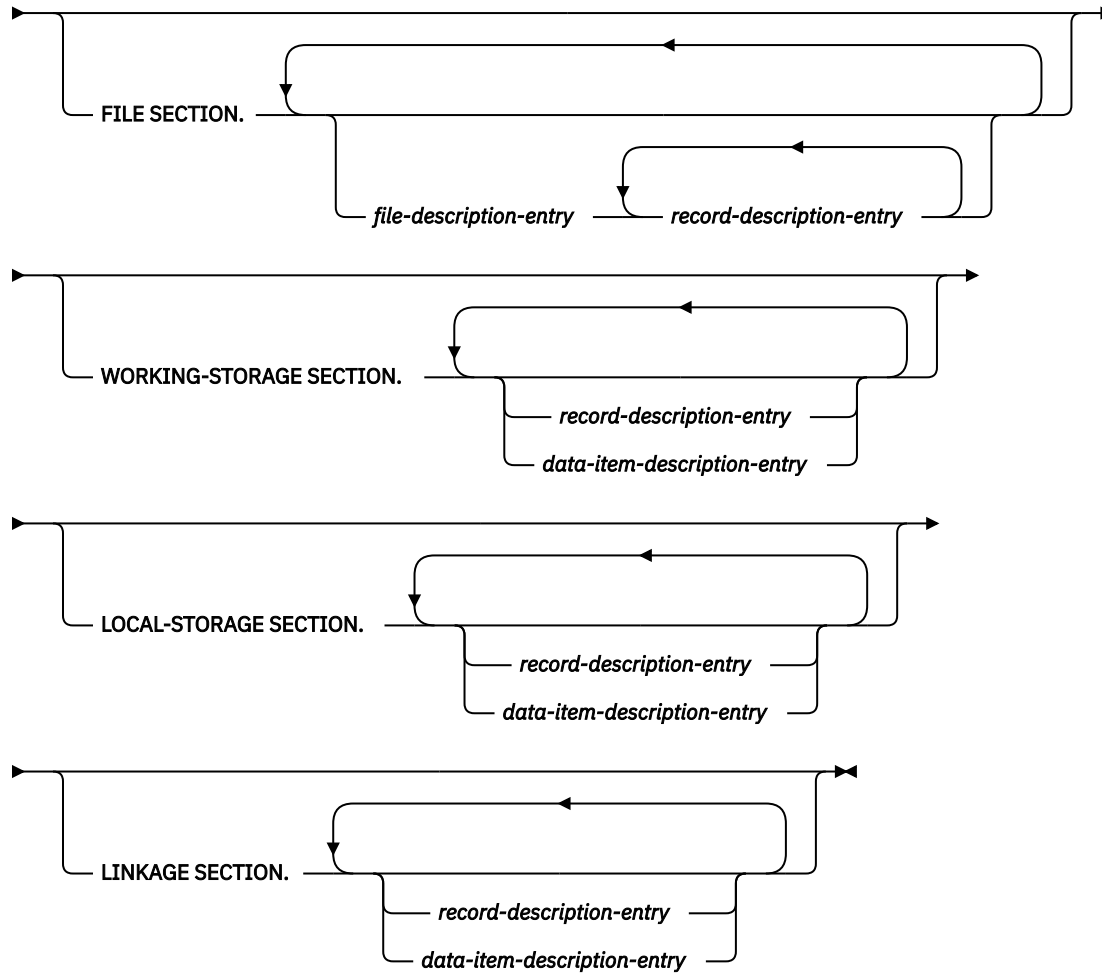
The DATA DIVISION of a user-defined function describes, in a structured manner, all the data to be processed by the function.

Function prototype data division

The function prototype data division contains data description entries in the LINKAGE SECTION that describe the parameters and returning item of the function.

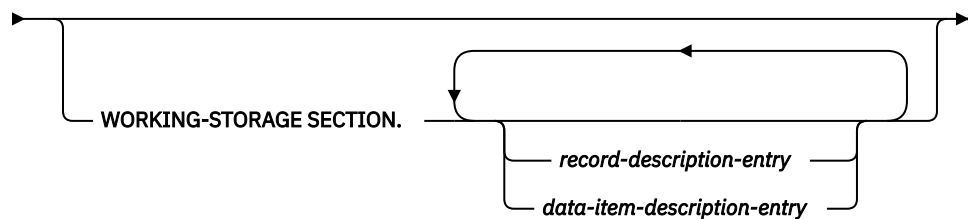
Format: program and method data division

➡ DATA DIVISION. ➡

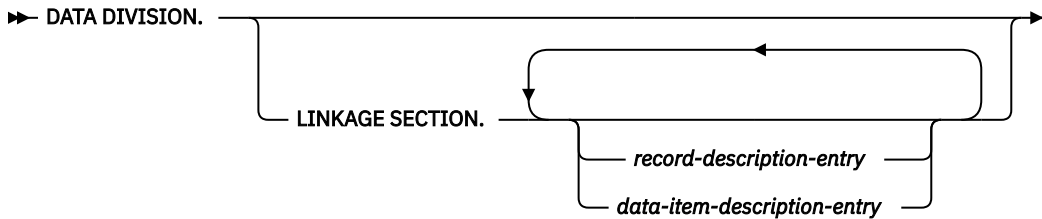


Format: object and factory data division

➡ DATA DIVISION. ➡



Format: function prototype data division



FILE SECTION

The FILE SECTION defines the structure of data files. The FILE SECTION must begin with the header FILE SECTION, followed by a separator period.

file-description-entry

Represents the highest level of organization in the FILE SECTION. It provides information about the physical structure and identification of a file, and gives the record-names associated with that file. For the format and the clauses required in a file description entry, see [Chapter 25, “DATA DIVISION--file description entries,”](#) on page 179.

record-description-entry

A set of data description entries (described in Chapter 26, “DATA DIVISION--data description entry,” on page 193) that describe the particular records contained within a particular file.

A record in the FILE SECTION must be described as an alphanumeric group item, a national group item, or an elementary data item of class alphabetic, alphanumeric, DBCS, national, or numeric.

More than one record description entry can be specified; each is an alternative description of the same record storage area.

Data areas described in the FILE SECTION are not available for processing unless the file that contains the data area is open.

A method FILE SECTION can define external files only. A single run-unit-level file connector is shared by all programs and methods that contain a definition of a given external file.

The FILE SECTION cannot be specified in a function prototype definition.

WORKING-STORAGE SECTION

The WORKING-STORAGE SECTION describes data records that are not part of data files but are developed and processed by a program or method. The WORKING-STORAGE SECTION also describes data items whose values are assigned in the source program or method and do not change during execution of the object program.

The WORKING-STORAGE SECTION must begin with the section header WORKING-STORAGE SECTION, followed by a separator period.

Program WORKING-STORAGE

The WORKING-STORAGE SECTION for programs, functions, and methods can also describe external data records, which are shared by programs and methods throughout the run unit. All clauses that are used in record descriptions in the FILE SECTION and also the VALUE and EXTERNAL clauses (which might not be specified in record description entries in the FILE SECTION) can be used in record descriptions in the WORKING-STORAGE SECTION.

User-defined function WORKING-STORAGE

Same as program WORKING-STORAGE.

Function prototype WORKING-STORAGE

The WORKING-STORAGE SECTION cannot be specified in a function prototype definition.

Method WORKING-STORAGE

A single copy of the WORKING-STORAGE for a method is statically allocated on the first invocation of the method and persists in a last-used state for the duration of the run unit. The same copy is used whenever the method is invoked regardless of which object instance the method is invoked upon.

If a VALUE clause is specified on a method WORKING-STORAGE data item, the data item is initialized to the VALUE clause value on the first invocation.

If the EXTERNAL clause is specified on a data description entry in a method WORKING-STORAGE SECTION, a single copy of the storage for that data item is allocated once for the duration of the run unit. That storage is shared by all programs and methods in the run unit that contain a definition for the external data item.

Object WORKING-STORAGE

The data described in the WORKING-STORAGE SECTION of an object paragraph is object instance data, usually called *instance data*. A separate copy of instance data is statically allocated for each object instance when the object is instantiated. Instance data persists in a last-used state until the object instance is freed by the Java runtime system.

Instance data can be initialized by VALUE clauses specified in data declarations or by logic specified in an instance method.

Factory WORKING-STORAGE

The data described in the WORKING-STORAGE SECTION of a factory paragraph is factory data. A single copy of factory data is statically allocated when the factory object for the class is created. Factory data persists in a last-used state for the duration of the run unit.

Factory data can be initialized by VALUE clauses specified in data declarations or by logic specified in a factory method.

Data storage location

By default, data items in WORKING-STORAGE are allocated beyond the 2 GB bar if the LP(64) compiler option is in effect; data items in WORKING-STORAGE are allocated under 2 GB bar if the LP(32) compiler option is in effect. You can use the DATA compiler directive to change this default to a different storage location. See [“DATA” on page 718](#) for more information.

For AMODE 31 programs, you can use the DATA compiler option to control whether WORKING-STORAGE data items are allocated under the 16 MB line (DATA(24)) or try to get storage beyond the 16 MB line (DATA(31)).

Note: All subordinate items belonging to the same group item will be allocated in the same data location.

The WORKING-STORAGE SECTION contains record description entries and data description entries for independent data items, called *data item description entries*.

record-description-entry

Data entries in the WORKING-STORAGE SECTION that bear a definite hierarchic relationship to one another must be grouped into records structured by level number. See [Chapter 26, “DATA DIVISION--data description entry,” on page 193](#) for more information.

data-item-description-entry

Independent items in the WORKING-STORAGE SECTION that bear no hierarchic relationship to one another need not be grouped into records provided that they do not need to be further subdivided. Instead, they are classified and defined as independent elementary items. Each is defined in a separate data-item description entry that begins with either the level number 77 or 01. See [Chapter 26, “DATA DIVISION--data description entry,” on page 193](#) for more information.

LOCAL-STORAGE SECTION

The LOCAL-STORAGE SECTION defines storage that is allocated and freed on a per-invocation basis.

On each invocation, data items defined in the LOCAL-STORAGE SECTION are reallocated. Each data item that has a VALUE clause is initialized to the value specified in that clause.

For nested programs, data items defined in the LOCAL-STORAGE SECTION are allocated upon each invocation of the containing outermost program. However, each data item is reinitialized to the value specified in its VALUE clause each time the nested program is invoked.

For methods, a separate copy of the data defined in LOCAL-STORAGE is allocated and initialized on each invocation of the method. The storage allocated for the data is freed when the method returns.

Data items defined in the LOCAL-STORAGE SECTION cannot specify the EXTERNAL clause.

The LOCAL-STORAGE SECTION must begin with the header LOCAL-STORAGE SECTION, followed by a separator period.

You can specify the LOCAL-STORAGE SECTION in recursive programs, nonrecursive programs, methods, and user-defined functions.

Method and user-defined function LOCAL-STORAGE content is the same as program LOCAL-STORAGE content except that the GLOBAL clause has no effect (because methods and user-defined functions cannot be nested).

The LOCAL-STORAGE SECTION cannot be specified in a function prototype definition.

LINKAGE SECTION

The LINKAGE SECTION describes data made available from another program or method.

record-description-entry

See [“WORKING-STORAGE SECTION” on page 163](#) for a description.

data-item-description-entry

See [“WORKING-STORAGE SECTION” on page 163](#) for a description.

Record description entries and data item description entries in the LINKAGE SECTION provide names and descriptions, but storage within the program or method is not reserved because the data area exists elsewhere.

Any data description clause, except for the EXTERNAL clause, can be used to describe items in the LINKAGE SECTION.

You can specify the GLOBAL clause in the LINKAGE SECTION. The GLOBAL clause has no effect for methods, however.

All level 01 or 77 data description entries in the LINKAGE SECTION of a function prototype definition must also be specified on the PROCEDURE DIVISION USING phrase or RETURNING phrase.

Data units

Data is grouped into the conceptual units as listed in the topic.

- File data
- Program data
- Method data
- Factory data
- Instance data
- User-defined function data
- Function prototype data

File data

File data is contained in files. A *file* is a collection of data records that exist on some input-output device. A file can be considered as a group of physical records; it can also be considered as a group of logical records. The DATA DIVISION describes the relationship between physical and logical records.

For more information, see [“FILE SECTION” on page 184](#).

A *physical record* is a unit of data that is treated as an entity when moved into or out of storage. The dimension of a physical record is determined by the particular input-output device on which it is stored. The dimension does not necessarily have a direct relationship to the amount or content of the logical information contained in the file.

A *logical record* is a unit of data whose subdivisions have a logical relationship. A logical record can itself be a physical record (that is, be contained completely within one physical unit of data); several logical records can be contained within one physical record, or one logical record can extend across several physical records.

File description entries specify the physical aspects of the data (such as the size relationship between physical and logical records, the size and names of the logical records, labeling information, and so forth).

Record description entries describe the logical records in the file (including the category and format of data within each field of the logical record), different values the data might be assigned, and so forth.

After the relationship between physical and logical records has been established, only logical records are made available to you. For this reason, a reference in this information to "records" means logical records, unless the term "physical records" is used.

Program data

Program data is created by a program instead of being read from a file.

The concept of logical records applies to program data as well as to file data. Program data can thus be grouped into logical records, and be defined by a series of record description entries. Items that need not be so grouped can be defined in independent data description entries (called *data item description entries*).

Method data

Method data is defined in the DATA DIVISION of a method and is processed by the procedural code in that method. Method data is organized into logical records and independent data description entries in the same manner as program data.

Factory data

Factory data is defined in the DATA DIVISION in the factory paragraph of a class definition and is processed by procedural code in the factory methods of that class. Factory data is organized into logical records and independent data description entries in the same manner as program data.

There is one factory object for a given class in a run unit, and therefore only one instance of factory data in a run unit for that class.

Instance data

Instance data is defined in the DATA DIVISION in the object paragraph of a class definition and is processed by procedural code in the instance methods of that class. Instance data is organized into logical records and independent data description entries in the same manner as program data.

There is one copy of instance data in each object instance of a given class. There can be many object instances for a given class. Each has its own separate copy of instance data.

User-defined function data

Same as [program data](#).

Function prototype data

Function prototype data is defined within the LINKAGE SECTION of the DATA DIVISION of a function prototype definition.

Function prototype data is organized into logical records and independent data description entries in the same manner as program data. Since no program object is produced when you compile a function prototype, function prototype data has no representation at runtime. Rather, the data description entries provide the information to the compiler in order to perform conformance checking between the invocation of a function and its prototype, as well as consistency checking between prototypes of the same name.

Data relationships

The relationships among all data to be used in a program are defined in the DATA DIVISION through a system of level indicators and level-numbers.

A *level indicator*, with its descriptive entry, identifies each file in a program. Level indicators represent the highest level of any data hierarchy with which they are associated. FD is the file description level indicator and SD is the sort-merge file description level indicator.

A *level-number*, with its descriptive entry, indicates the properties of specific data. Level-numbers can be used to describe a data hierarchy; they can indicate that this data has a special purpose. Although they can be associated with (and subordinate to) level indicators, they can also be used independently to describe internal data or data common to two or more programs. (See [“Level-numbers”](#) on page 194 for level-number rules.)

Levels of data

After a record has been defined, it can be subdivided to provide more detailed data references.

For example, in a customer file for a department store, one complete record could contain all data that pertains to one customer. Subdivisions within that record could be, for example, customer name, customer address, account number, department number of sale, unit amount of sale, dollar amount of sale, previous balance, and other pertinent information.

The basic subdivisions of a record (that is, those fields not further subdivided) are called *elementary items*. Thus a record can be made up of a series of elementary items or can itself be an elementary item.

It might be necessary to refer to a set of elementary items; thus, elementary items can be combined into *group items*. Groups can also be combined into a more inclusive group that contains one or more subgroups. Thus within one hierarchy of data items, an elementary item can belong to more than one group item.

A system of level-numbers specifies the organization of elementary and group items into records. Special level-numbers are also used to identify data items used for special purposes.

Levels of data in a record description entry

Each group and elementary item in a record requires a separate entry, and each must be assigned a level-number.

A level-number is a one-digit or two-digit integer between 01 and 49, or one of three special level-numbers: 66, 77, or 88. The following level-numbers are used to structure records:

01

This level-number specifies the record itself, and is the most inclusive level-number possible. A level-01 entry can be an alphanumeric group item, a national group item, or an elementary item. The level number must begin in Area A.

02 through 49

These level-numbers specify group and elementary items within a record. They can begin in Area A or Area B. Less inclusive data items are assigned higher (not necessarily consecutive) level-numbers in this series.

The relationship between level-numbers within a group item defines the hierarchy of data within that group.

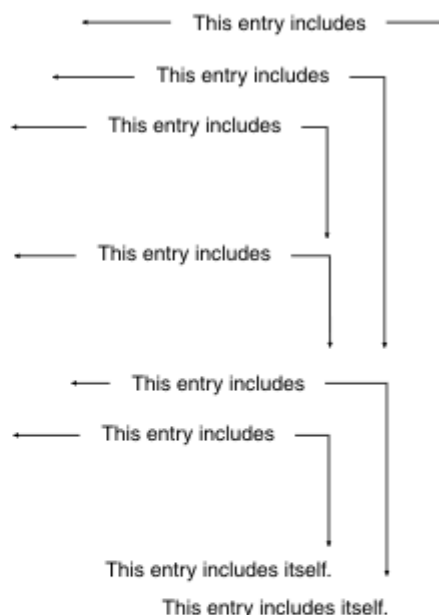
A group item includes all group and elementary items that follow it until a level-number less than or equal to the level-number of that group is encountered.

The following figure illustrates a group wherein all groups immediately subordinate to the level-01 entry have the same level-number.

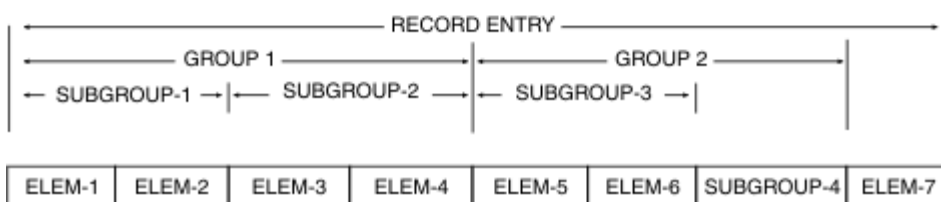
The COBOL record description entry written as follows:

```
01 RECORD-ENTRY.  
  05 GROUP-1.  
    10 SUBGROUP-1.  
      15 ELEM-1 PIC...  
      15 ELEM-2 PIC...  
    10 SUBGROUP-2.  
      15 ELEM-3 PIC...  
      15 ELEM-4 PIC...  
  05 GROUP-2.  
    15 SUBGROUP-3.  
      25 ELEM-5 PIC...  
      25 ELEM-6 PIC...  
    15 SUBGROUP-4 PIC...  
  05 ELEM-7 PIC...
```

is subdivided as indicated below:



The storage arrangement of the record description entry is illustrated below:



You can also define groups with subordinate items that have different level-numbers for the same level in the hierarchy. In the following example, 05 EMPLOYEE-NAME and 04 EMPLOYEE-ADDRESS in EMPLOYEE-RECORD define the same level in the hierarchy. The compiler renumbers the levels in a relative fashion, as shown in MAP output.

```
01 EMPLOYEE-RECORD.  
  05 EMPLOYEE-NAME.  
    10 FIRST-NAME PICTURE X(10).  
    10 LAST-NAME PICTURE X(10).  
  04 EMPLOYEE-ADDRESS.  
    08 STREET PICTURE X(10).  
    08 CITY PICTURE X(10).
```

The following record description entry defines the same data hierarchy as the preceding record description entry:

```
01  EMPLOYEE-RECORD.  
   02  EMPLOYEE-NAME.  
       03  FIRST-NAME PICTURE  X(10).  
       03  LAST-NAME  PICTURE  X(10).  
   02  EMPLOYEE-ADDRESS.  
       03  STREET     PICTURE  X(10).  
       03  CITY       PICTURE  X(10).
```

Elementary items can be specified at any level within the hierarchy.

Special level-numbers

Special level-numbers identify items that do not structure a record.

The special level-numbers are:

66

Identifies items that must contain a RENAME clause; such items regroup previously defined data items. (For details, see [“RENAME clause”](#) on page 228.)

77

Identifies data item description entries that are independent WORKING-STORAGE, LOCAL-STORAGE, or LINKAGE SECTION items; they are not subdivisions of other items and are not subdivided themselves. Level-77 items must begin in Area A.

88

Identifies any condition-name entry that is associated with a particular value of a conditional variable. (For details, see [“VALUE clause”](#) on page 245.)

Level-77 and level-01 entries in the WORKING-STORAGE, LOCAL-STORAGE, or LINKAGE SECTION that are referenced in a program or method must be given unique data-names because level-77 and level-01 entries cannot be qualified. Subordinate data-names that are referenced in the program or method must be either uniquely defined, or made unique through qualification. Unreferenced data-names need not be uniquely defined.

Indentation

Successive data description entries can begin in the same column as preceding entries, or can be indented.

Indentation is useful for documentation but does not affect the action of the compiler.

Classes and categories of group items

Enterprise COBOL has three types of groups: alphanumeric groups, national groups, and UTF-8 groups.

Groups that do not specify a GROUP-USAGE clause are alphanumeric groups. An alphanumeric group has class and category alphanumeric and is treated as though its usage were DISPLAY, regardless of the representation of the elementary data items that are contained within the group. In many operations, such as moves and compares, alphanumeric groups are treated as though they were elementary items of category alphanumeric, except that no editing or conversion of data representation takes place. In other operations, such as MOVE CORRESPONDING and ADD CORRESPONDING, the subordinate data items are processed as separate elementary items.

National groups are defined by a GROUP-USAGE clause with the NATIONAL phrase at the group level. All subordinate data items must be explicitly or implicitly described with usage NATIONAL, and subordinate groups must be explicitly or implicitly described with GROUP-USAGE NATIONAL.

Unless stated otherwise, a national group item is processed exactly as though it were an elementary data item of usage national, class and category national, described with PICTURE N(m), where m is the length of the group in national character positions. Because national groups contain only national characters,

data is converted as necessary for moves and compares. The compiler ensures proper truncation and padding. In other operations, such as MOVE CORRESPONDING and ADD CORRESPONDING, the subordinate data items are processed as separate elementary items. See [“GROUP-USAGE clause” on page 198](#) for details.

UTF-8 groups are defined by a GROUP-USAGE clause with the UTF-8 phrase at the group level. All subordinate data items must be explicitly or implicitly described with USAGE UTF-8 and must be defined with the BYTE-LENGTH phrase of the PICTURE clause. Subordinate groups must be explicitly or implicitly described with GROUP-USAGE UTF-8.

Unless stated otherwise, a UTF-8 group item is processed exactly as though it were an elementary data item of usage UTF-8, class and category UTF-8, described with PICTURE U BYTE-LENGTH *m*, where *m* is the length of the group in bytes. Because UTF-8 groups contain only UTF-8 characters, data is converted as necessary for moves and compares. The compiler ensures proper truncation and padding. In other operations, such as MOVE CORRESPONDING, the subordinate data items are processed as separate elementary items. See [“GROUP-USAGE clause” on page 198](#) for details.

The following table summarizes the classes and categories of group items.

<i>Table 7. Classes and categories of group items</i>				
Group description	Class of group	Category of group	USAGE of elementary items within a group	USAGE of a group
Without a GROUP-USAGE clause	Alphanumeric	Alphanumeric (even though the elementary items in the group can have any category)	Any	Treated as DISPLAY when usage is relevant
With explicit or implicit GROUP-USAGE clause	National	National	NATIONAL	NATIONAL
With explicit or implicit GROUP-USAGE clause	UTF-8	UTF-8	UTF-8	UTF-8

Classes and categories of data

Most data and all literals used in a COBOL program are divided into classes and categories. Data classes are groupings of data categories. Data categories are determined by the attributes of data description entries or function definitions.

For more information about data categories, see [“Category descriptions” on page 172](#).

The following elementary data items do not have a class and category:

- Index data items
- Items described with USAGE POINTER, USAGE FUNCTION-POINTER, USAGE PROCEDURE-POINTER, or USAGE OBJECT REFERENCE

All other types of elementary data items have a class and category as shown in [Table 8 on page 171](#).

A function references an elementary data item and belongs to the data class and category associated with the type of the function, as shown in [Table 9 on page 171](#).

Literals have a class and category as shown in [Table 10 on page 172](#). Figurative constants (except NULL) have a class and category that depends on the literal or value represented by the figurative constant in the context of its use. For details, see [“Figurative constants” on page 15](#).

All group items have a class and category, even if the subordinate elementary items belong to another class and category. For the classification of group items, see “Classes and categories of group items” on page 169.

<i>Table 8. Class, category, and usage of elementary data items</i>		
Class of elementary data items²	Category	Usage
Alphabetic	Alphabetic	DISPLAY
Alphanumeric	Alphanumeric	DISPLAY
	Alphanumeric-edited	DISPLAY
	Numeric-edited	DISPLAY
Date-Time ¹	Date ¹ Time ¹	DISPLAY PACKED-DECIMAL
	Timestamp ¹	DISPLAY
DBCS ¹	DBCS ¹	DISPLAY-1
National ¹	National ¹	NATIONAL
	National-edited ¹	NATIONAL
	Numeric-edited ¹	NATIONAL
UTF-8	UTF-8	UTF-8
Numeric	Numeric	DISPLAY (type zoned decimal)
		NATIONAL (type national decimal)
		PACKED-DECIMAL (type internal decimal)
		COMP-3 (type internal decimal)
		BINARY
		COMP
		COMP-4
		COMP-5
	Internal floating-point ¹	COMP-1
		COMP-2
	External floating-point ¹	DISPLAY
		NATIONAL

<i>Table 9. Classes and categories of functions</i>	
Function type	Class and category
Alphanumeric	Alphanumeric

<i>Table 9. Classes and categories of functions (continued)</i>	
Function type	Class and category
National	National
UTF-8	UTF-8
Integer	Numeric
Numeric	Numeric

<i>Table 10. Classes and categories of literals</i>	
Literal	Class and category
Alphanumeric (including hexadecimal formats)	Alphanumeric
DBCS	DBCS
National (including hexadecimal formats)	National
UTF-8 (including hexadecimal formats)	UTF-8
Numeric (fixed-point and floating-point)	Numeric

Category descriptions

The category of a data item is established by the attributes of its data description entry (such as its PICTURE character-string or USAGE clause) or by its function definition.

The meaning of each category is given in the following sections.

Alphabetic

A data item is described as category alphabetic by its PICTURE character-string. For PICTURE character-string details, see [“Alphabetic items” on page 213](#).

A data item of category alphabetic is referred to as an alphabetic data item.

Alphanumeric

Each of the following data items is of category alphanumeric:

- An elementary data item described as alphanumeric by its PICTURE character-string. For PICTURE character-string details, see [“Alphanumeric items” on page 213](#).
- An alphanumeric group item.
- An alphanumeric function.
- The following special registers:
 - DEBUG-ITEM
 - SHIFT-OUT

- SHIFT-IN
- SORT-CONTROL
- SORT-MESSAGE
- WHEN-COMPILED
- XML-EVENT
- XML-TEXT

Alphanumeric-edited

A data item is described as category alphanumeric-edited by its PICTURE character-string. For PICTURE character-string details, see [“Alphanumeric-edited items” on page 214](#).

A data item of category alphanumeric-edited is referred to as an alphanumeric-edited data item.

DBCS

A data item is described as category DBCS by its PICTURE character-string and the NSYMBOL(DBCS) compiler option or by an explicit USAGE DISPLAY-1 clause. For PICTURE character-string details, see [“DBCS items” on page 214](#).

A data item of category DBCS is referred to as a DBCS data item.

External floating-point

A data item is described as category external floating-point by its PICTURE character-string. For PICTURE character-string details, see [“External floating-point items” on page 214](#). An external floating-point data item can be described with USAGE DISPLAY or USAGE NATIONAL.

When the usage is DISPLAY, the item is referred to as a display floating-point data item.

When the usage is NATIONAL, the item is referred to as a national floating-point data item.

An external floating-point data item is of class numeric and, unless specifically excluded, is included in a reference to a numeric data item.

Internal floating-point

A data item is described as category internal floating-point by a USAGE clause with the COMP-1 or COMP-2 phrase.

A data item of category internal floating-point is referred to as an internal floating-point data item. An internal floating-point data item is of class numeric and, unless specifically excluded, is included in a reference to a numeric data item.

National

Each of the following data items is of category national:

- A data item that is described as category national by its PICTURE character-string and the NSYMBOL(NATIONAL) compiler option or by an explicit USAGE NATIONAL clause. For PICTURE character-string details, see [“National items” on page 215](#).
- A group item explicitly or implicitly described with a GROUP-USAGE NATIONAL clause.
- A national function.
- The special register XML-NTEXT.

National-edited

A data item is described as category national-edited by its PICTURE character-string. For PICTURE character-string details, see [“National-edited items” on page 216](#).

A data item of category national-edited is referred to as a national-edited data item.

UTF-8

Each of the following data items is of category UTF-8:

- A data item is described as category UTF-8 when its PICTURE character-string contains one or more U symbols. The explicit USAGE UTF-8 clause may be present for items containing U in their PICTURE character-string but is not required. For PICTURE character-string details, see [“UTF-8 items” on page 218](#).
- A group item explicitly or implicitly described with a GROUP-USAGE UTF-8 clause.
- A UTF-8 function.

Numeric

Each of the following data items is of category numeric:

- An elementary data item described as numeric by its PICTURE character-string and not described with a BLANK WHEN ZERO clause. For PICTURE character-string details, see [“Numeric items” on page 217](#).
- An elementary data item described with one of the following usages:
 - BINARY, COMPUTATIONAL, COMPUTATIONAL-4, COMPUTATIONAL-5, COMP, COMP-4, or COMP-5
 - PACKED-DECIMAL, COMPUTATIONAL-3, or COMP-3
- A special register of numeric type:
 - JSON-CODE
 - JSON-STATUS
 - LENGTH OF
 - LINAGE-COUNTER
 - RETURN-CODE
 - SORTCORE-SIZE
 - SORT-FILE-SIZE
 - SORT-MODE-SIZE
 - SORT-RETURN
 - TALLY
 - XML-CODE
- A numeric function.
- An integer function.

A data item of category numeric is referred to as a numeric data item.

Numeric-edited

Each of the following data items is of category numeric-edited:

- A data item described as numeric-edited by its PICTURE character-string. For PICTURE character-string details, see [“Numeric-edited items” on page 218](#).
- A data item described as numeric by its PICTURE character-string and described with a BLANK WHEN ZERO clause.

Alignment rules

The standard alignment rules for positioning data in an elementary item depend on the category of a receiving item.

A receiving item is an item into which the data is moved. For more details about a receiving item, see [“Elementary moves” on page 406](#).

Numeric

For numeric receiving items, the following rules apply:

1. The data is aligned on the assumed decimal point and, if necessary, truncated or padded with zeros. (An *assumed decimal point* is one that has logical meaning but that does not exist as an actual character in the data.)
2. If an assumed decimal point is not explicitly specified, the receiving item is treated as though an assumed decimal point is specified immediately to the right of the field. The data is then treated according to the preceding rule.

Numeric-edited

The data is aligned on the decimal point, and (if necessary) truncated or padded with zeros at either end except when editing causes replacement of leading zeros.

Internal floating-point

A decimal point is assumed immediately to the left of the field. The data is then aligned on the leftmost digit position that follows the decimal point, with the exponent adjusted accordingly.

External floating-point

The data is aligned on the leftmost digit position; the exponent is adjusted accordingly.

Alphanumeric, alphanumeric-edited, alphabetic, DBCS

For these receiving items, the following rules apply:

1. The data is aligned at the leftmost character position, and (if necessary) truncated or padded with spaces at the right.
2. If the JUSTIFIED clause is specified for this receiving item, the above rule is modified as described in [“JUSTIFIED clause” on page 198](#).

National, national-edited

For these receiving items, the following rules apply:

1. The data is aligned at the leftmost character position, and (if necessary) truncated or padded with default Unicode spaces (NX'0020') at the right. Truncation occurs at the boundary of a national character position.
2. If the JUSTIFIED clause is specified for this receiving item, the above rule is modified as described in [“JUSTIFIED clause” on page 198](#).

UTF-8

For these receiving items, the following rules apply:

1. The data is aligned at the leftmost character position, and (if necessary) truncated or padded with default UTF-8 spaces (UX'20') at the right. Truncation occurs at the boundary of a UTF-8 character position.
2. If the JUSTIFIED clause is specified for this receiving item, the above rule is modified as described in [“JUSTIFIED clause” on page 198](#).

Character-string and item size

For items described with a PICTURE clause, the size of an elementary item is expressed in source code by the number of character positions described in the PICTURE character-string and a SIGN clause (if applicable). Storage size, however, is determined by the actual number of bytes the item occupies as determined by the combination of its PICTURE character-string, SIGN IS SEPARATE clause (if specified), and USAGE clause.

For items described with USAGE DISPLAY (categories alphabetic, alphanumeric, alphanumeric-edited, numeric-edited, numeric, and external floating-point), 1 byte of storage is reserved for each character position described by the item's PICTURE character-string and SIGN IS SEPARATE clause (if applicable).

For items described with USAGE DISPLAY-1 (category DBCS), 2 bytes of storage are reserved for each character position described by the item's PICTURE character-string.

For items described with USAGE NATIONAL (categories national, national-edited, numeric-edited, numeric, and external floating-point), 2 bytes of storage are reserved for each character position described by the item's PICTURE character-string and SIGN IS SEPARATE clause (if specified).

For items described with USAGE UTF-8 (category UTF-8), when the BYTE-LENGTH phrase of the PICTURE clause and the DYNAMIC LENGTH clause are not specified in the item's definition, 4 bytes of storage are reserved for each character position described by the item's PICTURE character-string. Note, however, that due to the varying-length nature of UTF-8 characters, many UTF-8 strings of the specified character length can be represented using fewer bytes than the maximum reserved. In such a case, any unused bytes in the data item are padded with UTF-8 blanks.

For internal floating-point items, the size of the item in storage is determined by its USAGE clause. USAGE COMPUTATIONAL-1 reserves 4 bytes of storage for the item; USAGE COMPUTATIONAL-2 reserves 8 bytes of storage.

Normally, when an arithmetic item is moved from a longer field into a shorter one, the compiler truncates the data to the number of digits represented in the shorter item's PICTURE character-string by truncating leading digits. For example, if a sending field with PICTURE S99999 that contains the value +12345 is moved to a BINARY receiving field with PICTURE S99, the data is truncated to +45. For additional information, see [“USAGE clause”](#) on page 237.

The TRUNC compiler option can affect the value of a binary numeric item. For information about TRUNC, see *TRUNC* in the *Enterprise COBOL Programming Guide*.

Signed data

There are two categories of algebraic signs used in COBOL: operational signs and editing signs.

Operational signs

Operational signs are associated with signed numeric items, and indicate their algebraic properties.

The internal representation of an algebraic sign depends on the item's USAGE clause, its SIGN clause (if present), and the operating environment. (For further details about the internal representation, see *Examples: numeric data and internal representation* in the *Enterprise COBOL Programming Guide*.)

Editing signs

Editing signs are associated with numeric-edited items. Editing signs are PICTURE symbols that identify the sign of the item in edited output.

Dynamic-length items

Enterprise COBOL supports usage of dynamic-length items. Generally, dynamic-length items are items whose logical length might change at runtime.

Dynamic-length elementary items

A dynamic-length item is a data item whose logical length might change at runtime.

If a dynamic-length elementary item is used as a sending operand (which might be reference-modified), the item is treated as a fixed-length data item with a length that is the same as the current length of the dynamic-length elementary item. If a dynamic-length elementary item is used as a receiving operand and is not reference-modified, the sender contents are moved into the receiver's content buffer.

If the length of the sender is longer than the length of the receiver, then a receiver content buffer with more capacity might be allocated at runtime to contain the contents of the sender, before moving any data. The length of the receiver is then set to the length of the sender. If the length of the sender is zero, no data is moved and the length of the receiver is set to zero. If the sender is a figurative constant, the length of the operand is as specified in [“Figurative constants” on page 15](#).

If a dynamic-length elementary item is used as a receiving operand and is reference-modified, the item is treated as a fixed-length item with a length that is the same as the current length of the dynamic-length elementary item. The dynamic-length elementary item receiver buffer will not be allocated or reallocated if it is reference-modified.

Note: Do not use dynamic-length elementary items as reference-modified receiving items before they have been initialized. Doing so results in unpredictable behavior.

For items described with the DYNAMIC LENGTH clause, the number of bytes allocated for the item depends on the length of the item at program runtime.

The following statements support dynamic-length elementary items:

ACCEPT statement - Format 2 (See [“System date-related information transfer” on page 309](#))

[“CALL statement” on page 319](#)

[“CANCEL statement” on page 327](#)

[“DISPLAY statement” on page 333](#)

[“EVALUATE statement” on page 340](#)

[“MOVE statement” on page 405](#)

[“SET statement” on page 445](#)

[“STOP statement” on page 462](#)

[“STRING statement” on page 462](#)

[“UNSTRING statement” on page 470](#)

Dynamic-length group items

A dynamic-length group item is a group item that contains a subordinate dynamic-length elementary item and whose logical length might change at runtime.

A dynamic-length group item's data description entry contains one or more subordinate dynamic-length elementary items. A group item that is not a dynamic-length group item is considered to be a fixed-length group item, such as group items that do not contain subordinate dynamic-length elementary items. Fixed-length group items can contain variable length tables whose data description entry contains the OCCURS DEPENDING ON clause.

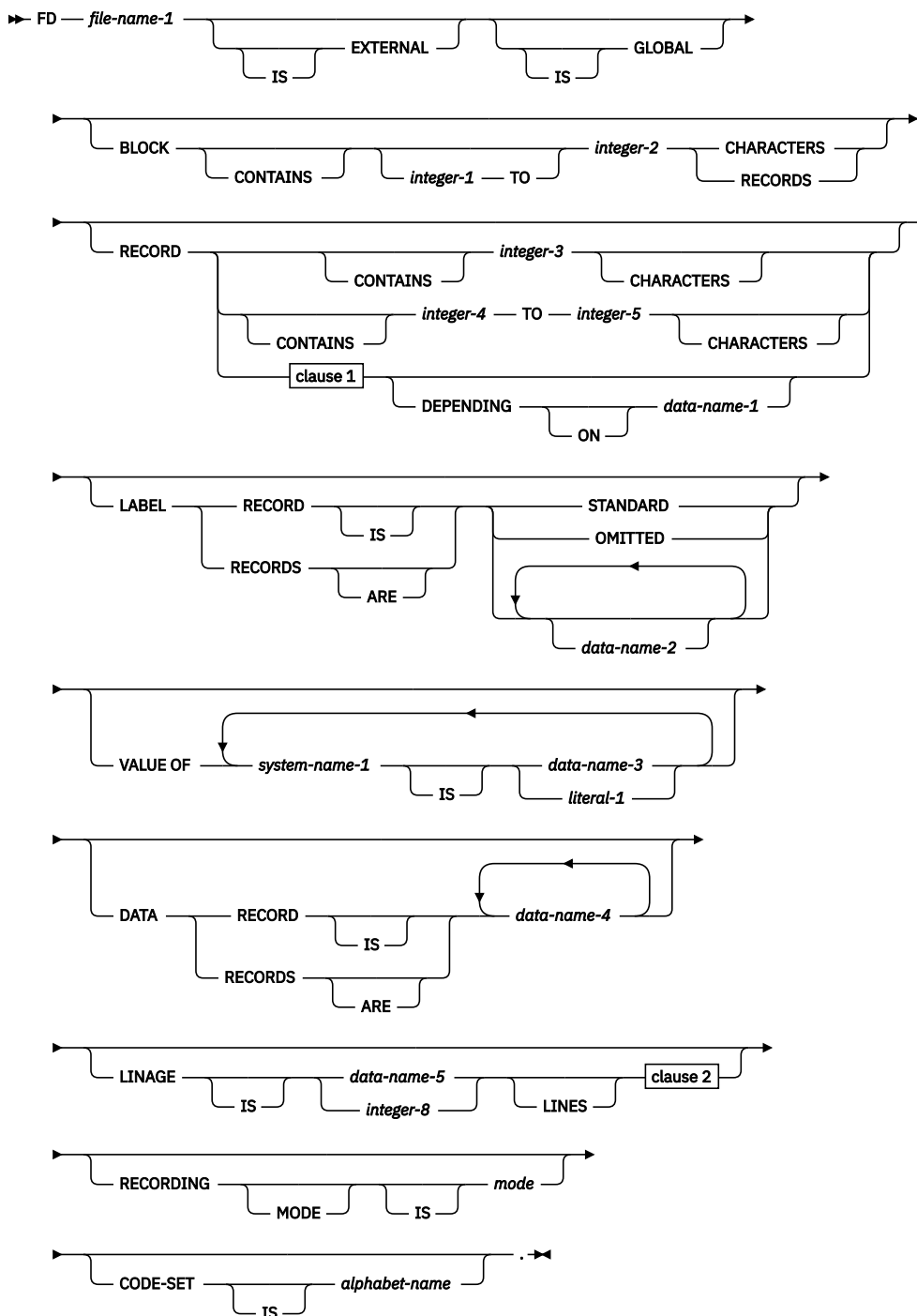
Dynamic-length group items may not be compared to, or moved to, other group items (dynamic-length or not).

Fixed-length group items are always compatible and comparable with other fixed-length group items.

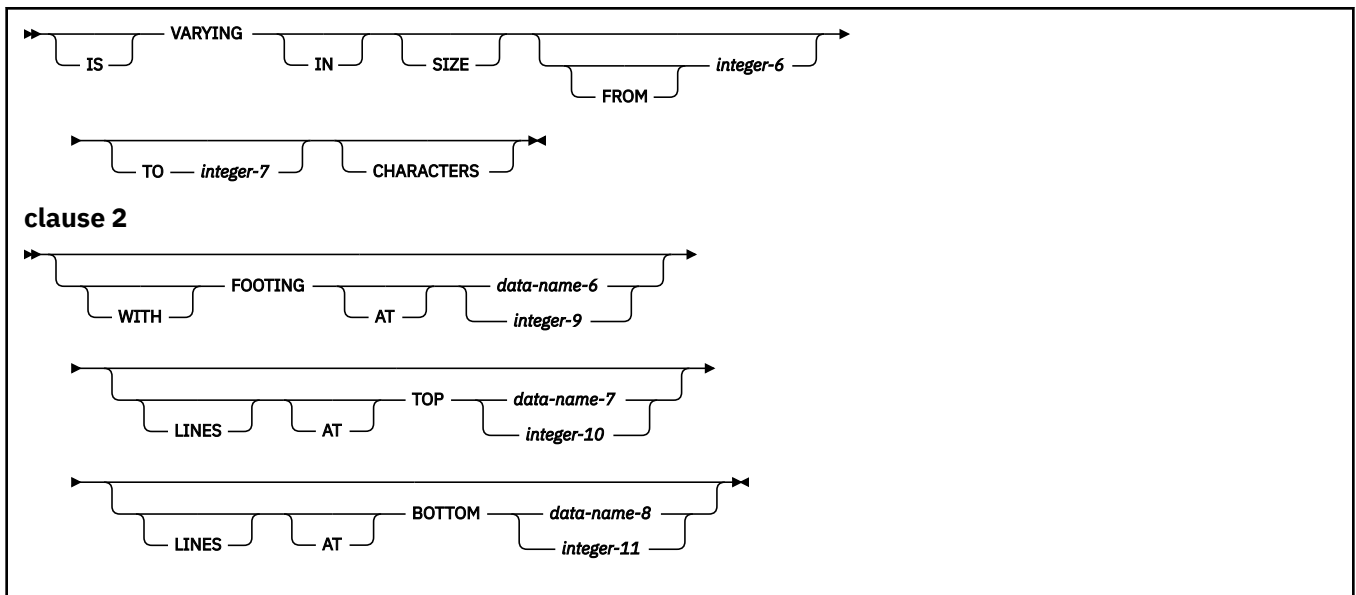
Chapter 25. DATA DIVISION--file description entries

In a COBOL program, the *File Description (FD) Entry* (or *Sort File Description (SD) Entry* for sort/merge files) represents the highest level of organization in the FILE SECTION. The order in which the optional clauses follow the FD or SD entry is not important.

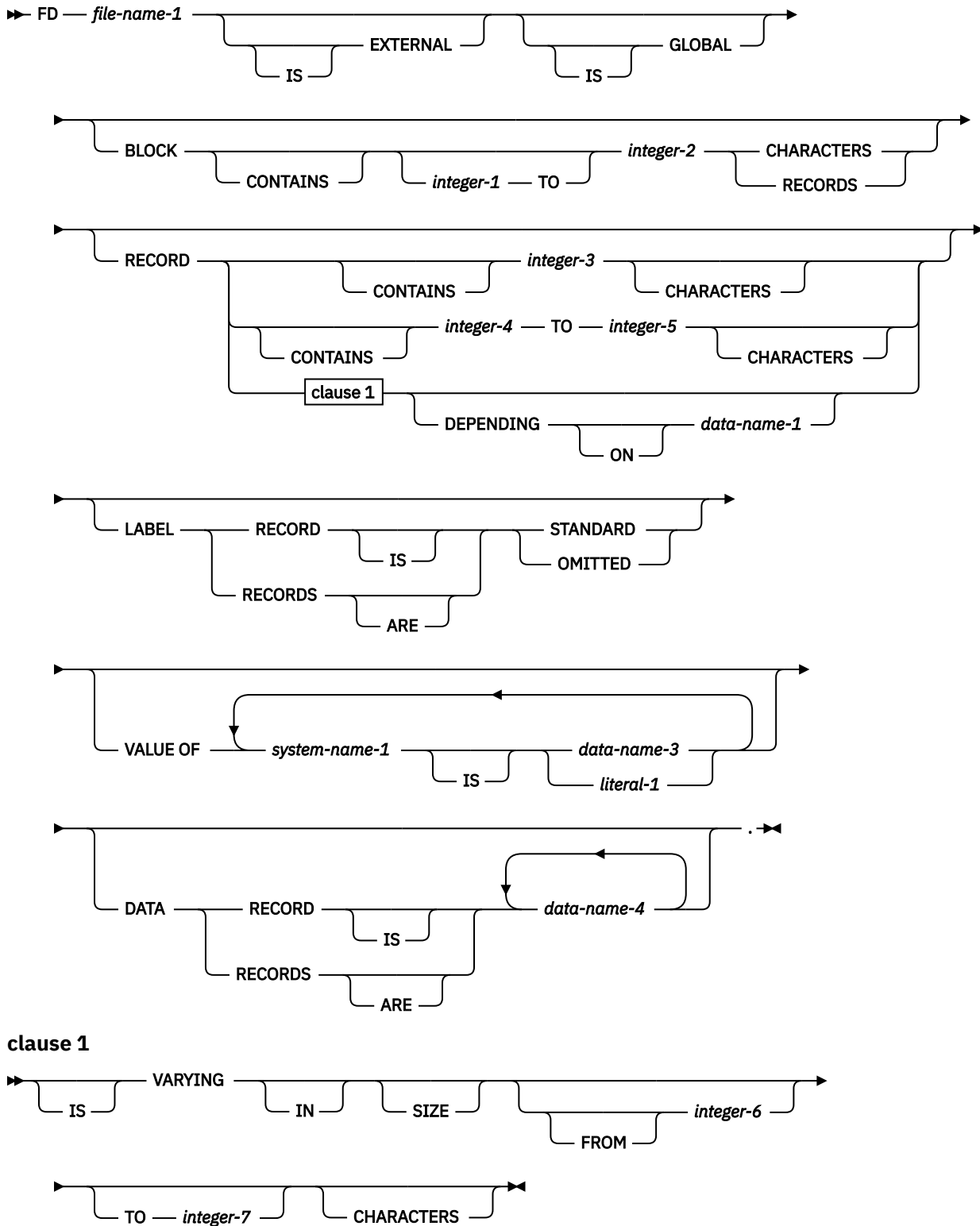
Format 1: sequential file description entry



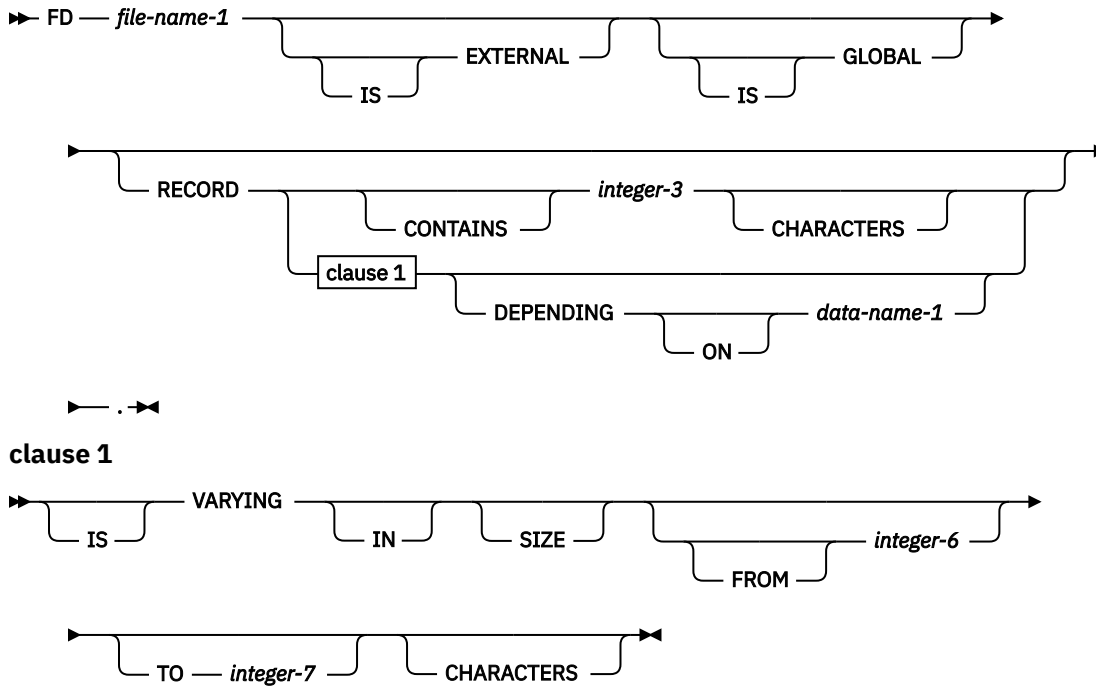
clause 1



Format 2: relative or indexed file description entry

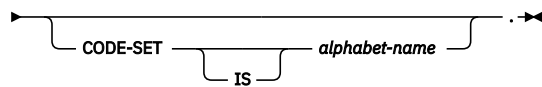
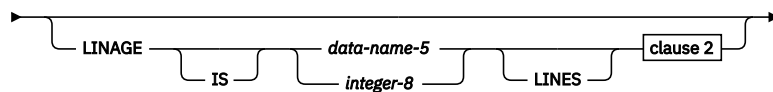
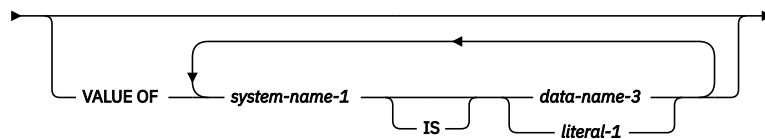
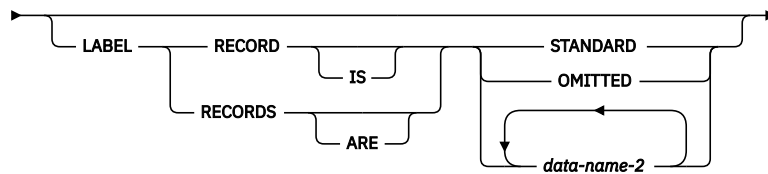
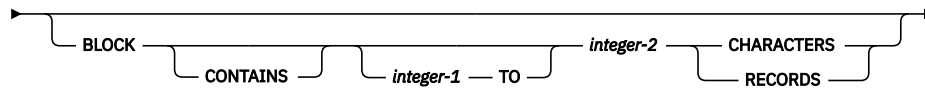
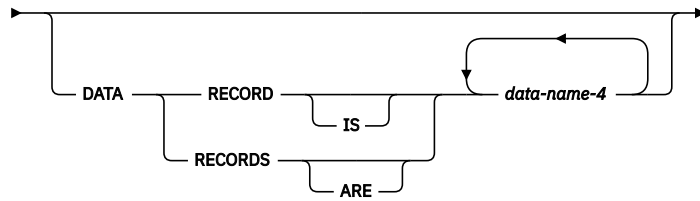
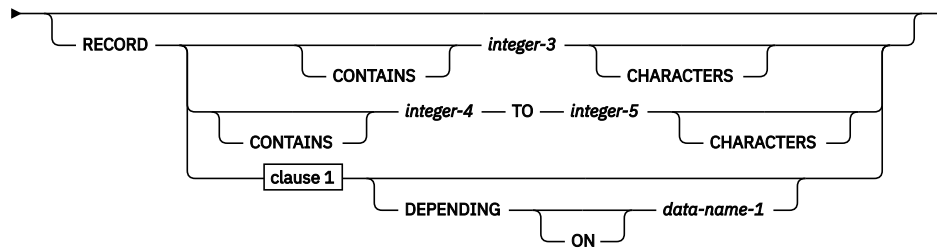


Format 3: line-sequential file description entry

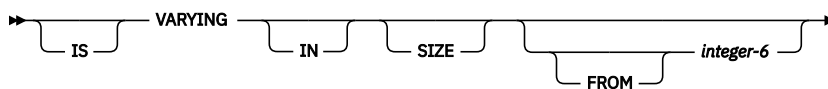


Format 4: sort/merge file description entry

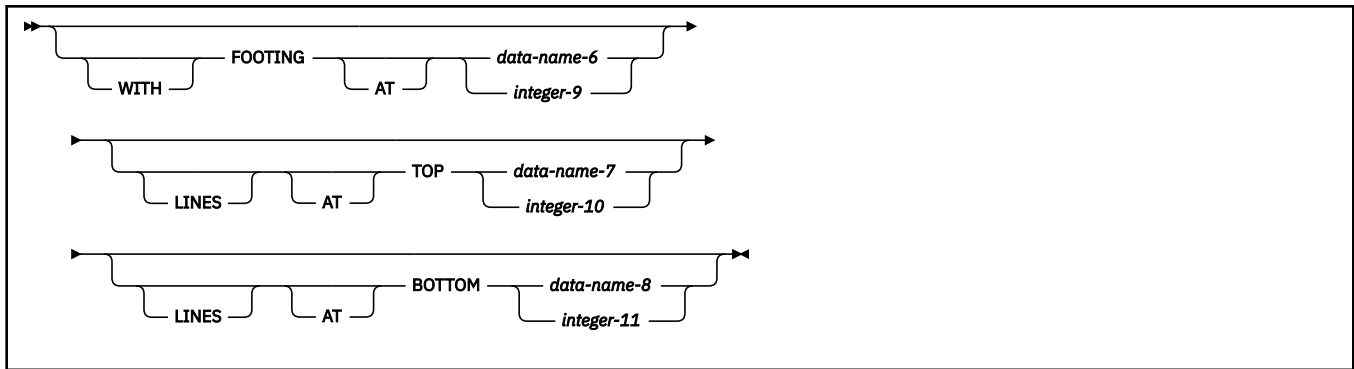
► SD — *file-name-1* →



clause 1



clause 2



FILE SECTION

The FILE SECTION must contain a level-indicator for each input and output file. For all files except sort or merge files, the FILE SECTION must contain an FD entry. For each sort or merge file, the FILE SECTION must contain an SD entry.

file-name

Must follow the level indicator (FD or SD), and must be the same as that specified in the associated SELECT clause. *file-name* must adhere to the rules of formation for a user-defined word; at least one character must be alphabetic. *file-name* must be unique within this program.

One or more record description entries must follow *file-name*. When more than one record description entry is specified, each entry implies a redefinition of the same storage area.

The clauses that follow *file-name* are optional, and they can appear in any order.

FD (formats 1, 2, and 3)

The last clause in the FD entry must be immediately followed by a separator period.

SD (format 4)

An SD entry must be written for each sort or merge file in the program. The last clause in the SD entry must be immediately followed by a separator period.

The following example illustrates the FILE SECTION entries needed for a sort or merge file:

```
SD  SORT-FILE.
01  SORT-RECORD  PICTURE X(80).
```

A record in the FILE SECTION must be described as an alphanumeric group item, a national group item, or an elementary item of class alphabetic, alphanumeric, DBCS, national, or numeric.

EXTERNAL clause

The EXTERNAL clause specifies that a file connector is external, and permits communication between two programs by the sharing of files.

A file connector is external if the storage associated with that file is associated with the run unit rather than with any particular program within the run unit. An external file can be referenced by any program in the run unit that describes the file. References to an external file from different programs that use separate descriptions of the file are always to the same file. In a run unit, there is only one representative of an external file.

In the FILE SECTION, the EXTERNAL clause can be specified only in file description entries.

The records appearing in the file description entry need not have the same name in corresponding external file description entries. In addition, the number of such records need not be the same in corresponding file description entries.

Use of the EXTERNAL clause does not imply that the associated file-name is a global name. See *Sharing data by using the EXTERNAL clause* in the *Enterprise COBOL Programming Guide* for specific information about the use of the EXTERNAL clause.

GLOBAL clause

The GLOBAL clause specifies that the file connector named by a file-name is a global name. A global file-name is available to the program that declares it and to every program that is contained directly or indirectly in that program.

A file-name is global if the GLOBAL clause is specified in the file description entry for that file-name. A record-name is global if the GLOBAL clause is specified in the record description entry by which the record-name is declared or, in the case of record description entries in the FILE SECTION, if the GLOBAL clause is specified in the file description entry for the file-name associated with the record description entry. For details on using the GLOBAL clause, see *Using data in input and output operations* and *Scope of names* in the *Enterprise COBOL Programming Guide*.

Two programs in a run unit can reference global file connectors in the following circumstances:

- An external file connector can be referenced from any program that describes that file connector.
- If a program is contained within another program, both programs can refer to a global file connector by referring to an associated global file-name either in the containing program or in any program that directly or indirectly contains the containing program.

BLOCK CONTAINS clause

The BLOCK CONTAINS clause specifies the size of the physical records.

The CHARACTERS phrase indicates that the integer specified in the BLOCK CONTAINS clause reflects the number of bytes in the record. For example, if you have a block with 10 DBCS characters or 10 national characters, the BLOCK CONTAINS clause should say `BLOCK CONTAINS 20 CHARACTERS`.

If the records in the file are not blocked, the BLOCK CONTAINS clause can be omitted. When it is omitted, the compiler assumes that records are not blocked. Even if each physical record contains only one complete logical record, coding `BLOCK CONTAINS 1 RECORD` would result in fixed blocked records.

The BLOCK CONTAINS clause can be omitted when the associated file-control entry specifies a VSAM file. The concept of blocking has no meaning for VSAM files. The BLOCK CONTAINS clause is syntax checked but has no effect on the execution of the program.

For external files, the value of all BLOCK CONTAINS clauses of corresponding external files must match within the run unit. This conformance is in terms of bytes and does not depend upon whether the value was specified as CHARACTERS or as RECORDS.

integer-1 , integer-2

Must be unsigned integers. They specify:

CHARACTERS

Specifies the number of bytes required to store the physical record, no matter what USAGE the data items have within the data record.

If only *integer-2* is specified, it specifies the exact number of bytes in the physical record. When *integer-1* and *integer-2* are both specified, they represent the minimum and maximum number of bytes in the physical record, respectively.

integer-1 and *integer-2* must include any control bytes and padding contained in the physical record. (Logical records do not include padding.)

The CHARACTERS phrase is the default. CHARACTERS must be specified when:

- The physical record contains padding.

integer-3

Must be an unsigned integer that specifies the number of bytes contained in each record in the file.

The RECORD CONTAINS 0 CHARACTERS clause can be specified for input QSAM files containing fixed-length records; the record size is determined at run time from the DD statement parameters or the data set label. If, at run time, the actual record is beyond the 01 record description, then only the 01 record length is available. If the actual record is shorter, then only the actual record length can be referred to. Otherwise, uninitialized data or an addressing exception can be produced.

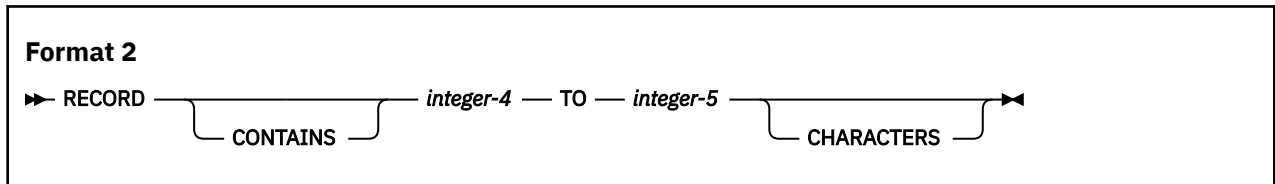
Usage note: If the RECORD CONTAINS 0 clause is specified, then the SAME AREA, SAME RECORD AREA, or APPLY WRITE-ONLY clauses cannot be specified.

Do not specify the RECORD CONTAINS 0 clause for an SD entry.

Format 2

Format 2 specifies the number of bytes for either fixed-length or variable-length records.

Fixed-length records are obtained when all 01 record description entry lengths are the same. The format-2 RECORD CONTAINS clause is never required, because the minimum and maximum record lengths are determined from the record description entries.

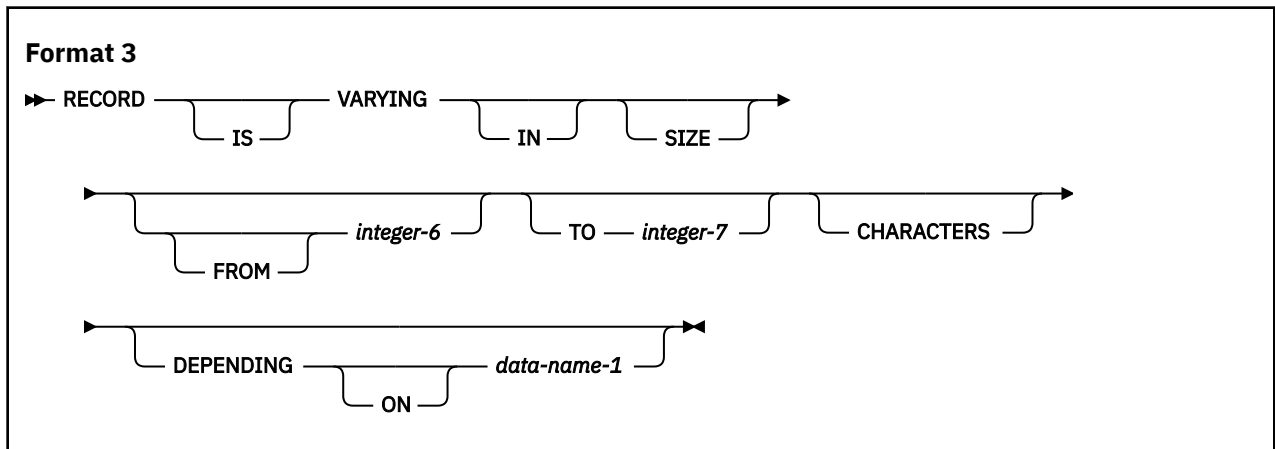


integer-4 , integer-5

Must be unsigned integers. *integer-4* specifies the size of the smallest data record, and *integer-5* specifies the size of the largest data record.

Format 3

Format 3 is used to specify variable-length records.



integer-6

Specifies the minimum number of bytes to be contained in any record of the file. If *integer-6* is not specified, the minimum number of bytes to be contained in any record of the file is equal to the least number of bytes described for a record in that file.

integer-7

Specifies the maximum number of bytes in any record of the file. If *integer-7* is not specified, the maximum number of bytes to be contained in any record of the file is equal to the greatest number of bytes described for a record in that file.

The number of bytes associated with a record description is determined by the sum of the number of bytes in all elementary data items (excluding redefinitions and renamings), plus any implicit FILLER due to synchronization. If a table is specified:

- The minimum number of table elements described in the record is used in the summation above to determine the minimum number of bytes associated with the record description.
- The maximum number of table elements described in the record is used in the summation above to determine the maximum number of bytes associated with the record description.

If *data-name-1* is specified:

- *data-name-1* must be an elementary unsigned integer.
- The number of bytes in the record must be placed into the data item referenced by *data-name-1* before any RELEASE, REWRITE, or WRITE statement is executed for the file.
- The execution of a DELETE, RELEASE, REWRITE, START, or WRITE statement or the unsuccessful execution of a READ or RETURN statement does not alter the content of the data item referenced by *data-name-1*.
- After the successful execution of a READ or RETURN statement for the file, the contents of the data item referenced by *data-name-1* indicate the number of bytes in the record just read.

During the execution of a RELEASE, REWRITE, or WRITE statement, the number of bytes in the record is determined by the following conditions:

- If *data-name-1* is specified, by the content of the data item referenced by *data-name-1*
- If *data-name-1* is not specified and the record does not contain a variable occurrence data item, by the number of bytes positions in the record
- If *data-name-1* is not specified and the record contains a variable occurrence data item, by the sum of the fixed position and that portion of the table described by the number of occurrences at the time of execution of the output statement

During the execution of a READ ... INTO or RETURN ... INTO statement, the number of bytes in the current record that participate as the sending data items in the implicit MOVE statement is determined by the following conditions:

- If *data-name-1* is specified, by the content of the data item referenced by *data-name-1*
- If *data-name-1* is not specified, by the value that would have been moved into the data item referenced by *data-name-1* had *data-name-1* been specified

LABEL RECORDS clause

For sequential, relative, or indexed files, and for sort/merge SDs, the LABEL RECORDS clause is syntax checked, but has no effect on the execution of the program.

The LABEL RECORDS clause documents the presence or absence of labels.

STANDARD

Labels conforming to system specifications exist for this file.

STANDARD is permitted for mass storage devices and tape devices.

OMITTED

No labels exist for this file.

OMITTED is permitted for tape devices.

data-name-2

User labels are present in addition to standard labels. *data-name-2* specifies the name of a user label record. *data-name-2* must appear as the subject of a record description entry associated with the file.

VALUE OF clause

The VALUE OF clause describes an item in the label records associated with the file.

data-name-3

Should be qualified when necessary, but cannot be subscripted. It must be described in the WORKING-STORAGE SECTION. It cannot be described with the USAGE IS INDEX clause.

literal-1

Can be numeric or alphanumeric, or a figurative constant of category numeric or alphanumeric. Cannot be a floating-point literal.

The VALUE OF clause is syntax checked, but has no effect on the execution of the program.

DATA RECORDS clause

The DATA RECORDS clause is syntax checked but serves only as documentation for the names of data records associated with the file.

data-name-4

The names of record description entries associated with the file.

The data-name need not have an associated 01 level number record description with the same name.

LINAGE clause

The LINAGE clause specifies the depth of a logical page in number of lines. Optionally, it also specifies the line number at which the footing area begins and the top and bottom margins of the logical page. (The logical page and the physical page cannot be of the same dimension.)

The LINAGE clause is effective for sequential files opened as OUTPUT or EXTEND.

All integers must be unsigned. All data-names must be described as unsigned integer data items.

data-name-5 , integer-8

The number of lines that can be written or spaced on this logical page. The area of the page that these lines represent is called the *page body*. The value must be greater than zero.

WITH FOOTING AT

integer-9 or the value of the data item in *data-name-6* specifies the first line number of the footing area within the page body. The footing line number must be greater than zero, and not greater than the last line of the page body. The footing area extends between those two lines.

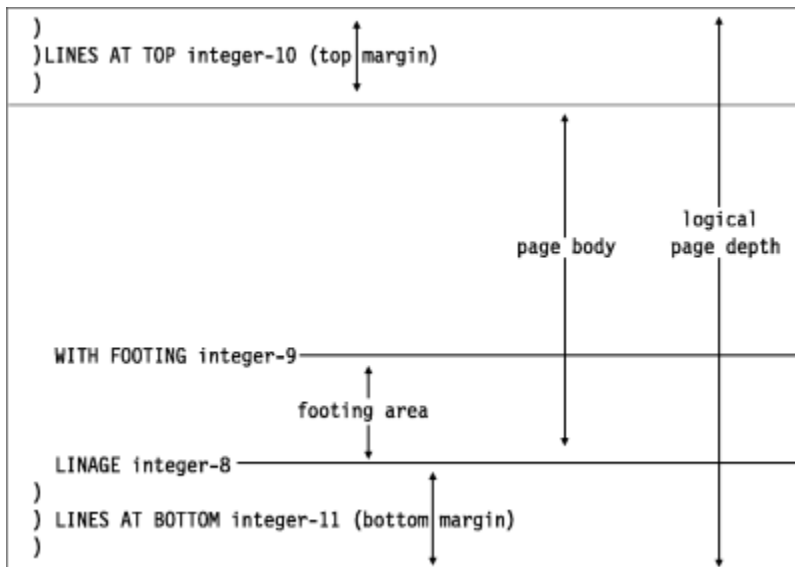
LINES AT TOP

integer-10 or the value of the data item in *data-name-7* specifies the number of lines in the top margin of the logical page. The value can be zero.

LINES AT BOTTOM

integer-11 or the value of the data item in *data-name-8* specifies the number of lines in the bottom margin of the logical page. The value can be zero.

The following figure illustrates the use of each phrase of the LINAGE clause.



The logical page dimension specified in the LINAGE clause is the sum of all values specified in each phrase except the FOOTING phrase. If the LINES AT TOP phrase is omitted, the assumed value for the top margin is zero. Similarly, if the LINES AT BOTTOM phrase is omitted, the assumed value for the bottom margin is zero. Each logical page immediately follows the preceding logical page, with no additional spacing provided.

If the FOOTING phrase is omitted, its assumed value is equal to that of the page body (*integer-8* or *data-name-5*).

At the time an OPEN OUTPUT statement is executed, the values of *integer-8*, *integer-9*, *integer-10*, and *integer-11*, if specified, are used to determine the page body, first footing line, top margin, and bottom margin of the logical page for this file. (See the figure mentioned earlier.) These values are then used for all logical pages printed for this file during a given execution of the program.

At the time an OPEN statement with the OUTPUT phrase is executed for the file, *data-name-5*, *data-name-6*, *data-name-7*, and *data-name-8* determine the page body, first footing line, top margin, and bottom margin for the first logical page only.

At the time a WRITE statement with the ADVANCING PAGE phrase is executed or a page overflow condition occurs, the values of *data-name-5*, *data-name-6*, *data-name-7*, and *data-name-8* if specified, are used to determine the page body, first footing line, top margin, and bottom margin for the next logical page.

If an external file connector is associated with this file description entry, all file description entries in the run unit that are associated with this file connector must have:

- A LINAGE clause, if any file description entry has a LINAGE clause
- The same corresponding values for *integer-8*, *integer-9*, *integer-10*, and *integer-11*, if specified
- The same corresponding external data items referenced by *data-name-5*, *data-name-6*, *data-name-7*, and *data-name-8*

See [“ADVANCING phrase” on page 478](#) for the behavior of carriage control characters in external files.

A LINAGE clause under an SD is syntax checked, but has no effect on the execution of the program.

LINAGE-COUNTER special register

For information about the LINAGE-COUNTER special register, see [“LINAGE-COUNTER” on page 23](#).

RECORDING MODE clause

The RECORDING MODE clause specifies the format of the physical records in a QSAM file. The clause is ignored for a VSAM file.

Permitted values for RECORDING MODE are:

Recording mode F (fixed)

All the records in a file are the same length and each is wholly contained within one block. Blocks can contain more than one record, and there is usually a fixed number of records for each block. In this mode, there are no record-length or block-descriptor fields.

Recording mode V (variable)

The records can be either fixed-length or variable-length, and each must be wholly contained within one block. Blocks can contain more than one record. Each data record includes a record-length field and each block includes a block-descriptor field. These fields are not described in the DATA DIVISION. They are each 4 bytes long and provision is automatically made for them. These fields are not available to you.

Recording mode U (fixed or variable)

The records can be either fixed-length or variable-length. However, there is only one record for each block. There are no record-length or block-descriptor fields.

You cannot use RECORDING MODE U if you are using the BLOCK CONTAINS clause.

Recording mode S (spanned)

The records can be either fixed-length or variable-length, and can be larger than a block. If a record is larger than the remaining space in a block, a segment of the record is written to fill the block. The remainder of the record is stored in the next block (or blocks, if required). Only complete records are made available to you. Each segment of a record in a block, even if it is the entire record, includes a segment-descriptor field, and each block includes a block-descriptor field. These fields are not described in the DATA DIVISION; provision is automatically made for them. These fields are not available to you.

When recording mode S is used, the BLOCK CONTAINS CHARACTERS clause must be used. Recording mode S is not allowed for ASCII files.

If the RECORDING MODE clause is not specified for a QSAM file, the Enterprise COBOL compiler determines the recording mode as follows:

F

The compiler determines the recording mode to be F if the largest level-01 record associated with the file is not greater than the block size specified in the BLOCK CONTAINS clause, and you do one of the following things:

- Use the RECORD CONTAINS *integer* clause. (For more information, see the *Enterprise COBOL Migration Guide*.)
- Omit the RECORD clause and make sure that all level-01 records associated with the file are the same size and none contains an OCCURS DEPENDING ON clause.

V

The compiler determines the recording mode to be V if the largest level-01 record associated with the file is not greater than the block size specified in the BLOCK CONTAINS clause, and you do one of the following things:

- Use the RECORD IS VARYING clause.
- Omit the RECORD clause and make sure that all level-01 records associated with the file are not the same size or some contain an OCCURS DEPENDING ON clause.
- Use the RECORD CONTAINS *integer-1* TO *integer-2* clause, with *integer-1* the minimum length and *integer-2* the maximum length of the level-01 records associated with the file. The two integers must be different, with values matching minimum and maximum length of either different length records or records with an OCCURS DEPENDING ON clause.

S

The compiler determines the recording mode to be S if the maximum block size is smaller than the largest record size.

U

Recording mode U is never obtained by default. The RECORDING MODE U clause must be explicitly specified to get recording mode U.

CODE-SET clause

The CODE-SET clause specifies the character code used to represent data on a magnetic tape file. When the CODE-SET clause is specified, an alphabet-name identifies the character code convention used to represent data on the input-output device.

alphabet-name must be defined in the SPECIAL-NAMES paragraph as STANDARD-1 (for ASCII-encoded files), STANDARD-2 (for ISO 7-bit encoded files), EBCDIC (for EBCDIC-encoded files), or NATIVE. When NATIVE is specified, the CODE-SET clause is syntax checked but has no effect on the execution of the program.

The CODE-SET clause also specifies the algorithm for converting the character codes on the input-output medium from and to the internal EBCDIC character set.

When the CODE-SET clause is specified for a file, all data in the file must have USAGE DISPLAY; and if signed numeric data is present, it must be described with the SIGN IS SEPARATE clause.

When the CODE-SET clause is omitted, the EBCDIC character set is assumed for the file.

If the associated file connector is an external file connector, all CODE-SET clauses in the run unit that are associated with the file connector must have the same character set.

The CODE-SET clause is valid only for magnetic tape files.

The CODE-SET clause is syntax checked but has no effect on the execution of the program when specified under an SD.

Chapter 26. DATA DIVISION--data description entry

A *data description entry* specifies the characteristics of a data item. In the sections that follow, sets of data description entries are called *record description entries*. The term *data description entry* refers to data and record description entries.

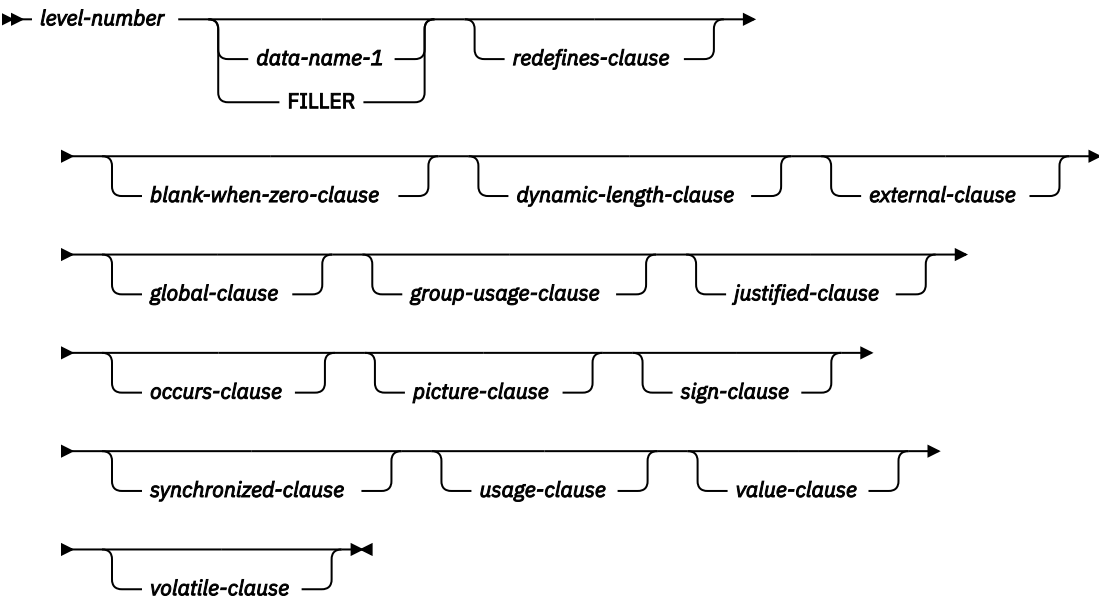
Data description entries that define independent data items do not make up a record. These entries are known as *data item description entries*.

Data description entries have three general formats, and all data description entries must end with a separator period.

Format 1

Format 1 is used for data description entries in all DATA DIVISION sections.

Format 1: data description entry



The clauses can be written in any order, with the following exceptions:

- *data-name-1* or *FILLER*, if specified, must immediately follow the level-number.
- When the *REDEFINES* clause is specified, it must immediately follow *data-name-1* or *FILLER*, if either is specified. If *data-name-1* or *FILLER* is not specified, the *REDEFINES* clause must immediately follow the level-number.

The level-number in format 1 can be any number in the range 01–49, or 77.

A space, a comma, or a semicolon must separate clauses.

Format 2

Format 2 regroups previously defined items.

Format 2: renames

► 66 — *data-name-1* — *renames-clause*. ►

A level-66 entry cannot rename another level-66 entry, nor can it rename a level-01, level-77, or level-88 entry.

All level-66 entries associated with one record must immediately follow the last data description entry in that record.

See [“RENAMES clause” on page 228](#) for further details.

Format 3

Format 3 describes condition-names.

Format 3: condition-name

➡ 88 — *condition-name-1* — *value-clause*. ➡

condition-name-1

A user-specified name that associates a value, a set of values, or a range of values with a conditional variable.

Level-88 entries must immediately follow the data description entry for the conditional variable with which the condition-names are associated.

Format 3 can be used to describe elementary items, national group items, or alphanumeric group items. Additional information about condition-name entries can be found under [“VALUE clause” on page 245](#) and [“Condition-name condition” on page 271](#).

Level-numbers

The level-number specifies the hierarchy of data within a record, and identifies special-purpose data entries. A level-number begins a data description entry, a renamed or redefined item, or a condition-name entry.

A level-number has an integer value between 1 and 49, inclusive, or one of the special level-number values 66, 77, or 88.

Format

➡ *level-number* — *data-name-1* — *FILLER*. ➡

level-number

01 and 77 must begin in Area A and be followed either by a separator period or by a space followed by its associated data-name, FILLER, or appropriate data description clause.

Level numbers 02 through 49 can begin in Areas A or B and must be followed by a space or a separator period.

Level numbers 66 and 88 can begin in Areas A or B and must be followed by a space.

Single-digit level-numbers 1 through 9 can be substituted for level-numbers 01 through 09.

Successive data description entries can start in the same column as the first entry or can be indented according to the level-number. Indentation does not affect the magnitude of a level-number.

When level-numbers are indented, each new level-number can begin any number of spaces to the right of Area A. The extent of indentation to the right is limited only by the width of Area B.

For more information, see [“Levels of data” on page 167](#).

data-name-1

Explicitly identifies the data being described.

data-name-1, if specified, identifies a data item used in the program. *data-name-1* must be the first word following the level-number.

The data item can be changed during program execution.

data-name-1 must be specified for level-66 and level-88 items. It must also be specified for any entry containing the GLOBAL or EXTERNAL clause, and for record description entries associated with file description entries that have the GLOBAL or EXTERNAL clauses.

FILLER

A data item that is not explicitly referred to in a program. The keyword FILLER is optional. If specified, FILLER must be the first word following the level-number.

The keyword FILLER can be used with a conditional variable if explicit reference is never made to the conditional variable but only to values that it can assume. FILLER cannot be used with a condition-name.

In a MOVE CORRESPONDING statement or in an ADD CORRESPONDING or SUBTRACT CORRESPONDING statement, FILLER items are ignored.

In an INITIALIZE statement:

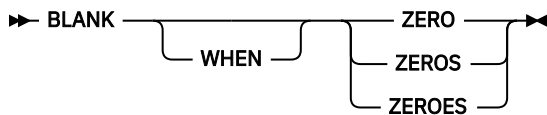
- When the FILLER phrase is not specified, elementary FILLER items are ignored.
- When the FILLER phrase is specified, the receiving elementary data items that have an explicit or implicit FILLER clause will be initialized.

If *data-name-1* or the FILLER clause is omitted, the data item being described is treated as though FILLER had been specified.

BLANK WHEN ZERO clause

The BLANK WHEN ZERO clause specifies that an item contains only spaces when its value is zero.

Format



The BLANK WHEN ZERO clause may be specified only for an elementary item described by its picture character string as category numeric-edited or numeric, without the picture symbol S or *. These items must be described, either implicitly or explicitly, as USAGE DISPLAY or USAGE NATIONAL.

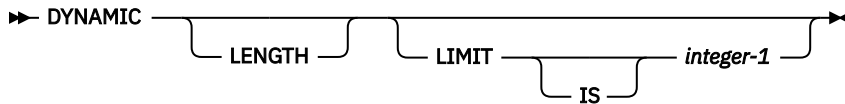
A BLANK WHEN ZERO clause that is specified for an item defined as numeric by its picture character string defines the item as category numeric-edited.

The BLANK WHEN ZERO clause does not affect initial settings as determined by the VALUE clause.

DYNAMIC LENGTH clause

The DYNAMIC LENGTH clause specifies a dynamic-length elementary item. The LENGTH keyword is optional.

Format



LIMIT

If the LIMIT phrase is not specified, then the limit is set to the maximum value according to the data item class. Attempts to receive into a dynamic-length elementary item more characters over the limit will result in truncation at a character boundary on the right.

Note: In practice, the maximum length of a dynamic-length elementary item might be limited by available runtime memory or other runtime environment limits.

integer-1

An integer specifies the maximum number of alphanumeric or national characters that the data item can contain.

When the PICTURE clause is PIC U, *integer-1* represents the maximum bytes that the data item can contain. *integer-1* must be an integer greater than or equal to 1, and less than or equal to 999999999 for alphanumeric class and 999999999 for UTF-8 class dynamic-length elementary items.

A dynamic-length elementary item is a data item whose length can vary at runtime. The length of a data item is the current number of characters contained by this data item.

A dynamic-length elementary item has a minimum length of zero, and a maximum length that is the smallest of the following limitations:

- *integer-1* characters of the LIMIT phrase
- 999999999 characters if the PICTURE clause is 'X' (alphanumeric class) or 999999999 bytes if the PICTURE clause is 'U' (UTF-8 class)
- The available runtime memory.

The PICTURE clause of a dynamic-length elementary item must be either PIC X or PIC U, making this a data item of class alphanumeric or UTF-8 respectively. No other PICTURE clause strings other than a single instance of 'X' or 'U' are allowed. PIC N is not supported in a dynamic-length elementary item.

Dynamic-length elementary items can be specified in the WORKING-STORAGE, LOCAL-STORAGE, or LINKAGE SECTION. For LINKAGE SECTION dynamic-length elementary items, the item must also be specified in either of the following ways:

- A PROCEDURE DIVISION USING item passed BY REFERENCE on the current program or function PROCEDURE DIVISION header.
- A PROCEDURE DIVISION RETURNING item on the current program PROCEDURE DIVISION header.

Dynamic-length elementary items as PROCEDURE DIVISION RETURNING items can be specified for programs only. Those programs can be called statically, dynamically, or via DLL. If called via DLL, the calling program must use the call-by-literal syntax (rather than the call-by-identifier syntax). Programs containing dynamic-length elementary items as PROCEDURE DIVISION RETURNING items cannot be called using a function-pointer or procedure-pointer.

Dynamic-length elementary items can be specified as subordinate items within a group. The existence of a dynamic-length elementary item within a group item will make it a dynamic-length group item (See [“Dynamic-length group items”](#) on page 177).

Dynamic-length elementary items cannot be variably located or a subordinate data item within a table containing the OCCURS DEPENDING ON phrase. When a dynamic-length elementary item appears within a group, the content of the item can be considered logically inline within this group, even if the physical location of the content is remotely located. Comparisons and moves between groups where one or both groups contains a subordinate dynamic-length elementary item is not allowed. Moving a dynamic-length

elementary item to a fixed-length group is allowed, and moving any group (dynamic-length or not) to a dynamic-length elementary item is allowed.

Usage note: Dynamic-length elementary items in the LOCAL-STORAGE section may not be freed when the program executes unstructured GOTOs such as an EXEC CICS® HANDLE statement or the Language Environment service CEEMRCE (Move resume cursor explicit).

EXTERNAL clause

The EXTERNAL clause specifies that the storage associated with a data item is associated with the run unit rather than with any particular program or method within the run unit.

An external data item can be referenced by any program or method in the run unit that describes the data item. References to an external data item from different programs or methods using separate descriptions of the data item are always to the same data item. In a run unit, there is only one representative of an external data item.

The EXTERNAL clause can be specified only on data description entries whose level-number is 01. It can be specified only on data description entries that are in the WORKING-STORAGE SECTION of a program or method. It cannot be specified in LINKAGE SECTION, LOCAL-STORAGE SECTION, or FILE SECTION data description entries. Any data item described by a data description entry subordinate to an entry that describes an external record also attains the external attribute. Indexes in an external data record do not possess the external attribute.

The data contained in the record named by the data-name clause is external and can be accessed and processed by any program or method in the run unit that describes and, optionally, redefines it. This data is subject to the following rules:

- If two or more programs or methods within a run unit describe the same external data record, each record-name of the associated record description entries must be the same, and the records must define the same number of bytes. However, a program or method that describes an external record can contain a data description entry including the REDEFINES clause that redefines the complete external record, and this complete redefinition need not occur identically in other programs or methods in the run unit.
- Use of the EXTERNAL clause does not imply that the associated data-name is a global name.

GLOBAL clause

The GLOBAL clause specifies that a data-name is available to every program contained within the program that defines it, as long as the contained program does not itself have a definition for that name. All data-names subordinate to or condition-names or indexes associated with a global name are global names.

A data-name is global if the GLOBAL clause is specified either in the data description entry by which the data-name is defined or in another entry to which that data description entry is subordinate. The GLOBAL clause can be specified in the WORKING-STORAGE SECTION, the FILE SECTION, the LINKAGE SECTION, and the LOCAL-STORAGE SECTION, but only in data description entries whose level-number is 01.

In the same DATA DIVISION, the data description entries for any two data items for which the same data-name is specified must not include the GLOBAL clause.

A statement in a program contained directly or indirectly within a program that describes a global name can reference that name without describing it again.

Two programs in a run unit can reference common data in the following circumstances:

- The data content of an external data record can be referenced from any program that describes the data record as external.
- If a program is contained within another program, both programs can refer to data that possesses the global attribute either in the containing program or in any program that directly or indirectly contains the containing program.

JUSTIFIED clause

The JUSTIFIED clause overrides standard positioning rules for receiving items of category alphabetic, alphanumeric, DBCS, or national.

Format



You can specify the JUSTIFIED clause only at the elementary level. JUST is an abbreviation for JUSTIFIED, and has the same meaning.

You cannot specify the JUSTIFIED clause:

- For data items of category numeric, numeric-edited, alphanumeric-edited, or national-edited
- For edited DBCS items
- For index data items
- For items described as USAGE FUNCTION-POINTER, USAGE POINTER, USAGE PROCEDURE-POINTER, or USAGE OBJECT REFERENCE
- For external floating-point or internal floating-point items
- With level-66 (RENAMES) and level-88 (condition-name) entries

When the JUSTIFIED clause is specified for a receiving item, the data is aligned at the rightmost character position in the receiving item. Also:

- If the sending item is larger than the receiving item, the leftmost character positions are truncated.
- If the sending item is smaller than the receiving item, the unused character positions at the left are filled with spaces. For a DBCS item, each unused position is filled with a DBCS space (X'4040'); for an item described with usage NATIONAL, each unused position is filled with the default UTF-16 space (NX'0020'); for an item described with USAGE UTF-8, each unused position is filled with the default UTF-8 space (UX'20'); otherwise, each unused position is filled with an alphanumeric space.

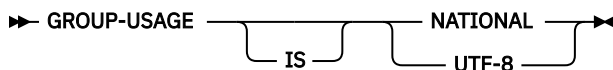
If you omit the JUSTIFIED clause, the rules for standard alignment are followed (see [“Alignment rules”](#) on page 175).

The JUSTIFIED clause does not affect initial settings as determined by the VALUE clause.

GROUP-USAGE clause

A GROUP-USAGE clause with the NATIONAL phrase specifies that the group item defined by the entry is a national group item. A national group item contains national characters in all subordinate data items and subordinate group items. A GROUP-USAGE clause with the UTF-8 phrase specifies that the group item defined by the entry is a UTF-8 group item. A UTF-8 group item contains UTF-8 characters in all subordinate data items and subordinate group items.

Format



When GROUP-USAGE NATIONAL is specified:

- The subject of the entry is a national group item. The class and category of a national group are national.
- A USAGE clause must not be specified for the subject of the entry. A USAGE NATIONAL clause is implied.

- A USAGE NATIONAL clause is implied for any subordinate elementary data items that are not described with a USAGE NATIONAL clause.
- All subordinate elementary data items must be explicitly or implicitly described with USAGE NATIONAL.
- Any signed numeric data items must be described with the SIGN IS SEPARATE clause.
- A GROUP-USAGE NATIONAL clause is implied for any subordinate group items that are not described with a GROUP-USAGE NATIONAL clause.
- All subordinate group items must be explicitly or implicitly described with a GROUP-USAGE NATIONAL clause.
- The JUSTIFIED clause must not be specified.

Unless stated otherwise, a national group item is processed as though it were an elementary data item of usage national, class and category national, described with PICTURE N(*m*), where *m* is the length of the group in national character positions.

Usage note: When you use national groups, the compiler can ensure proper truncation and padding of group items for statements such as MOVE and INSPECT. Groups defined without a GROUP-USAGE NATIONAL clause are alphanumeric groups. The content of alphanumeric groups, including any national characters, is treated as alphanumeric data, possibly leading to invalid truncation or mishandling of national character data.

The following table summarizes the cases where a national and UTF-8 group items are processed as a group item.

When GROUP-USAGE UTF-8 is specified:

- The subject of the entry is a UTF-8 group item. The class and category of a UTF-8 group are UTF-8.
- A USAGE clause must not be specified for the subject of the entry. A USAGE UTF-8 clause is implied.
- A USAGE UTF-8 clause is implied for any subordinate elementary data items that are not described with a USAGE UTF-8 clause.
- All subordinate elementary data items must be explicitly or implicitly described with USAGE UTF-8 and must be defined with either the BYTE-LENGTH phrase of the PICTURE clause or the DYNAMIC LENGTH clause.
- A GROUP-USAGE UTF-8 clause is implied for any subordinate group items that are not described with a GROUP-USAGE UTF-8 clause.
- All subordinate group items must be explicitly or implicitly described with a GROUP-USAGE UTF-8 clause.
- The JUSTIFIED clause must not be specified.

Unless stated otherwise, a UTF-8 group item is processed as though it were an elementary data item of usage UTF-8, class and category UTF-8, and defined with PICTURE U BYTE-LENGTH *m*, where *m* is the length of the group in bytes.

Usage note: When you use UTF-8 groups, the compiler can ensure proper truncation and padding of group items for statements such as MOVE. Groups defined without a GROUP-USAGE UTF-8 or GROUP-USAGE NATIONAL clause are alphanumeric groups. The content of alphanumeric groups, including any UTF-8 characters, is treated as alphanumeric data, possibly leading to invalid truncation or mishandling of UTF-8 character data.

The following table summarizes the cases where a national and UTF-8 group items are processed as a group item.

<i>Table 11. Where national and UTF-8 group items are processed as groups</i>	
Language feature	Processing of national group items
Name qualification	The names of national and UTF-8 group items can be used to qualify the names of elementary data items and subordinate group items in national and UTF-8 groups. The rules of qualification for national and UTF-8 groups are the same as the rules of qualification for an alphanumeric group.
RENAMES clause	The rules for national and UTF-8 group items specified in the THROUGH phrase are the same as the rules for an alphanumeric group item specified in the THROUGH phrase. The result is an alphanumeric group item.
CORRESPONDING phrase	National and UTF-8 group items are processed as a group in accordance with the rules of the CORRESPONDING phrase. Elementary data items within a national or UTF-8 group are processed the same as they would be if defined within an alphanumeric group.
INITIALIZE statement	National and UTF-8 group items are processed as a group in accordance with the rules of the INITIALIZE statement. Elementary items within the national or UTF-8 group are initialized the same as they would be if defined within an alphanumeric group.
XML GENERATE statement	A national group item specified in the FROM phrase is processed as a group in accordance with the rules of the XML GENERATE statement. Elementary items within the national group are processed the same as they would be if defined within an alphanumeric group.
JSON GENERATE statement	A national group item specified in the FROM phrase is processed as a group in accordance with the rules of the JSON GENERATE statement. Elementary items within the national group are processed the same as they would be if defined within an alphanumeric group.

OCCURS clause

The DATA DIVISION language elements used for table handling are the OCCURS clause and the INDEXED BY phrase.

For the INDEXED BY phrase description, see [“INDEXED BY phrase” on page 202](#).

The OCCURS clause specifies tables whose elements can be referred to by indexing or subscripting. It also eliminates the need for separate entries for repeated data items.

Formats for the OCCURS clause include fixed-length tables and variable-length tables.

The *subject* of an OCCURS clause is the data-name of the data item that contains the OCCURS clause. Except for the OCCURS clause itself, data description clauses used with the subject apply to each occurrence of the item described.

Whenever the subject of an OCCURS clause or any data-item subordinate to it is referenced, it must be subscripted or indexed, with the following exceptions:

- When the subject of the OCCURS clause is used as the subject of a SEARCH statement.
- When the subject of the OCCURS clause is used as the subject of a format 2 SORT statement.
- When the subject or a subordinate data item is the object of the ASCENDING/DESCENDING KEY phrase.
- When the subordinate data item is the object of the REDEFINES clause.
- When the subordinate data item is used as the subject of a LENGTH OF special register. For details, see [“LENGTH OF” on page 22](#).

When subscripted or indexed, the subject refers to one occurrence within the table, unless the ALL subscript is used in an intrinsic function.

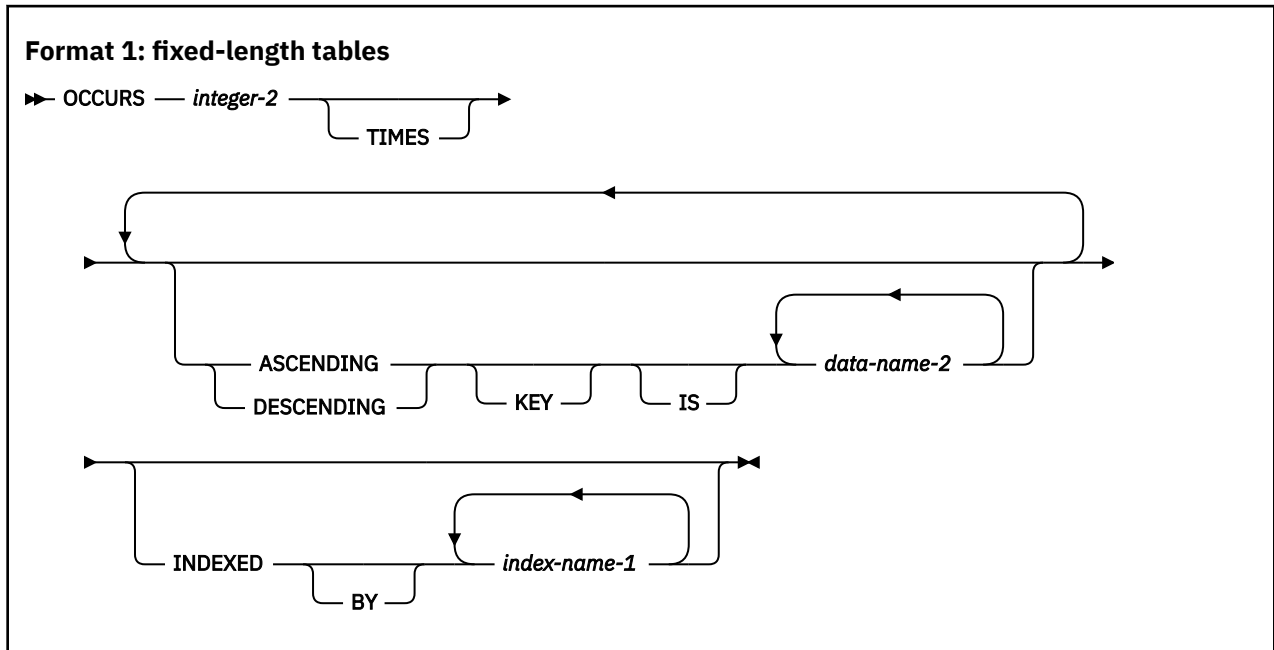
The OCCURS clause cannot be specified in a data description entry that:

- Has a level number of 01, 66, 77, or 88.
- Describes a redefined data item. (However, a redefined item can be subordinate to an item that contains an OCCURS clause.) See [“REDEFINES clause”](#) on page 225.

Fixed-length tables

Fixed-length tables are specified using the OCCURS clause.

Because seven subscripts or indexes are allowed, six nested levels and one outermost level of the format-1 OCCURS clause are allowed. The format-1 OCCURS clause can be specified as subordinate to the OCCURS DEPENDING ON clause. In this way, a table of up to seven dimensions can be specified.



integer-2

The exact number of occurrences. *integer-2* must be greater than zero.

ASCENDING KEY and DESCENDING KEY phrases

Data is arranged in ascending or descending order, depending on the keyword specified, according to the values contained in *data-name-2*. The data-names are listed in their descending order of significance.

The order is determined by the rules for comparison of operands (see [“Relation conditions”](#) on page 272). The ASCENDING KEY and DESCENDING KEY data items are used in OCCURS clauses, the SEARCH ALL statements for a binary search of the table element, and the format 2 SORT statements. As an alternative, keys can be specified with the format 2 SORT statements.

data-name-2

Must be the name of the subject entry or the name of an entry subordinate to the subject entry. *data-name-2* can be qualified.

If *data-name-2* names the subject entry, that entire entry becomes the ASCENDING KEY or DESCENDING KEY and is the only key that can be specified for this table element.

If *data-name-2* does not name the subject entry, then *data-name-2*:

- Must be subordinate to the subject of the table entry itself
- Must *not* be subordinate to, or follow, any other entry that contains an OCCURS clause
- Must not contain an OCCURS clause

data-name-2 must not have subordinate items that contain OCCURS DEPENDING ON clauses.

When the ASCENDING KEY or DESCENDING KEY phrase is specified, the following rules apply:

- Keys must be listed in decreasing order of significance.
- The total number of keys for a given table element must not exceed 12.
- The data in the table must be arranged in ascending or descending sequence according to the collating sequence in use.
- The key must be described with one of the following usages:
 - BINARY
 - DISPLAY
 - DISPLAY-1
 - NATIONAL
 - UTF-8
 - PACKED-DECIMAL
 - COMPUTATIONAL
 - COMPUTATIONAL-1
 - COMPUTATIONAL-2
 - COMPUTATIONAL-3
 - COMPUTATIONAL-4
 - COMPUTATIONAL-5
- A key described with usage NATIONAL can have one of the following categories: national, national-edited, numeric-edited, numeric, or external floating-point.
- The sum of the lengths of all the keys associated with one table element must not exceed 256.
- If a key is specified without qualifiers and it is not a unique name, the key will be implicitly qualified with the subject of the OCCURS clause and all qualifiers of the OCCURS clause subject.

The following example illustrates the specification of ASCENDING KEY data items:

```
WORKING-STORAGE SECTION.
01 TABLE-RECORD.
   05 EMPLOYEE-TABLE OCCURS 100 TIMES
      ASCENDING KEY IS WAGE-RATE EMPLOYEE-NO
      INDEXED BY A, B.
      10 EMPLOYEE-NAME PIC X(20).
      10 EMPLOYEE-NO PIC 9(6).
      10 WAGE-RATE PIC 9999V99.
      10 WEEK-RECORD OCCURS 52 TIMES
         ASCENDING KEY IS WEEK-NO INDEXED BY C.
         15 WEEK-NO PIC 99.
         15 AUTHORIZED-ABSENCES PIC 9.
         15 UNAUTHORIZED-ABSENCES PIC 9.
         15 LATE-ARRIVALS PIC 9.
```

The keys for EMPLOYEE-TABLE are subordinate to that entry, and the key for WEEK-RECORD is subordinate to that subordinate entry.

In the preceding example, records in EMPLOYEE-TABLE must be arranged in ascending order of WAGE-RATE, and in ascending order of EMPLOYEE-NO within WAGE-RATE. Records in WEEK-RECORD must be arranged in ascending order of WEEK-NO. If they are not, results of any SEARCH ALL statement are unpredictable.

INDEXED BY phrase

The INDEXED BY phrase specifies the indexes that can be used with a table. A table without an INDEXED BY phrase can be referred to through indexing by using an index-name associated with another table.

For more information about using indexing, see [“Subscripting using index-names \(indexing\)” on page 74.](#)

Indexes normally are allocated in static memory associated with the program that contains the table. Thus indexes are in the last-used state when a program is reentered. However, in the following cases, indexes are allocated on a per-invocation basis. Thus you must set the value of the index on every entry for indexes on tables in the following sections:

- The LOCAL-STORAGE SECTION
- The WORKING-STORAGE SECTION of:
 - A program with the IS INITIAL clause on the PROGRAM-ID phrase
 - A class definition (object instance variables)
- The LINKAGE SECTION of:
 - Methods
 - Programs compiled with the RECURSIVE clause
 - Programs compiled with the THREAD option

Indexes specified in an external data record do not possess the external attribute.

index-name-1

Each index-name specifies an index to be created by the compiler for use by the program. These index-names are *not* data-names and are not identified elsewhere in the COBOL program; instead, they can be regarded as private special registers for the use of this object program only. They are not data and are not part of any data hierarchy.

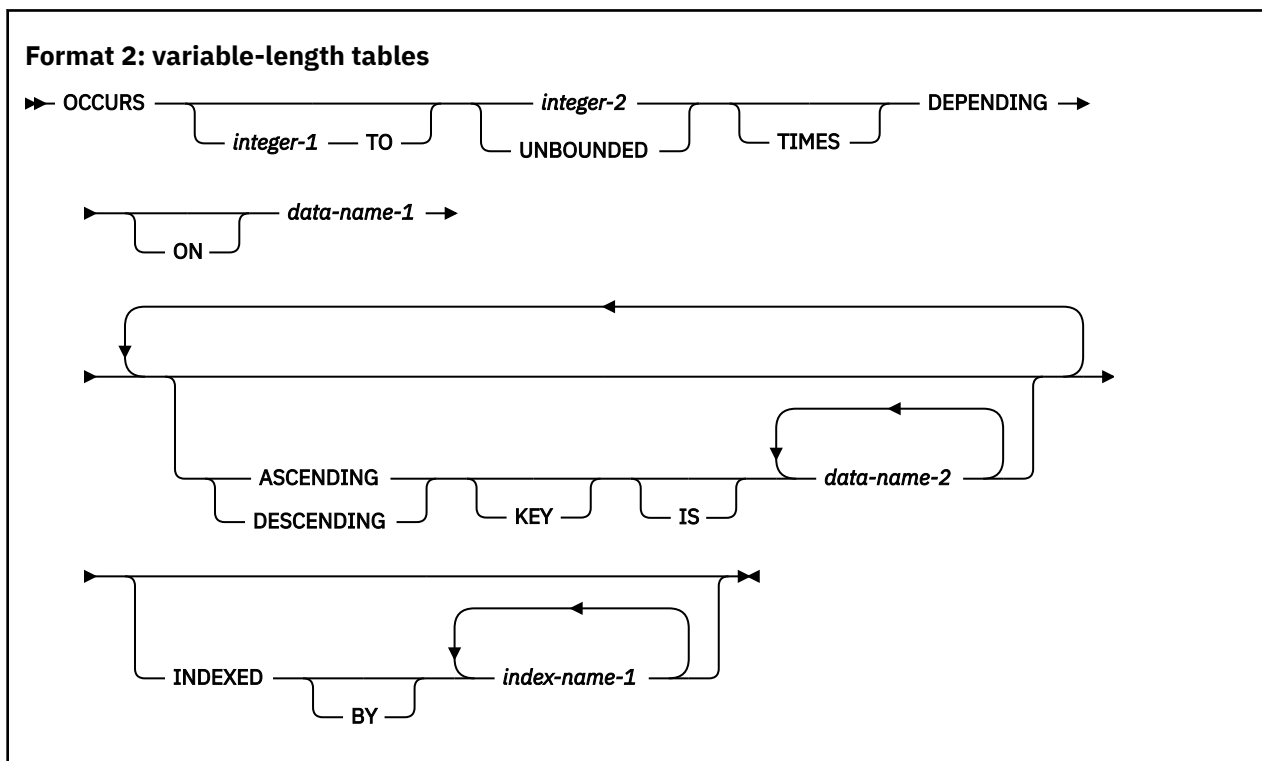
Unreferenced index names need not be uniquely defined.

In one table entry, up to 12 index-names can be specified.

If a data item that possesses the global attribute includes a table accessed with an index, that index also possesses the global attribute. Therefore, the scope of an index-name is the same as that of the data-name that names the table in which the index is defined.

Variable-length tables

You can specify variable-length tables by using the OCCURS DEPENDING ON clause.



integer-1

The minimum number of occurrences.

The value of *integer-1* must be greater than or equal to zero, and it must also be less than the value of *integer-2*.

If *integer-1* is omitted, a value of 1 is assumed and the keyword TO must also be omitted.

If *integer-1* is omitted, you will receive warning messages if the RULES (NOOMITODOMIN) compiler option is in effect. For more information, see *RULES* in the *Enterprise COBOL Programming Guide*.

integer-2

The maximum number of occurrences.

integer-2 must be greater than *integer-1*.

The *length* of the subject item is fixed. Only the *number of repetitions* of the subject item is variable.

UNBOUNDED

Unbounded maximum number of occurrences.

Unbounded table

A table with an OCCURS clause that specifies UNBOUNDED.

You can reference unbounded tables in COBOL syntax anywhere a table can be referenced.

Unbounded group

A group that contains at least one unbounded table.

You can define unbounded groups only in the LINKAGE SECTION. Either alphanumeric groups or national groups can be unbounded.

You can reference unbounded groups in COBOL syntax anywhere an alphanumeric or national group can be referenced, with the following exceptions:

- You cannot specify unbounded groups as a BY CONTENT argument in a CALL statement.

- You cannot specify unbounded groups as *data-name-2* on the PROCEDURE DIVISION RETURNING phrase.
- You cannot specify unbounded groups as arguments to intrinsic functions, except as an argument to the LENGTH intrinsic function.

The total dimension of an unbounded group at run time must be less than 999,999,999 bytes.

For unbounded tables and groups, the effect of the SSRANGE compiler option is limited. For more information, see *SSRANGE* in the *Enterprise COBOL Programming Guide*.

For references about working with unbounded tables and groups, see *Working with unbounded tables and groups* in the *Enterprise COBOL Programming Guide*.

OCCURS DEPENDING ON clause

The OCCURS DEPENDING ON clause specifies variable-length tables.

data-name-1

Identifies the *object* of the OCCURS DEPENDING ON clause; that is, the data item whose current value represents the current number of occurrences of the *subject* item. The contents of items whose occurrence numbers exceed the value of the object are undefined.

The object of the OCCURS DEPENDING ON clause (*data-name-1*) must describe an integer data item.

The object of the OCCURS DEPENDING ON clause must not occupy any storage position within the range of the table (that is, any storage position from the first character position in the table through the last character position in the table).

The object of the OCCURS DEPENDING ON clause cannot be variably located; the object cannot follow an item that contains an OCCURS DEPENDING ON clause.

If the OCCURS clause is specified in a data description entry included in a record description entry that contains the EXTERNAL clause, *data-name-1*, if specified, must reference a data item that possesses the external attribute. *data-name-1* must be described in the same DATA DIVISION as the subject of the entry.

If the OCCURS clause is specified in a data description entry subordinate to one that contains the GLOBAL clause, *data-name-1*, if specified, must be a global name. *data-name-1* must be described in the same DATA DIVISION as the subject of the entry.

All data-names used in the OCCURS clause can be qualified; they cannot be subscripted or indexed.

At the time that the group item, or any data item that contains a subordinate OCCURS DEPENDING ON item or that follows but is not subordinate to the OCCURS DEPENDING ON item, is referenced, the value of the object of the OCCURS DEPENDING ON clause must fall within the range *integer-1* through *integer-2*, if *integer-2* is specified (that is, if the table is not UNBOUNDED).

The behavior is undefined if the value of the object is outside of the range *integer-1* through *integer-2*.

When a group item that contains a subordinate OCCURS DEPENDING ON item is referred to, the part of the table area used in the operation is determined as follows:

- If the object is outside the group, only that part of the table area that is specified by the object at the start of the operation is used.
- If the object is included in the same group and the group data item is referenced as a sending item, only that part of the table area that is specified by the value of the object at the start of the operation is used in the operation.
- If the object is included in the same group and the group data item is referenced as a receiving item, the maximum length of the group item is used in the operation.

The following statements are affected by the maximum length rule:

- ACCEPT *identifier* (format 1 and 2)
- CALL ... USING BY REFERENCE *identifier*

- INVOKE ... USING BY REFERENCE *identifier*
- MOVE ... TO *identifier*
- READ ... INTO *identifier*
- RELEASE *identifier* FROM ...
- RETURN ... INTO *identifier*
- REWRITE *identifier* FROM ...
- STRING ... INTO *identifier*
- UNSTRING ... INTO *identifier* DELIMITER IN *identifier*
- WRITE *identifier* FROM ...

If a variable-length group item is not followed by a nonsubordinate item, the maximum length of the group is used when it appears as the identifier in CALL ... USING BY REFERENCE *identifier*. Therefore, the object of the OCCURS DEPENDING ON clause does not need to be set unless the group is variably located.

If the group item is followed by a nonsubordinate item, the actual length, rather than the maximum length, is used. At the time the subject of entry is referenced, or any data item subordinate or superordinate to the subject of entry is referenced, the object of the OCCURS DEPENDING ON clause must fall within the range *integer-1* through *integer-2*, if *integer-2* is specified.

Note:

The maximum length rule does not apply to unbounded groups. For unbounded groups, based on the current run time value of the OCCURS DEPENDING ON objects, the actual length of the group is used for all references to the group. Consequently, before any COBOL statement that references an unbounded group runs, you must set the OCCURS DEPENDING ON objects for that group.

Certain uses of the OCCURS DEPENDING ON clause result in *complex OCCURS DEPENDING ON* (ODO) items. The following items constitute complex ODO items:

- A data item described with an OCCURS DEPENDING ON clause that is followed by a nonsubordinate elementary data item, described with or without an OCCURS clause
- A data item described with an OCCURS DEPENDING ON clause that is followed by a nonsubordinate group item
- A group item that contains one or more subordinate items described with an OCCURS DEPENDING ON clause
- A data item described with an OCCURS clause or an OCCURS DEPENDING ON clause that contains a subordinate data item described with an OCCURS DEPENDING ON clause (a table that contains variable-length elements)
- An index-name associated with a table that contains variable-length elements

The object of an OCCURS DEPENDING ON clause cannot be a nonsubordinate item that follows a complex ODO item.

Any nonsubordinate item that follows an item described with an OCCURS DEPENDING ON clause is a *variably located item*. That is, its location is affected by the value of the OCCURS DEPENDING ON object.

When implicit redefinition is used in a File Description (FD) entry, subordinate level items can contain OCCURS DEPENDING ON clauses.

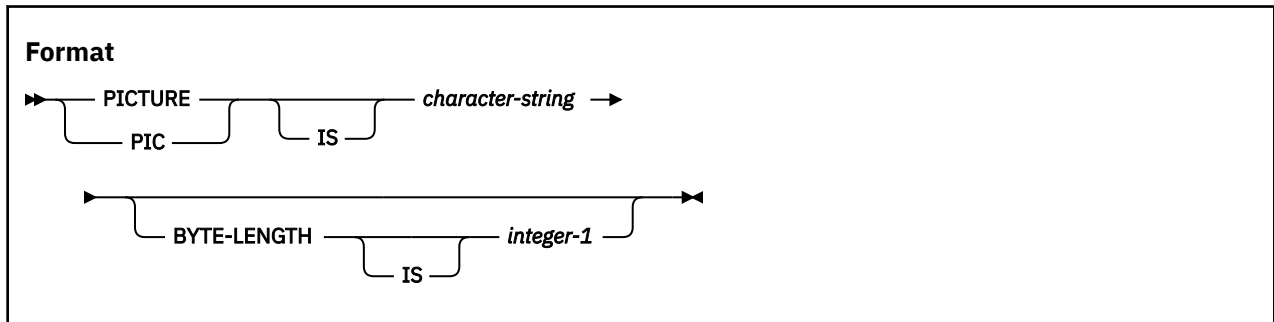
The INDEXED BY phrase can be specified for a table that has a subordinate item that contains an OCCURS DEPENDING ON clause.

For more information about complex OCCURS DEPENDING ON, see *Complex OCCURS DEPENDING ON* in the *Enterprise COBOL Programming Guide*.

The ASCENDING KEY phrase, the DESCENDING KEY phrase, and the INDEXED BY clause are described under [“Fixed-length tables”](#) on page 201.

PICTURE clause

The PICTURE clause specifies the general characteristics and editing requirements of an elementary item.



PICTURE or PIC

The PICTURE clause must be specified for every elementary item except the following ones:

- Index data items
- The subject of the RENAMES clause
- Items described with USAGE POINTER, USAGE FUNCTION-POINTER, USAGE PROCEDURE-POINTER, or USAGE OBJECT REFERENCE
- Internal floating-point data items

In these cases, use of the PICTURE clause is prohibited.

The PICTURE clause can be specified only at the elementary level.

PIC is an abbreviation for PICTURE and has the same meaning.

character-string

character-string is made up of certain COBOL characters used as picture symbols. The allowable combinations determine the category of the elementary data item.

character-string can contain a maximum of 50 characters.

BYTE-LENGTH *integer-1*

The BYTE-LENGTH phrase is only allowed for UTF-8 data items (i.e., data items defined with the U picture symbol) and indicates that the UTF-8 data item has a fixed byte-length but a varying number of characters. The number of bytes occupied by a UTF-8 data item defined with the BYTE-LENGTH phrase of the PICTURE clause is indicated by *integer-1*. The number of UTF-8 characters that can be stored in a data item defined with the BYTE-LENGTH phrase of the PICTURE clause depends on the size of each character. The maximum number of characters that can be stored is *integer-1*, which happens when each UTF-8 character is a single byte in length.

The BYTE-LENGTH phrase can only be specified when the picture string consists of a single 'U' symbol. The DYNAMIC LENGTH clause must not also be specified.

UTF-8 data items defined with the BYTE-LENGTH phrase of the PICTURE clause are truncated on UTF-8 character boundaries when truncation is needed and are always padded with UTF-8 spaces (x'20') to a byte length of *integer-1*.

Only UTF-8 data items defined with the BYTE-LENGTH phrase of the PICTURE clause can be used as Db2® host variables and only UTF-8 data items defined with the BYTE-LENGTH phrase of the PICTURE clause can be part of a group defined with the GROUP-USAGE UTF-8 clause.

UTF-8 data items defined with the BYTE-LENGTH phrase of the PICTURE clause are strongly recommended for UTF-8 items that need to function as a sort key or record key.

Symbols used in the PICTURE clause

Any punctuation character that appears within the PICTURE character-string is not considered a punctuation character, but rather is a PICTURE character-string symbol.

When specified in the SPECIAL-NAMES paragraph, DECIMAL-POINT IS COMMA exchanges the functions of the period and the comma in PICTURE character-strings and in numeric literals.

The lowercase letters that correspond to the uppercase letters that represent the following PICTURE symbols are equivalent to their uppercase representations in a PICTURE character-string:

A, B, E, G, N, P, S, U, V, X, Z, CR, DB

All other lowercase letters are not equivalent to their corresponding uppercase representations.

Table 12 on page 208 defines the meaning of each PICTURE clause symbol. The heading *Size* indicates how the item is counted in determining the number of character positions in the item. The type of the character positions depends on the USAGE clause specified for the item, as follows:

Usage	Type of character positions	Number of bytes per character
DISPLAY	Alphanumeric	1
DISPLAY-1	DBCS	2
NATIONAL	National	2
UTF-8	UTF-8	1 to 4 bytes
All others	Conceptual	Not applicable

Table 12. PICTURE clause symbol meanings		
Symbol	Meaning	Size
A	A character position that can contain only a letter of the Latin alphabet or a space.	Each 'A' is counted as one character position in the size of the data item.
B	For usage DISPLAY, a character position into which an alphanumeric space is inserted. For usage DISPLAY-1, a character position into which a DBCS space is inserted. For usage NATIONAL, a character position into which a national space is inserted.	Each 'B' is counted as one character position in the size of the data item.
E	Marks the start of the exponent in an external floating-point item. For additional details of external floating-point items, see “Data categories and PICTURE rules” on page 212.	Each 'E' is counted as one character position in the size of the data item.
G	A DBCS character position.	Each 'G' is counted as one character position in the size of the data item.

Table 12. PICTURE clause symbol meanings (continued)		
Symbol	Meaning	Size
N	A DBCS character position when specified with usage DISPLAY-1 or when usage is unspecified and the NSYMBOL(DBCS) compiler option is in effect. For category national, a national character position when specified with usage NATIONAL or when usage is unspecified and the NSYMBOL(NATIONAL) compiler option is in effect. For category national-edited, a national character position.	Each 'N' is counted as one character position in the size of the data item.
P	An assumed decimal scaling position. Used to specify the location of an assumed decimal point when the point is not within the number that appears in the data item. See “P symbol” on page 211 for further details.	Not counted in the size of the data item. Scaling position characters are counted in determining the maximum number of digit positions in numeric-edited items or in items that are used as arithmetic operands. The size of the value is the number of digit positions represented by the PICTURE character-string.
S	An indicator of the presence (but not the representation, and not necessarily the position) of an operational sign. An operational sign indicates whether the value of an item involved in an operation is positive or negative.	Not counted in the size of the elementary item, unless an associated SIGN clause specifies the SEPARATE CHARACTER phrase (which would be counted as one character position).
U	A UTF-8 character position. USAGE UTF-8 is assumed whenever a U symbol appears in the PICTURE character-string of a data item.	Each 'U' counts as one UTF-8 character position in the size of the data item. If the BYTE-LENGTH phrase of the PICTURE clause is specified, then only one 'U' symbol can be used in the PICTURE clause and the byte length of the data item is indicated in the BYTE-LENGTH phrase, and the number of characters that can be stored in the data item varies and depends on the size of each character.
V	An indicator of the location of the assumed decimal point. Does not represent a character position. When the assumed decimal point is to the right of the rightmost symbol in the string, the V is redundant.	Not counted in the size of the elementary item.
X	A character position that can contain any allowable character from the alphanumeric character set of the computer.	Each 'X' is counted as one character position in the size of the data item.
Z	A leading numeric character position. When that position contains a zero, a space character replaces the zero.	Each 'Z' is counted as one character position in the size of the data item.

Table 12. PICTURE clause symbol meanings (continued)		
Symbol	Meaning	Size
9	A character position that contains a numeral.	Each nine specifies one decimal digit in the value of the item. For usages DISPLAY and NATIONAL, each nine is counted as one character position in the size of the data item.
0	A character position into which the numeral zero is inserted.	Each zero is counted as one character position in the size of the data item.
/	A character position into which the slash character is inserted.	Each slash character is counted as one character position in the size of the data item.
,	A character position into which a comma is inserted.	Each comma is counted as one character position in the size of the data item.
.	An editing symbol that represents the decimal point for alignment purposes. In addition, it represents a character position into which a period is inserted.	Each period is counted as one character position in the size of the data item.
+ - CR DB	Editing sign control symbols. Each represents the character position into which the editing sign control symbol is placed.	Each character used in the editing sign symbol is counted as one character position in the size of the data item.
*	A check protect symbol: a leading numeric character position into which an asterisk is placed when that position contains a zero.	Each asterisk is counted as one character position in the size of the item.
cs	cs can be any valid currency symbol. A currency symbol represents a character position into which a currency sign value is placed. The default currency symbol is the character assigned the value X'5B' in the code page in effect at compile time. In this document, the default currency symbol is represented by the dollar sign (\$) and cs stands for any valid currency symbol. For details, see “Currency symbol” on page 212 .	The first occurrence of a currency symbol adds the number of characters in the currency sign value to the size of the data item. Each subsequent occurrence adds one character position to the size of the data item.

The following figure shows the sequences in which picture symbols can be specified to form picture character-strings. More detailed explanations of PICTURE clause symbols follow the figure.

FIRST SYMBOL	SECOND SYMBOL	Non-Floating Insertion Symbols										Floating Insertion Symbols										Other Symbols									
		B	0	/	.	-	+	CR	DB	CS	E	Z	+	-	CS	g	A	S	V	P	P	G	N								
NON-FLOATING INSERTION SYMBOLS	B	*	*	*	*	*	*				*	*	*	*	*	*	*	*	*	*	*	*	*								
	0	*	*	*	*	*	*				*	*	*	*	*	*	*	*	*	*	*	*	*								
	/	*	*	*	*	*	*				*	*	*	*	*	*	*	*	*	*	*	*	*								
	.	*	*	*	*	*	*				*	*	*	*	*	*	*	*	*	*	*	*	*								
	-	*	*	*	*	*	*				*	*	*	*	*	*	*	*	*	*	*	*	*								
	+	*	*	*	*	*	*				*	*	*	*	*	*	*	*	*	*	*	*	*								
	CR	*	*	*	*	*	*				*	*	*	*	*	*	*	*	*	*	*	*	*								
	DB	*	*	*	*	*	*				*	*	*	*	*	*	*	*	*	*	*	*	*								
	CS						*																								
	E				*	*											*		*												
FLOATING INSERTION SYMBOLS	Z	*	*	*	*	*	*				*																				
	+	*	*	*	*	*	*				*																				
	-	*	*	*	*	*	*				*																				
	CS	*	*	*	*	*	*				*																				
	g	*	*	*	*	*	*				*																				
	A	*	*	*	*	*	*				*																				
	S	*	*	*	*	*	*				*																				
	V	*	*	*	*	*	*				*																				
	P	*	*	*	*	*	*				*																				
	P	*	*	*	*	*	*				*																				
OTHER SYMBOLS	g	*	*	*	*	*	*				*						*	*	*	*	*	*	*								
	A	*	*	*	*	*	*				*						*	*	*	*	*	*	*								
	S	*	*	*	*	*	*				*						*	*	*	*	*	*	*								
	V	*	*	*	*	*	*				*						*	*	*	*	*	*	*								
	P	*	*	*	*	*	*				*						*	*	*	*	*	*	*								
	P	*	*	*	*	*	*				*						*	*	*	*	*	*	*								
G	*																							*							
N	*	*	*																						*						

Legend:

- Closed circle indicates that the symbol(s) at the top of the column can, in a given character-string, appear anywhere to the left of the symbol(s) at the left of the row.

{ }

Symbols that appear twice

Nonfloating insertion symbols + and -, floating insertion symbols Z, *, +, -, and CS, and the symbol P appear twice. The leftmost column and uppermost row for each symbol represents its use to the left of the decimal point position. The second appearance of the symbol in the table represents its use to the right of the decimal point position.

P symbol

The symbol P specifies a scaling position and implies an assumed decimal point (to the left of the Ps if the Ps are leftmost PICTURE characters; to the right of the Ps if the Ps are rightmost PICTURE characters).

The assumed decimal point symbol V is redundant as either the leftmost or rightmost character within such a PICTURE description.

The symbol P can be specified only as a continuous string of Ps in the leftmost or rightmost digit positions within a PICTURE character-string.

In certain operations that reference a data item whose PICTURE character-string contains the symbol P, the algebraic value of the data item is used rather than the actual character representation of the data item. This algebraic value assumes the decimal point in the prescribed location and zero in place of the digit position specified by the symbol P. The size of the value is the number of digit positions represented by the PICTURE character-string. These operations are any of the following ones:

- Any operation that requires a numeric sending operand
- A MOVE statement where the sending operand is numeric and its PICTURE character-string contains the symbol P
- A MOVE statement where the sending operand is numeric-edited and its PICTURE character-string contains the symbol P, and the receiving operand is numeric or numeric-edited
- A comparison operation where both operands are numeric

In all other operations, the digit positions specified with the symbol P are ignored and are not counted in the size of the operand.

Currency symbol

The currency symbol in a picture character-string is represented by the default currency symbol \$ or by a single character specified either in the CURRENCY compiler option or in the CURRENCY SIGN clause in the SPECIAL-NAMES paragraph of the ENVIRONMENT DIVISION.

Although the default currency symbol is represented by \$ in this document, the actual default currency symbol is the character with the value X'5B' in the EBCDIC code page in effect at compile time.

If the CURRENCY SIGN clause is specified, the CURRENCY and NOCURRENCY compiler options are ignored. If the CURRENCY SIGN clause is not specified and the NOCURRENCY compiler option is in effect, the dollar sign (\$) is used as the default currency sign value and currency symbol. For more information about the CURRENCY SIGN clause, see “CURRENCY SIGN clause” on page 129. For more information about the CURRENCY and NOCURRENCY compiler options, see *CURRENCY* in the *Enterprise COBOL Programming Guide*.

A currency symbol can be repeated within the PICTURE character-string to specify floating insertion. Different currency symbols must not be used in the same PICTURE character-string.

Unlike all other picture symbols, currency symbols are case sensitive. For example, 'D' and 'd' specify different currency symbols.

A currency symbol can be used only to define a numeric-edited item with USAGE DISPLAY.

Character-string representation

The topic lists symbols that can appear once or more than once in the PICTURE character-string.

Symbols that can appear more than once

The following symbols can appear more than once in one PICTURE character-string:

```
A B G N P X U Z 9 0 / , + - * cs
```

At least one of the symbols A, G, N, X, U, Z, 9, or *, or at least two of the symbols +, –, or cs must be present in a PICTURE string.

An unsigned nonzero integer enclosed in parentheses immediately following any of these symbols specifies the number of consecutive occurrences of that symbol.

Example: The following two PICTURE clause specifications are equivalent:

```
PICTURE IS $99999.99CR  
PICTURE IS $9(5).9(2)CR
```

Symbols that can appear only once

The following symbols can appear only once in one PICTURE character-string:

```
E S V . CR DB
```

Except for the PICTURE symbol V, each occurrence of any of the aforementioned symbols in a given PICTURE character-string represents an occurrence of that character or set of allowable characters in the data item.

Data categories and PICTURE rules

The allowable combinations of PICTURE symbols determine the data category of the item.

The data categories are:

- Alphabetic
- Alphanumeric

- Alphanumeric-edited
- DBCS
- External floating-point
- National
- National-edited
- Numeric
- Numeric-edited
- UTF-8

Note: Category internal floating point is defined by a USAGE clause that specifies the COMP-1 or COMP-2 phrase.

Alphabetic items

The PICTURE character-string can contain only the symbol A.

The content of the item must consist only of letters of the Latin alphabet and the space character.

Other clauses

USAGE DISPLAY must be specified or implied.

Any associated VALUE clause must specify an alphanumeric literal containing only alphabetic characters, SPACE, or a symbolic-character as the value of a figurative constant.

Do not include a single byte character in a DBCS data item.

When padding is required for a DBCS data item, the following rules apply:

- Padding is done using double-byte space characters until the data area is filled (based on the number of double-byte character positions allocated for the data item).
- Padding is done using single-byte space characters when the padding needed is not an even number of bytes (for example, when an alphanumeric group item is moved to a DBCS data item).

Alphanumeric items

The PICTURE character-string must consist of certain symbols.

The symbols are:

- One or more occurrences of the symbol X.
- Combinations of the symbols A, X, and 9. (A character-string containing all As or all 9s does not define an alphanumeric item.)

The item is treated as if the character-string contained only the symbol X.

The contents of the item in standard data format can be any allowable characters from the character set of the computer.

Other clauses

USAGE DISPLAY must be specified or implied.

Any associated VALUE clause must specify an alphanumeric literal or one of the following figurative constants:

- ZERO
- SPACE
- QUOTE
- HIGH-VALUE
- LOW-VALUE

- *symbolic-character*
- *ALL alphanumeric-literal*

Alphanumeric-edited items

The PICTURE character-string can contain the following symbols: A X 9 B 0 /.

The string must contain at least one A or X, and at least one B or 0 (zero) or /.

The contents of the item in standard data format must be two or more characters from the character set of the computer.

Other clauses

USAGE DISPLAY must be specified or implied.

Any associated VALUE clause must specify an alphanumeric literal or one of the following figurative constants:

- ZERO
- SPACE
- QUOTE
- HIGH-VALUE
- LOW-VALUE
- *symbolic-character*
- *ALL alphanumeric-literal*

The literal is treated exactly as specified; no editing is done.

DBCS items

The PICTURE character-string can contain the symbols G, B, or N. Each G, B, or N represents a single DBCS character position.

Any associated VALUE clause must contain a DBCS literal, the figurative constant SPACE, or the figurative constant *ALL DBCS-literal*.

Other clauses

When PICTURE symbol G is used, USAGE DISPLAY-1 must be specified. When PICTURE symbol N is used and the NSYMBOL(DBCS) compiler option is in effect, USAGE DISPLAY-1 is implied if the USAGE clause is omitted.

External floating-point items

A data item is described as category external floating-point by its PICTURE character-string.

The PICTURE character-string details are described here.

Format

→ + mantissa E + exponent →

+ or -

A sign character must immediately precede both the mantissa and the exponent.

A + sign indicates that a positive sign will be used in the output to represent positive values and that a negative sign will represent negative values.

A - sign indicates that a blank will be used in the output to represent positive values and that a negative sign will represent negative values.

Each sign position occupies one byte of storage.

mantissa

The mantissa can contain the symbols:

9 . V

An actual decimal point can be represented with a period (.) while an assumed decimal point is represented by a V.

Either an actual or an assumed decimal point must be present in the mantissa; the decimal point can be leading, embedded, or trailing.

The mantissa can contain from 1 to 16 numeric characters.

E

Indicates the exponent.

exponent

The exponent must consist of the symbol 99.

Example: Pic -9v9(9)E-99

The DISPLAY phrase of the USAGE clause and a floating-point picture character-string define the item as a *display floating-point data item*.

The NATIONAL phrase of the USAGE clause and a floating-point picture character-string define the item as a *national floating-point data item*.

For items defined with usage DISPLAY, each picture symbol except V defines one alphanumeric character position in the item.

For items defined with usage NATIONAL, each picture symbol except V defines one national character position in the item.

Other clauses

The DISPLAY phrase or the NATIONAL phrase of the USAGE clause must be specified or implied.

The OCCURS, REDEFINES, and RENAMES clauses can be associated with external floating-point items.

The SIGN clause is accepted as documentation and has no effect on the representation of the sign.

The SYNCHRONIZED clause is treated as documentation.

The following clauses are invalid with external floating-point items:

- BLANK WHEN ZERO
- JUSTIFIED
- VALUE

National items

The PICTURE character-string can contain one or more occurrences of the picture symbol N.

These rules apply when the NSYMBOL(NATIONAL) compiler option is in effect or the USAGE NATIONAL clause is specified. In the absence of a USAGE NATIONAL clause, if the NSYMBOL(DBCS) compiler option is in effect, picture symbol N represents a DBCS character and the rules of the PICTURE clause for a DBCS item apply.

Each N represents a single national character position.

Any associated VALUE clause must specify an alphanumeric literal, a national literal, or one of the following figurative constants:

- ZERO
- SPACE
- QUOTE
- HIGH-VALUE
- LOW-VALUE
- *symbolic-character*
- *ALL alphanumeric-literal*
- *ALL national-literal*

Other clauses

Only the NATIONAL phrase can be specified in the USAGE clause. When PICTURE symbol N is used and the NSYMBOL(NATIONAL) compiler option is in effect, USAGE NATIONAL is implied if the usage clause is omitted.

The following clauses can be used:

- JUSTIFIED
- EXTERNAL
- GLOBAL
- OCCURS
- REDEFINES
- RENAMES
- SYNCHRONIZED
- VOLATILE

The following clauses cannot be used:

- BLANK WHEN ZERO
- SIGN

National-edited items

The PICTURE character-string must contain at least one symbol N, and at least one instance of one of these symbols: B 0 (zero) or / (slash).

Each symbol represents a single national character position.

Any associated VALUE clause must specify an alphanumeric literal, a national literal, or one of the following figurative constants:

- ZERO
- SPACE
- QUOTE
- HIGH-VALUE
- LOW-VALUE
- *symbolic-character*
- *ALL alphanumeric-literal*
- *ALL national-literal*

The literal is treated exactly as specified; no editing is done.

The NSYMBOL(NATIONAL) compiler option has no effect on the definition of a data item of category national-edited.

Other clauses

USAGE NATIONAL must be specified or implied.

The following clauses can be used:

- JUSTIFIED
- EXTERNAL
- GLOBAL
- OCCURS
- REDEFINES
- RENAMES
- SYNCHRONIZED
- VOLATILE

The following clauses cannot be used:

- BLANK WHEN ZERO
- SIGN

Numeric items

There are several types of numeric items.

The types are:

- Binary
- Packed decimal (internal decimal)
- Zoned decimal (external decimal)
- National decimal (external decimal)

The type of a numeric item is defined by the usage clause as shown in the following table.

Table 13. Numeric types	
Type	USAGE clause
Binary	BINARY, COMP, COMP-4, or COMP-5
Internal decimal	PACKED-DECIMAL, COMP-3
Zoned decimal (external decimal)	DISPLAY
National decimal (external decimal)	NATIONAL

For all numeric fields, the PICTURE character-string can contain only the symbols 9, P, S, and V.

The symbol S can be written only as the leftmost character in the PICTURE character-string.

The symbol V can be written only once in a given PICTURE character-string.

For binary items, the number of digit positions must range from 1 through 18 inclusive. For packed decimal and zoned decimal items the number of digit positions must range from 1 through 18, inclusive, when the ARITH(COMPAT) compiler option is in effect, or from 1 through 31, inclusive, when the ARITH(EXTEND) compiler option is in effect.

If unsigned, the contents of the item in standard data format must contain a combination of the Arabic numerals 0-9. If signed, it can also contain a +, -, or other representation of the operational sign.

Examples of valid ranges

PICTURE	Valid range of values
9999	0 through 9999

S99	-99 through +99
S999V9	-999.9 through +999.9
PPP999	0 through .000999
S999PPP	-1000 through -999000 and +1000 through +999000 or zero

Other clauses

The USAGE of the item can be DISPLAY, NATIONAL, BINARY, COMPUTATIONAL, PACKED-DECIMAL, COMPUTATIONAL-3, COMPUTATIONAL-4, or COMPUTATIONAL-5.

For signed numeric items described with usage NATIONAL, the SIGN IS SEPARATE clause must be specified or implied.

The NUMPROC and TRUNC compiler options can affect the use of numeric data items. For details, see *NUMPROC* and *TRUNC* in the *Enterprise COBOL Programming Guide*.

Numeric-edited items

The PICTURE character-string can contain certain symbols.

The symbols are:

B P V Z 9 0 / , . + - CR DB * cs

The combinations of symbols allowed are determined from the PICTURE clause symbol order allowed (see the figure in “Symbols used in the PICTURE clause” on page 208), and the editing rules (see “PICTURE clause editing” on page 219).

The following rules apply:

- Either the BLANK WHEN ZERO clause must be specified for the item, or the string must contain at least one of the following symbols:

B / Z 0 , . * + - CR DB cs

- Only one of the following symbols can be written in a given PICTURE character-string:

+ - CR DB

- If the ARITH(COMPAT) compiler option is in effect, then the number of digit positions represented in the character-string must be in the range 1 through 18, inclusive. If the ARITH(EXTEND) compiler option is in effect, then the number of digit positions represented in the character-string must be in the range 1 through 31, inclusive.
- The total number of character positions in the string (including editing-character positions) must not exceed 249.
- The contents of those character positions representing digits in standard data format must be one of the 10 Arabic numerals.

Other clauses

USAGE DISPLAY or NATIONAL must be specified or implied.

If the usage of the item is DISPLAY, any associated VALUE clause must specify an alphanumeric literal or a figurative constant. The value is assigned without editing.

If the usage of the item is NATIONAL, any associated VALUE clause must specify an alphanumeric literal, a national literal, or a figurative constant. The value is assigned without editing.

UTF-8 items

The PICTURE character-string can contain one or more occurrences of the picture symbol U.

Each U represents a single UTF-8 character position.

Any associated VALUE clause must specify an alphanumeric literal, a UTF-8 literal, or one of the following figurative constants:

- ZERO
- SPACE
- QUOTE
- HIGH-VALUE
- LOW-VALUE
- *symbolic-character*
- ALL *utf-8-literal*

Other clauses

Only the UTF-8 phrase can be specified in the USAGE clause. USAGE UTF-8 is implied if the usage clause is omitted.

The following clauses can be used:

- JUSTIFIED
- EXTERNAL
- GLOBAL
- OCCURS
- REDEFINES
- RENAMES
- SYNCHRONIZED
- VOLATILE

The following clauses cannot be used:

- BLANK WHEN ZERO
- SIGN

PICTURE clause editing

There are two general methods of editing in a PICTURE clause, insertion editing, and suppression and replacement editing.

Insertion editing includes the following types of editing:

- Simple insertion
- Special insertion
- Fixed insertion
- Floating insertion

Suppression and replacement editing includes the following types of editing:

- Zero suppression and replacement with asterisks
- Zero suppression and replacement with spaces

The type of editing allowed for an item depends on its *data category*. The type of editing that is valid for each category is shown in the following table. *cs* indicates any valid currency symbol.

Table 14. <i>Data categories</i>		
Data category	Type of editing	Insertion symbol
Alphabetic	None	None

Table 14. **Data categories** (continued)

Data category	Type of editing	Insertion symbol
Alphanumeric	None	None
Alphanumeric-edited	Simple insertion	B 0 /
DBCS	Simple insertion	B
External floating-point	Special insertion	.
National	None	None
National-edited	Simple insertion	B 0 /
Numeric	None	None
Numeric-edited	Simple insertion Special insertion Fixed insertion Floating insertion Zero suppression Replacement	B 0 / , . cs + - CR DB cs + - Z * Z * + - cs
UTF-8	None	None

Types of editing are described in the following sections:

- [“Simple insertion editing” on page 220](#)
- [“Special insertion editing” on page 221](#)
- [“Fixed insertion editing” on page 221](#)
- [“Floating insertion editing” on page 222](#)
- [“Zero suppression and replacement editing” on page 224](#)

Simple insertion editing

This type of editing is valid for alphanumeric-edited, numeric-edited, and DBCS items.

Each insertion symbol is counted in the size of the item, and represents the position within the item where the equivalent character is to be inserted. For edited DBCS items, each insertion symbol (B) is counted in the size of the item and represents the position within the item where the DBCS space is to be inserted.

For example:

PICTURE	Value of data	Edited result
X(10)/XX	ALPHANUMER01	ALPHANUMER/01
X(5)BX(7)	ALPHANUMERIC	ALPHA NUMERIC
99,B999,B000	1234	01,b234,b000 ¹
99,999	12345	12,345
GGBBGG	D1D2D3D4	D1D2bbbbD3D4 ¹
Notes: 1. The symbol <i>b</i> represents a space.		

Special insertion editing

This type of editing is valid for either numeric-edited items or external floating-point items.

The period (.) is the special insertion symbol; it also represents the actual decimal point for alignment purposes.

Note: If the DECIMAL-POINT IS COMMA clause is specified, then a comma will be used in place of the period.

The period insertion symbol is counted in the size of the item, and represents the position within the item where the actual decimal point is inserted.

Either the actual decimal point or the symbol V as the assumed decimal point, but not both, must be specified in one PICTURE character-string.

For example:

PICTURE	Value of data	Edited result
999.99	1.234	001.23
999.99	12.34	012.34
999.99	123.45	123.45
999.99	1234.5	234.50
+999.99E+99	12345	+123.45E+02

Fixed insertion editing

Fixed insertion editing is valid only for numeric-edited items.

The following insertion symbols are used:

- CS
- + - CR DB (editing-sign control symbols)

In fixed insertion editing, only one currency symbol and one editing-sign control symbol can be specified in a PICTURE character-string.

Unless it is preceded by a + or - symbol, the currency symbol must be the first character in the character-string.

When either + or - is used as a symbol, it must be the first or last character in the character-string.

When CR or DB is used as a symbol, it must occupy the rightmost two character positions in the character-string. If these two character positions contain the symbols CR or DB, the uppercase letters are the insertion characters.

Editing sign control symbols produce results that depend on the value of the data item, as shown in the following table:

Editing symbol in PICTURE character-string	Result: data item positive or zero	Result: data item negative
+	+	-
-	space	-
CR	2 spaces	CR
DB	2 spaces	DB

For example:

PICTURE	Value of data	Edited result
999.99+	+6555.556	555.55+
+9999.99	-6555.555	-6555.55
9999.99	+1234.56	1234.56
\$999.99	-123.45	\$123.45
-\$999.99	-123.456	-\$123.45
-\$999.99	+123.456	\$123.45
\$9999.99CR	+123.45	\$0123.45
\$9999.99CR	-123.45	\$0123.45CR

Floating insertion editing

Floating insertion editing is valid only for numeric-edited items.

The following symbols are used:

CS + -

Within one PICTURE character-string, these symbols are mutually exclusive as floating insertion characters.

Floating insertion editing is specified by using a string of at least two of the allowable floating insertion symbols to represent leftmost character positions into which the actual characters can be inserted.

The leftmost floating insertion symbol in the character-string represents the leftmost limit at which the actual character can appear in the data item. The rightmost floating insertion symbol represents the rightmost limit at which the actual character can appear.

The second leftmost floating insertion symbol in the character-string represents the leftmost limit at which numeric data can appear within the data item. Nonzero numeric data can replace all characters at or to the right of this limit.

Any simple-insertion symbols (B 0 / ,) within or to the immediate right of the string of floating insertion symbols are considered part of the floating character-string. If the period (.) special-insertion symbol is included within the floating string, it is considered to be part of the character-string.

To avoid truncation, the minimum size of the PICTURE character-string must be:

- The number of character positions in the sending item, plus
- The number of nonfloating insertion symbols in the receiving item, plus
- One character position for the floating insertion symbol

Representing floating insertion editing

In a PICTURE character-string, there are two ways to represent floating insertion editing and thus two ways in which editing is performed:

1. Any or all leading numeric character positions to the left of the decimal point are represented by the floating insertion symbol. When editing is performed, a single floating insertion character is placed to the immediate left of the first nonzero digit in the data, or of the decimal point, whichever is farther to the left. The character positions to the left of the inserted character are filled with spaces.

If all numeric character positions in the PICTURE character-string are represented by the insertion character, then at least one of the insertion characters must be to the left of the decimal point.

2. All the numeric character positions are represented by the floating insertion symbol. When editing is performed, then:

- If the value of the data is zero, the entire data item will contain spaces.
- If the value of the data is nonzero, the result is the same as in rule 1.

For example:

PICTURE	Value of data	Edited result
\$\$\$\$.99	.123	\$.12
\$\$\$9.99	.12	\$0.12
,\$\$\$,999.99	-1234.56	\$1,234.56
+,+++,999.99	-123456.789	-123,456.78
\$\$,,\$\$,,\$\$.99CR	-1234567	\$1,234,567.00CR
++,+++,+++.+++	0000.00	

Zero suppression and replacement editing

Zero suppression and replacement editing is valid only for numeric-edited items.

In zero suppression editing, the symbols Z and * are used. These symbols are mutually exclusive in one PICTURE character-string.

The following symbols are mutually exclusive as floating replacement symbols in one PICTURE character-string:

Z * + - cs

Specify zero suppression and replacement editing with a string of one or more of the allowable symbols to represent leftmost character positions in which zero suppression and replacement editing can be performed.

Any simple insertion symbols (B 0 / ,) within or to the immediate right of the string of floating editing symbols are considered part of the string. If the period (.) special insertion symbol is included within the floating editing string, it is considered to be part of the character-string.

Representing zero suppression

In a PICTURE character-string, there are two ways to represent zero suppression, and two ways in which editing is performed:

- Any or all of the leading numeric character positions to the left of the decimal point are represented by suppression symbols. When editing is performed, the replacement character replaces any leading zero in the data that appears in the same character position as a suppression symbol. Suppression stops at the leftmost character:
 - That does not correspond to a suppression symbol
 - That contains nonzero data
 - That is the decimal point
- All the numeric character positions in the PICTURE character-string are represented by the suppression symbols. When editing is performed and the value of the data is nonzero, the result is the same as in the preceding rule. If the value of the data is zero, then:
 - If Z has been specified, the entire data item will contain spaces.
 - If * has been specified, the entire data item except the actual decimal point will contain asterisks.

For example:

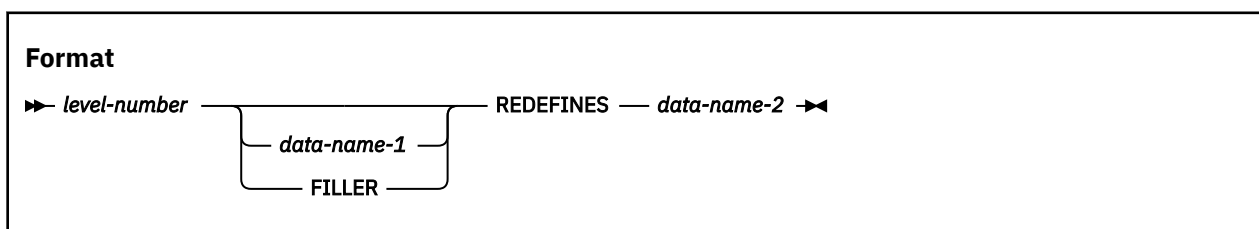
PICTURE	Value of data	Edited result
****. **	0000.00	****. **
ZZZZ.ZZ	0000.00	
ZZZZ.99	0000.00	.00
****.99	0000.00	****.00

PICTURE	Value of data	Edited result
ZZ99.99	0000.00	00.00
Z,ZZZ.ZZ+	+123.456	123.45+
*,***.***	-123.45	**123.45-
,,***.***	+12345678.9	12,345,678.90+
\$Z,ZZZ,ZZZ.ZZCR	+12345.67	\$ 12,345.67
\$B*,***,***.**BBDB	-12345.67	\$ ***12,345.67 DB

Do not specify both the asterisk (*) as a suppression symbol and the BLANK WHEN ZERO clause for the same entry.

REDEFINES clause

The REDEFINES clause allows you to use different data description entries to describe the same computer storage area.



(*level-number*, *data-name-1*, and FILLER are not part of the REDEFINES clause, and are included in the format only for clarity.)

When specified, the REDEFINES clause must be the first entry following *data-name-1* or FILLER. If *data-name-1* or FILLER is not specified, the REDEFINES clause must be the first entry following the level-number.

***data-name-1*, FILLER**

Identifies an alternate description for the data area identified by *data-name-2*; *data-name-1* is the *redefining* item or the REDEFINES *subject*.

Neither *data-name-1* nor any of its subordinate entries can contain a VALUE clause.

data-name-2

Identifies the *redefined* item or the REDEFINES *object*.

The data description entry for *data-name-2* can contain a REDEFINES clause.

The data description entry for *data-name-2* cannot contain an OCCURS clause. However, *data-name-2* can be subordinate to an item whose data description entry contains an OCCURS clause; in this case, the reference to *data-name-2* in the REDEFINES clause must not be subscripted.

Neither *data-name-1* nor *data-name-2* can contain an OCCURS DEPENDING ON clause.

data-name-1 and *data-name-2* must have the same level in the hierarchy; however, the level numbers need not be the same. Neither *data-name-1* nor *data-name-2* can be defined with level number 66 or 88.

data-name-1 and *data-name-2* can each be described with any usage.

Redefinition begins at *data-name-1* and ends when a level-number less than or equal to that of *data-name-1* is encountered. No entry that has a level-number numerically less than those of *data-name-1* and *data-name-2* can occur between these entries. In the following example:

```
05  A PICTURE X(6).
05  B REDEFINES A.
    10 B-1          PICTURE X(2).
    10 B-2          PICTURE 9(4).
05  C              PICTURE 99V99.
```

A is the redefined item, and B is the redefining item. Redefinition begins with B and includes the two subordinate items B-1 and B-2. Redefinition ends when the level-05 item C is encountered.

If the GLOBAL clause is used in the data description entry that contains the REDEFINES clause, only *data-name-1* (the redefining item) possesses the global attribute. For example, in the following description, only item B possesses the GLOBAL attribute:

```
05  A PICTURE X(6).
05  B REDEFINES A GLOBAL PICTURE X(4).
```

The EXTERNAL clause must not be specified in the same data description entry as a REDEFINES clause.

If the redefined data item (*data-name-2*) is declared to be an external data record, the size of the redefining data item (*data-name-1*) must not be greater than the size of the redefined data item. If the redefined data item is not declared to be an external data record, there is no such constraint.

The following example shows that the redefining item, B, can occupy more storage than the redefined item, A. The size of storage for the REDEFINED clause is determined in number of bytes. Item A occupies 6 bytes of storage and item B, a data item of category national, occupies 8 bytes of storage.

```
05  A PICTURE X(6).
05  B REDEFINES A GLOBAL PICTURE N(4).
```

One or more redefinitions of the same storage area are permitted. The entries that give the new descriptions of the storage area must immediately follow the description of the redefined area without intervening entries that define new character positions. Multiple redefinitions can, but need not, all use the data-name of the original entry that defined this storage area. For example:

```
05  A          PICTURE 9999.
05  B REDEFINES A PICTURE 9V999.
05  C REDEFINES A PICTURE 99V99.
```

Also, multiple redefinitions can use the name of the preceding definition as shown in the following example:

```
05  A          PICTURE 9999.
05  B REDEFINES A PICTURE 9V999.
05  C REDEFINES B PICTURE 99V99.
```

When more than one level-01 entry is written subordinate to an FD entry, a condition known as *implicit redefinition* occurs. That is, the second level-01 entry implicitly redefines the storage allotted for the first entry. In such level-01 entries, the REDEFINES clause must not be specified.

When the data item implicitly redefines multiple 01-level records in a file description (FD) entry, items subordinate to the redefining or redefined item can contain an OCCURS DEPENDING ON clause.

REDEFINES clause considerations

The topic lists considerations of using the REDEFINES clause.

When an area is redefined, all descriptions of the area are always in effect; that is, redefinition does not supersede a previous description. Thus, if B REDEFINES C has been specified, either of the two procedural statements MOVE X TO B or MOVE Y TO C could be executed at any point in the program. In the first case, the area described as B would receive the value and format of X. In the second case, the same physical area (described now as C) would receive the value and format of Y. Note that if the second statement is executed immediately after the first, the value of Y replaces the value of X in the one storage area.

The usage of a redefining data item need not be the same as that of a redefined item. This does not, however, cause any change in the format or content of existing data. For example:

```
05 B          PICTURE 99 USAGE DISPLAY VALUE 8.
05 C REDEFINES B PICTURE S99 USAGE COMPUTATIONAL-4.
05 A          PICTURE S99 USAGE COMPUTATIONAL-4.
```

Redefining B does not change the bit configuration of the data in the storage area. Therefore, the following two statements produce different results:

```
ADD B TO A
ADD C TO A
```

In the first case, the value 8 is added to A (because B has USAGE DISPLAY). In the second statement, the value -3848 is added to A (because C has USAGE COMPUTATIONAL-4), and the bit configuration of the storage area has the binary value -3848. This example demonstrates how the improper use of redefinition can give unexpected or incorrect results.

REDEFINES clause examples

The REDEFINES clause can be specified for an item within the scope of (subordinate to) an area that is redefined.

In the following example, WEEKLY-PAY redefines SEMI-MONTHLY-PAY (which is within the scope of REGULAR-EMPLOYEE, while REGULAR-EMPLOYEE is redefined by TEMPORARY-EMPLOYEE).

```
05 REGULAR-EMPLOYEE.
  10 LOCATION          PICTURE A(8).
  10 GRADE              PICTURE X(4).
  10 SEMI-MONTHLY-PAY  PICTURE 9999V99.
  10 WEEKLY-PAY REDEFINES SEMI-MONTHLY-PAY
                        PICTURE 999V999.
05 TEMPORARY-EMPLOYEE REDEFINES REGULAR-EMPLOYEE.
  10 LOCATION          PICTURE A(8).
  10 FILLER             PICTURE X(6).
  10 HOURLY-PAY         PICTURE 99V99.
```

The REDEFINES clause can also be specified for an item subordinate to a redefining item, as shown for CODE-H REDEFINES HOURLY-PAY in the following example:

```
05 REGULAR-EMPLOYEE.
  10 LOCATION          PICTURE A(8).
  10 GRADE              PICTURE X(4).
  10 SEMI-MONTHLY-PAY  PICTURE 999V99.
05 TEMPORARY-EMPLOYEE REDEFINES REGULAR-EMPLOYEE.
  10 LOCATION          PICTURE A(8).
  10 FILLER             PICTURE X(6).
  10 HOURLY-PAY         PICTURE 99V99.
  10 CODE-H REDEFINES HOURLY-PAY PICTURE 9999.
```

Data items within an area can be redefined without changing their lengths. For example:

```
05 NAME-2.  
 10 SALARY          PICTURE XXX.  
 10 SO-SEC-NO       PICTURE X(9).  
 10 MONTH           PICTURE XX.  
05 NAME-1 REDEFINES NAME-2.  
 10 WAGE            PICTURE XXX.  
 10 EMP-NO          PICTURE X(9).  
 10 YEAR            PICTURE XX.
```

Data item lengths and types can also be respecified within an area. For example:

```
05 NAME-2.  
 10 SALARY          PICTURE XXX.  
 10 SO-SEC-NO       PICTURE X(9).  
 10 MONTH           PICTURE XX.  
05 NAME-1 REDEFINES NAME-2.  
 10 WAGE            PICTURE 999V999.  
 10 EMP-NO          PICTURE X(6).  
 10 YEAR            PICTURE XX.
```

Data items can also be respecified with a length that is greater than the length of the redefined item. For example:

```
05 NAME-2.  
 10 SALARY          PICTURE XXX.  
 10 SO-SEC-NO       PICTURE X(9).  
 10 MONTH           PICTURE XX.  
05 NAME-1 REDEFINES NAME-2.  
 10 WAGE            PICTURE 999V999.  
 10 EMP-NO          PICTURE X(6).  
 10 YEAR            PICTURE X(4).
```

This does not change the length of the redefined item NAME-2.

Undefined results

Undefined results can occur in the conditions as listed in the topic.

- A redefining item is moved to a redefined item (that is, if B REDEFINES C and the statement MOVE B TO C is executed).
- A redefined item is moved to a redefining item (that is, if B REDEFINES C and the statement MOVE C TO B is executed).

RENAMES clause

The RENAMES clause specifies alternative and possibly overlapping groupings of elementary data items.

Format

►► 66 — *data-name-1* — RENAMES — *data-name-2* — *THROUGH* — *data-name-3* —
 THRU

The special level-number 66 must be specified for data description entries that contain the RENAMES clause. (Level-number 66 and *data-name-1* are not part of the RENAMES clause, and are included in the format only for clarity.)

One or more RENAMES entries can be written for a logical record. All RENAMES entries associated with one logical record must immediately follow the last data description entry of that record.

data-name-1

Identifies an alternative grouping of data items.

A level-66 entry cannot rename a level-01, level-77, level-88, or another level-66 entry.

data-name-1 cannot be used as a qualifier; it can be qualified only by the names of level indicator entries or level-01 entries.

data-name-2, data-name-3

Identify the original grouping of elementary data items; that is, they must name elementary or group items within the associated level-01 entry and must not be the same data-name. Both data-names can be qualified.

data-name-2 and *data-name-3* can each reference any of the following items:

- An elementary data item
- An alphanumeric group item
- A national group item

When *data-name-2* or *data-name-3* references a national group item, the referenced item is processed as a group (not as an elementary data item of category national).

The OCCURS clause must not be specified in the data entries for *data-name-2* and *data-name-3*, or for any group entry to which they are subordinate. In addition, the OCCURS DEPENDING clause must not be specified for any item defined between *data-name-2* and *data-name-3*.

The keywords THROUGH and THRU are equivalent.

When the THROUGH phrase is specified:

- *data-name-1* defines an alphanumeric group item that includes all the elementary items that:
 - Start with *data-name-2* if it is an elementary item, or the first elementary item within *data-name-2* if it is a group item
 - End with *data-name-3* if it is an elementary item, or the last elementary item within *data-name-3* if it is an alphanumeric group item or national group item
- The storage area occupied by the starting item through the ending item becomes the storage area occupied by *data-name-1*.

Usage note: The group defined with the THROUGH phrase can include data items of usage NATIONAL.

The leftmost character position in *data-name-3* must not precede the leftmost character position in *data-name-2*, and the rightmost character position in *data-name-3* must not precede the rightmost character position in *data-name-2*. This means that *data-name-3* cannot be totally subordinate to *data-name-2*.

When the THROUGH phrase is not specified:

- The storage area occupied by *data-name-2* becomes the storage area occupied by *data-name-1*.
- All of the data attributes of *data-name-2* become the data attributes for *data-name-1*. That is:
 - When *data-name-2* is an alphanumeric group item, *data-name-1* is an alphanumeric group item.
 - When *data-name-2* is a national group item, *data-name-1* is a national group item.
 - When *data-name-2* is an elementary item, *data-name-1* is an elementary item.

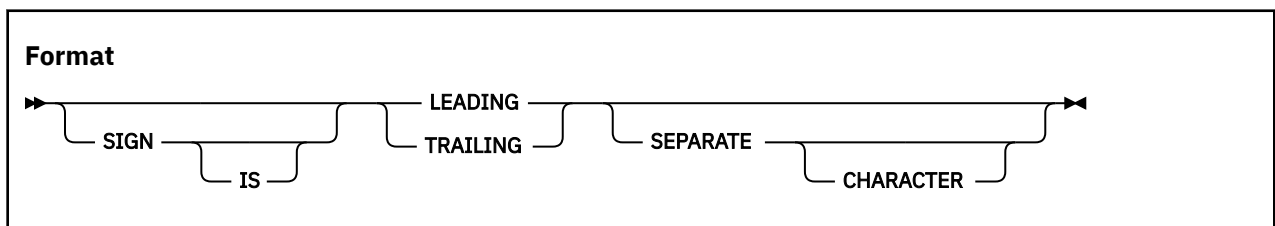
The following figure illustrates valid and invalid RENAMES clause specifications.

COBOL Specifications	Storage Layouts
Example 1 (Valid) 01 RECORD-I. 05 DN-1... . 05 DN-2... . 05 DN-3... . 05 DN-4... . 66 DN-6 RENAMES DN-1 THROUGH DN-3.	
Example 2 (Valid) 01 RECORD-II. 05 DN-1. 10 DN-2... . 10 DN-2A... . 05 DN-1A REDEFINES DN-1. 10 DN-3A... . 10 DN-3... . 10 DN-3B... . 05 DN-5... . 66 DN-6 RENAMES DN-2 THROUGH DN-3.	
Example 3 (Invalid) 01 RECORD-III. 05 DN-2. 10 DN-3... . 10 DN-4... . 05 DN-5... . 66 DN-6 RENAMES DN-2 THROUGH DN-3. DN-6 is indeterminate	
Example 4 (Invalid) 01 RECORD-IV. 05 DN-1. 10 DN-2A... . 10 DN-2B... . 10 DN-2C REDEFINES DN-2B. 15 DN-2CA... . 15 DN-2D... . 05 DN-3... . 66 DN-4 RENAMES DN-1 THROUGH DN-2CA. DN-4 is indeterminate	

SIGN clause

The SIGN clause specifies the position and mode of representation of the operational sign for the signed numeric item to which it applies.

The SIGN clause is required only when an explicit description of the properties or position of the operational sign is necessary.



The SIGN clause can be specified only for the following items:

- An elementary numeric data item of usage DISPLAY or NATIONAL that is described with an S in its picture character string, or
- A group item that contains at least one such elementary entry as a subordinate item

When the SIGN clause is specified at the group level, that SIGN clause applies only to subordinate signed numeric elementary data items of usage DISPLAY or NATIONAL. Such a group can also contain items that are not affected by the SIGN clause. If the SIGN clause is specified for a group or elementary entry that

is subordinate to a group item that has a SIGN clause, the SIGN clause for the subordinate entry takes precedence for that subordinate entry.

The SIGN clause is treated as documentation for external floating-point items.

When the SIGN clause is specified without the SEPARATE phrase, USAGE DISPLAY must be specified explicitly or implicitly. When SIGN IS SEPARATE is specified, either USAGE DISPLAY or USAGE NATIONAL can be specified.

If you specify the CODE-SET clause in an FD entry, any signed numeric data description entries associated with that file description entry must be described with the SIGN IS SEPARATE clause.

If the SEPARATE CHARACTER phrase is not specified, then:

- The operational sign is presumed to be associated with the LEADING or TRAILING digit position, whichever is specified, of the elementary numeric data item. (In this instance, specification of SIGN IS TRAILING is the equivalent of the standard action of the compiler.)
- The character S in the PICTURE character string is not counted in determining the size of the item (in terms of standard data format characters).

If the SEPARATE CHARACTER phrase is specified, then:

- The operational sign is presumed to be the LEADING or TRAILING character position, whichever is specified, of the elementary numeric data item. This character position is not a digit position.
- The character S in the PICTURE character string is counted in determining the size of the data item (in terms of standard data format characters).
- + is the character used for the positive operational sign.
- - is the character used for the negative operational sign.

SYNCHRONIZED clause

The SYNCHRONIZED clause specifies the alignment of an elementary item on a natural boundary in storage.



SYNC is an abbreviation for SYNCHRONIZED and has the same meaning.

The SYNCHRONIZED clause is never required, but can improve performance on some systems for binary items used in arithmetic.

The SYNCHRONIZED clause can be specified for elementary items and for level-01 group items, in which case every elementary item within the group item is synchronized.

LEFT

Specifies that the elementary item is to be positioned so that it will begin at the left character position of the natural boundary in which the elementary item is placed.

RIGHT

Specifies that the elementary item is to be positioned such that it will terminate on the right character position of the natural boundary in which it has been placed.

When specified, the LEFT and the RIGHT phrases are syntax checked but have no effect on the execution of the program.

The length of an elementary item is not affected by the SYNCHRONIZED clause.

The following table lists the effect of the SYNCHRONIZE clause on other language elements.

Table 15. SYNCHRONIZE clause effect on other language elements

Language element	Comments
OCCURS clause	When specified for an item within the scope of an OCCURS clause, each occurrence of the item is synchronized.
USAGE DISPLAY or PACKED-DECIMAL	Each item is syntax checked, but the SYNCHRONIZED clause has no effect on execution.
USAGE NATIONAL	Each item is syntax checked, but the SYNCHRONIZED clause has no effect on execution.
USAGE BINARY or COMPUTATIONAL	When the item is the first elementary item subordinate to an item that contains a REDEFINES clause, the item must not require the addition of unused character positions. When the synchronized clause is not specified for a subordinate data item (one with a level number of 02 through 49): • The item is aligned at a displacement that is a multiple of 2 relative to the beginning of the record if its USAGE is BINARY and its PICTURE is in the range of S9 through S9(4). • The item is aligned at a displacement that is a multiple of 4 relative to the beginning of the record if its USAGE is BINARY and its PICTURE is in the range of S9(5) through S9(18), or its USAGE is INDEX. When SYNCHRONIZED is not specified for binary items, no space is reserved for slack bytes.
USAGE POINTER, PROCEDURE-POINTER, FUNCTION-POINTER, OBJECT REFERENCE, INDEX	The data is aligned on a fullword boundary when the LP(32) compiler option is in effect. The data is aligned on a doubleword boundary when the LP(64) compiler option is in effect.
USAGE COMPUTATIONAL-1	The data is aligned on a fullword boundary.
USAGE COMPUTATIONAL-2	The data is aligned on a doubleword boundary.
USAGE COMPUTATIONAL-3	The data is treated the same as the SYNCHRONIZED clause for a PACKED-DECIMAL item.
USAGE COMPUTATIONAL-4	The data is treated the same as the SYNCHRONIZED clause for a COMPUTATIONAL item.
USAGE COMPUTATIONAL-5	The data is treated the same as the SYNCHRONIZED clause for a COMPUTATIONAL item.
USAGE UTF-8	Each item is syntax checked, but the SYNCHRONIZED clause has no effect on execution.
DBCS and external floating-point items	Each item is syntax checked, but the SYNCHRONIZED clause has no effect on execution.

Table 15. **SYNCHRONIZE** clause effect on other language elements (continued)

Language element	Comments
REDEFINES clause	For an item that contains a REDEFINES clause, the data item that is redefined must have the proper boundary alignment for the data item that redefines it. For example, if you write the following, be sure that data item A begins on a fullword boundary: <pre> 02 A PICTURE X(4) . 02 B REDEFINES A PICTURE S9(9) BINARY SYNC . </pre>

In the FILE SECTION, the compiler assumes that all level-01 records that contain SYNCHRONIZED items are aligned on doubleword boundaries in the buffer. You must provide the necessary slack bytes between records to ensure alignment when there are multiple records in a block.

In the WORKING-STORAGE SECTION, the compiler aligns all level-01 entries on a doubleword boundary.

For the purposes of aligning binary items in the LINKAGE SECTION, all level-01 items are assumed to begin on doubleword boundaries. Therefore, if you issue a CALL statement, such operands of any USING phrase within it must be aligned correspondingly.

Slack bytes

There are two types of slack bytes.

- Slack bytes *within* records: unused character positions that precede each synchronized item in the record
- Slack bytes *between* records: unused character positions added between blocked logical records

If the RULES=(NOSLACKBYTES) option is in effect, warning messages are issued for any SYNCHRONIZED data items that cause the compiler to add slack bytes, either slack bytes within records or slack bytes between records. For details about the RULES option, see *RULES* in the *Enterprise COBOL Programming Guide*.

Slack bytes within records

For any data description that has binary items that are not on their natural boundaries, the compiler inserts slack bytes within a record to ensure that all SYNCHRONIZED items are on their proper boundaries.

Because it is important that you know the length of the records in a file, you need to determine whether slack bytes are required and, if so, how many bytes the compiler will add. The algorithm that the compiler uses is as follows:

- The total number of bytes occupied by all elementary data items that precede the binary item are added together, including any slack bytes that are previously added.
- This sum is divided by m , where:
 - $m = 2$ for binary items of four-digit length or less
 - $m = 4$ for binary items of five-digit length or more and for COMPUTATIONAL-1 data items
 - $m = 4$ for data items described with USAGE INDEX, USAGE POINTER, USAGE PROCEDURE-POINTER, USAGE OBJECT REFERENCE, or USAGE FUNCTION-POINTER
 - $m = 8$ for COMPUTATIONAL-2 data items
- If the remainder (r) of this division is equal to zero, no slack bytes are required. If the remainder is not equal to zero, the number of slack bytes that must be added is equal to $m - r$.

These slack bytes are added to each record immediately following the elementary data item that precedes the binary item. They are defined as if they constitute an item with a level-number equal to

that of the elementary item that immediately precedes the SYNCHRONIZED binary item, and are included in the size of the group that contains them.

For example:

```

01 FIELD-A.
  05 FIELD-B                PICTURE X(5) .
  05 FIELD-C.
    10 FIELD-D              PICTURE XX.
    [10 SLACK-BYTES         PICTURE X.  INSERTED BY COMPILER]
    10 FIELD-E COMPUTATIONAL PICTURE S9(6) SYNC.
01 FIELD-L.
  05 FIELD-M                PICTURE X(5) .
  05 FIELD-N                PICTURE XX.
  [05 SLACK-BYTES           PICTURE X.  INSERTED BY COMPILER]
  05 FIELD-O.
    10 FIELD-P COMPUTATIONAL PICTURE S9(6) SYNC.

```

Slack bytes can also be added by the compiler when a group item is defined with an OCCURS clause and contains within it a SYNCHRONIZED binary data item. To determine whether slack bytes are to be added, the following action is taken:

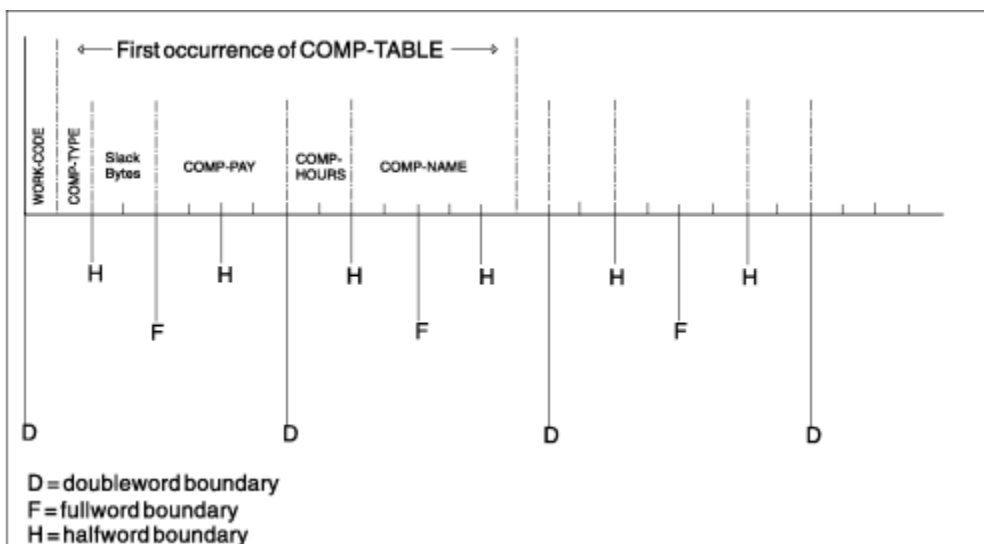
- The compiler calculates the size of the group, including all the necessary slack bytes within a record.
- This sum is divided by the largest m required by any elementary item within the group.
- If r is equal to zero, no slack bytes are required. If r is not equal to zero, $m - r$ slack bytes must be added.

The slack bytes are inserted at the end of each occurrence of the group item that contains the OCCURS clause. For example, a record defined as follows appears in storage, as shown, in the figure after the record:

```

01 WORK-RECORD.
  05 WORK-CODE              PICTURE X.
  05 COMP-TABLE OCCURS 10 TIMES.
    10 COMP-TYPE            PICTURE X.
    [10 SLACK-BYTES         PIC XX.  INSERTED BY COMPILER]
    10 COMP-PAY             PICTURE S9(4)V99 COMP SYNC.
    10 COMP-HOURS           PICTURE S9(3) COMP SYNC.
    10 COMP-NAME            PICTURE X(5).

```



In order to align COMP-PAY and COMP-HOURS on their proper boundaries, the compiler added 2 slack bytes within the record.

In the previous example, without further adjustment, the second occurrence of COMP-TABLE would begin 1 byte before a doubleword boundary, and the alignment of COMP-PAY and COMP-HOURS would not be

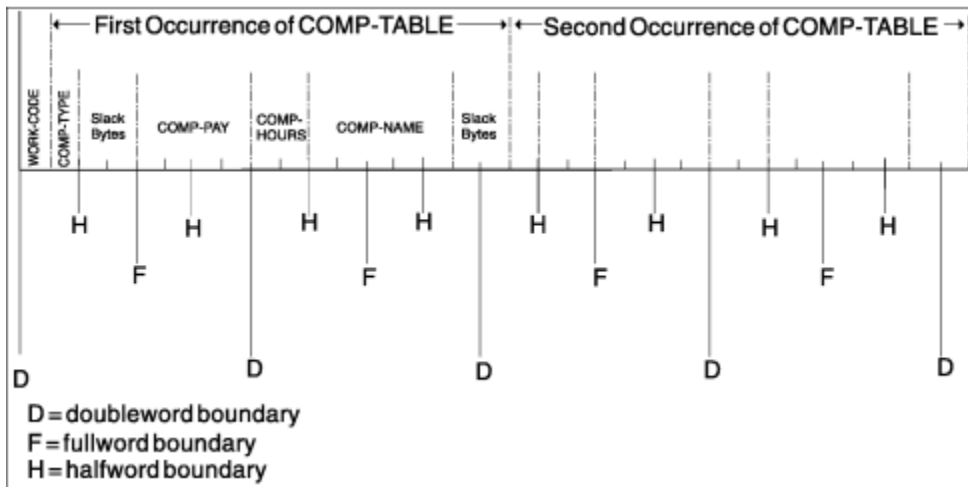
valid for any occurrence of the table after the first. Therefore, the compiler must add slack bytes at the end of the group, as though the record had been written as follows:

```

01  WORK-RECORD.
   05  WORK-CODE          PICTURE X.
   05  COMP-TABLE OCCURS 10 TIMES.
       10  COMP-TYPE      PICTURE X.
       [10  SLACK-BYTES    PIC XX.  INSERTED BY COMPILER]
       10  COMP-PAY       PICTURE S9(4)V99 COMP SYNC.
       10  COMP-HOURS     PICTURE S9(3)  COMP SYNC.
       10  COMP-NAME      PICTURE X(5).
       [10  SLACK-BYTES    PIC XX.  INSERTED BY COMPILER]

```

In this example, the second and each succeeding occurrence of COMP-TABLE begins 1 byte beyond a doubleword boundary. The storage layout for the first occurrence of COMP-TABLE now appears as shown in the following figure:



Each succeeding occurrence within the table will now begin at the same relative position as the first.

Slack bytes between records

If the file contains blocked logical records that are to be processed in a buffer, and any of the records contain binary entries for which the SYNCHRONIZED clause is specified, you can improve performance by adding any needed slack bytes between records for proper alignment.

The lengths of all the elementary data items in the record, including all slack bytes, are added. (For variable-length records, it is necessary to add an additional 4 bytes for the count field.) The total is then divided by the highest value of m for any one of the elementary items in the record.

If r (the remainder) is equal to zero, no slack bytes are required. If r is not equal to zero, $m - r$ slack bytes are required. These slack bytes can be specified by writing a level-02 FILLER at the end of the record.

Consider the following record description:

```

01  COMP-RECORD.
   05  A-1    PICTURE X(5).
   05  A-2    PICTURE X(3).
   05  A-3    PICTURE X(3).
   05  B-1    PICTURE S9999  USAGE COMP SYNCHRONIZED.
   05  B-2    PICTURE S99999 USAGE COMP SYNCHRONIZED.
   05  B-3    PICTURE S9999  USAGE COMP SYNCHRONIZED.

```

The number of bytes in A-1, A-2, and A-3 totals 11. B-1 is a four-digit COMPUTATIONAL item and 1 slack byte must therefore be added before B-1. With this byte added, the number of bytes that precede B-2 totals 14. Because B-2 is a COMPUTATIONAL item of five digits in length, 2 slack bytes must be added before it. No slack bytes are needed before B-3.

The revised record description entry now appears as:

```
01  COMP-RECORD.
    05  A-1          PICTURE X(5).
    05  A-2          PICTURE X(3).
    05  A-3          PICTURE X(3).
    [05  SLACK-BYTE-1 PICTURE X.  INSERTED BY COMPILER]
    05  B-1          PICTURE S9999 USAGE COMP SYNCHRONIZED.
    [05  SLACK-BYTE-2 PICTURE XX. INSERTED BY COMPILER]
    05  B-2          PICTURE S99999 USAGE COMP SYNCHRONIZED.
    05  B-3          PICTURE S9999  USAGE COMP SYNCHRONIZED.
```

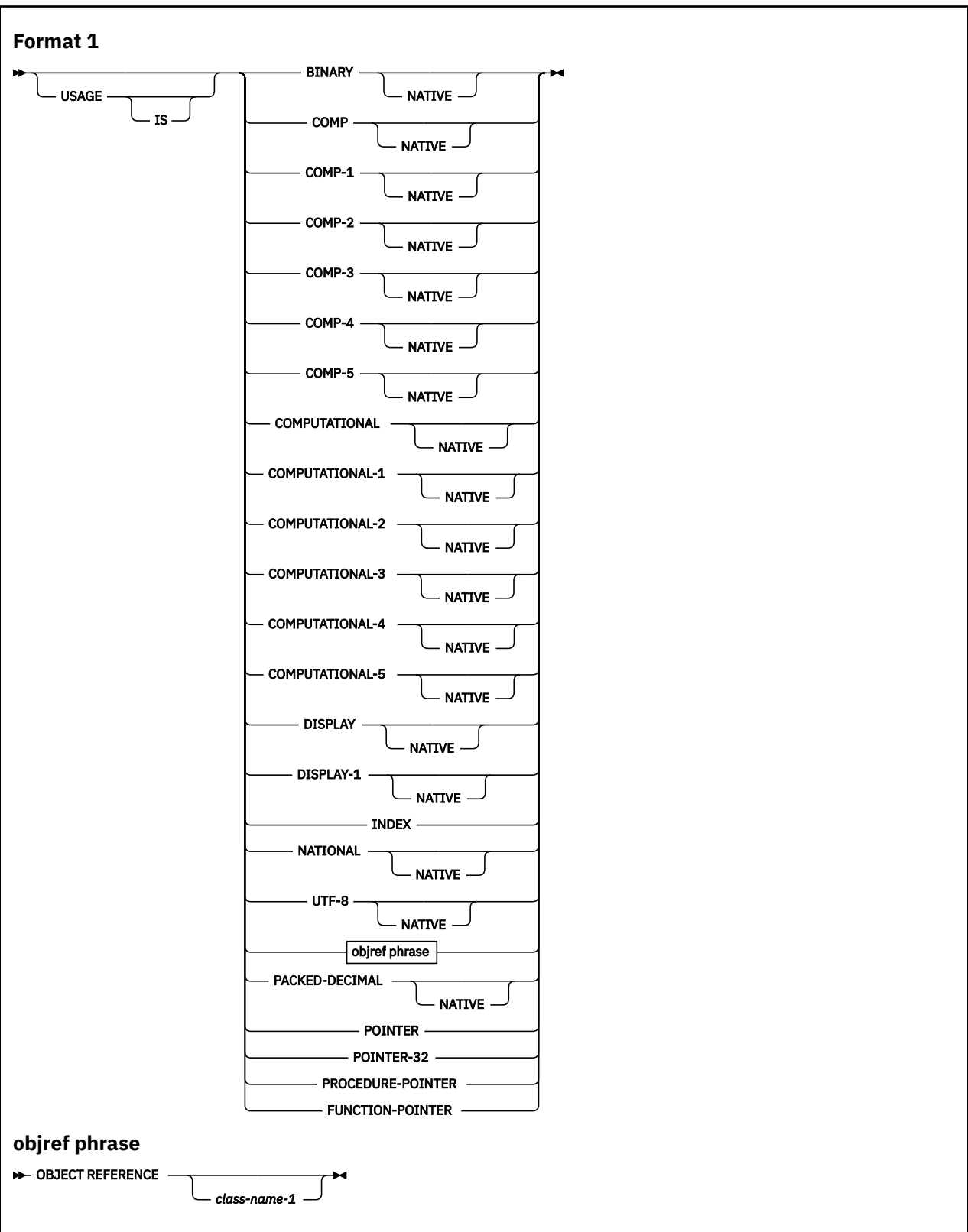
There is a total of 22 bytes in COMP-RECORD, but from the rules mentioned earlier, it appears that $m = 4$ and $r = 2$. Therefore, to attain proper alignment for blocked records, you must add 2 slack bytes at the end of the record.

The final record description entry appears as:

```
01  COMP-RECORD.
    05  A-1          PICTURE X(5).
    05  A-2          PICTURE X(3).
    05  A-3          PICTURE X(3).
    [05  SLACK-BYTE-1 PICTURE X.  INSERTED BY COMPILER]
    05  B-1          PICTURE S9999 USAGE COMP SYNCHRONIZED.
    [05  SLACK-BYTE-2 PICTURE XX. INSERTED BY COMPILER]
    05  B-2          PICTURE S99999 USAGE COMP SYNCHRONIZED.
    05  B-3          PICTURE S9999  USAGE COMP SYNCHRONIZED.
    05  FILLER       PICTURE XX.  [SLACK BYTES YOU ADD]
```


USAGE clause

The USAGE clause specifies the format in which data is represented in storage.



Note: NATIVE is treated as a comment in all phrases for which NATIVE is shown in the USAGE clause.

The USAGE clause can be specified for a data description entry with any level-number other than 66 or 88.

When specified at the group level, the USAGE clause applies to each elementary item in the group. The usage of elementary items must not contradict the usage of a group to which the elementary items belongs.

A USAGE clause must not be specified in a group level entry for which a GROUP-USAGE NATIONAL clause is specified.

When a GROUP-USAGE NATIONAL clause is specified or implied for a group level entry, USAGE NATIONAL must be specified or implied for every elementary item within the group. For details, see [“GROUP-USAGE clause”](#) on page 198.

When the USAGE clause is not specified at either the group or elementary level, a usage clause is implied with:

- Usage DISPLAY when the PICTURE clause contains only symbols other than G or N
- Usage NATIONAL when the PICTURE clause contains only one or more of the symbol N and the NSYMBOL(NATIONAL) compiler option is in effect
- Usage DISPLAY-1 when the PICTURE clause contains one or more of the symbol N and the NSYMBOL(DBCS) compiler option is in effect

Computational items

A computational item is a value used in arithmetic operations. It must be numeric. If a group item is described with a computational usage, the elementary items within the group have that usage.

The maximum length of a computational item is 18 decimal digits, except for a PACKED-DECIMAL item. If the ARITH(COMPAT) compiler option is in effect, then the maximum length of a PACKED-DECIMAL item is 18 decimal digits. If the ARITH(EXTEND) compiler option is in effect, then the maximum length of a PACKED-DECIMAL item is 31 decimal digits.

The PICTURE of a computational item can contain only:

- 9**
One or more numeric character positions
- S**
One operational sign
- V**
One implied decimal point
- P**
One or more decimal scaling positions

COMPUTATIONAL-1 and COMPUTATIONAL-2 items (internal floating-point) cannot have PICTURE strings.

BINARY

Specified for binary data items. Such items have a decimal equivalent consisting of the decimal digits 0 through 9, plus a sign. Negative numbers are represented as the two's complement of the positive number with the same absolute value.

The amount of storage occupied by a binary item depends on the number of decimal digits defined in its PICTURE clause:

Digits in PICTURE clause	Storage occupied
1 through 4	2 bytes (halfword)
5 through 9	4 bytes (fullword)
10 through 18	8 bytes (doubleword)

Binary data is *big-endian*: the operational sign is contained in the leftmost bit.

BINARY, COMPUTATIONAL, and COMPUTATIONAL-4 data items can be affected by the TRUNC compiler option. For information about the effect of this compiler option, see *TRUNC* in the *Enterprise COBOL Programming Guide*.

PACKED-DECIMAL

Specified for internal decimal items. Such an item appears in storage in packed decimal format. There are two digits for each character position, except for the trailing character position, which is occupied by the low-order digit and the sign. Such an item can contain any of the digits 0 through 9, plus a sign, representing a value not exceeding 18 decimal digits, unless the ARTIH(EXTEND) compiler option is in effect, in which case up to 31 digits might be represented.

The sign representation uses the same bit configuration as the 4-bit sign representation in zoned decimal fields. For details, see *Sign representation of zoned and packed-decimal data* in the *Enterprise COBOL Programming Guide*.

The most efficient use of packed-decimal data items is to define them with an odd number of digits, to use all of the bits. This can also result in more efficient generated code. To find out if you have packed-decimal data items with an even number of digits in your program, you can use the RULES(NOEVENPACK) compiler option. For details, see *RULES* in the *Enterprise COBOL Programming Guide*.

COMPUTATIONAL or COMP (binary)

This is the equivalent of BINARY. The COMPUTATIONAL phrase is synonymous with BINARY.

COMPUTATIONAL-1 or COMP-1 (floating-point)

Specified for internal floating-point items (single precision). COMP-1 items are 4 bytes long.

COMPUTATIONAL-2 or COMP-2 (long floating-point)

Specified for internal floating-point items (double precision). COMP-2 items are 8 bytes long.

COMPUTATIONAL-3 or COMP-3 (internal decimal)

This is the equivalent of PACKED-DECIMAL.

COMPUTATIONAL-4 or COMP-4 (binary)

This is the equivalent of BINARY.

COMPUTATIONAL-5 or COMP-5 (native binary)

These data items are represented in storage as binary data. The data items can contain values up to the capacity of the native binary representation (2, 4, or 8 bytes), rather than being limited to the value implied by the number of nines in the picture for the item (as is the case for USAGE BINARY data). When numeric data is moved or stored into a COMP-5 item, truncation occurs at the binary field size rather than at the COBOL picture size limit. When a COMP-5 item is referenced, the full binary field size is used in the operation.

When a usage COMP-5 binary sender is involved in an operation with a receiver that is not numeric, the sender is truncated to the number of digits in its PICTURE clause.

The TRUNC(BIN) compiler option causes all binary data items (USAGE BINARY, COMP, COMP-4) to be handled as if they were declared USAGE COMP-5.

The following table shows several picture character strings, the resulting storage representation, and the range of values for data items described with USAGE COMP-5.

Picture	Storage representation	Numeric values
S9(1) through S9(4)	Binary halfword (2 bytes)	-32768 through +32767
S9(5) through S9(9)	Binary fullword (4 bytes)	-2,147,483,648 through +2,147,483,647
S9(10) through S9(18)	Binary doubleword (8 bytes)	-9,223,372,036,854,775,808 through +9,223,372,036,854,775,807
9(1) through 9(4)	Binary halfword (2 bytes)	0 through 65535

Picture	Storage representation	Numeric values
9(5) through 9(9)	Binary fullword (4 bytes)	0 through 4,294,967,295
9(10) through 9(18)	Binary doubleword (8 bytes)	0 through 18,446,744,073,709,551,615

The picture for a COMP-5 data item can specify a scaling factor (that is, decimal positions or implied integer positions). In this case, the maximal capacities listed in the aforementioned table must be scaled appropriately. For example, a data item described with PICTURE S99V99 COMP-5 is represented in storage as a binary halfword, and supports a range of values from -327.68 to +327.67.

USAGE NOTE: When the ON SIZE ERROR phrase is used on an arithmetic statement and a receiver is defined with USAGE COMP-5, the maximum value that the receiver can contain is the value implied by the item's decimal PICTURE character-string. Any attempt to store a value larger than this maximum will result in a size error condition.

DISPLAY phrase

The data item is stored in character form, one character for each 8-bit byte. This corresponds to the format used for printed output. DISPLAY can be explicit or implicit.

USAGE IS DISPLAY is valid for the following types of items:

- Alphabetic
- Alphanumeric
- Alphanumeric-edited
- Numeric-edited
- External floating-point
- External decimal

Alphabetic, alphanumeric, alphanumeric-edited, and numeric-edited items are discussed in [“Data categories and PICTURE rules” on page 212](#).

External decimal items with USAGE DISPLAY are sometimes referred to as *zoned decimal* items. Each digit of a number is represented by a single byte. The 4 high-order bits of each byte are zone bits; the 4 high-order bits of the low-order byte represent the sign of the item. The 4 low-order bits of each byte contain the value of the digit.

If the ARITH(COMPAT) compiler option is in effect, then the maximum length of an external decimal item is 18 digits. If the ARITH(EXTEND) compiler option is in effect, then the maximum length of an external decimal item is 31 digits.

The PICTURE character-string of an external decimal item can contain only:

- One or more of the symbol 9
- The operational-sign, S
- The assumed decimal point, V
- One or more of the symbol P

DISPLAY-1 phrase

The DISPLAY-1 phrase defines an item as DBCS. The data item is stored in character form, with each character occupying 2 bytes of storage.

FUNCTION-POINTER phrase

The FUNCTION-POINTER phrase defines an item as a *function-pointer data item*. A function-pointer data item can contain the address of a descriptor for a procedure entry point.

A function-pointer is a 4-byte elementary item or an 8-byte elementary item depending on whether the LP(32) or LP(64) is in effect. If LP(32) is in effect, 4 bytes are allocated for the item; otherwise, 8 bytes are allocated for the item. Function-pointers have the same capabilities as procedure-pointers. Function-pointers are thus more easily interoperable with C function pointers.

A function-pointer can point to a function descriptor for one of the following or can contain NULL:

- The primary entry point of a COBOL program, defined by the PROGRAM-ID paragraph of the outermost program
- An alternate entry point of a COBOL program, defined by a COBOL ENTRY statement
- An entry point in a non-COBOL program

A VALUE clause for a function-pointer data item can contain only NULL or NULLS.

The GLOBAL, EXTERNAL, OCCURS and VOLATILE clauses can be used with USAGE IS FUNCTION-POINTER.

A function-pointer can be used in the same contexts as a procedure-pointer, as defined in [“PROCEDURE-POINTER phrase”](#) on page 244.

INDEX phrase

A data item defined with the INDEX phrase is an *index data item*.

An *index data item* is a 4-byte elementary item or an 8-byte elementary item depending on whether the LP(32) or LP(64) compiler option is in effect, that can be used to save index-name values for future reference. An *index data item* is not necessarily connected with any specific table. Through a SET statement, an index data item can be assigned an index-name value. Such a value corresponds to the occurrence number in a table.

Direct references to an index data item can be made only in a SEARCH statement, a SET statement, a relation condition, the USING phrase of the PROCEDURE DIVISION header, or the USING phrase of the CALL or ENTRY statement.

An index data item can be part of an alphanumeric group item that is referenced in a MOVE statement or an input/output statement.

An index data item saves values that represent table occurrences, yet is not necessarily defined as part of any table. There is no conversion of values when an index data item is referenced in the following circumstances:

- directly in a SEARCH or SET statement
- indirectly in a MOVE statement
- indirectly in an input or output statement

An index data item cannot be a conditional variable.

The JUSTIFIED, PICTURE, BLANK WHEN ZERO, or VALUE clauses cannot be used to describe a group item or elementary items described with the USAGE IS INDEX clause.

SYNCHRONIZED can be used with USAGE IS INDEX to obtain efficient use of the index data item.

NATIONAL phrase

The NATIONAL phrase defines an item whose content is represented in storage in UTF-16 (CCSID 1200). The class and category of the data item depend on the picture symbols that are specified in the associated PICTURE clause.

OBJECT REFERENCE phrase

A data item defined with the OBJECT REFERENCE phrase is an *object reference*. An object reference data item is a 4-byte elementary item or an 8-byte elementary item depending on whether the LP(32) or LP(64) is in effect. If LP(32) is in effect, 4 bytes are allocated for the item; otherwise, 8 bytes are allocated for the item.

class-name-1

An optional class name.

You must define *class-name-1* in the REPOSITORY paragraph in the configuration section of the containing class or outermost program.

If specified, *class-name-1* indicates that *data-name-1* always refers to an object-instance of class *class-name-1* or a class derived from *class-name-1*.

Important: The programmer must ensure that the referenced object meets this requirement; violations are not diagnosed.

If *class-name-1* is not specified, the object reference can refer to an object of any class. In this case, *data-name-1* is a *universal* object reference.

You can specify *data-name-1* within an alphanumeric group item without affecting the semantics of the group item. There is no conversion of values or other special handling of the object references when statements are executed that operate on the group. The group continues to behave as an alphanumeric group item.

An object reference can be defined in any section of the DATA DIVISION of a factory definition, object definition, method, or program. An object-reference data item can be used in only:

- A SET statement (format 7 only)
- A relation condition
- An INVOKE statement
- The USING or RETURNING phrase of an INVOKE statement
- The USING or RETURNING phrase of a CALL statement
- A program procedure division or ENTRY statement USING or RETURNING phrase
- A method procedure division USING or RETURNING phrase

Object-reference data items:

- Are ignored in CORRESPONDING operations
- Are unaffected by INITIALIZE statements
- Can be the subject or object of a REDEFINES clause
- Cannot be a conditional variable
- Can be written to a file (but upon subsequent reading of the record the content of the object reference is undefined)

A VALUE clause for an object-reference data item can contain only NULL or NULLS.

You can use the SYNCHRONIZED clause with the USAGE OBJECT REFERENCE clause to obtain efficient alignment of the object-reference data item.

The JUSTIFIED, PICTURE, and BLANK WHEN ZERO clauses cannot be used to describe group or elementary items defined with the USAGE OBJECT REFERENCE clause.

POINTER phrase

A data item defined with USAGE IS POINTER is a *pointer data item* or *data-pointer*. A pointer data item is a 4-byte elementary item or an 8-byte elementary item depending on whether the LP(32) or LP(64) is

in effect. If LP(32) is in effect, 4 bytes are allocated for the item; otherwise, 8 bytes are allocated for the item.

You can use pointer data items to accomplish limited base addressing. Pointer data items can be compared for equality or moved to other pointer items.

A pointer data item can be used only:

- In an ALLOCATE statement
- In a FREE statement
- In a SET statement (format 5 only)
- In a relation condition
- In the USING phrase of a CALL statement, an ENTRY statement, or the PROCEDURE DIVISION header

A POINTER data item can be set to a POINTER-32 data item, and vice versa. When the LP(32) compiler option is in effect, USAGE POINTER and USAGE POINTER-32 are synonyms. When the LP(64) compiler option is in effect, the following statements apply:

Note: The size of the POINTER data item is 8 bytes in the following cases.

- A POINTER data item can be SET to a value from a POINTER-32 data item. The high-order word of the POINTER data item is cleared to zero.
- Pointer data items can be part of an alphanumeric group that is referred to in a MOVE statement or an input/output statement. However, if a pointer data item is part of a group, there is no conversion of values when the statement is executed.
- A pointer data item can be the subject or object of a REDEFINES clause.
- SYNCHRONIZED can be used with USAGE IS POINTER to obtain efficient use of the pointer data item.
- A VALUE clause for a pointer data item can contain only NULL or NULLS.
- A pointer data item cannot be a conditional variable.
- A pointer data item does not belong to any class or category.

The following table lists clauses that can or cannot be used to describe group or elementary items defined with the USAGE IS POINTER.

Table 16. Clauses that can or cannot be used with USAGE IS POINTER	
Can be used with USAGE IS POINTER	Cannot be used with USAGE IS POINTER
GLOBAL clause EXTERNAL clause OCCURS clause VOLATILE clause	JUSTIFIED clause PICTURE clause BLANK WHEN ZERO clause

Pointer data items are ignored in the processing of a CORRESPONDING phrase.

A pointer data item can be written to a data set, but upon subsequent reading of the record that contains the pointer, the address contained might no longer represent a valid pointer.

USAGE IS POINTER is implicitly specified for the ADDRESS OF special register. For more information, see *Using tables (arrays) and pointers* in the *Enterprise COBOL Programming Guide*.

POINTER-32 phrase

A data item defined with USAGE IS POINTER-32 is a *pointer data item* or *data-pointer*. A pointer data item is a 4-byte elementary item regardless of the LP compiler option setting.

You can use POINTER-32 data items to accomplish limited base addressing. Pointer data items can be compared for equality or moved to other pointer items.

A POINTER-32 data item can be used only:

- In an ALLOCATE statement
- In a FREE statement
- In a SET statement (format 5 only)
- In a relation condition
- In the USING phrase of a CALL statement, an ENTRY statement, or the PROCEDURE DIVISION header

A POINTER-32 data item can be set to a POINTER data item, and vice versa. When the LP(32) compiler option is in effect, USAGE POINTER and USAGE POINTER-32 are synonyms. When the LP(64) compiler option is in effect, the following statements apply:

- A POINTER-32 data item can be SET to a value from a POINTER data item. Only the low-order word of the POINTER data item is used in this case. It is a programming error if the high-order word of the 64-bit pointer data item is not zero.
- A POINTER-32 data item can be compared with a POINTER data item in a relation condition. The bit representation in the POINTER-32 data item is extended to 64-bit if the high-order word is zero before the comparison.
- POINTER-32 data items can be part of an alphanumeric group that is referred to in a MOVE statement or an input/output statement. However, if a pointer data item is part of a group, there is no conversion of values when the statement is executed.
- A POINTER-32 data item can be the subject or object of a REDEFINES clause.
- SYNCHRONIZED can be used with USAGE IS POINTER-32 to obtain efficient use of the pointer data item.
- A VALUE clause for a POINTER-32 data item can contain only NULL or NULLS.
- A POINTER-32 data item cannot be a conditional variable.
- A POINTER-32 data item does not belong to any class or category.

The following table lists clauses that can or cannot be used to describe group or elementary items defined with the USAGE IS POINTER-32.

Table 17. Clauses that can or cannot be used with USAGE IS POINTER-32	
Can be used with USAGE IS POINTER-32	Cannot be used with USAGE IS POINTER
GLOBAL clause EXTERNAL clause OCCURS clause VOLATILE clause	JUSTIFIED clause PICTURE clause BLANK WHEN ZERO clause

POINTER-32 data items are ignored in the processing of a CORRESPONDING phrase.

A POINTER-32 data item can be written to a data set, but upon the subsequent reading of the record that contains the pointer, the address contained might no longer represent a valid pointer.

PROCEDURE-POINTER phrase

The PROCEDURE-POINTER phrase defines an item as a *procedure-pointer data item*. A procedure-pointer data item can contain the address of a descriptor for a procedure entry point.

A procedure-pointer data item is an 8-byte elementary item regardless of the LP compiler option setting.

A procedure-pointer can point to a function descriptor for one of the following or can contain NULL:

- The primary entry point of a COBOL program as defined by the program-ID paragraph of the outermost program of a compilation unit
- An alternate entry point of a COBOL program as defined by a COBOL ENTRY statement
- An entry point in a non-COBOL program

A procedure-pointer data item can be used only:

- In a SET statement (format 6 only)
- In a CALL statement (from a high-level language or LE-conforming assembler program)
- In a relation condition
- In the USING phrase of an ENTRY statement or the PROCEDURE DIVISION header

Procedure-pointer data items can be compared for equality or moved to other procedure-pointer data items.

Procedure-pointer data items can be part of a group that is referred to in a MOVE statement or an input/output statement. However, there is no conversion of values when the statement is executed. If a procedure-pointer data item is written to a data set, subsequent reading of the record that contains the procedure-pointer can result in an invalid value in the procedure-pointer.

A procedure-pointer data item can be the subject or object of a REDEFINES clause.

SYNCHRONIZED can be used with USAGE IS PROCEDURE-POINTER to obtain efficient alignment of the procedure-pointer data item.

The GLOBAL, EXTERNAL, OCCURS, and VOLATILE clauses can be used with USAGE IS PROCEDURE-POINTER.

A VALUE clause for a procedure-pointer data item can contain only NULL or NULLS.

The JUSTIFIED, PICTURE, and BLANK WHEN ZERO clauses cannot be used to describe group or elementary items defined with the USAGE IS PROCEDURE-POINTER clause.

A procedure-pointer data item cannot be a conditional variable.

A procedure-pointer data item does not belong to any class or category.

Procedure-pointer data items are ignored in CORRESPONDING operations.

NATIVE phrase

The NATIVE phrase is syntax checked, but has no effect on the execution of the program.

UTF-8 phrase

The UTF-8 phrase defines an item whose content is represented in storage in UTF-8 (CCSID 1208). The class and category of the data item are UTF-8.

VALUE clause

The VALUE clause specifies the initial contents of a data item or the values associated with a condition-name. The use of the VALUE clause differs depending on the DATA DIVISION section in which it is specified.

In the WORKING-STORAGE SECTION and the LOCAL-STORAGE SECTION, the VALUE clause can be used in condition-name entries or in specifying the initial value of any data item. The data item assumes the specified value at the beginning of program execution. If the initial value is not explicitly specified, the value is unpredictable.

Format 1

Format 1 specifies the initial value of a data item. Initialization is independent of any BLANK WHEN ZERO or JUSTIFIED clause that is specified.

Format 1: literal value

➡ VALUE literal ➡
 IS

A format-1 VALUE clause specified in a data description entry that contains or is subordinate to an OCCURS clause causes every occurrence of the associated data item to be assigned the specified value. Each structure that contains the DEPENDING ON phrase of the OCCURS clause is assumed to contain the maximum number of occurrences for the purposes of VALUE initialization.

The VALUE clause must not be specified for a data description entry that contains or is subordinate to an entry that contains either an EXTERNAL or a REDEFINES clause. This rule does not apply to condition-name entries.

A format-1 VALUE clause can be specified for an elementary data item or for a group item. When the VALUE clause is specified at the group level, the group area is initialized without consideration for the subordinate entries within the group. In addition, a VALUE clause must not be specified for subordinate entries within the group.

For group items, the VALUE clause must not be specified if any subordinate entries contain a JUSTIFIED or SYNCHRONIZED clause.

If the VALUE clause is specified for an alphanumeric group, all subordinate items must be explicitly or implicitly described with USAGE DISPLAY.

The VALUE clause must not conflict with other clauses in the data description entry or in the data description of that entry's hierarchy.

The functions of the editing characters in a PICTURE clause are ignored in determining the initial value of the item described. However, editing characters are included in determining the size of the item. Therefore, any editing characters must be included in the literal. For example, if the item is defined as PICTURE +999.99 and the value is to be +12.34, then the VALUE clause should be specified as VALUE "+012.34".

A VALUE clause cannot be specified for external floating-point items.

A data item cannot contain a VALUE clause if the prior data item contains an OCCURS clause with the DEPENDING ON phrase.

Rules for literal values

- Wherever a literal is specified, a figurative constant can be substituted, in accordance with the rules specified in [“Figurative constants” on page 15](#).
- If the item is class numeric, the VALUE clause literal must be numeric. If the literal defines the value of a WORKING-STORAGE item or LOCAL-STORAGE item, the literal is aligned according to the rules for numeric moves, with one additional restriction: The literal must not have a value that requires truncation of nonzero digits. If the literal is signed, the associated PICTURE character-string must contain a sign symbol.
- With some exceptions, numeric literals in a VALUE clause must have a value within the range of values indicated by the PICTURE clause for the item. For example, for PICTURE 99PPP, the literal must be zero or within the range 1000 through 99000. For PICTURE PPP99, the literal must be within the range 0.00000 through 0.00099.

The exceptions are the following ones:

- Data items described with usage COMP-5 that do not have a picture symbol P in their PICTURE clause.
- When the TRUNC(BIN) compiler option is in effect, data items described with usage BINARY, COMP, or COMP-4 that do not have a picture symbol P in their PICTURE clause.

A VALUE clause for these items can have a value up to the capacity of the native binary representation.

- If the VALUE clause is specified for an elementary alphabetic, alphanumeric, alphanumeric-edited, or numeric-edited item described with usage DISPLAY, the VALUE clause literal must be an alphanumeric literal or a figurative constant. The literal is aligned according to the alphanumeric alignment rules, with one additional restriction: the number of characters in the literal must not exceed the size of the item.

- If the VALUE clause is specified for an elementary national, national-edited, or numeric-edited item described with usage NATIONAL, the VALUE clause literal must be a national or alphanumeric literal or a figurative constant as specified in [“Figurative constants” on page 15](#). The value of an alphanumeric literal is converted from its source code representation to UTF-16 representation. The literal is aligned according to the national alignment rules, with one additional restriction: the number of characters in the literal must not exceed the size, in character positions, of the item.
- If the VALUE clause is specified for an elementary UTF-8 item, the VALUE clause literal must be a UTF-8 or an alphanumeric literal or a figurative constant as specified in [“Figurative constants” on page 15](#). The value of an alphanumeric or national literal is converted from its source code representation to UTF-8 representation. The literal is aligned according to the UTF-8 alignment rules, with one additional restriction: if the UTF-8 item is a fixed character-length UTF-8 item (i.e., it was not defined with the BYTE-LENGTH phrase of the PICTURE clause or the DYNAMIC LENGTH clause), then the number of characters in the literal must not exceed the size, in character positions, of the item. Otherwise, the number of bytes of the literal must not exceed the maximum byte length allowed by the item.
- If the VALUE clause is specified at the group level for an alphanumeric group, the literal must be an alphanumeric literal or a figurative constant as specified in [“Figurative constants” on page 15](#), other than ALL *national-literal* or ALL *utf-8-literal*. The size of the literal must not exceed the size of the group item.
- If the VALUE clause is specified at the group level for a national group, the literal can be an alphanumeric literal, a national literal, or one of the figurative constants ZERO, SPACE, QUOTES, HIGH-VALUE, LOW-VALUE, *symbolic character*, ALL *national-literal*, or ALL *-literal*. The value of an alphanumeric literal is converted from its source code representation to UTF-16 representation. Each figurative constant represents a national character value. The size of the literal must not exceed the size of the group item.
- If the VALUE clause is specified at the group level for a UTF-8 group, the literal can be an alphanumeric literal, a UTF-8 literal, or one of the figurative constants ZERO, SPACE, QUOTES, HIGH-VALUE, LOW-VALUE, *symbolic character*, ALL *utf-8-literal*, or ALL *-literal*. The value of an alphanumeric literal is converted from its source code representation to UTF-8 representation. Each figurative constant represents a UTF-8 character value. The size of the literal must not exceed the size of the group item.
- A VALUE clause associated with a DBCS item must contain a DBCS literal, the figurative constant SPACE, or the figurative constant ALL *DBCS-literal*. The length of the literal must not exceed the size indicated by the data item's PICTURE clause.
- A VALUE clause that specifies a national literal can be associated only with a data item of class national.
- A VALUE clause that specifies a UTF-8 literal can be associated only with a data item of class UTF-8.
- A VALUE clause that specifies a DBCS literal can be associated only with a data item of class DBCS.
- A VALUE clause associated with a COMPUTATIONAL-1 or COMPUTATIONAL-2 (internal floating-point) item must specify a floating-point literal. In addition, the figurative constant ZERO and both integer and decimal forms of the zero literal can be specified in a floating-point VALUE clause.

You cannot specify a floating-point format numeric literal in the VALUE clause of a fixed-point numeric item.

For information about floating-point literal values, see [“Rules for floating-point literal values” on page 45](#).

Example

The following example illustrates the use of a format-1 VALUE clause:

```
IDENTIFICATION DIVISION.
PROGRAM-ID. HELLO.

DATA DIVISION.
    WORKING-STORAGE SECTION.
        01 WS-A PIC 9(2)V9 VALUE 5.6.
        01 WS-B PIC A(15) VALUE 'Hello world'.
        01 WS-C PIC 99 VALUE ZERO.

PROCEDURE DIVISION.
```


When *literal-1* and *literal-2* are NATIONAL literals, the range of national character values specified by the THROUGH phrase is based on the binary collating sequence of the hexadecimal values of the national characters represented by the literals.

When *literal-1* and *literal-2* are UTF-8 literals, the range of UTF-8 character values specified by the THROUGH phrase is based on the binary collating sequence of the hexadecimal values of the UTF-8 characters represented by the literals.

If the associated conditional variable is of class DBCS, *literal-1* and *literal-2* must be DBCS literals. The figurative constant SPACE or the figurative constant ALL *DBCS-literal* can be specified.

If the associated conditional variable is of class NATIONAL, *literal-1* and *literal-2* must be either both national literals or both alphanumeric literals for a given condition-name. The figurative constants ZERO, SPACE, QUOTE, HIGH-VALUE, LOW-VALUE, *symbolic-character*, ALL *national-literal*, or ALL *literal* can be specified.

If the associated conditional variable is of class UTF-8, *literal-1* and *literal-2* must be either both UTF-8 literals or both alphanumeric literals for a given condition-name. The figurative constants ZERO, SPACE, QUOTE, HIGH-VALUE, LOW-VALUE, *symbolic-character*, ALL *utf-8-literal*, or ALL *literal* can be specified.

WHEN SET TO FALSE

Allows specification of a FALSE condition value. This value is moved to the associated conditional variable when the SET TO FALSE statement is executed for the associated condition-name.

literal-3

When a condition-name is referenced in a SET TO FALSE statement, the value of *literal-3* from the FALSE phrase is placed in the associated conditional variable.

Note: The true values of a conditional variable are all the values associated with its condition-names, and all other values are false values. The WHEN SET TO FALSE phrase specifies just one of possibly many false values. The purpose is to define a single value to be used by the SET TO FALSE statement. When a condition-name condition is tested, all the non-true values give a false result, not just the one false value defined by the WHEN SET TO FALSE phrase.

Rules for condition-name entries

There are certain rules for condition-name entries.

The rules are:

- The VALUE clause is required in a condition-name entry, and must be the only clause in the entry. Each condition-name entry is associated with a preceding conditional variable. Thus every level-88 entry must always be preceded either by the entry for the conditional variable or by another level-88 entry when several condition-names apply to one conditional variable. Each such level-88 entry implicitly has the PICTURE characteristics of the conditional variable.
- A space, a separator comma, or a separator semicolon must separate successive operands.

Each entry must end with a separator period.

- The keywords THROUGH and THRU are equivalent.
- The condition-name entries associated with a particular conditional variable must immediately follow the conditional variable entry. The conditional variable can be any elementary data description entry except the following ones:
 - Another condition-name
 - A RENAME clause (level-66 item)
 - An item described with USAGE IS INDEX
 - An item described with USAGE POINTER, USAGE PROCEDURE-POINTER, USAGE FUNCTION-POINTER, or USAGE OBJECT REFERENCE
- Condition-names can be specified both at the group level and at subordinate levels within an alphanumeric group, national group or UTF-8 group.

- When the condition-name is specified for an alphanumeric group data description entry:
 - The value of *literal-1* (or *literal-1* and *literal-2*) must be specified as an alphanumeric literal or figurative constant.
 - The group can contain items of any usage.
- When the condition-name is specified for a national group data description entry:
 - The value of *literal-1* (or *literal-1* and *literal-2*) must be specified as an alphanumeric literal, a national literal, or a figurative constant.
 - The group can contain only items of usage national, as specified for the [“GROUP-USAGE clause” on page 198](#).
- When the condition-name is specified for a UTF-8 group data description entry:
 - The value of *literal-1* (or *literal-1* and *literal-2*) must be specified as an alphanumeric literal, a UTF-8 literal, or a figurative constant.
 - The group can contain only items of usage UTF-8, as specified for the [“GROUP-USAGE clause” on page 198](#).
- When the condition-name is associated with an alphanumeric group data description entry or a national group data description entry or a UTF-8 group data description entry:
 - The size of each literal value must not exceed the sum of the sizes of all the elementary items within the group.
 - No element within the group can contain a JUSTIFIED or SYNCHRONIZED clause.
- Relation tests implied by the definition of a condition-name are performed in accordance with the rules referenced in the following table.

<i>Table 18. Relation test references for condition-names</i>	
Type of conditional variable	Relation condition rules
Alphanumeric group item	“Group comparisons” on page 279
National group item (treated as elementary data item of class national)	“National comparisons” on page 278
Elementary data item of class alphanumeric	“Alphanumeric comparisons” on page 276
Elementary data item of class national	“National comparisons” on page 278
Elementary data item of class numeric	“Numeric comparisons” on page 279
Elementary data item of class DBCS	“DBCS comparisons” on page 277
Elementary data item of class UTF-8	“UTF-8 comparisons” on page 279

- A VALUE clause that specifies a national literal can be associated with a condition-name defined only for a data item of class national.
- A VALUE clause that specifies a DBCS literal can be associated with a condition-name defined only for a data item of class DBCS.
- The literals in a condition-name entry for an elementary data item of class national or a national group item must be either national literals or alphanumeric literals, and *literal-1* and *literal-2* must be of the same class. For alphanumeric groups or elementary data items of other classes, the type of literal must be consistent with the data type of the conditional variable. In the following example:
 - CITY-COUNTY-INFO, COUNTY-NO, and CITY are conditional variables.
 The PICTURE associated with COUNTY-NO limits the condition-name value to a two-digit numeric literal.
 The PICTURE associated with CITY limits the condition-name value to a three-character alphanumeric literal.

- The associated condition-names are level-88 entries.

Any values for the condition-names associated with CITY-COUNTY-INFO cannot exceed five characters.

Because this is an alphanumeric group item, the literal must be alphanumeric.

```

05 CITY-COUNTY-INFO.
   88 BRONX VALUE "03NYC".
   88 BROOKLYN VALUE "24NYC".
   88 MANHATTAN VALUE "31NYC".
   88 QUEENS VALUE "41NYC".
   88 STATEN-ISLAND VALUE "43NYC".
10 COUNTY-NO PICTURE 99.
   88 DUTCHESS VALUE 14.
      WHEN FALSE IS 99.
   88 KINGS VALUE 24.
   88 NEW-YORK VALUE 31.
   88 RICHMOND VALUE 43.
10 CITY PICTURE X(3).
   88 BUFFALO VALUE "BUF".
   88 NEW-YORK-CITY VALUE "NYC".
   88 POUGHKEEPSIE VALUE "POK".
05 POPULATION...
```

Example

The following example illustrates the use of a format-2 VALUE clause. This example verifies whether the input year is between 2000 and 2023 or now.

```

IDENTIFICATION DIVISION.
PROGRAM-ID. LEVEL88.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 YEAR.
   05 YEAR-INIT PIC 9(4).
   88 YEAR-VALID VALUE 2000 THRU 2023.
   88 YEAR-INVALID VALUE 0001 THRU 1999
      2024 THRU 9999.

PROCEDURE DIVISION.
   DISPLAY 'Enter a year:'.
   ACCEPT YEAR.

   IF YEAR-VALID
      DISPLAY 'YEAR:' YEAR
   ELSE
      DISPLAY 'INVALID YEAR'
   END-IF.

   STOP RUN.
```

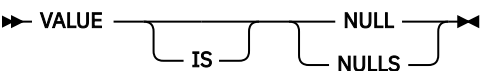
The following is the output, if the input for YEAR is 1999:

```
INVALID YEAR
```

Format 3

This format assigns an invalid address as the initial value of an item defined as USAGE POINTER, USAGE PROCEDURE-POINTER, or USAGE FUNCTION-POINTER. It also assigns an invalid object reference as the initial value of an item defined as USAGE OBJECT REFERENCE.

Format 3: NULL value



VALUE IS NULL can be specified only for elementary items described implicitly or explicitly as USAGE POINTER, USAGE PROCEDURE-POINTER, USAGE FUNCTION-POINTER, or USAGE OBJECT REFERENCE.

VOLATILE clause

The VOLATILE clause indicates that a data item's value can be modified or referenced in ways that the compiler cannot detect, such as by a Language Environment (LE) condition handler routine or by some other asynchronous process or thread. Thus, optimization is restricted for the data item.

Format

➡ VOLATILE ➡

In particular, the compiler will enforce the following restrictions:

- A volatile data item is loaded from memory each time it is referenced and stored to memory each time it is modified.
- Loads and stores to the data item are never reordered or eliminated.
- Storage is always allocated for the data item and initialized where necessary, even when no references to the data item are in the compilation unit.

Note: The STGOPT option is ignored for data items that have the VOLATILE clause.

The VOLATILE clause can be specified on data items that are defined in the FILE SECTION, WORKING-STORAGE SECTION, LOCAL-STORAGE SECTION, and LINKAGE SECTION. This clause can be specified together with any other clauses. For example, VOLATILE can be specified on tables, group data items, elementary data items, record descriptions and variably located data items.

There are additional considerations for groups:

- When a group item is explicitly defined with the VOLATILE clause, all items subordinate to the group item are treated as volatile by the compiler.
- When a group item has one or more subordinate items that are explicitly defined with the VOLATILE clause, the group item is treated as volatile by the compiler.

The VOLATILE clause cannot be specified on level-66 or level-88 data items.

It is not possible to indicate that all memory associated with a class instance is volatile. However, individual members of a class can be defined with the VOLATILE clause.

Example of using VOLATILE with groups:

Consider the following group definition:

```
01 DATA-COLLECTION.  
  03 DATA-ITEMS-A VOLATILE.  
    05 DATA-A1 PIC S9(9) BINARY.  
    05 DATA-A2 PIC S9(9) BINARY.  
  03 DATA-ITEMS-B.  
    05 DATA-B1 PIC S9(9).  
    05 DATA-B2 PIC S9(9) VOLATILE.  
  03 DATA-ITEMS-C.  
    05 DATA-C1 PIC S9(9).  
    05 DATA-C2 PIC S9(9).
```

In this example:

- DATA-ITEMS-A and DATA-B2 are considered volatile because they are defined with the VOLATILE clause.
- DATA-A1 and DATA-A2 are treated as volatile because they are both subordinate to a group item (DATA-ITEMS-A) that has the VOLATILE clause.
- DATA-COLLECTION and DATA-ITEMS-B are treated as volatile because they are group items that have subordinates that are defined with the VOLATILE clause. For example:


```
MOVE DATA-ITEMS-B TO DATA-ITEMS-C.
```

In this case, by treating DATA-ITEMS-B as volatile, the compiler ensures that the latest value of its subordinate member DATA-B2 is used in the memory copy operation.

In the following LE condition handler scenario, it is necessary to specify the "STEP" data item with the VOLATILE clause to achieve correct results. In particular, if the VOLATILE clause is not used, the compiler might assume that "STEP" is never referenced between the assignment of "2" to "STEP" and the assignment of "3" to "STEP" and might therefore decide to eliminate the first assignment during optimization. Unfortunately, this could result in a problem because if a divide-by-zero condition occurs during execution of the subsequent line of code, the condition handler will execute and reference the external variable "STEP", which might have the incorrect value.

```
Main program:
IDENTIFICATION DIVISION.
PROGRAM-ID. MAIN.
DATA DIVISION.
WORKING-STORAGE SECTION.
77 USER-HANDLER PROCEDURE-POINTER.
77 TOKEN    PIC S9(9) COMP.
01 QTY      PIC 9(8)  BINARY.
01 DIVISOR  PIC 9(8)  BINARY VALUE 0.
01 ANSWER   PIC 9(8)  BINARY.
01 STEP     PIC 9(8)  BINARY VALUE 0 EXTERNAL VOLATILE.
:
SET USER-HANDLER TO ENTRY 'HANDLER'
CALL 'CEEHDLR' USING USER-HANDLER, TOKEN, NULL
COMPUTE STEP = 2                *> Compiler thinks this store has no purpose and may remove it
COMPUTE ANSWER = NUMBER / DIVISOR *> Divide-by-zero exception occurs here, handler is invoked,
                                *> and reference to 'STEP' is made but hidden from compiler
DISPLAY 'ANSWER = ' ANSWER
COMPUTE STEP = 3
DISPLAY 'STEP = ' STEP
COMPUTE ANSWER = QTY + 2
:
Condition handler program:
IDENTIFICATION DIVISION.
PROGRAM-ID. HANDLER.
DATA DIVISION.
WORKING-STORAGE SECTION.
01 STEP PIC 9(8) BINARY EXTERNAL.
PROCEDURE DIVISION.
:
DISPLAY 'ERROR: A PROBLEM WAS ENCOUNTERED IN STEP ' STEP.
```

Part 6. PROCEDURE DIVISION

Chapter 27. Procedure division structure

The PROCEDURE DIVISION is an optional division.

Program procedure division

The program procedure division consists of optional declaratives, and procedures that contain sections, paragraphs, sentences, and statements.

Factory procedure division

The factory procedure division contains only factory method definitions.

Object procedure division

The object procedure division contains only object method definitions.

Method procedure division

A method procedure division consists of optional declaratives, and procedures that contain sections, paragraphs, sentences, and statements. A method can INVOKE other methods, be recursively invoked, and issue a CALL to a program. A method procedure division cannot contain nested programs or methods.

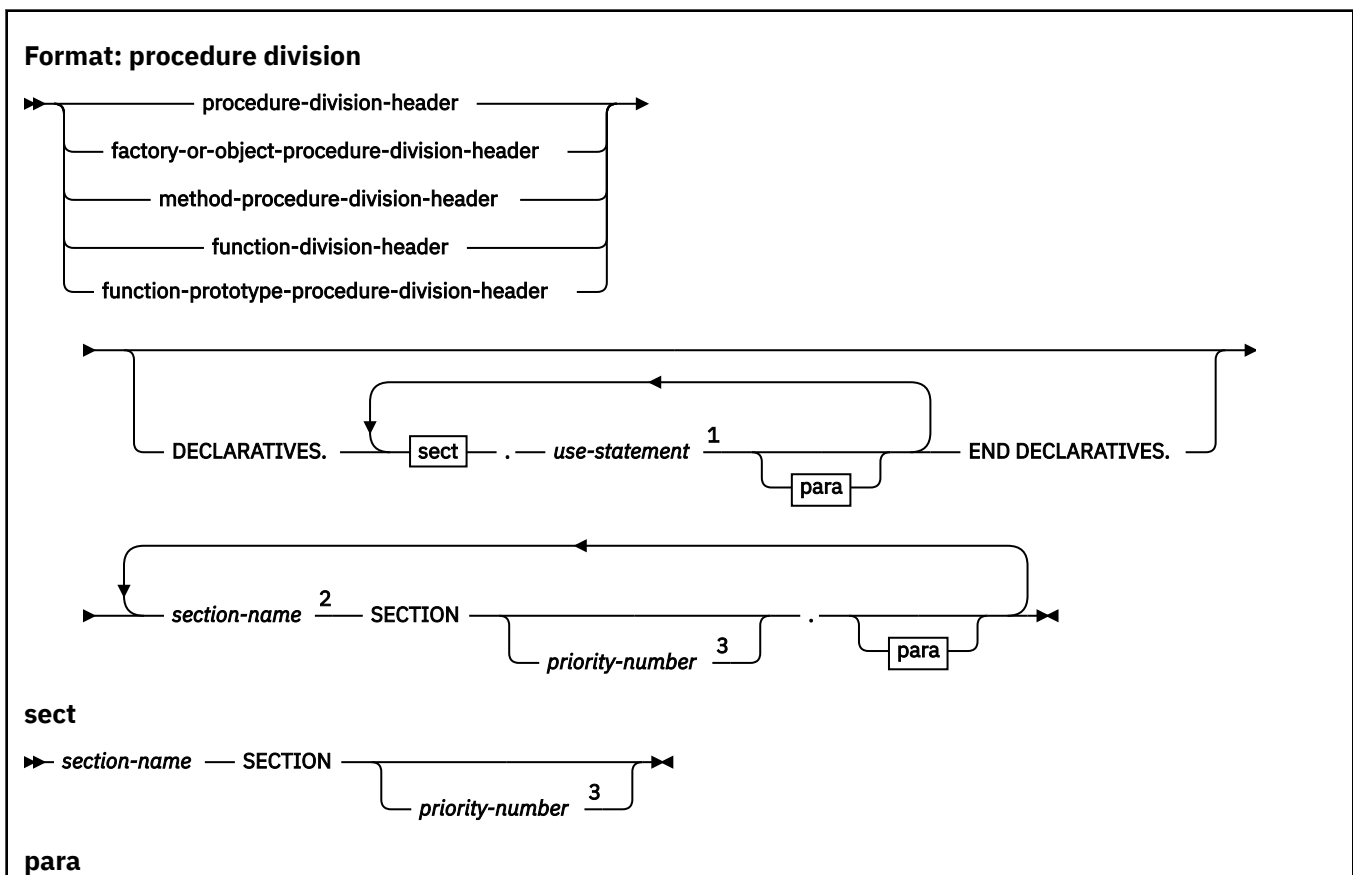
For additional details on a method procedure division, see [“Requirements for a method procedure division”](#) on page 258.

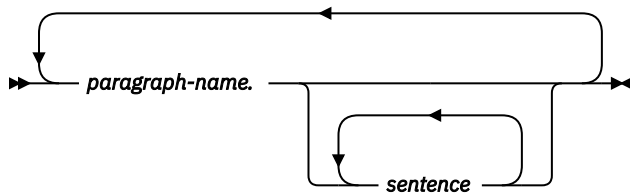
Function procedure division

The function procedure division consists of optional declaratives, and procedures that contain sections, paragraphs, sentences, and statements.

Function prototype procedure division

The function prototype procedure division consists only of the function prototype division header. No declaratives, sections, paragraphs, sentences, or statements are allowed. A function prototype procedure division cannot contain nested programs or functions. The function prototype procedure division is required in the definition of a function prototype.





Notes:

- ¹ The USE statement is described under [“USE statement” on page 713.](#)
- ² *Section-name* can be omitted. If you omit *section-name*, *paragraph-name* can be omitted.
- ³ Priority-numbers are not valid for methods, recursive programs, or programs compiled with the THREAD option.

Requirements for a method procedure division

There are specific requirements when you code a method procedure division.

The requirements are:

- You can use the EXIT METHOD statement or the GOBACK statement to return control to the invoking method or program. An implicit EXIT METHOD statement is generated as the last statement of every method procedure division.

For details on the EXIT METHOD statement, see [“Format 3 \(method\)” on page 344.](#)

- You can use the STOP RUN statement (which terminates the run unit) in a method.
- You can use the RETURN-CODE special register within a method procedure division to access return codes from subprograms that are called with the CALL statement, but the RETURN-CODE value is not returned to the invoker of the current method. Use the procedure division RETURNING data name to return a value to the invoker of the current method. For details, see the discussion of RETURNING *data-name-2* under [“The PROCEDURE DIVISION header” on page 258.](#)

You cannot specify the following statements or clauses in a method procedure division:

- ALTER
- ENTRY
- EXIT PROGRAM
- GO TO without a specified procedure name
- SEGMENT-LIMIT
- USE FOR DEBUGGING

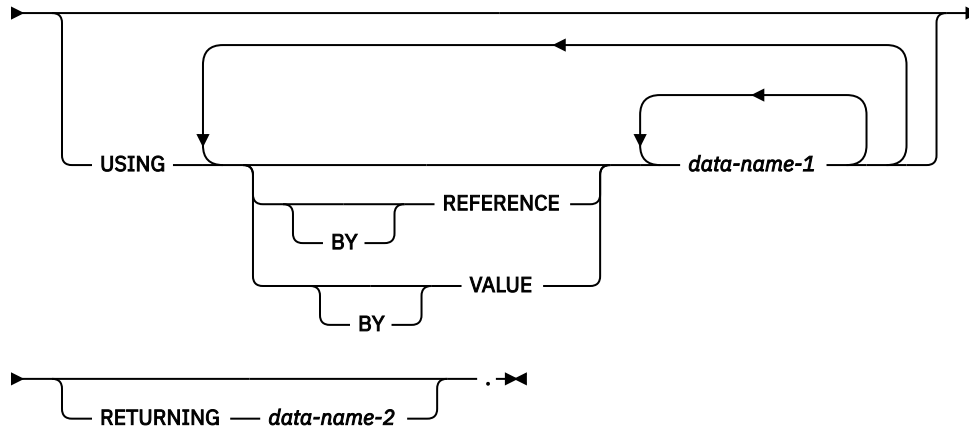
The PROCEDURE DIVISION header

The PROCEDURE DIVISION, if specified, is identified by one of the following headers, depending on whether you are specifying a program, a factory definition, an object definition, a method definition, a function definition, or a function prototype definition.

The following syntax diagram shows the format for a PROCEDURE DIVISION header in a program.

Format: program procedure division header

➤➤ PROCEDURE DIVISION ➔



The following syntax diagram shows the format for a PROCEDURE DIVISION header in a factory paragraph or object paragraph.

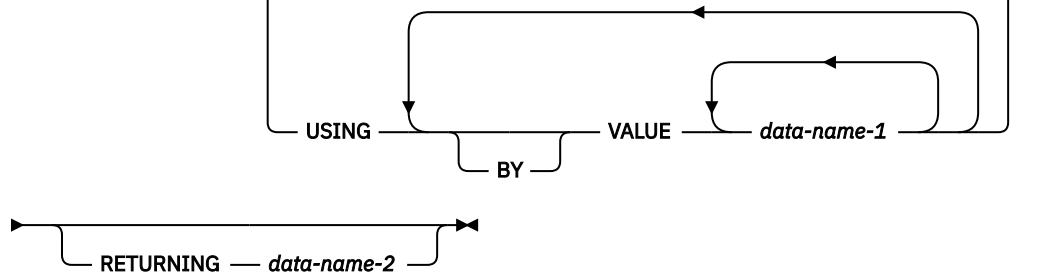
Format: factory and object procedure division header

➤➤ PROCEDURE DIVISION. ➤➤

The following syntax diagram shows the format for a PROCEDURE DIVISION header in a method.

Format: method procedure division header

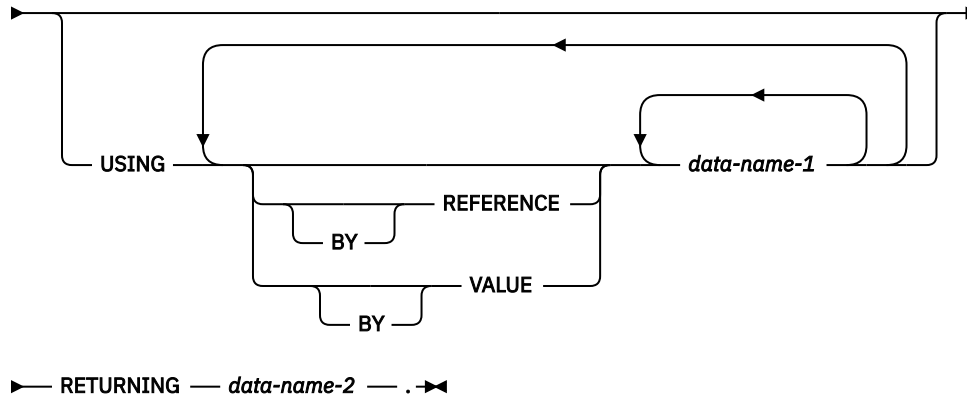
➤➤ PROCEDURE DIVISION



The following syntax diagram shows the format for a PROCEDURE DIVISION header in a user-defined function.

Format: function procedure division header

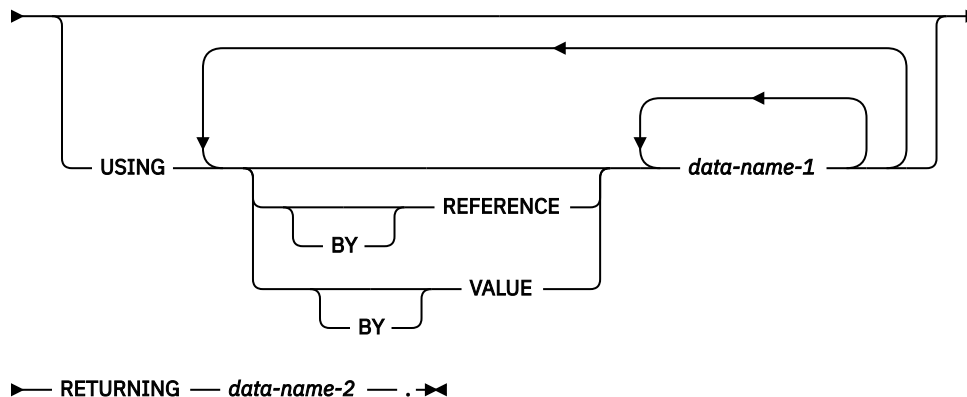
►► PROCEDURE DIVISION →



The following syntax diagram shows the format for a PROCEDURE DIVISION header in a function prototype.

Format: function prototype procedure division header

►► PROCEDURE DIVISION →



USING phrase

The USING phrase specifies the parameters that a program, method, or user-defined function receives when it is called or invoked. For function prototypes, the USING phrase specifies the parameters that will be used for conformance checking during function invocation.

The USING phrase is valid in the PROCEDURE DIVISION header of a called subprogram, invoked method entered at the beginning of the nondeclaratives portion, invoked user-defined function, or in the PROCEDURE DIVISION header of a function prototype definition. Each USING identifier must be defined as a level-01 or level-77 item in the LINKAGE SECTION of the called subprogram, invoked method, or invoked function.

Data items specified as arguments to user-defined function or function prototype invocations must follow the rules in [“Conformance of parameters for user-defined functions and function prototypes”](#) on page 262.

In a called subprogram entered at the first executable statement following an ENTRY statement, the USING phrase is valid in the ENTRY statement. Each USING identifier must be defined as a level-01 or level-77 item in the LINKAGE SECTION of the called subprogram.

However, a data item specified in the USING phrase of the CALL statement can be a data item of any level in the DATA DIVISION of the calling COBOL program, method, or function. A data item specified in the USING phrase of an INVOKE statement can be a data item of any level in the DATA DIVISION of the invoking COBOL program, method, or function. Likewise, a data item specified as an argument to a user-defined function invocation can be a data item of any level in the DATA DIVISION of the calling COBOL program, method, or function.

A data item in the USING phrase of the header can have a REDEFINES clause in its data description entry.

It is possible to call COBOL programs from non-COBOL programs or to pass user parameters from a system command to a COBOL main program. COBOL methods can be invoked only from Java or COBOL.

The order of appearance of USING identifiers in both calling and called subprograms, or invoking methods or programs and invoked methods, determines the correspondence of single sets of data available to both. The correspondence is positional and not by name. For calling and called subprograms, corresponding identifiers must contain the same number of bytes although their data descriptions need not be the same.

The order of appearance of arguments in a function invocation of a user-defined function or a function prototype, and USING identifiers in a function definition with the same name determines the correspondence of argument to formal parameter. That is, the order is positional and the first argument corresponds to the first formal parameter, etc. Each argument of a user-defined function or function prototype invocation must conform to each of its corresponding formal parameters according to the rules in [“Conformance of parameters for user-defined functions and function prototypes” on page 262](#). The number of arguments must match the number of formal parameters.

For index-names, no correspondence is established. Index-names in a caller (that can be a program, method, or function) and index-names in a called routine (that can be a program, method, or function) always refer to separate indexes.

The identifiers specified in a CALL USING or INVOKE USING statement name the data items available to the calling program or invoking method or program that can be referred to in the called program or invoked method. These items can be defined in any DATA DIVISION section. Likewise, identifiers specified as arguments in a user-defined function invocation name the data items available to the calling program, method, or function that can be referred to in the invoked function. These items can also be defined in any DATA DIVISION section.

A given identifier can appear more than once in a USING phrase. The last value passed to it by a CALL statement, INVOKE statement, or user-defined function invocation is used.

The BY REFERENCE or BY VALUE phrase applies to all parameters that follow until overridden by another BY REFERENCE or BY VALUE phrase.

BY REFERENCE (for programs and methods)

When an argument is passed BY CONTENT or BY REFERENCE, BY REFERENCE must be specified or implied for the corresponding formal parameter on the PROCEDURE or ENTRY USING phrase.

BY REFERENCE is the default if neither BY REFERENCE nor BY VALUE is specified.

If the reference to the corresponding data item in the CALL statement declares the parameter to be passed BY REFERENCE (explicit or implicit), the program executes as if each reference to a USING identifier in the called subprogram is replaced by a reference to the corresponding USING identifier in the calling program.

If the reference to the corresponding data item in the CALL statement declares the parameter to be passed BY CONTENT, the value of the item is moved when the CALL statement is executed and placed into a system-defined storage item that possesses the attributes declared in the LINKAGE SECTION for *data-name-1*. The data description of each parameter in the BY CONTENT phrase of the CALL statement must be the same, meaning no conversion or extension or truncation, as the data description of the corresponding parameter in the USING phrase of the header.

BY REFERENCE (for functions only)

When an argument is received BY REFERENCE in a function definition, BY REFERENCE must be specified or implied for the corresponding formal parameter on the PROCEDURE DIVISION USING phrase.

To pass an argument effectively BY CONTENT on a user-defined function invocation, use the CONTENT-OF intrinsic function. For details about this intrinsic function, see [Chapter 41, “CONTENT-OF,”](#) on page 543. Also find an example in *Passing arguments BY CONTENT to user-defined functions* in the *Enterprise COBOL Programming Guide*.

BY REFERENCE is the default if neither BY REFERENCE nor BY VALUE is specified.

If the reference to the corresponding argument in the function invocation declares the parameter to be passed BY REFERENCE explicitly or implicitly, the program executes as if each reference to a USING identifier in the invoked function was replaced by a reference to the corresponding argument in the calling program, method, or function.

BY VALUE (for programs and methods)

When an argument is passed BY VALUE, the value of the argument is passed, not a reference to the sending data item. The receiving subprogram or method has access only to a temporary copy of the sending data item. Any modifications made to the formal parameters that correspond to an argument passed BY VALUE do not affect the argument.

Parameters specified in the USING phrase of a method procedure division header must be passed to the method BY VALUE.

See *Passing data* in the *Enterprise COBOL Programming Guide* for examples that illustrate these concepts.

BY VALUE (for functions only)

When an argument is received BY VALUE in a function definition, the value of the argument is passed, not a reference to the sending data item. The receiving function has access to only a temporary copy of the sending data item. Any modifications made to the formal parameters that correspond to an argument passed BY VALUE do not affect the argument in the invoking program, method, or function.

data-name-1

data-name-1 must be a level-01 or level-77 item in the LINKAGE SECTION.

When *data-name-1* is an object reference in a method procedure division header, an explicit class-name must be specified in the data description entry for that object reference; that is, *data-name-1* must not be a universal object reference.

For methods, the parameter data types are restricted to the data types that are interoperable between COBOL and Java, as listed in [“Interoperable data types for OO COBOL and Java”](#) on page 366.

Conformance of parameters for user-defined functions and function prototypes

In the following section, the argument of a user-defined function or function prototype invocation is considered the sending item, and the corresponding formal parameter defined in the function is considered the receiving item.

The number of arguments for a user-defined function or function prototype invocation must be the same as the number of formal parameters specified in the user-defined function definition.

If both an argument and its corresponding formal parameter are elementary items, the conformance rules for elementary items apply. Otherwise, the conformance rules for group items apply.

Group items

If either the sender or the receiver is an alphanumeric, national, or UTF-8 group item, and:

- If the parameter is passed BY REFERENCE, then the formal parameter (the receiver) must be described with the same or fewer number of bytes as the corresponding argument (the sender). For variable-

length data items such as those described with the OCCURS DEPENDING ON clause, the maximum length is used. No compile-time checking is done for unbounded groups.

- Group items may not be passed BY VALUE.

Elementary items passed BY REFERENCE

For elementary items passed BY REFERENCE, the definition of the formal parameter and the definition of the argument must have the same BLANK WHEN ZERO, DYNAMIC LENGTH, JUSTIFIED, PICTURE, SIGN, and USAGE clauses, with the following exceptions:

- Currency symbols match only if the corresponding currency strings are the same.
- Period and comma picture symbols match only if the DECIMAL-POINT IS COMMA clause is in effect for both the calling program, method, or function and the invoked function, or for neither of them.
- For the PICTURE clause of Alphanumeric (PICTURE X), National (PICTURE N), or UTF-8 (PICTURE U) category items, the argument definition can include the same number of picture clause symbols or more than the definition of the formal parameter.

Elementary items passed BY VALUE

The formal parameter must be one of the following items:

- Binary (USAGE BINARY, COMP, COMP-4, or COMP-5)
- Floating point (USAGE COMP-1 or COMP-2)
- Function-pointer (USAGE FUNCTION-POINTER)
- Pointer (USAGE POINTER)
- Procedure-pointer (USAGE PROCEDURE-POINTER)
- One single-byte alphanumeric character (such as PIC X or PIC A)
- One national character (PIC N) that is described as an elementary data item of category national

If the formal parameter is of class pointer, the conformance rules will be as if a SET statement were performed with the argument as the sending operand and the corresponding formal parameter as the receiving operand.

If the formal parameter is of class numeric, the conformance rules will be as if a COMPUTE statement were performed with the argument as the sending operand and the corresponding formal parameter as the receiving operand.

Otherwise, the conformance rules will be as if a MOVE statement were performed with the argument as the sending operand and the corresponding formal parameter as the receiving operand.

RETURNING phrase

The RETURNING phrase specifies a data item that is to receive the program, method, or function result. For function prototypes, the RETURNING phrase specifies the kind of function result.

data-name-2

data-name-2 is the RETURNING data item. *data-name-2* must be a level-01 or level-77 item in the LINKAGE SECTION.

Note: An unbounded group cannot be specified as *data-name-2*.

In a method procedure division header, the data type of *data-name-2* must be one of the types supported for Java interoperation, as listed in [“Interoperable data types for OO COBOL and Java” on page 366](#).

The RETURNING data item is an output-only parameter. On entry to the method, the initial state of the RETURNING data item has an undefined and unpredictable value. You must initialize the PROCEDURE DIVISION RETURNING data item before you reference its value. The value that is returned to the invoking routine is the value that the data item has at the point of exit from the method. See [“RETURNING phrase” on page 364](#) for further details on conformance requirements for the INVOKE RETURNING identifier and the method RETURNING data item.

The PROCEDURE DIVISION RETURNING phrase must be specified for user-defined functions and function prototype definitions.

Do not use the PROCEDURE DIVISION RETURNING phrase in:

- Programs that contain the ENTRY statement.
- Nested programs.
- Main programs: Results of specifying PROCEDURE DIVISION RETURNING on a main program are undefined. You should specify the PROCEDURE DIVISION RETURNING phrase only on called subprograms. For main programs, use the RETURN-CODE special register to return a value to the operating environment.

References to items in the LINKAGE SECTION

Data items defined in the LINKAGE SECTION of the called program, invoked function, or invoked method can be referenced within the PROCEDURE DIVISION of that program if and only if they satisfy one of the conditions as listed in the topic.

- They are operands of the USING phrase of the PROCEDURE DIVISION header or the ENTRY statement.
- They are operands of SET ADDRESS OF, ALLOCATE, CALL ... BY REFERENCE ADDRESS OF, or INVOKE ... BY REFERENCE ADDRESS OF.
- They are defined with a REDEFINES or RENAMES clause, the object of which satisfies the above conditions.
- They are items subordinate to any item that satisfies the condition in the rules above.
- They are condition-names or index-names associated with data items that satisfy any of the above conditions.

Declaratives

Declaratives provide one or more special-purpose sections that are executed when an exceptional condition occurs.

When declarative sections are specified, they must be grouped at the beginning of the procedure division and the entire PROCEDURE DIVISION must be divided into sections.

Each declarative section starts with a USE statement that identifies the section's function. The series of procedures that follow specify the actions that are to be taken when the exceptional condition occurs. Each declarative section ends with another section-name followed by a USE statement, or with the keywords END DECLARATIVES.

The entire group of declarative sections is preceded by the keyword DECLARATIVES written on the line after the PROCEDURE DIVISION header. The group is followed by the keywords END DECLARATIVES. The keywords DECLARATIVES and END DECLARATIVES must each begin in Area A and be followed by a separator period. No other text can appear on the same line.

In the declaratives part of the PROCEDURE DIVISION, each section header must be followed by a separator period, and must be followed by a USE statement followed by a separator period. No other text can appear on the same line.

The USE statement has the following formats:

- [“EXCEPTION/ERROR declarative” on page 714](#)
- [“DEBUGGING declarative” on page 715](#)

The USE statement itself is never executed; instead, the USE statement defines the conditions that execute the succeeding procedural paragraphs, which specify the actions to be taken. After the procedure is executed, control is returned to the routine that activated it.

A declarative procedure can be performed from a nondeclarative procedure.

A nondeclarative procedure can be performed from a declarative procedure.

A declarative procedure can be referenced in a GO TO statement in a declarative procedure.

A nondeclarative procedure can be referenced in a GO TO statement in a declarative procedure.

You can include a statement that executes a previously called USE procedure that is still in control. However, to avoid an infinite loop, you must be sure there is an eventual exit at the bottom.

The declarative procedure is exited when the last statement in the procedure is executed.

Procedures

Within the PROCEDURE DIVISION, a *procedure* consists of a *section* or a group of sections, and a *paragraph* or group of paragraphs.

A *procedure-name* is a user-defined name that identifies a section or a paragraph.

Section

A *section-header* optionally followed by one or more paragraphs.

Section-header

A *section-name* followed by the keyword SECTION, optionally followed by a *priority-number*, followed by a separator period.

Section-headers are optional after the keywords END DECLARATIVES or if there are no declaratives.

Section-name

A user-defined word that identifies a section. A referenced section-name, because it cannot be qualified, must be unique within the program in which it is defined.

Priority-number

An integer or a positive signed numeric literal ranging in value from 0 through 99. *Priority-number* identifies a fixed segment or an independent segment that is to contain the section.

Sections in the declaratives portion must contain priority numbers in the range of 0 through 49.

You cannot specify priority-numbers:

- In a method definition
- In a program that is declared with the RECURSIVE attribute
- In a program compiled with the THREAD compiler option

A section ends immediately before the next section header, or at the end of the PROCEDURE DIVISION, or, in the declaratives portion, at the keywords END DECLARATIVES.

Segments

A segment consists of all sections in a program that have the same priority-number. Priority-number determines whether a section is stored in a fixed segment or an independent segment at run time.

Segments with a priority-number of 0 through 49 are fixed segments. Segments with a priority-number of 50 through 99 are independent segments.

The type of segment (fixed or independent) controls the segmentation feature.

In fixed segments, procedures are always in last-used state. In independent segments, procedures are in initial state each time the segment receives control from a segment with a different priority-number, except when the transfer of control results from the execution of a GOBACK or EXIT PROGRAM statement. Restrictions on the use of ALTER, SORT, and MERGE statements in independent segments are described under those statements.

Enterprise COBOL does not support the overlay feature of the 85 COBOL Standard segmentation module.

Paragraph

A *paragraph-name* followed by a separator period, optionally followed by one or more sentences.

Paragraphs must be preceded by a period because paragraphs always follow either the IDENTIFICATION DIVISION header, a section, or another paragraph, all of which must end with a period.

Paragraph-name

A user-defined word that identifies a paragraph. A paragraph-name, because it can be qualified, need not be unique.

If there are no declaratives (format 2), a paragraph-name is not required in the PROCEDURE DIVISION.

A paragraph ends immediately before the next paragraph-name or section header, or at the end of the PROCEDURE DIVISION, or, in the declaratives portion, at the keywords END DECLARATIVES.

Paragraphs need not all be contained within sections, even if one or more paragraphs are so contained.

Sentence

One or more *statements* terminated by a separator period.

Statement

A syntactically valid combination of *identifiers* and symbols (literals, relational-operators, and so forth) beginning with a COBOL statement.

Identifier

The word or words necessary to make unique reference to a data item, optionally including qualification, subscripting, indexing, and reference-modification. In any PROCEDURE DIVISION reference (except the class test), the contents of an identifier must be compatible with the class specified through its PICTURE clause, otherwise results are unpredictable.

Execution begins with the first statement in the PROCEDURE DIVISION, excluding declaratives. Statements are executed in the order in which they are presented for compilation, unless the statement rules dictate some other order of execution.

The end of the PROCEDURE DIVISION is indicated by one of the following items:

- An IDENTIFICATION DIVISION header that indicates the start of a nested source program
- An END PROGRAM, END METHOD, END FACTORY, or END OBJECT marker
- The physical end of a program; that is, the physical position in a source program after which no further source program lines occur

Arithmetic expressions

Arithmetic expressions are used as operands of certain conditional and arithmetic statements.

An arithmetic expression can consist of any of the following items:

1. An identifier described as a numeric elementary item (including numeric functions)
2. A numeric literal
3. The figurative constant ZERO
4. Identifiers and literals, as defined in items 1, 2, and 3, separated by arithmetic operators
5. Two arithmetic expressions, as defined in items 1, 2, 3, or 4, separated by an arithmetic operator
6. An arithmetic expression, as defined in items 1, 2, 3, 4, or 5, enclosed in parentheses

Any arithmetic expression can be preceded by a unary operator.

Identifiers and literals that appear in arithmetic expressions must represent either numeric elementary items or numeric literals on which arithmetic can be performed.

If an exponential expression is evaluated as both a positive and a negative number, the result is always the positive number. For example, the square root of 4:

```
4 ** 0.5
```

is evaluated as +2 and -2. Enterprise COBOL always returns +2.

If the value of an expression to be raised to a power is zero, the exponent must have a value greater than zero. Otherwise, the size error condition exists. In any case where no real number exists as the result of an evaluation, the size error condition exists.

Arithmetic operators

Five binary arithmetic operators and two unary arithmetic operators can be used in arithmetic expressions. These operators are represented by specific characters that must be preceded and followed by a space.

These binary and unary arithmetic operators are shown in [Table 19 on page 267](#).

<i>Table 19. Binary and unary operators</i>			
Binary operator	Meaning	Unary operator	Meaning
+	Addition	+	Multiplication by +1
-	Subtraction	-	Multiplication by -1
*	Multiplication		
/	Division		
**	Exponentiation		

Limitation: Exponents in fixed-point exponential expressions cannot contain more than nine digits. The compiler will truncate any exponent with more than nine digits. In the case of truncation, the compiler will issue a diagnostic message if the exponent is a literal or constant; if the exponent is a variable or data-name, a diagnostic message is issued at run time.

Parentheses can be used in arithmetic expressions to specify the order in which elements are to be evaluated.

Expressions within parentheses are evaluated first. When expressions are contained within nested parentheses, evaluation proceeds from the least inclusive to the most inclusive set.

When parentheses are not used, or parenthesized expressions are at the same level of inclusiveness, the following hierarchic order is implied:

1. Unary operator
2. Exponentiation
3. Multiplication and division
4. Addition and subtraction

Parentheses either eliminate ambiguities in logic where consecutive operations appear at the same hierarchic level, or modify the normal hierarchic sequence of execution when this is necessary. When the order of consecutive operations at the same hierarchic level is not completely specified by parentheses, the order is from left to right.

An arithmetic expression can begin only with a left parenthesis, a unary operator, or an operand (that is, an identifier or a literal). It can end only with a right parenthesis or an operand. An arithmetic expression must contain at least one reference to an identifier or a literal.

There must be a one-to-one correspondence between left and right parentheses in an arithmetic expression, with each left parenthesis placed to the left of its corresponding right parenthesis.

If the first operator in an arithmetic expression is a unary operator, it must be immediately preceded by a left parenthesis if that arithmetic expression immediately follows an identifier or another arithmetic expression.

The following table shows permissible arithmetic symbol pairs. An arithmetic symbol pair is the combination of two such symbols in sequence. In the table:

Yes

Indicates a permissible pairing.

No

Indicates that the pairing is not permitted.

<i>Table 20. Valid arithmetic symbol pairs</i>					
	Identifier or literal second symbol	* / ** + - second symbol	Unary + or unary - second symbol	(second symbol	) second symbol
Identifier or literal first symbol	No	Yes	No	No	Yes
* / ** + - first symbol	Yes	No	Yes	Yes	No
Unary + or unary - first symbol	Yes	No	No	Yes	No
(first symbol	Yes	No	Yes	Yes	No
) first symbol	No	Yes	No	No	Yes

Conditional expressions

A conditional expression causes the object program to select alternative paths of control, depending on the truth value of a test. Conditional expressions are specified in EVALUATE, IF, PERFORM, and SEARCH statements.

A conditional expression can be specified in either simple conditions or complex conditions. Both simple and complex conditions can be enclosed within any number of paired parentheses; the parentheses do not change whether the condition is simple or complex.

Simple conditions

There are five simple conditions.

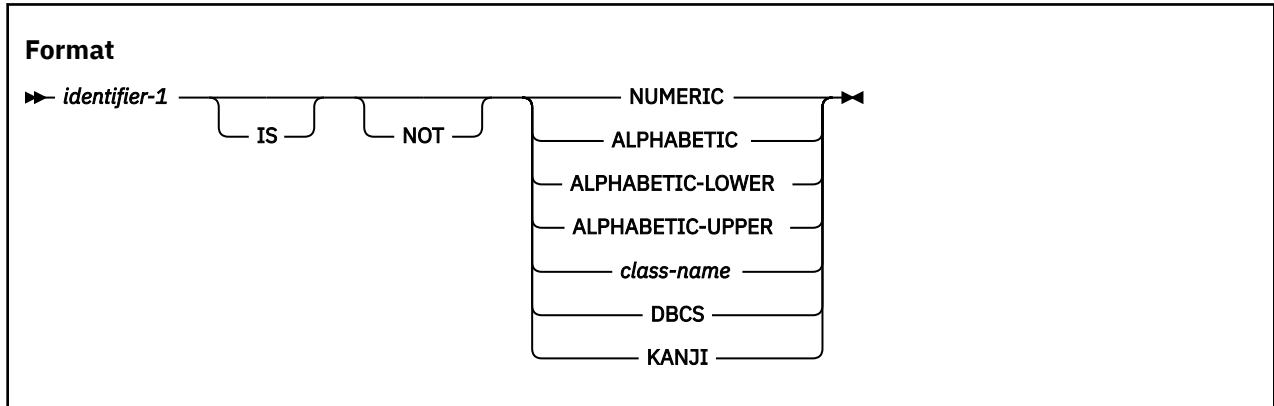
The simple conditions are:

- Class condition
- Condition-name condition
- Relation condition
- Sign condition
- Switch-status condition

A simple condition has a truth value of either true or false.

Class condition

The class condition determines whether the content of a data item is alphabetic, alphabetic-lower, alphabetic-upper, numeric, DBCS, KANJI, or contains only the characters in the set of characters specified by the CLASS clause as defined in the SPECIAL-NAMES paragraph of the ENVIRONMENT DIVISION.



identifier-1

Must reference a data item described with one of the following usages:

- DISPLAY, NATIONAL, COMPUTATIONAL-3, or PACKED-DECIMAL when NUMERIC is specified
- DISPLAY-1 when DBCS or KANJI is specified
- DISPLAY or NATIONAL when ALPHABETIC, ALPHABETIC-UPPER, or ALPHABETIC-LOWER is specified
- DISPLAY when class-name is specified

Must not be of class alphabetic when NUMERIC is specified.

Must not be of class numeric when ALPHABETIC, ALPHABETIC-UPPER, or ALPHABETIC-LOWER is specified.

Table 21 on page 270 lists the forms of class condition that are valid for each type of identifier.

If *identifier-1* is a function-identifier, it must reference an alphanumeric or national function.

An alphanumeric group item can be used in a class condition where an elementary alphanumeric item can be used, *except* that the NUMERIC class condition cannot be used if the group contains one or more signed elementary items.

When *identifier-1* is described with usage NATIONAL, the class-condition tests for the national character representation of the characters associated with the specified character class. For example, specifying a class condition of the form IF *national-item* IS ALPHABETIC is a test for the lowercase and uppercase letters Latin capital letter A through Latin capital letter Z and the space, as represented in national characters. Specifying IF *national-item* IS NUMERIC is a test for the characters 0 through 9.

When *identifier-1* is described with usage UTF-8, the class-condition tests for the UTF-8 character representation of the characters associated with the specified character class. For example, specifying a class condition of the form IF *utf8-item* IS ALPHABETIC is a test for the lowercase and uppercase letters Latin capital letter A through Latin capital letter Z and the space, as represented by UTF-8 characters.

NOT

When used, NOT and the next keyword define the class test to be executed for truth value. For example, NOT NUMERIC is a truth test for determining that the result of a NUMERIC class test is false (in other words, the item contains data that is nonnumeric).

NUMERIC

identifier-1 consists entirely of the characters 0 through 9, with or without an operational sign.

If its PICTURE does not contain an operational sign, the identifier being tested is determined to be numeric only if the contents are numeric and an operational sign is not present.

If its PICTURE does contain an operational sign, the identifier being tested is determined to be numeric only if the item is an elementary item, the contents are numeric, and a valid operational sign is present.

Usage note: Valid operational signs are determined from the setting of the NUMCLS installation option and the NUMPROC compiler option. For more information, see *Checking for incompatible data (numeric class test)* in the *Enterprise COBOL Programming Guide*.

ALPHABETIC

identifier-1 consists entirely of any combination of the lowercase or uppercase Latin alphabetic characters A through Z and the space.

ALPHABETIC-LOWER

identifier-1 consists entirely of any combination of the lowercase Latin alphabetic characters a through z and the space.

ALPHABETIC-UPPER

identifier-1 consists entirely of any combination of the uppercase Latin alphabetic characters A through Z and the space.

class-name

identifier-1 consists entirely of the characters listed in the definition of class-name in the SPECIAL-NAMES paragraph.

DBCS

identifier-1 consists entirely of DBCS characters.

A range check is performed on the item for valid character representation. The valid range is X'41' through X'FE' for both bytes of each DBCS character and X'4040' for the DBCS blank.

KANJI

identifier-1 consists entirely of DBCS characters.

A range check is performed on the item for valid character representation. The valid range is from X'41' through X'7E' for the first byte, from X'41' through X'FE' for the second byte, and X'4040' for the DBCS blank.

Table 21. Valid forms of the class condition for different types of data items		
Type of data item referenced by <i>identifier-1</i>	Valid forms of the class condition	
Alphabetic	ALPHABETIC ALPHABETIC-LOWER ALPHABETIC-UPPER <i>class-name</i>	NOT ALPHABETIC NOT ALPHABETIC-LOWER NOT ALPHABETIC-UPPER NOT <i>class-name</i>
Alphanumeric, alphanumeric-edited, or numeric-edited	ALPHABETIC ALPHABETIC-LOWER ALPHABETIC-UPPER NUMERIC <i>class-name</i>	NOT ALPHABETIC NOT ALPHABETIC-LOWER NOT ALPHABETIC-UPPER NOT NUMERIC NOT <i>class-name</i>
External-decimal or internal-decimal	NUMERIC	NOT NUMERIC

Table 21. **Valid forms of the class condition for different types of data items** (continued)

Type of data item referenced by <i>identifier-1</i>	Valid forms of the class condition	
DBCS	DBCS KANJI	NOT DBCS NOT KANJI
National	NUMERIC ALPHABETIC ALPHABETIC-LOWER ALPHABETIC-UPPER	NOT NUMERIC NOT ALPHABETIC NOT ALPHABETIC-LOWER NOT ALPHABETIC-UPPER
Numeric	NUMERIC <i>class-name</i>	NOT NUMERIC NOT <i>class-name</i>

Condition-name condition

A condition-name condition tests a conditional variable to determine whether its value is equal to any values that are associated with the condition-name.

Format

► *condition-name-1* ◄

A condition-name is used in conditions as an abbreviation for the relation condition. The rules for comparing a conditional variable with a condition-name value are the same as those specified for relation conditions.

If *condition-name-1* has been associated with a range of values (or with several ranges of values), the conditional variable is tested to determine whether its value falls within the ranges, including the end values. The result of the test is true if one of the values that corresponds to the condition-name equals the value of its associated conditional variable.

Condition-names are allowed for alphanumeric, DBCS, national, UTF-8, and floating-point data items, as well as others, as defined for the condition-name format of the VALUE clause.

The following example illustrates the use of conditional variables and condition-names:

```
01 AGE-GROUP      PIC 99.
   88 INFANT      VALUE 0.
   88 BABY        VALUE 1, 2.
   88 CHILD       VALUE 3 THRU 12.
   88 TEENAGER    VALUE 13 THRU 19.
```

AGE-GROUP is the conditional variable; INFANT, BABY, CHILD, and TEENAGER are condition-names. For individual records in the file, only one of the values specified in the condition-name entries can be present.

The following IF statements can be added to the above example to determine the age group of a specific record:

```
IF INFANT...      (Tests for value 0)
IF BABY...        (Tests for values 1, 2)
IF CHILD...       (Tests for values 3 through 12)
IF TEENAGER...    (Tests for values 13 through 19)
```

Depending on the evaluation of the condition-name condition, alternative paths of execution are taken by the object program.

Relation conditions

A relation condition specifies the comparison of two operands. The relational operator that joins the two operands specifies the type of comparison. The relation condition is true if the specified relation exists between the two operands; the relation condition is false if the specified relation does not exist.

Comparisons are defined for the following cases:

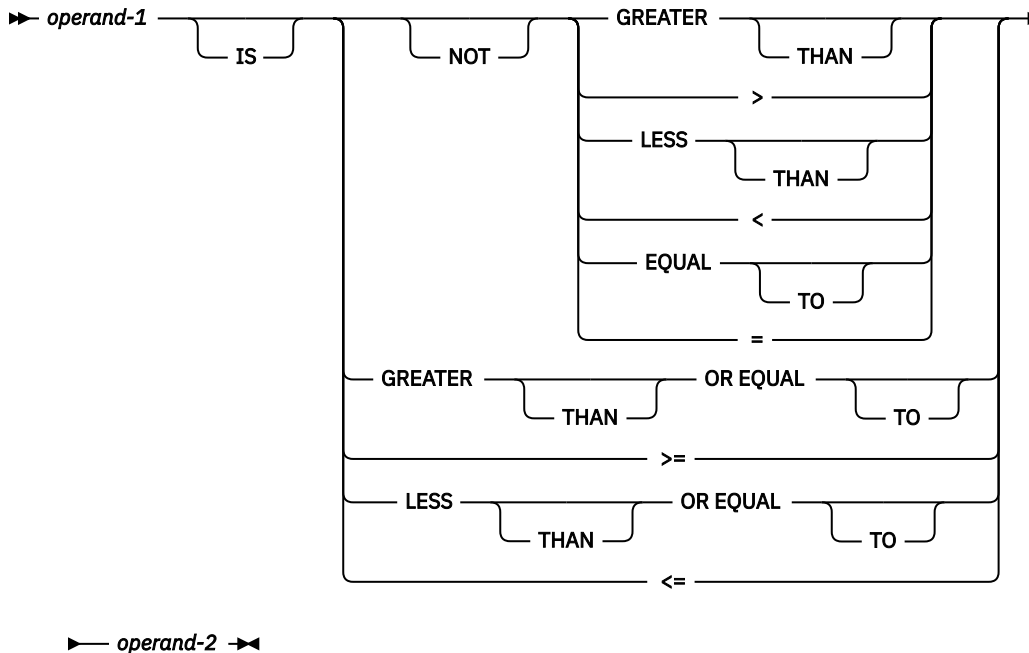
- Two operands of class alphabetic
- Two operands of class alphanumeric
- Two operands of class DBCS
- Two operands of class national
- Two operands of class numeric
- Two operands of class UTF-8.
- Two operands of different classes where each operand is one of the classes alphabetic, alphanumeric, national or UTF-8
- Two operands where one is a numeric integer and the other is class alphanumeric or national
- Two operands where one is class DBCS and the other is class national
- Comparisons involving indexes or index data items
- Two data pointer operands
- Two procedure pointer operands
- Two function pointer operands
- Two object reference operands
- An alphanumeric group and any operand that has usage DISPLAY, DISPLAY-1, NATIONAL, or UTF-8.

The following relation condition formats are defined:

- A general relation condition, for comparisons that involve only data items, literals, index-names, or index data items. For details, see [“General relation conditions” on page 272](#).
- A data pointer relation condition. For details, see [“Data pointer relation conditions” on page 280](#).
- A program pointer relation condition, for comparison of procedure pointers or function pointers. For details, see [“Procedure-pointer and function-pointer relation conditions” on page 281](#).
- An object-reference relation condition. For details, see [“Object-reference relation conditions” on page 282](#).

General relation conditions

A general relation condition compares two operands, either of which can be an identifier, literal, arithmetic expression, or index-name.

Format 1: general relation condition**operand-1**

The subject of the relation condition. Can be an identifier, literal, function-identifier, arithmetic expression, or index-name.

operand-2

The object of the relation condition. Can be an identifier, literal, function-identifier, arithmetic expression, or index-name.

An alphanumeric literal can be enclosed in parentheses within a relation condition.

The relation condition must contain at least one reference to an identifier.

The relational operators, shown in Table 22 on page 273, specify the type of comparison to be made. Each relational operator must be preceded and followed by a space. The two characters of the relational operators **>=** and **<=** must not have a space between them.

Table 22. Relational operators and their meanings

Relational operator	Can be written	Meaning
IS GREATER THAN	IS >	Greater than
IS NOT GREATER THAN	IS NOT >	Not greater than
IS LESS THAN	IS <	Less than
IS NOT LESS THAN	IS NOT <	Not less than
IS EQUAL TO	IS =	Equal to
IS NOT EQUAL TO	IS NOT =	Not equal to
IS GREATER THAN OR EQUAL TO	IS >=	Is greater than or equal to
IS LESS THAN OR EQUAL TO	IS <=	Is less than or equal to

In a general relation condition, data items, literals, and figurative constants of class alphabetic, alphanumeric, DBCS, national, UTF-8, and numeric are compared using the following comparison types:

Comparison type	Meaning
Alphanumeric	Comparison of the alphanumeric character value of two operands
DBCS	Comparison of the DBCS character value of two operands
National	Comparison of the national character value of two operands
UTF-8	Comparison of the UTF-8 character value of two operands
Numeric	Comparison of the algebraic value of two operands
Group	Comparison of the alphanumeric character value of two operands, where one or both operands is an alphanumeric group item

Table 23 on page 275 and Table 24 on page 276 show the permissible pairs for comparisons with different types of operands. The comparison type is indicated at the row and column intersection for permitted comparisons, using the following key:

Alph

Comparison of alphanumeric characters (further described in [“Alphanumeric comparisons”](#) on page 276)

DBCS

Comparison of DBCS characters (further described in [“DBCS comparisons”](#) on page 277)

Nat

Comparison of national characters (further described in [“National comparisons”](#) on page 278)

UTF-8

Comparison of UTF-8 characters (further described in [“UTF-8 comparisons”](#) on page 279)

Num

Comparison of algebraic value (further described in [“Numeric comparisons”](#) on page 279)

Group

Comparison of alphanumeric characters involving an alphanumeric group (further described in [“Group comparisons”](#) on page 279)

(Int)

Integer items only (combined with comparison type Alph, Nat, Num, or Group)

Blank

Comparison is not allowed

For rules and restrictions for comparisons involving index-names and index data items, see [“Comparison of index-names and index data items”](#) on page 280.

Introduction to Table 23 on page 275: This table is organized in the following manner:

- In the first column, under "Type of data item or literal", each row identifies a type of operand. In some cases, the type of operand references a grouping of operands that have common properties for comparison. For example, the row for "Alphanumeric character items" references all the types of operands that are listed in the cell, as follows:
 - Data items of category:
 - Alphanumeric
 - Alphanumeric- edited
 - Numeric-edited with usage DISPLAY
 - Alphanumeric functions
- Subsequent column headings refer to the type of an operand or a grouping of operands. For example, the column heading "Alphabetic and alphanumeric character items" refers to the types of operands identified as "Alphabetic data items" and all the types of operands that are grouped under the operand titled "Alphanumeric character items".

- Literals are listed as a type of operand only in the first column. They do not appear as column headings because literals cannot be used as both operands of a relation condition.

Table 23. <i>Comparisons involving data items and literals</i>										
Type of data item or literal	Alpha-numeric group items	Alphabetic and alphanumeric character items	Zoned decimal items	Native numeric items	Alpha-numeric floating-point items	National character items	National decimal items	National floating-point items	DBCS items	UTF-8 character items
Alphanumeric group item	Group	Group	Group (Int)		Group	Group	Group (Int)	Group	Group	Group
Alphabetic data items	Group	Alph	Alph (Int)		Alph	Nat	Alph (Int)	Nat		
Alphanumeric character items: • Data items of category: – Alphanumeric – Alphanumeric- edited – Numeric-edited with usage DISPLAY • Alphanumeric functions	Group	Alph	Alph (Int)		Alph	Nat	Alph (Int)	Nat		UTF-8
Alphanumeric literals	Group	Alph	Alph (Int)		Alph	Nat	Alph (Int)	Nat		UTF-8
Numeric literals	Group (Int)	Alph (Int)	Num	Num	Num	Nat (Int)	Num	Num		
Zoned decimal data items	Group (Int)	Alph (Int)	Num	Num	Num	Nat (Int)	Num	Num		
Native numeric items: • Binary • Arithmetic expression • Internal decimal • Internal floating point Numeric and integer intrinsic functions			Num	Num	Num		Num	Num		
Display floating-point items	Group	Alph	Num	Num	Num	Nat	Num	Num		
Floating-point literals			Num	Num	Num		Num	Num		
National character items: • Data items of category: – National – National- edited – Numeric- edited with usage NATIONAL • National intrinsic functions • National groups (treated as elementary item)	Group	Nat	Nat (Int)		Nat	Nat	Nat (Int)	Nat	Nat	Group
National literals	Group	Nat	Nat (Int)		Nat	Nat	Nat (Int)	Nat	Nat	UTF-8
National decimal items	Group (Int)	Alph (Int)	Num	Num	Num	Nat (Int)	Num	Num		
National floating-point items	Group	Nat	Num	Num	Num	Nat	Num	Num		

Table 23. Comparisons involving data items and literals (continued)										
Type of data item or literal	Alpha-numeric group items	Alpha-betic and alpha-numeric character items	Zoned decimal items	Native numeric items	Alpha-numeric floating-point items	National character items	National decimal items	National floating-point items	DBCS items	UTF-8 character items
DBCS data items	Group					Nat			DBCS	
DBCS literals	Group					Nat			DBCS	
UTF-8 character items	Group	UTF-8				UTF-8				UTF-8
UTF-8 literals	Group	UTF-8				UTF-8				UTF-8
UTF-8 literals	Group	UTF-8				UTF-8				UTF-8

Table 24. Comparisons involving figurative constants										
Figurative constant	Alpha-numeric group items	Alpha-betic and alpha-numeric character items	Zoned decimal items	Native numeric items	Alpha-numeric floating point items	National character items	National decimal items	National floating point items	DBCS items	UTF-8 data items
ZERO	Group	Alph	Num	Num	Num	Nat	Num	Num		UTF-8
SPACE	Group	Alph	Alph (Int)		Alph	Nat	Nat (Int)	Nat	DBCS	UTF-8
HIGH-VALUE, LOW-VALUE QUOTE	Group	Alph	Alph (Int)		Alph	Nat	Nat (Int)	Nat		UTF-8
Symbolic character	Group	Alph	Alph (Int)		Alph	Nat	Nat (Int)	Nat		UTF-8
ALL alphanumeric literal	Group	Alph	Alph (Int)		Alph	Nat	Nat (Int)	Nat		UTF-8
ALL national literal	Group	Nat	Nat (Int)		Nat	Nat	Nat (Int)	Nat	Nat	UTF-8
ALL UTF-8 literal	UTF-8	UTF-8	UTF-8	UTF-8	UTF-8	UTF-8	UTF-8	UTF-8		UTF-8
ALL DBCS literal	Group					Nat			DBCS	

Alphanumeric comparisons

An alphanumeric comparison is a comparison of the single-byte character values of two operands.

When one of the operands is neither class alphanumeric nor class alphabetic, that operand is processed as follows:

- A display floating-point data item is treated as though it were a data item of category alphanumeric, rather than as a numeric value.
- A zoned decimal integer operand is treated as though it were moved to a temporary elementary data item of category alphanumeric with a length the same as the total number of digits in the number, according to the rules of the MOVE statement.

When the ZWB compiler option is in effect, the unsigned value of the integer operand is moved to the temporary data item. When the NOZWB compiler option is specified, the signed value is moved to the temporary data item. See *ZWB* in the *Enterprise COBOL Programming Guide* for more details about the ZWB (NOZWB) compiler option.

Comparison then proceeds with the temporary data item of category alphanumeric.

Comparison of two alphanumeric operands

Alphanumeric comparisons are made with respect to the collating sequence of the character set in use as follows:

- For the EBCDIC character set, the EBCDIC collating sequence is used.
- For the ASCII character set, the ASCII collating sequence is used. (See [Appendix E, “EBCDIC and ASCII collating sequences,”](#) on page 763.)
- When the PROGRAM COLLATING SEQUENCE clause is specified in the object-computer paragraph, the collating sequence used is the one associated in the special-names paragraph with the specified alphabet-name.

The size of each operand is the total number of character positions in that operand; the size affects the result of the comparison. There are two cases to consider:

Operands of equal size

Characters in corresponding positions of the two operands are compared, beginning with the leftmost character and continuing through the rightmost character.

If all pairs of characters through the last pair evaluate as equal, the operands are equal.

If a pair of unequal characters is encountered, the characters are tested to determine their relative positions in the collating sequence. The operand that contains the character higher in the sequence is considered the greater operand.

Operands of unequal size

If the operands are of unequal size, the comparison is made as though the shorter operand were extended to the right with enough spaces to make the operands equal in size.

The higher collating value is determined using the hexadecimal value of characters.

Standard comparison

A standard comparison is any comparison that is not based on a locale. The standard comparison method depends on whether the operands to be compared are of equal length or unequal length.

If the operands are of unequal length, the comparison proceeds as though the shorter operand were padded on the right with appropriate space characters to make the operands of equal length. The comparison then proceeds according to the rules for the comparison of operands of equal length.

If the operands are of equal length, the comparison proceeds by comparing corresponding character positions in the two operands, starting from the leftmost position, until either unequal characters are encountered or the rightmost character position is reached, whichever comes first. The operands are determined to be equal if all corresponding characters are equal.

The first-encountered unequal character in the operands is compared to determine the relation of the operands. The operand that contains the character with the higher collating value is the greater operand.

DBCS comparisons

A DBCS comparison is a comparison of two DBCS operands.

The following rules apply to a DBCS comparison:

- If the DBCS operands are not the same length, the comparison is made as though the shorter operand were padded on the right with DBCS spaces to the length of the longer operand.
- The comparison is based on the binary collating sequence of the hexadecimal values of the DBCS characters.

National comparisons

A national comparison is a comparison of the national character value of two operands of class national.

When the relation condition specifies an operand that is neither class national nor class UTF-8, that operand is converted to a data item of category national before the comparison. When a UTF-8 item is compared with a national item, the comparison is always done in UTF-8. The following list describes the conversion of operands to category national.

DBCS

A DBCS operand is treated as though it were moved to a temporary data item of category national of the same length as the DBCS operand. DBCS characters are converted to the corresponding national characters. The source code page used for the conversion is the one in effect for the CODEPAGE compiler option when the source code was compiled.

Alphabetic, alphanumeric, alphanumeric-edited, and numeric-edited with usage DISPLAY and UTF-8

The operand is treated as though it were moved to a temporary data item of category national of the length needed to represent the number of character positions in that operand. Alphanumeric characters are converted to the corresponding national characters. The source code page used for the conversion is the one in effect for the CODEPAGE compiler option when the source code was compiled.

Numeric integer

A numeric integer operand is treated as though it were moved to a temporary data item of category alphanumeric of a length the same as the number of digits in the integer. The unsigned value is used. The resulting temporary data item is then converted as an alphanumeric operand.

External floating-point

A display floating-point item is treated as though it were a data item of category alphanumeric, rather than as a numeric value, and then converted as an alphanumeric operand.

A national floating-point item is treated as though it were a data item of category national, rather than as a numeric value.

The implicit moves for the conversions are carried out in accordance with the rules of the MOVE statement.

The resulting category national data item is used in the comparison of two national operands.

Comparison of two national operands

If the operands are of unequal length, the comparison proceeds as though the shorter operand were padded on the right with the default national space character (NX'0020') to make the operands of equal length. The comparison then proceeds according to the rules for the comparison of operands of equal length.

If the operands are of equal length, the comparison proceeds by comparing corresponding national character positions in the two operands, starting from the leftmost position, until either unequal national characters are encountered or the rightmost national character position is reached, whichever comes first. The operands are determined to be equal if all corresponding national characters are equal.

The first-encountered unequal national character in the operands is compared to determine the relation of the operands. The operand that contains the national character with the higher collating value is the greater operand.

The higher collating value is determined using the hexadecimal value of characters.

The PROGRAM COLLATING SEQUENCE clause has no effect on comparisons of national operands.

UTF-8 comparisons

A UTF-8 comparison is a comparison of two operands of class UTF-8. When the relation condition specifies an operand that is not class UTF-8, that operand is converted to a data item of category UTF-8 before the comparison.

Alphabetic, alphanumeric, alphanumeric-edited, numeric-edited with usage DISPLAY, national, national-edited and numeric-edited with usage NATIONAL

The operand is treated as though it were moved to a temporary data item of category UTF-8 of the length needed to represent the number of character positions in that operand. Alphanumeric characters are converted to the corresponding UTF-8 characters. The source code page used for the conversion is the one in effect for the CODEPAGE compiler option when the source code was compiled.

The implicit moves for the conversions are carried out in accordance with the rules of the MOVE statement.

The resulting category UTF-8 data item is used in the comparison of two UTF-8 operands.

Comparison of two UTF-8 operands

If the operands are of unequal length, the comparison proceeds as though the shorter operand were padded on the right with the default UTF-8 space character (UX'20') to make the operands of equal length. The comparison then proceeds according to the rules for the comparison of operands of equal length.

If the operands are of equal length, the comparison proceeds by comparing corresponding UTF-8 character positions in the two operands, starting from the leftmost position, until either unequal UTF-8 characters are encountered or the rightmost UTF-8 character position is reached, whichever comes first. The operands are determined to be equal if all corresponding UTF-8 characters are equal.

The first-encountered unequal UTF-8 character in the operands is compared to determine the relation of the operands. The operand that contains the UTF-8 character with the higher collating value is the greater operand.

Note: The higher collating value is determined using the hexadecimal value of characters, and the PROGRAM COLLATING SEQUENCE clause has no effect on comparisons of UTF-8 operands.

Numeric comparisons

A numeric comparison is a comparison of the algebraic value of two operands of class numeric.

When the algebraic values of numeric operands are compared:

- The length (number of digits) of the operands is not significant.
- The usage of the operands is not significant.
- Unsigned numeric operands are considered positive.
- All zero values compare equal; the presence or absence of a sign does not affect the result.

The behavior of numeric comparisons depends on the settings of the NUMPROC and INVDATA compiler options. For details, see *NUMPROC* and *INVDATA* in the *Enterprise COBOL Programming Guide*.

Group comparisons

A group comparison is a comparison of the alphanumeric character values of two group item operands.

For the comparison operation between fixed-length groups, each operand is treated as though it were an elementary data item of category alphanumeric of the same size as the operand, in bytes. The comparison then proceeds as for two elementary operands of category alphanumeric, as described in [“Alphanumeric comparisons”](#) on page 276.

Comparing a dynamic-length group with another group is not allowed.

Usage note: There is no conversion of data for group comparisons. The comparison operates on bytes of data without regard to data representation. The result of comparing an elementary item or literal operand to an alphanumeric group item is predictable when that operand and the content of the group item have the same data representation.

Comparison of index-names and index data items

Comparisons involving index-names, index data items, or both conform to rules.

The rules for comparisons are:

- The comparison of two index-names is actually the comparison of the corresponding occurrence numbers.
- In the comparison of an index-name with a data item (other than an index data item), or in the comparison of an index-name with a literal, the occurrence number that corresponds to the value of the index-name is compared with the data item or literal.
- In the comparison of an index-name with an arithmetic expression, the occurrence number that corresponds to the value of the index-name is compared with the arithmetic expression.

Because an integer function can be used wherever an arithmetic expression can be used, you can compare an index-name to an integer or numeric function.

- In the comparison of an index data item with an index-name or another index data item, the actual values are compared without conversion. Results of any other comparison involving an index data item are undefined.

Valid comparisons for index-names and index data items are shown in the following table.

<i>Table 25. Comparisons for index-names and index data items</i>					
Operands compared	Index-name	Index data item	Data-name (numeric integer only)	Literal (numeric integer only)	Arithmetic Expression
Index-name	Compare occurrence number	Compare without conversion	Compare occurrence number with content of referenced data item	Compare occurrence number with literal	Compare occurrence number with arithmetic expression
Index data item	Compare without conversion	Compare without conversion	Invalid	Invalid	Invalid

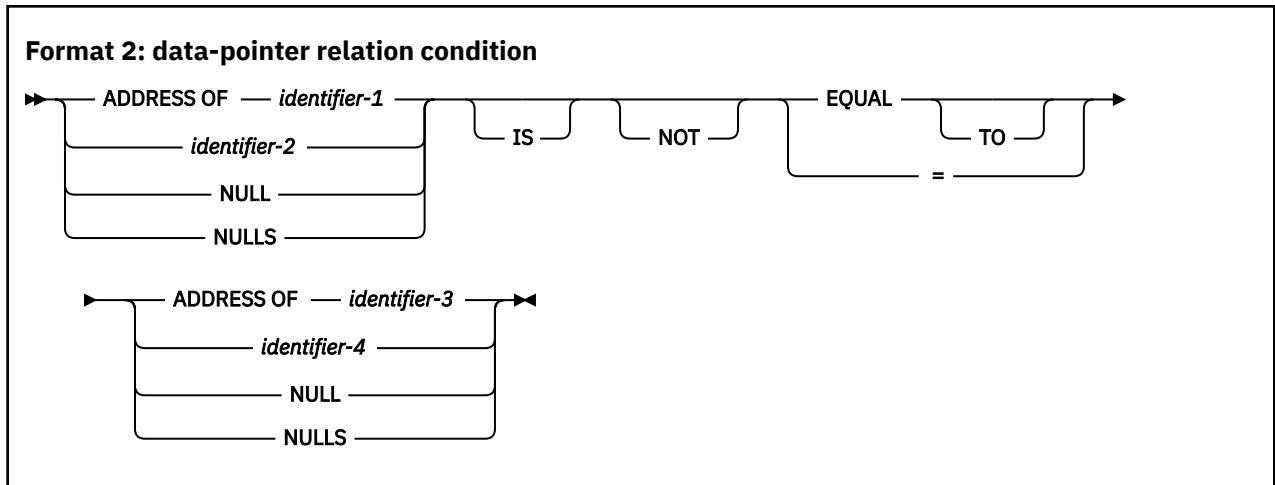
Data pointer relation conditions

Only EQUAL and NOT EQUAL are allowed as relational operators when specifying pointer data items.

Pointer data items are items defined explicitly as USAGE POINTER, or are ADDRESS OF special registers, which are implicitly defined as USAGE POINTER.

The operands are equal if the two addresses used in the comparison would both result in the same storage location.

This relation condition is allowed in IF, PERFORM, EVALUATE, and SEARCH format-1 statements. It is not allowed in SEARCH format-2 (SEARCH ALL) statements because there is no meaningful ordering that can be applied to pointer data items.



identifier-1 , identifier-3

Can specify any level item defined in the LINKAGE SECTION, except 66 and 88.

identifier-2 , identifier-4

Must be described as USAGE POINTER.

NULL, NULLS

Can be used only if the other operand is defined as USAGE POINTER. That is, NULL=NULL is not allowed.

The following table summarizes the permissible comparisons for USAGE POINTER, NULL, and ADDRESS OF.

<i>Table 26. Permissible comparisons for USAGE POINTER, NULL, and ADDRESS OF</i>			
Permissible comparisons	USAGE POINTER second operand	ADDRESS OF second operand	NULL or NULLS second operand
USAGE POINTER first operand	Yes	Yes	Yes
ADDRESS OF first operand	Yes	Yes	Yes
NULL/NULLS first operand	Yes	Yes	No
Yes Comparison allowed only for EQUAL, NOT EQUAL No No comparison allowed			

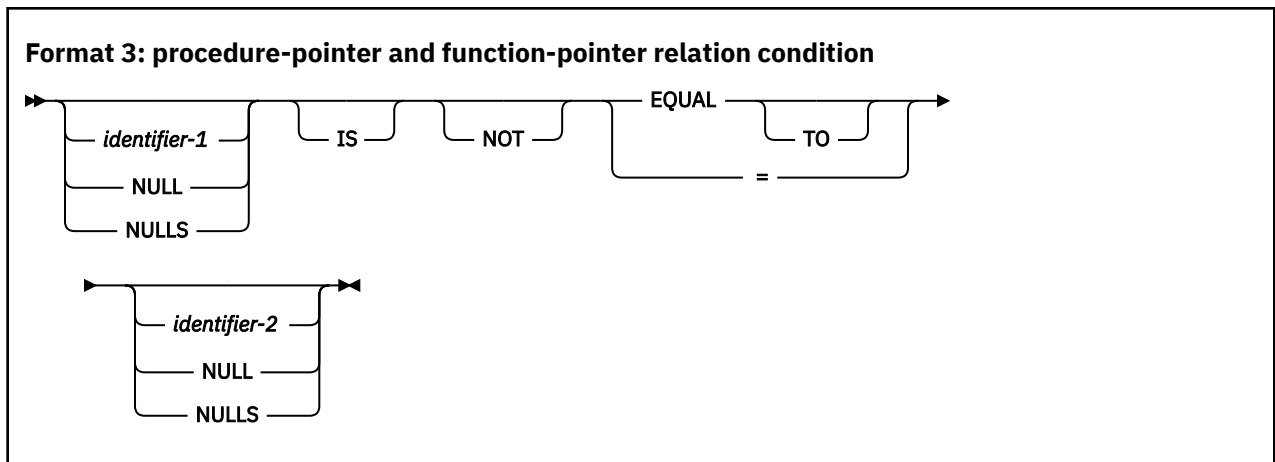
Procedure-pointer and function-pointer relation conditions

Only EQUAL and NOT EQUAL are allowed as relational operators when specifying procedure-pointer or function-pointer data items in a relation condition.

Procedure-pointer data items are defined explicitly as USAGE PROCEDURE-POINTER. Function-pointer data items are defined explicitly as USAGE FUNCTION-POINTER.

The operands are equal if the two addresses used in the comparison would both result in the same storage location.

This relation condition is allowed in IF, PERFORM, EVALUATE, and SEARCH format-1 statements. It is not allowed in SEARCH format-2 (SEARCH ALL) statements, because there is no meaningful ordering that can be applied to procedure-pointer data items.



identifier-1 , identifier-2

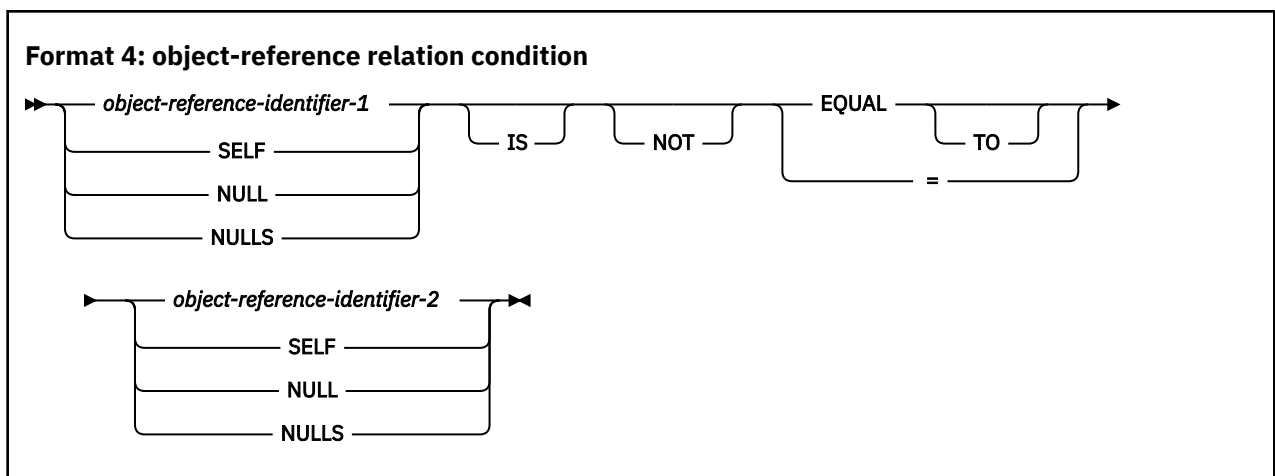
Must be described as USAGE PROCEDURE-POINTER or USAGE FUNCTION-POINTER. *identifier-1* and *identifier-2* need not be described the same.

NULL, NULLS

Can be used only if the other operand is defined as USAGE FUNCTION-POINTER or USAGE PROCEDURE-POINTER. That is, NULL=NULL is not allowed.

Object-reference relation conditions

A data item of usage OBJECT REFERENCE can be compared for equality or inequality with another data item of usage OBJECT REFERENCE or with NULL, NULLS, or SELF.

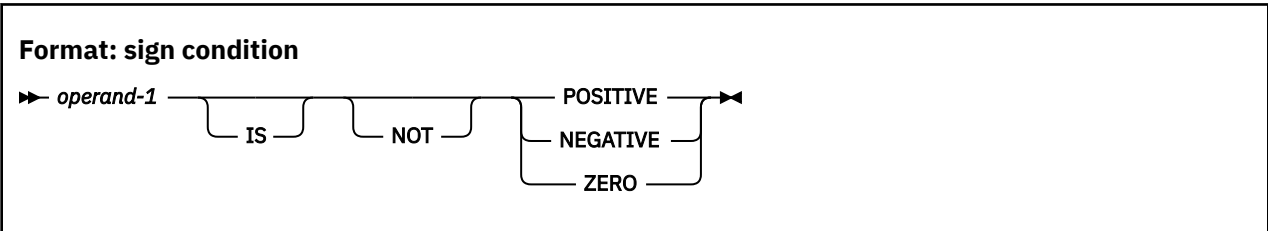


A comparison with SELF is allowed only in a method.

Two object-references compare equal only if the data items identify the same object.

Sign condition

The sign condition determines whether the algebraic value of a numeric operand is greater than, less than, or equal to zero.



operand-1

Must be defined as a numeric identifier, or as an arithmetic expression that contains at least one reference to a variable. *operand-1* can be defined as a floating-point identifier.

The operand is:

- POSITIVE if its value is greater than zero
- NEGATIVE if its value is less than zero
- ZERO if its value is equal to zero

An unsigned operand is either POSITIVE or ZERO.

NOT

One algebraic test is executed for the truth value of the sign condition. For example, NOT ZERO is regarded as true when the operand tested is positive or negative in value.

The results of the sign condition test depend on the setting of the NUMPROC compiler option. For details, see *NUMPROC* in the *Enterprise COBOL Programming Guide*.

Switch-status condition

The switch-status condition determines the on or off status of a UPSI switch.



condition-name

Must be defined in the special-names paragraph as associated with the on or off value of an UPSI switch. (See [“SPECIAL-NAMES paragraph” on page 124.](#))

The switch-status condition tests the value associated with *condition-name*. (The value is considered to be alphanumeric.) The result of the test is true if the UPSI switch is set to the value (0 or 1) corresponding to *condition-name*.

Complex conditions

A complex condition is formed by combining simple conditions, combined conditions, or complex conditions with logical operators, or negating those conditions with logical negation.

Each logical operator must be preceded and followed by a space. The following table shows the logical operators and their meanings.

Table 27. **Logical operators and their meanings**

Logical operator	Name	Meaning
AND	Logical conjunction	The truth value is true when both conditions are true.
OR	Logical inclusive OR	The truth value is true when either or both conditions are true.
NOT	Logical negation	Reversal of truth value (the truth value is true if the condition is false).

Unless modified by parentheses, the following list is the order of precedence (from highest to lowest):

1. Arithmetic operations
2. Simple conditions
3. NOT
4. AND
5. OR

The truth value of a complex condition (whether parenthesized or not) is the truth value that results from the interaction of all the stated logical operators on either of the following options:

- The individual truth values of simple conditions
- The intermediate truth values of conditions logically combined or logically negated

A complex condition can be either of the following options:

- A negated simple condition
- A combined condition (which can be negated)

Negated simple conditions

A simple condition is negated through the use of the logical operator NOT.

Format

➡ NOT — *condition-1* ➡

The negated simple condition gives the opposite truth value of the simple condition. That is, if the truth value of the simple condition is true, then the truth value of that same negated simple condition is false, and vice versa.

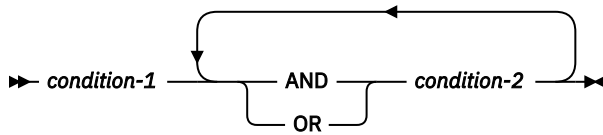
Placing a negated simple condition within parentheses does not change its truth value. That is, the following two statements are equivalent:

```
NOT A IS EQUAL TO B.
NOT (A IS EQUAL TO B).
```


Combined conditions

Two or more conditions can be logically connected to form a combined condition.

Format



The condition to be combined can be any of the following ones:

- A simple-condition
- A negated simple-condition
- A combined condition
- A negated combined condition (that is, the NOT logical operator followed by a combined condition enclosed in parentheses)
- A combination of the preceding conditions that is specified according to the rules in the following table

Combined condition element	Left most	When not leftmost, can be immediately preceded by:	Right most	When not rightmost, can be immediately followed by:
simple-condition	Yes	OR NOT AND (	Yes	OR AND)
OR AND	No	simple-condition)	No	simple-condition NOT (
NOT	Yes	OR AND (	No	simple-condition (
(	Yes	OR NOT AND (	No	simple-condition NOT (
)	No	simple-condition)	Yes	OR AND)

Parentheses are never needed when either ANDs or ORs (but not both) are used exclusively in one combined condition. However, parentheses might be needed to modify the implicit precedence rules to maintain the correct logical relation of operators and operands.

There must be a one-to-one correspondence between left and right parentheses, with each left parenthesis to the left of its corresponding right parenthesis.

The following table illustrates the relationships between logical operators and conditions C1 and C2.

<i>Table 29. Logical operators and evaluation results of combined conditions</i>							
Value for C1	Value for C2	C1 AND C2	C1 OR C2	NOT (C1 AND C2)	NOT C1 AND C2	NOT (C1 OR C2)	NOT C1 OR C2
True	True	True	True	False	False	False	True
False	True	False	True	True	True	False	True
True	False	False	True	True	False	False	False
False	False	False	False	True	False	True	True

Order of evaluation of conditions

Parentheses, both explicit and implicit, define the level of inclusiveness within a complex condition. Two or more conditions connected by only the logical operators AND or OR at the same level of inclusiveness establish a hierarchical level within a complex condition. Therefore an entire complex condition is a nested structure of hierarchical levels, with the entire complex condition being the most inclusive hierarchical level.

Within this context, the evaluation of the conditions within an entire complex condition begins at the left of the condition. The constituent connected conditions within a hierarchical level are evaluated in order from left to right, and evaluation of that hierarchical level terminates as soon as a truth value for it is determined, regardless of whether all the constituent connected conditions within that hierarchical level have been evaluated.

Values are established for arithmetic expressions and functions if and when the conditions that contain them are evaluated. Similarly, negated conditions are evaluated if and when it is necessary to evaluate the complex condition that they represent. For example:

```
NOT A IS GREATER THAN B OR A + B IS EQUAL TO C AND D IS POSITIVE
```

is evaluated as if parenthesized as follows:

```
(NOT (A IS GREATER THAN B)) OR  
(((A + B) IS EQUAL TO C) AND (D IS POSITIVE))
```

Order of evaluation:

1. (NOT (A IS GREATER THAN B)) is evaluated, giving some intermediate truth value, *t1*. If *t1* is true, the combined condition is true, and no further evaluation takes place. If *t1* is false, evaluation continues as follows.
2. (A + B) is evaluated, giving some intermediate result, *x*.
3. (*x* IS EQUAL TO C) is evaluated, giving some intermediate truth value, *t2*. If *t2* is false, the combined condition is false, and no further evaluation takes place. If *t2* is true, the evaluation continues as follows.
4. (D IS POSITIVE) is evaluated, giving some intermediate truth value, *t3*. If *t3* is false, the combined condition is false. If *t3* is true, the combined condition is true.

Example

The following example depicts an IF statement with two conditions combined with the logical AND operator.

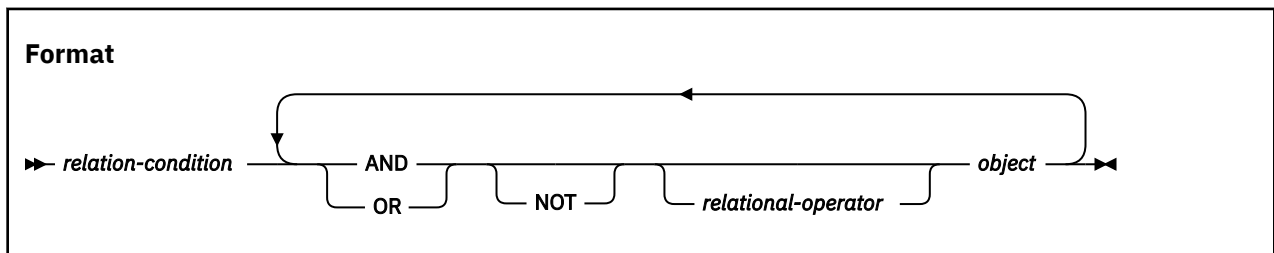
```
IDENTIFICATION DIVISION.  
PROGRAM-ID. COMBINE.  
  
DATA DIVISION.  
WORKING-STORAGE SECTION.  
01 N1 PIC 9(2) VALUE 30.  
01 N2 PIC 9(2) VALUE 45.  
01 N3 PIC 9(2) VALUE 30.  
  
PROCEDURE DIVISION.  
A000-FIRST-PARA.  
  
    IF N1 IS LESS THAN N2 AND N1 = N3 THEN  
        DISPLAY 'BOTH CONDITIONS OK'  
    ELSE  
        DISPLAY 'EITHER OR BOTH CONDITIONS FAILED'  
    END-IF.  
  
    STOP RUN.
```

The example produces the following output:

```
BOTH CONDITIONS OK
```

Abbreviated combined relation conditions

When relation-conditions are written consecutively, any relation-condition after the first can be abbreviated by omission of the subject, or by omission of the subject and relational operator.



In any consecutive sequence of relation-conditions, both forms of abbreviation can be specified. The abbreviated condition is evaluated as if:

1. The last stated subject is the missing subject.
2. The last stated relational operator is the missing relational operator.

The resulting combined condition must comply with the rules for element sequences in combined conditions, as shown in [“Combined conditions” on page 285](#).

If NOT is immediately followed by GREATER THAN, >, LESS THAN, <, EQUAL TO, or =, then the NOT participates as part of the relational operator. NOT in any other position is considered a logical operator (and thus results in a negated relation condition).

Using parentheses

You can use parentheses in combined relation conditions to specify an intended order of evaluation. Using parentheses can also help improve the readability of conditional expressions.

The following rules govern the use of parentheses in abbreviated combined relation conditions:

1. Parentheses can be used to change the order of evaluation of the logical operators AND and OR.
2. The word NOT participates as part of the relational operator when it is immediately followed by GREATER THAN, >, LESS THAN, <, EQUAL TO, or =.

3. NOT in any other position is considered a logical operator and thus results in a negated relation condition. If you use NOT as a logical operator, only the relation condition immediately following the NOT is negated; the negation is not propagated through the abbreviated combined relation condition along with the subject and relational operator.
4. The logical NOT operator can appear within a parenthetical expression that immediately follows a relational operator.
5. When a left parenthesis appears immediately after the relational operator, the relational operator is distributed to all objects enclosed in the parentheses. In the case of a "distributed" relational operator, the subject and relational operator remain current after the right parenthesis which ends the distribution. The following three restrictions apply to cases where the relational operator is distributed throughout the expression:
 - a. A simple condition cannot appear within the scope of the distribution.
 - b. Another relational operator cannot appear within the scope of the distribution.
 - c. The logical operator NOT cannot appear immediately after the left parenthesis, which defines the scope of the distribution.
6. Evaluation proceeds from the least to the most inclusive condition.
7. There must be a one-to-one correspondence between left and right parentheses, with each left parenthesis to the left of its corresponding right parenthesis. If the parentheses are unbalanced, the compiler inserts a parenthesis and issues an E-level message. However, if the compiler-inserted parenthesis results in the truncation of the expression, you will receive an S-level diagnostic message.
8. The last stated subject is inserted in place of the missing subject.
9. The last stated relational operator is inserted in place of the missing relational operator.
10. Insertion of the omitted subject or relational operator ends when:
 - a. Another simple condition is encountered.
 - b. A condition-name is encountered.
 - c. A right parenthesis is encountered that matches a left parenthesis that appears to the left of the subject.
11. In any consecutive sequence of relation conditions, you can use both abbreviated relation conditions that contain parentheses and those that do not.
12. Consecutive logical NOT operators cancel each other and result in an S-level message. Note, however, that an abbreviated combined relation condition can contain two consecutive NOT operators when the second NOT is part of a relational operator. For example, you can abbreviate the first condition as the second condition listed below.

```
A = B and not A not = C
A = B and not not = C
```

The following table summarizes the rules for forming an abbreviated combined relation condition.

<i>Table 30. Abbreviated combined conditions: permissible element sequences</i>				
Combined condition element	Left- most	When not leftmost, can be immediately preceded by:	Right- most	When not rightmost, can be immediately followed by:
Subject	Yes	NOT (	No	Relational operator

<i>Table 30. Abbreviated combined conditions: permissible element sequences (continued)</i>				
Combined condition element	Left- most	When not leftmost, can be immediately preceded by:	Right- most	When not rightmost, can be immediately followed by:
Object	No	Relational operator AND OR NOT (	Yes	AND OR)
Relational operator	No	Subject AND OR NOT	No	Object (
AND OR	No	Object)	No	Object Relational operator NOT (
NOT	Yes	AND OR (	No	Subject Object Relational operator (
(	Yes	Relational operator AND OR NOT (	No	Subject Object NOT (
)	No	Object)	Yes	AND OR)

The following table shows examples of abbreviated combined relation conditions, with and without parentheses, and their unabbreviated equivalents.

<i>Table 31. Abbreviated combined conditions: unabbreviated equivalents</i>	
Abbreviated combined relation condition	Equivalent
A = B AND NOT < C OR D	((A = B) AND (A NOT < C)) OR (A NOT < D)
A NOT > B OR C	(A NOT > B) OR (A NOT > C)
NOT A = B OR C	(NOT (A = B)) OR (A = C)
NOT (A = B OR < C)	NOT ((A = B) OR (A < C))
NOT (A NOT = B AND C AND NOT D)	NOT (((A NOT = B) AND (A NOT = C)) AND (NOT (A NOT = D))))

Statement categories

There are four categories of COBOL statements: imperative statements, conditional statements, delimited scope statements, and compiler-directing statements. See the following topics for more details.

Imperative statements

An *imperative statement* either specifies an unconditional action to be taken by the program, or is a conditional statement terminated by its explicit scope terminator.

A series of imperative statements can be specified wherever an imperative statement is allowed. A conditional statement that is terminated by its explicit scope terminator is also classified as an imperative statement.

For more information about explicit scope terminator, see [“Delimited scope statements” on page 293](#).

The following lists contain the COBOL imperative statements.

Arithmetic

- ADD¹
- COMPUTE¹
- DIVIDE¹
- MULTIPLY¹
- SUBTRACT¹

1. Without the ON SIZE ERROR or the NOT ON SIZE ERROR phrase.

Data movement

- ACCEPT (DATE, DAY, DAY-OF-WEEK, TIME)
- INITIALIZE
- INSPECT
- JSON GENERATE³
- JSON PARSE³
- MOVE
- SET
- STRING²
- UNSTRING²
- XML GENERATE³
- XML PARSE³

2. Without the ON OVERFLOW or the NOT ON OVERFLOW phrase.

3. Without the ON EXCEPTION or NOT ON EXCEPTION phrase.

Ending

- STOP RUN
- EXIT PROGRAM
- EXIT METHOD
- GOBACK

Input-output

- ACCEPT *identifier*
- CLOSE
- DELETE⁴
- DISPLAY
- OPEN
- READ⁵
- REWRITE⁴
- START⁴
- STOP *literal*
- WRITE⁶

4. Without the INVALID KEY or the NOT INVALID KEY phrase.

5. Without the AT END or NOT AT END, and INVALID KEY or NOT INVALID KEY phrases.

6. Without the INVALID KEY or NOT INVALID KEY, and END-OF-PAGE or NOT END-OF-PAGE phrases.

Ordering

- ALLOCATE
- Format 1 SORT
- FREE
- MERGE
- RELEASE
- RETURN⁷

7. Without the AT END or NOT AT END phrase.

Procedure-branching

- ALTER
- CONTINUE
- Format 1 EXIT
- GO TO
- PERFORM

Program or method linkage

- CALL⁸
- CANCEL
- INVOKE

8. Without the ON OVERFLOW phrase, and without the ON EXCEPTION or NOT ON EXCEPTION phrase.

Table-handling

- Format 2 SORT (table SORT)
- SET

Conditional statements

A *conditional statement* specifies that the truth value of a condition is to be determined and that the subsequent action of the object program is dependent on this truth value.

For more information about conditional expressions, see [“Conditional expressions” on page 268](#).

The following lists contain COBOL statements that become conditional when a *condition* (for example, ON SIZE ERROR or ON OVERFLOW) is included and when the statement is not terminated by its explicit scope terminator.

Arithmetic

- ADD ... ON SIZE ERROR
- ADD ... NOT ON SIZE ERROR
- COMPUTE ... ON SIZE ERROR
- COMPUTE ... NOT ON SIZE ERROR
- DIVIDE ... ON SIZE ERROR
- DIVIDE ... NOT ON SIZE ERROR
- MULTIPLY ... ON SIZE ERROR
- MULTIPLY ... NOT ON SIZE ERROR
- SUBTRACT ... ON SIZE ERROR
- SUBTRACT ... NOT ON SIZE ERROR

Data movement

- JSON GENERATE ... ON EXCEPTION
- JSON GENERATE ... NOT ON EXCEPTION
- JSON PARSE ... ON EXCEPTION
- JSON PARSE ... NOT ON EXCEPTION
- STRING ... ON OVERFLOW
- STRING ... NOT ON OVERFLOW
- UNSTRING ... ON OVERFLOW
- UNSTRING ... NOT ON OVERFLOW
- XML GENERATE ... ON EXCEPTION
- XML GENERATE ... NOT ON EXCEPTION
- XML PARSE ... ON EXCEPTION
- XML PARSE ... NOT ON EXCEPTION

Decision

- IF
- EVALUATE

Input-output

- DELETE ... INVALID KEY
- DELETE ... NOT INVALID KEY
- READ ... AT END
- READ ... NOT AT END
- READ ... INVALID KEY

- READ ... NOT INVALID KEY
- REWRITE ... INVALID KEY
- REWRITE ... NOT INVALID KEY
- START ... INVALID KEY
- START ... NOT INVALID KEY
- WRITE ... AT END-OF-PAGE
- WRITE ... NOT AT END-OF-PAGE
- WRITE ... INVALID KEY
- WRITE ... NOT INVALID KEY

Ordering

- RETURN ... AT END
- RETURN ... NOT AT END

Program or method linkage

- CALL ... ON OVERFLOW
- CALL ... ON EXCEPTION
- CALL ... NOT ON EXCEPTION
- INVOKE ... ON EXCEPTION
- INVOKE ... NOT ON EXCEPTION

Table-handling

- SEARCH

Delimited scope statements

In general, a DELIMITED SCOPE statement uses an explicit scope terminator to turn a conditional statement into an imperative statement.

The resulting imperative statement can then be nested. Explicit scope terminators can also be used to terminate the scope of an imperative statement. Explicit scope terminators are provided for all COBOL statements that can have conditional phrases.

Unless explicitly specified otherwise, a delimited scope statement can be specified wherever an imperative statement is allowed by the rules of the language.

Explicit scope terminators

An *explicit scope terminator* marks the end of certain PROCEDURE DIVISION statements.

A conditional statement that is delimited by its explicit scope terminator is considered an imperative statement and must follow the rules for imperative statements.

These are the explicit scope terminators:

- END-ADD
- END-CALL
- END-COMPUTE
- END-DELETE
- END-DIVIDE
- END-EVALUATE

- END-IF
- END-INVOKE
- END-JSON
- END-MULTIPLY
- END-PERFORM
- END-READ
- END-RETURN
- END-REWRITE
- END-SEARCH
- END-START
- END-STRING
- END-SUBTRACT
- END-UNSTRING
- END-WRITE
- END-XML

Implicit scope terminators

At the end of any sentence, an *implicit scope terminator* is a separator period that terminates the scope of all previous statements not yet terminated.

An unterminated conditional statement cannot be contained by another statement.

Except for nesting conditional statements within IF statements, nested statements must be imperative statements and must follow the rules for imperative statements. You should not nest conditional statements.

Related references

RULES (Enterprise COBOL Programming Guide)

Compiler-directing statements

A compiler-directing statement is a statement that causes the compiler to take a specific action during compilation.

For more information about statements that direct the compiler to take a specified action, see [Chapter 114, “Compiler-directing statements,”](#) on page 693.

Statement operations

The topic shows types of operations performed by COBOL statements.

COBOL statements perform the following types of operations:

- Arithmetic
- Data manipulation
- Input/output
- Procedure branching

There are several phrases common to arithmetic and data manipulation statements, such as:

- CORRESPONDING phrase
- GIVING phrase
- ROUNDED phrase
- SIZE ERROR phrases

CORRESPONDING phrase

The CORRESPONDING (CORR) phrase causes ADD, SUBTRACT, and MOVE operations to be performed on elementary data items of the same name if the alphanumeric group item or national group item to which they belong is specified.

A national group is processed as a group item when the CORRESPONDING phrase is used.

Both identifiers that follow the keyword CORRESPONDING must name group items. In this discussion, these identifiers are referred to as *identifier-1* and *identifier-2*. *identifier-1* references the sending group item. *identifier-2* references the receiving group item.

Two subordinate data items, one from *identifier-1* and one from *identifier-2*, correspond if the following conditions are true:

- In an ADD or SUBTRACT statement, both of the data items are elementary numeric data items. Other data items are ignored.
- In a MOVE statement, at least one of the data items is an elementary item, and the move is permitted by the move rules.
- The two subordinate items have the same name and the same qualifiers up to but not including *identifier-1* and *identifier-2*.
- The subordinate items are not identified by the keyword FILLER.
- Neither *identifier-1* nor *identifier-2* is described as a level 66, 77, or 88 item, and neither is described as an index data item. Neither *identifier-1* nor *identifier-2* can be reference-modified.
- Neither *identifier-1* nor *identifier-2* is described with USAGE POINTER, USAGE FUNCTION-POINTER, USAGE PROCEDURE-POINTER, or USAGE OBJECT REFERENCE.
- The subordinate items do not include a REDEFINES, RENAMES, OCCURS, USAGE INDEX, USAGE POINTER, USAGE PROCEDURE-POINTER, USAGE FUNCTION-POINTER, or USAGE OBJECT REFERENCE clause in their descriptions.

However, *identifier-1* and *identifier-2* themselves can contain or be subordinate to items that contain a REDEFINES or OCCURS clause in their descriptions.

- The name of each subordinate data item that satisfies these conditions is unique after application of implicit qualifiers.

identifier-1, *identifier-2*, or both can be subordinate to a FILLER item.

For example, consider two data hierarchies defined as follows:

```
05 ITEM-1 OCCURS 6.  
  10 ITEM-A PIC S9(3).  
  10 ITEM-B PIC +99.9.  
  10 ITEM-C PIC X(4).  
  10 ITEM-D REDEFINES ITEM-C PIC 9(4).  
  10 ITEM-E USAGE COMP-1.  
  10 ITEM-F USAGE INDEX.  
05 ITEM-2.  
  10 ITEM-A PIC 99.  
  10 ITEM-B PIC +9V9.  
  10 ITEM-C PIC A(4).  
  10 ITEM-D PIC 9(4).  
  10 ITEM-E PIC 9(9) USAGE COMP.  
  10 ITEM-F USAGE INDEX.
```

If ADD CORR ITEM-2 TO ITEM-1(x) is specified, ITEM-A and ITEM-A(x), ITEM-B and ITEM-B(x), and ITEM-E and ITEM-E(x) are considered to be corresponding and are added together. ITEM-C and ITEM-C(x) are not included because they are not numeric. ITEM-D and ITEM-D(x) are not included because ITEM-D(x) includes a REDEFINES clause in its data description. ITEM-F and ITEM-F(x) are not included because they are index data items. Note that ITEM-1 is valid as either *identifier-1* or *identifier-2*.

If any of the individual operations in the ADD CORRESPONDING statement produces a size error condition, *imperative-statement-1* in the ON SIZE ERROR phrase is not executed until all of the individual additions are completed.

GIVING phrase

For arithmetic statements, the value of the identifier that follows the word GIVING is set equal to the calculated result of the arithmetic operation. Because this identifier is not involved in the computation, it can be a numeric-edited item.

ROUNDED phrase

After decimal point alignment, the number of places in the fraction of the result of an arithmetic operation is compared with the number of places provided for the fraction of the resultant identifier.

When the size of the fractional result exceeds the number of places provided for its storage, truncation occurs unless ROUNDED is specified. When ROUNDED is specified, the least significant digit of the resultant identifier is increased by 1 whenever the most significant digit of the excess is greater than or equal to 5.

When the resultant identifier is described by a PICTURE clause that contains rightmost Ps and when the number of places in the calculated result exceeds the number of integer positions specified, rounding or truncation occurs relative to the rightmost integer position for which storage is allocated.

In a floating-point arithmetic operation, the ROUNDED phrase has no effect; the result of a floating-point operation is always rounded. For more information on floating-point arithmetic expressions, see *Fixed-point contrasted with floating-point arithmetic* in the *Enterprise COBOL Programming Guide*.

When the ARITH(EXTEND) compiler option is in effect, the ROUNDED phrase is not supported for arithmetic receivers with 31 digit positions to the right of the decimal point. For example, neither X nor Y below is valid as a receiver with the ROUNDED phrase:

```
01 X PIC V31.
01 Y PIC P(30)9(1).

  COMPUTE X ROUNDED = A + B
  COMPUTE Y ROUNDED = A - B
```

Otherwise, the ROUNDED phrase is fully supported for extended-precision arithmetic statements.

SIZE ERROR phrases

A size error condition can occur in different ways.

- When the absolute value of the result of an arithmetic evaluation, after decimal point alignment, exceeds the largest value that can be contained in the result field.
- When division by zero occurs.
- In an exponential expression, as indicated in the following table:

Table 32. <i>Exponentiation size error conditions</i>		
Size error	Action taken when a SIZE ERROR clause is present	Action taken when a SIZE ERROR clause is not present
Zero raised to zero power	The SIZE ERROR imperative is executed.	The value returned is 1, and a message is issued.
Zero raised to a negative number	The SIZE ERROR imperative is executed.	The program is terminated abnormally.

Table 32. <i>Exponentiation size error conditions</i> (continued)		
Size error	Action taken when a SIZE ERROR clause is present	Action taken when a SIZE ERROR clause is not present
A negative number raised to a fractional power	The SIZE ERROR imperative is executed.	The absolute value of the base is used, and a message is issued.

The size error condition applies only to final results, not to any intermediate results.

If the resultant identifier is defined with usage BINARY, COMPUTATIONAL, COMPUTATIONAL-4, or COMPUTATIONAL-5, the largest value that the resultant data item can contain is the value implied by the item's decimal PICTURE character-string, regardless of the TRUNC compiler option in effect.

If the ROUNDED phrase is specified, rounding takes place before size error checking.

When a size error occurs, the subsequent action of the program depends on whether the ON SIZE ERROR phrase is specified.

If the ON SIZE ERROR phrase is not specified and a size error condition occurs, truncation rules apply and the value of the affected resultant identifier is computed.

If the ON SIZE ERROR phrase is specified and a size error condition occurs, the value of the resultant identifier affected by the size error is not altered; that is, the error results are not placed in the receiving identifier. After completion of the execution of the arithmetic operation, the imperative statement in the ON SIZE ERROR phrase is executed, control is transferred to the end of the arithmetic statement, and the NOT ON SIZE ERROR phrase, if specified, is ignored.

For ADD CORRESPONDING and SUBTRACT CORRESPONDING statements, if an individual arithmetic operation causes a size error condition, the ON SIZE ERROR imperative statement is not executed until all the individual additions or subtractions have been completed.

If the NOT ON SIZE ERROR phrase has been specified and, after execution of an arithmetic operation, a size error condition does not exist, the NOT ON SIZE ERROR phrase is executed.

When both the ON SIZE ERROR and NOT ON SIZE ERROR phrases are specified, and the statement in the phrase that is executed does not contain any explicit transfer of control, then if necessary an implicit transfer of control is made after execution of the phrase to the end of the arithmetic statement.

Arithmetic statements

The arithmetic statements are used for computations. Individual operations are specified by the ADD, SUBTRACT, MULTIPLY, and DIVIDE statements. These individual operations can be combined symbolically in a formula that uses the COMPUTE statement.

Arithmetic statement operands

The data descriptions of operands in an arithmetic statement need not be the same. Throughout the calculation, the compiler performs any necessary data conversion and decimal point alignment.

Size of operands

If the ARITH(COMPAT) compiler option is in effect, the maximum size of each operand is 18 decimal digits. If the ARITH(EXTEND) compiler option is in effect, the maximum size of each operand is 31 decimal digits.

The *composite of operands* is a hypothetical data item resulting from aligning the operands at the decimal point and then superimposing them on one another.

If the ARITH(COMPAT) compiler option is in effect, the composite of operands can be a maximum of 30 digits. If the ARITH(EXTEND) compiler option is in effect, the composite of operands can be a maximum of 31 digits.

The following table shows how the composite of operands is determined for arithmetic statements:

<i>Table 33. How the composite of operands is determined</i>	
Statement	Determination of the composite of operands
SUBTRACT ADD	Superimposing all operands in a given statement except those following the word GIVING
MULTIPLY	Superimposing all receiving data items
DIVIDE	Superimposing all receiving data items except the REMAINDER data item
COMPUTE	Restriction does not apply

For example, assume that each item is defined as follows in the DATA DIVISION:

```
A PICTURE 9(7)V9(5).
B PICTURE 9(11)V99.
C PICTURE 9(12)V9(3).
```

If the following statement is executed, the composite of operands consists of 17 decimal digits:

```
ADD A B TO C
```

It has the following implicit description:

```
COMPOSITE-OF-OPERANDS PICTURE 9(12)V9(5).
```

In the ADD and SUBTRACT statements, if the composite of operands is 30 digits or less with the ARITH(COMPAT) compiler option, or 31 digits or less with the ARITH(EXTEND) compiler option, the compiler ensures that enough places are carried so that no significant digits are lost during execution.

In all arithmetic statements, it is important to define data with enough digits and decimal places to ensure the required accuracy in the final result. For more information, see *Appendix A. Intermediate results and arithmetic precision* in the *Enterprise COBOL Programming Guide*.

Overlapping operands

When operands in an arithmetic statement share part of their storage (that is, when the operands overlap), the result of the execution of such a statement is unpredictable.

Multiple results

When an arithmetic statement has multiple results, execution conceptually proceeds as follows:

1. The statement performs all arithmetic operations to find the result to be placed in the receiving items, and stores that result in a temporary location.
2. A sequence of statements transfers or combines the value of this temporary result with each single receiving field. The statements are considered to be written in the same left-to-right order in which the multiple results are listed.

For example, executing the following statement:

```
ADD A, B, C, TO C, D(C), E.
```

is equivalent to executing the following series of statements:

```
ADD A, B, C GIVING TEMP.
ADD TEMP TO C.
```

```
ADD TEMP TO D(C) .  
ADD TEMP TO E.
```

In the above example, TEMP is a compiler-supplied temporary result field. When the addition operation for D(C) is performed, the subscript C contains the new value of C.

Data manipulation statements

The following COBOL statements move and inspect data: ACCEPT, INITIALIZE, INSPECT, JSON GENERATE, JSON PARSE, MOVE, READ, RELEASE, RETURN, REWRITE, SET, STRING, UNSTRING, WRITE, XML PARSE, and XML GENERATE.

Overlapping operands

When the sending and receiving fields of a data manipulation statement share a part of their storage (that is, when the operands overlap), the result of the execution of such a statement is unpredictable.

Input-output statements

COBOL input-output statements transfer data to and from files stored on external media, and also control low-volume data that is obtained from or sent to an input/output device.

In COBOL, the unit of file data made available to the program is a record. You need only be concerned with such records. Provision is automatically made for such operations as the movement of data into buffers, internal storage, validity checking, error correction (where feasible), blocking and deblocking, and volume-switching procedures.

The description of the file in the ENVIRONMENT DIVISION and the DATA DIVISION governs which input-output statements are allowed in the PROCEDURE DIVISION. Permissible statements for sequential files are shown in [Table 53 on page 416](#) , and permissible statements for indexed files and relative files are shown in [Table 54 on page 417](#). Permissible statements for line sequential files are shown in [Table 55 on page 417](#).

Common processing facilities

Several common processing facilities apply to more than one input-output statement.

The common processing facilities provided are:

- “File status key” on [page 299](#)
- “Invalid key condition” on [page 303](#)
- “INTO and FROM phrases” on [page 304](#)
- “File position indicator” on [page 305](#)

Discussions in the following sections use the terms *volume* and *reel*. The term *volume* refers to all non-unit-record input-output devices. The term *reel* applies only to tape devices. Treatment of direct-access devices in the sequential access mode is logically equivalent to the treatment of tape devices.

File status key

If the FILE STATUS clause is specified in the file-control entry, a value is placed in the specified file status key (the two-character data item named in the FILE STATUS clause) during execution of any request on that file; the value indicates the status of that request.

The value is placed in the file status key before execution of any EXCEPTION/ERROR declarative, INVALID KEY phrase, or AT END phrase associated with the request.

There are two file status key data-names. One is described by *data-name-1* in the FILE STATUS clause of the file-control entry. This is a two-character data item with the first character known as file status key 1 and the second character known as file status key 2. The combinations of possible values and their meanings are shown in [Table 34 on page 300](#).

The other file status key is described by *data-name-8* in the FILE STATUS clause of the file-control entry. *data-name-8* does not apply to QSAM files. For more information about *data-name-8*, see “[FILE STATUS clause](#)” on page 153.

Table 34. <i>File status key values and meanings</i>			
High-order digit	Meaning	Low-order digit	Meaning
0	Successful completion	0	No further information
		2	This file status value applies only to indexed files with alternate keys that allow duplicates. The input-output statement was successfully executed, but a duplicate key was detected. For a READ statement, the key value for the current key of reference was equal to the value of the same key in the next record within the current key of reference. For a REWRITE or WRITE statement, the record just written created a duplicate key value for at least one alternate record key for which duplicates are allowed.
		4	A READ statement was successfully executed, but the number of character positions that were read was less than the minimum size or was greater than the maximum size specified by the record description entries associated with the FD for the file. Note: If the VLR(COMPAT) option is in effect, you will get the status value of 00 when READ statements encounter a record length conflict. For details about the VLR option, see <i>VLR</i> in the <i>Enterprise COBOL Programming Guide</i> .
		5	An OPEN statement was successfully executed, but the referenced optional file was unavailable at the time the OPEN statement was executed. The file had been created if the open mode was I-O or EXTEND. This does not apply to VSAM sequential files.
		7	For a CLOSE statement with the NO REWIND, REEL/UNIT, or FOR REMOVAL phrase or for an OPEN statement with the NO REWIND phrase, the referenced file was on a non-reel/unit medium.
1	At-end condition	0	A sequential READ statement was attempted and no next logical record existed in the file because the end of the file had been reached. Or the first READ was attempted on an optional input file that was unavailable.
		4	A sequential READ statement was attempted for a relative file, and the number of significant digits in the relative record number was larger than the size of the relative key data item described for the file.

Table 34. *File status key values and meanings (continued)*

High-order digit	Meaning	Low-order digit	Meaning
2	Invalid key condition	1	A sequence error exists for a sequentially accessed indexed file. The prime record key value was changed by the program between the successful execution of a READ statement and the execution of the next REWRITE statement for that file. Or the ascending requirements for successive record key values were violated.
		2	An attempt was made to write a record that would create a duplicate key in a relative file. Or an attempt was made to write or rewrite a record that would create a duplicate prime record key or a duplicate alternate record key without the DUPLICATES phrase in an indexed file.
		3	An attempt was made to randomly access a record that does not exist in the file. Or a START or random READ statement was attempted on an optional input file that was unavailable.
		4	An attempt was made to write beyond the externally defined boundaries of a relative or indexed file. Or a sequential WRITE statement was attempted for a relative file and the number of significant digits in the relative record number was larger than the size of the relative key data item described for the file.
3	Permanent error condition	0	No further information
		4	A permanent error exists because of a boundary violation; an attempt was made to write beyond the externally defined boundaries of a sequential file.
		5	An OPEN statement with the INPUT, I-O, or EXTEND phrase was attempted on a nonoptional file that was unavailable.
		7	An OPEN statement was attempted on a file that would not support the open mode specified in the OPEN statement. Possible violations are: • The EXTEND or OUTPUT phrase was specified but the file would not support write operations. • The I-O phrase was specified but the file would not support the input and output operations permitted. • The INPUT phrase was specified but the file would not support read operations.
		8	An OPEN statement was attempted on a file previously closed with lock.
		9	The OPEN statement was unsuccessful because a conflict was detected between the fixed file attributes and the attributes specified for that file in the program. These attributes include the organization of the file (sequential, relative, or indexed), the prime record key, the alternate record keys, the code set, the maximum record size, the record type (fixed or variable), and the blocking factor.

Table 34. *File status key values and meanings (continued)*

High-order digit	Meaning	Low-order digit	Meaning
4	Logic error condition	1	An OPEN statement was attempted for a file in the open mode.
		2	A CLOSE statement was attempted for a file not in the open mode.
		3	For a mass storage file in the sequential access mode, the last input-output statement executed for the associated file prior to the execution of a REWRITE statement was not a successfully executed READ statement. For relative and indexed files in the sequential access mode, the last input-output statement executed for the file prior to the execution of a DELETE or REWRITE statement was not a successfully executed READ statement.
		4	A boundary violation exists because an attempt was made to rewrite a record to a file and the record was not the same size as the record being replaced. Or an attempt was made to write or rewrite a record that was larger than the largest or smaller than the smallest record allowed by the RECORD IS VARYING clause of the associated file-name.
		6	A sequential READ statement was attempted on a file open in the input or I-O mode and no valid next record had been established because: • The preceding READ statement was unsuccessful but did not cause an at-end condition. • The preceding READ statement caused an at-end condition.
		7	The execution of a READ statement was attempted on a file not open in the input or I-O mode.
		8	The execution of a WRITE statement was attempted on a file not open in the I-O, output, or extend mode.
		9	The execution of a DELETE or REWRITE statement was attempted on a file not open in the I-O mode.

Table 34. <i>File status key values and meanings</i> (continued)			
High-order digit	Meaning	Low-order digit	Meaning
9	Implementor-defined condition	0	- For multithreading only: A CLOSE of a VSAM or QSAM file was attempted on a thread that did not open the file. - Without multithreading: For VSAM only: See the information about VSAM return codes in <i>Using VSAM status codes (VSAM files only)</i> in the <i>Enterprise COBOL Programming Guide</i>. - QSAM files: No further information available. See the DFSMS error message for more information.
		1	For VSAM only: Password failure
		2	Logic error
		3	For all files, except QSAM: Resource unavailable
		5	For all files except QSAM: Invalid or incomplete file information
		6	For VSAM file: An OPEN statement with the OUTPUT phrase was attempted, or an OPEN statement with the I-O or EXTEND phrase was attempted for an optional file but no DD statement was specified for the file. For QSAM file: An OPEN statement with the OUTPUT phrase was attempted, or an OPEN statement with the I-O or EXTEND phrase was attempted for an optional file but no DD statement was specified for the file and the CBLQDA(OFF) runtime option was specified.
		7	For VSAM only: OPEN statement execution successful: File integrity verified Note: If the VSAMOPENFS(SUCC) option is in effect, you will get the status value of 00 when a VSAM OPEN statement is successfully verified. For details about the VSAMOPENFS option, see <i>VSAMOPENFS</i> in the <i>Enterprise COBOL Programming Guide</i>.
		8	Open failed due to the invalid contents of an environment variable specified in a SELECT ... ASSIGN clause or due to dynamic allocation failure. For more information about the contents of environment variables, see “ASSIGN clause” on page 142.

Invalid key condition

The invalid key condition can occur during execution of a START, READ, WRITE, REWRITE, or DELETE statement. When an invalid key condition occurs, the input-output statement that caused the condition is unsuccessful.

When the invalid key condition is recognized, actions are taken in the following order:

1. If the FILE STATUS clause is specified in the file-control entry, a value is placed into the file status key to indicate an invalid key condition, as shown in [Table 34 on page 300](#).
2. If the INVALID KEY phrase is specified in the statement that caused the condition, control is transferred to the INVALID KEY imperative statement. Any EXCEPTION/ERROR declarative procedure specified for this file is not executed. Execution then continues according to the rules for each statement specified in the imperative statement.

3. If the INVALID KEY phrase is not specified in the input-output statement for a file and an applicable EXCEPTION/ERROR procedure exists, that procedure is executed. The NOT INVALID KEY phrase, if specified, is ignored.

Both the INVALID KEY phrase and the EXCEPTION/ERROR procedure can be omitted.

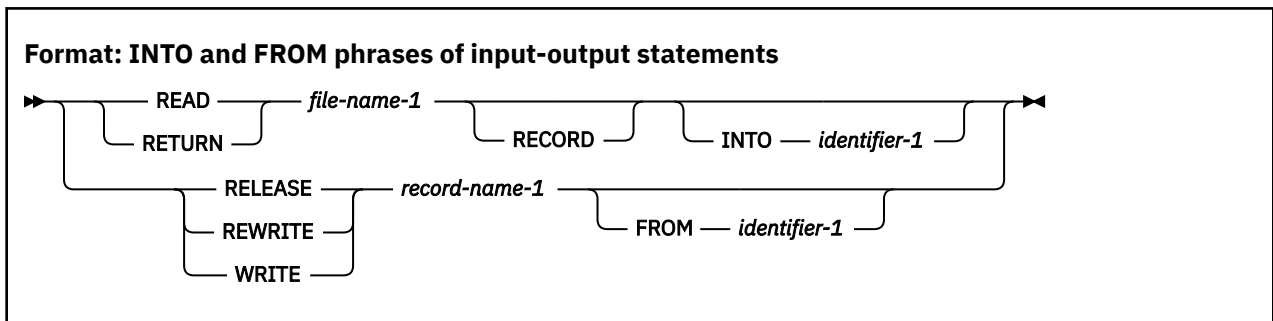
If the invalid key condition does not exist after execution of the input-output operation, the INVALID KEY phrase is ignored, if specified, and the following actions are taken:

- If an exception condition that is not an invalid key condition exists, control is transferred according to the rules of the USE statement following the execution of any USE AFTER EXCEPTION procedure.
- If no exception condition exists, control is transferred to the end of the input-output statement or the imperative statement specified in the NOT INVALID KEY phrase, if it is specified.

INTO and FROM phrases

The INTO and FROM phrases are valid for READ, RETURN, RELEASE, REWRITE, and WRITE statements.

You must specify an identifier that is the name of an entry in the WORKING-STORAGE SECTION or the LINKAGE SECTION, or of a record description for another previously opened file.



- *record-name-1* and *identifier-1* must not refer to the same storage area.
- If *record-name-1* or *identifier-1* refers to a national group item, the item is processed as an elementary data item of category national.
- The INTO phrase can be specified in a READ or RETURN statement.

The result of the execution of a READ or RETURN statement with the INTO phrase is equivalent to the application of the following rules in the order specified:

- The execution of the same READ or RETURN statement without the INTO phrase.
- The current record is moved from the record area to the area specified by *identifier-1* according to the rules for the MOVE statement without the CORRESPONDING phrase. The size of the current record is determined by rules specified in the RECORD clause. If the file description entry contains a RECORD IS VARYING clause, the implied move is a group move. The implied MOVE statement does not occur if the execution of the READ or RETURN statement was unsuccessful. Any subscripting or reference-modification associated with *identifier-1* is evaluated after the record has been read or returned and immediately before it is moved to the data item. The record is available in both the record area and the data item referenced by *identifier-1*.

- The FROM phrase can be specified in a RELEASE, REWRITE, or WRITE statement.

The result of the execution of a RELEASE, REWRITE, or WRITE statement with the FROM phrase is equivalent to the execution of the following statements in the order specified:

1. MOVE *identifier-1* TO *record-name-1*
2. The same RELEASE, REWRITE, or WRITE statement without the FROM phrase

After the execution of the RELEASE, REWRITE or WRITE statement is complete, the information in the area referenced by *identifier-1* is available even though the information in the area referenced by *record-name-1* is unavailable, except as specified by the SAME RECORD AREA clause.

File position indicator

The file position indicator is a conceptual entity used in this document to facilitate exact specification of the next record to be accessed within a given file during certain sequences of input-output operations.

The setting of the file position indicator is affected only by the OPEN, CLOSE, READ and START statements. The concept of a file position indicator has no meaning for a file opened in the output or extend mode.

Related references

[COBOL application receiving file status key 90](#)

Chapter 28. PROCEDURE DIVISION statements

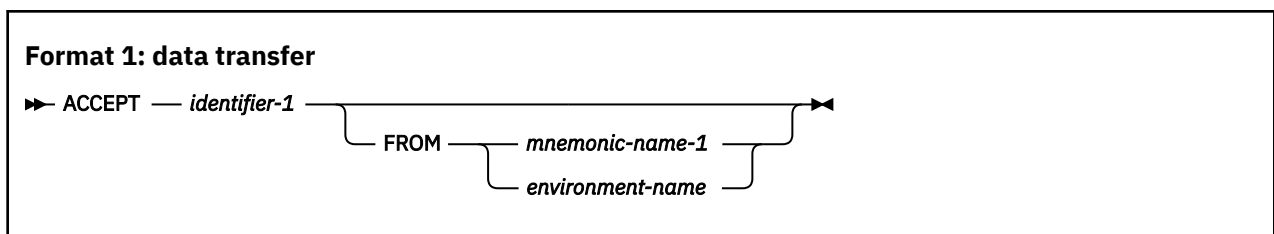
Statements, sentences, and paragraphs in the PROCEDURE DIVISION are executed sequentially except when a procedure branching statement such as EXIT, GO TO, PERFORM, GOBACK, or STOP is used.

ACCEPT statement

The ACCEPT statement transfers data or system date-related information into the data area referenced by the specified identifier. There is no editing or error checking of the incoming data.

Data transfer

Format 1 transfers data from an input source into the data item referenced by *identifier-1* (the receiving area). When the FROM phrase is omitted, the system input device is assumed.



Format 1 is useful for exceptional situations in a program when operator intervention (to supply a given message, code, or exception indicator) is required. The operator must of course be supplied with the appropriate messages with which to reply.

identifier-1

The receiving area. Can be:

- An alphanumeric group item
- A national group item
- An elementary data item of usage DISPLAY, DISPLAY-1, or NATIONAL

A national group item is processed as an elementary data item of category national.

identifier-1 must not be a dynamic-length group item or a dynamic-length elementary item.

mnemonic-name-1

Specifies the input device. *mnemonic-name-1* must be associated in the SPECIAL-NAMES paragraph with an environment-name. See [“SPECIAL-NAMES paragraph” on page 124](#).

- System input device

The length of a data transfer is the same as the length of the record on the input device, with a maximum of 32,760 bytes.

The system input device is read until the receiving area is filled or EOF is encountered. If the length of the receiving area is not an even multiple of the system input device record length, the final record will be truncated as required. If EOF is encountered after data has been moved and before the receiving area has been filled, the receiving area is padded with spaces of the appropriate representation for the receiving area. If EOF is encountered before any data has been moved to the

receiving area, padding will not take place and the contents of the receiving area are unchanged. Each input record is concatenated with the previous input record.

If the input record is of a fixed-length format, the entire input record is used. No editing is performed to remove trailing or leading blanks.

If the input record is of the variable-length format, the actual record length is used to determine the amount of data received. With variable-format records, the Record Definition Word (RDW) is removed from the beginning of the input record. Only the actual input data is transferred to *identifier-1*.

If the data item referenced by *identifier-1* is of usage national, data is transferred without conversion and without checking for validity. The input data is assumed to be in UTF-16 format.

- Console

1. A system-generated message code is automatically displayed, followed by the literal AWAITING REPLY.

The maximum length of an input message is 114 characters.

2. Execution is suspended.

3. After the message code (the same code as in item 1) is entered from the console and recognized by the system, ACCEPT statement execution is resumed. The message is moved to the receiving area and left-justified regardless of its PICTURE clause.

If *identifier-1* references a data item of usage NATIONAL, the message is converted from the native code page representation to national character representation. The native code page is the one that was specified by the CODEPAGE compiler option when the source code was compiled.

The ACCEPT statement is terminated if any of the following conditions occurs:

- No data is received from the console; for example, if the operator hits the Enter key. The target data item does not receive a new value and retains the value it had prior to the ACCEPT.
- The receiving data item is filled with data.
- Fewer than 114 characters of data are entered.

If 114 bytes of data are entered and the receiving area is still not filled with data, more requests for data are issued to the console.

If more than 114 characters of data are entered, only the first 114 characters will be recognized by the system.

If the receiving area is longer than the incoming message, the rightmost characters are padded with spaces of the appropriate representation for the receiving area.

If the incoming message is longer than the receiving area, the character positions beyond the length of the receiving area are truncated.

For information about obtaining ACCEPT input from a z/OS UNIX file or stdin, see *Assigning input from a screen or file (ACCEPT)* in the *Enterprise COBOL Programming Guide*.

environment-name

Identifies the source of input data. An environment-name from the names given in [Table 5 on page 126](#) can be specified.

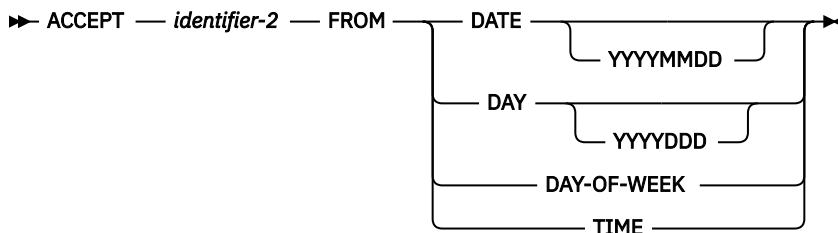
If the device is the same as that used for READ statements for a LINE SEQUENTIAL file, results are unpredictable.

System date-related information transfer

System information contained in the specified conceptual data items DATE, DATE YYYYMMDD, DAY, DAY YYYYDDD, DAY-OF-WEEK, or TIME, can be transferred into the data item referenced by *identifier-2*. The transfer must follow the rules for the MOVE statement without the CORRESPONDING phrase.

For more information, see [“MOVE statement” on page 405](#).

Format 2: system information transfer



identifier-2

The receiving area. Can be:

- An alphanumeric group item
- A national group item
- An elementary data item of one of the following categories:
 - alphanumeric
 - alphanumeric-edited
 - numeric-edited (with usage DISPLAY or NATIONAL)
 - national
 - national-edited
 - numeric
 - internal floating-point
 - external floating-point (with usage DISPLAY or NATIONAL)

A national group item is processed as an elementary data item of category national.

identifier-2 might not be a dynamic-length group item, but can be a dynamic-length elementary item.

Format 2 accesses the current date in two formats: the day of the week or the time of day as carried by the system (which can be useful in identifying when a particular run of an object program was executed). You can also use format 2 to supply the date in headings and footings.

The current date and time can also be accessed with the intrinsic function CURRENT-DATE, which also supports four-digit year values and provides additional information (see [Chapter 42, “CURRENT-DATE,” on page 545](#)).

DATE, DATE YYYYMMDD, DAY, DAY YYYYDDD, DAY-OF-WEEK, and TIME

The conceptual data items DATE, DATE YYYYMMDD, DAY, DAY YYYYDDD, DAY-OF-WEEK, and TIME implicitly have USAGE DISPLAY. Because these are conceptual data items, they cannot be described in the COBOL program.

The content of the conceptual data items is moved to the receiving area using the rules of the MOVE statement. If the receiving area is of usage NATIONAL, the data is converted to national character representation.

DATE

Has the implicit PICTURE 9(6).

The sequence of data elements (from left to right) is:

```
Two digits for the year  
Two digits for the month  
Two digits for the day
```

Thus 27 April 2003 is expressed as 030427.

DATE YYYYMMDD

Has the implicit PICTURE 9(8).

The sequence of data elements (from left to right) is:

```
Four digits for the year  
Two digits for the month  
Two digits for the day
```

Thus 27 April 2003 is expressed as 20030427.

DAY

Has the implicit PICTURE 9(5).

The sequence of data elements (from left to right) is:

```
Two digits for the year  
Three digits for the day
```

Thus 27 April 2003 is expressed as 03117.

DAY YYYYDDD

Has the implicit PICTURE 9(7).

The sequence of data elements (from left to right) is:

```
Four digits for the year  
Three digits for the day
```

Thus 27 April 2003 is expressed as 2003117.

DAY-OF-WEEK

Has the implicit PICTURE 9(1).

The single data element represents the day of the week according to the following values:

1 represents Monday	5 represents Friday
2 represents Tuesday	6 represents Saturday
3 represents Wednesday	7 represents Sunday
4 represents Thursday	

Thus Wednesday is expressed as 3.

TIME

Has the implicit PICTURE 9(8).

The sequence of data elements (from left to right) is:

```
Two digits for hour of day  
Two digits for minute of hour  
Two digits for second of minute  
Two digits for hundredths of second
```

Thus 2:41 PM is expressed as 14410000.

Example of the ACCEPT statement

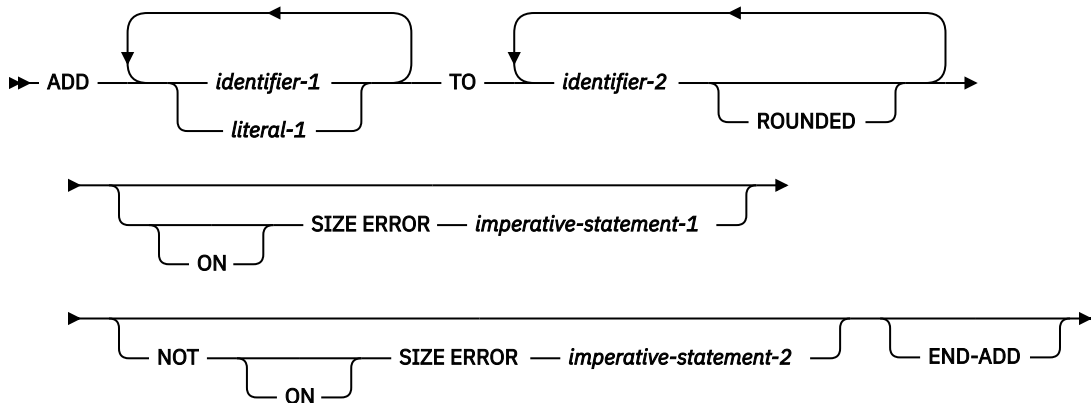
This topic lists an example for the ACCEPT statement.

```
//COBOL.SYSIN DD *
IDENTIFICATION DIVISION.
PROGRAM-ID. ACCPTST.
ENVIRONMENT DIVISION.
DATA DIVISION.
WORKING-STORAGE SECTION.
01 AGE PIC 9(3).
01 GENDER PIC X(1).
PROCEDURE DIVISION.
ACCEPT AGE.
ACCEPT GENDER.
EVALUATE TRUE ALSO TRUE
    WHEN AGE > 60 ALSO GENDER = 'M'
        DISPLAY 'THE MAN IS RETIRED '
    WHEN AGE > 60 ALSO GENDER = 'F'
        DISPLAY 'THE WOMAN IS RETIRED '
    WHEN AGE <= 60 ALSO GENDER = 'M'
        DISPLAY 'THE MAN IS NOT RETIRED '
    WHEN AGE <= 60 ALSO GENDER = 'F'
        DISPLAY 'THE WOMAN IS NOT RETIRED '
    WHEN OTHER
        DISPLAY 'INVALID INPUT '
        DISPLAY 'AGE =' AGE ' and GENDER =' GENDER
END-EVALUATE.
STOP RUN.
/*
//GO.SYSIN DD *
64
M
//*
```

ADD statement

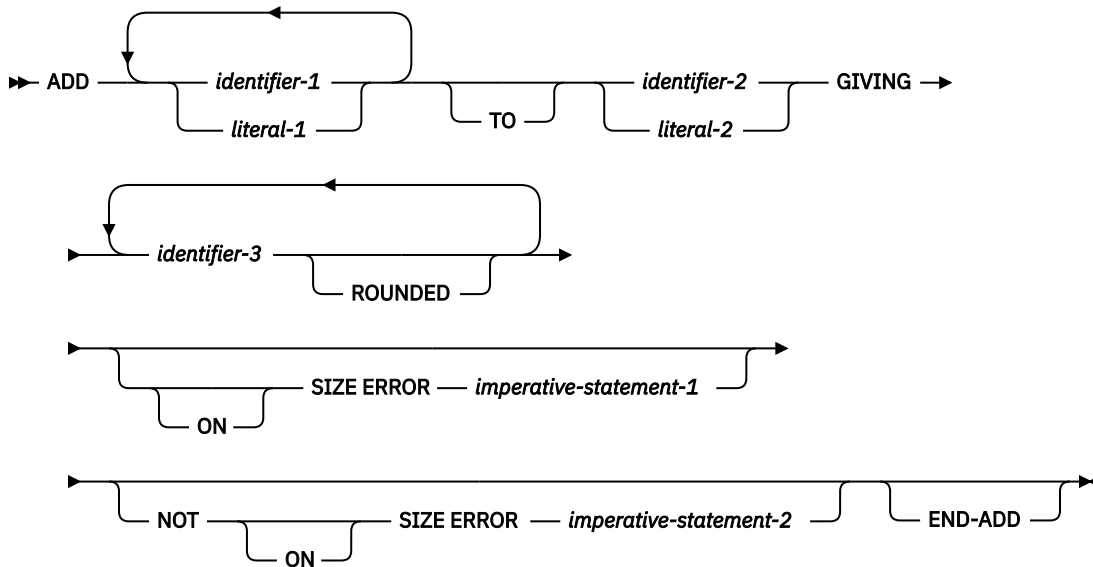
The ADD statement sums two or more numeric operands and stores the result.

Format 1: ADD statement



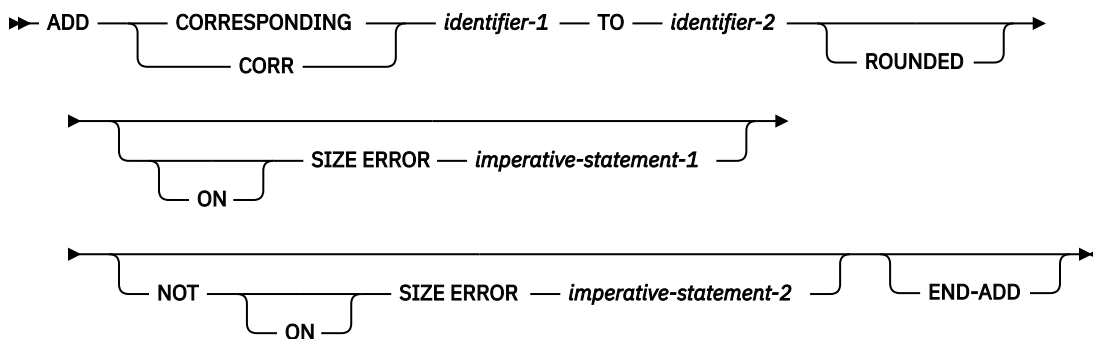
All identifiers or literals that precede the keyword TO are added together, and this sum is added to and stored in *identifier-2*. This process is repeated for each successive occurrence of *identifier-2* in the left-to-right order in which *identifier-2* is specified.

Format 2: ADD statement with GIVING phrase



The values of the operands that precede the word GIVING are added together, and the sum is stored as the new value of each data item referenced by *identifier-3*.

Format 3: ADD statement with CORRESPONDING phrase



Elementary data items within *identifier-1* are added to and stored in the corresponding elementary items within *identifier-2*.

For all formats:

identifier-1, identifier-2

In format 1, must name an elementary numeric item.

In format 2, must name an elementary numeric item except when following the word GIVING. Each identifier that follows the word GIVING must name an elementary numeric or numeric-edited item.

In format 3, must name an alphanumeric group item or national group item.

literal

Must be a numeric literal.

Floating-point data items and literals can be used anywhere that a numeric data item or literal can be specified.

When the ARITH(COMPAT) compiler option is in effect, the composite of operands can contain a maximum of 30 digits. When the ARITH(EXTEND) compiler option is in effect, the composite of operands can contain a maximum of 31 digits. For more information, see [“Arithmetic statement operands”](#) on page

297 and the details on arithmetic intermediate results in *Appendix A. Intermediate results and arithmetic precision* in the *Enterprise COBOL Programming Guide*.

ROUNDED phrase

For formats 1, 2, and 3, see [“ROUNDED phrase” on page 296](#).

SIZE ERROR phrases

For formats 1, 2, and 3, see [“SIZE ERROR phrases” on page 296](#).

CORRESPONDING phrase (format 3)

See [“CORRESPONDING phrase” on page 295](#).

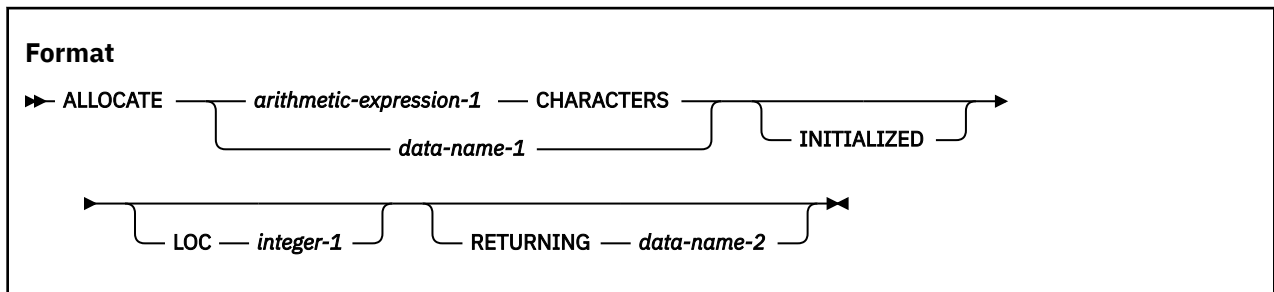
END-ADD phrase

This explicit scope terminator serves to delimit the scope of the ADD statement. END-ADD permits a conditional ADD statement to be nested in another conditional statement. END-ADD can also be used with an imperative ADD statement.

For more information, see [“Delimited scope statements” on page 293](#).

ALLOCATE statement

The ALLOCATE statement obtains dynamic storage from the Language Environment heap.



If *data-name-1* is specified, the address of the data item is set to the address of the obtained storage, as if the "SET ADDRESS OF *data-name-1* TO *address*" statement was used. If a RETURNING data item is also specified, the pointer data item will contain that address.

If *arithmetic-expression-1* CHARACTERS is specified, the RETURNING *data-item-2* will be set to the address of the obtained storage.

data-name-1

Must be a level-01 or level-77 item defined in the LINKAGE SECTION.

If *data-name-1* is specified, the RETURNING phrase can be omitted. Otherwise, the RETURNING phrase must be specified.

Cannot be reference modified.

Cannot be a group item that contains an unbounded table.

integer-1

Must be an unsigned integer with value of 24, 31 or 64. The default for LP(64) is 64, and for LP(32) is 31. The value 64 is not supported in LP(32).

data-name-2

Must be defined as USAGE POINTER or USAGE POINTER-32.

Can be qualified or subscripted.

arithmetic-expression-1

Specifies a number of bytes of storage to be allocated:

- If *arithmetic-expression-1* does not evaluate to an integer, the result is rounded up to the next whole number.
- If *arithmetic-expression-1* evaluates to 0 or a negative value, the data item referenced by *data-name-2* is set to the predefined address NULL.

INITIALIZED phrase

The INITIALIZED phrase initializes the allocated storage:

- If the INITIALIZED phrase is not specified, the content of the allocated storage is undefined.
- If both *arithmetic-expression-1* and the INITIALIZED phrase are specified, all bytes of the allocated storage are initialized to binary zeros.
- If both *data-name-1* and the INITIALIZED phrase are specified, the allocated storage is initialized as if an INITIALIZE *data-name-1* WITH FILLER ALL TO VALUE THEN TO DEFAULT statement were executed.

LOC phrase

The LOC phrase controls how ALLOCATE acquires storage:

- LOC 24 causes ALLOCATE to acquire storage from below the 16 MB line, regardless of the setting of the DATA compiler option.
- LOC 31 causes ALLOCATE to attempt to acquire storage from above the 16 MB line, regardless of the setting of the DATA compiler option.

Note: It is still possible that storage is acquired below the 16 MB line with LOC 31 if storage above the 16 MB line is exhausted.

- LOC 64 causes ALLOCATE to attempt to acquire storage from above the 2 GB bar, regardless of the setting of the DATA compiler option.

When the LOC phrase is not specified:

- LOC 31 is assumed to be specified whenever the DATA(31) compiler option is in effect.
- LOC 24 is assumed to be specified whenever the DATA(24) compiler option is in effect.
- LOC 64 is assumed to be specified whenever the LP(64) compiler option is in effect.

Note: It is recommended to use the default value of LOC unless dynamic storage below the 16 MB line is required when the DATA(31) option is in effect, or dynamic storage above the 16 MB line is desired when the DATA(24) option is in effect.

If LOC 64 is specified and LP(32) is in effect, a diagnostic message will be issued.

If *data-name-1* is specified, the amount of storage to be allocated is the number of bytes required to hold an item as described by *data-name-1*. If a data description entry that is subordinate to *data-name-1* contains an OCCURS DEPENDING ON clause, the maximum length of the record is allocated.

When *data-name-2* references a USAGE POINTER or USAGE POINTER-32 data item, the allocated storage depends on the LP and DATA compiler options, as well as the LOC phrase settings. The following table shows how storage is allocated based on these settings:

Table 35. Location of allocated storage				
data-name-2 USAGE	LP compiler option	DATA compiler option	ALLOCATE LOC phrase	Location of allocated storage
POINTER/ POINTER-32	LP(32)	DATA(31)/Not specified	Not specified	Below the bar

<i>Table 35. Location of allocated storage (continued)</i>				
data-name-2 USAGE	LP compiler option	DATA compiler option	ALLOCATE LOC phrase	Location of allocated storage
POINTER/ POINTER-32	LP(32)	DATA(24)	Not specified	Below the line
POINTER/ POINTER-32	LP(32)	DATA(24)/ DATA(31)/Not specified	LOC 24	Below the line
POINTER/ POINTER-32	LP(32)	DATA(24)/ DATA(31)/Not specified	LOC 31	Below the bar
POINTER/ POINTER-32	LP(32)	DATA(24)/ DATA(31)/Not specified	LOC 64	N/A ¹
POINTER	LP(64)	DATA(31)/Not specified	Not specified/LOC 64	Above the bar
POINTER/ POINTER-32	LP(64)	DATA(31)/Not specified	LOC 24	Below the line
POINTER/ POINTER-32	LP(64)	DATA(31)/Not specified	LOC 31	Below the bar
POINTER-32	LP(64)	DATA(31)/Not specified	Not specified	Below the bar
POINTER-32	LP(64)	DATA(31)/Not specified	LOC 64	N/A ¹
POINTER/ POINTER-32	LP(64)	DATA(24)	LOC 24/LOC 31/LOC 64/Not specified	N/A ¹
Note: 1. A diagnostic message will be issued, and no storage will be allocated.				

If the specified amount of storage is available for allocation:

- If the RETURNING phrase is specified, the data item referenced by *data-name-2* is set to the address of that storage.
- If *data-name-1* is specified, the address of the 01 or 77 LINKAGE SECTION data item referenced by *data-name-1* is set to the address of that storage, as if the "SET ADDRESS OF *data-name-1* TO *address-of-obtained-storage*" statement was used.

If the specified amount of storage is not available for allocation:

- If the RETURNING phrase is specified, the data item referenced by *data-name-2* is set to the predefined address NULL.
- If *data-name-1* is specified, the address of the 01 or 77 LINKAGE SECTION data item referenced by *data-name-1* is set to the predefined address NULL.

The allocated storage persists until explicitly released with a FREE statement or the run unit is terminated, whichever occurs first.

You can use ALLOCATE to dynamically increase the size of an UNBOUNDED table, see [“Example: ALLOCATE and FREE storage for UNBOUNDED tables” on page 316](#).

Related references

[DATA directive](#)

[“FREE statement” on page 345](#)

[“INITIALIZE statement” on page 350](#)

[POINTER phrase](#)

[POINTER-32 phrase](#)

[DATA compiler option \(Enterprise COBOL Programming Guide\)](#)

[Storage and its addressability](#)

[\(Enterprise COBOL Programming Guide\)](#)

Example: ALLOCATE and FREE storage for UNBOUNDED tables

This example illustrates one way to manage an UNBOUNDED table that needs to be dynamically increased in size by using the ALLOCATE and FREE statements.

```
*-----
* ALLOC: An example using the ALLOCATE and FREE statements to
*       allocate and resize an unbounded table.
*A-1-B-+---2---+---3---+---4---+---5---+---6---+---7-|
identification division.
program-id. ALLOC.
*
environment division.
data division.

working-storage section.

77 k                pic 9(4) binary.
77 move-size        pic 9(4) binary.
77 num-elements     pic 9(4) binary.
*
77 vargrp-ptr       pointer.
77 vargrp-size      pic 9(4) binary.
77 vargrp-old-ptr   pointer.
77 vargrp-old-size  pic 9(4) binary.

linkage section.

01 vargrp.
  02 vartab-bound    pic 9(4) comp.
  02 vartab-group.
    03 vartab        occurs 1 to unbounded
                      depending on vartab-bound.
      04 t1          pic 9(4).
      04 t2          pic x(8).
      04 t3          pic 9(4) comp.

01 vargrp-old       pic x(999999999).

/*****
* main
*****/
procedure division.

    display "Start testcase ALLOC"

    *> allocate a table with 20 elements
    compute num-elements = 20
    perform vargrp-alloc

    *> Set some test values to validate re-allocated table
    compute t1(12) = 9999
    move "HI MOM" to t2(17)
    move "END-VGRP" to t2(20)
    perform vargrp-display

    *> allocate a bigger table and show content
    compute num-elements = 30
    perform vargrp-realloc
    perform vargrp-display

    *> allocate a smaller table and show content
    compute num-elements = 17
    perform vargrp-realloc
    perform vargrp-display
```



```

display " "
display "End testcase ALLOC"

goback.

/*****
* vargrp-alloc(num-elements)
*****/
vargrp-alloc.

    compute vargrp-size = length of vartab-bound
    + length of vartab * num-elements
    allocate vargrp-size characters
    initialized
    returning vargrp-ptr
    set address of vargrp to vargrp-ptr
    move num-elements to vartab-bound

    exit.

/*****
* vargrp-realloc(num-elements)
*****/
vargrp-realloc.

    *> save the address/length of the current table
    set vargrp-old-ptr to vargrp-ptr
    compute vargrp-old-size = vargrp-size

    *> allocate the new copy
    perform vargrp-alloc

    *> copy the old data to the new area
    compute move-size =
        function min(vargrp-old-size, vargrp-size)
    set address of vargrp-old to vargrp-old-ptr
    move vargrp-old(1:move-size)
        to vargrp(1:move-size)
    set address of vargrp-old to null
    move num-elements to vartab-bound

    *> free the old area
    free vargrp-old-ptr

    exit.

/*****
* vargrp-display
*****/
vargrp-display.

    display "VARGRP is at 0x"
    function hex-of(vargrp-ptr)
    " with " vartab-bound " elements,"
    " size " vargrp-size " bytes."
    perform varying k from 1 by 1 until k > vartab-bound
    display "vartab(" k ") =" vartab(k)
    end-perform
    display " "

    exit.

end program alloc.

```

Related references

[“ALLOCATE statement” on page 313](#)

[“FREE statement” on page 345](#)

[“MOVE statement” on page 405](#)

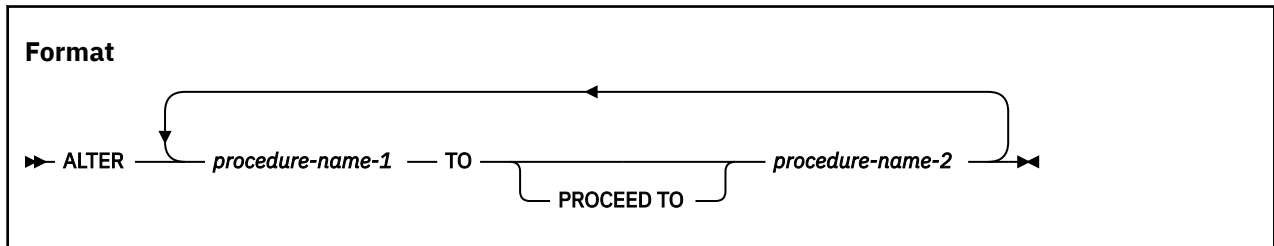
Working with unbounded tables and groups
(*Enterprise COBOL Programming Guide*)

ALTER statement

The ALTER statement changes the transfer point specified in a GO TO statement.

The ALTER statement encourages the use of unstructured programming practices; the EVALUATE statement provides the same function as the ALTER statement but helps to ensure that a program is well-structured.

Note: When the LP(64) compiler option is in effect, a compiler diagnostic message will be issued if the ALTER statement is specified. The EVALUATE statement should be used instead.



The ALTER statement modifies the GO TO statement in the paragraph named by *procedure-name-1*. Subsequent executions of the modified GO TO statement transfer control to *procedure-name-2*.

procedure-name-1

Must name a PROCEDURE DIVISION paragraph that contains only one sentence: a GO TO statement without the DEPENDING ON phrase.

procedure-name-2

Must name a PROCEDURE DIVISION section or paragraph.

Before the ALTER statement is executed, when control reaches the paragraph specified in *procedure-name-1*, the GO TO statement transfers control to the paragraph specified in the GO TO statement. After execution of the ALTER statement however, the next time control reaches the paragraph specified in *procedure-name-1*, the GO TO statement transfers control to the paragraph specified in *procedure-name-2*.

The ALTER statement acts as a program switch, allowing, for example, one sequence of execution during initialization and another sequence during the bulk of file processing.

Altered GO TO statements in programs with the INITIAL attribute are returned to their initial states each time the program is entered.

Do not use the ALTER statement in programs that have the RECURSIVE attribute, in methods, or in programs compiled with the THREAD option.

Segmentation considerations

A GO TO statement that is coded in an independent segment must not be referenced by an ALTER statement in a segment with a different *priority-number*. All other uses of the ALTER statement are valid and are performed even if the GO TO referenced by the ALTER statement is in a fixed segment.

Altered GO TO statements in independent segments are returned to their initial state when control is transferred to the independent segment that contains the ALTERED GO TO from another independent segment with a different *priority-number*.

This transfer of control can take place because of:

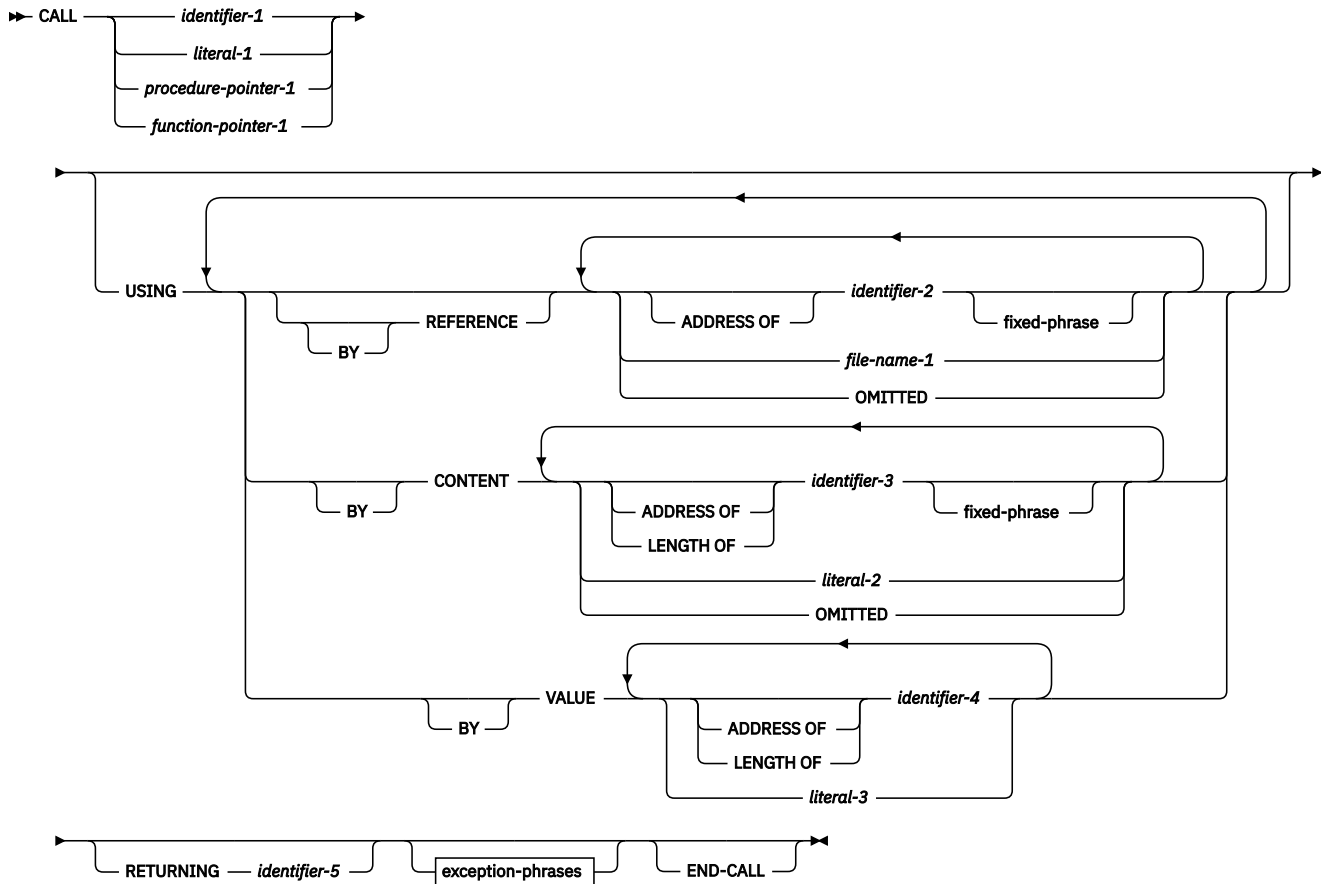
- The effect of previous statements
- An explicit transfer of control with a PERFORM or GO TO statement
- A sort or merge statement with the INPUT or OUTPUT phrase specified

CALL statement

The CALL statement transfers control from one program to another within a run unit.

The program containing the CALL statement is the calling program; the program identified in the CALL statement is the called subprogram. Called programs can contain CALL statements; however, only a program defined with the RECURSIVE clause can execute a CALL statement that directly or indirectly calls itself.

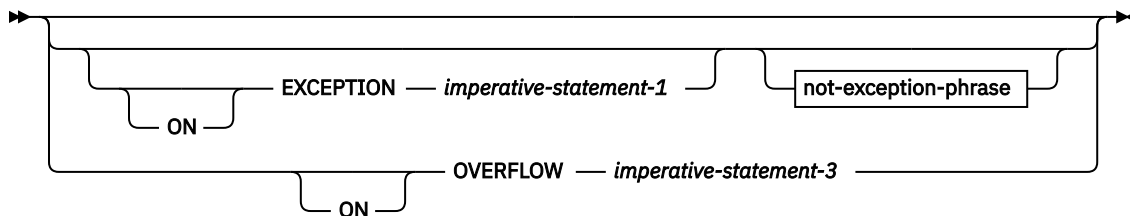
Format



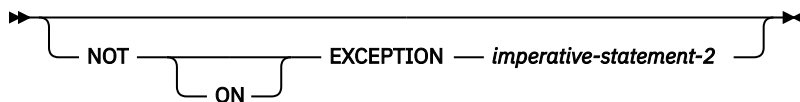
fixed-phrase

➤ AS FIXED LENGTH — integer-4 ➤

exception-phrases



not-exception-phrase



identifier-1, literal-1

literal-1 must be an alphanumeric literal. *identifier-1* must be an alphanumeric, alphabetic, or numeric data item described with USAGE DISPLAY such that its value can be a program-name.

The rules of formation for program-names are dependent on the PGMNAME compiler option. For details, see the discussion of program-names in Chapter 15, “PROGRAM-ID paragraph,” on page 101 and also the description of PGMNAME in the *Enterprise COBOL Programming Guide*.

If the value of *literal-1* is of the form 'Java.java-class-name-1.java-static-method-name-1', then the call will be interpreted as a call to the static Java method with class name 'java-class-name-1' and method name 'java-static-method-name-1'. For details, see [“Calling static Java methods from COBOL” on page 325](#).

Usage note: Do not specify the name of a class or method in the CALL statement.

procedure-pointer-1

Must be defined with USAGE IS PROCEDURE-POINTER and must be set to a valid program entry point; otherwise, the results of the CALL statement are undefined.

After a program has been canceled by COBOL, released by PL/I or C, or deleted by assembler, any procedure-pointers that had been set to that program's entry point are no longer valid.

function-pointer-1

Must be defined with USAGE IS FUNCTION-POINTER and must be set to a valid function or program entry point; otherwise, the results of the CALL statement are undefined.

After a program has been canceled by COBOL, released by PL/I or C, or deleted by the assembler, any function-pointers that had been set to that function or program's entry point are no longer valid.

When the called subprogram is to be entered at the beginning of the PROCEDURE DIVISION, *literal-1* or the contents of *identifier-1* must specify the program-name of the called subprogram.

When the called subprogram is entered through an ENTRY statement, *literal-1* or the contents of *identifier-1* must be the same as the name specified in the called subprogram's ENTRY statement.

AMODE 64 considerations

For AMODE 64 COBOL programs, both static and dynamic calls support calling other AMODE 64 Language Environment conforming programs.

AMODE 64 COBOL programs cannot be called by non-Language Environment conforming programs. CALL using file-name is not supported with LP(64). An assembler program using LOAD and then branch to an entry point of an LP(64) COBOL subprogram will not work. Instead, use the LE macro CEEFETCH to fetch and call AMODE 64 COBOL programs.

Parameter passing convention is XPLINK.

The AS FIXED LENGTH phrase is not currently supported for programs compiled with LP(64).

For information about using dynamic call in an environment with mixed AMODE 31 and AMODE 64 support, see Dynamic call between AMODE 31 and AMODE 64 programs in the *Enterprise COBOL Programming Guide*.

USING phrase

The USING phrase specifies arguments that are passed to the target program.

Include the USING phrase in the CALL statement only if there is a USING phrase in the PROCEDURE DIVISION header or the ENTRY statement through which the called program is run. The number of operands in each USING phrase must be identical.

For more information about the USING phrase, see [“The PROCEDURE DIVISION header” on page 258](#).

The sequence of the operands in the USING phrase of the CALL statement and in the corresponding USING phrase in the called subprogram's PROCEDURE DIVISION header or ENTRY statement determines

the correspondence between the operands used by the calling and called programs. This correspondence is positional.

The values of the parameters referenced in the USING phrase of the CALL statement are made available to the called subprogram at the time the CALL statement is executed. The description of the data items in the called program must describe the same number of character positions as the description of the corresponding data items in the calling program.

The BY CONTENT, BY REFERENCE, and BY VALUE phrases apply to parameters that follow them until another BY CONTENT, BY REFERENCE, or BY VALUE phrase is encountered. BY REFERENCE is assumed if you do not specify a BY CONTENT, BY REFERENCE, or BY VALUE phrase prior to the first parameter.

BY REFERENCE phrase

If the BY REFERENCE phrase is either specified or implied for a parameter, the corresponding data item in the calling program occupies the same storage area as the data item in the called program.

identifier-2

Can be any data item of any level in the DATA DIVISION. *identifier-2* cannot be a function-identifier.

Note: *identifier-2* can only be a dynamic length elementary item if the AS FIXED LENGTH phrase is specified.

If it is defined in the LINKAGE SECTION or FILE SECTION, you must have already provided addressability for *identifier-2* prior to invocation of the CALL statement. You can do this by coding either one of the following: SET ADDRESS OF *identifier-2* TO pointer or PROCEDURE/ENTRY USING.

file-name-1

A file-name for a QSAM file. See *Passing data* in the *Enterprise COBOL Programming Guide* for details on using file-name with the CALL statement. This phrase is not supported with AMODE 64 (LP(64)).

ADDRESS OF *identifier-2*

identifier-2 must be a level-01 or level-77 item defined in the LINKAGE SECTION.

OMITTED

Indicates that no argument is passed.

AS FIXED LENGTH phrase

If the AS FIXED LENGTH phrase is specified, then the corresponding data item must be a dynamic-length elementary item.

integer-4

The value of *integer-4* represents the number of characters for items of class alphanumeric or national, or bytes for items of class UTF-8. *integer-4* must be an integer greater than zero, and less than the value implied or specified on the LIMIT phrase of the dynamic-length elementary item.

If the item is of class alphanumeric, national, or UTF-8, then the padding character is EBCDIC blanks, UTF-16 blanks, or UTF-8 blanks respectively.

When the AS FIXED LENGTH phrase is specified, the address of the dynamic-length elementary item is passed to the callee as if the equivalent fixed-length alphanumeric, national, or UTF-8 class data item would be whose length is *integer-4*. If *integer-4* is larger than the current length of the dynamic-length elementary item, then the item is extended and padded on the right with blanks to either *integer-4* characters. After execution of the CALL statement the current length of the dynamic-length elementary item remains as it was before the execution of the CALL statement.

When the BY CONTENT phrase is specified or implied for a dynamic-length elementary item parameter, then the AS FIXED LENGTH phrase must be specified.

BY CONTENT phrase

If the BY CONTENT phrase is specified or implied for a parameter, the called program cannot change the value of this parameter as referenced in the CALL statement's USING phrase, though the called program can change the value of the data item referenced by the corresponding data-name in the called program's PROCEDURE DIVISION header. Changes to the parameter in the called program do not affect the corresponding argument in the calling program.

identifier-3

Can be any data item of any level in the DATA DIVISION. *identifier-3* cannot be a function identifier or an unbounded group.

If defined in the LINKAGE SECTION or FILE SECTION, you must have already provided addressability for *identifier-3* prior to invocation of the CALL statement. You can do this by coding one of the following phrases:

- SET ADDRESS OF *identifier-3* TO pointer
- PROCEDURE DIVISION USING
- ENTRY USING

literal-2

Can be:

- An alphanumeric literal
- A figurative constant (except ALL *literal* or NULL/NULLS)
- A DBCS literal
- A national literal

LENGTH OF special register

For information about the LENGTH OF special register, see [“LENGTH OF” on page 22](#).

ADDRESS OF *identifier-3*

identifier-3 must be a data item of any level except 66 or 88 defined in the LINKAGE SECTION, the WORKING-STORAGE SECTION, or the LOCAL-STORAGE SECTION.

OMITTED

Indicates that no argument is passed.

For alphanumeric literals, the called subprogram should describe the parameter as PIC X(*n*) USAGE DISPLAY, where *n* is the number of characters in the literal.

For DBCS literals, the called subprogram should describe the parameter as PIC G(*n*) USAGE DISPLAY-1, or PIC N(*n*) with implicit or explicit USAGE DISPLAY-1, where *n* is the length of the literal.

For national literals, the called subprogram should describe the parameter as PIC N(*n*) with implicit or explicit USAGE NATIONAL, where *n* is the length of the literal.

BY VALUE phrase

The BY VALUE phrase applies to all arguments that follow until overridden by another BY REFERENCE or BY CONTENT phrase.

If the BY VALUE phrase is specified or implied for an argument, the value of the argument is passed, not a reference to the sending data item. The called program can modify the formal parameter that corresponds to the BY VALUE argument, but any such changes do not affect the argument because the called program has access to a temporary copy of the sending data item.

Although BY VALUE arguments are primarily intended for communication with non-COBOL programs (such as C), they can also be used for COBOL-to-COBOL invocations. In this case, BY VALUE must be specified or implied for both the argument in the CALL USING phrase and the corresponding formal parameter in the PROCEDURE DIVISION USING phrase.

identifier-4

Must be an elementary data item in the DATA DIVISION. It must be one of the following items:

- Binary (USAGE BINARY, COMP, COMP-4, or COMP-5)
- Floating point (USAGE COMP-1 or COMP-2)
- Function-pointer (USAGE FUNCTION-POINTER)
- Pointer (USAGE POINTER)
- Procedure-pointer (USAGE PROCEDURE-POINTER)
- Object reference (USAGE OBJECT REFERENCE)
- One single-byte alphanumeric character (such as PIC X or PIC A)
- One national character (PIC N), described as an elementary data item of category national.

The following items can also be passed BY VALUE:

- Reference-modified item of USAGE DISPLAY and length 1
- Reference-modified item of USAGE NATIONAL and length 1
- SHIFT-IN and SHIFT-OUT special registers
- LINAGE-COUNTER special register when it is USAGE BINARY

ADDRESS OF *identifier-4*

identifier-4 must be a data item of any level except 66 or 88 defined in the LINKAGE SECTION, the WORKING-STORAGE SECTION, or the LOCAL-STORAGE SECTION.

LENGTH OF special register

A LENGTH OF special register passed BY VALUE is treated as a PIC 9(9) binary. For information about the LENGTH OF special register, see [“LENGTH OF” on page 22](#).

literal-3

Must be of one of the following types:

- A numeric literal
- A figurative constant ZERO
- A one-character alphanumeric literal
- A one-character national literal
- A symbolic character
- A single-byte figurative constant
 - SPACE
 - QUOTE
 - HIGH-VALUE
 - LOW-VALUE

ZERO is treated as a numeric value; a fullword binary zero is passed.

If *literal-3* is a fixed-point numeric literal, it must have a precision of nine or fewer digits. In this case, a fullword binary representation of the literal value is passed.

If *literal-3* is a floating-point numeric literal, an 8-byte internal floating-point (COMP-2) representation of the value is passed.

literal-3 must not be a DBCS literal.

RETURNING phrase

identifier-5

The RETURNING data item, which can be any data item defined in the DATA DIVISION. The return value of the called program is implicitly stored into *identifier-5*.

You can specify the RETURNING phrase for calls to functions written in COBOL, C, or in other programming languages that use C linkage conventions. If you specify the RETURNING phrase on a CALL to a COBOL subprogram:

- The called subprogram must specify the RETURNING phrase on its PROCEDURE DIVISION header.
- *identifier-5* and the corresponding PROCEDURE DIVISION RETURNING identifier in the target program must have the same PICTURE, USAGE, SIGN, SYNCHRONIZE, JUSTIFIED, and BLANK WHEN ZERO clauses (except that PICTURE clause currency symbols can differ, and periods and commas can be interchanged due to the DECIMAL POINT IS COMMA clause).

When the target returns, its return value is assigned to *identifier-5* using the rules for the SET statement if *identifier-6* is of usage INDEX, POINTER, FUNCTION-POINTER, PROCEDURE-POINTER, or OBJECT REFERENCE. When *identifier-5* is of any other usage, the rules for the MOVE statement are used.

The CALL ... RETURNING data item is an output-only parameter. On entry to the called program, the initial state of the PROCEDURE DIVISION RETURNING data item has an undefined and unpredictable value. You must initialize the PROCEDURE DIVISION RETURNING data item in the called program before you reference its value. The value that is passed back to the calling program is the final value of the PROCEDURE DIVISION RETURNING data item when the called program returns.

Note: If a COBOL program returns a doubleword binary item via a PROCEDURE DIVISION RETURNING header to a calling COBOL program with a CALL ... RETURNING statement, an issue occurs if only one of the programs is recompiled with Enterprise COBOL V6. Both the called and calling programs must be recompiled with Enterprise COBOL V6 together, so that the linkage convention for the RETURNING item is consistent.

If an EXCEPTION or OVERFLOW occurs, *identifier-5* is not changed. *identifier-5* must not be reference-modified.

The RETURN-CODE special register is not set by execution of CALL statements that include the RETURNING phrase.

ON EXCEPTION phrase

An exception condition occurs under the following two conditions:

- When the target of the CALL statement is a literal of the form 'Java.java-class-name.java-static-method-name' and a Java exception occurs during any part of the call as determined by the JNI function ExceptionOccurred.
- When the called subprogram cannot be made available.

When one of above exception conditions is true, one of the following two actions will occur:

1. If the ON EXCEPTION phrase is specified, control is transferred to *imperative-statement-1*. Execution then continues according to the rules for each statement specified in *imperative-statement-1*. If a procedure branching or conditional statement that causes explicit transfer of control is executed, control is transferred in accordance with the rules for that statement. Otherwise, upon completion of the execution of *imperative-statement-1*, control is transferred to the end of the CALL statement and the NOT ON EXCEPTION phrase, if specified, is ignored.
2. If the ON EXCEPTION phrase is not specified in the CALL statement, the NOT ON EXCEPTION phrase, if specified, is ignored.

Note: In the case of an exception during a call to a static Java method, the exception could occur while executing the JNI functions that facilitate the call to Java, or the exception could be an exception thrown in application-level Java code. In all cases, before control is transferred to user exception handling statements, the Java exception is automatically cleared via the JNI function ExceptionClear(). In *imperative-statement-1*, the Java exception object can be referenced from special register IGY-JAVAIOP-CALL-EXCEPTION. See [“IGY-JAVAIOP-CALL-EXCEPTION”](#) on page 20 special register for details.

NOT ON EXCEPTION phrase

If an exception condition does not occur, control is transferred to the called program. After control is returned from the called program, control is transferred to:

- *imperative-statement-2*, if the NOT ON EXCEPTION phrase is specified.
- The end of the CALL statement in any other case. (If the ON EXCEPTION phrase is specified, it is ignored.)

If control is transferred to *imperative-statement-2*, execution continues according to the rules for each statement specified in *imperative-statement-2*. If a procedure branching or conditional statement that causes explicit transfer of control is executed, control is transferred in accordance with the rules for that statement. Otherwise, upon completion of the execution of *imperative-statement-2*, control is transferred to the end of the CALL statement.

ON OVERFLOW phrase

The ON OVERFLOW phrase has the same effect as the ON EXCEPTION phrase.

END-CALL phrase

This explicit scope terminator serves to delimit the scope of the CALL statement. END-CALL permits a conditional CALL statement to be nested in another conditional statement. END-CALL can also be used with an imperative CALL statement.

For more information, see [“Delimited scope statements”](#) on page 293.

Related references

[“Mapping between COBOL and Java data types for non-OO COBOL/Java interoperability”](#) on page 733
Compiling, linking, and running non-OO COBOL applications that interoperate with Java (*Enterprise COBOL Programming Guide*)

Running the cjbuild utility to build a DLL of Java stub programs (*Enterprise COBOL Programming Guide*)

JAVAIOP (*Enterprise COBOL Programming Guide*)

PARMCHECK (*Enterprise COBOL Programming Guide*)

Calling static Java methods from COBOL

In a CALL statement where the called subprogram is identified by a literal of the form '*Java.java-class-name-1.java-static-method-name-1*', the call is interpreted as a call from COBOL to a static Java method with class name '*java-class-name-1*' and method name '*java-static-method-name-1*', where '*java-class-name-1*' is a Java class name that must be fully-qualified with its package name if it belongs to a package.

Note:

- The literal prefix "Java." is not case sensitive, so prefixes like 'JAVA.' and 'jaVA.' are accepted. However, the remaining portion of the string is case sensitive and the fully-qualified class name and static method name must match what is used in Java.
- If the static method being called is part of one or more nested classes, then the name of each nested class involved must be introduced in the literal with a \$ character. For example, to call static method "dumpData()" in class Util which is nested inside an outer class TestApp, the CALL statement should be as follows:

```
CALL 'Java.TestApp$Util.dumpData' USING ...
```

If TestApp is part of a package named com.acme, the CALL statement should be as follows:

```
CALL 'Java.com.acme.TestApp$Util.dumpData' USING ....
```

In this scenario, a Java call stub program that is automatically generated by the COBOL compiler will serve as an interface between the COBOL calling program and the static Java method that is the intended target of the call. At run time, when the CALL statement is executed, the call stub program is executed

first, and the call stub program then uses Java Native Interface (JNI) routines to make the call to the specified static Java method.

The generated Java method call stub program is written to a z/OS UNIX directory that is specified by the JAVAIOP(OUTPATH(zos-unix-directory)) option. If that option is not in effect, the default output location is the current directory if the compiler is being run from the cob2 utility; otherwise, the output location is the home directory of the userid under which the compiler is running.

The name of the Java call stub program that is generated will be of the following form:

```
Java.java-class-name-1.java-method_name-1.cbl
```

Handling parameters and returned values

- If the CALL statement contains the USING phrase, indicating that arguments are being passed to the static Java method, the Java call stub program that is generated by the compiler automatically handles conversions between outgoing COBOL argument types and their corresponding Java types before calling the Java method. The conversions are performed according to the [COBOL/Java type mapping defined in Legal COBOL types for Java interoperability and corresponding Java types](#). The conversion code is generated accordingly.
- If the CALL statement contains a RETURNING phrase, then after the Java static method is called, the Java call stub program will automatically convert the returned Java value to the type of the COBOL receiver indicated in the RETURNING phrase. This conversion is performed according to the mapping defined in [Legal COBOL types for Java interoperability and corresponding Java types](#).

Note: There is no type conformance checking at compile time or run time between outgoing COBOL arguments and Java method parameters or between incoming Java returned value and the type of the corresponding COBOL receiver. If outgoing COBOL arguments or incoming Java returned values are not in the correct format for the corresponding Java or COBOL receiver, unpredictable results, including program abend, can occur at run time.

Building Java call stub programs

Java call stub programs must be compiled along with any other stub programs for the application into a DLL using the cjbuild utility. See for details.

If it is possible for this call to Java to be executed in the application before any Java code is executed in the application, then it will be the Java call stub program for this call that is responsible for starting the Java virtual machine (JVM) for the enclave in these scenarios. To specify that non-default JVM initialization options are to be used when starting the JVM at run time, in addition to any other options used to compile the program making the call, specify option "JAVIOP(JVMINITOPTIONS(jvm-init-string))" as well. Alternatively, the COBJVMINITOPTIONS environment variable can be set at run time for the application.

Restrictions

- Literals of any kind are not permitted as arguments in calls to static Java methods.
- A single Java call stub program is generated for all calls to a particular static Java method in an application, so the static Java method cannot be overloaded and all calls to the static method must have the same argument types across the entire application; otherwise there may be unpredictable results at run time. It is the user's responsibility to ensure that this condition is met because the compiler cannot easily enforce this across all files of the application.
- All Java interoperability-related stub programs generated by the compiler for an application, including all Java call stub programs, must be built into a DLL using the cjbuild utility. If cjbuild is instructed to output the DLL to the z/OS UNIX file system, then the location of this DLL must be reflected in your LIBPATH environment variable at run time. On the other hand, if cjbuild is instructed to output the DLL to an MVS data set, then that data set must be included in your STEPLIB at run time. Furthermore, the CALL statement in the user COBOL program that calls the static Java method is always treated as a DLL call, regardless of the current value of the DYNAM and DLL compiler options in effect or any CALLINTERFACE directive that may be in effect at the time of the call.

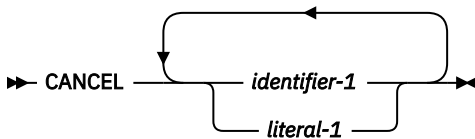
Related references

“Mapping between COBOL and Java data types for non-OO COBOL/Java interoperability” on page 733
Compiling, linking, and running non-OO COBOL applications that interoperate with Java (*Enterprise COBOL Programming Guide*)
Running the cjbuild utility to build a DLL of Java stub programs (*Enterprise COBOL Programming Guide*)
JAVAIOP (*Enterprise COBOL Programming Guide*)
PARMCHECK (*Enterprise COBOL Programming Guide*)

CANCEL statement

The CANCEL statement ensures that the referenced subprogram is entered in initial state the next time that it is called.

Format



identifier-1, *literal-1*

literal-1 must be an alphanumeric literal. *identifier-1* must be an alphanumeric, alphabetic, or zoned decimal data item such that its value can be a program-name. The rules of formation for program-names are dependent on the PGMNAME compiler option. For details, see the discussion of program-names in Chapter 15, “PROGRAM-ID paragraph,” on page 101 and the description of *PGMNAME* in the *Enterprise COBOL Programming Guide*.

literal-1 or the contents of *identifier-1* must be the same as a literal or the contents of an identifier specified in an associated CALL statement.

Do not specify the name of a class or a method in the CANCEL statement.

After a CANCEL statement for a called subprogram has been executed, that subprogram no longer has a logical connection to the program. The contents of data items in external data records described by the subprogram are not changed when that subprogram is canceled. If a CALL statement is executed later by any program in the run unit naming the same subprogram, that subprogram is entered in its initial state.

When a CANCEL statement is executed, all programs contained within the program referenced in the CANCEL statement are also canceled. The result is the same as if a valid CANCEL were executed for each contained program in the reverse order in which the programs appear in the separately compiled program.

A CANCEL statement frees all allocated buffers for dynamic-length elementary items that are associated with the named program, including buffers for dynamic-length elementary items in its nested programs.

A CANCEL statement closes all open files that are associated with an internal file connector in the program named in an explicit CANCEL statement. USE procedures associated with those files are not executed.

You can cancel a called subprogram in any of the following ways:

- By referencing it as the operand of a CANCEL statement
- By terminating the run unit of which the subprogram is a member
- By executing an EXIT PROGRAM statement or a GOBACK statement in the called subprogram if that subprogram possesses the initial attribute

No action is taken when a CANCEL statement is executed if the specified program:

- Has not been dynamically called in this run unit by another COBOL program
- Has been called and subsequently canceled

In a multithreaded environment, a program cannot execute a CANCEL statement naming a program that is active on any thread. The named program must be completely inactive.

Called subprograms can contain CANCEL statements. However, a called subprogram must not execute a CANCEL statement that directly or indirectly cancels the calling program itself or that cancels any program higher than itself in the calling hierarchy. In such a case, the run unit is terminated.

A program named in a CANCEL statement must be a program that has been called and has executed an EXIT PROGRAM statement or a GOBACK statement.

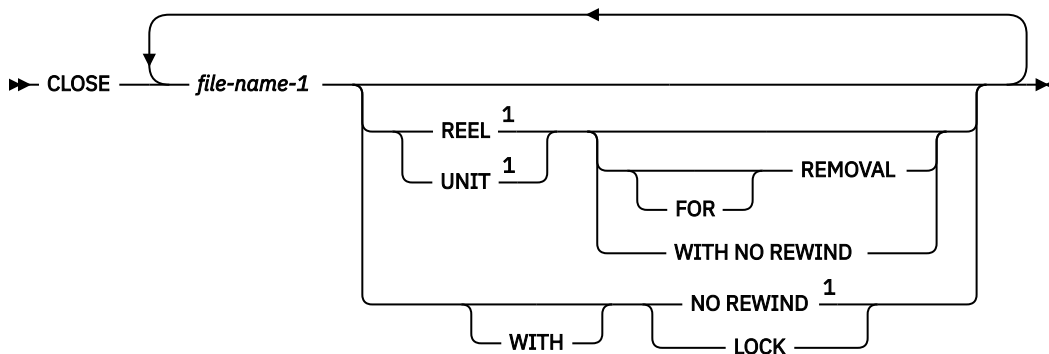
A program can cancel a program that it did not call, provided that, in the calling hierarchy, the program that executes the CANCEL statement is higher than or equal to the program it is canceling. For example:

A calls B and B calls C (When A receives control, it can cancel C.)
A calls B and A calls C (When C receives control, it can cancel B.)

CLOSE statement

The CLOSE statement terminates the processing of volumes and files.

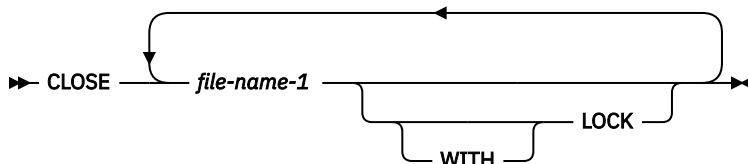
Format 1: CLOSE statement for sequential files



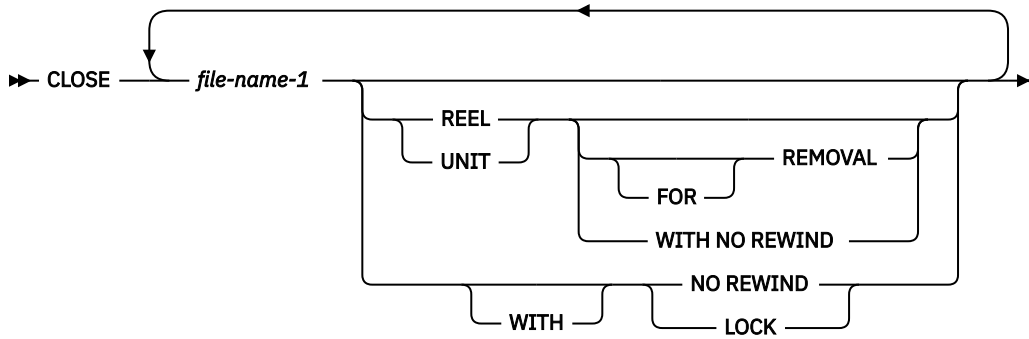
Notes:

¹ The REEL, UNIT, and NO REWIND phrases are not valid for VSAM files.

Format 2: CLOSE statement for indexed and relative files



Format 3: CLOSE statement for line-sequential files



file-name-1

Designates the file upon which the CLOSE statement is to operate. If more than one file-name is specified, the files need not have the same organization or access. *file-name-1* must not be a sort or merge file.

REEL and UNIT

You can specify these phrases only for QSAM multivolume or single volume files. The terms REEL and UNIT are interchangeable.

WITH NO REWIND and FOR REMOVAL

These phrases apply only to QSAM tape files. If they are specified for storage devices to which they do not apply, the close operation is successful and a status key value is set to indicate the file was on a non-reel medium.

A CLOSE statement can be executed only for a file in an open mode. After successful execution of a CLOSE statement (without the REEL/UNIT phrase if using format 1):

- The record area associated with the file-name is no longer available. Unsuccessful execution of a CLOSE statement leaves availability of the record data undefined.
- An OPEN statement for the file must be executed before any other input/output statement can be executed for the file and before data is moved to a record description entry associated with the file.

If the FILE STATUS clause is specified in the file-control entry, the associated file status key is updated when the CLOSE statement is executed.

If the file is in an open status and the execution of a CLOSE statement is unsuccessful, the EXCEPTION/ERROR procedure (if specified) for this file is executed.

Effect of CLOSE statement on file types

If the SELECT OPTIONAL clause is specified in the file-control entry for a file, and the file is not available at run time, standard end-of-file processing is not performed. For QSAM files, the file position indicator and current volume pointer are unchanged.

Files are divided into the following types:

Non-reel/unit

A file whose input or output medium is such that rewinding, reels, and units have no meaning. All VSAM files are of non-reel/unit file types. QSAM files can be of non-reel/unit file types.

Sequential single volume

A sequential file that is contained entirely on one volume. More than one file can be contained on this volume. All VSAM files are single volume. QSAM files can be single volume.

Sequential multivolume

A sequential file that is contained on more than one volume. QSAM files are the only files that can be multivolume. The concept of volume has no meaning for VSAM files.

The permissible combinations of CLOSE statement phrases are shown in the following tables:

- For sequential files: [Sequential files and CLOSE statement phrases](#)
- For indexed and relative files: [Table 37 on page 330](#)
- For line-sequential files: [Table 38 on page 330](#)

The meaning of each key letter is shown in [Table 39 on page 330](#).

<i>Table 36. Sequential files and CLOSE statement phrases</i>			
CLOSE statement phrases	Non-reel/ unit	Sequential single-volume	Sequential multivolume
CLOSE	C	C, G	A, C, G
CLOSE REEL/UNIT	F	F, G	F, G
CLOSE REEL/UNIT WITH NO REWIND	F	B, F	B, F
CLOSE REEL/UNIT FOR REMOVAL	D	D	D
CLOSE WITH NO REWIND	C, H	B, C	A, B, C
CLOSE WITH LOCK	C, E	C, E, G	A, C, E, G

<i>Table 37. Indexed and relative file types and CLOSE statement phrases</i>	
CLOSE statement phrases	Action
CLOSE	C
CLOSE WITH LOCK	C,E

<i>Table 38. Line-sequential file types and CLOSE statement phrases</i>	
CLOSE statement phrases	Action
CLOSE	C
CLOSE WITH LOCK	C,E

<i>Table 39. Meanings of key letters for sequential file types</i>	
Key	Actions taken
A	Previous volumes unaffected Input and input-output files: Standard volume-switch processing is performed for all previous volumes (except those controlled by a previous CLOSE REEL/UNIT statement). Any subsequent volumes are not processed. Output files: Standard volume-switch processing is performed for all previous volumes (except those controlled by a previous CLOSE REEL/UNIT statement).
B	No rewinding of current reel: The current volume is left in its current position.
C	Close file Standard system closing procedures are performed.
D	Volume removal: Treated as a comment.
E	File lock: The compiler ensures that this file cannot be opened again during this execution of the object program. If the file is a tape unit, it will be rewound and unloaded.

Table 39. **Meanings of key letters for sequential file types** (continued)

Key	Actions taken
F	Close volume Input and input-output files: If the current reel/unit is the last or only reel/unit for the file or if the reel is on a non-reel/unit medium, no volume switching is performed. If another reel/unit exists for the file, the following operations are performed: a volume switch, and the first record on the new volume is made available for reading. If no data records exist for the current volume, another volume switch occurs. Output (reel/unit media) files: The following operations are performed: a volume switch. The next executed WRITE statement places the next logical record on the next direct access volume available. A close statement with the REEL phrase does not close the output file; only an end-of-volume condition occurs. Output (non-reel/unit media) files: Execution of the CLOSE statement is considered successful. The file remains in the open mode and no action takes place except that the value of the I-O status associated with the file is updated.
G	Rewind: The current volume is positioned at its physical beginning.
H	Optional phrases ignored: The CLOSE statement is executed as if none of the optional phrases were present.

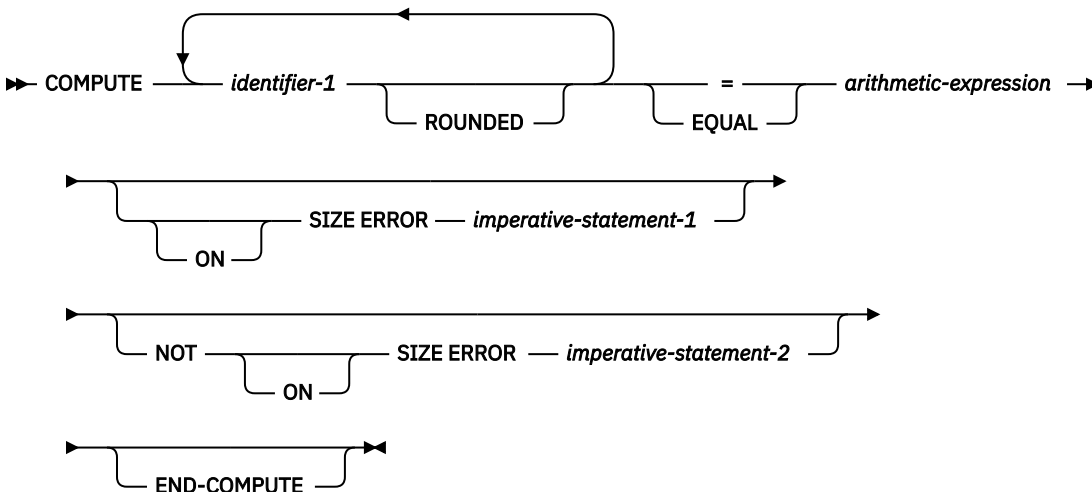
COMPUTE statement

The COMPUTE statement assigns the value of an arithmetic expression to one or more data items.

With the COMPUTE statement, arithmetic operations can be combined without the restrictions on receiving data items imposed by the rules for the ADD, SUBTRACT, MULTIPLY, and DIVIDE statements.

When arithmetic operations are combined, the COMPUTE statement can be more efficient than the separate arithmetic statements written in a series.

Format



identifier-1

Must name an elementary numeric item or an elementary numeric-edited item.

Can name an elementary floating-point data item.

arithmetic-expression

Can be any arithmetic expression, as defined in [“Arithmetic expressions”](#) on page 266.

When the COMPUTE statement is executed, the value of *arithmetic expression* is calculated and stored as the new value of each data item referenced by *identifier-1*.

An arithmetic expression consisting of a single identifier, numeric function, or literal allows the user to set the value of the data items that are referenced by *identifier-1* equal to the value of that identifier, function, or literal.

ROUNDED phrase

For a discussion of the ROUNDED phrase, see [“ROUNDED phrase” on page 296](#).

SIZE ERROR phrases

For a discussion of the SIZE ERROR phrases, see [“SIZE ERROR phrases” on page 296](#).

END-COMPUTE phrase

This explicit scope terminator serves to delimit the scope of the COMPUTE statement. END-COMPUTE permits a conditional COMPUTE statement to be nested in another conditional statement. END-COMPUTE can also be used with an imperative COMPUTE statement.

For more information, see [“Delimited scope statements” on page 293](#).

CONTINUE statement

The CONTINUE statement is a no operation statement. CONTINUE indicates that no executable instruction is present.

Format

►► CONTINUE ►◄

DELETE statement

The DELETE statement removes a record from an indexed or relative file. For indexed files, the key can then be reused for record addition. For relative files, the space is then available for a new record with the same RELATIVE KEY value.

When the DELETE statement is executed, the associated file must be open in I-O mode.

Format

►► DELETE — *file-name-1* ———►
 └── RECORD ─┘

└── INVALID ─┘ ——— *imperative-statement-1* —►
 └── KEY ─┘

└── NOT INVALID ─┘ ——— *imperative-statement-2* —┘ └── END-DELETE ►◄
 └── KEY ─┘

file-name-1

Must be defined in an FD entry in the DATA DIVISION and must be the name of an indexed or relative file.

After successful execution of a DELETE statement, the record is removed from the file and can no longer be accessed.

Execution of the DELETE statement does not affect the contents of the record area associated with *file-name-1* or the content of the data item referenced by the data-name specified in the DEPENDING ON phrase of the RECORD clause associated with *file-name-1*.

If the FILE STATUS clause is specified in the file-control entry, the associated file status key is updated when the DELETE statement is executed.

The file position indicator is not affected by execution of the DELETE statement.

Sequential access mode

For a file in sequential access mode, the previous input/output statement must be a successfully executed READ statement. When the DELETE statement is executed, the system removes the record that was retrieved by that READ statement.

For a file in sequential access mode, the INVALID KEY and NOT INVALID KEY phrases must not be specified. An EXCEPTION/ERROR procedure can be specified.

Random or dynamic access mode

In random or dynamic access mode, DELETE statement execution results depend on the file organization: indexed or relative.

When the DELETE statement is executed, the system removes the record identified by the contents of the prime RECORD KEY data item for indexed files, or the RELATIVE KEY data item for relative files. If the file does not contain such a record, an INVALID KEY condition exists. (See [“Invalid key condition”](#) on page 303.)

Both the INVALID KEY phrase and an applicable EXCEPTION/ERROR procedure can be omitted.

Transfer of control after the successful execution of a DELETE statement, with the NOT INVALID KEY phrase specified, is to the imperative statement associated with the phrase.

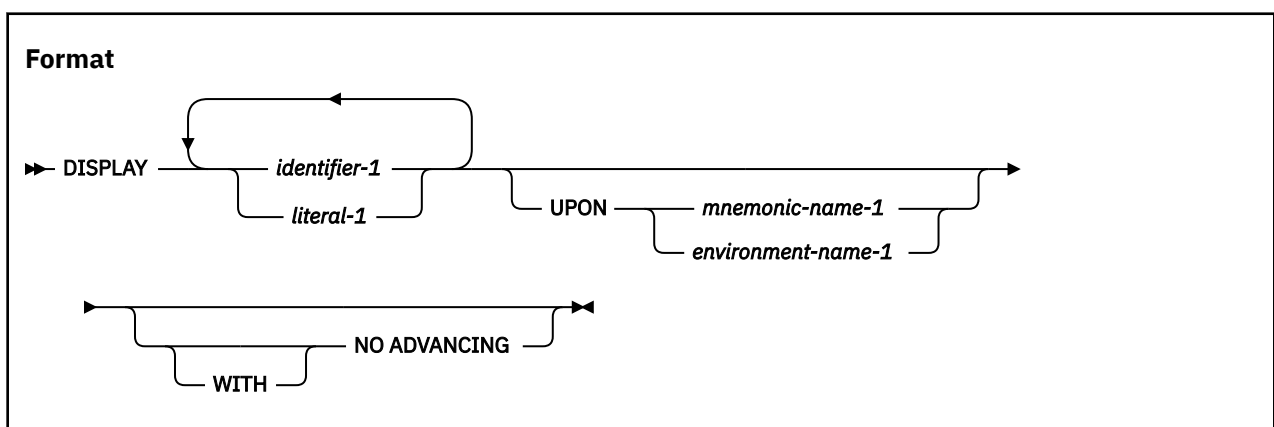
END-DELETE phrase

This explicit scope terminator serves to delimit the scope of the DELETE statement. END-DELETE permits a conditional DELETE statement to be nested in another conditional statement. END-DELETE can also be used with an imperative DELETE statement.

For more information, see [“Delimited scope statements”](#) on page 293.

DISPLAY statement

The DISPLAY statement transfers the contents of each operand to the output device. The contents are displayed on the output device in the order, left to right, in which the operands are listed.



identifier-1

Identifier-1 references the data that is to be displayed. *Identifier-1* can reference any data item except an item of usage PROCEDURE-POINTER, FUNCTION-POINTER, OBJECT REFERENCE, or INDEX. *Identifier-1* cannot be an index-name.

If *identifier-1* is a binary, internal decimal, or internal floating-point data item, *identifier-1* is converted automatically to external format as follows:

- Binary and internal decimal items are converted to zoned decimal. Negative signed values cause a low-order sign overpunch. This can cause unreadable output if the DISPSIGN(COMPAT) compiler option is used. If the DISPSIGN(SEP) compiler option is in effect, the sign is displayed separately from the data, as if SIGN IS SEPARATE was specified. For details, see *DISPSIGN* in the *Enterprise COBOL Programming Guide*.
- Internal floating-point numbers are converted to external floating-point numbers for display such that:
 - A COMP-1 item will display as if it had an external floating-point PICTURE clause of -.9(8)E-99.
 - A COMP-2 item will display as if it had an external floating-point PICTURE clause of -.9(17)E-99.

Data items defined with USAGE POINTER are converted to a zoned decimal number that has an implicit PICTURE clause of PIC 9(10).

If the output is directed to CONSOLE, data items described with usage NATIONAL are converted from national character representation to EBCDIC. The conversion uses the EBCDIC code page that was specified in the CODEPAGE compiler option when the source code was compiled. National characters without EBCDIC counterparts are converted to default substitution characters; no exception condition is indicated or raised.

If the output is not directed to CONSOLE, data items described with usage NATIONAL are written without conversion and without data validation.

No other categories of data require conversion.

DBCS data items, explicitly or implicitly defined as USAGE DISPLAY-1, are transferred to the sending field of the output device. For proper results, the output device must have the capability to recognize DBCS shift-out and shift-in control characters.

Both DBCS and non-DBCS operands can be specified in a single DISPLAY statement.

identifier-1 must not be a dynamic-length group item.

literal-1

Can be any literal or any figurative constant as specified in “[Figurative constants](#)” on [page 15](#). When a figurative constant is specified, only a single occurrence of that figurative constant is displayed.

UPON

environment-name-1 or the environment name associated with *mnemonic-name-1* must be associated with an output device. See “[SPECIAL-NAMES paragraph](#)” on [page 124](#).

A default logical record size is assumed for each device, as follows:

The system logical output device

120 characters

The system punch device

80 characters

The console

100 characters

A maximum logical record size is allowed for each device, as follows:

The system logical output device

255 characters

The system punch device

255 characters

The console

100 characters

On the system punch device, the last eight characters are used for PROGRAM-ID name.

When the UPON phrase is omitted, the system's logical output device is assumed. The list of valid environment-names in a DISPLAY statement is shown in [Table 5 on page 126](#).

For details on routing DISPLAY output to stdout, see *Displaying values on a screen or in a file (DISPLAY)* in the *Enterprise COBOL Programming Guide*.

WITH NO ADVANCING

When specified, the positioning of the output device will not be changed in any way following the display of the last operand.

If the WITH NO ADVANCING phrase is not specified, after the last operand has been transferred to the output device, the positioning of the output device will be reset to the leftmost position of the next line of the device.

Enterprise COBOL does not support output devices that are capable of positioning to a specific character position. See *Displaying values on a screen or in a file (DISPLAY)* in the *Enterprise COBOL Programming Guide* for more information about the DISPLAY statement.

The DISPLAY statement transfers the data in the sending field to the output device. The size of the sending field is the total byte count of all operands listed. If the output device is capable of receiving data of the same size as the data item being transferred, then the data item is transferred. If the output device is not capable of receiving data of the same size as the data item being transferred, then one of the following applies:

- If the total count is less than the device maximum, the remaining rightmost positions are padded with spaces.
- If the total count exceeds the maximum, as many records are written as are needed to display all operands. Any operand being printed or displayed when the end of a record is reached is continued in the next record.

If a DBCS operand must be split across multiple records, it will be split only on a double-byte boundary.

Shift code insertion is required for splitting DBCS items. That is, when a DBCS operand is split across multiple records, the shift-in character is inserted at the end of the current record, and the shift-out character is inserted at the beginning of the next record. A space is padded after the shift-in character, if necessary. These inserted shift codes and spaces are included in the total byte count of the sending data items.

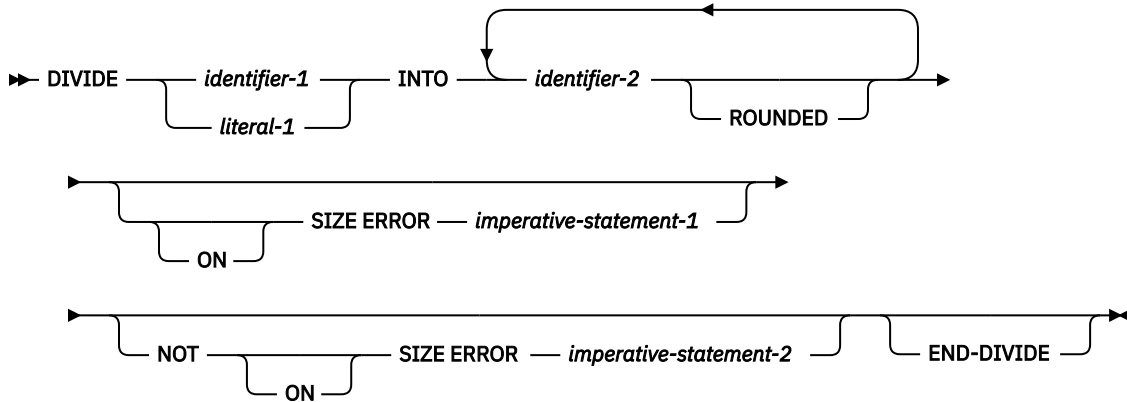
After the last operand has been transferred to the output device, the device is reset to the leftmost position of the next line of the device.

If a DBCS data item or literal is specified in a DISPLAY statement, the size of the sending field is the total byte count of all operands listed, with each DBCS character counted as two bytes, plus the necessary shift codes and spaces for DBCS.

DIVIDE statement

The DIVIDE statement divides one numeric data item into or by others and sets the values of data items equal to the quotient and remainder.

Format 1: DIVIDE statement



In format 1, the value of *identifier-1* or *literal-1* is divided into the value of *identifier-2*, and the quotient is then stored in *identifier-2*. For each successive occurrence of *identifier-2*, the division takes place in the left-to-right order in which *identifier-2* is specified.

Format 1 example:

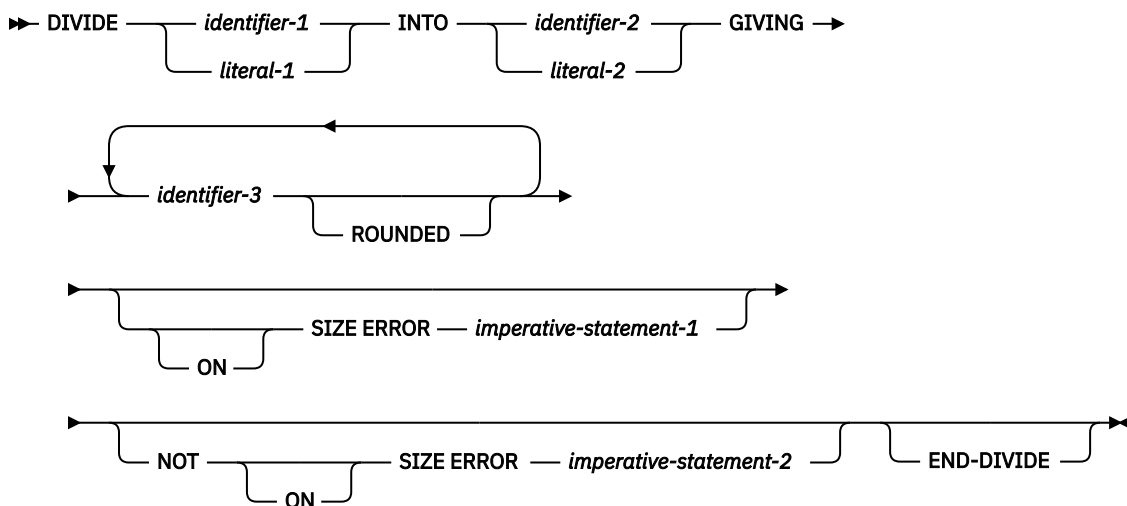
```
DIVIDE A INTO B
```

The value in A is divided into the value in B and the result is stored in B. The value in A is unchanged.

```
DIVIDE C INTO D E
```

The value in C is divided into the value in D, storing the answer in D. The value of C is also divided into E, storing the value in E. The value in C is unchanged.

Format 2: DIVIDE statement with INTO and GIVING phrases



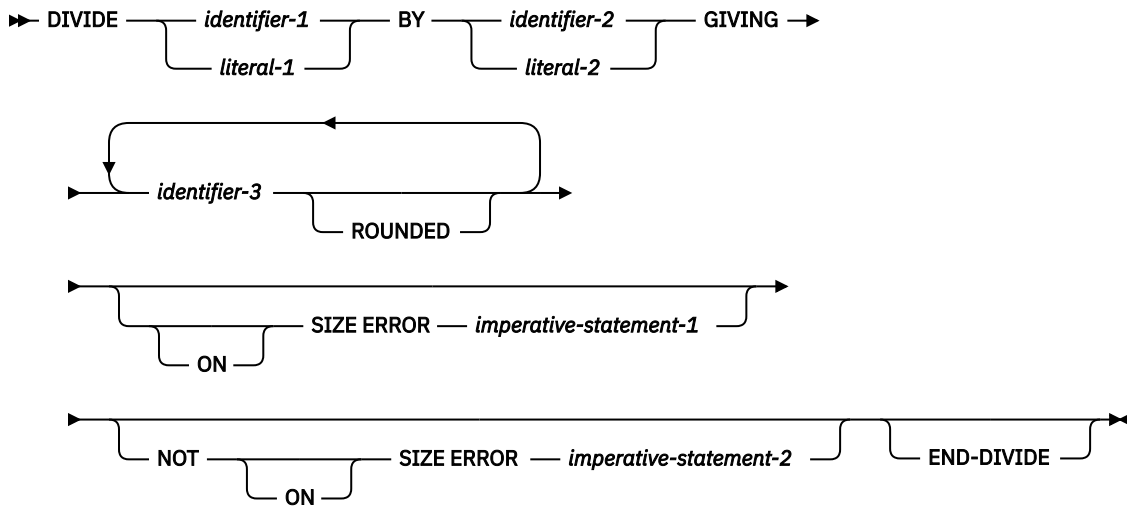
In format 2, the value of *identifier-1* or *literal-1* is divided into the value of *identifier-2* or *literal-2*. The value of the quotient is stored in each data item referenced by *identifier-3*.

Format 2 example:

```
DIVIDE A INTO B GIVING C
```

The value in A is divided into the value in B and the result is stored in C. The values in A and B are unchanged.

Format 3: DIVIDE statement with BY and GIVING phrases



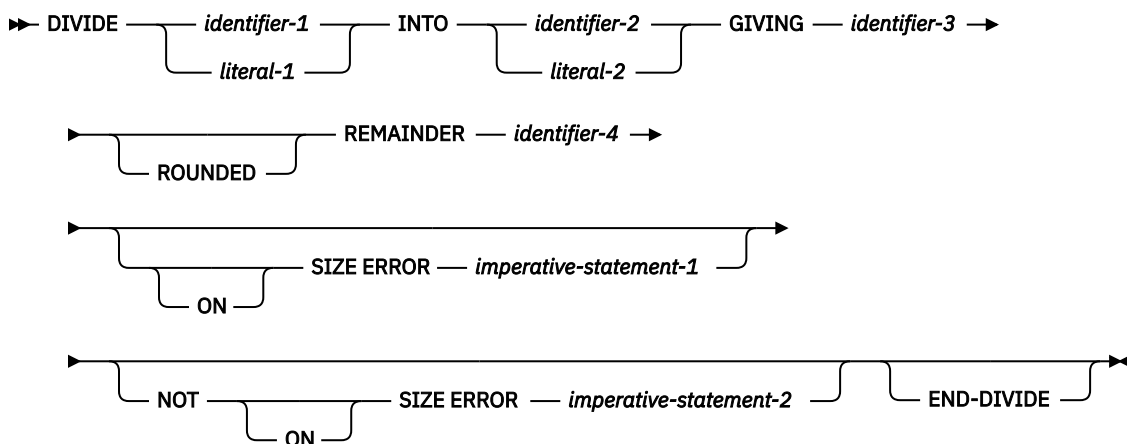
In format 3, the value of *identifier-1* or *literal-1* is divided by the value of *identifier-2* or *literal-2*. The value of the quotient is stored in each data item referenced by *identifier-3*.

Format 3 example:

```
DIVIDE A BY B GIVING C
```

The value in A is divided by the value in B and the result is stored in C. The values in A and B are unchanged.

Format 4: DIVIDE statement with INTO and REMAINDER phrases



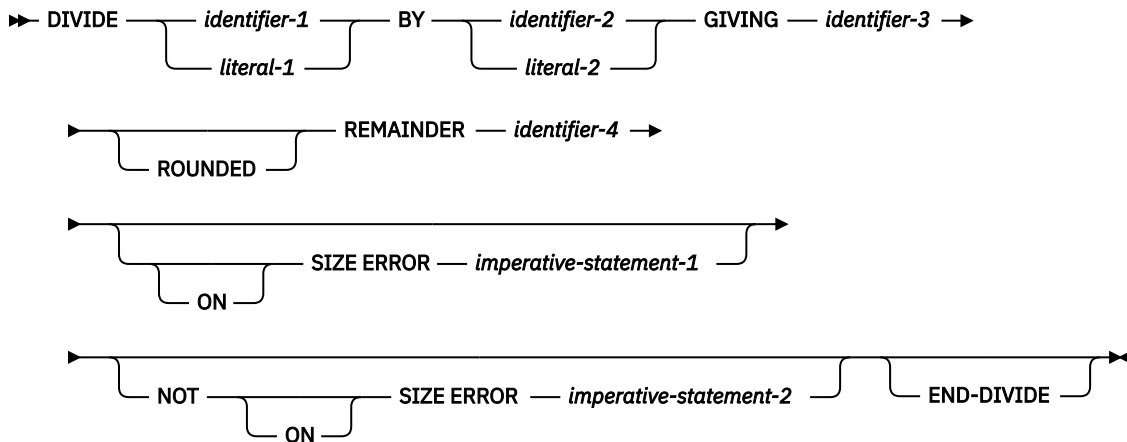
In format 4, the value of *identifier-1* or *literal-1* is divided into *identifier-2* or *literal-2*. The value of the quotient is stored in *identifier-3*, and the value of the remainder is stored in *identifier-4*.

Format 4 example:

```
DIVIDE A INTO B GIVING C REMAINDER D
```

The value in A is divided into the value in B and the results stored in C with the remainder being stored in D. The values in A and B are unchanged.

Format 5: DIVIDE statement with BY and REMAINDER phrases



In format 5, the value of *identifier-1* or *literal-1* is divided by *identifier-2* or *literal-2*. The value of the quotient is stored in *identifier-3*, and the value of the remainder is stored in *identifier-4*.

Format 5 example:

```
DIVIDE A BY B GIVING C REMAINDER D
```

The value in A is divided by the value in B and the result is stored in C with the remainder being stored in D. The values in A and B are unchanged.

For all formats:

identifier-1, identifier-2

Must name an elementary numeric data item.

identifier-3, identifier-4

Must name an elementary numeric or numeric-edited item.

literal-1, literal-2

Must be a numeric literal.

In formats 1, 2, and 3, floating-point data items and literals can be used anywhere that a numeric data item or literal can be specified.

In formats 4 and 5, floating-point data items or literals cannot be used.

ROUNDED phrase

For formats 1, 2, and 3, see [“ROUNDED phrase” on page 296](#).

For formats 4 and 5, the quotient used to calculate the remainder is in an intermediate field. The value of the intermediate field is truncated rather than rounded.

REMAINDER phrase

The result of subtracting the product of the quotient and the divisor from the dividend is stored in *identifier-4*. If *identifier-3*, the quotient, is a numeric-edited item, the quotient used to calculate the remainder is an intermediate field that contains the unedited quotient.

The REMAINDER phrase is invalid if the receiver or any of the operands is a floating-point item.

Any subscripts for *identifier-4* in the REMAINDER phrase are evaluated after the result of the divide operation is stored in *identifier-3* of the GIVING phrase.

SIZE ERROR phrases

For formats 1, 2, and 3, see “SIZE ERROR phrases” on page 296.

For formats 4 and 5, if a size error occurs in the quotient, no remainder calculation is meaningful. Therefore, the contents of the quotient field (*identifier-3*) and the remainder field (*identifier-4*) are unchanged.

If size error occurs in the remainder, the contents of the remainder field (*identifier-4*) are unchanged.

In either of these cases, you must analyze the results to determine which situation has actually occurred.

For information about the NOT ON SIZE ERROR phrase, see “SIZE ERROR phrases” on page 296.

END-DIVIDE phrase

This explicit scope terminator serves to delimit the scope of the DIVIDE statement. END-DIVIDE turns a conditional DIVIDE statement into an imperative statement that can be nested in another conditional statement. END-DIVIDE can also be used with an imperative DIVIDE statement.

For more information, see “Delimited scope statements” on page 293.

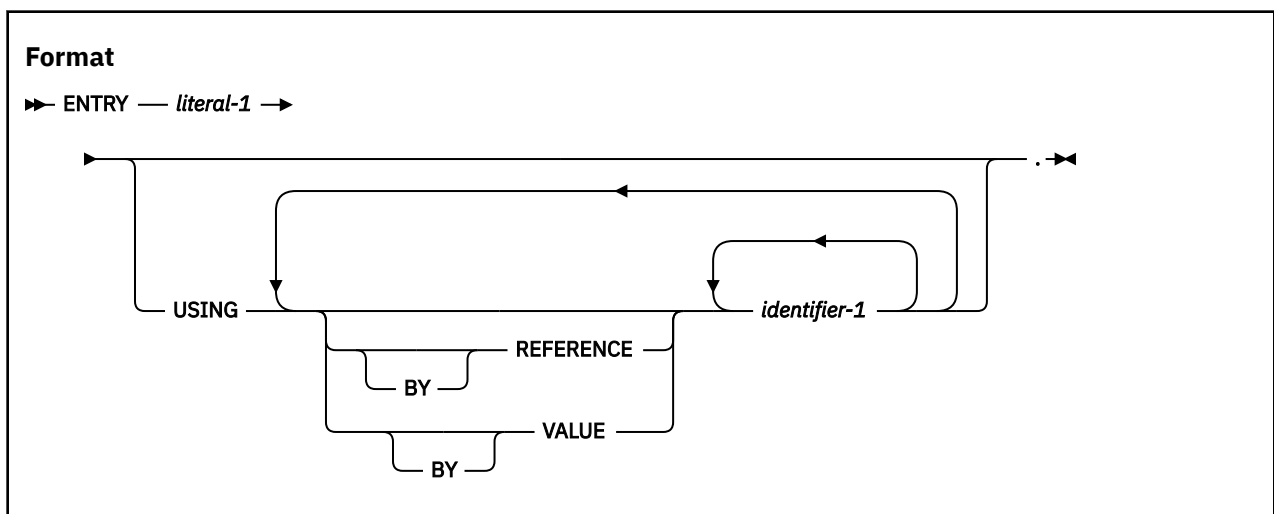
ENTRY statement

The ENTRY statement establishes an alternate entry point into a COBOL called subprogram.

The ENTRY statement cannot be used in:

- Programs that specify a return value using the PROCEDURE DIVISION RETURNING phrase. For details, see the discussion of the RETURNING phrase under “The PROCEDURE DIVISION header” on page 258.
- Nested program. See “Nested programs” on page 85 for a description of nested programs.
- Dynamic call or procedure-pointer call under AMODE 64. The programs must be compiled with LP(32) (the default) and run under AMODE(31) if there are ENTRY statements entered using dynamic call or via procedure pointer.

When a CALL statement that specifies the alternate entry point is executed in a calling program, control is transferred to the next executable statement following the ENTRY statement.



literal-1

Must be an alphanumeric literal that conform to the rules for the formation of a program-name in an outermost program (see [Chapter 15, “PROGRAM-ID paragraph,”](#) on page 101).

Must not match the program-ID or any other ENTRY literal in this program.

Must not be a figurative constant.

Execution of the called program begins at the first executable statement following the ENTRY statement whose literal corresponds to the literal or identifier specified in the CALL statement.

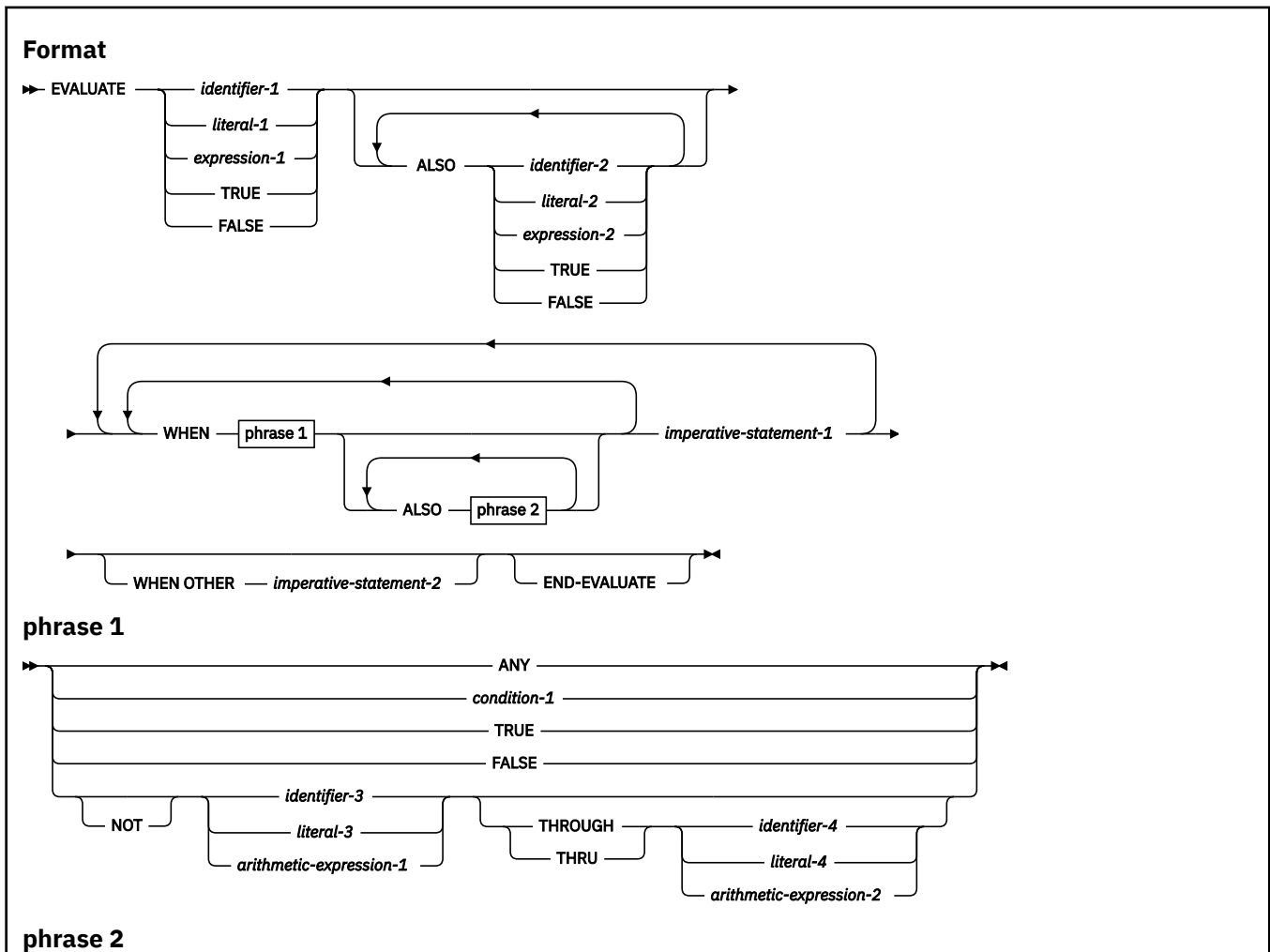
The entry point name on the ENTRY statement can be affected by the PGMNAME compiler option. For details, see *PGMNAME* in the *Enterprise COBOL Programming Guide*.

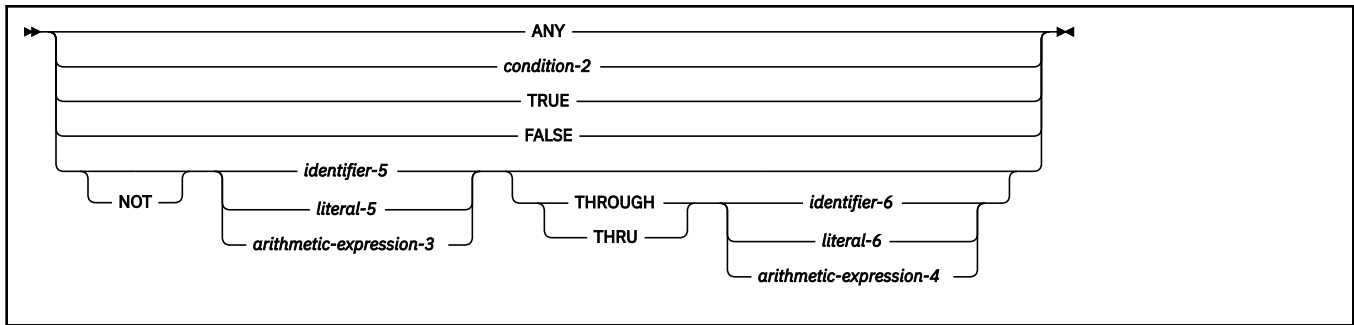
USING phrase

For a discussion of the USING phrase, see [“The PROCEDURE DIVISION header”](#) on page 258.

EVALUATE statement

The EVALUATE statement provides a shorthand notation for a series of nested IF statements. The EVALUATE statement can evaluate multiple conditions. The subsequent action depends on the results of these evaluations.





Operands before the WHEN phrase

Are interpreted in one of two ways, depending on how they are specified:

- Individually, they are called selection *subjects*.
- Collectively, they are called a *set* of selection subjects.

Operands in the WHEN phrase

Are interpreted in one of two ways, depending on how they are specified:

- Individually, they are called selection *objects*
- Collectively, they are called a *set* of selection objects.

ALSO

Separates selection subjects within a set of selection subjects; separates selection objects within a set of selection objects.

THROUGH and THRU

Are equivalent.

All identifiers in the EVALUATE statement must not be dynamic-length group items.

Two operands connected by a THRU phrase must be of the same class. The two operands thus connected constitute a single selection object.

The number of selection objects within each set of selection objects must be equal to the number of selection subjects.

Each selection object within a set of selection objects must correspond to the selection subject having the same ordinal position within the set of selection subjects, according to the following rules:

- Identifiers, literals, or arithmetic expressions appearing within a selection object must be valid operands for comparison to the corresponding operand in the set of selection subjects.
- *condition-1*, *condition-2*, or the word TRUE or FALSE appearing as a selection object must correspond to a conditional expression or the word TRUE or FALSE in the set of selection subjects.
- The word ANY can correspond to a selection subject of any type.

END-EVALUATE phrase

This explicit scope terminator serves to delimit the scope of the EVALUATE statement. END-EVALUATE permits a conditional EVALUATE statement to be nested in another conditional statement.

For more information, see [“Delimited scope statements” on page 293](#).

Determining values

The execution of the EVALUATE statement operates as if each selection subject and selection object were evaluated and assigned a numeric, alphanumeric, DBCS, or national character value; a range of numeric, alphanumeric, DBCS, or national character values; or a truth value.

These values are determined as follows:

- Any selection subject specified by *identifier-1*, *identifier-2*, ... and any selection object specified by *identifier-3* or *identifier-5* without the NOT or THRU phrase are assigned the value and class of the data item that they reference.
- Any selection subject specified by *literal-1*, *literal-2*, ... and any selection object specified by *literal-3* or *literal-5* without the NOT or THRU phrase are assigned the value and class of the specified literal. If *literal-3* or *literal-5* is the figurative constant ZERO, QUOTE, or SPACE, the figurative constant is assigned the class of the corresponding selection subject.
- Any selection subject in which *expression-1*, *expression-2*, ... is specified as an *arithmetic* expression, and any selection object without the NOT or THRU phrase in which *arithmetic-expression-1* or *arithmetic-expression-3* is specified, are assigned numeric values according to the rules for evaluating an arithmetic expression. (See [“Arithmetic expressions”](#) on page 266.)
- Any selection subject in which *expression-1*, *expression-2*, ... is specified as a *conditional* expression, and any selection object in which *condition-1* or *condition-2* is specified, are assigned a truth value according to the rules for evaluating conditional expressions. (See [“Conditional expressions”](#) on page 268.)
- Any selection subject or any selection object specified by the words TRUE or FALSE is assigned a truth value. The truth value "true" is assigned to those items specified with the word TRUE, and the truth value "false" is assigned to those items specified with the word FALSE.
- Any selection object specified by the word ANY is not further evaluated.
- If the THRU phrase is specified for a selection object without the NOT phrase, the range of values includes all values that, when compared to the selection subject, are greater than or equal to the first operand and less than or equal to the second operand according to the rules for comparison. If the first operand is greater than the second operand, there are no values in the range.
- If the NOT phrase is specified for a selection object, the values assigned to that item are all values not equal to the value, or range of values, that would have been assigned to the item had the NOT phrase been omitted.

Comparing selection subjects and objects

The execution of the EVALUATE statement then proceeds as if the values assigned to the selection subjects and selection objects were compared to determine whether any WHEN phrase satisfies the set of selection subjects.

This comparison proceeds as follows:

1. Each selection object within the set of selection objects for the first WHEN phrase is compared to the selection subject having the same ordinal position within the set of selection subjects. One of the following conditions must be satisfied if the comparison is to be satisfied:
 - a. If the items being compared are assigned numeric, alphanumeric, DBCS, or national character values, or a range of numeric, alphanumeric, DBCS, or national character values, the comparison is satisfied if the value, or one value in the range of values, assigned to the selection object is equal to the value assigned to the selection subject according to the rules for comparison.
 - b. If the items being compared are assigned truth values, the comparison is satisfied if the items are assigned identical truth values.
 - c. If the selection object being compared is specified by the word ANY, the comparison is always satisfied, regardless of the value of the selection subject.
2. If the above comparison is satisfied for every selection object within the set of selection objects being compared, the WHEN phrase containing that set of selection objects is selected as the one satisfying the set of selection subjects.
3. If the above comparison is not satisfied for every selection object within the set of selection objects being compared, that set of selection objects does not satisfy the set of selection subjects.
4. This procedure is repeated for subsequent sets of selection objects in the order of their appearance in the source text, until either a WHEN phrase satisfying the set of selection subjects is selected or until all sets of selection objects are exhausted.

Executing the EVALUATE statement

After the comparison operation is completed, execution of the EVALUATE statement proceeds.

- If a WHEN phrase is selected, execution continues with the first *imperative-statement-1* following the selected WHEN phrase. Note that multiple WHEN statements are allowed for a single *imperative-statement-1*.
- If no WHEN phrase is selected and a WHEN OTHER phrase is specified, execution continues with *imperative-statement-2*.
- If no WHEN phrase is selected and no WHEN OTHER phrase is specified, execution continues with the next executable statement following the scope delimiter.
- The scope of execution of the EVALUATE statement is terminated when execution reaches the end of the scope of the selected WHEN phrase or WHEN OTHER phrase, or when no WHEN phrase is selected and no WHEN OTHER phrase is specified.

EXIT statement

The EXIT statement provides a common end point for a series of procedures. It also provides a way to exit from a section, a paragraph, or an inline PERFORM statement.

Note: Enterprise COBOL does not yet support the format 4 EXIT statement, EXIT FUNCTION.

Format 1 (simple)

The format 1 EXIT statement provides a common end point for a series of procedures.

Format 1

➤ *paragraph-name* — . — EXIT ➤

The format 1 EXIT statement enables you to assign a procedure-name to a given point in a program.

The format 1 EXIT statement is treated as a CONTINUE statement. Any statements following the EXIT statement are executed.

Format 2 (program)

The EXIT PROGRAM statement specifies the end of a called program and returns control to the calling program.

You can specify EXIT PROGRAM only in the PROCEDURE DIVISION of a program. EXIT PROGRAM must not be used in a declarative procedure in which the GLOBAL phrase is specified.

Format 2

➤ EXIT PROGRAM ➤

If control reaches an EXIT PROGRAM statement in a program that does not possess the INITIAL attribute while operating under the control of a CALL statement (that is, the CALL statement is active), control returns to the point in the calling routine (program or method) immediately following the CALL statement. The state of the calling routine is identical to that which existed at the time it executed the CALL statement. The contents of data items and the contents of data files shared between the calling and called routine could have been changed. The state of the called program or method is not altered except that the ends of the ranges of all executed PERFORM statements are considered to have been reached.

The execution of an EXIT PROGRAM statement in a called program that possesses the INITIAL attribute is equivalent also to executing a CANCEL statement referencing that program.

If control reaches an EXIT PROGRAM statement, and no CALL statement is active, control passes through the exit point to the next executable statement.

If a subprogram specifies the PROCEDURE DIVISION RETURNING phrase, the value in the data item referred to by the RETURNING phrase becomes the result of the subprogram invocation.

The EXIT PROGRAM statement should be the last statement in a sequence of imperative statements. When it is not, statements following the EXIT PROGRAM will not be executed if a CALL statement is active.

When there is no next executable statement in a called program, an implicit EXIT PROGRAM statement is executed.

Format 3 (method)

The EXIT METHOD statement specifies the end of an invoked method.

Format 3

➡ EXIT METHOD ➡

You can specify EXIT METHOD only in the PROCEDURE DIVISION of a method. EXIT METHOD causes the executing method to terminate, and control returns to the invoking statement. If the containing method specifies the PROCEDURE DIVISION RETURNING phrase, the value in the data item referred to by the RETURNING phrase becomes the result of the method invocation.

If you need method-specific data to be in the *last-used* state on each invocation, define it in method WORKING-STORAGE. If you need method-specific data to be in the *initial* state on each invocation, define it in method LOCAL-STORAGE.

If control reaches an EXIT METHOD statement in a method definition, control returns to the point that immediately follows the INVOKE statement in the invoking program or method. The state of the invoking program or method is identical to that which existed at the time it executed the INVOKE statement.

The contents of data items and the contents of data files shared between the invoking program or method and the invoked method could have changed. The state of the invoked method is not altered except that the end of the ranges of all PERFORM statements executed by the method are considered to have been reached.

The EXIT METHOD statement does not have to be the last statement in a sequence of imperative statements, but the statements following the EXIT METHOD will not be executed.

When there is no next executable statement in an invoked method, an implicit EXIT METHOD statement is executed.

Format 5 (inline-perform)

The EXIT PERFORM statement controls the exit from an inline PERFORM without using a GO TO statement or a PERFORM ... THROUGH statement.

Format 5

➡ EXIT PERFORM ——— ➡
 └── CYCLE ─┘

An EXIT PERFORM statement specified outside of an inline PERFORM statement is not allowed.

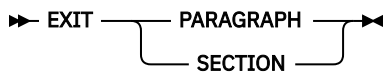
When an EXIT PERFORM statement without the CYCLE phrase is executed, control is passed to an implicit CONTINUE statement. This implicit CONTINUE statement immediately follows the END-PERFORM phrase that matches the most closely preceding and unterminated inline PERFORM statement.

When an EXIT PERFORM statement with the CYCLE phrase is executed, control is passed to an implicit CONTINUE statement. This implicit CONTINUE statement immediately precedes the END-PERFORM phrase that matches the most closely preceding and unterminated inline PERFORM statement.

Format 6 (procedure)

The EXIT PARAGRAPH statement controls the exit from the middle of a paragraph without executing any following statements within the paragraph. The EXIT SECTION statement controls the exit from a section without executing any following statements within the section.

Format 6



EXIT PARAGRAPH

When an EXIT PARAGRAPH statement is executed, control is passed to an implicit CONTINUE statement that immediately follows the last explicit statement of the current paragraph. This return mechanism supersedes any other return mechanisms that are associated with language elements, such as PERFORM, SORT, and USE for that paragraph.

EXIT SECTION

The EXIT SECTION statement can be specified only in a section.

When an EXIT SECTION statement is executed, control is passed to an unnamed empty paragraph that immediately follows the last paragraph of the current section. This return mechanism supersedes any other return mechanisms that are associated with language elements, such as PERFORM, SORT, and USE for that section.

FREE statement

The FREE statement releases dynamic storage that was previously obtained with an ALLOCATE statement from the Language Environment heap.

Format



data-name-1

Must be defined as USAGE POINTER or USAGE POINTER-32.

Can be qualified or subscripted.

The FREE statement is processed as follows:

- If the pointer referenced by *data-name-1* identifies the start of storage that is currently allocated by an ALLOCATE statement, that storage is released and the pointer referenced by *data-name-1* is set to NULL, the length of the released storage is the length of the storage obtained by the ALLOCATE statement, and the contents of any data items located within the released storage area become undefined.

- If the pointer referenced by *data-name-1* contains the predefined address NULL or the address of storage that is not acquired by the ALLOCATE statement (for example, using a GETMAIN or STORAGE macro), no storage will be freed. The pointer *data-name-1* will be kept unchanged and the behavior is undefined.

If more than one *data-name-1* is specified in a FREE statement, the result of executing this FREE statement is the same as if a separate FREE statement had been written for each *data-name-1* in the same order as specified in the FREE statement.

Related references

[“ALLOCATE statement” on page 313](#)

[“Example: ALLOCATE and FREE storage for UNBOUNDED tables” on page 316](#)

[POINTER phrase](#)

[POINTER-32 phrase](#)

GOBACK statement

The GOBACK statement functions like the EXIT PROGRAM statement when it is coded as part of a called program (or the EXIT METHOD statement when GOBACK is coded as part of an invoked method) and like the STOP RUN statement when coded in a main program.

The GOBACK statement specifies the logical end of a called program or invoked method.

Format

➤ GOBACK ➤

A GOBACK statement should appear as the only statement or as the last of a series of imperative statements in a sentence because any statements following the GOBACK are not executed. GOBACK must not be used in a declarative procedure in which the GLOBAL phrase is specified.

If control reaches a GOBACK statement while a CALL statement is active, control returns to the point in the calling program or method immediately following the CALL statement, as in the EXIT PROGRAM statement.

If control reaches a GOBACK statement while an INVOKE statement is active, control returns to the point in the invoking program or method immediately following the INVOKE statement, as in the EXIT METHOD statement.

In addition, the execution of a GOBACK statement in a called program that possesses the INITIAL attribute is equivalent to executing a CANCEL statement referencing that program.

The table below shows the action taken for the GOBACK statement in a main program, a subprogram, and an invoked method.

Termination statement	Main program	Subprogram	Invoked method
GOBACK	Returns to the calling program. (Can be the system, which causes the application to end.)	Returns to the calling program.	Returns to the calling method.

GO TO statement

The GO TO statement transfers control from one part of the PROCEDURE DIVISION to another.

The types of GO TO statements are:

- Unconditional
- Conditional

- A program compiled with the THREAD compiler option

An ALTER statement referring to the paragraph that contains the altered GO TO statement should be executed before the GO TO statement is executed. Otherwise, the GO TO statement acts like a CONTINUE statement.

Format 3: altered GO TO statement

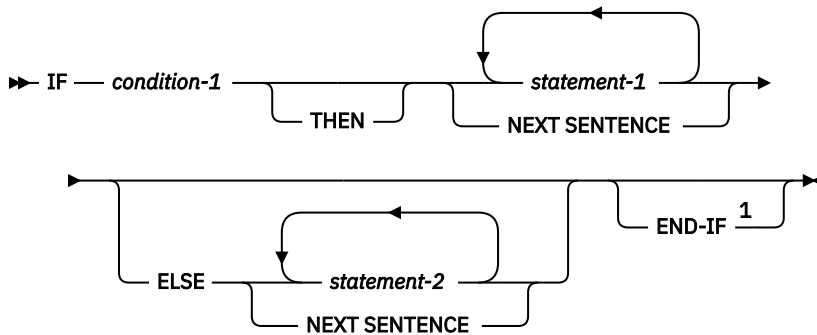
➤ *paragraph-name* — . — GO — TO — ➤

When an ALTER statement refers to a paragraph, the paragraph can consist only of the paragraph-name followed by an unconditional or altered GO TO statement.

IF statement

The IF statement evaluates a condition and provides for alternative actions in the object program, depending on the evaluation.

Format



Notes:

¹ END-IF can be specified with *statement-2* or NEXT SENTENCE.

condition-1

Can be any simple or complex condition, as described in [“Conditional expressions”](#) on page 268.

statement-1, statement-2

Can be any one of the following options:

- An imperative statement
- A conditional statement
- An imperative statement followed by a conditional statement

NEXT SENTENCE

The NEXT SENTENCE phrase transfers control to an implicit CONTINUE statement immediately following the next *separator period*.

When NEXT SENTENCE is specified with END-IF, control does not pass to the statement following the END-IF. Instead, control passes to the statement after the closest following period.

END-IF phrase

This explicit scope terminator serves to delimit the scope of the IF statement. END-IF permits a conditional IF statement to be nested in another conditional statement. For more information about explicit scope terminators, see [“Delimited scope statements” on page 293](#).

The scope of an IF statement can be terminated by any of the following options:

- An END-IF phrase at the same level of nesting
- A separator period
- If nested, by an ELSE phrase associated with an IF statement at a higher level of nesting

Transferring control

The topic describes the actions to take when conditions tested is true or false.

If the condition tested is true, one of the following actions takes place:

- If *statement-1* is specified, *statement-1* is executed. If *statement-1* contains a procedure branching or conditional statement, control is transferred according to the rules for that statement. If *statement-1* does not contain a procedure-branching statement, the ELSE phrase, if specified, is ignored, and control passes to the next executable statement after the corresponding END-IF or separator period.
- If NEXT SENTENCE is specified, control passes to an implicit CONTINUE statement immediately following the next separator period.

If the condition tested is false, one of the following actions takes place:

- If ELSE *statement-2* is specified, *statement-2* is executed. If *statement-2* contains a procedure-branching or conditional statement, control is transferred, according to the rules for that statement. If *statement-2* does not contain a procedure-branching or conditional statement, control is passed to the next executable statement after the corresponding END-IF or separator period.
- If ELSE NEXT SENTENCE is specified, control passes to an implicit CONTINUE STATEMENT immediately preceding the next separator period.
- If neither ELSE *statement-2* nor ELSE NEXT SENTENCE is specified, control passes to the next executable statement after the corresponding END-IF or separator period.

When the ELSE phrase is omitted, all statements following the condition and preceding the corresponding END-IF or the separator period for the sentence are considered to be part of *statement-1*.

Nested IF statements

When an IF statement appears as *statement-1* or *statement-2*, or as part of *statement-1* or *statement-2*, that IF statement is *nested*.

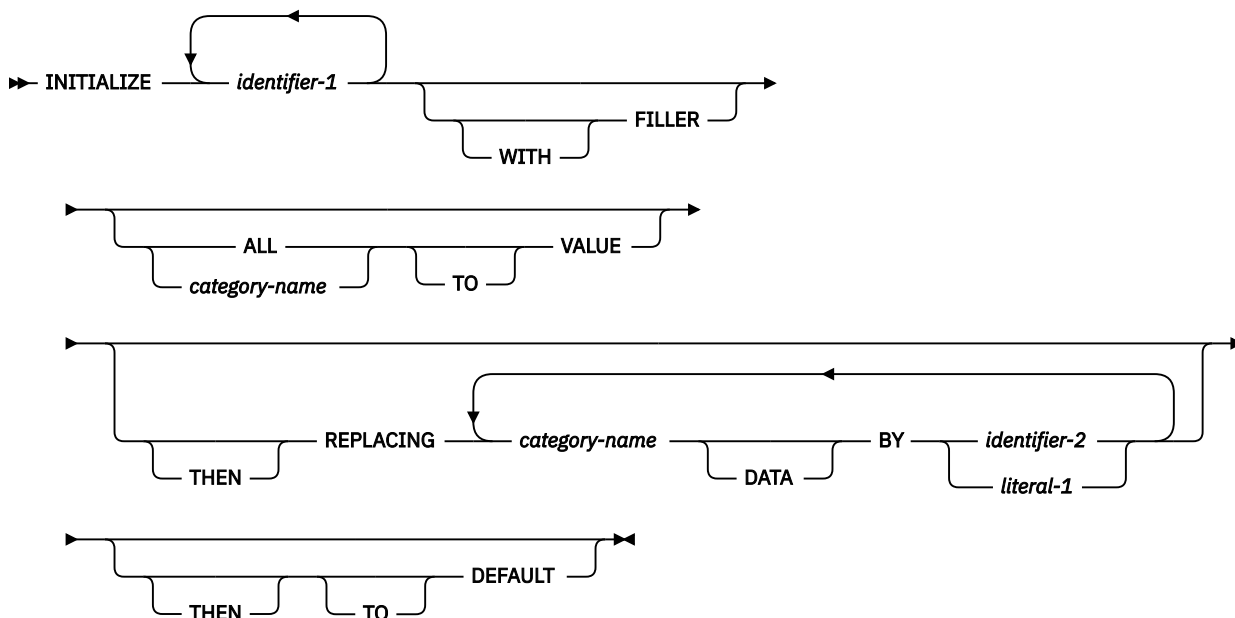
When an IF statement appears as *statement-1* or *statement-2*, or as part of *statement-1* or *statement-2*, that IF statement is *nested*.

Nested IF statements are considered to be matched IF, ELSE, and END-IF combinations proceeding from left to right. Thus, any ELSE encountered is matched with the nearest preceding IF that either has not been already matched with an ELSE or has not been implicitly or explicitly terminated. Any END-IF encountered is matched with the nearest preceding IF that has not been implicitly or explicitly terminated.

INITIALIZE statement

The INITIALIZE statement sets selected categories of data fields to predetermined values. The INITIALIZE statement is functionally equivalent to one or more MOVE statements.

Format



Where *category-name* is:

- ALPHABETIC
- ALPHANUMERIC
- ALPHANUMERIC-EDITED
- DBCS
- EGCS
- NATIONAL
- NATIONAL-EDITED
- NUMERIC
- NUMERIC-EDITED
- UTF-8

identifier-1

Receiving areas.

identifier-1 must reference one of the following items:

- An alphanumeric group item
- A national group item
- A UTF-8 group item
- An elementary data item of one of the following categories:
 - Alphabetic
 - Alphanumeric
 - Alphanumeric-edited
 - DBCS

- External floating-point
- Internal floating-point
- National
- National-edited
- Numeric
- Numeric-edited
- UTF-8
- A special register that is valid as a receiving operand in a MOVE statement with *identifier-2* or *literal-1* as the sending operand.

identifier-1 references an elementary item or a group item. The effect of the execution of an INITIALIZE statement is as if a series of implicit MOVE statements, each of which has an elementary data item as its receiving operand, were executed.

When *identifier-1* references a national or UTF-8 group item, *identifier-1* is processed as a group item.

identifier-1 cannot be a dynamic-length elementary item.

identifier-2, literal-1

Sending areas.

When *identifier-2* references a national or UTF-8 group item, *identifier-2* is processed as an elementary data item of category national or UTF-8, respectively.

identifier-2 must reference an elementary data item (or a national or UTF-8 group item treated as elementary) that is valid as a sending operand in a MOVE statement with *identifier-1* as the receiving operand.

literal-1 must be a literal that is valid as a sending operand in a MOVE statement with *identifier-1* as the receiving operand.

A subscripted item can be specified for *identifier-1*. A complete table can be initialized only by specifying *identifier-1* as a group that contains the complete table.

Usage note: The data description entry for *identifier-1* can contain the DEPENDING phrase of the OCCURS clause. However, you cannot use the INITIALIZE statement to initialize a variably-located item or a variable-length item.

The data description entry for *identifier-1* must not contain a RENAME clause.

Special registers can be specified for *identifier-1* and *identifier-2* only if they are valid receiving fields or sending fields, respectively, for the implied MOVE statements.

FILLER phrase

When the FILLER phrase is specified, the receiving elementary data items that have an explicit or implicit FILLER clause will be initialized.

VALUE phrase

When the VALUE phrase is specified:

- If ALL is specified in the VALUE phrase, it is as if all of the categories listed in *category-name* were specified.
- The same category cannot be repeated in a VALUE phrase.

REPLACING phrase

When the REPLACING phrase is specified:

- *identifier-2* must reference an item of a category that is valid as a sending operand in a MOVE statement to an item of the corresponding category specified in the REPLACING phrase.
- *literal-1* must be of a category that is valid as a sending operand in a MOVE statement to an item of the corresponding category specified in the REPLACING phrase.
- A floating-point literal, a data item of category internal floating-point, or a data item of category external floating point is treated as if it were in the NUMERIC category.
- The same category cannot be repeated in a REPLACING phrase.

With the exception of EGCS, the keyword after the word REPLACING corresponds to a category of data shown in [“Classes and categories of data” on page 170](#).

EGCS in the REPLACING phrase is synonymous with DBCS.

Related references

[“ALLOCATE statement” on page 313](#)

INITIALIZE statement rules

The effect of the execution of an INITIALIZE statement is as if a series of implicit MOVE statements, each of which has an elementary data item as its receiving operand, were executed. The receiving operands of these implicit statements are defined in [rule 1](#) and the sending operands are defined in [rule 2](#).

1. The receiving operand in each implicit MOVE statement is determined by applying the rules a, b, and c in the order they appear below. Note that if a data item is not excluded as a receiver by a particular rule, it may be excluded as a receiver when a subsequent rule is applied. For example, if a data item is not excluded by rule a, that data item may still be excluded by rule b or rule c.
 - a. First, the following data items are excluded as receiving operands:
 - Any identifiers that are not valid receiving operands of a MOVE statement.
 - Elementary data items that have an explicit or implicit FILLER clause if the FILLER phrase is not specified.
 - Any elementary data item subordinate to *identifier-1* whose data description entry contains a REDEFINES or RENAMES clause or is subordinate to a data item whose data description entry contains a REDEFINES clause. However, *identifier-1* might itself have a REDEFINES clause or be subordinate to a data item with a REDEFINES clause.
 - b. Second, an elementary data item is a possible receiving item in either of the following cases:
 - It is explicitly referenced by *identifier-1*.
 - It is contained within the group data item referenced by *identifier-1*. If the elementary data item is a table element, each occurrence of the elementary data item is a possible receiving operand.
 - c. Finally, each possible receiving operand is a receiving operand if at least one of the following conditions is true:
 - The VALUE phrase is specified, the category of the elementary data item is one of the categories specified or implied in the VALUE phrase, and either of the following conditions is true:
 - A data-item format VALUE clause is specified in the data description entry of the elementary data item.
 - A table format VALUE clause is specified in the data description entry of the elementary item and that VALUE clause specifies a value for the particular occurrence of the elementary data item.
 - The REPLACING phrase is specified and the category of the elementary data item is one of the categories specified in the REPLACING phrase.
 - The DEFAULT phrase is specified.
 - Neither the REPLACING phrase nor the VALUE phrase is specified.
2. The sending operand in each implicit MOVE statement is determined as follows:

- If the data item qualifies as a receiving operand because of the VALUE phrase, the sending operand is determined by the literal in the VALUE clause specified in the data description entry of the data item. If the data item is a table element, the literal in the VALUE clause that corresponds to the occurrence being initialized determines the sending operand. The actual sending operand is a literal that, when moved to the receiving operand with a MOVE statement, produces the same result as the initial value of the data item as produced by the application of the VALUE clause.
- If the data item does not qualify as a receiving operand because of the VALUE phrase, but does qualify because of the REPLACING phrase, the sending operand is the *literal-1* or *identifier-2* associated with the category specified in the REPLACING phrase.
- If the data item does not qualify in accordance with the preceding two rules, the sending operand used depends on the category of the receiving operand as follows:
 - SPACE is the implied sending item for receiving items of category alphabetic, alphanumeric, alphanumeric-edited, DBCS, EGCS, national, national-edited, or UTF-8.
 - ZERO is the implied sending item for receiving items of category numeric or numeric-edited.

INSPECT statement

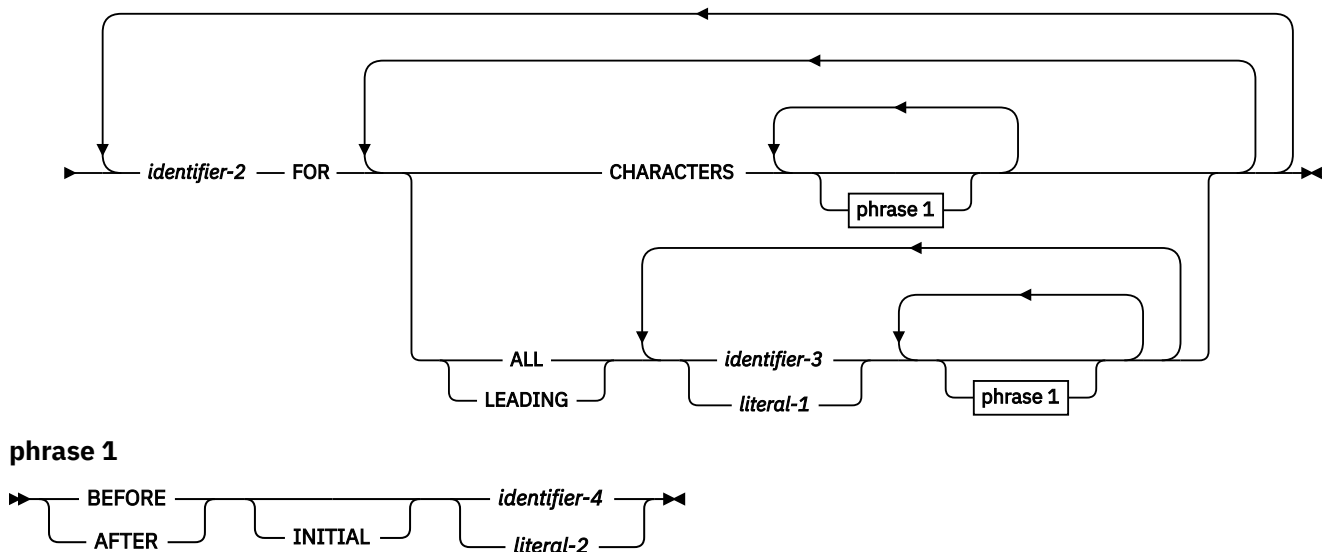
The INSPECT statement examines characters or groups of characters in a data item.

The INSPECT statement does the following tasks:

- Counts the occurrences of a specific character (alphanumeric, DBCS, or national) in a data item (formats 1 and 3).
- Counts the occurrences of specific characters and fills all or portions of a data item with specified characters, such as spaces or zeros (formats 2 and 3).
- Converts all occurrences of specific characters in a data item to user-supplied replacement characters (format 4).

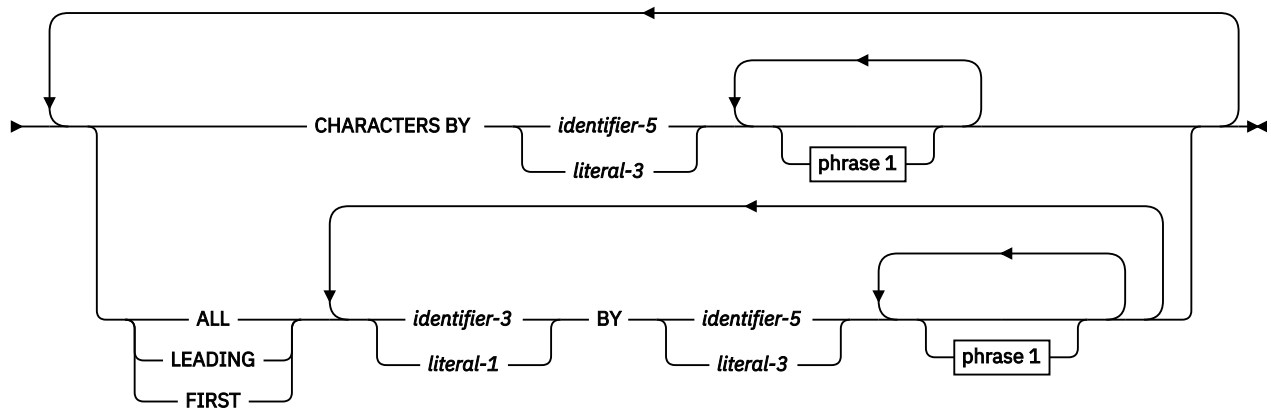
Format 1: INSPECT statement with TALLYING phrase

►► INSPECT — *identifier-1* — TALLYING —►

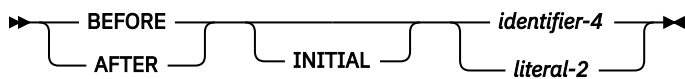


Format 2: INSPECT statement with REPLACING phrase

►► INSPECT — *identifier-1* — REPLACING →

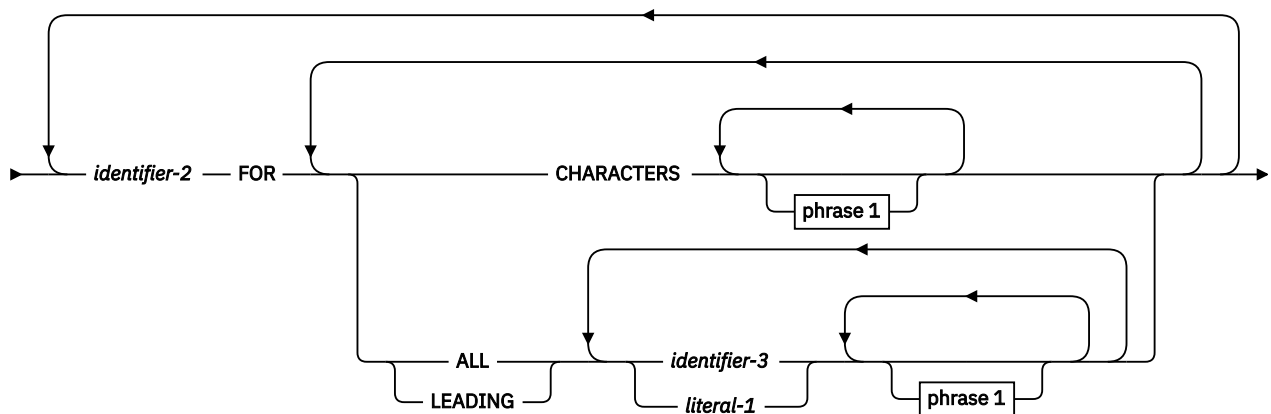


phrase 1

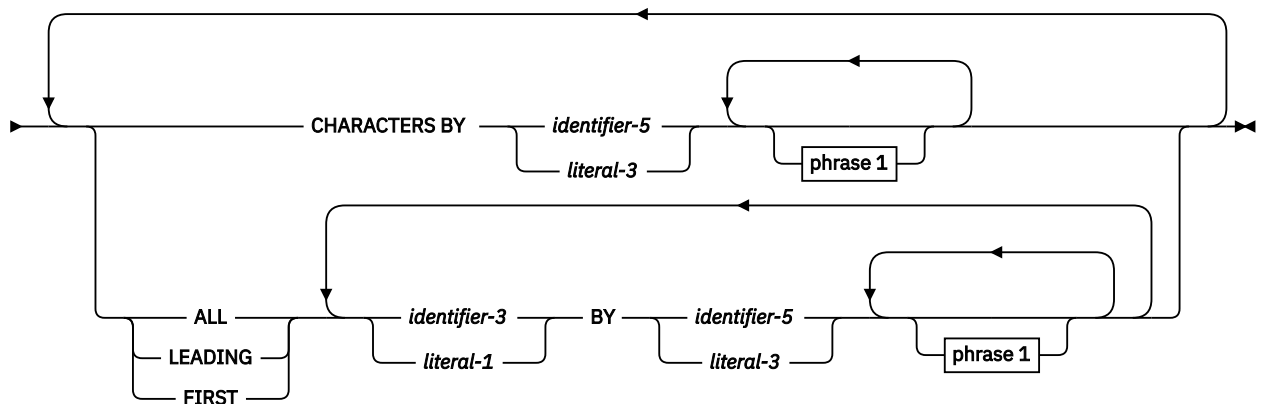


Format 3: INSPECT statement with TALLYING and REPLACING phrases

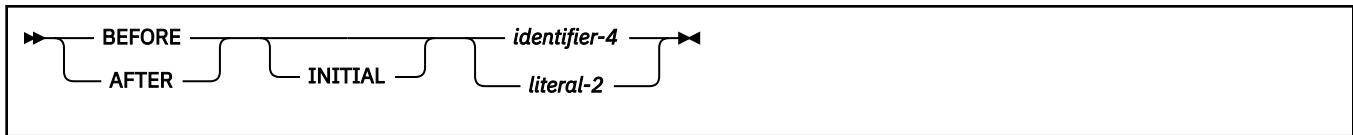
►► INSPECT — *identifier-1* — TALLYING →



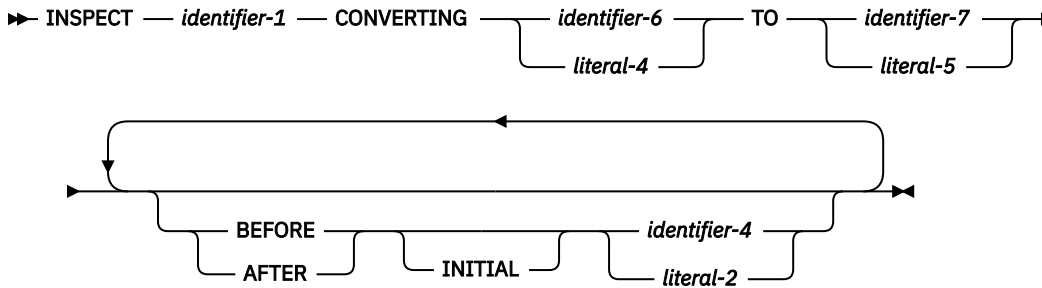
►► REPLACING →



phrase 1



Format 4: INSPECT statement with CONVERTING phrase



identifier-1

Is the *inspected item* and can be any of the following items:

- An alphanumeric group item or a national group item
- An elementary data item described explicitly or implicitly with usage DISPLAY, DISPLAY-1, or NATIONAL. The item can have any category that is valid for the selected usage.

identifier-3 , identifier-4 , identifier-5 , identifier-6 , identifier-7

Must reference an elementary data item described explicitly or implicitly with usage DISPLAY, DISPLAY-1, or NATIONAL.

literal-1 , literal-2 , literal-3 , literal-4

Must be of category alphanumeric, DBCS, or national.

When *identifier-1* is of usage NATIONAL, literals must be of category national.

When *identifier-1* is of usage DISPLAY-1, literals must be of category DBCS.

When *identifier-1* is of usage DISPLAY, literals must be of category alphanumeric.

When *identifier-1* is of usage DISPLAY-1 (DBCS) literals may be the figurative constant SPACE.

When *identifier-1* is of usage DISPLAY or NATIONAL, literals can be any figurative constant that does not begin with the word ALL, as specified in “Figurative constants” on page 15. The figurative constant is treated as a one-character alphanumeric literal when *identifier-1* is of usage DISPLAY, and as a one-character national literal when *identifier-1* is of usage NATIONAL.

All identifiers (except *identifier-2*) must have the same usage as *identifier-1*. All literals must have category alphanumeric, DBCS, or national when *identifier-1* has usage DISPLAY, DISPLAY-1, or NATIONAL, respectively.

All identifiers may not be dynamic-length group or dynamic-length elementary items.

TALLYING phrase (formats 1 and 3)

This phrase counts the occurrences of a specific character or special character in a data item.

When *identifier-1* is a DBCS data item, DBCS characters are counted; when *identifier-1* is a data item of usage national, national characters (encoding units) are counted; otherwise, alphanumeric characters (bytes) are counted.

identifier-2

Is the *count field*, and must be an elementary integer item defined without the symbol P in its PICTURE character-string.

identifier-2 cannot be of category external floating-point.

You must initialize *identifier-2* before execution of the INSPECT statement begins.

Usage note: The count field can be an integer data item defined with usage NATIONAL.

identifier-3 or literal-1

Is the *tallying field* (the item whose occurrences will be tallied).

CHARACTERS

When CHARACTERS is specified and neither the BEFORE nor AFTER phrase is specified, the count field (*identifier-2*) is increased by 1 for each character (including the space character) in the inspected item (*identifier-1*). Thus, execution of an INSPECT statement with the TALLYING phrase increases the value in the count field by the number of character positions in the inspected item.

ALL

When ALL is specified and neither the BEFORE nor AFTER phrase is specified, the count field (*identifier-2*) is increased by 1 for each nonoverlapping occurrence of the tallying comparand (*identifier-3 or literal-1*) in the inspected item (*identifier-1*), beginning at the leftmost character position and continuing to the rightmost.

LEADING

When LEADING is specified and neither the BEFORE nor AFTER phrase is specified, the count field (*identifier-2*) is increased by 1 for each contiguous nonoverlapping occurrence of the tallying comparand in the inspected item (*identifier-1*), provided that the leftmost such occurrence is at the point where comparison began in the first comparison cycle for which the tallying comparand is eligible to participate.

FIRST (format 3 only)

When FIRST is specified and neither the BEFORE nor AFTER phrase is specified, the substitution field replaces the leftmost occurrence of the subject field in the inspected item (*identifier-1*).

TALLYING phrase example

```
WORKING-STORAGE SECTION.  
77 CNTR PIC 9(3) COMP.  
77 CHARS PIC X(18).  
PROCEDURE DIVISION.  
.....  
    MOVE 'In order to form a' To CHARS  
    MOVE 0 To CNTR  
    INSPECT CHARS TALLYING CNTR FOR ALL SPACES  
    DISPLAY 'Number of spaces = ' CNTR
```

The example output:

```
Number of spaces = 004
```

REPLACING phrase (formats 2 and 3)

This phrase fills all or portions of a data item with specified characters, such as spaces or zeros.

identifier-3 or literal-1

Is the *subject field*, which identifies the characters to be replaced.

identifier-5 or literal-3

Is the *substitution field* (the item that replaces the subject field).

The subject field and the substitution field must be the same length.

CHARACTERS BY

When the CHARACTERS BY phrase is used, the substitution field must be one character position in length.

When CHARACTERS BY is specified and neither the BEFORE nor AFTER phrase is specified, the substitution field replaces each character in the inspected item (*identifier-1*), beginning at the leftmost character position and continuing to the rightmost.

ALL

When ALL is specified and neither the BEFORE nor AFTER phrase is specified, the substitution field replaces each nonoverlapping occurrence of the subject field in the inspected item (*identifier-1*), beginning at the leftmost character position and continuing to the rightmost.

LEADING

When LEADING is specified and neither the BEFORE nor AFTER phrase is specified, the substitution field replaces each contiguous nonoverlapping occurrence of the subject field in the inspected item (*identifier-1*), provided that the leftmost such occurrence is at the point where comparison began in the first comparison cycle for which this substitution field is eligible to participate.

FIRST

When FIRST is specified and neither the BEFORE nor AFTER phrase is specified, the substitution field replaces the leftmost occurrence of the subject field in the inspected item (*identifier-1*).

When both the TALLYING and REPLACING phrases are specified (format 3), the INSPECT statement is executed as if an INSPECT TALLYING statement (format 1) were specified, immediately followed by an INSPECT REPLACING statement (format 2).

The following replacement rules apply:

- When the subject field is a figurative constant, the one-character substitution field replaces each character in the inspected item that is equivalent to the figurative constant.
- When the substitution field is a figurative constant, the substitution field replaces each nonoverlapping occurrence of the subject field in the inspected item.
- When the subject and substitution fields are character-strings, the character-string specified in the substitution field replaces each nonoverlapping occurrence of the subject field in the inspected item.
- After replacement has occurred in a given character position in the inspected item, no further replacement for that character position is made in this execution of the INSPECT statement.

REPLACING phrase example

```
WORKING-STORAGE SECTION.  
77 CNTR PIC 9(3) COMP.  
77 CHARS PIC X(18).  
PROCEDURE DIVISION.  
    . . .  
    MOVE 'more,perfect,union' To CHARS  
    MOVE 0 To CNTR  
  
    INSPECT CHARS TALLYING CNTR FOR ALL ','  
    REPLACING ALL ',' BY SPACES  
  
    DISPLAY 'Number of commas replaced = ' CNTR  
    DISPLAY 'CHARS is now = ' CHARS
```

The example result:

```
Number of commas replaced = 002  
CHARS is now = more perfect union
```

BEFORE and AFTER phrases (all formats)

This phrase narrows the set of items being tallied or replaced.

No more than one BEFORE phrase and one AFTER phrase can be specified for any one ALL, LEADING, CHARACTERS, FIRST or CONVERTING phrase.

identifier-4 or *literal-2*

Is the *delimiter*.

Delimiters are not counted or replaced.

INITIAL

The first occurrence of a specified item.

The BEFORE and AFTER phrases change how counting and replacing are done:

- When BEFORE is specified, counting or replacing of the inspected item (*identifier-1*) begins at the leftmost character position and continues until the first occurrence of the delimiter is encountered. If no delimiter is present in the inspected item, counting or replacing continues toward the rightmost character position.
- When AFTER is specified, counting or replacing of the inspected item (*identifier-1*) begins with the first character position to the right of the delimiter and continues toward the rightmost character position in the inspected item. If no delimiter is present in the inspected item, no counting or replacement takes place.

BEFORE and AFTER phrases example

```
WORKING-STORAGE SECTION.  
77 CNTR1 PIC 9(3) COMP.  
77 CNTR2 PIC 9(3) COMP.  
77 CNTR3 PIC 9(3) COMP.  
77 CHARS PIC X(50).  
  
PROCEDURE DIVISION.  
  
    MOVE '$some.confusing_text with.hyphens-periods.spaces'  
      To CHARS  
    MOVE 0 To CNTR1 CNTR2 CNTR3  
    INSPECT CHARS  
      TALLYING CNTR1 FOR CHARACTERS AFTER '$' BEFORE '_'  
            CNTR2 FOR ALL '.' AFTER 'h'  
            CNTR3 FOR ALL '.' BEFORE INITIAL 'a'  
      REPLACING ALL '.' BY SPACES BEFORE 'h'  
            ALL ' ' BY SPACES AFTER 's'  
            ALL '-' BY SPACES  
    DISPLAY 'Number of characters after $ and before _ =' CNTR1  
    DISPLAY 'Number of periods after h =' CNTR2  
    DISPLAY 'Number of periods before initial a =' CNTR3  
    DISPLAY 'CHARS is now = ' CHARS
```

The example result:

Note: For each BEFORE and AFTER phrase, the scanning continues after the previous phrase is done rather than starting at the beginning of CHARS.

```
Number of characters after $ and before _ =014  
Number of periods after h =002  
Number of periods before initial a =000  
CHARS is now = $some confusing text with.hyphens-periods.spaces
```

CONVERTING phrase (format 4)

This phrase converts all occurrences of a specific character or string of characters in a data item (*identifier-1*) to user-supplied replacement characters.

identifier-6 or *literal-4*

Specifies the character string to be *replaced*.

The same character must not appear more than once in either *literal-4* or *identifier-6*.

identifier-7 or *literal-5*

Specifies the *replacing* character string.

The replacing character string (*identifier-7* or *literal-5*) must be the same size as the replaced character string (*identifier-6* or *literal-4*).

A format-4 INSPECT statement is interpreted and executed as if a format-2 INSPECT statement had been written with a series of ALL phrases (one for each character of *literal-4*), specifying the same *identifier-1*. The effect is as if each single character of *literal-4* were referenced as *literal-1*, and the corresponding

single character of *literal-5* referenced as *literal-3*. Correspondence between the characters of *literal-4* and the characters of *literal-5* is by ordinal position within the data item.

If *identifier-4*, *identifier-6*, or *identifier-7* occupies the same storage area as *identifier-1*, the result of the execution of this statement is undefined, even if they are defined by the same data description entry.

The following table describes the treatment of data items that can be used as an operand in the INSPECT statement:

Table 40. Treatment of the content of data items	
When referenced by any identifier except <i>identifier-2</i>, the content of each item of category ...	Is treated ...
Alphanumeric or alphabetic	As an alphanumeric character string
DBCS	As a DBCS character string
National	As a national character string
Alphanumeric-edited, numeric-edited with usage DISPLAY, or numeric with usage DISPLAY (unsigned, external decimal)	As if redefined as category alphanumeric, with the INSPECT statement referring to an alphanumeric character string
National-edited, numeric-edited with usage NATIONAL or numeric with usage NATIONAL (unsigned, external decimal)	As if redefined as category national, with the INSPECT statement referring to a national character string
Numeric with usage DISPLAY (signed, external decimal)	As if moved to an unsigned external decimal item of usage DISPLAY with the same length as the identifier and then redefined as category alphanumeric, with the INSPECT statement referring to an alphanumeric character string If the sign is a separate character, the byte containing the sign is not examined and, therefore, not replaced. If the referenced item is <i>identifier-1</i>, the string that results from any replacing or converting action is copied back to <i>identifier-1</i>.
Numeric with usage NATIONAL (signed, external decimal)	As if moved to an unsigned external decimal item of usage NATIONAL with the same length as the identifier and then redefined as category national, with the INSPECT statement referring to a national character string If the sign is a separate character, the byte containing the sign is not examined and, therefore, not replaced. If the referenced item is <i>identifier-1</i>, the string that results from any replacing or converting action is copied back to <i>identifier-1</i>.
External floating-point with usage DISPLAY	As if redefined as category alphanumeric, with the INSPECT statement referring to an alphanumeric character-string
External floating-point with usage NATIONAL	As if redefined as category national, with the INSPECT statement referring to a national character-string

CONVERTING phrase example

The following example shows the use of INSPECT CONVERTING with AFTER and BEFORE phrases to examine and replace characters in data item DATA-4. All characters that follow the first instance of the character / but that precede the first instance of the character ? (if any) are translated from lowercase to uppercase.

```
01 DATA-4          PIC X(11).  
  . . .  
    INSPECT DATA-4  
      CONVERTING  
        "abcdefghijklmnopqrstuvwxyz" TO  
        "ABCDEFGHIJKLMNOPQRSTUVWXYZ"  
      AFTER INITIAL "/"  
      BEFORE INITIAL "?"
```

Table 41. CONVERTING example result	
DATA-4 before converting	DATA-4 after converting
a/five/?six	a/FIVE/?six
r/Rexx/RRRr	r/REXX/RRRR
zfour?inspe	zfour?inspe

Data flow

Except when the BEFORE or AFTER phrase is specified, inspection begins at the leftmost character position of the inspected item (*identifier-1*) and proceeds character-by-character to the rightmost position.

The comparands of the following phrases are compared in the left-to-right order in which they are specified in the INSPECT statement:

- TALLYING (*literal-1* or *identifier-3*, ...)
- REPLACING (*literal-3* or *identifier-5*, ...)

If any identifier is subscripted or reference modified, or is a function-identifier, the subscript, reference-modifier, or function is evaluated only once as the first operation in the execution of the INSPECT statement.

For examples of TALLYING and REPLACING, see *Tallying and replacing data items (INSPECT)* in the *Enterprise COBOL Programming Guide*.

Comparison cycle

The comparison cycle consists of the actions as described in this topic.

1. The first comparand is compared with an equal number of leftmost contiguous character positions in the inspected item. The comparand matches the inspected characters only if both are equal, character-for-character.

If the CHARACTERS phrase is specified, an implied one-character comparand is used. The implied character is always considered to match the inspected character in the inspected item.
2. If no match occurs for the first comparand and there are more comparands, the comparison is repeated for each successive comparand until either a match is found or all comparands have been acted upon.
3. Depending on whether a match is found, these actions are taken:
 - If a match is found, tallying or replacing takes place as described in the TALLYING and REPLACING phrase descriptions.

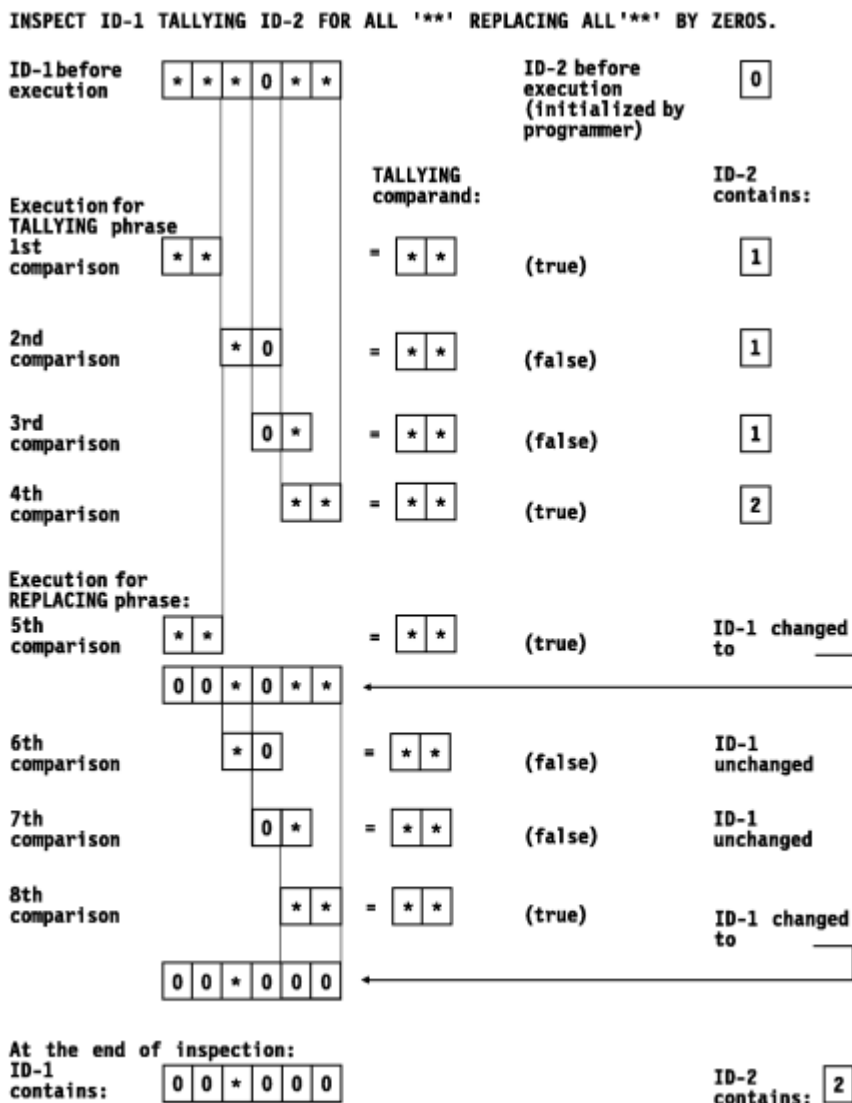
If there are more character positions in the inspected item, the first character position following the rightmost matching character is now considered to be in the leftmost character position. The process described in actions 1 and 2 is then repeated.

- If no match is found and there are more character positions in the inspected item, the first character position following the leftmost inspected character is now considered to be in the leftmost character position. The process described in actions 1 and 2 is then repeated.
4. Actions 1 through 3 are repeated until the rightmost character position in the inspected item either has been matched or has been considered as being in the leftmost character position.

When the BEFORE or AFTER phrase is specified, the comparison cycle is modified, as described in [“BEFORE and AFTER phrases \(all formats\)”](#) on page 357.

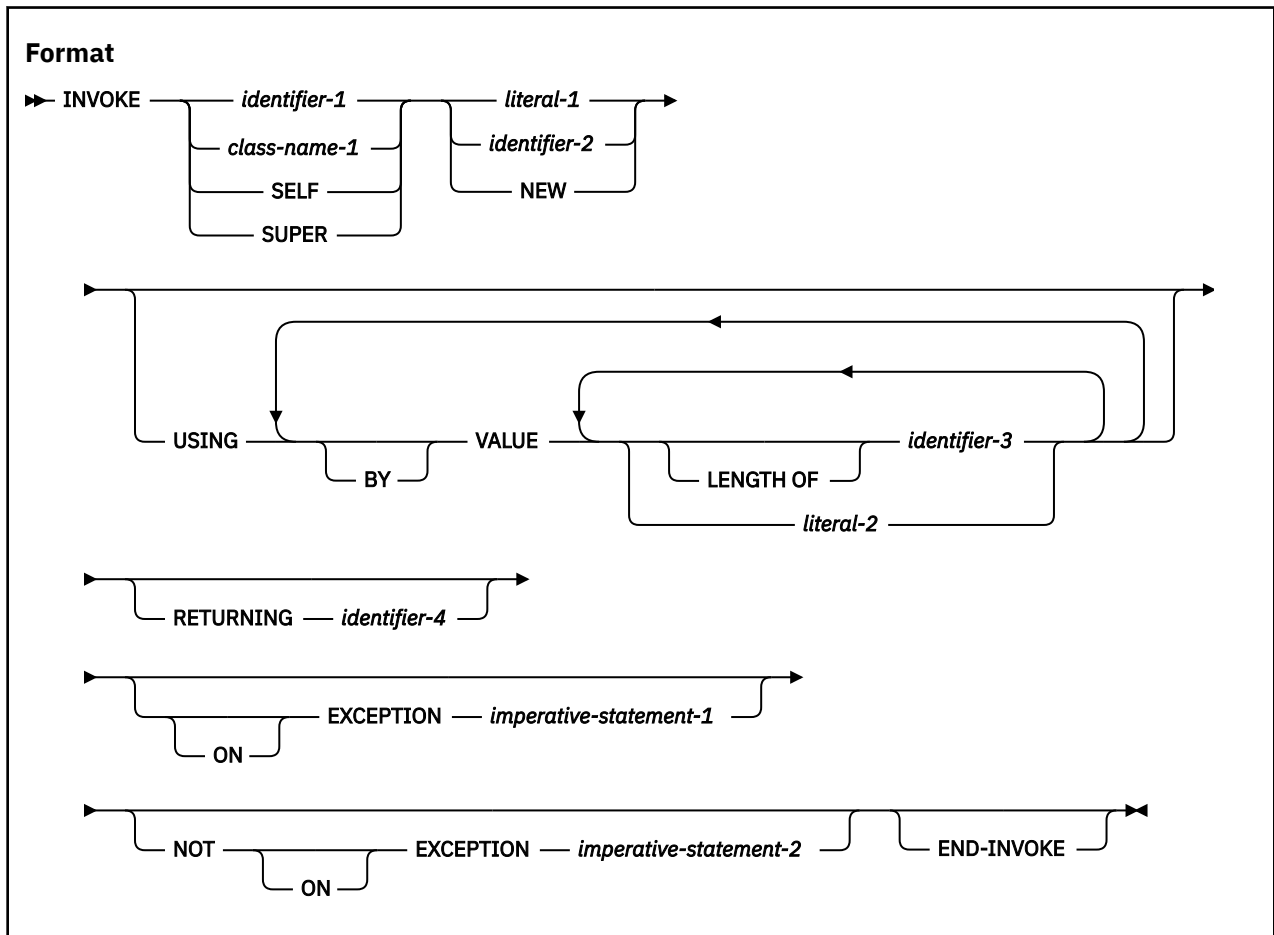
Example of the INSPECT statement

The topic shows an example of INSPECT statement results.



INVOKE statement

The INVOKE statement can create object instances of a COBOL or Java class and can invoke a method defined in a COBOL or Java class.



identifier-1

Must be defined as USAGE OBJECT REFERENCE. The contents of *identifier-1* specify the object on which a method is invoked.

When *identifier-1* is specified, either *literal-1* or *identifier-2* must be specified, identifying the name of the method to be invoked.

It must not be a dynamic-length elementary item or a dynamic-length group item.

The results of the INVOKE statement are undefined if either:

- *identifier-1* does not contain a valid reference to an object.
- *identifier-1* contains NULL.

class-name-1

When *class-name-1* is specified together with *literal-1* or *identifier-2*, the INVOKE statement invokes a static or factory method of the class referenced by *class-name-1*. *literal-1* or *identifier-2* specifies the name of the method that is to be invoked. The method must be a static method if *class-name-1* is a Java class; the method must be a factory method if *class-name-1* is a COBOL class.

When *class-name-1* is specified together with NEW, the INVOKE statement creates a new object that is an instance of class *class-name-1*.

You must specify *class-name-1* in the REPOSITORY paragraph of the configuration section of the class or program that contains the INVOKE statement.

SELF

An implicit reference to the object used to invoke the currently executing method. When SELF is specified, the INVOKE statement must appear within the PROCEDURE DIVISION of a method.

SUPER

An implicit reference to the object that was used to invoke the currently executing method. The resolution of the method to be invoked will ignore any methods declared in the class definition of the currently executing method and methods defined in any class derived from that class; thus the method invoked will be one that is inherited from an ancestor class.

literal-1

The value of *literal-1* is the name of the method to be invoked. The referenced object must support the method identified by *literal-1*.

literal-1 must be an alphanumeric literal or a national literal.

literal-1 is interpreted in a case-sensitive manner. The method name, the number of arguments, and the data types of the arguments in the USING phrase of the INVOKE statement are used to select the method with matching signature that is supported by the object. The method can be overloaded.

identifier-2

A data item of category alphabetic, alphanumeric, or national that at run time contains the name of the method to be invoked. The referenced object must support the method identified by *identifier-2*.

If *identifier-2* is specified, *identifier-1* must be defined as USAGE OBJECT REFERENCE without any optional phrases; that is, *identifier-1* must be a universal object reference.

It must not be a dynamic-length elementary item or a dynamic-length group item.

The content of *identifier-2* is interpreted in a case-sensitive manner. The method name, the number of arguments, and the data types of the arguments in the USING phrase of the INVOKE statement are used to select the method with matching signature that is supported by the object. The method can be overloaded.

NEW

The NEW operand specifies that the INVOKE statement is to create a new object instance of the class *class-name-1*. *class-name-1* must be specified.

When *class-name-1* is implemented in Java, the USING phrase of the INVOKE statement can be specified. The number of arguments and the data types of the arguments in the USING phrase of the INVOKE statement are used to select the Java constructor with matching signature that is supported by the class. An object instance of class *class-name-1* is allocated, the selected constructor (or the default constructor) is executed, and a reference to the created object is returned.

When *class-name-1* is implemented in COBOL, the USING phrase of the INVOKE statement must not be specified. An object instance of class *class-name-1* is allocated, instance data items are initialized to the values specified in associated VALUE clauses, and a reference to the created object is returned.

When NEW is specified, you must also specify a RETURNING phrase as described in [“RETURNING phrase” on page 364](#).

USING phrase

The USING phrase specifies arguments that are passed to the target method. The argument data types and argument linkage conventions are restricted to those supported by Java. See [“BY VALUE phrase” on page 363](#) for details.

BY VALUE phrase

Arguments specified in an INVOKE statement must be passed BY VALUE.

The BY VALUE phrase specifies that the value of the argument is passed, not a reference to the sending data item. The invoked method can modify the formal parameter that corresponds to an argument passed

by value, but changes do not affect the argument because the invoked method has access only to a temporary copy of the sending data item.

identifier-3

Must be an elementary data item in the DATA DIVISION. The data type of *identifier-3* must be one of the types supported for Java interoperation, as listed in [“Interoperable data types for OO COBOL and Java”](#) on page 366. Miscellaneous cases that are also supported as *identifier-3* are listed in [“Miscellaneous argument types for COBOL and Java”](#) on page 368, with their corresponding Java type.

See [Conformance requirements for arguments](#) for additional requirements that apply to *identifier-3*.

It must not be a dynamic-length elementary item or a dynamic-length group item.

literal-2

Must be of a type suitable for Java interoperation and must exactly match the type of the corresponding parameter in the target method. Supported literal forms are listed in [“Miscellaneous argument types for COBOL and Java”](#) on page 368, with their corresponding Java type.

literal-2 must not be a DBCS literal.

LENGTH OF *identifier-3*

Specifies that the length of *identifier-3* is passed as an argument in the LENGTH OF special register. A LENGTH OF special register passed BY VALUE is treated as a PIC 9(9) binary value. For information about the LENGTH OF special register, see [“LENGTH OF”](#) on page 22.

Conformance requirements for arguments

When *identifier-3* is an object reference, certain rules apply.

The rules are:

- A class-name must be specified in the data description entry for that object reference. That is, *identifier-3* must not be a universal object reference.
- The specified class-name must reference a class that is exactly the class of the corresponding parameter in the invoked method. That is, the class of *identifier-3* must not be a subclass or a superclass of the corresponding parameter's class.

When *identifier-3* is not an object reference, the following rules apply:

- If the target method is implemented in COBOL, the description of *identifier-3* must exactly match the description of the corresponding formal parameter in the target method.
- If the target method is implemented in Java, the description of *identifier-3* must correspond to the Java type of the formal parameter in the target method, as specified in [“Interoperable data types for OO COBOL and Java”](#) on page 366.

Usage note: Adherence to conformance requirements for arguments is the responsibility of the programmer. Conformance requirements are not verified by the compiler.

RETURNING phrase

The RETURNING phrase specifies a data item that will contain the value returned from the invoked method. You can specify the RETURNING phrase on the INVOKE statement when invoking methods that are written in COBOL or Java.

identifier-4

The RETURNING data item, *identifier-4*:

- Must be defined in the DATA DIVISION
- Must not be reference-modified
- Is not changed if an EXCEPTION occurs

The data type of *identifier-4* must be one of the types supported for Java interoperation, as listed in [“Interoperable data types for OO COBOL and Java”](#) on page 366.

See [Conformance requirements for the RETURNING item](#) for additional requirements that apply to *identifier-4*.

It must not be a dynamic-length elementary item or a dynamic-length group item.

If *identifier-4* is specified and the target method is written in COBOL, the target method must have a RETURNING phrase in its PROCEDURE DIVISION header. When the target method returns, its return value is assigned to *identifier-4* using the rules for the SET statement if *identifier-4* is described with USAGE OBJECT REFERENCE; otherwise, the rules for the MOVE statement are used.

The RETURNING data item is an output-only parameter. On entry to the called method, the initial state of the PROCEDURE DIVISION RETURNING data item has an undefined and unpredictable value. You must initialize the PROCEDURE DIVISION RETURNING data item in the invoked method before you reference its value. The value that is passed back to the invoker is the final value of the PROCEDURE DIVISION RETURNING data item when the invoked method returns.

See *Managing local and global references* in the *Enterprise COBOL Programming Guide* for discussion of local and global object references as defined in Java. These attributes affect the life-time of object references.

Usage note: The RETURN-CODE special register is not set by execution of INVOKE statements.

Conformance requirements for the RETURNING item

For INVOKE statements that specify *class-name-1* NEW, the RETURNING phrase is required.

The returning item must be one of the following ones:

- A universal object reference
- An object reference specifying *class-name-1*
- An object reference specifying a superclass of *class-name-1*

For INVOKE statements without the NEW phrase, the RETURNING item specified in the method invocation and in the corresponding target method must satisfy the following requirements:

- The presence or absence of a return value must be the same on the INVOKE statement and in the target method.
- If the RETURNING item is not an object reference, the following rules apply:
 - If the target method is implemented in COBOL, the returning item in the INVOKE statement and the RETURNING item in the target method must have an identical data description entry.
 - If the target method is implemented in Java, the returning item in the INVOKE statement must correspond to the Java type of the method result, as described in [“Interoperable data types for OO COBOL and Java”](#) on page 366.
- If the RETURNING item is an object reference, the RETURNING item specified in the INVOKE statement must be an object reference typed exactly to the class of the returning item specified in the target method. That is, the class of *identifier-4* must not be a subclass or a superclass of the class of the returning item in the target method.

Usage note: Adherence to conformance requirements for returning items is the responsibility of the programmer. Conformance requirements are not verified by the compiler.

ON EXCEPTION phrase

An exception condition occurs when the identified object or class does not support a method with a signature that matches the signature of the method specified in the INVOKE statement. When an exception condition occurs, one of the following actions occurs:

- If the ON EXCEPTION phrase is specified, control is transferred to *imperative-statement-1*.

- If the ON EXCEPTION phrase is not specified, a severity-3 Language Environment condition is raised at run time.

NOT ON EXCEPTION phrase

If an exception condition does not occur (that is, the identified method is supported by the specified object), control is transferred to the invoked method. After control is returned from the invoked method, control is then transferred:

1. To *imperative-statement-2*, if the NOT ON EXCEPTION phrase is specified.
2. To the end of the INVOKE statement if the NOT ON EXCEPTION phrase is not specified.

END-INVOKE phrase

This explicit scope terminator serves to delimit the scope of the INVOKE statement. An INVOKE statement that is terminated by END-INVOKE, along with its contained statements, becomes a unit that is treated as though it were an imperative statement. It can be specified as an imperative statement in a conditional statement; for example, in the exception phrase of another statement.

Restrictions

- The INVOKE statement is not supported when the LP(64) compiler option is in effect.
- The INVOKE statement does not support calling AMODE 64 Java methods.

Interoperable data types for OO COBOL and Java

A subset of COBOL data types can be used for interoperation between OO COBOL and Java.

Note: This section provides information about COBOL/Java data type compatibility for OO COBOL programs. For information about COBOL/Java data type compatibility for non-OO COBOL/Java interoperability, see [“Mapping between COBOL and Java data types for non-OO COBOL/Java interoperability” on page 733](#).

You can specify the interoperable data types as arguments in COBOL INVOKE statements and as the RETURNING item in COBOL INVOKE statements. Similarly, you can pass these types as arguments from a Java method invocation expression and receive them as parameters in the USING phrase or as the RETURNING item in the PROCEDURE DIVISION header of a COBOL method.

The following table lists the primitive Java types and the COBOL data types that are supported for interoperation and the correspondence between them.

Table 42. <i>Interoperable Java and COBOL data types</i>	
Java data type	COBOL data type
boolean ¹	Conditional variable and two condition-names of the form: <pre> level-number data-name PIC X. 88 data-name-false VALUE X'00'. 88 data-name-true VALUE X'01' THROUGH X'FF'.</pre>
byte	Single-byte alphanumeric, PIC X or PIC A
short	USAGE BINARY, COMP, COMP-4, or COMP-5, with a PICTURE clause of the form S9(n), where 1 <= n <= 4
int	USAGE BINARY, COMP, COMP-4, or COMP-5, with a PICTURE clause of the form S9(n), where 5 <= n <= 9
long	USAGE BINARY, COMP, COMP-4, or COMP-5, with a PICTURE clause of the form S9(n), where 10 <= n <= 18

Table 42. Interoperable Java and COBOL data types (continued)	
Java data type	COBOL data type
float ²	USAGE COMP-1
double ²	USAGE COMP-2
char	Single-character national: PIC N USAGE NATIONAL (an elementary data item of category national)
class types (object references)	USAGE OBJECT REFERENCE <i>class-name</i>
1. Enterprise COBOL interprets a PIC X argument or parameter as the Java boolean type only when the PIC X data item is followed by exactly two condition-names of the form shown. In all other cases, a PIC X argument or parameter is interpreted as the Java byte type. 2. Java floating-point data is represented in IEEE binary floating-point, while Enterprise COBOL uses the IBM hexadecimal floating-point representation. The representations are automatically converted as necessary when Java methods are invoked from COBOL and when COBOL methods are invoked from Java.	

In addition to the primitive types, Java Strings and arrays of Java primitive types can interoperate with COBOL. This requires specialized mechanisms provided by the COBOL runtime system and the Java Native Interface (JNI).

In a Java program, to pass array data to COBOL or to receive array data from COBOL, you declare the array types using the usual Java syntax. In the COBOL program, you declare the array as an object reference that contains an instance of one of the special classes provided for array support. Conversion between the Java and COBOL types is automatic at the time of method invocation.

In a Java program, to pass String data to COBOL or to receive String data from COBOL, you declare the array types using the usual Java syntax. In the COBOL program, you declare the String as an object reference that contains an instance of the special jstring class. Conversion between the Java and COBOL types is automatic at the time of method invocation. The following table lists the Java array and String data types and the corresponding special COBOL data types.

Table 43. Interoperable COBOL and Java array and String data types	
Java data type	COBOL data type
boolean[]	object reference jbooleanArray
byte[]	object reference jbyteArray
short[]	object reference jshortArray
int[]	object reference jintArray
long[]	object reference jlongArray
char[]	object reference jcharArray
Object[]	object reference jobjectArray
String	object reference jstring

The following java array types are not currently supported:

Java data type	COBOL data type
float[]	object reference jfloatArray
double[]	object reference jdoubleArray

You must code an entry in the repository paragraph for each special class that you want to use, just as you do for other classes. For example, to use jstring, code the following entry:

```
Configuration Section.
Repository.
    Class jstring is "jstring".
```

Alternatively, for the String type, the COBOL repository entry can specify an external class name of java.lang.String:

```
Repository.
    Class jstring is "java.lang.String".
```

Callable services are provided by the Java Native Interface (JNI) for manipulating the COBOL objects of these types in COBOL. For example, callable services can be used to set COBOL alphanumeric or national data into a jstring object or to extract data from a jstring object. For details on use of JNI callable services for these and other purposes, see *Accessing JNI services* in the *Enterprise COBOL Programming Guide*.

For details on repository entries for class definitions, see “[REPOSITORY paragraph](#)” on page 132. For examples, see *Example: external class-names and Java packages* in the *Enterprise COBOL Programming Guide*.

Miscellaneous argument types for COBOL and Java

There are miscellaneous cases of COBOL items that can be used as arguments in an INVOKE statement. The COBOL miscellaneous argument types and the corresponding Java types are listed in the following table.

<i>Table 44. COBOL miscellaneous argument types and corresponding Java types</i>	
COBOL argument	Corresponding Java data type
Reference-modified item of usage display with length one	byte
Reference-modified item of usage national with length one (either an elementary data item of usage national or a national group item)	char
SHIFT-IN and SHIFT-OUT special registers	byte
LINAGE-COUNTER special register when its usage is binary	int
LENGTH OF special register	int

The following table lists COBOL literal types that can be used as arguments in an INVOKE statement, with the corresponding Java type.

<i>Table 45. COBOL literal argument types and corresponding Java types</i>	
COBOL literal argument	Corresponding Java data type
Fixed-point numeric literal with no decimal positions and with nine digits or less	int

Table 45. **COBOL literal argument types and corresponding Java types** (continued)

COBOL literal argument	Corresponding Java data type
Floating-point numeric literal	double
Figurative constant ZERO	int
One-character alphanumeric literal	byte
One-character national literal	char
Symbolic character	byte
Figurative constants SPACE, QUOTE, HIGH-VALUE, or LOW-VALUE	byte

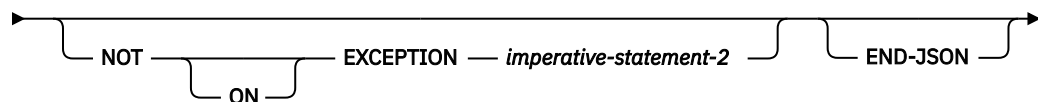
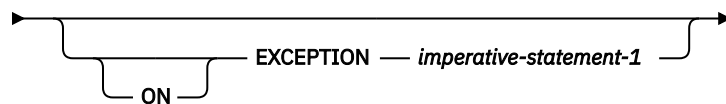
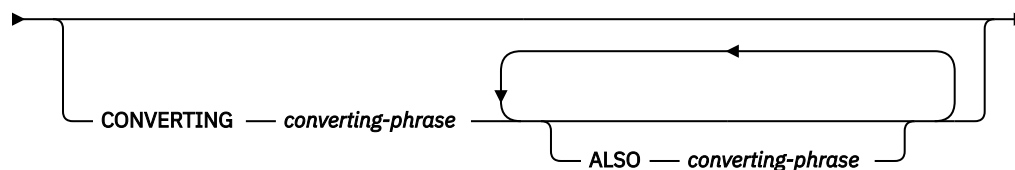
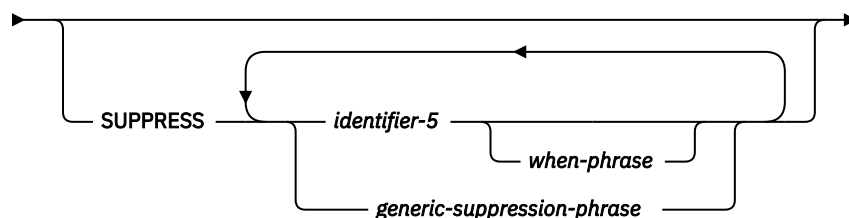
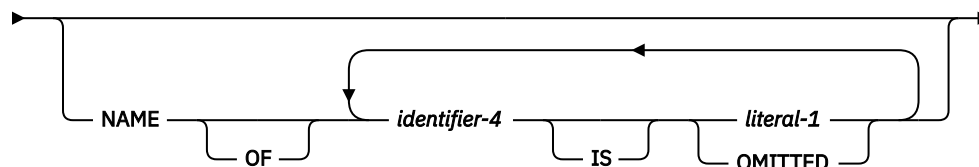
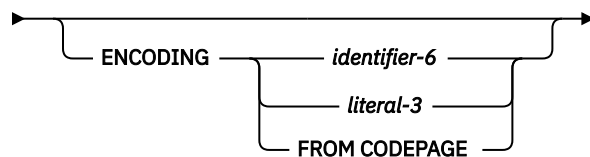
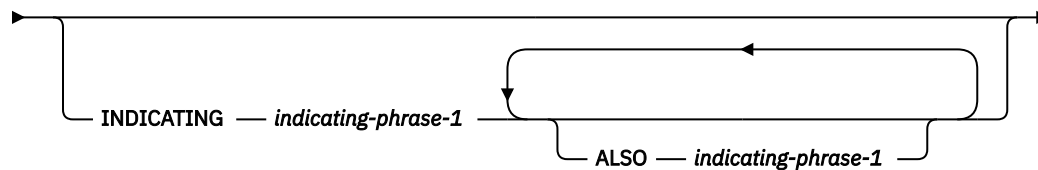
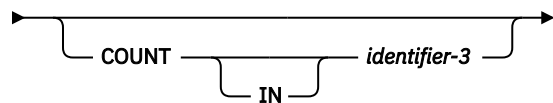
JSON GENERATE statement

The JSON GENERATE statement converts data to JSON format.

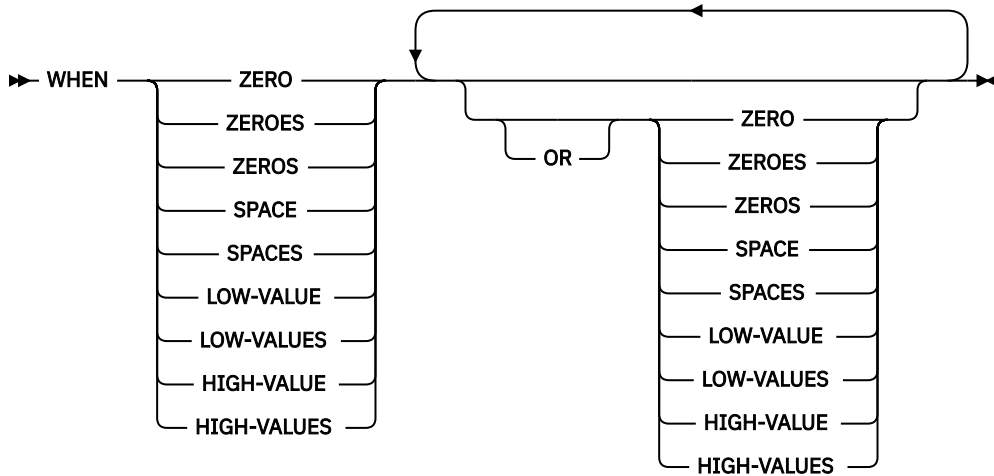
You can watch this [video](#) to get an overview of the JSON support in Enterprise COBOL 6.

Format

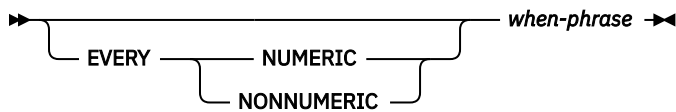
►► JSON GENERATE — *identifier-1* — FROM — *identifier-2* —►



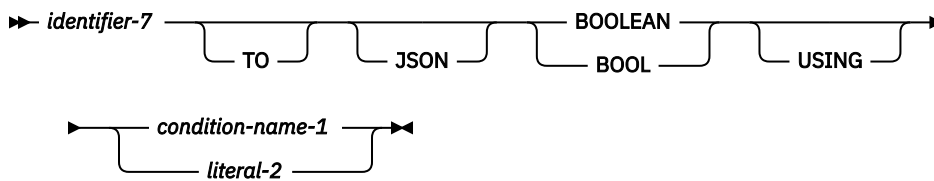
***when-phrase* Format**



***generic-suppression-phrase* Format**



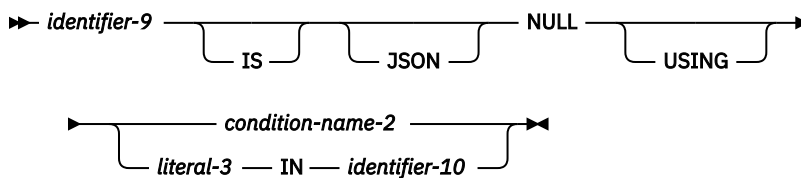
***converting-phrase* Format 1**



***converting-phrase* Format 2**



***indicating-phrase-1* format**



identifier-1

The receiving area for the generated JSON text. *identifier-1* must reference one of the following items:

- An elementary data item of category alphanumeric
- An alphanumeric group item

- An elementary data item of category national
- A national group item
- An elementary data item of category UTF-8
- A UTF-8 group item

When *identifier-1* references an alphanumeric group item, *identifier-1* is treated as though it were an elementary data item of category alphanumeric. When *identifier-1* references a national group item, *identifier-1* is processed as an elementary data item of category national. When *identifier-1* references a UTF-8 group item, *identifier-1* is processed as an elementary data item of category UTF-8.

identifier-1 must not be defined with the JUSTIFIED clause, and cannot be a function identifier. *identifier-1* can be subscripted or reference modified.

identifier-1 must not overlap *identifier-2* or *identifier-3*.

identifier-1 can be a dynamic-length elementary item of category alphanumeric or UTF-8. *identifier-1* cannot be a dynamic-length group item of any category, nor an elementary item of category national.

The generated JSON text is encoded in UTF-8 (CCSID 1208), except in the following scenarios:

- *identifier-1* is of category national, in which case it is encoded in UTF-16 Big-Endian (CCSID1200)
- *identifier-1* is of category alphanumeric and the ENCODING phrase is specified with an EBCDIC CCSID

Conversion of the data values and NAME phrase literals is done according to the CODEPAGE compiler option in effect for the compilation. Conversion of original data names is always done using CCSID 1140. For details, see CODEPAGE in the *Enterprise COBOL Programming Guide*.

identifier-1 must be large enough to contain the generated JSON text. Typically, it should be from 2 to 3 times the size of *identifier-2*, depending on the lengths of the data-names within *identifier-2*. If *identifier-1* is not large enough, an exception condition exists at the end of the JSON GENERATE statement. If the COUNT phrase is specified, *identifier-3* contains the number of character encoding units that were actually generated.

identifier-2

The group or elementary data item to be converted to JSON format. *identifier-2* must reference one of the following items:

- An elementary data item of category alphanumeric
- An alphanumeric group item
- An elementary data item of category national
- A national group item
- An elementary data item of category UTF-8
- A UTF-8 group item

identifier-2 cannot be a function identifier or be reference modified, but it can be subscripted.

identifier-2 must not overlap *identifier-1* or *identifier-3*.

identifier-2 can be a dynamic-length group item or a dynamic-length elementary item of category alphanumeric or UTF-8. *identifier-2* cannot be a dynamic-length group or elementary item of category national.

The data description entry for *identifier-2* must not contain a RENAMES clause.

The following data items that are specified by *identifier-2* are ignored by the JSON GENERATE statement:

- Any subordinate unnamed elementary data items or elementary FILLER data items
- Any slack bytes inserted for SYNCHRONIZED items
- Any data item subordinate to *identifier-2* that is defined with the REDEFINES clause or that is subordinate to such a redefining item

- Any data item subordinate to *identifier-2* that is defined with the RENAME clause
- Any group data item whose subordinate data items are all ignored

All data items specified by *identifier-2* that are not ignored according to the previous rules must satisfy the following conditions:

- Each elementary data item must have a USAGE other than POINTER, FUNCTION-POINTER, PROCEDURE-POINTER, or OBJECT REFERENCE.
- There must be at least one such elementary data item.
- Each non-FILLER data-name must have a unique identifier within *identifier-2*.

Example 1

For example, consider the following data declaration:

```
01 STRUCT.
  02 STAT PIC X(4).
  02 IN-AREA PIC X(100).
  02 OK-AREA REDEFINES IN-AREA.
    03 FLAGS PIC X.
    03 PIC X(3).
    03 COUNTER USAGE COMP-5 PIC S9(9).
    03 ASFNPTR REDEFINES COUNTER USAGE FUNCTION-POINTER.
    03 UNREFERENCED PIC X(92).
  02 NG-AREA1 REDEFINES IN-AREA.
    03 FLAGS PIC X.      03 PIC X(3).
    03 PTR USAGE POINTER.
    03 ASNUM REDEFINES PTR USAGE COMP-5 PIC S9(9).
    03 PIC X(92).
  02 NG-AREA2 REDEFINES IN-AREA.
    03 FN-CODE PIC X.
    03 UNREFERENCED PIC X(3).
    03 QTYONHAND USAGE BINARY PIC 9(5).
    03 DESC USAGE NATIONAL PIC N(40).
    03 UNREFERENCED PIC X(12).
```

The following data items from the example can be specified as *identifier-2*:

- STRUCT, of which subordinate data items STAT and IN-AREA would be converted to JSON format. (OK-AREA, NG-AREA1, and NG-AREA2 are ignored because they specify the REDEFINES clause.)
- OK-AREA, of which subordinate data items FLAGS, COUNTER, and UNREFERENCED would be converted. (The item whose data description entry specifies 03 PIC X(3) is ignored because it is an elementary FILLER data item. ASFNPTR is ignored because it specifies the REDEFINES clause.)
- Any of the elementary data items that are subordinate to STRUCT except:
 - ASFNPTR or PTR (disallowed usage)
 - UNREFERENCED OF NG-AREA2 (nonunique names for data items that are otherwise eligible)
 - Any FILLER data items

The following data items cannot be specified as *identifier-2*:

- NG-AREA1, because subordinate data item PTR specifies USAGE POINTER but does not specify the REDEFINES clause. (PTR would be ignored if it is defined with the REDEFINES clause.)
- NG-AREA2, because subordinate elementary data items have the nonunique name UNREFERENCED.

Example 2

The following example shows the UTF-8 and dynamic-length support for JSON GENERATE:

```
01 GRP.
  05 Ac-No PIC AA9999 value 'SX1234'.
  05 MORE.
    10 Stuff PIC S99V9 OCCURS 2.
  05 SSN PIC 999/99/9999 value SPACE.
01 d pic u dynamic limit 250.
  MOVE 7.8 TO stuff(1), MOVE -9 TO stuff(2)
  JSON GENERATE d FROM grp COUNT i
```

The output of the JSON GENERATE statement would be the UTF-8 encoded text in *d* containing the following JSON text:

```
{ "GRP": { "Ac-No": "SX1234", "MORE": { "Stuff": [7.8, -9.0] }, "SSN": "  " }
```

COUNT phrase

If the COUNT phrase is specified, *identifier-3* contains (after successful execution of the JSON GENERATE statement) the count of generated JSON character code points. If *identifier-1* (the receiver) is of category national, the count is in double-bytes. If *identifier-1* is of category UTF-8, the count is in Unicode code points. Otherwise, the count is in bytes.

identifier-3

The data count field. Must be an integer data item defined without the symbol P in its picture string.

identifier-3 must not overlap *identifier-1*, *identifier-2*, *identifier-4*, or *identifier-5*.

INDICATING phrase

The INDICATING phrase is specified with an indicated item, *identifier-9*, and an indicator item. The indicator item is specified either directly via *identifier-10*, or indirectly as the parent elementary item of *condition-name-2*.

The INDICATING phrase allows you to generate JSON null values for the indicated item when the indicator item contains the values specified in *condition-name-2* or *literal-3*. In other words, the indicator item is a satellite data item that informs the JSON GENERATE statement if a JSON null value shall be generated.

Both the indicator item and the indicated item must be subordinate to *identifier-2*.

The indicator item must be a single byte alphanumeric elementary data item whose data definition entry contains PICTURE X.

condition-name-2 and *literal-3* represent values of the indicator item. When the indicator item contains one of those values, the indicated item (in *identifier-9*) will be generated as a JSON null value. All other values of the indicator item will result in the indicated item being generated into JSON according to [“Operation of JSON GENERATE” on page 380](#). The indicator item is otherwise ignored by the JSON GENERATE statement because the indicator item is not subject to conversion to JSON text.

condition-name-2 must be a level-88 item and can be specified with multiple values or value ranges. *literal-2* must be a single byte alphanumeric literal.

The INDICATING phrase can be specified multiple times by using the ALSO keyword.

Below are three examples of the INDICATING phrase:

- Example 1:

```
01 DOCX PIC X(1000).
01 MY-RECORD.
   02 DATA-1-IS-NULL PIC X.
   02 DATA-1 PIC X(100).

MOVE 'Y' TO DATA-1-IS-NULL
JSON GENERATE DOCX FROM MY-RECORD
  INDICATING DATA-1 IS JSON NULL USING 'Y' IN DATA-1-IS-NULL
END-JSON
```

The output in *docx* encoded in UTF-8 is as follows:

```
{ "MY-RECORD" : { "DATA-1" : null } }
```

- Example 2:

```
01 DOCX PIC X(1000).
01 MY-RECORD.
   02 GRP OCCURS 2.
     03 DATA-1-IS-NULL PIC X.
     03 DATA-1 PIC X(100).
MOVE 'VAL1' to DATA-1(1)
MOVE 'VAL2' to DATA-1(2)
```

```

MOVE 'Y' TO DATA-1-IS-NULL(1)
MOVE 'N' TO DATA-1-IS-NULL(2)
JSON GENERATE DOCX FROM MY-RECORD
  INDICATING DATA-1 IS JSON NULL USING 'Y' IN DATA-1-IS-NULL
END-JSON

```

The output in *docx* encoded in UTF-8 is as follows:

```
{ "MY-RECORD": { "GRP": [ { "DATA-1": null }, { "DATA-1": "VAL2" } ] } }
```

- Example 3:

```

01 DOCX PIC X(1000).
01 MY-RECORD.
  02 GRP.
    03 DATA-1-IS-NULL PIC X OCCURS 2.
    03 DATA-1 PIC X(100) OCCURS 2.
  MOVE 'VAL1' to DATA-1(1)
  MOVE 'VAL2' to DATA-1(2)
  MOVE 'Y' TO DATA-1-IS-NULL(1)
  MOVE 'N' TO DATA-1-IS-NULL(2)
  JSON GENERATE DOCX FROM MY-RECORD
    INDICATING DATA-1 IS JSON NULL USING 'Y' IN DATA-1-IS-NULL
  END-JSON

```

The output in *docx* encoded in UTF-8 is as follows:

```
{ "MY-RECORD": { "GRP": { "DATA-1": [ null, "VAL2" ] } } }
```

For a summary of the ways to generate JSON null values, see "Generating JSON null values".

ENCODING phrase

The ENCODING phrase, if specified, determines the encoding of the generated JSON document. The ENCODING phrase must follow these rules:

- *identifier-6* must be an unsigned integer data item.
- *literal-3* must be an unsigned integer literal.
- If *identifier-1* is alphanumeric:
 - The value of *identifier-6* or *literal-3* can be either an EBCDIC coded character set identifier (CCSID) from the below [Table 46 on page 375. "Single byte EBCDIC coded character sets for JSON documents"](#) or 1208.
 - If FROM CODEPAGE was specified, the CCSID of the CODEPAGE compiler option is assumed and must be an EBCDIC CCSID from the below [Table 46 on page 375. "Single byte EBCDIC coded character sets for JSON documents"](#).
- If *identifier-1* is national or UTF-8:
 - The value of *identifier-6* or *literal-3* must be 1200 or 1208 respectively.
 - FROM CODEPAGE cannot be specified.

If the ENCODING phrase is omitted and:

- If *identifier-1* is alphanumeric or UTF-8, the JSON document is encoded in Unicode UTF-8 (CCSID 1208).
- If *identifier-1* is national, the JSON document is encoded in Unicode UTF-16 (CCSID 1200).

Table 46. Single byte EBCDIC coded character sets for JSON documents	
CCSID	Description
1047	Latin 1/Open Systems
1140, 37	USA, Canada,...Euro Country Extended Code Page (ECECP), Country Extended Code Page (CECP)

Table 46. Single byte EBCDIC coded character sets for JSON documents (continued)	
CCSID	Description
1141, 273	Austria, Germany ECECP, CECF
1142, 277	Denmark, Norway ECECP, CECF
1143, 278	Finland, Sweden ECECP, CECF
1144, 280	Italy ECECP, CECF
1145, 284	Spain, Latin America (Spanish) ECECP, CECF
1146, 285	UK ECECP, CECF
1147, 297	France ECECP, CECF
1148, 500	International ECECP, CECF
1149, 871	Iceland ECECP, CECF

If the ENCODING phrase specifies a single byte EBCDIC CCSID, *literal-1* of the JSON GENERATE NAME phrase must be an alphanumeric literal.

NAME phrase

Allows you to override the default JSON names derived from *identifier-2* or its subordinate data items.

identifier-4 must reference *identifier-2* or one of its subordinate data items. It cannot be a function identifier and cannot be reference modified or subscripted. It must not specify any data item which is ignored by the JSON GENERATE statement. For more information about *identifier-2*, see the description of *identifier-2*. If *identifier-4* is specified more than once in the NAME phrase, the last specification is used.

literal-1 must be an alphanumeric or national literal containing the name to be generated in the JSON text corresponding to *identifier-4*. With PTF for APAR PH18641 installed, alternatively, you can specify OMITTED to generate an anonymous JSON object or array, whose top-level parent name is not generated. For examples of generating JSON anonymous arrays, see "Generating JSON anonymous arrays" in the *Programming Guide*. With PTF for APAR PH58384 installed, OMITTED can be also used to generate only a JSON value (string, number, true, false, or null), whose JSON name is not generated in JSON name/value pairs.

When you specify OMITTED, *identifier-4* must reference *identifier-2*.

If *literal-1* is a NATIONAL or UTF-8 literal, *identifier-1* must reference a data item of category NATIONAL or UTF-8, or the ENCODING phrase must specify a CCSID of 1208 for Unicode UTF-8.

SUPPRESS phrase

Allows you to identify and exclude items that are subordinate to *identifier-2*, and thus selectively generate output for the JSON GENERATE statement. If the SUPPRESS phrase is specified, *identifier-1* must be large enough to contain the generated JSON document before any suppression.

With the *generic-suppression-phrase*, elementary items subordinate to *identifier-2* that are not otherwise ignored by JSON GENERATE operations are identified generically for potential suppression. Either items of class numeric, if the NUMERIC keyword is specified, or items that are not of class numeric, if the NONNUMERIC keyword is specified, or both if neither is specified, might be suppressed.

If multiple *generic-suppression-phrase* are specified, the effect is cumulative.

If the *generic-suppression-phrase* is specified, data items are selected for potential suppression according to the following rules:

- If ZERO, ZEROES, or ZEROS is specified in the WHEN phrase, all data items except those that are defined with USAGE DISPLAY-1 are selected.
- If SPACE or SPACES is specified in the WHEN phrase, data items of USAGE DISPLAY, DISPLAY-1, or NATIONAL are selected. For zoned or national decimal items, only integers are selected.

- If LOW-VALUE, LOW-VALUES, HIGH-VALUE, or HIGH-VALUES is specified in the WHEN phrase, data items of USAGE DISPLAY or NATIONAL are selected. For zoned or national decimal items, only integers are selected.

identifier-5 explicitly identifies items for potential suppression. *identifier-5* must reference a data item that is subordinate to *identifier-2* and that is not otherwise ignored by the operation of the JSON GENERATE statement. *identifier-5* cannot be a function identifier and cannot be reference modified or subscripted. If the WHEN phrase is specified, *identifier-5* must reference an elementary data item. If the WHEN phrase is omitted, *identifier-5* may be a group item. If *identifier-5* specifies a group data item, that group data item and all data items that are subordinate to the group item are excluded. Duplicate specifications of *identifier-5* are permitted.

If *identifier-5* is specified, the following rules apply to it:

- If ZERO, ZEROES, or ZEROS is specified in the WHEN phrase, *identifier-5* must not be of USAGE DISPLAY-1.
- If SPACE or SPACES is specified in the WHEN phrase, *identifier-5* must be of USAGE DISPLAY, DISPLAY-1, or NATIONAL. If *identifier-5* is a zoned or national decimal item, it must be an integer.
- If LOW-VALUE, LOW-VALUES, HIGH-VALUE, or HIGH-VALUES is specified in the WHEN phrase, *identifier-5* must be of USAGE DISPLAY or NATIONAL. If *identifier-5* is a zoned or national decimal item, it must be an integer.

The comparison operation that determines whether an item will be suppressed is a relation condition as shown in the table of Comparisons involving figurative constants. That is, the comparison is a numeric comparison if the value specified is ZERO, ZEROS, or ZEROES, and the item is of class numeric. For all other cases, the comparison operation is an alphanumeric, DBCS, national, or UTF-8 comparison, depending on whether the item is of usage DISPLAY, DISPLAY-1, NATIONAL, or UTF-8, respectively.

When the SUPPRESS phrase is specified, a group item subordinate to *identifier-2* is excluded from the generated JSON text if all the eligible items subordinate to the group item are excluded. The outermost object that corresponds to *identifier-2* itself is always generated, even if all the items subordinate to *identifier-2* are excluded. In this case, the value generated for *identifier-2* is an empty object, as follows:

```
{"identifier-2":{}}
```

For example, consider the following data declaration:

```
1 a.
2 b.
3 c occurs 0 to 2 depending j.
4 d pic x.
2 e pic x.
```

If the ODO object *j* contains the value 2 and group *a* is populated with all ‘_’, the statement JSON GENERATE *x* FROM *a* (without the SUPPRESS phrase) produces the following JSON text:

```
{"a":{"b":{"c":[{"d":"_"}, {"d":"_"}]}, "e":"_"}}
```

Group item *b* is eliminated from the output if a SUPPRESS phrase specifies any one of data items *b*, *c* or *d*, resulting in the following JSON text:

```
{"a":{"e":"_"}}
```

As an example of complete recursive suppression, the statement JSON GENERATE *x* FROM *a* SUPPRESS *b* *e* produces:

```
{"a":{}}
```

JSON has an explicit representation of a table with no elements:

```
{"table-name":[]}
```

which is thus retained in the generated JSON text unless explicitly suppressed.

For example if ODO object *j* contains the value 0 and thus table *d* has no occurrences, and group *a* is populated with all '_', the statement `JSON GENERATE x FROM a` produces the following JSON text:

```
{ "a": { "b": { "c": [] }, "e": "_" }
```

Components of a zero-occurrence group table do not contribute any JSON text to the output. As a result, a `SUPPRESS` phrase that specified only *d* would have no effect on this generated output.

Suppressing data items *b* or *c*, and *e*, which do contribute JSON text, has the same result as for non-zero occurrences of table *c*, illustrated above.

CONVERTING phrase

Allows you to specify items that will be generated as either JSON BOOLEAN or JSON null name/value pairs.

converting-phrase Format 1

Use *Format 1* to specify items that will be generated as JSON boolean name/value pairs.

identifier-7 must be a single-byte alphanumeric elementary data item whose data definition entry contains PICTURE *X*.

condition-name-1 and *literal-2* represent values of *identifier-7* that will be generated as a JSON BOOLEAN true value. All other values of *identifier-7* will be generated as a JSON BOOLEAN false value. *condition-name-1* must be a level-88 item directly subordinate to *identifier-7* and can be specified with multiple values or value ranges. *literal-2* must be a single-byte alphanumeric literal.

The CONVERTING phrase can be specified with multiple items to be generated as JSON BOOLEAN name/value pairs by using the `ALSO` keyword.

For example, consider the following COBOL structure and statements:

```
01 myrecord.  
  02 data-a PIC X.  
  02 data-b PIC X.  
    88 data-b-flag VALUE 'a' THRU 'z'.
```

```
MOVE 'F' TO data-a  
MOVE 'b' TO data-b
```

```
JSON GENERATE docx FROM myrecord  
  CONVERTING data-a TO BOOLEAN USING 'T'  
    ALSO data-b TO BOOLEAN USING data-b-flag
```

The output of the JSON GENERATE statement would be the UTF-8 encoded text in *docx* containing the following JSON text:

```
{ "myrecord" : { "data-a" : false, "data-b" : true } }
```

converting-phrase Format 2

Use *Format 2* to specify items that might be generated as JSON null name/value pairs.

identifier-8 must be a group or elementary item that references *identifier-2* or is subordinate to *identifier-2*.

fig-con-1 represents one of the figurative constants from the list below:

- SPACE, SPACES
- ZERO, ZEROES, ZEROS
- LOW-VALUE, LOW-VALUES
- HIGH-VALUE, HIGH-VALUES

During execution of the JSON GENERATE statement, *fig-con-1* will be compared to *identifier-8* in an `IS EQUAL` relation condition. If the result of the comparison is true, the value of *identifier-8*

will be JSON null. Otherwise the value of *identifier-8* will be generated as per the general rules of JSON GENERATE. *fig-con-1* must be a permissible pair for comparison according to [Table 24 on page 276](#). "Comparisons involving figurative constants".

The following example shows a COBOL structure and statements generating JSON null values using the CONVERTING phrase:

```
01 docx pic x(100).
01 my-record.
  02 data-a pic 9999.
  02 data-b pic x(10).

MOVE zero TO data-a
MOVE low-values TO data-b

JSON GENERATE docx FROM my-record
  CONVERTING data-a TO NULL USING zero
  ALSO data-b TO NULL USING low-values
END-JSON
```

The output of the JSON GENERATE statement would be the UTF-8 encoded text in *docx* containing the following JSON text:

```
{ "myrecord" : { "data-a" : null, "data-b" : null } }
```

Note: *identifier-7* of Format 1 and *identifier-8* might reference the same data item, which might be useful when you want to convert the item to JSON true, false, or null.

ON EXCEPTION phrase

An exception condition exists when an error occurs during generation of the JSON document, for example if *identifier-1* is not large enough to contain the generated JSON document. In this case, JSON generation stops and the content of the receiver, *identifier-1*, is undefined. If the COUNT phrase is specified, *identifier-3* contains the number of character positions that were actually generated.

If the ON EXCEPTION phrase is specified, control is transferred to *imperative-statement-1*. If the ON EXCEPTION phrase is not specified, the NOT ON EXCEPTION phrase, if any, is ignored, and control is transferred to the end of the JSON GENERATE statement. Special register JSON-CODE contains an exception code, as detailed in *JSON GENERATE exceptions* in the *Enterprise COBOL Programming Guide*.

NOT ON EXCEPTION phrase

If an exception condition does not occur during generation of the JSON document, control is passed to *imperative-statement-2*, if specified, otherwise to the end of the JSON GENERATE statement. The ON EXCEPTION phrase, if specified, is ignored. Special register JSON-CODE contains zero after execution of the JSON GENERATE statement.

END-JSON phrase

This explicit scope terminator delimits the scope of the JSON GENERATE or JSON PARSE statements. END-JSON permits a conditional JSON GENERATE or JSON PARSE statement (that is, a JSON GENERATE or JSON PARSE statement that specifies the ON EXCEPTION or NOT ON EXCEPTION phrase) to be nested in another conditional statement.

The scope of a conditional JSON GENERATE or JSON PARSE statement can be terminated by:

- An END-JSON phrase at the same level of nesting
- A separator period

END-JSON can also be used with a JSON GENERATE or JSON PARSE statement that does not specify either the ON EXCEPTION or the NOT ON EXCEPTION phrase.

For more information about explicit scope terminators, see [“Delimited scope statements” on page 293](#).

Nested JSON GENERATE or JSON PARSE statements

When a given JSON GENERATE or JSON PARSE statement appears in *imperative-statement-1* or *imperative-statement-2* of another JSON GENERATE or JSON PARSE statement, that given JSON GENERATE or JSON PARSE statement is a *nested* JSON GENERATE or JSON PARSE statement.

Nested JSON GENERATE or JSON PARSE statements are considered to be matched JSON GENERATE and END-JSON combinations, or JSON PARSE and END-JSON combinations, proceeding from left to right. Thus, any END-JSON phrase that is encountered is matched with the nearest preceding JSON GENERATE or JSON PARSE statement that has not been implicitly or explicitly terminated.

Operation of JSON GENERATE

The content of each eligible elementary data item within *identifier-2* that has not been excluded from JSON generation according to a SUPPRESS phrase, is converted to character format in the resulting JSON text.

Only the first definition of each storage area is processed. Redefinitions of data items are not included. Data items that are effectively defined by the RENAMES clause are also not included.

A table item that has zero element occurrences is included in the JSON text as an empty array, such as `{"name" : []}`.

Note: if the zero-occurrence table is a group item, its subordinate items do not appear in the JSON text and thus specifying them in the SUPPRESS phrase has no effect. See [the description of the SUPPRESS phrase](#) for more information.

For information about the format conversion of elementary data, see [“Format conversion of elementary data” on page 381](#) and [“Trimming of generated JSON data” on page 382](#).

The JSON names are obtained from the NAME phrase if specified; otherwise by default they are derived from the data-names within *identifier-2* as described in [“JSON name formation” on page 382](#). The names of group items that contain the selected elementary items are retained as the names of parent JSON objects.

No extra white space (new lines, indentation, and so forth) is inserted to make the generated JSON text more readable.

If the receiving area specified by *identifier-1* is not large enough to contain the resulting JSON text, an exception condition exists. See the description of the ON EXCEPTION phrase above for details.

If *identifier-1* is longer than the generated JSON text, only that part of *identifier-1* in which JSON is generated is changed. The rest of *identifier-1* contains the data that was present before this execution of the JSON GENERATE statement.

To avoid referring to that data, either initialize *identifier-1* to spaces before the JSON GENERATE statement or specify the COUNT phrase.

If the COUNT phrase is specified, *identifier-3* contains (after successful execution of the JSON GENERATE statement) the total number of character positions (UTF-16 code points, UTF-8 code points or bytes) that were generated. You can use *identifier-3* as a reference modification length field to refer to the part of *identifier-2* that contains the generated JSON text.

After execution of the JSON GENERATE statement, special register JSON-CODE contains either zero, which indicates successful completion, or a nonzero exception code. For details, see *JSON GENERATE exceptions* in the *Enterprise COBOL Programming Guide*.

The JSON PARSE statement also uses special register JSON-CODE. Therefore if you code a JSON GENERATE statement in the processing procedure of a JSON PARSE statement, save the value of JSON-CODE before that JSON GENERATE statement executes and restore the saved value after the JSON GENERATE statement terminates.

A byte order mark is not generated for JSON texts.

Format conversion of elementary data

Elementary data items within *identifier-2* are converted in the sequence of the following steps. Some of these steps are optional.

Conversion to character format:

Elementary data items are converted to character format depending on the type of the data item:

- Data items of category alphabetic, alphanumeric, alphanumeric-edited, DBCS, external floating-point, national, national-edited, and numeric-edited are not converted, except as required to the correct Unicode encoding form.
 - Fixed-point numeric data items other than COMPUTATIONAL-5 (COMP-5) binary data items or binary data items compiled with the TRUNC(BIN) compiler option are converted as if they were moved to a numeric-edited item that has:
 - As many integer positions as the numeric item has, but with at least one integer position, possibly zero (0)
 - An explicit decimal point, if the numeric item has at least one decimal position; the decimal point is represented by a period regardless of whether the DECIMAL-POINT IS COMMA clause is specified in the SPECIAL-NAMES paragraph
 - The same number of decimal positions as the numeric item has
 - A leading '-' picture symbol if the data item is signed (has an S in its PICTURE clause)
 - COMPUTATIONAL-5 (COMP-5) binary data items or binary data items compiled with the TRUNC(BIN) compiler option are converted in the same way as the other fixed-point numeric items, except for the number of integer positions. The number of integer positions is computed depending on the number of '9' symbols in the picture character string as follows:
 - 5 minus the number of decimal places, if the data item has 1 to 4 '9' picture symbols
 - 10 minus the number of decimal places, if the data item has 5 to 9 '9' picture symbols
 - 20 minus the number of decimal places, if the data item has 10 to 18 '9' picture symbols
 - Internal floating-point data items are converted as if they were moved to a data item as follows:
 - For COMP-1: an external floating-point data item with PICTURE -9.9(8)E+99
 - For COMP-2: an external floating-point data item with PICTURE -9.9(17)E+99 (illegal because of the number of digit positions)
 - External floating-point data items are converted as if they were moved to another external floating-point data item with the same precision and scale, and with:
 - A - sign for the mantissa.
 - An actual decimal point, indicated by a . PICTURE symbol.
 - A + sign for the exponent.
- For example, an external floating-point item defined with PICTURE -9(3)V9(5)E-99 would be converted as if it were moved to an external floating-point item defined with PICTURE -9(3).9(5)E+99.
- Index data items are converted as if they were declared USAGE COMP-5 PICTURE S9(9).

Trimming:

After any conversion to character format, leading and trailing spaces and leading zeroes are eliminated, as described under [“Trimming of generated JSON data” on page 382](#).

Conversion to the target encoding:

If *identifier-1* is a data item of category national, any nonnational values are converted to national format. Otherwise, all values are represented in UTF-8. The conversion is done according to the compiler CODEPAGE option in effect for the compilation.

Escaping characters:

The characters NX'0022' (") and NX'005C' (\) are escaped as \" and \\, respectively.

For compactness and appearance, other common characters are represented by a two-character escape sequence as follows:

```
NX'0008' \b - backspace
NX'0009' \t - tab
NX'000A' \n - line feed
NX'000C' \f - form feed
NX'000D' \r - carriage return
NX'0085' \x - next line
```

The remaining characters in the range NX'0000' through NX'001F' are escaped as \uhhhh, where "h" represents a hexadecimal digit 0 through F.

Representation of out-of-range Unicode characters:

Any remaining Unicode character that has a Unicode scalar value greater than NX'FFFF' is represented by a surrogate pair for national output, or a four-byte sequence for UTF-8 output. For example, the musical symbol G clef (U+1D11E) is represented in UTF-16 by the surrogate pair NX'D834' NX'DD1E', and in UTF-8 by the byte sequence x'F09D849E'.

Trimming of generated JSON data

Trimming is performed on data values after any conversion to character format.

For more information about the conversion, see [“Format conversion of elementary data” on page 381](#).

For values converted from signed numeric values, the leading space is removed if the value is positive.

For values converted from numeric items, leading zeroes (after any initial minus sign) up to but not including the digit immediately before the actual or implied decimal point are eliminated. Trailing zeroes after a decimal point are retained. For example:

- -012.340 becomes -12.340
- 0000.45 becomes 0.45
- 0013 becomes 13
- 0000 becomes 0

Character values from data items of class alphabetic, alphanumeric, DBCS, and national have either trailing or leading spaces removed, depending on whether the corresponding data items have left (default) or right justification, respectively. That is, trailing spaces are removed from values whose corresponding data items do not specify the JUSTIFIED clause. Leading spaces are removed from values whose data items do specify the JUSTIFIED clause. If a character value consists solely of spaces, one space remains as the value after trimming is finished.

JSON name formation

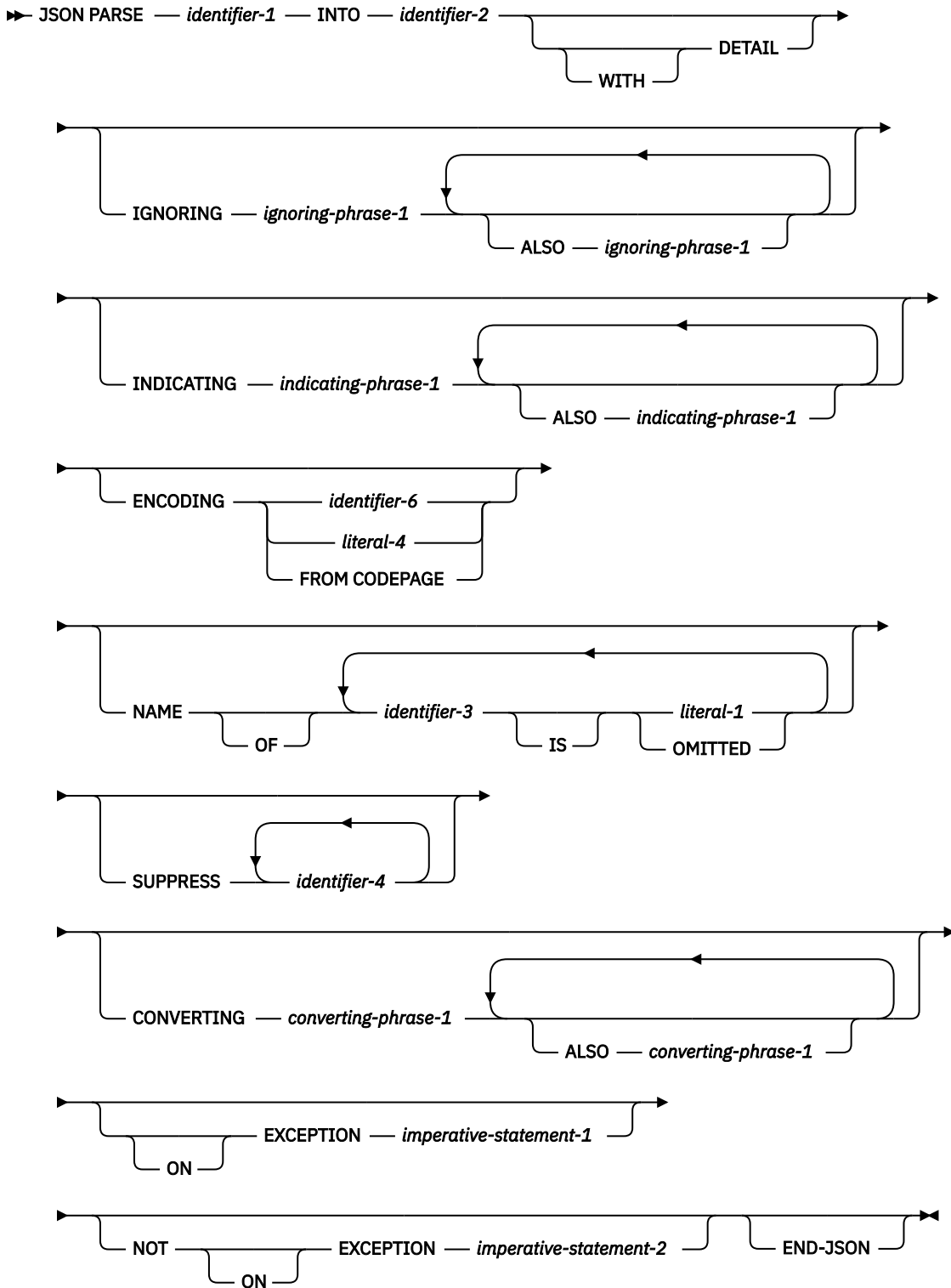
In the JSON text that is generated from *identifier-2*, the names are obtained from the NAME phrase if specified; otherwise they are derived from the name of the data item specified by *identifier-2* and from any eligible data-names that are subordinate to *identifier-2*. The exact mixed-case spelling of data-names from the data description entry is retained. The spellings from any references to data items (for example, in an OCCURS DEPENDING ON clause) are not used.

JSON PARSE statement

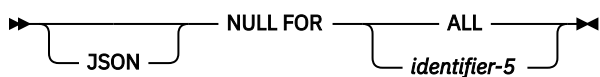
The JSON PARSE statement converts JSON text to COBOL data formats.

You can watch this [video](#) to get an overview of the JSON support in Enterprise COBOL 6.

Format



ignoring-phrase-1 Format



***indicating-phrase-1* Format**



***converting-phrase-1* Format 1**

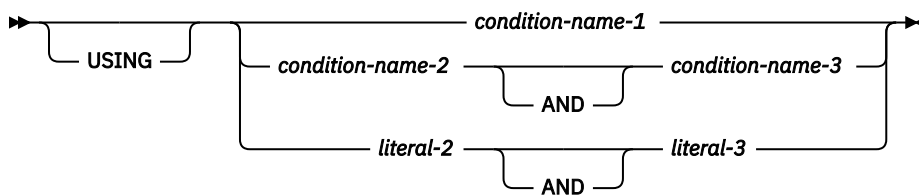


Note: *indicating-phrase-1* reuses the existing *using-phrase-2*. See the *using-phrase-2* syntax diagram below.

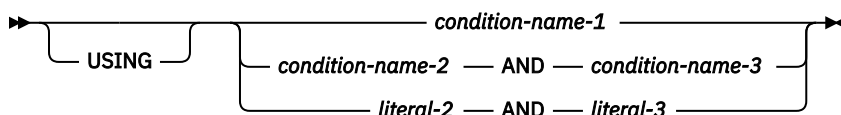
***converting-phrase-1* Format 2**



***using-phrase-1* Format**



***using-phrase-2* Format**



Note: To use the JSON PARSE statement, the CODEPAGE compiler option must specify a single-byte EBCDIC CCSID.

identifier-1

The data item that contains the JSON text. *identifier-1* must reference one of the following items:

- An elementary data item of category alphanumeric
- An alphanumeric group item
- An elementary data item of category national
- A national group item
- An elementary data item of category UTF-8
- A UTF-8 group item

When *identifier-1* references an alphanumeric group item, *identifier-1* is treated as though it were an elementary data item of category alphanumeric. When *identifier-1* references a national group item, *identifier-1* is processed as an elementary data item of category national. When *identifier-1* references a UTF-8 group item, *identifier-1* is processed as an elementary data item of category UTF-8.

identifier-1 must not be defined with the JUSTIFIED clause, and cannot be a function identifier. *identifier-1* can be subscripted or reference modified.

identifier-1 must not overlap *identifier-2*.

identifier-1 can be a dynamic-length elementary item of category alphanumeric or UTF-8. *identifier-1* cannot be a dynamic-length group item of any category, nor an elementary item of category national.

The JSON text is assumed to be encoded in UTF-8 (CCSID 1208) unless the ENCODING phrase is specified.

All Unicode control characters (U+0000 through U+001F), the quotation mark (U+0022), and the reverse solidus (U+005C) must be escaped. For information on EBCDIC control characters that must be escaped, see [Control character mapping table between UTF-8 and EBCDIC](#).

All the escaped character sequences defined in the JSON specification are accepted. Also accepted is the sequence “\x”, which is generated by JSON GENERATE, and which represents the EBCDIC NL (newline) control character X'15', equivalent to the Unicode NEXT LINE control character, NX'0085'.

Conversion from Unicode of the JSON names and values is done according to the compiler CODEPAGE option in effect for the compilation.

identifier-2

The group or elementary data item to be populated from the JSON text. *identifier-2* must reference one of the following items:

- An elementary data item of category alphanumeric
- An alphanumeric group item
- An elementary data item of category national
- A national group item
- An elementary data item of category UTF-8
- A UTF-8 group item

identifier-2 cannot be a function identifier or be reference modified, but it can be subscripted.

identifier-2 must not overlap *identifier-1*.

identifier-2 and its subordinate data items must not contain the UNBOUNDED clause.

identifier-2 can be a dynamic-length group item or a dynamic-length elementary item of category alphanumeric or UTF-8. *identifier-2* cannot be a dynamic-length group or elementary item of category national.

The data description entry for *identifier-2* must not contain a RENAMES clause.

The following data items that are specified by *identifier-2* are ignored by the JSON PARSE statement:

- Any subordinate unnamed elementary data items or elementary FILLER data items
- Any slack bytes inserted for SYNCHRONIZED items
- Any data item subordinate to *identifier-2* that is defined with the REDEFINES clause or that is subordinate to such a redefining item
- Any data item subordinate to *identifier-2* that is defined with the RENAMES clause
- Any group data item all of whose subordinate data items are ignored

All data items specified by *identifier-2* that are not ignored according to the previous rules must satisfy the following conditions:

- Each elementary data item must have a USAGE other than DISPLAY-1, FUNCTION-POINTER, INDEX, OBJECT REFERENCE, POINTER, or PROCEDURE-POINTER.
- There must be at least one such elementary data item.
- Each non-FILLER data-name must have a unique identifier within *identifier-2*.
- If (the data declaration of) *identifier-2* or any subordinate data item contains the OCCURS DEPENDING ON clause, then the object(s) of the OCCURS DEPENDING ON clause(s) must not be subordinate to *identifier-2*. Thus, any objects of OCCURS DEPENDING ON clauses will not be updated by the JSON PARSE statement.

The following example shows the UTF-8 and dynamic-length support for JSON PARSE:

```
01 GRP.
   05 Ac-No PIC AA9999.
   05 MORE.
      10 Stuff PIC S99V9 OCCURS 2.
   05 SSN PIC 999/99/9999.
01 UTF8DYN PIC U DYNAMIC.
   MOVE '{ "GRP": {"Ac-No": "SX1234", "MORE": {"Stuff": [7.8, -9.0]}, "SSN": "- '987654321"} }' TO
   UTF8DYN.
   JSON PARSE UTF8DYN INTO GRP.
```

The JSON PARSE statement is used to extract and decode the JSON data within *UTF8DYN* and populate the following structure in *GRP*:

```
GRP:
Ac-No: SX1234
MORE:
Stuff(1): 07H (Signed Overpunch for +7.8)
Stuff(2): 09H (Signed Overpunch for -9.0)
SSN: 987/65/4321
```

IGNORING phrase

Allows you to indicate that JSON null values shall be ignored, either for all items or the specified list of items.

Note: When JSON null values are not ignored, which is the default behavior of JSON PARSE, and null values appear within the parsed JSON text, the JSON-STATUS special register is set to 32. And, if the WITH DETAIL phrase is specified, a runtime informational message is issued.

If the ALL keyword is specified, JSON null values will be ignored for all items except for the following items:

- Items that appear as *identifier-8* in any specified CONVERTING phrase, in which case the JSON null values are converted to the specified value for those items.
- Items that appear as *identifier-9* in any specified INDICATING phrase, in which case the JSON null values are indicated in the respective null indicator items.

If *identifier-5* is specified, JSON null values are ignored for that item. *identifier-5* must not be simultaneously specified as neither *identifier-8* nor *identifier-9* and *identifier-5* must be an item that references *identifier-2* or is subordinate to *identifier-2*.

If multiple IGNORING phrases are specified, their effects are cumulative.

INDICATING phrase

Allows you to parse JSON null values for the indicated item, *identifier-9*, while having a value moved into an indicator item that informs you whether a JSON null value was encountered or not. The indicator item is specified either directly via *identifier-10*, or indirectly as the parent elementary item of *condition-name-1* or *condition-name-2*.

identifier-10 must, and can only be specified when *literal-2* and *literal-3* are specified.

The indicator item must reference a single byte elementary alphanumeric data item whose data definition entry contains PICTURE X.

The values moved into the indicator item can be as follows:

- The VALUE of *condition-name-1* if a null was found or the FALSE value of *condition-name-1* if a null was not found. The first VALUE clause literal will be used if multiple VALUE clause literals are specified.
- The VALUE of *condition-name-2* if a null was found and *condition-name-3* if a null was not found. The first VALUE clause literal will be used if multiple VALUE clause literals are specified.
- *literal-2* if a null was found or *literal-3* if a null was not found.

The indicator item must appear within the same dimension as the indicated item.

The indicator item is otherwise ignored by the JSON PARSE statement and the JSON text in *identifier-2* shall not contain name/value pairs matching any indicator item.

The indicated item must be subordinate to *identifier-2*.

condition-name-1 must be a level-88 item specified with both the VALUE clause and the WHEN SET TO FALSE phrase.

condition-name-2 and *condition-name-3* must be level-88 items with the same direct parent.

literal-2 and *literal-3* must be single byte alphanumeric literals.

The INDICATING phrase can be specified multiple times using the ALSO keyword.

For example, assume the item *docx* contains the following UTF-8 encoded text:

```
{ "myrecord" : { "data-1" : null } }
```

Consider the following COBOL structure and statements:

```
01 DOCX PIC X(1000).
01 MY-RECORD.
    02 DATA-1-IS-NULL PIC X.
    02 DATA-1 PIC X(100).

JSON PARSE DOCX INTO MY-RECORD
    INDICATING DATA-1 IS JSON NULL USING 'Y' AND 'N' IN DATA-1-IS-NULL
END-JSON

DISPLAY 'DATA-1-IS-NULL: ' DATA-1-IS-NULL
```

The output of the program would be as follows:

```
DATA-1-IS-NULL: Y
```

For a summary of the ways to parse JSON null values, see "Handling JSON null values".

ENCODING phrase

The ENCODING phrase specifies the encoding assumed for the source JSON document in *identifier-1*. The ENCODING phrase must follow these rules:

- *identifier-6* must be an unsigned integer data item.
- *literal-4* must be an unsigned integer literal.
- If *identifier-1* is alphanumeric:
 - The value of *identifier-6* or *literal-4* can be either an EBCDIC coded character set identifier (CCSID) from Table 46 on page 375. ["Single byte EBCDIC coded character sets for JSON documents"](#) or 1208.
 - If FROM CODEPAGE was specified, the CCSID of the CODEPAGE compiler option is assumed and must be an EBCDIC CCSID from Table 46 on page 375. ["Single byte EBCDIC coded character sets for JSON documents"](#).
- If *identifier-1* is national or UTF-8:
 - The value of *identifier-6* or *literal-4* must be 1200 or 1208 respectively.
 - FROM CODEPAGE cannot be specified.

If the ENCODING phrase is omitted and:

- If *identifier-1* is alphanumeric or UTF-8, the JSON document is assumed to be encoded in Unicode UTF-8 (CCSID 1208).
- If *identifier-1* is national, the JSON document is assumed to be encoded in Unicode UTF-16 (CCSID 1200).

If the ENCODING phrase specifies a single byte EBCDIC CCSID, *literal-1* of the JSON PARSE NAME phrase must be an alphanumeric literal. For more information, see [Table 46 on page 375. "Single byte EBCDIC coded character sets for JSON documents"](#).

WITH DETAIL phrase

The WITH DETAIL phrase causes the JSON PARSE statement to emit runtime messages for any nonexception and exception conditions encountered during parsing.

NAME phrase

For the purpose of matching the name of a JSON name/value pair, the NAME phrase allows you to effectively change the name of a data item to the specified literal during the execution of the JSON PARSE statement.

You can specify OMITTED to parse an anonymous JSON object or array, whose top parent name is not specified. For examples of parsing JSON anonymous arrays, see "Parsing JSON anonymous arrays" in the *Programming Guide*.

Notes:

- Only anonymous JSON object or array can be parsed using OMITTED and the rest of anonymous JSON types (string, number, true, false, and null) cannot.
- Any JSON value can be anonymous. For example, an anonymous JSON object is one whose name is omitted in a JSON name/value pair. Given the following JSON text:

```
{ "top" : { "A": "value1", "B": "value2" } }
```

"top" is a name and { "A": "value1", "B": "value2" } is its JSON object value pair. When the name "top" is omitted, { "A": "value1", "B": "value2" } becomes an anonymous JSON object because it does not have its corresponding name.

identifier-3 must reference *identifier-2* or one of its subordinate data items. It cannot be a function identifier and cannot be reference modified or subscripted. It must not specify any data item which is ignored by the JSON PARSE statement. For more information about *identifier-2*, see the description of *identifier-2*. If *identifier-3* is specified more than once in the NAME phrase, the last specification is used. If OMITTED is specified, *identifier-3* must refer to *identifier-2*.

literal-1 must be an alphanumeric or national literal containing the JSON name to be associated with *identifier-3*. The literal is case-sensitive and must match the JSON name exactly.

The NAME phrase in aggregate must not result in an ambiguous name specification. For example, given the following data declarations and JSON text:

```
01 G.  
  05 H.  
    10 A pic x(10).  
    10 3_ pic 9.  
    10 C-C pic x(10).  
  
'{"g": {"H": {"A": "Eh?", "3_": 5, "C-C": "See"}}}'.
```

Then, if it were allowed, specifying the NAME phrase as:

```
NAME of A is 'C-C'
```

would result in no data item receiving the value "Eh?", and an ambiguity about which data item should receive the value "See", effectively defining the declaration of group G as:

```
01 G.  
  05 H.  
    10 C-C pic x(10).  
    10 3_ pic 9.  
    10 C-C pic x(10).
```

which would be illegal if referenced as *identifier-2* in a JSON PARSE statement. Specifying the NAME phrase as:

```
NAME of A is 'C-C' C-C is 'A'
```

is not ambiguous, and would simply swap the assignments to data items A and C-C.

Given the following data declaration and JSON text:

```
01 top1.  
  02 A pic x(20).  
  02 B pic x(20).  
  
'{"A":"value1","B":"value2"}'
```

This anonymous JSON object can be parsed using OMITTED:

```
NAME top1 IS OMITTED
```

If OMITTED is not specified, the JSON object would need to contain a parent name called "top":

```
'{"top1":{"A":"value1","B":"value2"}}'
```

If *literal-1* is a NATIONAL or UTF-8 literal, *identifier-1* must reference a data item of category NATIONAL or UTF-8, or the ENCODING phrase must specify a CCSID of 1208 for Unicode UTF-8.

SUPPRESS phrase

Allows you to identify and unconditionally exclude items that are subordinate to *identifier-2* from assignment by the JSON PARSE statement.

identifier-4 must reference a data item that is subordinate to *identifier-2* and that is not otherwise ignored by the operation of the JSON PARSE statement. *identifier-4* cannot be a function identifier and cannot be reference modified or subscripted. *identifier-4* can reference an entire table.

If *identifier-4* specifies a group data item, that group data item and all data items that are subordinate to the group item are excluded.

Duplicate specifications of *identifier-4* are permitted.

A data item that is specified in the SUPPRESS phrase, is suppressed even if the same data item is also specified in the NAME phrase.

CONVERTING phrase

Allows you to specify items that will be parsed as either JSON BOOLEAN or JSON null name/value pairs.

phrase-1 Format 1

Use Format 1 to specify items that will be parsed as JSON boolean name/value pairs.

identifier-7 must be a single-byte alphanumeric elementary data item whose data definition entry contains PICTURE X.

The USING phrase provides various methods of specifying the values that shall be effectively moved into *identifier-7* when a JSON BOOLEAN true or false value is encountered during parsing.

condition-name-1 must be a level-88 item directly subordinate to *identifier-7* and must be specified with both the VALUE clause and the WHEN SET TO FALSE phrase. The first VALUE clause literal (of possibly many values and ranges) will be used to populate *identifier-7* when parsing a JSON BOOLEAN true value. The FALSE value will be used to populate *identifier-7* when parsing a JSON BOOLEAN false value.

condition-name-2 and *condition-name-3* must be level-88 items directly subordinate to *identifier-7* whose VALUE clauses are used to populate *identifier-7* when a JSON BOOLEAN true or false value is parsed respectively. The first VALUE clause literal will be used in both cases.

literal-2 and *literal-3* must be single-byte alphanumeric literals. *literal-2* and *literal-3* are used to populate *identifier-7* when a JSON BOOLEAN true or false value is parsed respectively.

The CONVERTING phrase can be specified with multiple items to be parsed as JSON BOOLEAN name/value pairs by using the ALSO keyword.

Example: CONVERTING phrase with all three formats of the USING phrase

Consider the following COBOL structure and statements:

```
01 docx pic x(1000).
01 myrecord.
  02 data-a pic x.
    88 data-a-flag value 'T' false 'F'.
  02 data-b pic x.
    88 data-b-true value '1'.
    88 data-b-false value '0'.
  02 data-c pic x.

JSON PARSE docx INTO myrecord
  CONVERTING data-a FROM BOOLEAN USING data-a-flag
    ALSO data-b FROM BOOLEAN USING data-b-true AND data-b-false
    ALSO data-c FROM BOOLEAN USING 'a' AND 'z'
DISPLAY data-a
DISPLAY data-b
DISPLAY data-c
```

Assume *docx* contains the following UTF-8 encoded JSON text:

```
{ "myrecord" :
  { "data-a" : true,
    "data-b" : false,
    "data-c" : true
  }
}
```

The output of the program would be:

```
T
0
a
```

phrase-1 Format 2

Use Format 2 to specify items that might be parsed as JSON null name/value pairs.

identifier-8 must be a group or elementary item that references *identifier-2* or is subordinate to *identifier-2*.

fig-con-1 represents one of the figurative constants from the list below:

- SPACE, SPACES
- ZERO, ZEROES, ZEROS
- LOW-VALUE, LOW-VALUES
- HIGH-VALUE, HIGH-VALUES

The figurative constant is moved to *identifier-8* when a JSON null value is encountered for that name. *fig-con-1* must be a legal sender in the implicit move with *identifier-8* as the receiver according to the rules in [“MOVE statement” on page 405](#).

The following example shows JSON text containing null values being parsed into a COBOL data item using the CONVERTING phrase. Assume the item *docx* contains the following UTF-8 encoded text:

```
{ "my-record" : { "data-a" : null, "data-b" : null } }
```

```
01 my-record.
  02 data-a pic 9999.
  02 data-b pic x(10).

MOVE 1234 to data-a
MOVE "0123456789" to data-b

JSON PARSE docx INTO my-record
  CONVERTING data-a FROM NULL USING ZERO
    ALSO data-b FROM NULL USING SPACES
END-JSON
```

```
DISPLAY data-a  
DISPLAY "''" data-b "''"
```

The output of the program would be as follows:

```
0000  
, ,
```

ON EXCEPTION phrase

An exception condition exists when an error occurs during parsing of the JSON text, for example an ill-formed JSON value, or during assignment of a value to a COBOL data item. In such cases, JSON parsing stops and the receiver, *identifier-2*, might be partially modified.

Special register JSON-CODE contains an exception code, as detailed in JSON PARSE conditions and associated codes and runtime messages in the Enterprise COBOL Programming Guide. Special register JSON-STATUS might also contain a nonzero status value, representing one or more non-exception conditions that occurred prior to the exception condition. For more details, see *Operation of JSON PARSE*.

If the ON EXCEPTION phrase is specified, control is transferred to *imperative-statement-1*. If the ON EXCEPTION phrase is not specified, the NOT ON EXCEPTION phrase, if any, is ignored, and control is transferred to the end of the JSON PARSE statement.

NOT ON EXCEPTION phrase

If an exception condition does not occur during parsing of the JSON text, control is passed to *imperative-statement-2*, if specified, otherwise to the end of the JSON PARSE statement. The ON EXCEPTION phrase, if specified, is ignored. Special register JSON-CODE contains zero after execution of the JSON PARSE statement.

Nonexception conditions that can occur during execution of the JSON PARSE statement might result in special register JSON-STATUS being set to a nonzero status value, and the receiver *identifier-2* being partially modified.

For more details, see “Operation of JSON PARSE” and JSON PARSE conditions and associated codes and runtime messages in the Enterprise COBOL Programming Guide.

END-JSON phrase

This explicit scope terminator delimits the scope of JSON GENERATE or JSON PARSE statements. END-JSON permits a conditional JSON GENERATE or JSON PARSE statement (that is, a JSON GENERATE or JSON PARSE statement that specifies the ON EXCEPTION or NOT ON EXCEPTION phrase) to be nested in another conditional statement.

The scope of a conditional JSON GENERATE or JSON PARSE statement can be terminated by:

- An END-JSON phrase at the same level of nesting
- A separator period

END-JSON can also be used with a JSON GENERATE or JSON PARSE statement that does not specify either the ON EXCEPTION or the NOT ON EXCEPTION phrase.

For more information about explicit scope terminators, see [“Delimited scope statements” on page 293](#).

Examples of JSON text and JSON PARSE statements

Given the following COBOL data declaration:

```
01 TOP1.  
  02 A PIC X(20) OCCURS 2.  
  02 B OCCURS 2.  
    03 C PIC 9(2).  
    03 D PIC 9(2).
```

Below are examples of JSON text and its corresponding JSON PARSE statement that can parse the JSON text:

```
{ "TOP1" :  
  { "A" : ["VALUE1", "VALUE2"],  
    "B" : [{ "C":11, "D":22},  
            { "C":33, "D":44}]    } }
```

```
JSON PARSE JSON-TEXT INTO TOP1
```

```
{ "A" : ["VALUE1", "VALUE2"],  
  "B" : [{ "C":11, "D":22},  
          { "C":33, "D":44}]    } }
```

```
JSON PARSE JSON-TEXT INTO TOP1  
NAME TOP1 IS OMITTED
```

```
{ "A" : ["VALUE1", "VALUE2"] }
```

```
JSON PARSE JSON-TEXT INTO A
```

```
["VALUE1", "VALUE2"]
```

```
JSON PARSE JSON-TEXT INTO A  
NAME A IS OMITTED
```

```
{ "B" : [{ "C":11, "D":22},  
          { "C":33, "D":44}] }
```

```
JSON PARSE JSON-TEXT INTO B
```

```
[{ "C":11, "D":22},  
  { "C":33, "D":44}]
```

```
JSON PARSE JSON-TEXT INTO B  
NAME B IS OMITTED
```

Video resource

Watch the [video](#) to get an overview of the JSON support in Enterprise COBOL 6.

Nested JSON GENERATE or JSON PARSE statements

When a given JSON GENERATE or JSON PARSE statement appears in *imperative-statement-1* or *imperative-statement-2* of another JSON GENERATE or JSON PARSE statement, that given JSON GENERATE or JSON PARSE statement is a *nested* JSON GENERATE or JSON PARSE statement.

Nested JSON GENERATE or JSON PARSE statements are considered to be matched JSON GENERATE and END-JSON combinations, or JSON PARSE and END-JSON combinations, proceeding from left to right. Thus, any END-JSON phrase that is encountered is matched with the nearest preceding JSON GENERATE or JSON PARSE statement that has not been implicitly or explicitly terminated.

Operation of JSON PARSE

The JSON text must be valid. Otherwise the statement terminates with an exception condition, and *identifier-2* might be partially modified.

Parsing is guided by a name-matching algorithm in which the containment structure of the names must match exactly, except that omissions of complete elementary or "group" levels are tolerated in the JSON text. For both the JSON name and the COBOL data name, each lowercase single-byte alphabetic letter, a through z, is considered to be equivalent to its corresponding single-byte uppercase alphabetic letter, A through Z. In other words, the name-matching between JSON names and COBOL data names is case insensitive. In the case of an applicable NAME phrase specification, the NAME phrase literal, after being

converted by the compiler from the codepage in effect by the CODEPAGE compiler option to Unicode UTF-8, must match the JSON name exactly in a case sensitive match.

For each matching JSON name and data item name, the matching JSON value is assigned to the corresponding data item and occurrence in accordance with the table of “Valid and invalid elementary moves” on page 395, and with the same semantics as the equivalent COBOL MOVE statement.

Whitespace characters (SP, HT, LF, and CR) are ignored, except within strings, and are illegal within numbers.

If, for each elementary data item subordinate to *identifier-2*, there is exactly one matching appropriately qualified JSON name/value pair, regardless of the order in the JSON text, then the JSON PARSE statement terminates without an exception, and with special registers JSON-STATUS and JSON-CODE containing zero.

A number of other conditions encountered during execution of the JSON PARSE statement might result in a nonzero JSON-STATUS value, or an exception and a nonzero JSON-CODE value.

For example, if any combination of JSON value and COBOL data type is invalid, the statement terminates with an exception condition. If any matching JSON value results in one or more substitution characters when translated from Unicode to the CCSID specified by the CODEPAGE compiler option, the value is accepted but results in a nonzero JSON-STATUS value and possibly one or more runtime messages, under control of the WITH DETAIL phrase.

Superfluous JSON name/value pairs that do not match any data item names in *identifier-2* are tolerated, but result in setting special register JSON-STATUS to a condition code, and possibly in one or more runtime messages, under control of the WITH DETAIL phrase.

If there are no matching items, the statement terminates with an exception condition, and *identifier-2* is unmodified.

The special value null is interpreted as an instruction to skip assignment of the corresponding data item or occurrence.

This could be useful for tables. For example parsing the JSON text fragment:

```
{"myTable": [31, null, 37, null, 41]}
```

would result in setting only the first, third, and fifth occurrences of data item “myTable”.

Each JSON array should have the same cardinality as the associated table data item. If the JSON array has fewer elements than the table item, the additional table elements are not modified. If the JSON array has more values than the matching table item, the additional values are ignored. Both kinds of mismatch result in a nonzero JSON-STATUS setting.

For a complete description of JSON PARSE conditions, see *JSON PARSE conditions and associated codes and runtime messages* in the *Enterprise COBOL Programming Guide*.

Examples of matched and mismatched data definitions and JSON text

The following data definitions and JSON text are considered an exact match.

Data definitions:

```
01 G.  
  05 h.  
    10 a pic x(10).  
    10 3_ pic 9.  
    10 C-c pic x(10).
```

JSON text:

```
{"g": {"H": {"A": "Eh?", "3_": 5, "c-C": "See"}}}
```

The JSON text subsequently generated from this structure would also be an exact match for the JSON text input to the JSON PARSE statement:

```
{"G": {"h": {"a": "Eh?", "3_": 5, "C-c": "See"}}}
```

Omission in the JSON text of names corresponding with elementary items or complete substructures is tolerated, but the level (“qualification”) must match. For example, the following data definitions and JSON text are compatible, but data-items “3_” and “etCetera” would not be modified by the corresponding JSON PARSE statement, and would result in a JSON-STATUS value of 1 at statement termination and a runtime message under control of the WITH DETAIL phrase:

```
01 G.  
  05 h.  
    10 a pic x(10).  
    10 3_ pic 9.  
    10 C-c pic x(10).  
  05 etCetera.  
    10 etCetera pic x(10).  
  
{"g": {"H": {"A": "Eh?", "c-C": "See"}}}
```

Superfluous items in the JSON are tolerated, but result in nonzero JSON-STATUS codes, and runtime messages under control of the WITH DETAIL phrase. For example, the following data definitions and JSON text are compatible, and would result in completely populating group G, and terminating without an exception. However, because JSON name/value pair "B": "Bee" would not be used, special register JSON-STATUS would be set to a reason code of 2:

```
01 G.  
  05 h.  
    10 a pic x(10).  
    10 3_ pic 9.  
    10 C-c pic x(10).  
  
{"G": {"h": {"A": "Eh?", "B": "Bee", "3_": 5, "c-C": "See"}}}
```

The following data definitions and JSON text are fully compatible, despite the name order mismatch.

```
01 G.  
  05 h.  
    10 a pic x(10).  
    10 3_ pic 9.  
    10 C-c pic x(10).  
  
{"g": {"H": {"3_": 5, "A": "Eh?", "c-C": "See"}}}
```

The following data definitions and JSON text are not compatible, because of the omitted name level (qualification by “H”), and, because no data items would be changed, would result in an exception condition:

```
01 G.  
  05 h.  
    10 a pic x(10).  
    10 3_ pic 9.  
    10 C-c pic x(10).  
  
{"g": {"A": "Eh?", "3_": 5, "c-C": "See"}}
```

The following data definitions and JSON text are not compatible, because the JSON values are all incompatible with the corresponding data items, and would result in an exception condition:

```
01 G.  
  02 h.  
    10 a pic a(10).  
    10 3_ pic 9.  
    10 C-c pic 99.  
  
{"g": {"H": {"A": 42, "3_": "x", "c-C": "abc"}}}
```

Count of table elements set by JSON PARSE

JSON texts do not necessarily include a count of values in an array. To determine how many table elements are set by a given JSON PARSE statement, preset all occurrences of the table to a special value.

The preset value should be known to be absent from the incoming JSON array values, then after execution of the JSON PARSE statement, search the table for the first occurrence of the special value. The preceding values are those that were set by the JSON PARSE statement.

Valid and invalid elementary moves

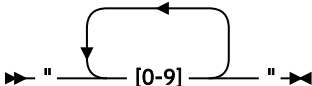
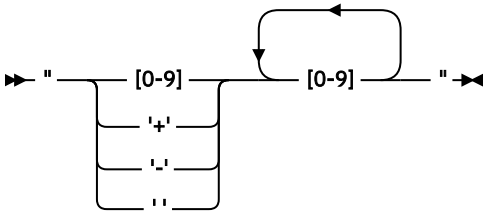
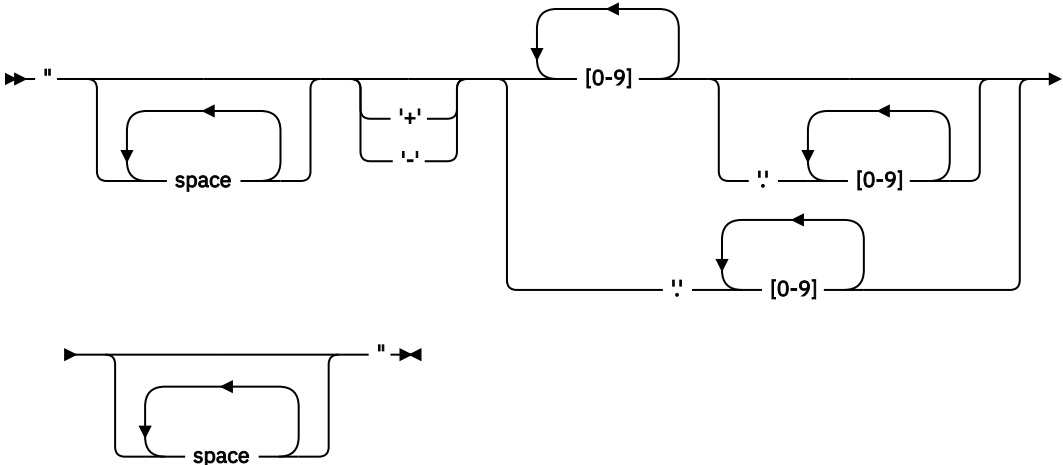
The table shows valid and invalid elementary moves for each category.

In the table:

- YES = Move is valid.
- NO = Move is invalid.
- Column headings indicate receiving item categories; row headings indicate JSON values.

Table 47. Valid and invalid elementary moves								
Valid and invalid elementary moves	Alpha-betic	Alpha-numeric	Alpha-numeric edited	Numeric	Numeric-edited	External floating-point	Internal floating-point	National, national-edited
String	Yes	Yes	Yes	Yes ^{1, 2, 3}	Yes ^{1, 2, 3}	Yes ^{1, 2, 3}	Yes ^{1, 2, 3}	Yes
Number (fixed-point integer)	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number (fixed-point non-integer or float)	No	No	No	Yes	Yes	Yes	Yes	No

Table 47. Valid and invalid elementary moves (continued)

Valid and invalid elementary moves	Alpha-betic	Alpha-numeric	Alpha-numeric edited	Numeric	Numeric-edited	External floating-point	Internal floating-point	National, national-edited
1. The string must consist only of decimal digits, and is treated as an integer.  2. If the COBOL program is recompiled with compiler APAR PH65097 and Runtime APAR PH65057, the JSON string is treated as an integer if it conforms to the following diagram.  For example: "123", "+123", "-123", and " 123" would be strings treated as integers. The string must follow the format dictated in the following syntax diagram. 3. If the COBOL program is recompiled with compiler APAR PH65097 and Runtime APAR PH65883, the JSON string conforms to the following syntax diagram, it is either treated as an integer or fixed-point non-integer depending on its contents. If a period or decimal point is present before or between decimal digits, the string is treated as a fixed-point non-integer.  Note: If DECIMAL-POINT IS COMMA clause is specified in the SPECIAL-NAMES paragraph, it exchanges the functions of the period '.' and the comma ',' in JSON string.								

DECIMAL-POINT IS COMMA clause behavior in JSON Strings and Numbers

For JSON Strings, when the DECIMAL -POINT IS COMMA clause is specified in the SPECIAL -NAMES paragraph, the roles of the period '.' and comma ',' are exchanged. In addition, period and comma cannot be present together regardless of if DECIMAL -POINT IS COMMA clause is specified.

Valid JSON string examples:


```
{ "TOP": "12,34" } or { "TOP": " -1,234 " } or { "TOP": " +123,4" }
```

Note: In the valid JSON strings, the comma ‘,’ represents the decimal point.

For JSON Numbers, regardless of whether DECIMAL-POINT IS COMMA clause is specified, a period ‘.’ represents the decimal point in JSON number value. The comma ‘,’ represents a key-value pair separator.

Valid JSON number examples:

```
{ "TOP": 12.34 } or { "TOP": -1.234 }
```

Invalid JSON number examples:

```
{ "TOP": 12,34 } or { "TOP": -1,234 } or { "TOP": +123,4 }
```

Note: In the first invalid JSON Number example, “TOP” is a key and 12 is value. The comma ‘,’ acts as a separator and a key is expected after this, therefore 34 is considered a key.

The following table illustrates the JSON PARSE output when decimal point is represented by a period ‘.’ or a comma ‘,’.

JSON value	PIC	DECIMAL POINT IS COMMA ‘,’ (OUTPUT)	DECIMAL POINT IS PERIOD ‘.’ (OUTPUT)
"26,97"	99,99	26,97	Parse Fail ¹
	99.99	0.26 ³	Parse Fail ¹
"56.12"	99,99	Parse Fail ²	00,12 ³
	99.99	Parse Fail ²	56.12
"8951,58"	9.999,99	8.951,58	Parse Fail ¹
	9999.99	1580.00 ³	Parse Fail ¹
"6171.82"	9.999,99	Parse Fail ²	1.820,00 ³
	9999.99	Parse Fail ²	6171.82

JSON value	PIC	DECIMAL POINT IS COMMA ‘,’ (OUTPUT)	DECIMAL POINT IS PERIOD ‘.’ (OUTPUT)
13,75	99,99	Parse Fail ⁴	Parse Fail ⁴
	99.99	Parse Fail ⁴	Parse Fail ⁴
56.12	99,99	56,12	00,56 ³
	99.99	0.56 ³	56.12
4289.32	9.999,99	4.289,32	9.320,00 ³
	9999.99	9320.00 ³	4289.32

1. The JSON string is invalid, the JSON string can only use comma ‘,’ as decimal point since DECIMAL-POINT IS COMMA clause is specified.

2. The JSON string is invalid, the JSON string can only use period ' . ' as decimal point since DECIMAL-POINT IS COMMA clause is NOT specified.
3. There is loss of significance, depending on if DECIMAL-POINT IS COMMA clause is specified, the period or comma in the PIC are treated as a decimal point. This causes certain values to lose significance when the decimal point position in input JSON value and PICTURE clause do not align.
4. The JSON number value is invalid, the comma ' , ' represents a key-value pair separator. A period ' . ' must be used to represent decimal point regardless of whether DECIMAL-POINT IS COMMA clause is specified or not.

Example: JSON Input document:

```
{ "BANK": { "A": "26,97", "B": "26,97" } }
```

Example: COBOL Program

```
IDENTIFICATION DIVISION.
PROGRAM-ID. TEST1.
ENVIRONMENT DIVISION.
CONFIGURATION SECTION.
SPECIAL-NAMES.
    DECIMAL-POINT IS COMMA.
DATA DIVISION.
WORKING-STORAGE SECTION.

01 BANK.
    05 A PIC +99,99.
    05 B PIC +99.99.

01 JSON-WORK-FIELDS.
    05 JSON-LEN PIC 9(4) COMP-5 value 46.
    05 JSON-UTF8 PIC U(100).

PROCEDURE DIVISION.
MAIN-LOGIC SECTION.
    PERFORM EXAMPLE-2
    STOP RUN.

EXAMPLE-2 SECTION.
    MOVE '{"BANK":{"A":"26,97","B":"26,97"}}' TO JSON-UTF8
    INITIALIZE BANK
    JSON PARSE JSON-UTF8 (1 : JSON-LEN)
        INTO BANK
        WITH DETAIL
        ON EXCEPTION
            DISPLAY "JSON PARSE FAILED"
    END-JSON

    DISPLAY 'A: ' A.
    DISPLAY 'B: ' B.

EXIT.
```

In this COBOL program, the JSON string provides values for two fields: A and B, with comma defined as a decimal point. The resulting values from JSON Parse is shown in the following example:

```
IGZ0328I During execution of the JSON PARSE statement on line 26 of program TEST1, assignment
of the value of the JSON name/value pair at offset 25 to data item
B resulted in loss of significance ("SIZE ERROR").
A: +26,97
B: +00.26
```

Parsing JSON values that result in a loss of significance or information

JSON values may exceed the size, length, or precision of the data items they are being assigned to by the JSON PARSE statement. When such a situation occurs, the JSON PARSE statement will truncate the value similarly to the equivalent MOVE statement and populate the receiving data item with a loss of significance or information, and may set a reason code of 128 or 256 in the JSON-STATUS special register. If the user also specifies the WITH DETAIL phrase then one or more runtime messages may also be emitted. The type of truncation that occurs depends on the type of the receiving data item, and the type of

the JSON value. Some types of truncation such as loss of precision due to assignment to a floating-point receiver, or from a floating-point sender, will not be recognized by the JSON PARSE statement.

The assignment of a JSON value to its corresponding data item follows the [Alignment Rules](#).

JSON-STATUS special register values 128 and 256

The JSON-STATUS special register is set to 128 when a loss of significance of a fixed-point integer, fixed-point noninteger, or floating-point JSON value occurs similar to a COMPUTE SIZE ERROR condition. That is, when a JSON value, after decimal point alignment, exceeds the largest value that can be contained in the receiver. This also applies to JSON string values that contain only digits.

The JSON-STATUS special register is set to 256 when a loss of information of a JSON string value occurs.

Control character mapping table between UTF-8 and EBCDIC

The control characters listed in the mapping table must be escaped in JSON text.

All EBCDIC CCSIDs listed in [Single byte EBCDIC coded character sets for JSON documents](#) are applicable.

Table 50. Control character mapping table between UTF-8 and EBCDIC				
GCGID	Description	UTF-8 (CCSID) 1208		EBCDIC
		Code Point	Encoding (HEX value)	Encoding (HEX value)
NUL	Null	U+0000	00	00
SOH	Start of Heading	U+0001	01	01
STX	Start of Text	U+0002	02	02
ETX	End of Text	U+0002	03	03
EOT	End of Transmission	U+0004	04	37
ENQ	Enquiry	U+0005	05	2D
ACK	Acknowledge	U+0006	06	2E
BEL	Bell	U+0007	07	2F
BS	Backspace	U+0008	08	16
HT	Horizontal Tabulation	U+0009	09	05
LF	Line Feed	U+000A	0A	25
VT	Vertical Tabulation	U+000B	0B	0B
FF	Form Feed	U+000C	0C	0C
CR	Carriage Return	U+000D	0D	0D
SO	Locking Shift One (Shift Out)	U+000E	0E	0E
SI	Locking Shift Zero (Shift In)	U+000F	0F	0F
DLE	Data Link Escape	U+0010	10	10
DC1	Device Control One	U+0011	11	11
DC2	Device Control Two	U+0012	12	12

Table 50. Control character mapping table between UTF-8 and EBCDIC (continued)

GCGID	Description	UTF-8 (CCSID) 1208		EBCDIC
		Code Point	Encoding (HEX value)	Encoding (HEX value)
DC3	Device Control Three	U+0013	13	13
DC4	Device Control Four	U+0014	14	3C
NAK	Negative Acknowledge	U+0015	15	3D
SYN	Synchronous Idle	U+0016	16	32
ETB	End of Transmission Block	U+0017	17	26
CAN	Cancel	U+0018	18	18
EM	End of Medium	U+0019	19	19
SUB	Substitute	U+001A	1A	3F
ESC	Escape	U+001B	1B	27
IFS	Information File Separator	U+001C	1C	1C
IGS	Group Separator	U+001D	1D	1D
IRS	Record Separator	U+001E	1E	1E
IUS	Unit Separator	U+001F	1F	1F

MERGE statement

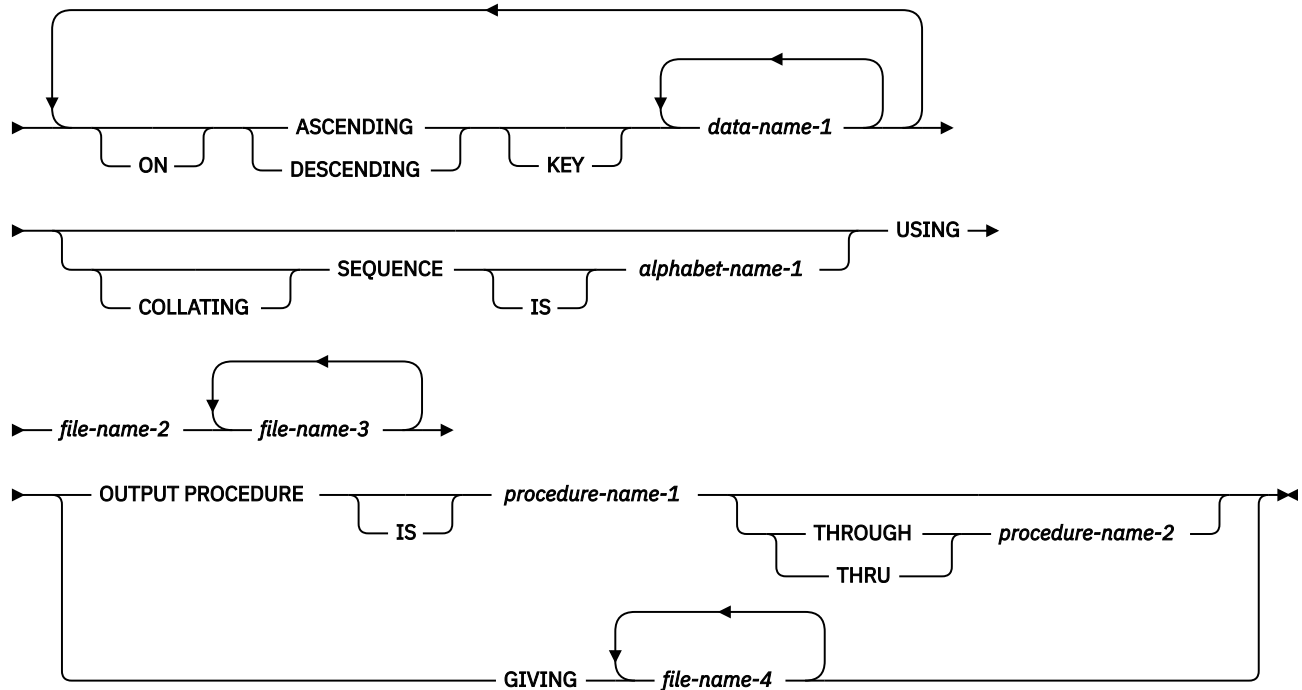
The MERGE statement combines two or more identically sequenced files (that is, files that have already been sorted according to an identical set of ascending or descending keys) on one or more keys and makes records available in merged order to an output procedure or output file.

A MERGE statement can appear anywhere in the PROCEDURE DIVISION except in a declarative section.

The MERGE statement is not supported for programs compiled with the THREAD compiler option.

Format

►► MERGE — *file-name-1* →



file-name-1

The name given in the SD entry that describes the records to be merged.

No file-name can be repeated in the MERGE statement.

No pair of file-names in a MERGE statement can be specified in the same SAME AREA, SAME SORT AREA, or SAME SORT-MERGE AREA clause. However, any file-names in the MERGE statement can be specified in the same SAME RECORD AREA clause.

When the MERGE statement is executed, all records contained in *file-name-2*, *file-name-3*, ... , are accepted by the merge program and then merged according to the keys specified.

ASCENDING/DESCENDING KEY phrase

This phrase specifies that records are to be processed in an ascending or descending sequence (depending on the phrase specified), based on the specified merge keys.

data-name-1

Specifies a KEY data item on which the merge will be based. Each such data-name must identify a data item in a record associated with *file-name-1*. The data-names following the word KEY are listed from left to right in the MERGE statement in order of decreasing significance without regard to how they are divided into KEY phrases. The leftmost data-name is the major key, the next data-name is the next most significant key, and so forth.

The following rules apply:

- A specific key data item must be physically located in the same position and have the same data format in each input file. However, it need not have the same data-name.
- If *file-name-1* has more than one record description, the KEY data items need be described in only one of the record descriptions.

- If *file-name-1* contains variable-length records, all of the KEY data-items must be contained within the first *n* character positions of the record, where *n* equals the minimum records size specified for *file-name-1*.
- KEY data items must not contain an OCCURS clause or be subordinate to an item that contains an OCCURS clause.
- KEY data items cannot be:
 - Variably located
 - Group items that contain variable-occurrence data items
 - Category numeric described with usage NATIONAL (national decimal type)
 - Category external floating-point described with usage NATIONAL (national floating-point)
 - Category DBCS
 - Dynamic-length elementary items
 - Dynamic-length group items
- KEY data items can be qualified.
- KEY data items can be any of the following data categories:
 - Alphabetic, alphanumeric, alphanumeric-edited
 - Numeric (except numeric with usage NATIONAL)
 - Numeric-edited (with usage DISPLAY or NATIONAL)
 - Internal floating-point or display floating-point
 - National or national-edited

The direction of the merge operation depends on the specification of the ASCENDING or DESCENDING keywords as follows:

- When ASCENDING is specified, the sequence is from the lowest key value to the highest key value.
- When DESCENDING is specified, the sequence is from the highest key value to the lowest key value.

If the KEY data item is described with usage NATIONAL, the sequence of the KEY values is based on the binary values of the national characters.

When the COLLATING SEQUENCE phrase is not specified, the key comparisons are performed according to the rules for comparison of operands in a relation condition. For details, see [“General relation conditions”](#) on page 272.

When the COLLATING SEQUENCE phrase is specified, the indicated collating sequence is used for key data items of alphabetic, alphanumeric, alphanumeric-edited, external floating-point, and numeric-edited categories. For all other key data items, the comparisons are performed according to the rules for comparison of operands in a relation condition.

COLLATING SEQUENCE phrase

This phrase specifies the collating sequence to be used in alphanumeric comparisons for the KEY data items in this merge operation.

The COLLATING SEQUENCE phrase has no effect for keys that are not alphabetic or alphanumeric.

alphabet-name-1

Must be specified in the ALPHABET clause of the SPECIAL-NAMES paragraph. Any one of the alphabet-name clause phrases can be specified, with the following results:

STANDARD-1

The ASCII collating sequence is used for all alphanumeric comparisons. (The ASCII collating sequence is shown in [“US English ASCII code page”](#) on page 766.)

STANDARD-2

The 7-bit code defined in the International Reference Version of *ISO/IEC 646, 7-bit coded character set for information interchange* is used for all alphanumeric comparisons.

NATIVE

The EBCDIC collating sequence is used for all alphanumeric comparisons. (The EBCDIC collating sequence is shown in [“EBCDIC collating sequence”](#) on page 763.)

EBCDIC

The EBCDIC collating sequence is used for all alphanumeric comparisons. (The EBCDIC collating sequence is shown in [“EBCDIC collating sequence”](#) on page 763.)

literal

The collating sequence established by the specification of literals in the ALPHABET-NAME clause is used for all alphanumeric comparisons.

When the COLLATING SEQUENCE phrase is omitted, the PROGRAM COLLATING SEQUENCE clause (if specified) in the OBJECT-COMPUTER paragraph identifies the collating sequence to be used. When both the COLLATING SEQUENCE phrase of the MERGE statement and the PROGRAM COLLATING SEQUENCE clause of the OBJECT-COMPUTER paragraph are omitted, the EBCDIC collating sequence is used.

USING phrase***file-name-2 , file-name-3 , ...***

Specifies the input files.

During the MERGE operation, all the records on *file-name-2*, *file-name-3*, ... (that is, the input files) are transferred to *file-name-1*. At the time the MERGE statement is executed, these files must not be open. The input files are automatically opened, read, and closed. If DECLARATIVE procedures are specified for these files for input operations, the declaratives will be driven for errors if errors occur.

All input files must specify sequential or dynamic access mode and be described in FD entries in the DATA DIVISION.

If *file-name-1* contains variable-length records, the size of the records contained in the input files (*file-name-2*, *file-name-3*, ...) must be neither less than the smallest record nor greater than the largest record described for *file-name-1*. If *file-name-1* contains fixed-length records, the size of the records contained in the input files must not be greater than the largest record described for *file-name-1*. For more information, see *Sorting and merging files* in the *Enterprise COBOL Programming Guide*.

GIVING phrase***file-name-4 , ...***

Specifies the output files.

When the GIVING phrase is specified, all the merged records in *file-name-1* are automatically transferred to the output files (*file-name-4*, ...).

All output files must specify sequential or dynamic access mode and be described in FD entries in the DATA DIVISION.

If the output files (*file-name-4*, ...) contain variable-length records, the size of the records contained in *file-name-1* must be neither less than the smallest record nor greater than the largest record described for the output files. If the output files contain fixed-length records, the size of the records contained in *file-name-1* must not be greater than the largest record described for the output files. For more information, see *Sorting and merging files* in the *Enterprise COBOL Programming Guide*.

At the time the MERGE statement is executed, the output files (*file-name-4*, ...) must not be open. The output files are automatically opened, written to, and closed. If DECLARATIVE procedures are specified for these files for output operations, the declaratives will be driven for errors if errors occur.

OUTPUT PROCEDURE phrase

This phrase specifies the name of a procedure that is to select or modify output records from the merge operation.

procedure-name-1

Specifies the first (or only) section or paragraph in the OUTPUT PROCEDURE.

procedure-name-2

Identifies the last section or paragraph of the OUTPUT PROCEDURE.

The OUTPUT PROCEDURE can consist of any procedure needed to select, modify, or copy the records that are made available one at a time by the RETURN statement in merged order from the file referenced by *file-name-1*. The range includes all statements that are executed as the result of a transfer of control by CALL, EXIT, GO TO, PERFORM, and XML PARSE statements in the range of the output procedure. The range also includes all statements in declarative procedures that are executed as a result of the execution of statements in the range of the output procedure. The range of the output procedure must not cause the execution of any MERGE, RELEASE, or format 1 SORT statement.

If an output procedure is specified, control passes to it after the file referenced by *file-name-1* has been sequenced by the MERGE statement. The compiler inserts a return mechanism at the end of the last statement in the output procedure and when control passes the last statement in the output procedure, the return mechanism provides the termination of the merge and then passes control to the next executable statement after the MERGE statement. Before entering the output procedure, the merge procedure reaches a point at which it can select the next record in merged order when requested. The RETURN statements in the output procedure are the requests for the next record.

The OUTPUT PROCEDURE phrase is similar to a basic PERFORM statement. For example, if you name a procedure in an OUTPUT PROCEDURE, that procedure is executed during the merging operation just as if it were named in a PERFORM statement. As with the PERFORM statement, execution of the procedure is terminated after the last statement completes execution. The last statement in an OUTPUT PROCEDURE can be the EXIT statement (see [“EXIT statement” on page 343](#)).

MERGE special registers

The topic describes special registers of the MERGE statement.

SORT-CONTROL special register

You identify the sort control file (through which you can specify additional options to the sort/merge function) with the SORT-CONTROL special register.

If you use a sort control file to specify control statements, the values specified in the sort control file take precedence over those in the other SORT special registers.

For information, see [“SORT-CONTROL” on page 25](#).

SORT-MESSAGE special register

For information, see [“SORT-MESSAGE” on page 26](#). The special register SORT-MESSAGE is equivalent to an option control statement keyword in the sort control file.

SORT-RETURN special register

For information, see [“SORT-RETURN” on page 27](#).

Segmentation considerations

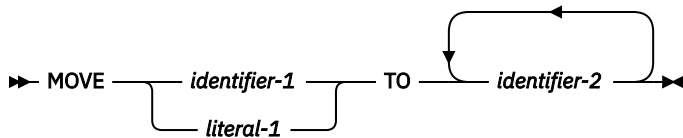
If a MERGE statement is coded in a fixed segment, any output procedure referenced by that MERGE statement must be either totally within a fixed segment or wholly contained in a single independent segment.

If a MERGE statement is coded in an independent segment, any output procedure referenced by that MERGE statement must be either totally within a fixed segment or wholly contained within the same independent segment as that MERGE statement.

MOVE statement

The MOVE statement transfers data from one area of storage to one or more other areas.

Format 1: MOVE statement



Format 2: MOVE statement with CORRESPONDING phrase



CORR is an abbreviation for, and is equivalent to, CORRESPONDING.

identifier-1 , *literal-1*

The sending area.

identifier-2

The receiving areas. *identifier-2* must not reference an intrinsic function.

When format 1 is specified:

- All identifiers can reference alphanumeric group items, national group items, UTF-8 group items, or elementary items.
- When one of *identifier-1* or *identifier-2* references a national group item and the other operand references an alphanumeric group item, the national group is processed as a group item; in all other cases, the national group item is processed as an elementary data item of category national.
- When one of *identifier-1* or *identifier-2* references a UTF-8 group item and the other operand references an alphanumeric group item, the UTF-8 group is processed as a group item; in all other cases, the UTF-8 group item is processed as an elementary data item of category UTF-8.
- The data in the sending area is moved into the data item referenced by each *identifier-2* in the order in which the *identifier-2* data items are specified in the MOVE statement. See [“Elementary moves” on page 406](#) and [“Group moves” on page 410](#) below.

When format 2 is specified:

- Both identifiers must be group items.
- A national group item is processed as a group item (and not as an elementary data item of category national).
- A UTF-8 group item is processed as a group item (and not as an elementary data item of category UTF-8).

- Selected items in *identifier-1* are moved to *identifier-2* according to the rules for the “[CORRESPONDING phrase](#)” on page 295. The results are the same as if each pair of CORRESPONDING identifiers were referenced in a separate MOVE statement.

Data items described with the following types of usage cannot be specified in a MOVE statement:

- INDEX
- POINTER
- FUNCTION-POINTER
- PROCEDURE-POINTER
- OBJECT REFERENCE

A data item defined with a usage of INDEX, POINTER, FUNCTION-POINTER, PROCEDURE-POINTER, or OBJECT REFERENCE can be part of an alphanumeric group item that is referenced in a MOVE CORRESPONDING statement; however, no movement of data from those data items takes place.

The evaluation of the length of the sending or receiving area can be affected by the DEPENDING ON phrase of the OCCURS clause (see “[OCCURS clause](#)” on page 200).

If the sending field (*identifier-1*) is reference-modified or subscripted, or is an alphanumeric or national function-identifier, the reference-modifier, subscript, or function is evaluated only once, immediately before data is moved to the first of the receiving operands.

Any length evaluation, subscripting, or reference-modification associated with a receiving field (*identifier-2*) is evaluated immediately before the data is moved into that receiving field.

For example, the result of the statement:

```
MOVE A(B) TO B, C(B).
```

is equivalent to:

```
MOVE A(B) TO TEMP.  
MOVE TEMP TO B.  
MOVE TEMP TO C(B).
```

where TEMP is defined as an intermediate result item. The subscript B has changed in value between the time that the first move took place and the time that the final move to C (B) is executed.

For further information about intermediate results, see *Appendix A. Intermediate results and arithmetic precision* in the *Enterprise COBOL Programming Guide*.

After execution of a MOVE statement, the sending fields contain the same data as before execution.

Usage note: Overlapping operands in a MOVE statement can cause unpredictable results.

Elementary moves

An elementary move is one in which the receiving item is an elementary data item and the sending item is an elementary data item or a literal.

Valid operands belong to one of the following categories:

- **Alphabetic:** includes data items of category alphabetic and the figurative constant SPACE
- **Alphanumeric:** includes the following items:
 - Data items of category alphanumeric
 - Alphanumeric functions
 - Alphanumeric literals
 - The figurative constant *ALL alphanumeric-literal* and all other figurative constants (except NULL) when used in a context that requires an alphanumeric sending item

- **Alphanumeric-edited:** includes data items of category alphanumeric-edited
- **DBCS:** includes data items of category DBCS, DBCS literals, and the figurative constant ALL DBCS-literal.
- **External floating-point:** includes data items of category external floating point (described with USAGE DISPLAY or USAGE NATIONAL) and floating-point literals.
- **Internal floating-point:** includes data items of category internal floating-point (defined as USAGE COMP-1 or USAGE COMP-2)
- **National:** includes the following items:
 - National group items (treated as elementary item of category national)
 - Data items of category national
 - National literals
 - National functions
 - Figurative constants ZERO, SPACE, QUOTE, and ALL *national-literal* when used in a context that requires a national sending item
- **National-edited:** includes data items of category national-edited
- **Numeric:** includes the following items:
 - Data items of category numeric
 - Numeric literals
 - The figurative constant ZERO (when ZERO is moved to a numeric or numeric-edited item).
- **Numeric-edited:** includes data items of category numeric-edited.
- **UTF-8:** includes data items of category UTF-8.

Elementary move rules

Any necessary conversion of data from one form of internal representation to another takes place during the move, along with any specified editing in, or de-editing implied by, the receiving item. The code page used for conversion to or from alphanumeric characters is the one in effect for the CODEPAGE compiler option when the source code was compiled.

The following rules outline the execution of valid elementary moves. When the receiving field is:

Alphabetic:

- Alignment and any necessary space filling or truncation occur as described under [“Alignment rules” on page 175](#).
- If the size of the sending item is greater than the size of the receiving item, excess characters on the right are truncated after the receiving item is filled.

Alphanumeric or alphanumeric-edited:

- If the sending item is a national decimal integer item, the sending data is converted to usage DISPLAY and treated as though it were moved to a temporary data item of category alphanumeric with the same number of character positions as the sending item. The resulting alphanumeric data item is treated as the sending item.
- Alignment and any necessary space filling or truncation take place, as described under [“Alignment rules” on page 175](#).
- If the receiving item is a dynamic-length elementary item, the length of the receiver is set to the minimum of the length of the sender or the specified or implied value of the LIMIT phrase. If the length of the sending item is greater than the length of the receiving item, then excess characters on the right are truncated after the receiving item is filled.
- For items that are not dynamic-length elementary items, if the size of the sending item is greater than the size of the receiving item, excess characters on the right are truncated after the receiving item is filled.

- If the initial sending item has an operational sign, the unsigned value is used. If the operational sign occupies a separate character, that character is not moved, and the size of the sending item is considered to be one less character than the actual size.

DBCS:

- If the sending and receiving items are not the same size, the sending data is either truncated on the right or padded with DBCS spaces on the right.

External floating-point:

- For a floating-point sending item, the floating-point value is converted to the usage of the receiving external floating-point item (if different from the sending item's representation).
- For other sending items, the numeric value is treated as though that value were converted to internal floating-point and then converted to the usage of the receiving external floating-point item.

Internal floating-point:

- When the category of the sending operand is not internal floating-point, the numeric value of the sending item is converted to internal floating-point format.

National or national-edited:

- If the representation of the sending item is not national characters, the sending data is converted to national characters and treated as though it were moved to a temporary data item of category national of a length not to cause truncation or padding. The resulting category national data item is treated as the sending data item.
- If the representation of the sending item is national characters, the sending data is used without conversion.
- Alignment and any necessary space filling or truncation take place as described under [“Alignment rules” on page 175](#). The programmer is responsible for ensuring that multiple encoding units that together form a graphic character are not split by truncation.
- If the sending item has an operational sign, the unsigned value is used. If the operational sign occupies a separate character, that character is not moved, and the size of the sending item is considered to be one less character than the actual size.

UTF-8:

- If the representation of the sending item is not UTF-8 characters, the sending data is converted to UTF-8 characters and treated as though it were moved to a temporary data item of category UTF-8 of a length not to cause truncation or padding. The resulting category UTF-8 data item is treated as the sending data item.
- If the representation of the sending item is UTF-8 characters, the sending data is used without conversion.
- Alignment and any necessary space filling or truncation take place as described under [“Alignment rules” on page 175](#). The programmer is responsible for ensuring that multiple encoding units that together form a graphic character are not split by truncation.
- If the sending item has an operational sign, the unsigned value is used. If the operational sign occupies a separate character, that character is not moved, and the size of the sending item is considered to be one less character than the actual size.

Numeric or numeric-edited:

- Except when zeros are replaced because of editing requirements, alignment by decimal point and any necessary zero filling take place, as described under [“Alignment rules” on page 175](#).
- If the receiving item is signed, the sign of the sending item is placed in the receiving item, with any necessary sign conversion. If the sending item is unsigned, a positive operational sign is generated for the receiving item.
- If the receiving item is unsigned, no operational sign is generated for the receiving item and the absolute value of the sending item is used in the move.

- When the category of the sending item is alphanumeric, alphanumeric-edited, national, or national-edited, the data is moved as if the sending item were described as an unsigned integer.
- When the sending item is floating-point, the data is first converted to either a binary or internal decimal representation and is then moved.
- When the receiving item is numeric-edited, editing takes place as defined by the picture character string or BLANK WHEN ZERO clause associated with the receiving item.
- When the sending item is numeric-edited, the compiler de-edits the sending data to establish the unedited value of the numeric-edited item (this value can be signed). The unedited numeric value is used in the move to the receiving numeric or numeric-edited data item.

Usage notes:

1. If the receiving item is of category alphanumeric, alphanumeric-edited, numeric-edited, national, or national-edited and the sending field is numeric, any digit positions described with picture symbol P in the sending item are considered to have the value zero. Each P is counted in the size of the sending item.
2. If the receiving item is numeric and the sending field is an alphanumeric literal, a national literal, or an ALL literal, all characters of the literal must be numeric characters.

Valid and invalid elementary moves

The table shows valid and invalid elementary moves for each category.

In the table:

- YES = Move is valid.
- NO = Move is invalid.
- Column headings indicate receiving item categories; row headings indicate sending item categories.

Valid and invalid elementary moves	Alpha-betic	Alpha-numeric	Alpha-numeric edited	Numeric	Numeric-edited	External floating-point	Internal floating-point	DBCS ¹	National	National-edited	UTF-8
Alphabetic and SPACE sending item	Yes	Yes	Yes	No	No	No	No	No	Yes	Yes	Yes
Alphanumeric sending item ²	Yes	Yes	Yes	Yes ³	Yes ³	Yes ⁸	Yes ⁸	No	Yes	Yes	Yes
Alphanumeric-edited sending item	Yes	Yes	Yes	No	No	No	No	No	Yes	Yes	Yes
Numeric integer and ZERO sending item ⁴	No	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	No
Numeric noninteger sending item ⁵	No	No	No	Yes	Yes	Yes	Yes	No	No	No	No
Numeric-edited sending item	No	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes
Floating-point sending item ⁶	No	No	No	Yes	Yes	Yes	Yes	No	No	No	No
DBCS sending item ⁷	No	No	No	No	No	No	No	Yes	Yes	Yes	No
National sending item ⁹	No	No	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes
National-edited sending item	No	No	No	No	No	No	No	No	Yes	Yes	Yes
UTF-8 sending item ¹⁰	No	No	No	No	No	No	No	No	Yes	Yes	Yes

1. Includes DBCS data items.

2. Includes alphanumeric literals.

3. Figurative constants and alphanumeric literals must consist only of numeric characters and will be treated as numeric integer fields.

4. Includes integer numeric literals.

5. Includes noninteger numeric literals.

6. Includes floating-point literals, external floating-point data items (USAGE DISPLAY or USAGE NATIONAL), and internal floating-point data items (USAGE COMP-1 or USAGE COMP-2).

7. Includes DBCS data-items, DBCS literals, and figurative constant SPACE.

8. Figurative constants and alphanumeric literals must consist only of numeric characters and will be treated as numeric integer fields. The ALL literal cannot be used as a sending item.

9. Includes national data items, national literals, national functions, and figurative constants ZERO, SPACE, QUOTE, and ALL national literal.

10. Includes UTF-8 data items, UTF-8 literals, UTF-8 functions, and figurative constants ZERO, SPACE, QUOTE, and ALL UTF-8 literal.

Moves involving file record areas

The successful execution of an OPEN statement for a given file makes the record area for that file available. You can move data to or from the record description entries associated with a file only when the file is in the open status.

Execution of an implicit or explicit CLOSE statement removes a file from open status and makes the record area unavailable.

Group moves

A group move can be any move in which an alphanumeric group item is a sending item or a receiving item, or both.

The group moves are:

- A move to an alphanumeric group item from one of the following items:
 - any elementary data item that is valid as a sending item in the MOVE statement
 - a national group item
 - a UTF-8 group item
 - a literal
 - a figurative constant
- A move from an alphanumeric group item to the following items:
 - any elementary data item that is valid as a receiving item in the MOVE statement
 - a national group item
 - a UTF-8 group item
 - an alphanumeric group item
- A move from a UTF-8 group item to the following items:
 - a national group item
- A move from a national group item to the following items:
 - a UTF-8 group item

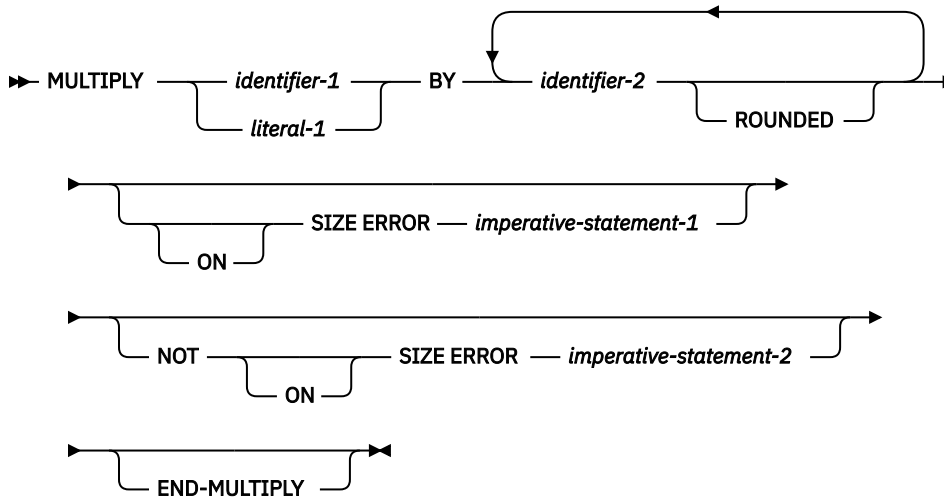
A group move is treated as though it were an alphanumeric-to-alphanumeric elementary move, except that there is no conversion of data from one form of internal representation to another. In a group move, the receiving area is filled without consideration for the individual elementary items contained within either the sending area or the receiving area, except as noted in the OCCURS clause. (See [“OCCURS clause”](#) on page 200.)

Dynamic-length group items can not be moved to, or from, another group item.

MULTIPLY statement

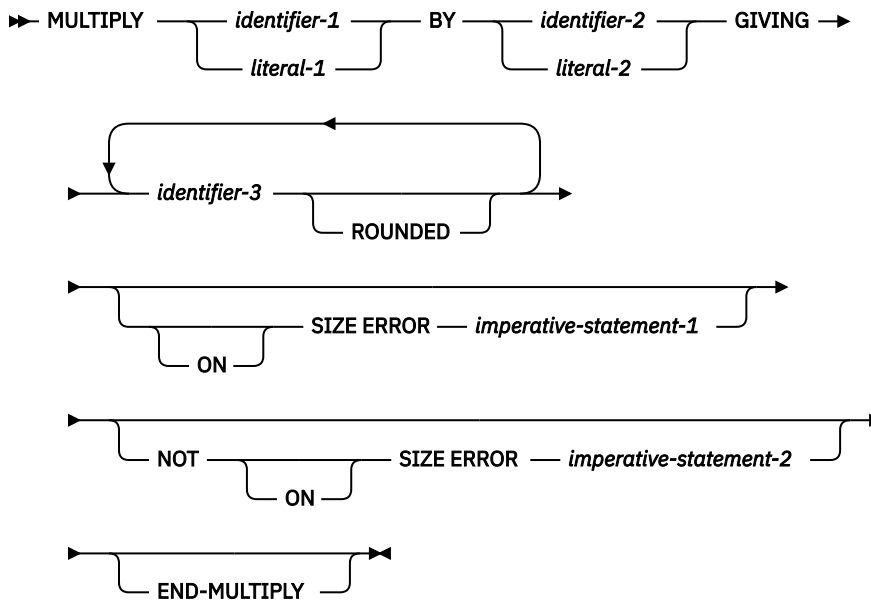
The MULTIPLY statement multiplies numeric items and sets the values of data items equal to the results.

Format 1: MULTIPLY statement



In format 1, the value of *identifier-1* or *literal-1* is multiplied by the value of *identifier-2*; the product is then placed in *identifier-2*. For each successive occurrence of *identifier-2*, the multiplication takes place in the left-to-right order in which *identifier-2* is specified.

Format 2: MULTIPLY statement with GIVING phrase



In format 2, the value of *identifier-1* or *literal-1* is multiplied by the value of *identifier-2* or *literal-2*. The product is then stored in the data items referenced by *identifier-3*.

For all formats:

identifier-1* , *identifier-2

Must name an elementary numeric item.

literal-1 , literal-2

Must be a numeric literal.

For format-2:**identifier-3**

Must name an elementary numeric or numeric-edited item.

Floating-point data items and literals can be used anywhere a numeric data item or literal can be specified.

When the ARITH(COMPAT) compiler option is in effect, the composite of operands can contain a maximum of 30 digits. When the ARITH(EXTEND) compiler option is in effect, the composite of operands can contain a maximum of 31 digits. For more information, see [“Arithmetic statement operands” on page 297](#) and the details on arithmetic intermediate results, *Appendix A. Intermediate results and arithmetic precision* in the *Enterprise COBOL Programming Guide*.

ROUNDED phrase

For formats 1 and 2, see [“ROUNDED phrase” on page 296](#).

SIZE ERROR phrases

For formats 1 and 2, see [“SIZE ERROR phrases” on page 296](#).

END-MULTIPLY phrase

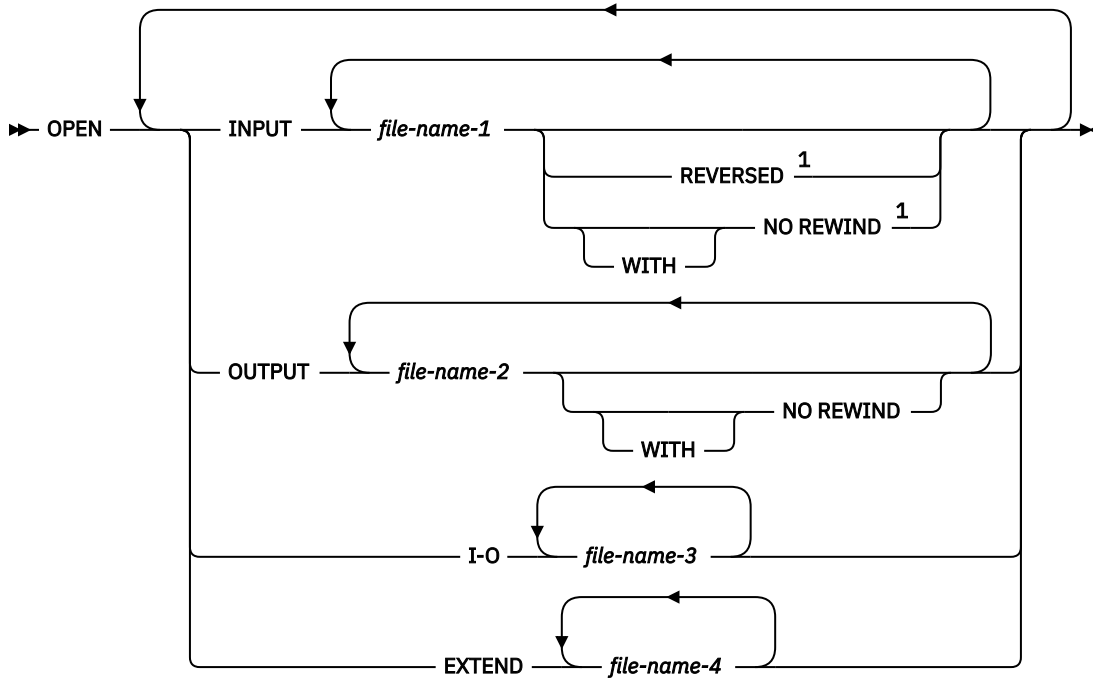
This explicit scope terminator serves to delimit the scope of the MULTIPLY statement. END-MULTIPLY permits a conditional MULTIPLY statement to be nested in another conditional statement. END-MULTIPLY can also be used with an imperative MULTIPLY statement.

For more information, see [“Delimited scope statements” on page 293](#).

OPEN statement

The OPEN statement initiates the processing of files. It also checks or writes labels, or both.

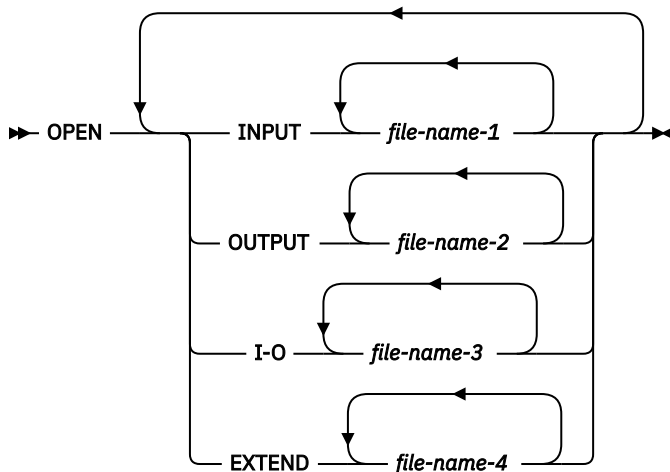
Format 1: OPEN statement for sequential files



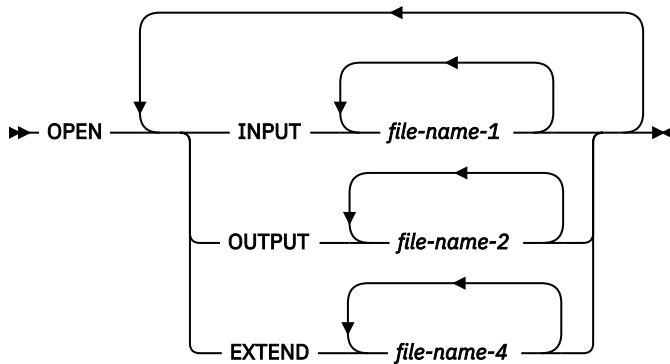
Notes:

¹ The REVERSED and WITH NO REWIND phrases are not valid for VSAM files.

Format 2: OPEN statement for indexed and relative files



Format 3: OPEN statement for line-sequential files



The phrases INPUT, OUTPUT, I-O, and EXTEND specify the mode to be used for opening the file. At least one of the phrases INPUT, OUTPUT, I-O, or EXTEND must be specified with the OPEN keyword. The INPUT, OUTPUT, I-O, and EXTEND phrases can appear in any order.

INPUT

Permits input operations.

OUTPUT

Permits output operations. This phrase can be specified when the file is being created.

Do not specify OUTPUT for files that:

- Contain records. The file will be replaced by new data.

If the OUTPUT phrase is specified for a file that already contains records, the data set must be defined as reusable and cannot have an alternate index. The records in the file will be replaced by the new data and any ALTERNATE RECORD KEY clause in the SELECT statement will be ignored.

- Are defined with a DD dummy card. Unpredictable results can occur.

I-O

Permits both input and output operations. The I-O phrase can be specified only for files assigned to direct access devices.

The I-O phrase is not valid for line-sequential files.

EXTEND

Permits output operations that append to or create a file.

The EXTEND phrase is allowed for sequential access files only if the new data is written in ascending sequence. The EXTEND phrase is allowed for files that specify the LINAGE clause.

For QSAM files, do not specify the EXTEND phrase for a multiple file reel.

If you want to append to a file, but are unsure if the file exists, use the SELECT OPTIONAL clause before opening the file in EXTEND mode. The file will be created or appended to, depending on whether the file exists.

file-name-1, file-name-2, file-name-3, file-name-4

Designate a file upon which the OPEN statement is to operate. If more than one file is specified, the files need not have the same organization or access mode. Each file-name must be defined in an FD entry in the DATA DIVISION and must not name a sort or merge file. The FD entry must be equivalent to the information supplied when the file was defined.

REVERSED

Valid only for sequential single-reel files. REVERSED is not valid for VSAM files.

If the concept of reels has no meaning for the storage medium (for example, a direct access device), the REVERSED and NO REWIND phrases do not apply.

NO REWIND

Valid only for sequential single-reel files. It is not valid for VSAM files.

For information on file sizes, see [Appendix D, “Compiler limits,”](#) on page 757.

General rules

The topic shows general rules of the OPEN statement.

- If a file opened with the INPUT phrase is an optional file that is not available, the OPEN statement sets the file position indicator to indicate that an optional input file is not available.
- Execution of an OPEN INPUT or OPEN I-O statement sets the file position indicator:
 - For indexed files, to the characters with the lowest ordinal position in the collating sequence associated with the file.
 - For sequential and relative files, to 1.
- When the EXTEND phrase is specified, the OPEN statement positions the file immediately after the last record written in the file. (The record with the highest prime record key value for indexed files or relative key value for relative files is considered the last record.) Subsequent WRITE statements add records as if the file were opened OUTPUT. The EXTEND phrase can be specified when a file is being created; it can also be specified for a file that contains records, or that has contained records that have been deleted. For more information, see note 1 in the [“OPEN statement notes”](#) on page 415 and SELECT OPTIONAL in the [“SELECT clause”](#) on page 142.
- For VSAM files, if no records exist in the file, the file position indicator is set so that the first format 1 READ statement executed results in an AT END condition.
- When NO REWIND is specified, the OPEN statement execution does not reposition the file; prior to OPEN statement execution, the file must be positioned at its beginning. When the NO REWIND phrase is specified (or when both the NO REWIND and REVERSE phrases are omitted), file positioning is specified with the LABEL parameter of the DD statement.
- When REVERSED is specified, OPEN statement execution positions the QSAM file at its end. Subsequent READ statements make the data records available in reversed order, starting with the last record.

When OPEN REVERSED is specified, the record format must be fixed.

- When the REVERSED, NO REWIND, or EXTEND phrases are not specified, OPEN statement execution positions the file at its beginning.

If the PASSWORD clause is specified in the file-control entry, the password data item must contain a valid password before the OPEN statement is executed. If a valid password is not present, OPEN statement execution is unsuccessful.

OPEN statement notes

The topic provides notes for the OPEN statement.

The notes are:

1. The successful execution of an OPEN statement determines the availability of the file and results in that file being in open mode. A QSAM file is *available* if it has a DD allocation and is physically present. A VSAM file is *available* if it has a DD allocation, has been defined using VSAM access method services, and contains records or has previously contained records. For more information regarding file availability, see *Opening a file (ESDS, KSDS, or RRDS)* in the *Enterprise COBOL Programming Guide*. The following table shows the results of opening available and unavailable files.

Table 52. Availability of a file		
Opened as	File is available	File is unavailable
INPUT	Normal open	Open is unsuccessful. (file status 35)
INPUT (optional file)	Normal open	Normal open; the first read causes the at end condition or the invalid key condition. (file status 05)
I-O	Normal open	Open is unsuccessful. (file status 35)
I-O (optional file)	Normal open	Open causes the file to be created. (file status 05)
OUTPUT	Normal open; the file contains no records	Open causes the file to be created.
EXTEND	Normal open	Open is unsuccessful. (file status 35)
EXTEND (optional file)	Normal open	Open causes the file to be created. (file status 05)

2. The successful execution of the OPEN statement places the file in open status and makes the associated record area available to the program.
3. The OPEN statement does not obtain or release the first data record.
4. You can move data to or from the record area only when the file is in open status.
5. An OPEN statement must be successfully executed prior to the execution of any of the permissible input-output statements, except a SORT or MERGE statement with the USING or GIVING phrase. In the following table, an 'X' indicates that the specified statement can be used with the open mode given at the top of the column.
6. The VSAMOPENFS option affects the user file status reported from successful OPEN statements on VSAM files.
7. The CBLQDA Language Environment (LE) runtime option affects the user file status reported from OPEN statements on QSAM files that are not found:
 - With CBLQDA(ON), LE would create a temporary data set and the OPEN would be successful.
 - With CBLQDA(OFF), LE would not create a temporary data set and the OPEN would fail.

For details about the CBLQDA runtime option, see CBLQDA in the *z/OS Language Environment Programming Reference*.

Table 53. Permissible statements for sequential files				
Statement	Input open mode	Output open mode	I-O open mode	Extend open mode
READ	X		X	
WRITE		X		X
REWRITE			X	

In the following table, an 'X' indicates that the specified statement, used in the access mode given for that row, can be used with the open mode given at the top of the column.

<i>Table 54. Permissible statements for indexed and relative files</i>					
File access mode	Statement	Input open mode	Output open mode	I-O open mode	Extend open mode
Sequential	READ	X		X	
	WRITE		X		X
	REWRITE			X	
	START	X		X	
	DELETE			X	
Random	READ	X		X	
	WRITE		X	X	
	REWRITE			X	
	START				
	DELETE			X	
Dynamic	READ	X		X	
	WRITE		X	X	
	REWRITE			X	
	START	X		X	
	DELETE			X	

In the following table, an 'X' indicates that the specified statement can be used with the open mode given at the top of the column.

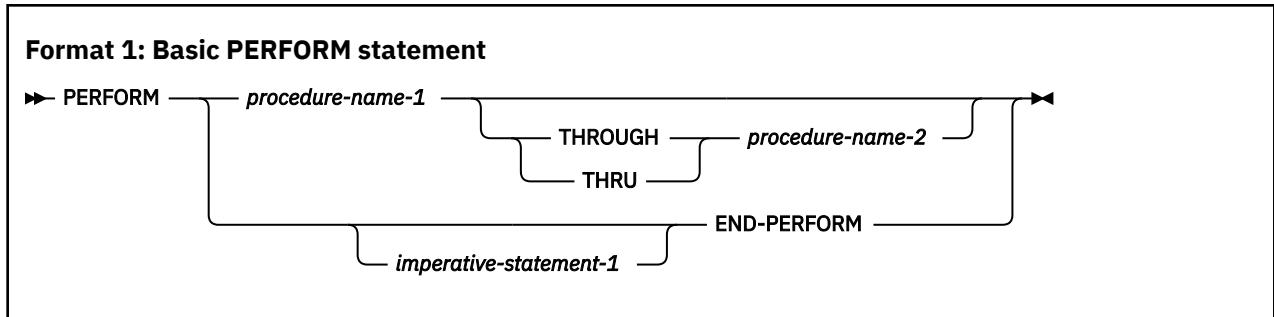
<i>Table 55. Permissible statements for line-sequential files</i>				
Statement	Input open mode	Output open mode	I-O open mode	Extend open mode
READ	X			
WRITE		X		X
REWRITE				

1. A file can be opened for INPUT, OUTPUT, I-O, or EXTEND (sequential and line-sequential files only) in the same program. After the first OPEN statement execution for a given file, each subsequent OPEN statement execution must be preceded by a successful CLOSE file statement execution without the REEL or UNIT phrase (for QSAM files only), or the LOCK phrase.
2. If the FILE STATUS clause is specified in the file-control entry, the associated file status key is updated when the OPEN statement is executed.
3. If an OPEN statement is executed for a file that is already open, the EXCEPTION/ERROR procedure (if specified) for this file is run.

PERFORM statement

The PERFORM statement transfers control explicitly to one or more procedures and implicitly returns control to the next executable statement after execution of the specified procedures is completed. The

PERFORM statement is also used to control execution of one or more imperative statements that are within the scope of that PERFORM statement.



procedure-name-1* , *procedure-name-2

Must name a section or paragraph in the procedure division.

When both *procedure-name-1* and *procedure-name-2* are specified, if either is a procedure-name in a declarative procedure, both must be procedure-names in the same declarative procedure.

If *procedure-name-1* is specified, *imperative-statement-1* and the END-PERFORM phrase must not be specified.

If *procedure-name-1* is omitted, *imperative-statement-1* and the END-PERFORM phrase must be specified.

imperative-statement-1

The statements to be executed for an in-line PERFORM

Inline and out-of-line PERFORM statements

The PERFORM statement is an inline PERFORM statement, when *procedure-name-1* is omitted.

The PERFORM statement is an out-of-line PERFORM statement, when *procedure-name-1* is specified.

An inline PERFORM must be delimited by the END-PERFORM phrase.

The inline and out-of-line formats cannot be combined. For example, if *procedure-name-1* is specified, imperative statements and the END-PERFORM phrase must not be specified.

You can use the EXIT PERFORM statement to exit from an inline PERFORM without using a GO TO statement or a PERFORM ... THROUGH statement. For details, see [“Format 5 \(inline-perform\)” on page 344](#).

You can use the INLINE directive to decide whether a procedure referenced by PERFORM statements is eligible for inlining. For details, see [“INLINE” on page 718](#).

END-PERFORM

Delimits the scope of the in-line PERFORM statement. Execution of an in-line PERFORM is completed after the last statement contained within it has been executed.

Basic PERFORM statement

The procedures referenced in the basic PERFORM statement are executed once, and control then passes to the next executable statement following the PERFORM statement.



Attention: A PERFORM statement must not cause itself to be executed. This constitutes a recursive PERFORM, which can cause unpredictable results. Therefore, you must not specify recursive PERFORM statements.

An in-line PERFORM statement functions according to the same general rules as an otherwise identical out-of-line PERFORM statement, except that statements contained within the in-line PERFORM are executed in place of the statements contained within the range of *procedure-name-1* (through *procedure-name-2*, if specified). Unless specifically qualified by the word *in-line* or the word *out-of-line*, all the rules that apply to the out-of-line PERFORM statement also apply to the in-line PERFORM.

Whenever an out-of-line PERFORM statement is executed, control is transferred to the first statement of the procedure named *procedure-name-1*. Control is always returned to the statement following the PERFORM statement. The point from which this control is returned is determined as follows:

- If *procedure-name-1* is a paragraph name and *procedure-name-2* is not specified, the return is made after the execution of the last statement of the *procedure-name-1* paragraph.
- If *procedure-name-1* is a section name and *procedure-name-2* is not specified, the return is made after the execution of the last statement of the last paragraph in the *procedure-name-1* section.
- If *procedure-name-2* is specified and it is a paragraph name, the return is made after the execution of the last statement of the *procedure-name-2* paragraph.
- If *procedure-name-2* is specified and it is a section name, the return is made after the execution of the last statement of the last paragraph in the *procedure-name-2* section.

The only necessary relationship between *procedure-name-1* and *procedure-name-2* is that a consecutive sequence of operations is executed, beginning at the procedure named by *procedure-name-1* and ending with the execution of the procedure named by *procedure-name-2*.

PERFORM statements can be specified within the performed procedure. If there are two or more logical paths to the return point, then *procedure-name-2* can name a paragraph that consists only of an EXIT statement; all the paths to the return point must then lead to this paragraph.

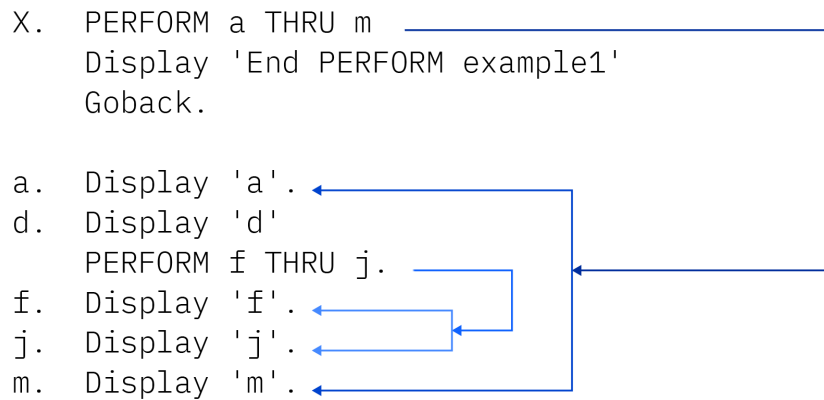
When the performed procedures include another PERFORM statement, the sequence of procedures associated with the embedded PERFORM statement must be totally included in or totally excluded from the performed procedures of the first PERFORM statement. That is, an active PERFORM statement whose execution point begins within the range of performed procedures of another active PERFORM statement must not allow control to pass through the exit point of the other active PERFORM statement. However, two or more active PERFORM statements can have a common exit.

When control passes to the sequence of procedures by means other than a PERFORM statement, control passes through the exit point to the next executable statement, as if no PERFORM statement referred to these procedures.

The following figures illustrate valid sequences of execution for PERFORM statements.

Example 1

Display 'Start PERFORM example1'.

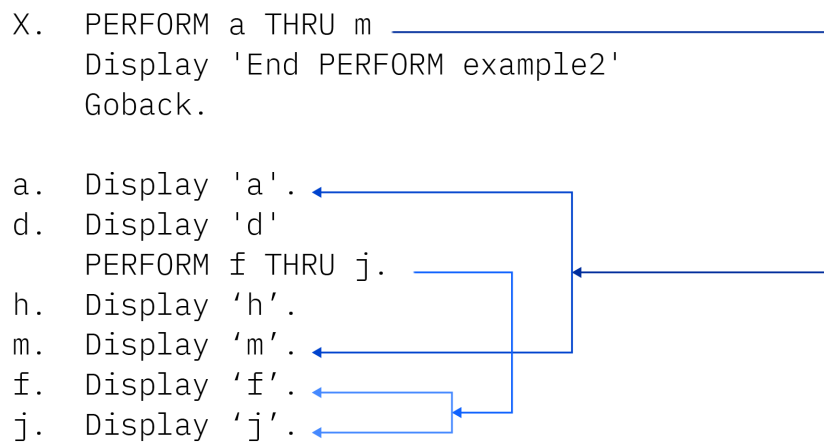


The output for example 1 looks like this:

```
OUTPUT:
Start PERFORM example1
a
d
f
j
j
f
j
m
End PERFORM example1
```

Example 2

Display 'Start PERFORM example2'.



The output for example 2 looks like this:

```
OUTPUT:
Start PERFORM example2
a
d
f
```



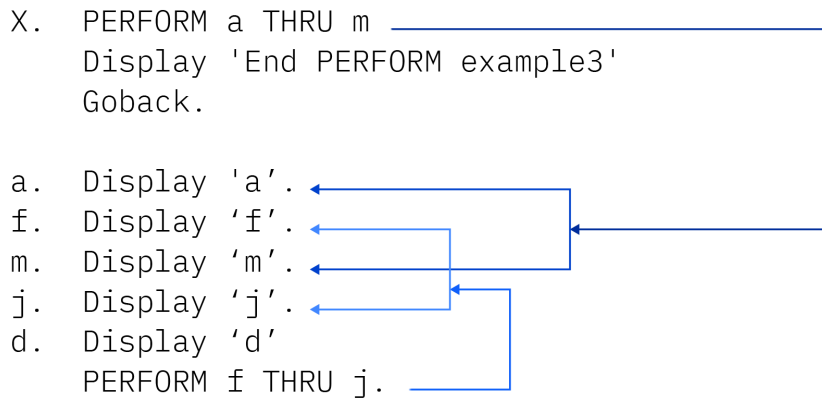
```

j
h
m
End   PERFORM example2

```

Example 3

Display 'Start PERFORM example3'.



The output of example 3 looks like this:

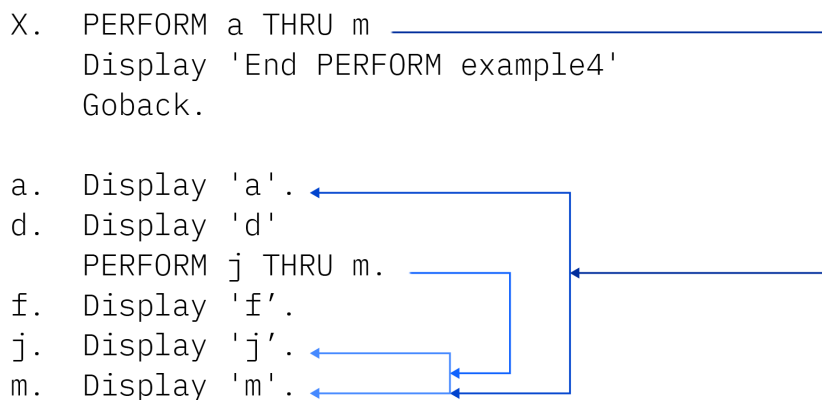
```

OUTPUT:
Start PERFORM example3
a
f
m
End   PERFORM example3

```

Example 4

Display 'Start PERFORM example4'.



The output for example 4 looks like this:

```

OUTPUT:

```

```

Start PERFORM example4
a
d
j
m
f
j
m
End   PERFORM example4

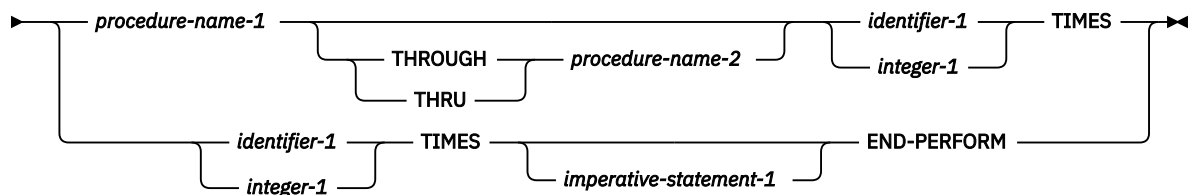
```

PERFORM with TIMES phrase

The procedures referred to in the TIMES phrase of the PERFORM statement are executed the number of times specified by the value in *identifier-1* or *integer-1*, up to a maximum of 999,999,999 times. Control then passes to the next executable statement following the PERFORM statement.

Format 2: PERFORM statement with TIMES phrase

►► PERFORM ►



If *procedure-name-1* is specified, *imperative-statement-1* and the END-PERFORM phrase must not be specified.

identifier-1

Must name an integer item.

If *identifier-1* is zero or a negative number at the time the PERFORM statement is initiated, control passes to the statement following the PERFORM statement.

After the PERFORM statement has been initiated, any change to *identifier-1* has no effect in varying the number of times the procedures are initiated.

integer-1

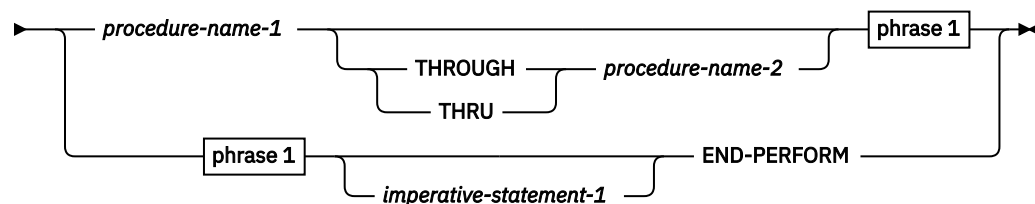
Can be a positive signed integer.

PERFORM with UNTIL phrase

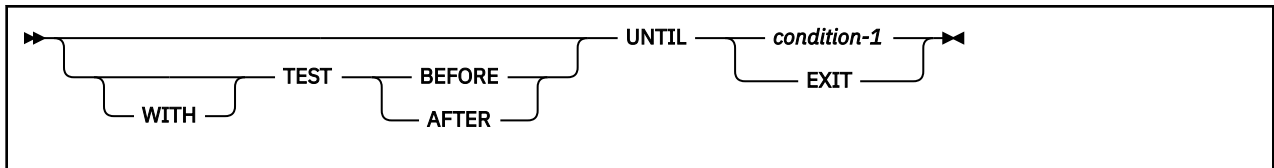
In the UNTIL phrase format, the procedures referred to are performed *until* the condition specified by the UNTIL phrase is true or for UNTIL EXIT, whenever control is escaped to avoid an infinite loop. For UNTIL with *condition-1*, control is then passed to the next executable statement following the PERFORM statement.

Format 3: PERFORM statement with UNTIL phrase

►► PERFORM ►



phrase 1



If *procedure-name-1* is specified, *imperative-statement-1* and the END-PERFORM phrase must not be specified.

condition-1

Can be any condition described under “Conditional expressions” on page 268. If the condition is true at the time the PERFORM statement is initiated, the specified procedures are not executed.

Any subscripting associated with the operands specified in *condition-1* is evaluated each time the condition is tested.

EXIT

If the UNTIL phrase with the EXIT reserved word is specified, execution proceeds exactly as if the same PERFORM statement were coded with *condition-1* specified, except that *condition-1* never evaluates as true.

Note: When UNTIL EXIT is specified, ensure that an escape from the PERFORM loop will be reached. For an inline PERFORM statement, this can be done by an EXIT PERFORM (but not EXIT PERFORM CYCLE) statement. For an out-of-line PERFORM statement, this can be done by a GOBACK or STOP statement. Make sure the escape statement used does escape the PERFORM loop. Several statements might appear to do so, but don't actually escape the loop. For example, an EXIT PARAGRAPH (from a performed paragraph) or an EXIT SECTION (from a performed section) do not escape a PERFORM with the UNTIL EXIT phrase.

If the TEST BEFORE phrase is specified or assumed, the condition is tested before any statements are executed.

If the TEST AFTER phrase is specified, the statements to be performed are executed at least once before the condition is tested.

In either case, if the condition is true, control is transferred to the next executable statement following the end of the PERFORM statement. If neither the TEST BEFORE nor the TEST AFTER phrase is specified, the TEST BEFORE phrase is assumed.

The UNTIL EXIT phrase must not be specified with the TEST BEFORE or TEST AFTER phrase nor with the PERFORM with VARYING phrase.

PERFORM with VARYING phrase

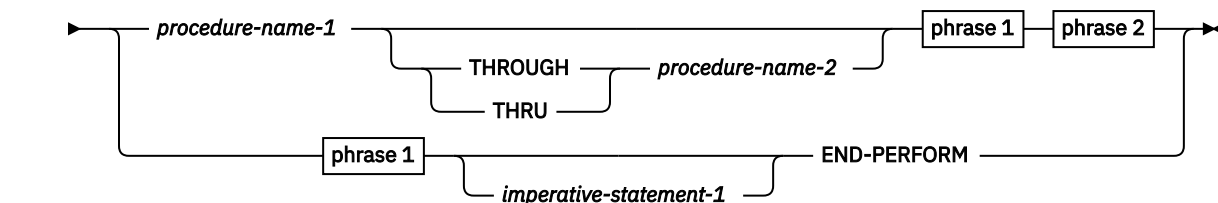
The VARYING phrase increases or decreases the value of one or more identifiers or index-names, according to certain rules.

For more information, see “Varying phrase rules” on page 428.

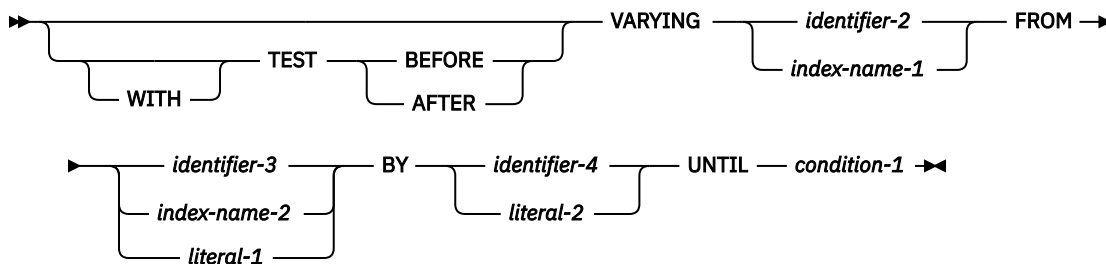
The format-4 VARYING phrase PERFORM statement can serially search an entire seven-dimensional table.

Format 4: PERFORM statement with VARYING phrase

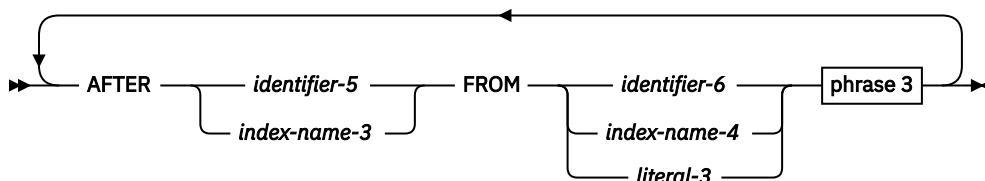
➤➤ PERFORM ➔



phrase 1



phrase 2



phrase 3



If *procedure-name-1* is specified, *imperative-statement-1* and the END-PERFORM phrase must not be specified. If *procedure-name-1* is omitted, the AFTER phrase must not be specified.

identifier-2 through identifier-7

Must name a numeric elementary item.

literal-1 through literal-4

Must represent a numeric literal.

condition-1, condition-2

Can be any condition described under “Conditional expressions” on page 268. If the condition is true at the time the PERFORM statement is initiated, the specified procedures are not executed.

After the conditions specified in the UNTIL phrase are satisfied, control is passed to the next executable statement following the PERFORM statement.

If any of the operands specified in *condition-1* or *condition-2* is subscripted, reference modified, or is a function-identifier, the subscript, reference-modifier, or function is evaluated each time the condition is tested.

Floating-point data items and literals can be used anywhere a numeric data item or literal can be specified.

When TEST BEFORE is indicated, all specified conditions are tested before the first execution, and the statements to be performed are executed, if at all, only when *all* specified tests fail. When TEST AFTER is indicated, the statements to be performed are executed at least once, before any condition is tested.

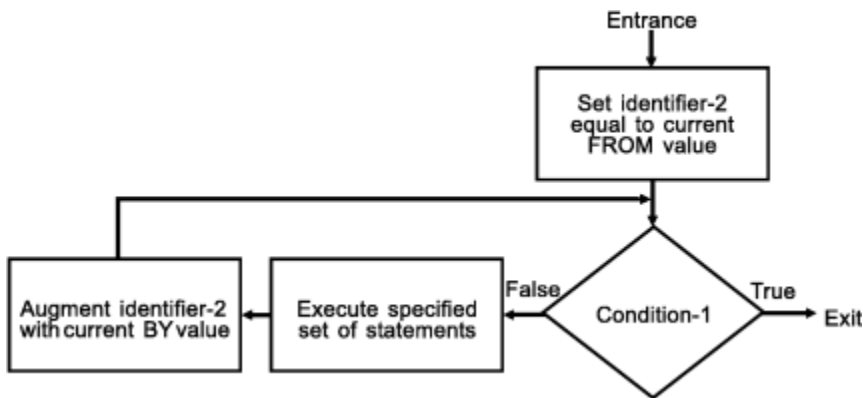
If neither the TEST BEFORE nor the TEST AFTER phrase is specified, the TEST BEFORE phrase is assumed.

Varying identifiers

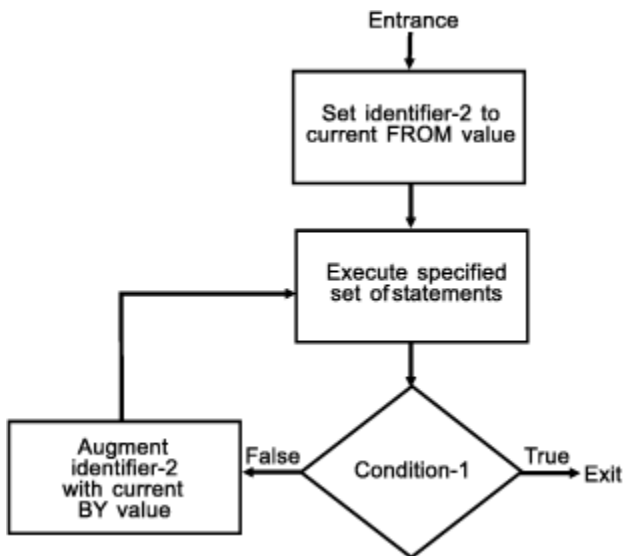
The way in which operands are increased or decreased depends on the number of variables specified. In the discussion, every reference to *identifier-n* refers equally to *index-name-n* (except when *identifier-n* is the object of the BY phrase).

If *identifier-2* or *identifier-5* is subscripted, the subscripts are evaluated each time the content of the data item referenced by the identifier is set or augmented. If *identifier-3*, *identifier-4*, *identifier-6*, or *identifier-7* is subscripted, the subscripts are evaluated each time the content of the data item referenced by the identifier is used in a setting or an augmenting operation.

The following figure illustrates the logic of the PERFORM statement when an identifier is varied with TEST BEFORE.



The following figure illustrates the logic of the PERFORM statement when an identifier is varied with TEST AFTER.



Varying two identifiers

The topic lists steps of varying two identifiers.

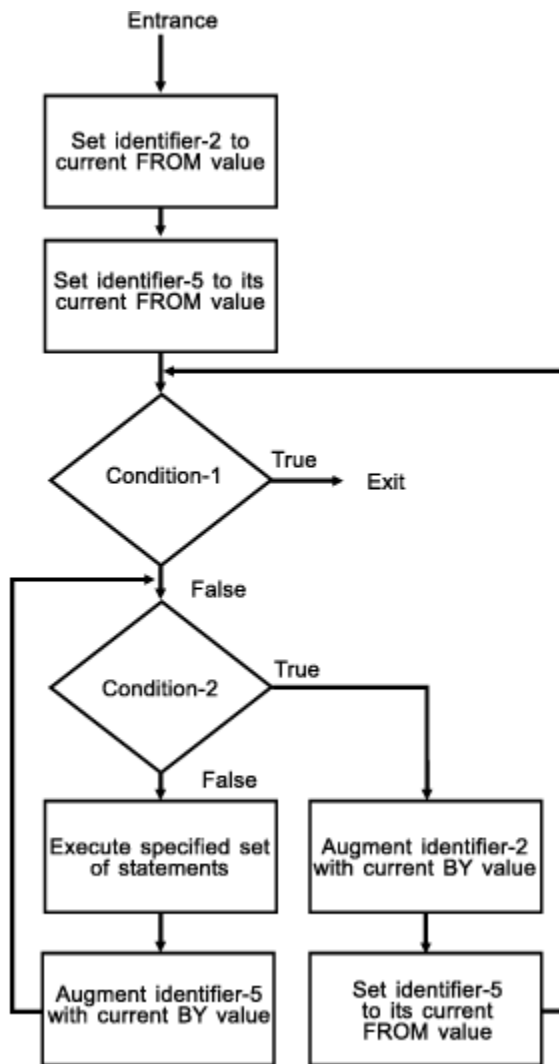
```
PERFORM PROCEDURE-NAME-1 THROUGH PROCEDURE-NAME-2
  VARYING IDENTIFIER-2 FROM IDENTIFIER-3
  BY IDENTIFIER-4 UNTIL CONDITION-1
  AFTER IDENTIFIER-5 FROM IDENTIFIER-6
  BY IDENTIFIER-7 UNTIL CONDITION-2
```

1. *identifier-2* and *identifier-5* are set to their initial values, *identifier-3* and *identifier-6*, respectively.
2. *condition-1* is evaluated as follows:
 - a. If it is false, steps 3 through 7 are executed.
 - b. If it is true, control passes directly to the statement following the PERFORM statement.
3. *condition-2* is evaluated as follows:
 - a. If it is false, steps 4 through 6 are executed.
 - b. If it is true, *identifier-2* is augmented by *identifier-4*, *identifier-5* is set to the current value of *identifier-6*, and step 2 is repeated.
4. *procedure-name-1* and *procedure-name-2* are executed once (if specified).
5. *identifier-5* is augmented by *identifier-7*.
6. Steps 3 through 5 are repeated until *condition-2* is true.
7. Steps 2 through 6 are repeated until *condition-1* is true.

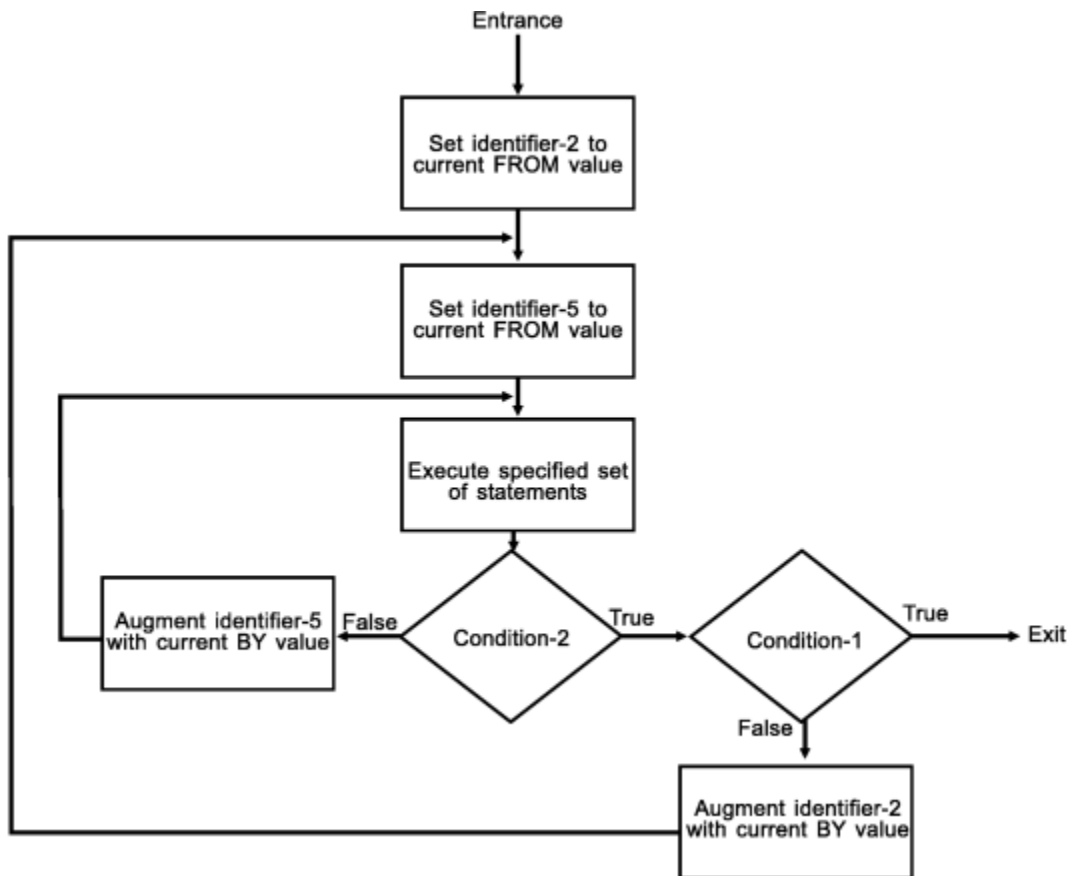
At the end of PERFORM statement execution:

- *identifier-5* contains the current value of *identifier-6*.
- *identifier-2* has a value that exceeds the last-used setting by the increment or decrement value (unless *condition-1* was true at the beginning of PERFORM statement execution, in which case, *identifier-2* contains the current value of *identifier-3*).

The following figure illustrates the logic of the PERFORM statement when two identifiers are varied with TEST BEFORE.



The following figure illustrates the logic of the PERFORM statement when two identifiers are varied with TEST AFTER.



Varying three identifiers

The topic lists steps of varying three identifiers.

```

PERFORM PROCEDURE-NAME-1 THROUGH PROCEDURE-NAME-2
  VARYING IDENTIFIER-2 FROM IDENTIFIER-3
    BY IDENTIFIER-4 UNTIL CONDITION-1
  AFTER IDENTIFIER-5 FROM IDENTIFIER-6
    BY IDENTIFIER-7 UNTIL CONDITION-2
  AFTER IDENTIFIER-8 FROM IDENTIFIER-9
    BY IDENTIFIER-10 UNTIL CONDITION-3
  
```

The actions are the same as those for two identifiers, except that *identifier-8* goes through the complete cycle each time that *identifier-5* is augmented by *identifier-7*, which, in turn, goes through a complete cycle each time that *identifier-2* is varied.

At the end of PERFORM statement execution:

- *identifier-5* and *identifier-8* contain the current values of *identifier-6* and *identifier-9*, respectively.
- *identifier-2* has a value exceeding its last-used setting by one increment/decrement value (unless *condition-1* was true at the beginning of PERFORM statement execution, in which case *identifier-2* contains the current value of *identifier-3*).

Varying more than three identifiers

You can produce analogous PERFORM statement actions to the previous example with the addition of up to four AFTER phrases.

Varying phrase rules

There are certain rules that apply to this phrase, no matter how many variables are specified.

The rules are:

- In the VARYING or AFTER phrases, when an index-name is specified:

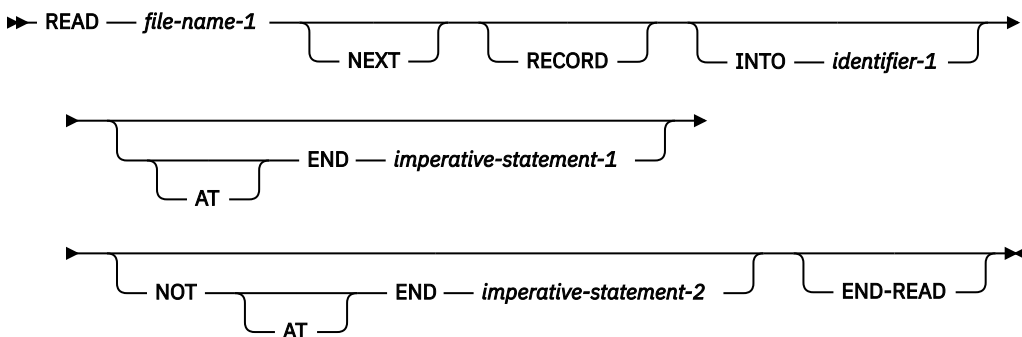
- The index-name is initialized and incremented or decremented according to the rules under “INDEX phrase” on page 241. (See also “SET statement” on page 445.)
- In the associated FROM phrase, an identifier must be described as an integer and have a positive value; a literal must be a positive integer.
- In the associated BY phrase, an identifier must be described as an integer; a literal must be a nonzero integer.
- In the FROM phrase, when an index-name is specified:
 - In the associated VARYING or AFTER phrase, an identifier must be described as an integer. It is initialized as described in the SET statement.
 - In the associated BY phrase, an identifier must be described as an integer and have a nonzero value; a literal must be a nonzero integer.
- In the BY phrase, identifiers and literals must have nonzero values.
- Changing the values of identifiers or index-names in the VARYING, FROM, and BY phrases during execution changes the number of times the procedures are executed.

READ statement

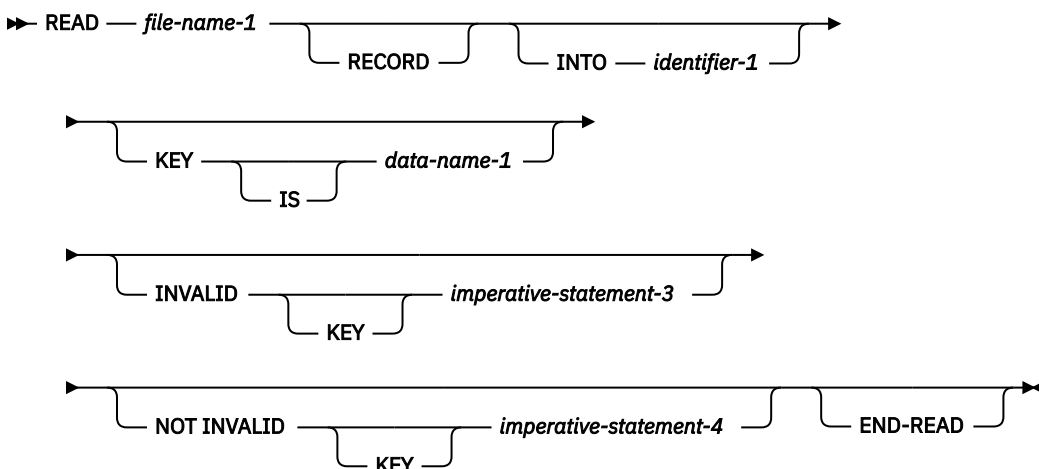
For sequential access, the READ statement makes the next logical record from a file available to the object program. For random access, the READ statement makes a specified record from a direct-access file available to the object program.

When the READ statement is executed, the associated file must be open in INPUT or I-O mode.

Format 1: READ statement for sequential retrieval



Format 2: READ statement for random retrieval



file-name-1

Must be defined in a DATA DIVISION FD entry.

NEXT RECORD

Reads the next record in the logical sequence of records. NEXT is optional when the access mode is sequential, and has no effect on READ statement execution.

You must specify the NEXT RECORD phrase to retrieve records sequentially from files in dynamic access mode.

INTO *identifier-1*

identifier-1 is the receiving field.

identifier-1 must be a valid receiving field for the selected sending record description entry in accordance with the rules of the MOVE statement.

The record areas associated with *file-name-1* and *identifier-1* must not be the same storage area.

When there is only one record description associated with *file-name-1* or all the records and the data item referenced by *identifier-1* describe an elementary alphanumeric item or an alphanumeric group item, the result of the execution of a READ statement with the INTO phrase is equivalent to the application of the following rules in the order specified:

- The execution of the same READ statement without the INTO phrase.
- The current record is moved from the record area to the area specified by *identifier-1* according to the rules for the MOVE statement without the CORRESPONDING phrase. The size of the current record is determined by rules specified for the RECORD clause. If the file description entry contains a RECORD IS VARYING clause, the implied move is a group move. The implied MOVE statement does not occur if the execution of the READ statement was unsuccessful. Any subscripting or reference modification associated with *identifier-1* is evaluated after the record has been read and immediately before it is moved to the data item. The record is available in both the record area and the data item referenced by *identifier-1*.

identifier-1 must not be a dynamic-length group item or a dynamic-length elementary item.

When there are multiple record descriptions associated with *file-name-1* and they do not all describe an alphanumeric group item or elementary alphanumeric item, the following rules apply:

1. If the file referenced by *file-name-1* is described as containing variable-length records, or as a QSAM file with RECORDING MODE 'S' or 'U', a group move will take place.
2. If the file referenced by *file-name-1* is described as containing fixed-length records, a move will take place according to the rules for a MOVE statement using, as a sending field description, the record that specifies the largest number of character positions. If more than one such record exists, the sending field record selected will be the one among those records that appears first under the description of *file-name-1*.

KEY IS phrase

The KEY IS phrase can be specified only for indexed files. *data-name-1* must identify a record key associated with *file-name-1*. *data-name-1* can be qualified; it cannot be subscripted.

AT END phrases

For sequential access, both the AT END phrase and an applicable EXCEPTION/ERROR procedure can be omitted.

For information about at-end condition processing, see [AT END condition](#).

INVALID KEY phrases

Both the INVALID KEY phrase and an applicable EXCEPTION/ERROR procedure can be omitted.

For information about INVALID KEY phrase processing, see [“Invalid key condition”](#) on page 303.

END-READ phrase

This explicit scope terminator serves to delimit the scope of the READ statement. END-READ permits a conditional READ statement to be nested in another conditional statement. END-READ can also be used with an imperative READ statement. For more information, see [“Delimited scope statements”](#) on page 293.

Processing files with variable-length records or multiple record descriptions

If more than one record description entry is associated with *file-name-1*, those records automatically share the same storage area; that is, they are implicitly redefined. After a READ statement is executed, only those data items within the range of the current record are replaced; data items stored beyond that range are undefined. The following example illustrates this concept.

If the length of the current record that is read is less than the minimum size specified by the record description entries for *file-name-1*, the portion of the record area which is to the right of the last valid character read is undefined. If the length of the current record exceeds the record description entries for *file-name-1*, the record is truncated on the right to the maximum record definition size.

In either of the previous cases, the READ statement is successful, and the I-O status is set to either 00 (hiding the record length conflict condition) or 04 (indicating that a record length conflict has occurred), depending on the VLR compiler option setting. If compiler option VLR(COMPAT) is in effect, the I-O status would be set to 00.

For more information about the VLR compiler option, see *VLR* in the *Enterprise COBOL Programming Guide*.

The following example shows two record areas of different sizes in an FD. When a shorter record is read, the content of the remaining record area is undefined.

```
FD INPUT-FILE LABEL RECORD OMITTED.  
01   RECORD-1 PICTURE X(30).  
01   RECORD-2 PICTURE X(20).
```

Content of input area when READ statement is executed:

```
ABCDEFGHIJKLMNOPQRSTUVWXYZ1234
```

Content of record being read in (RECORD-2):

```
01234567890123456789
```

Content of input area after READ statement is executed:

```
01234567890123456789??????????
```

The "?" characters are undefined characters in input area.

Sequential access mode

Format 1 must be used for all files in sequential access mode.

Execution of a format-1 READ statement retrieves the next logical record from the file. The next record accessed is determined by the file organization.

Sequential files

The NEXT RECORD is the next record in a logical sequence of records. The NEXT phrase need not be specified; it has no effect on READ statement execution.

If SELECT OPTIONAL is specified in the file-control entry for this file, and the file is unavailable during this execution of the object program, execution of the first READ statement causes an at-end condition; however, since no file is available, the system-defined end-of-file processing is not performed.

AT END condition

If the file position indicator indicates that no next logical record exists, or that an optional input file is not available, at-end condition processing occurs in a specific order.

The order is:

1. A value derived from the setting of the file position indicator is placed into the I-O status associated with *file-name-1* to indicate the at-end condition.
2. If the AT END phrase is specified in the statement causing the condition, control is transferred to *imperative-statement-1* in the AT END phrase. Any USE AFTER STANDARD EXCEPTION procedure associated with *file-name-1* is not executed.
3. If the AT END phrase is not specified and an applicable USE AFTER STANDARD EXCEPTION procedure exists, the procedure is executed. Return from that procedure is to the next executable statement following the end of the READ statement.

Both the AT END phrase and an applicable EXCEPTION/ERROR procedure can be omitted.

When the at-end condition occurs, execution of the READ statement is unsuccessful. The contents of the associated record area are undefined and the file position indicator is set to indicate that no valid next record has been established.

For QSAM files, attempts to access or move data into the record area after an unsuccessful read can result in a protection exception.

If an at-end condition does not occur during the execution of a READ statement, the AT END phrase is ignored, if specified, and the following actions occur:

1. The file position indicator is set and the I-O status associated with *file-name-1* is updated.
2. If an exception condition that is not an at-end condition exists, control is transferred to the end of the READ statement after the execution of any USE AFTER STANDARD EXCEPTION procedure applicable to *file-name-1*.

If no USE AFTER STANDARD EXCEPTION procedure is specified, control is transferred to the end of the READ statement or to *imperative-statement-2*, if specified.

3. If no exception condition exists, the record is made available in the record area and any implicit move resulting from the presence of an INTO phrase is executed. Control is transferred to the end of the READ statement or to *imperative-statement-2*, if specified. In the latter case, execution continues according to the rules for each statement specified in *imperative-statement-2*. If a procedure branching or conditional statement which causes explicit transfer of control is executed, control is transferred in accordance with the rules for that statement; otherwise, upon completion of the execution of *imperative-statement-2*, control is transferred to the end of the READ statement.

After the unsuccessful execution of a READ statement, the contents of the associated record area are undefined and the file position indicator is set to indicate that no valid next record has been established. Attempts to access or move data into the record area after an unsuccessful read can result in a protection exception.

Indexed or relative files

The NEXT RECORD is the next logical record in the key sequence.

For indexed files, the key sequence is the sequence of ascending values of the current key of reference. For relative files, the key sequence is the sequence of ascending values of relative record numbers for records that exist in the file.

Before the READ statement is executed, the file position indicator must have been set by a successful OPEN, START, or READ statement. When the READ statement is executed, the record indicated by the file position indicator is made available if it is still accessible through the path indicated by the file position indicator.

If the record is no longer accessible (because it has been deleted, for example), the file position indicator is updated to point to the next existing record in the file, and that record is made available.

For files in sequential access mode, the NEXT phrase need not be specified.

For files in dynamic access mode, the NEXT phrase must be specified for sequential record retrieval.

AT END condition

This condition exists when the file position indicator indicates that no next logical record exists or that an optional input file is not available. The same procedure occurs as for sequential files (see [AT END condition](#)).

If neither an at-end nor an invalid key condition occurs during the execution of a READ statement, the AT END or the INVALID KEY phrase is ignored, if specified. The same actions occur as when the at-end condition does not occur with sequential files (see [AT END condition](#)).

Sequentially accessed indexed files

When an ALTERNATE RECORD KEY with DUPLICATES is the key of reference, file records with duplicate key values are made available in the order in which they were placed in the file.

Sequentially accessed relative files

If the RELATIVE KEY clause is specified for this file, READ statement execution updates the RELATIVE KEY data item to indicate the relative record number of the record being made available.

Random access mode

Format 2 must be specified for indexed and relative files in random access mode, and also for files in the dynamic access mode when record retrieval is random.

Execution of the READ statement depends on the file organization, as explained in the following sections.

Indexed files

Execution of a format-2 READ statement causes the value of the key of reference to be compared with the value of the corresponding key data item in the file records, until the first record having an equal value is found. The file position indicator is positioned to this record, which is then made available. If no record can be so identified, an INVALID KEY condition exists, and READ statement execution is unsuccessful. (See [“Invalid key condition”](#) on page 303 for details of the invalid key condition.)

If the KEY phrase is not specified, the prime RECORD KEY becomes the key of reference for this request. When dynamic access is specified, the prime RECORD KEY is also used as the key of reference for subsequent executions of sequential READ statements, until a different key of reference is established.

When the KEY phrase is specified, *data-name-1* becomes the key of reference for this request. When dynamic access is specified, this key of reference is used for subsequent executions of sequential READ statements, until a different key of reference is established.

Relative files

Execution of a format-2 READ statement sets the file position indicator pointer to the record whose relative record number is contained in the RELATIVE KEY data item, and makes that record available.

If the file does not contain such a record, the INVALID KEY condition exists, and READ statement execution is unsuccessful. (See “Invalid key condition” on page 303 for details of the invalid key condition).

The KEY phrase must not be specified for relative files.

Dynamic access mode

For files with indexed or relative organization, dynamic access mode can be specified in the file-control entry. In dynamic access mode, either sequential or random record retrieval can be used, depending on the format used.

Format 1 with the NEXT phrase must be specified for sequential retrieval. All other rules for sequential access apply.

READ statement notes

This topic provides notes on the READ statement.

- If the FILE-STATUS clause is specified in the file-control entry, the associated file status key is updated when the READ statement is executed.
- After unsuccessful READ statement execution, the contents of the associated record area and the value of the file position indicator are undefined. Attempts to access or move data into the record area after an unsuccessful read can result in a protection exception.
- If the number of character positions in a record is less than the minimum size specified by the record description entries for *file-name-1*, after a READ statement is executed, the portion of the record area that is to the right of the last valid character read is undefined. If the number of character positions in a record exceed the maximum size specified by the record description entries for *file-name-1*, after a READ statement is executed, the record is truncated on the right to the maximum size.

In either of these cases, the READ statement is successful and the I-O status is set to either 00 (hiding the record length conflict condition) or 04 (indicating that a record length conflict has occurred), depending on the VLR compiler option setting.

- When the VLR(COMPAT) compiler option is in effect, the status value of 00 is set.
- When the VLR(STANDARD) compiler option is in effect, the status value of 04 is set.

For more information about the VLR compiler option, see VLR in the *Enterprise COBOL Programming Guide*.

RELEASE statement

The RELEASE statement transfers records from an input/output area to the initial phase of a sorting operation.

The RELEASE statement can be used only within the range of an INPUT PROCEDURE associated with a SORT statement.

Format: RELEASE

►► RELEASE — *record-name-1* — FROM — *identifier-1* —►

Within an INPUT PROCEDURE, at least one RELEASE statement must be specified.

When the RELEASE statement is executed, the current contents of *record-name-1* are placed in the sort file. This makes the record available to the initial phase of the sorting operation.

record-name-1

Must specify the name of a logical record in a sort-merge file description entry (SD). *record-name-1* can be qualified.

FROM phrase

The result of the execution of the RELEASE statement with the FROM *identifier-1* phrase is equivalent to the execution of the following statements in the order specified.

```
MOVE identifier-1 to record-name-1.  
RELEASE record-name-1.
```

The MOVE is performed according to the rules for the MOVE statement without the CORRESPONDING phrase.

identifier-1

identifier-1 must reference one of the following items:

- An entry in the WORKING-STORAGE SECTION, the LOCAL-STORAGE SECTION, or the LINKAGE SECTION
- A record description for another previously opened file
- An alphanumeric or national function.

identifier-1 must be a valid sending item with *record-name-1* as the receiving item in accordance with the rules of the MOVE statement.

identifier-1 and *record-name-1* must not refer to the same storage area.

identifier-1 must not be a dynamic-length group item or a dynamic-length elementary item.

After the RELEASE statement is executed, the information is still available in *identifier-1*. (See [“INTO and FROM phrases”](#) on page 304 under “Common processing facilities”).

If the RELEASE statement is executed without specifying the SD entry for *file-name-1* in a SAME RECORD AREA clause, the information in *record-name-1* is no longer available.

If the SD entry is specified in a SAME RECORD AREA clause, *record-name-1* is still available as a record of the other files named in that clause.

When FROM *identifier-1* is specified, the information is still available in *identifier-1*.

When control passes from the INPUT PROCEDURE, the sort file consists of all those records placed in it by execution of RELEASE statements.

Restriction: If a RELEASE statement appears in a nested program, it cannot be for a SORT statement in a program that contains the nested program.

RETURN statement

The RETURN statement transfers records from the final phase of a sorting or merging operation to an OUTPUT PROCEDURE.

The RETURN statement can be used only within the range of an OUTPUT PROCEDURE associated with a SORT or MERGE statement.

Diagram illustrating the structure of a RETURN statement in a programming language. The diagram shows three horizontal timelines representing the execution flow.

The first timeline starts with a double arrow pointing to **RETURN**, followed by a horizontal line labeled *file-name-1*. Below this line, two vertical lines mark the start and end of a block labeled **RECORD**. Further right, another vertical line marks the start of a block labeled **INTO**, followed by a horizontal line labeled *identifier-1*.

The second timeline starts with a single arrow pointing to a horizontal line, followed by a vertical line marking the start of a block labeled **END**, which is followed by the text *imperative-statement-1*. Below this line, a vertical line marks the start of a block labeled **AT**.

The third timeline starts with a single arrow pointing to a horizontal line, followed by a vertical line marking the start of a block labeled **NOT**. Below this line, a vertical line marks the start of a block labeled **END**, which is followed by the text *imperative-statement-2*. Further right, a vertical line marks the start of a block labeled **END-RETURN**.

When the RETURN statement is executed, the next record from *file-name-1* is made available for processing by the OUTPUT PROCEDURE.

file-name-1

If more than one record description is associated with *file-name-1*, those records automatically share the same storage; that is, the area is implicitly redefined. After RETURN statement execution, only the contents of the current record are available. If any data items lie beyond the length of the current record, their contents are undefined.

When there is only one record description associated with *file-name-1* or all the records and the data item referenced by *identifier-1* describe an elementary alphanumeric item or an alphanumeric group item, the result of the execution of a RETURN statement with the INTO phrase is equivalent to the application of the following rules in the order specified:

- When there are multiple record descriptions associated with *file-name-1* and they do not all describe an alphanumeric group item or elementary alphanumeric item, the following rules apply:

- The record areas associated with *file-name-1* and *identifier-1* must not be the same storage area.

AT END phrases

The imperative-statement specified on the AT END phrase executes after all records have been returned from *file-name-1*. No more RETURN statements can be executed as part of the current output procedure.

If an at-end condition does not occur during the execution of a RETURN statement, then after the record is made available and after executing any implicit move resulting from the presence of an INTO phrase, control is transferred to the imperative statement specified by the NOT AT END phrase. If an at-end condition does occur, control is transferred to the end of the RETURN statement.

END-RETURN phrase

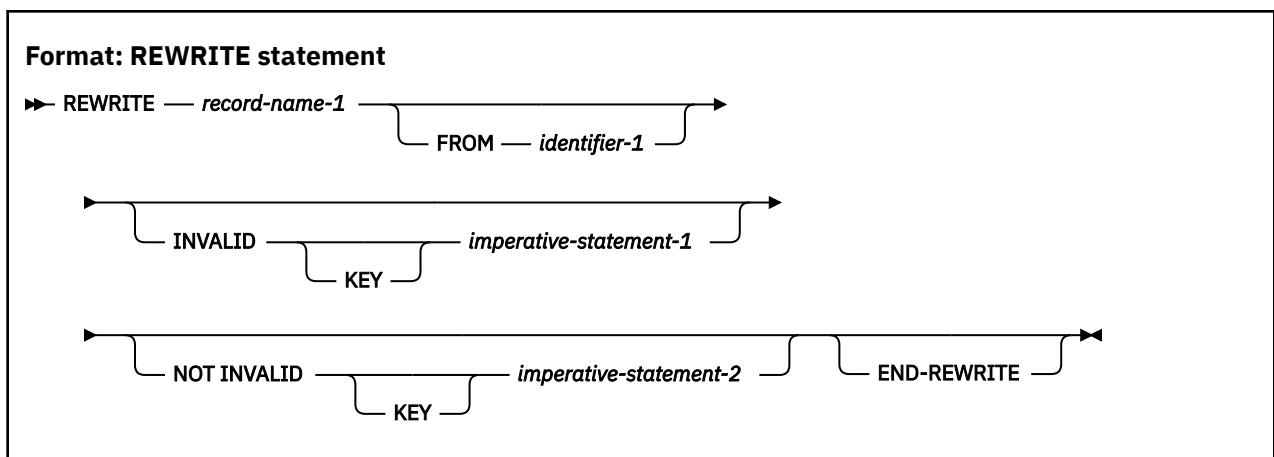
This explicit scope terminator serves to delimit the scope of the RETURN statement. END-RETURN permits a conditional RETURN statement to be nested in another conditional statement. END-RETURN can also be used with an imperative RETURN statement.

For more information, see [“Delimited scope statements” on page 293](#).

REWRITE statement

The REWRITE statement logically replaces an existing record in a direct-access file. When the REWRITE statement is executed, the associated direct-access file must be open in I-O mode.

The REWRITE statement is not supported for line-sequential files.



record-name-1

Must be the name of a logical record in a DATA DIVISION FD entry. The record-name can be qualified.

FROM phrase

The result of the execution of the REWRITE statement with the FROM *identifier-1* phrase is equivalent to the execution of the following statements in the order specified.

```
MOVE identifier-1 TO record-name-1.
REWRITE record-name-1
```

The MOVE is performed according to the rules for the MOVE statement without the CORRESPONDING phrase.

identifier-1

identifier-1 can reference one of the following items:

- A record description for another previously opened file
- An alphanumeric or national function
- A data item defined in the WORKING-STORAGE SECTION, the LOCAL-STORAGE SECTION, or the LINKAGE SECTION

identifier-1 must be a valid sending item with *record-name-1* as the receiving item in accordance with the rules of the MOVE statement.

identifier-1 and *record-name-1* must not refer to the same storage area.

identifier-1 must not be a dynamic-length group item or a dynamic-length elementary item.

After the REWRITE statement is executed, the information is still available in *identifier-1* ([“INTO and FROM phrases”](#) on page 304 under “Common processing facilities”).

INVALID KEY phrases

An INVALID KEY condition exists when:

- The access mode is sequential, and the value contained in the prime RECORD KEY of the record to be replaced does not equal the value of the prime RECORD KEY data item of the last-retrieved record from the file
- The value contained in the prime RECORD KEY does not equal that of any record in the file
- The value of an ALTERNATE RECORD KEY data item for which DUPLICATES is not specified is equal to that of a record already in the file

For details of invalid key processing, see [Invalid key condition](#).

END-REWRITE phrase

This explicit scope terminator serves to delimit the scope of the REWRITE statement. END-REWRITE permits a conditional REWRITE statement to be nested in another conditional statement. END-REWRITE can also be used with an imperative REWRITE statement.

For more information, see [“Delimited scope statements”](#) on page 293.

Reusing a logical record

After successful execution of a REWRITE statement, the logical record is no longer available in *record-name-1* unless the associated file is named in a SAME RECORD AREA clause (in which case, the record is also available as a record of the other files named in the SAME RECORD AREA clause).

The file position indicator is not affected by execution of the REWRITE statement.

If the FILE STATUS clause is specified in the file-control entry, the associated file status key is updated when the REWRITE statement is executed.

Sequential files

For files in the sequential access mode, the last prior input/output statement executed for this file must be a successfully executed READ statement. When the REWRITE statement is executed, the record retrieved by that READ statement is logically replaced.

The number of character positions in *record-name-1* must equal the number of character positions in the record being replaced.

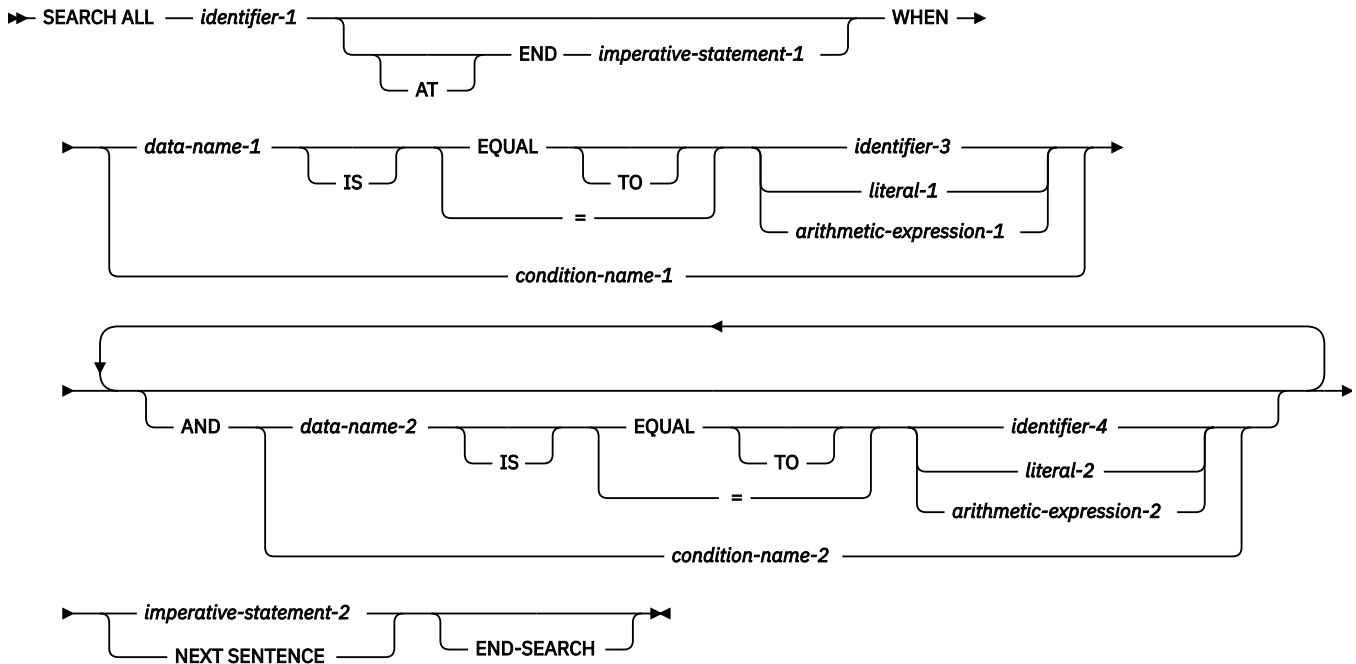
The INVALID KEY phrase must not be specified for a file with sequential organization. An EXCEPTION/ERROR procedure can be specified.

Indexed files

The number of character positions in *record-name-1* can be different from the number of character positions in the record being replaced.

When the access mode is sequential, the record to be replaced is specified by the value contained in the prime RECORD KEY. When the REWRITE statement is executed, this value must equal the value of the prime record key data item in the last record read from this file.

Format 2: SEARCH statement for binary search



Use format 1 (serial search) when the table that you want to search has not been sorted. Use format 1 to search a sorted table when you want to search serially through the table or you want to control subscripts or indexes.

Use format 2 (binary search) when you want to efficiently search across all occurrences in a table. The table must previously have been sorted, and you can sort the table with the format 2 SORT statement.

AT END and WHEN phrases

After *imperative-statement-1* or *imperative-statement-2* is executed, control passes to the end of the SEARCH statement, unless *imperative-statement-1* or *imperative-statement-2* ends with a GO TO statement.

The function of the AT END phrase is the same for a serial search and a binary search.

NEXT SENTENCE

NEXT SENTENCE transfers control to the first statement following the closest separator period.

When NEXT SENTENCE is specified with END-SEARCH, control does not pass to the statement following the END-SEARCH. Instead, control passes to the statement after the closest following period.

For the format-2 SEARCH ALL statement, neither *imperative-statement-2* nor NEXT SENTENCE is required. Without them, the SEARCH statement sets the index to the value in the table that matched the condition.

The function of the NEXT SENTENCE phrase is the same for a serial search and a binary search.

END-SEARCH phrase

This explicit scope terminator delimits the scope of the SEARCH statement. END-SEARCH permits a conditional SEARCH statement to be nested in another conditional statement.

For more information, see [“Delimited scope statements” on page 293](#).

The function of END-SEARCH is the same for a serial search and a binary search.

Serial search

The topic provides information of using the SEARCH statement for serial search.

***identifier-1* (serial search)**

identifier-1 identifies the table that is to be searched. *identifier-1* references all occurrences within that table.

The data description entry for *identifier-1* must contain an OCCURS clause.

The data description entry for *identifier-1* should contain an OCCURS clause with the INDEXED BY phrase, but a table can be searched using an alternate index defined for an appropriately defined alternate table. The element length of the table being searched and the element length of the alternate table to which the alternate index is associated should match. You must initialize the index (defined in the INDEXED BY phrase) for *identifier-1* to a valid value for the table being searched, even when the index of the table is not referenced or used in the serial search.

identifier-1 can reference a data item that is subordinate to a data item that is described with an OCCURS clause (that is, *identifier-1* can be a subordinate table within a multidimensional table). In this case, the data description entries must specify an INDEXED BY phrase for each dimension of the table.

identifier-1 must not be subscripted or reference-modified.

identifier-1 must not be a dynamic-length group item or a dynamic-length elementary item.

AT END

The condition that exists when the search operation terminates without satisfying the condition specified in any of the associated WHEN phrases.

Before executing a serial search, you must set the value of the first (or only) index associated with *identifier-1* (the search index) to indicate the starting occurrence for the search.

Before using a serial search on a multidimensional table, you must also set the value of the index for each superordinate dimension.

The SEARCH statement modifies only the value in the search index, and, if the VARYING phrase is specified, the value in *index-name-1* or *identifier-2*. Therefore, to search an entire two-dimensional to seven-dimensional table, you must execute a SEARCH statement for each dimension. In the WHEN phrases, you must specify the indexes for all dimensions. Before the execution of each SEARCH statement, you must initialize the associated indexes with SET statements.

The SEARCH statement executes a serial search beginning at the current setting of the search index.

When the search begins, if the value of the index associated with *identifier-1* is not greater than the highest possible occurrence number, the following actions take place:

- The conditions in the WHEN phrase are evaluated in the order in which they are written.
- If none of the conditions is satisfied, the index for *identifier-1* is increased to correspond to the next table element, and step 1 is repeated.
- If upon evaluation one of the WHEN conditions is satisfied, the search is terminated immediately, and the *imperative-statement-2* associated with that condition is executed. The index points to the table element that satisfied the condition. If NEXT SENTENCE is specified, control passes to the statement following the closest period.
- If the end of the table is reached (that is, the value of the incremented index is greater than the highest possible occurrence number) without the WHEN condition being satisfied, the search is terminated.

If, when the search begins, the value of the index-name associated with *identifier-1* is greater than the highest possible occurrence number, the search terminates immediately.

When the search terminates, if the AT END phrase is specified, *imperative-statement-1* is executed. If the AT END phrase is omitted, control passes to the next statement after the SEARCH statement.

Example: multidimensional serial search

The following code fragment shows a search of the inner dimension (table C) in the third occurrence within the superordinate table (table R):

```
. . .
Working-storage section.
1 G.
  2 R occurs 10 indexed by Rindex.
    3 C occurs 10 ascending key X indexed by Cindex.
      4 X pic 99.
1 Arg pic 99 value 34.
Procedure division.
. . .
* To search within occurrence 3 of table R, set its index to 3
* To search table C beginning at occurrence 1, set its index to 1
  Set Rindex to 3
  Set Cindex to 1
* In the SEARCH statement, specify C without indexes
  Search C
* Specify indexes for both dimensions in the WHEN phrase
  when X(Rindex Cindex) = Arg
    display "Found " X(Rindex Cindex)
End-search
. . .
```

VARYING phrase

index-name-1

One of the following actions applies:

- If *index-name-1* is an index for *identifier-1*, this index is used for the search. Otherwise, the first (or only) index-name is used.
- If *index-name-1* is an index for another table element, then the first (or only) index-name for *identifier-1* is used for the search; the occurrence number represented by *index-name-1* is increased by the same amount as the search index-name and at the same time.

When the VARYING *index-name-1* phrase is omitted, the first (or only) index-name for *identifier-1* is used for the search.

If indexing is used to search a table without an INDEXED BY phrase, correct results are ensured only if both the table defined with the index and the table defined without the index have table elements of the same length and with the same number of occurrences.

When the object of the VARYING phrase is an index-name for another table element, one serial SEARCH statement steps through two table elements at once.

identifier-2

Must be either an index data item or an elementary integer item. *identifier-2* cannot be subscripted by the first (or only) index-name specified for *identifier-1*. During the search, one of the following actions applies:

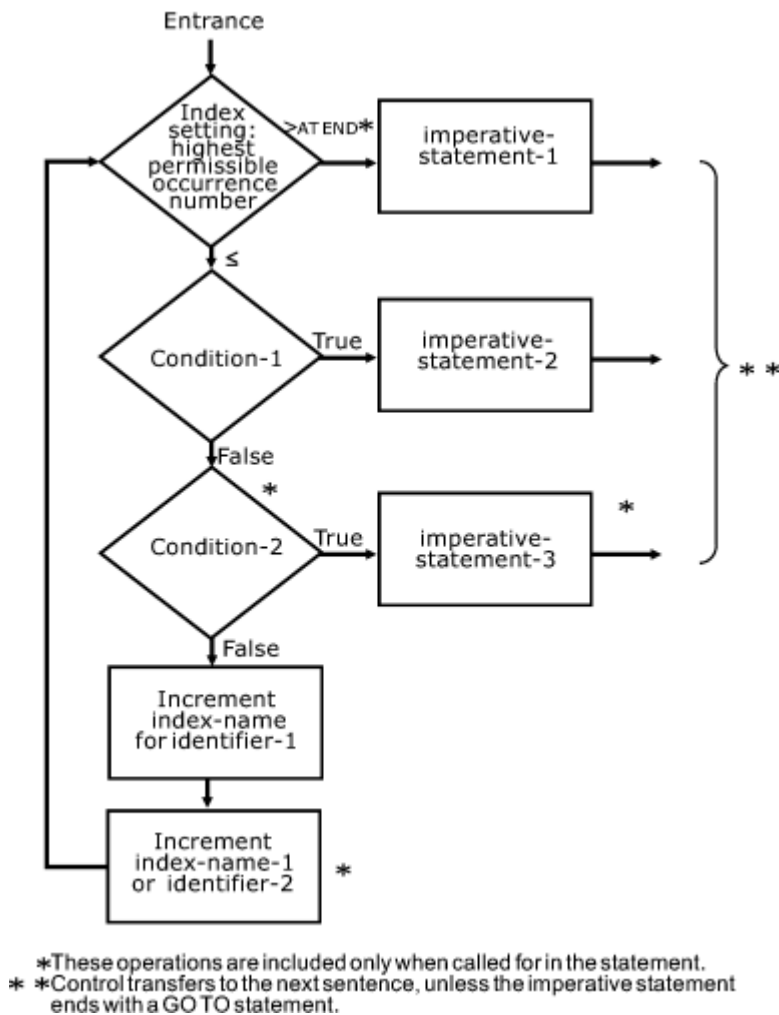
- If *identifier-2* is an index data item, then, whenever the search index is increased, the specified index data item is simultaneously increased by the same amount.
- If *identifier-2* is an integer data item, then, whenever the search index is increased, the specified data item is simultaneously increased by 1.

WHEN phrase (serial search)

condition-1

Can be any condition described under “Conditional expressions” on page 268.

The following figure illustrates a format-1 SEARCH operation containing two WHEN phrases.



Binary search

The topic provides information of using the SEARCH statement for binary search.

identifier-1 (binary search)

identifier-1 identifies the table that is to be searched. *identifier-1* references all occurrences within that table.

The data description entry for *identifier-1* must contain an OCCURS clause with the INDEXED BY and KEY IS phrases.

identifier-1 can reference a data item that is subordinate to a data item that contains an OCCURS clause (that is, *identifier-1* can be a subordinate table within a multidimensional table). In this case, the data description entry must specify an INDEXED BY phrase for each dimension of the table.

identifier-1 must not be subscripted or reference-modified.

AT END

The condition that exists when the search operation terminates without satisfying the conditions specified in the WHEN phrase.

The SEARCH ALL statement executes a binary search. The index associated with *identifier-1* (the search index) need not be initialized by SET statements. The index used is always that associated with the first index-name specified in the OCCURS clause.

Before using a binary search on a multidimensional table, you must execute SET statements to set the value of the index for each superordinate dimension.

The SEARCH statement modifies only the value in the search index. Therefore, to search an entire two-dimensional to seven-dimensional table, you must execute a SEARCH statement for each dimension. In the WHEN phrases, you must specify the indexes for all dimensions.

The results of a SEARCH ALL operation are predictable only when:

- The data in the table is ordered in ASCENDING KEY or DESCENDING KEY order
- The contents of the ASCENDING or DESCENDING keys specified in the WHEN clause provide a unique table reference.

WHEN phrase (binary search)

If a relation condition is specified in the WHEN phrase, the evaluation of the relation is based on the USAGE of the data item referenced by *data-name-1*. The search argument is moved to a temporary data item with the same USAGE as *data-name-1*, and this temporary data item is used for the compare operations associated with the SEARCH.

If the WHEN phrase cannot be satisfied for any setting of the index within this range, the search is unsuccessful. Control is passed to *imperative-statement-1* of the AT END phrase, when specified, or to the next statement after the SEARCH statement. In either case, the final setting of the index is not predictable.

If the WHEN phrase can be satisfied, control passes to *imperative-statement-2*, if specified, or to the next executable sentence if the NEXT SENTENCE phrase is specified. The index contains the value indicating the occurrence that allowed the WHEN conditions to be satisfied.

After *imperative-statement-2* is executed, control passes to the end of the SEARCH statement, unless *imperative-statement-2* ends with a GO TO statement.

condition-name-1* , *condition-name-2

Each condition-name specified must have only a single value, and each must be associated with an ASCENDING KEY or DESCENDING KEY data item for this table element.

data-name-1* , *data-name-2

Must specify an ASCENDING KEY or DESCENDING KEY data item in the table element referenced by *identifier-1* and must be subscripted by the first index-name associated with *identifier-1*. Each data-name can be qualified.

data-name-1 must be a valid operand for comparison with *identifier-3*, *literal-1*, or *arithmetic-expression-1* according to the rules of comparison.

data-name-2 must be a valid operand for comparison with *identifier-4*, *literal-2*, or *arithmetic-expression-2* according to the rules of comparison.

data-name-1 and *data-name-2* cannot reference:

- Floating-point data items
- Group items containing variable-occurrence data items

identifier-3* , *identifier-4

Must not be an ASCENDING KEY or DESCENDING KEY data item for *identifier-1* or an item that is subscripted by the first index-name for *identifier-1*.

identifier-3 and *identifier-4* cannot be data items defined with any of the usages POINTER, FUNCTION-POINTER, PROCEDURE-POINTER, or OBJECT REFERENCE.

If *identifier-3* or *literal-1* is of class national, *data-name-1* must be of class national.

If *identifier-4* or *literal-2* is of class national, *data-name-2* must be of class national.

literal-1* , *literal-2

literal-1 or *literal-2* must be a valid operand for comparison with *data-name-1* or *data-name-2*, respectively.

arithmetic-expression

Can be any of the expressions defined under “[Arithmetic expressions](#)” on page 266, with the following restriction: Any identifier in *arithmetic-expression* must not be an ASCENDING KEY or DESCENDING KEY data item for *identifier-1* or an item that is subscripted by the first index-name for *identifier-1*.

When an ASCENDING KEY or DESCENDING KEY data item is specified, explicitly or implicitly, in the WHEN phrase, all preceding ASCENDING KEY or DESCENDING KEY data-names for *identifier-1* must also be specified.

Search statement considerations

The topic lists considerations of using the SEARCH statement.

Index data items cannot be used as subscripts, because of the restrictions on direct reference to them.

To ensure correct execution of a SEARCH statement for a variable-length table, make sure the object of the OCCURS DEPENDING ON clause (*data-name-1*) contains a value that specifies the current length of the table.

The scope of a SEARCH statement can be terminated by any of the following items:

- An END-SEARCH phrase at the same level of nesting
- A separator period
- An ELSE or END-IF phrase associated with a previous IF statement

SET statement

The SET statement is used to perform an operation as described in this topic.

The operations are:

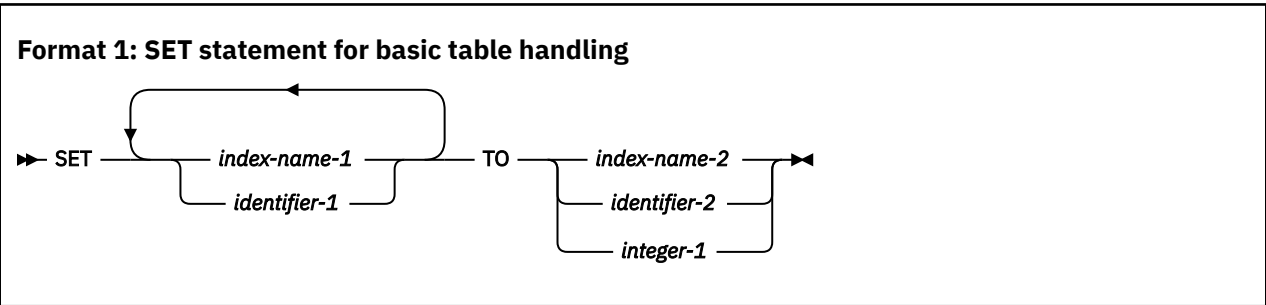
- Placing values associated with table elements into indexes associated with index-names
- Incrementing or decrementing an occurrence number
- Setting the status of an external switch to ON or OFF
- Moving data to condition names to make conditions true
- Setting USAGE POINTER data items to a data address
- Setting USAGE PROCEDURE-POINTER data items to an entry address
- Setting USAGE FUNCTION-POINTER data items to an entry address
- Setting USAGE OBJECT REFERENCE data items to refer to an object instance
- Setting the length of dynamic-length elementary items

Index-names are related to a given table through the INDEXED BY phrase of the OCCURS clause; they are not further defined in the program.

When the sending and receiving fields in a SET statement share part of their storage (that is, the operands overlap), the result of the execution of that SET statement is undefined.

Format 1: SET for basic table handling

When this form of the SET statement is executed, the current value of the receiving field is replaced by the value of the sending field (with conversion).



index-name-1

Receiving field.

Must name an index that is specified in the INDEXED BY phrase of an OCCURS clause.

identifier-1

Receiving field.

Must name either an index data item or an elementary numeric integer item.

index-name-2

Sending field.

Must name an index that is specified in the INDEXED BY phrase of an OCCURS clause. The value of the index before the SET statement is executed must correspond to an occurrence number of its associated table.

identifier-2

Sending field.

Must name either an index data item or an elementary numeric integer item.

integer-1

Sending field.

Must be a positive integer.

The following table shows valid combinations of sending and receiving fields in a format-1 SET statement.

Table 56. <i>Sending and receiving fields for format-1 SET statement</i>			
Sending field	Index-name receiving field	Index data item receiving field	Integer data item receiving field
Index-name*	Valid	Valid**	Valid
Index data item*	Valid**	Valid**	Invalid
Integer data item	Valid	Invalid	Invalid
Integer literal	Valid	Invalid	Invalid
*An index-name refers to an index named in the INDEXED BY phrase of an OCCURS clause. An index data item is defined with the USAGE IS INDEX clause.			
**No conversion takes place.			

Receiving fields are acted upon in the left-to-right order in which they are specified. Any subscripting or indexing associated with *identifier-1* is evaluated immediately before that receiving field is acted upon.

The value used for the sending field is the value at the beginning of SET statement execution.

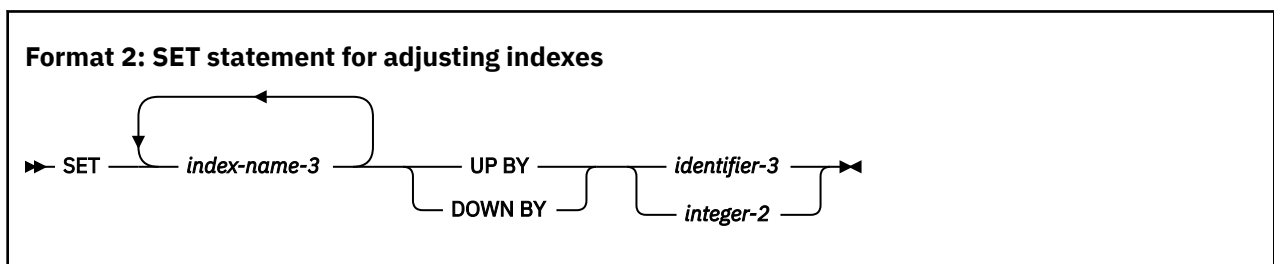
The value of an index after execution of a SEARCH or PERFORM statement can be undefined; therefore, use a format-1 SET statement to reinitialize such indexes before you attempt other table-handling operations.

If *index-name-2* is for a table that has a subordinate item that contains an OCCURS DEPENDING ON clause, then undefined values can be received into *identifier-1*.

For more information about complex OCCURS DEPENDING ON, see *Complex OCCURS DEPENDING ON* in the *Enterprise COBOL Programming Guide*.

Format 2: SET for adjusting indexes

When this form of the SET statement is executed, the value of the receiving index is increased (UP BY) or decreased (DOWN BY) by a value that corresponds to the value in the sending field.



The *receiving field* is an index specified by *index-name-3*. The index value both before and after the SET statement execution must correspond to an occurrence number in an associated table.

The *sending field* can be specified as *identifier-3*, which must be an elementary integer data item, or as *integer-2*, which must be a nonzero integer.

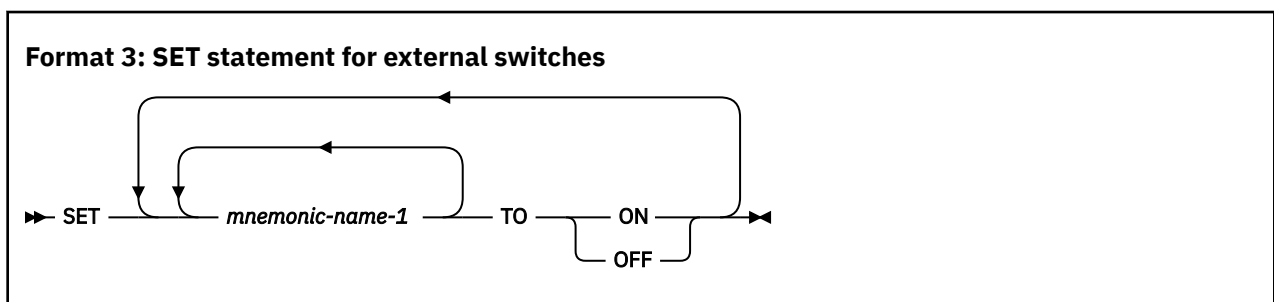
When the format-2 SET statement is executed, the contents of the receiving field are increased (UP BY) or decreased (DOWN BY) by a value that corresponds to the number of occurrences represented by the value of *identifier-3* or *integer-2*. Receiving fields are acted upon in the left-to-right order in which they are specified. The value of the incrementing or decrementing field at the beginning of SET statement execution is used for all receiving fields.

If *index-name-3* is for a table that has a subordinate item that contains an OCCURS DEPENDING ON clause, and if the ODO object is changed before executing a format-2 SET Statement, then *index-name-3* cannot contain a value that corresponds to an occurrence number of its associated table.

For more information about complex OCCURS DEPENDING ON, see *Complex OCCURS DEPENDING ON* in the *Enterprise COBOL Programming Guide*.

Format 3: SET for external switches

When this form of the SET statement is executed, the status of each external switch associated with the specified mnemonic-name is turned ON or OFF.

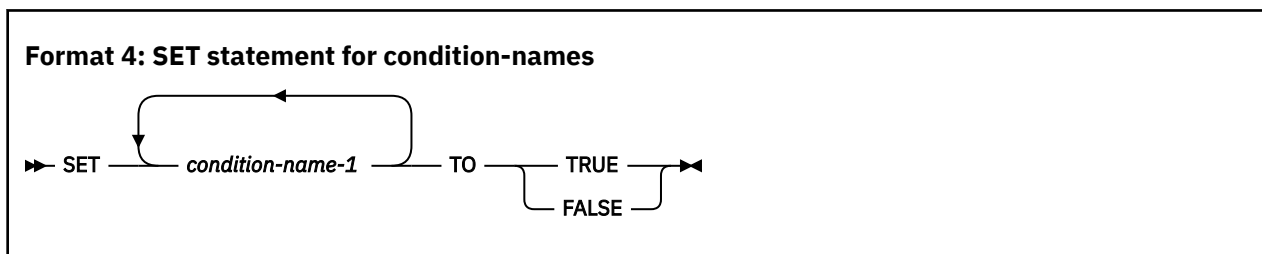


mnemonic-name-1

Must be associated with an external switch, the status of which can be altered.

Format 4: SET for condition-names

When this form of the SET statement is executed, the value associated with a condition-name is placed in its conditional variable according to the rules of the VALUE clause.

***condition-name-1***

Must be associated with a conditional variable.

If more than one literal is specified in the VALUE clause of *condition-name-1*, its associated conditional variable is set equal to the first literal.

If multiple condition-names are specified, the results are the same as if a separate SET statement had been written for each condition-name in the same order in which they are specified in the SET statement.

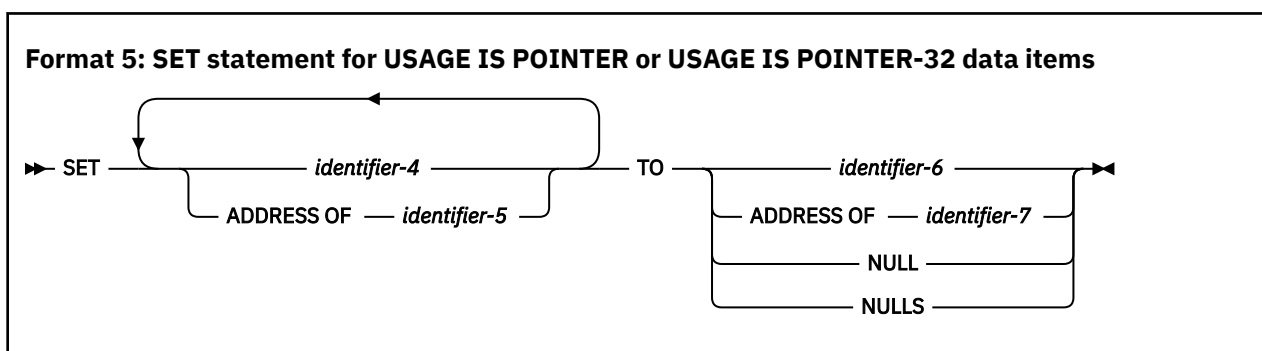
If SET *condition-name-1* TO FALSE is specified, there must be a corresponding WHEN SET TO FALSE phrase defined for *condition-name-1*.

Related references

Format 2 VALUE clause: condition-name value

Format 5: SET for USAGE IS POINTER data items

When this form of the SET statement is executed, the current value of the receiving field is replaced by the address value contained in the sending field.

***identifier-4***

Receiving field(s).

Must be defined as USAGE IS POINTER or USAGE IS POITNER-32.

ADDRESS OF *identifier-5*

Receiving field(s).

identifier-5 must be level-01 or level-77 items defined in the LINKAGE SECTION. The addresses of these items are set to the value of the operand specified in the TO phrase.

identifier-5 must not be reference-modified.

identifier-6

Sending field.

Must be defined as USAGE IS POINTER or USAGE IS POITNER-32.

ADDRESS OF identifier-7

Sending field.

identifier-7 must name an item of any level except 66 or 88 in the LINKAGE SECTION, the WORKING-STORAGE SECTION, or the LOCAL-STORAGE SECTION. It cannot name a dynamic-length elementary item or a dynamic-length group item. ADDRESS OF *identifier-7* contains the address of the identifier, and not the content of the identifier.

NULL, NULLS

Sending field.

Sets the receiving field to contain the value of an invalid address.

The following table shows valid combinations of sending and receiving fields in a format-5 SET statement.

<i>Table 57. Sending and receiving fields for format-5 SET statement</i>				
Sending field	USAGE IS POINTER receiving field	USAGE IS POINTER-32 receiving field	ADDRESS OF receiving field	NULL/NULLS receiving field
USAGE IS POINTER	Valid	Valid	Valid	Invalid
USAGE IS POINTER-32	Valid	Valid	Valid	Invalid
ADDRESS OF	Valid	Valid	Valid	Invalid
NULL/NULLS	Valid	Valid	Valid	Invalid

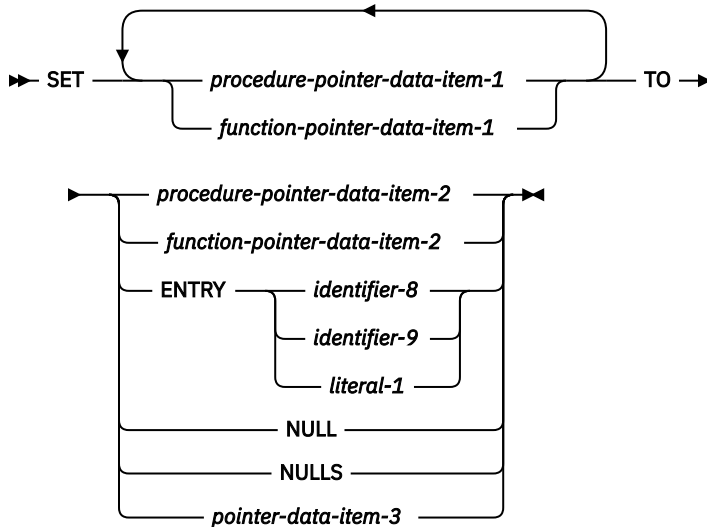
Format 6: SET for procedure-pointer and function-pointer data items

When this format of the SET statement is executed, the current value of the receiving field is replaced by the address value specified by the sending field.

At run time, function-pointers and procedure-pointers can reference the address of the primary entry point of a COBOL program, an alternate entry point in a COBOL program, or an entry point in a non-COBOL program; or they can be NULL.

COBOL function-pointers are more easily used than procedure-pointers for interoperation with C functions.

Format 6: SET statement for procedure-pointers and function-pointers



procedure-pointer-data-item-1 , procedure-pointer-data-item-2

Must be described as USAGE IS PROCEDURE-POINTER. *procedure-pointer-data-item-1* is a receiving field; *procedure-pointer-data-item-2* is a sending field.

function-pointer-data-item-1 , function-pointer-data-item-2

Must be described as USAGE IS FUNCTION-POINTER. *function-pointer-data-item-1* is a receiving field; *function-pointer-data-item-2* is a sending field.

identifier-8

Must be defined as an alphabetic or alphanumeric item such that the value can be a program name. For more information, see [Chapter 15, "PROGRAM-ID paragraph," on page 101](#). For entry points in non-COBOL programs, *identifier-8* can contain the characters @, #, and \$.

identifier-9

Must be defined as a user-defined function with a returning item described as USAGE IS PROCEDURE-POINTER or USAGE IS FUNCTION-POINTER.

literal-1

Must be alphanumeric and must conform to the rules for formation of program-names. For details on formation rules, see the discussion of program-name under [Chapter 15, "PROGRAM-ID paragraph," on page 101](#).

identifier-8 or *literal-1* must refer to one of the following types of entry points:

- The primary entry point of a COBOL program as defined by the PROGRAM-ID paragraph. The PROGRAM-ID must reference the outermost program of a compilation unit; it must not reference a nested program.
- An alternate entry point of a COBOL program as defined by a COBOL ENTRY statement.
- An entry point in a non-COBOL program.

The program-name referenced by the SET ... TO ENTRY statement can be affected by the PGMNAME compiler option. For details, see *PGMNAME* in the *Enterprise COBOL Programming Guide*.

NULL, NULLS

Sets the receiving field to contain the value of an invalid address.

pointer-data-item-3

Must be defined with USAGE POINTER. You must set *pointer-data-item-3* in a non-COBOL program to point to a valid program entry point.

Example of SET for procedure-pointer and function-pointer data items

The following example shows how to use the SET statement to assign values to a function-pointer (FP) and a procedure-pointer (PP), allowing for dynamic calls to different subprograms ("subp1" and "subp2") based on the assigned pointer references.

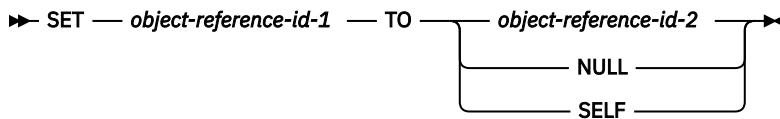
```
IDENTIFICATION DIVISION.
PROGRAM-ID DEMO.
DATA DIVISION.
WORKING-STORAGE SECTION.
01 FP USAGE FUNCTION-POINTER.
01 PP USAGE PROCEDURE-POINTER.

PROCEDURE DIVISION.
  SET FP to Entry "subp1"
  CALL FP
  SET PP to Entry "subp2"
  CALL PP
```

Format 7: SET for USAGE OBJECT REFERENCE data items

When this format of the SET statement is executed, the value in the receiving item is replaced by the value in the sending item.

Format 7: SET statement for object references



object-reference-id-1 and *object-reference-id-2* must be defined as USAGE OBJECT REFERENCE. *object-reference-id-1* is the receiving item and *object-reference-id-2* is the sending item. If *object-reference-id-1* is defined as an object reference of a certain class (defined as "USAGE OBJECT REFERENCE class-name"), *object-reference-id-2* must be an object reference of the same class or a class derived from that class.

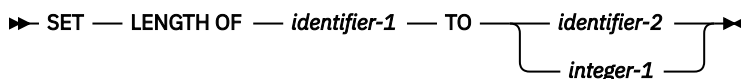
If the figurative constant NULL is specified, the receiving *object-reference-id-1* is set to the NULL value.

If SELF is specified, the SET statement must appear in the PROCEDURE DIVISION of a method. *object-reference-id-1* is set to reference the object upon which the currently executing method was invoked.

Format 8: SET for length of dynamic-length elementary items

When this format of the SET statement is executed, the length of the dynamic-length elementary item in the receiver is set to the value of the sending item.

Format 8: SET for length of dynamic-length elementary items



identifier-1 must be a dynamic-length elementary item.

identifier-2 must be an elementary numeric integer item.

integer-1 must be an integer greater than or equal to zero.

If *identifier-2* is less than zero then the length of *identifier-1* is set to zero.

If *identifier-2* or *integer-1* is less than the current length of *identifier-1*, then the length of *identifier-1* is set to the specified value and *identifier-1* is truncated on the right.

If *identifier-2* or *integer-1* is longer than the current length of *identifier-1*, then the length of *identifier-1* is set to the specified value, no padding occurs, and the newly available character positions on the right are not initialized.

SORT statement

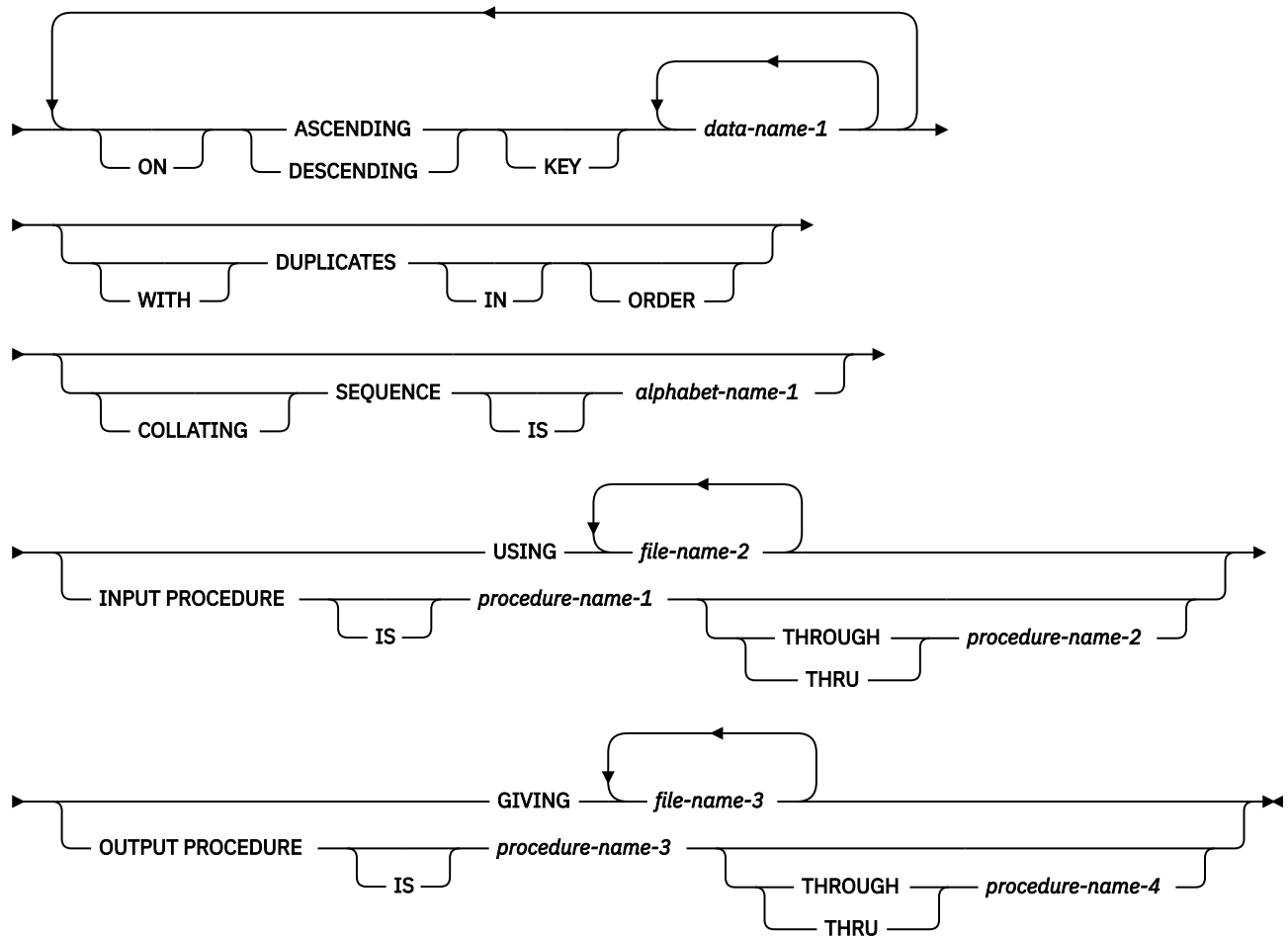
The SORT statement causes a set of records or table elements to be arranged in a user-specified sequence.

For sorting files, the SORT statement accepts records from one or more files, sorts them according to the specified keys, and makes the sorted records available either through an output procedure or in an output file.

For sorting tables, the SORT statement sorts table elements according to specified table keys.

Format 1: SORT statement

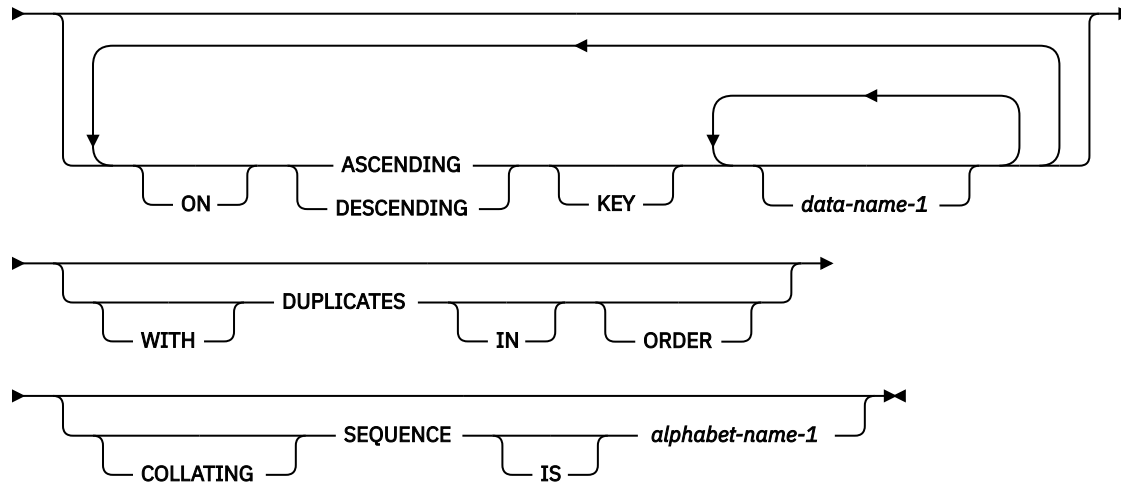
►► SORT — *file-name-1* ►►



Format 1 SORT statements can appear anywhere in the PROCEDURE DIVISION except in the declarative portion. This format of the SORT statement is not supported for programs that are compiled with the THREAD option. See also [“MERGE statement”](#) on page 400.

Format 2: Table SORT statement

►► SORT — *data-name-2* ►



Format 2 SORT statements can appear anywhere in the PROCEDURE DIVISION. This format of the SORT statement can be used with programs that are compiled with the THREAD option.

file-name-1

The name given in the SD entry that describes the records to be sorted.

No pair of file-names in a SORT statement can be specified in the same SAME SORT AREA clause or the SAME SORT-MERGE AREA clause. File-names associated with the GIVING clause (*file-name-3*, ...) cannot be specified in the SAME AREA clause; however, they can be associated with the SAME RECORD AREA clause.

data-name-2

Specifies a table data-name that is subject to the following rules:

- *data-name-2* must have an OCCURS clause in the data description entry.
- *data-name-2* can be qualified.
- *data-name-2* can be subscripted. The rightmost or only subscript of the table must be omitted or replaced with the word ALL.

The number of occurrences of table elements that are referenced by *data-name-2* is determined by the rules in the OCCURS clause. The sorted table elements are placed in the same table that is referenced by *data-name-2*.

ASCENDING KEY and DESCENDING KEY phrases (format 1)

This phrase specifies that records are to be processed in ascending or descending sequence (depending on the phrase specified), based on the specified sort keys.

data-name-1

Specifies a KEY data item on which the SORT statement will be based. Each such data-name must identify a data item in a record associated with *file-name-1*. The data-names following the word KEY are listed from left to right in the SORT statement in order of decreasing significance without regard to how they are divided into KEY phrases. The leftmost data-name is the major key, the next data-name is the next most significant key, and so forth. The following rules apply:

- A specific KEY data item must be physically located in the same position and have the same data format in each input file. However, it need not have the same data-name.
- If *file-name-1* has more than one record description, the KEY data items need be described in only one of the record descriptions.

- If *file-name-1* contains variable-length records, all of the KEY data-items must be contained within the first *n* character positions of the record, where *n* equals the minimum records size specified for *file-name-1*.
- KEY data items must not contain an OCCURS clause or be subordinate to an item that contains an OCCURS clause.
- KEY data items cannot be:
 - Variably located
 - Group items that contain variable-occurrence data items
 - Category numeric described with usage NATIONAL (national decimal item)
 - Category external floating-point described with usage NATIONAL (national floating-point item)
 - Category DBCS
 - Dynamic-length elementary items
 - Dynamic-length group items
- KEY data items can be qualified.
- KEY data items can belong to any of the following data categories:
 - Alphabetic, alphanumeric, alphanumeric-edited
 - Numeric (except numeric with usage NATIONAL)
 - Numeric-edited (with usage DISPLAY or NATIONAL)
 - Internal floating-point or display floating-point
 - National or national-edited

If *file-name-3* references an indexed file, the first specification of *data-name-1* must be associated with an ASCENDING phrase and the data item referenced by that *data-name-1* must occupy the same character positions in this record as the data item associated with the prime record key for that file.

The direction of the sorting operation depends on the specification of the ASCENDING or DESCENDING keywords as follows:

- When ASCENDING is specified, the sequence is from the lowest key value to the highest key value.
- When DESCENDING is specified, the sequence is from the highest key value to the lowest.
- If the KEY data item is described with usage NATIONAL, the sequence of the KEY values is based on the binary values of the national characters.
- If the KEY data item is internal floating point, the sequence of key values will be in numeric order.
- When the COLLATING SEQUENCE phrase is not specified, the key comparisons are performed according to the rules for comparison of operands in a relation condition. See [“General relation conditions” on page 272](#).
- When the COLLATING SEQUENCE phrase is specified, the indicated collating sequence is used for key data items of alphabetic, alphanumeric, alphanumeric-edited, external floating-point, and numeric-edited categories. For all other key data items, the comparisons are performed according to the rules for comparison of operands in a relation condition.

ASCENDING KEY and DESCENDING KEY phrases (format 2)

This phrase specifies that table elements are to be processed in ascending or descending sequence, based on the specified phrase and sort keys.

data-name-1

Specifies a KEY data name that is subject to the following rules:

- The data item that is identified by a key data-name must be the same as, or subordinate to, the data item that is referenced by *data-name-2*.
- KEY data items can be qualified.

- KEY data items can belong to any of the following data categories:
 - Alphabetic, alphanumeric, alphanumeric-edited
 - Numeric (except numeric with usage NATIONAL)
 - Numeric-edited (with usage DISPLAY or NATIONAL)
 - Internal floating-point or display floating-point
 - National or national-edited
- KEY data items cannot be:
 - Variably located
 - Group items that contain variable-occurrence data items
 - Category numeric that is described with usage NATIONAL (national decimal item)
 - Category external floating-point that is described with usage NATIONAL (national floating-point item)
 - Category DBCS
 - Class object or pointer
 - USAGE OBJECT, USAGE POINTER, USAGE PROCEDURE-POINTER, or USAGE FUNCTION-POINTER
 - Subscripted
 - Dynamic-length elementary items
 - Dynamic-length group items
- If the data item that is identified by a KEY data-name is subordinate to *data-name-2*, the following rules apply:
 - The data item cannot be described with an OCCURS clause.
 - The data item cannot be subordinate to an entry that is also subordinate to *data-name-2* and that contains an OCCURS clause.

The KEY phrase can be omitted only if the description of the table that is referenced by *data-name-2* contains a KEY phrase.

The words ASCENDING and DESCENDING are transitive across all occurrences of *data-name-1* until another word ASCENDING or DESCENDING is encountered.

The data items that are referenced by *data-name-1* are key data items, and these data items determine the order in which the sorted table elements are stored. The order of significance of the keys is the order in which data items are specified in the SORT statement, without regard to the association with ASCENDING or DESCENDING phrases.

The SORT statement sorts the table that is referenced by *data-name-2* and presents the sorted table in *data-name-2*. The sorting order is determined by either the ASCENDING and DESCENDING phrases (if specified), or by the KEY phrase that is associated with *data-name-2*.

The direction of the sorting operation depends on the specification of the ASCENDING or DESCENDING keywords:

- When ASCENDING is specified, the sequence is from the lowest key value to the highest one.
- When DESCENDING is specified, the sequence is from the highest key value to the lowest one.
- If the KEY data item is described with usage NATIONAL, the sequence of the KEY values is based on the binary values of the national characters.
- If the KEY data item is internal floating-point, the sequence of key values is in the numeric order.
- When the COLLATING SEQUENCE phrase is not specified, the EBCDIC sequence is used for key data items of alphabetic, alphanumeric, alphanumeric-edited, external floating-point, and numeric-edited categories. For all the other key data items, the comparisons are performed according to the rules for comparison of operands in a relation condition.

- When the COLLATING SEQUENCE phrase is specified, the indicated collating sequence is used for key data items of alphabetic, alphanumeric, alphanumeric-edited, external floating-point, and numeric-edited categories. For all the other key data items, the comparisons are performed according to the rules for comparison of operands in a relation condition.

To determine the relative order in which table elements are stored, the contents of corresponding key data items are compared according to the rules for comparison of operands in a relation condition. The sorting starts with the most significant key data item with the following rules:

- If the contents of the corresponding key data items are not equal and the key is associated with the ASCENDING phrase, the table element that contains the key data item with the lower value has the lower occurrence number.
- If the contents of the corresponding key data items are not equal and the key is associated with the DESCENDING phrase, the table element that contains the key data item with the higher value has the lower occurrence number.
- If the contents of the corresponding key data items are equal, the determination is based on the contents of the next most significant key data item.

If the KEY phrase is not specified, the sequence is determined by the KEY phrase in the data description entry of the table that is referenced by *data-name-2*.

If the KEY phrase is specified, it overrides any KEY phrase specified in the data description entry of the table that is referenced by *data-name-2*.

If *data-name-1* is omitted, the data item that is referenced by *data-name-2* is the key data item.

DUPLICATES phrase (format 1)

If the DUPLICATES phrase is specified, and the contents of all the key elements associated with one record are equal to the corresponding key elements in one or more other records, the order of return of these records is as follows:

- The order of the associated input files as specified in the SORT statement. Within a given file the order is that in which the records are accessed from that file.
- The order in which these records are released by an input procedure, when an input procedure is specified.

If the DUPLICATES phrase is not specified, the order of these records is undefined.

DUPLICATES phrase (format 2)

When both of the following conditions are met, the contents of table elements are in the relative order that is the same as the order before sorting operation:

- The DUPLICATES phrase is specified.
- The contents of all the key data items that are associated with one table element are equal to the contents of corresponding key data items that are associated with one or more other table elements.

If the DUPLICATES phrase is not specified and the second condition exists, the relative order of the contents of these table elements is undefined.

COLLATING SEQUENCE phrase (both formats)

This phrase specifies the collating sequence to be used in alphanumeric comparisons for the KEY data items in this sorting operation.

The COLLATING SEQUENCE phrase has no effect for keys that are not alphabetic or alphanumeric.

alphabet-name-1

Must be specified in the ALPHABET clause of the SPECIAL-NAMES paragraph. *alphabet-name-1* can be associated with any one of the ALPHABET clause phrases, with the following results:

STANDARD-1

The ASCII collating sequence is used for all alphanumeric comparisons. (The ASCII collating sequence is shown in [Appendix E, “EBCDIC and ASCII collating sequences,”](#) on page 763.)

STANDARD-2

The International Reference Version of *ISO/IEC 646, 7-bit coded character set for information processing interchange* is used for all alphanumeric comparisons.

NATIVE

The EBCDIC collating sequence is used for all alphanumeric comparisons. (The EBCDIC collating sequence is shown in [Appendix E, “EBCDIC and ASCII collating sequences,”](#) on page 763.)

EBCDIC

The EBCDIC collating sequence is used for all alphanumeric comparisons. (The EBCDIC collating sequence is shown in [Appendix E, “EBCDIC and ASCII collating sequences,”](#) on page 763.)

literal

The collating sequence established by the specification of literals in the alphabet-name clause is used for all alphanumeric comparisons.

When the COLLATING SEQUENCE phrase is omitted, the PROGRAM COLLATING SEQUENCE clause (if specified) in the OBJECT-COMPUTER paragraph specifies the collating sequence to be used. When both the COLLATING SEQUENCE phrase and the PROGRAM COLLATING SEQUENCE clause are omitted, the EBCDIC collating sequence is used.

USING phrase

file-name-2 , ...

The input files.

When the USING phrase is specified, all the records in *file-name-2 , ...*, (that is, the input files) are transferred automatically to *file-name-1*. At the time the SORT statement is executed, these files must not be open. The compiler opens, reads, makes records available, and closes these files automatically. If EXCEPTION/ERROR procedures are specified for these files, the compiler makes the necessary linkage to these procedures.

All input files must be described in FD entries in the DATA DIVISION.

If the USING phrase is specified and if *file-name-1* contains variable-length records, the size of the records contained in the input files (*file-name-2 , ...*) must be neither less than the smallest record nor greater than the largest record described for *file-name-1*. If *file-name-1* contains fixed-length records, the size of the records contained in the input files must not be greater than the largest record described for *file-name-1*. For more information, see *Describing the input to sorting or merging* in the *Enterprise COBOL Programming Guide*.

INPUT PROCEDURE phrase

This phrase specifies the name of a procedure that is to select or modify input records before the sorting operation begins.

procedure-name-1

Specifies the first (or only) section or paragraph in the input procedure.

procedure-name-2

Identifies the last section or paragraph of the input procedure.

The input procedure can consist of any procedure needed to select, modify, or copy the records that are made available one at a time by the RELEASE statement to the file referenced by *file-name-1*.

The range includes all statements that are executed as the result of a transfer of control by CALL, EXIT, GO TO, PERFORM, and XML PARSE statements in the range of the input procedure, as well as all statements in declarative procedures that are executed as a result of the execution of statements in the range of the input procedure. The range of the input procedure must not cause the execution of any MERGE, RETURN, or format 1 SORT statement.

If an input procedure is specified, control is passed to the input procedure before the file referenced by *file-name-1* is sequenced by the SORT statement. The compiler inserts a return mechanism at the end of the last statement in the input procedure. When control passes the last statement in the input procedure, the records that have been released to the file referenced by *file-name-1* are sorted.

GIVING phrase

file-name-3 , ...

The output files.

When the GIVING phrase is specified, all the sorted records in *file-name-1* are automatically transferred to the output files (*file-name-3, ...*).

All output files must be described in FD entries in the DATA DIVISION.

If the output files (*file-name-3, ...*) contain variable-length records, the size of the records contained in *file-name-1* must be neither less than the smallest record nor greater than the largest record described for the output files. If the output files contain fixed-length records, the size of the records contained in *file-name-1* must not be greater than the largest record described for the output files. For more information, see *Describing the output from sorting or merging* in the *Enterprise COBOL Programming Guide*.

At the time the SORT statement is executed, the output files (*file-name-3, ...*) must not be open. For each of the output files, the execution of the SORT statement causes the following actions to be taken:

- The processing of the file is initiated. The initiation is performed as if an OPEN statement with the OUTPUT phrase had been executed.
- The sorted logical records are returned and written onto the file. Each record is written as if a WRITE statement without any optional phrases had been executed.

For a relative file, the relative key data item for the first record returned contains the value '1'; for the second record returned, the value '2'. After execution of the SORT statement, the content of the relative key data item indicates the last record returned to the file.

- The processing of the file is terminated. The termination is performed as if a CLOSE statement without optional phrases had been executed.

These implicit functions are performed such that any associated USE AFTER EXCEPTION/ERROR procedures are executed; however, the execution of such a USE procedure must not cause the execution of any statement manipulating the file referenced by, or accessing the record area associated with, *file-name-3*. On the first attempt to write beyond the externally defined boundaries of the file, any USE AFTER STANDARD EXCEPTION/ERROR procedure specified for the file is executed. If control is returned from that USE procedure or if no such USE procedure is specified, the processing of the file is terminated.

OUTPUT PROCEDURE phrase

This phrase specifies the name of a procedure that is to select or modify output records from the sorting operation.

procedure-name-3

Specifies the first (or only) section or paragraph in the output procedure.

procedure-name-4

Identifies the last section or paragraph of the output procedure.

The output procedure can consist of any procedure needed to select, modify, or copy the records that are made available one at a time by the RETURN statement in sorted order from the file referenced by *file-name-1*. The range includes all statements that are executed as the result of a transfer of control by CALL, EXIT, GO TO, PERFORM, and XML PARSE statements in the range of the output procedure. The range also includes all statements in declarative procedures that are executed as a result of the execution of statements in the range of the output procedure. The range of the output procedure must not cause the execution of any MERGE, RELEASE, or format 1 SORT statement.

If an output procedure is specified, control passes to it after the file referenced by *file-name-1* has been sequenced by the SORT statement. The compiler inserts a return mechanism at the end of the last statement in the output procedure and when control passes the last statement in the output procedure, the return mechanism provides the termination of the sort and then passes control to the next executable statement after the SORT statement. Before entering the output procedure, the sort procedure reaches a point at which it can select the next record in sorted order when requested. The RETURN statements in the output procedure are the requests for the next record.

The INPUT PROCEDURE and OUTPUT PROCEDURE phrases are similar to those for a basic PERFORM statement. For example, if you name a procedure in an output procedure, that procedure is executed during the sorting operation just as if it were named in a PERFORM statement. As with the PERFORM statement, execution of the procedure is terminated after the last statement completes execution. The last statement in an input or output procedure can be the EXIT statement (see [“EXIT statement”](#) on page 343).

SORT special registers

The special registers, SORT-CORE-SIZE, SORT-MESSAGE, and SORT-MODE-SIZE, are equivalent to option control statement keywords in the sort control file. You define the sort control data set with the SORT-CONTROL special register.

Usage notes:

- The SORT special registers are not applicable to sorting a table by using the format 2 SORT statement.
- If you use a sort control file to specify control statements, the values specified in the sort control file take precedence over those in the special register.

SORT-MESSAGE special register

See [“SORT-MESSAGE”](#) on page 26.

SORT-CORE-SIZE special register

See [“SORT-CORE-SIZE”](#) on page 26.

SORT-FILE-SIZE special register

See [“SORT-FILE-SIZE”](#) on page 26.

SORT-MODE-SIZE special register

See [“SORT-MODE-SIZE”](#) on page 27.

SORT-CONTROL special register

See [“SORT-CONTROL”](#) on page 25.

SORT-RETURN special register

See [“SORT-RETURN”](#) on page 27.

Segmentation considerations

The topic lists considerations of using the SORT statement.

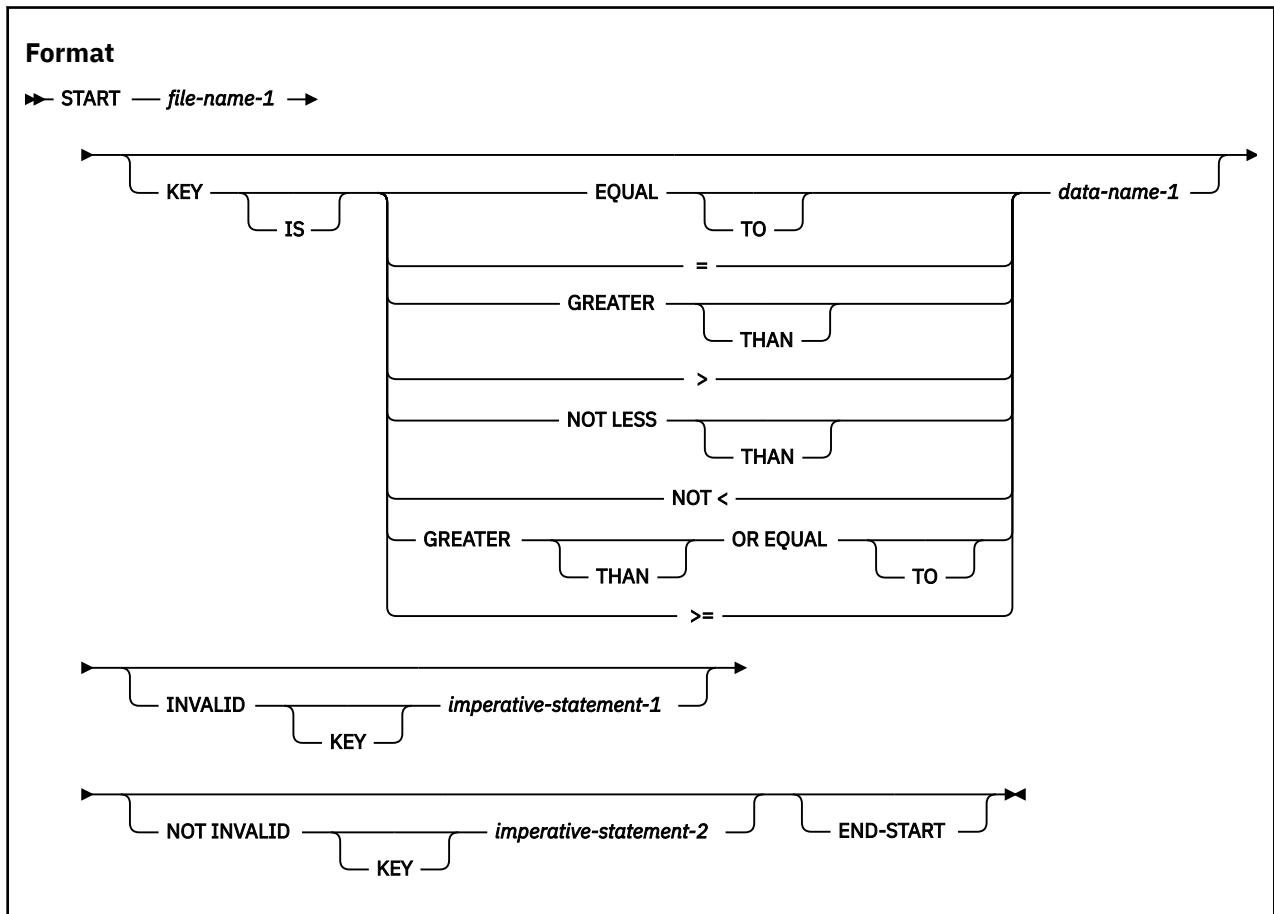
If a SORT statement is coded in a fixed segment, any input or output procedure referenced by that SORT statement must be either totally within a fixed segment or wholly contained in a single independent segment.

If a SORT statement is coded in an independent segment, any input or output procedure referenced by that SORT statement must be either totally within a fixed segment or wholly contained within the same independent segment as that SORT statement.

START statement

The START statement provides a means of positioning within an indexed or relative file for subsequent sequential record retrieval.

When the START statement is executed, the associated indexed or relative file must be open in either INPUT or I-O mode.



file-name-1

Must name a file with sequential or dynamic access. *file-name-1* must be defined in an FD entry in the DATA DIVISION and must not name a sort file.

KEY phrase

When the KEY phrase is specified, the file position indicator is positioned at the logical record in the file whose key field satisfies the comparison.

When the KEY phrase is not specified, KEY IS EQUAL (to the prime record key) is implied.

data-name-1

Can be qualified; it cannot be subscripted.

When the START statement is executed, a comparison is made between the current value in the key *data-name* and the corresponding key field in the file's index.

If the FILE STATUS clause is specified in the file-control entry, the associated file status key is updated when the START statement is executed (See [“File status key”](#) on page 299).

INVALID KEY phrases

If the comparison is not satisfied by any record in the file, an invalid key condition exists; the position of the file position indicator is undefined, and (if specified) the INVALID KEY imperative-statement is executed. (See [“INTO and FROM phrases”](#) on page 304 under "Common processing facilities".)

Both the INVALID KEY phrase and an applicable EXCEPTION/ERROR procedure can be omitted.

END-START phrase

This explicit scope terminator serves to delimit the scope of the START statement. END-START permits a conditional START statement to be nested in another conditional statement. END-START can also be used with an imperative START statement.

For more information, see [“Delimited scope statements”](#) on page 293.

Indexed files

When the KEY phrase is specified, the key data item used for the comparison is *data-name-1*.

When the KEY phrase is not specified, the key data item used for the EQUAL TO comparison is the prime record key.

When START statement execution is successful, the RECORD KEY or ALTERNATE RECORD KEY with which *data-name-1* is associated becomes the key of reference for subsequent READ statements.

data-name-1

Can be any of the following items:

- The prime RECORD KEY.
- Any ALTERNATE RECORD KEY.
- A data item within a record description for a file whose leftmost character position corresponds to the leftmost character position of that record key; it can be qualified. The size of the data item must be less than or equal to the length of the record key for the file.

Regardless of its category, *data-name-1* is treated as an alphanumeric item for purposes of the comparison operation.

Note: If your key is numeric, you must specify the EQUAL TO condition, otherwise, unexpected results can happen.

The file position indicator points to the first record in the file whose key field satisfies the comparison. If the operands in the comparison are of unequal lengths, the comparison proceeds as if the longer field were truncated on the right to the length of the shorter field. All other numeric and alphanumeric comparison rules apply, except that the PROGRAM COLLATING SEQUENCE clause, if specified, has no effect.

When START statement execution is successful, the RECORD KEY with which *data-name-1* is associated becomes the key of reference for subsequent READ statements.

When START statement execution is unsuccessful, the key of reference is undefined.

Relative files

When the KEY phrase is specified, *data-name-1* must specify the RELATIVE KEY.

Whether or not the KEY phrase is specified, the key data item used in the comparison is the RELATIVE KEY data item. Numeric comparison rules apply.

The file position indicator points to the logical record in the file whose key satisfies the specified comparison.

STOP statement

The STOP statement halts execution of the object program either permanently or temporarily.



literal

Can be a fixed-point numeric literal (signed or unsigned) or an alphanumeric literal. It can be any figurative constant except ALL *literal*.

When STOP *literal* is specified, the literal is communicated to the operator, and object program execution is suspended. Program execution is resumed only after operator intervention, and continues at the next executable statement in sequence.

The STOP *literal* statement is useful for special situations when operator intervention is needed during program execution; for example, when a special tape or disk must be mounted or a specific daily code must be entered. However, the ACCEPT and DISPLAY statements are preferred when operator intervention is needed.

Do not use the STOP *literal* statement in programs compiled with the THREAD compiler option.

When STOP RUN is specified, execution is terminated and control is returned to the system, using the current value of the RETURN-CODE special register (modulo 4096) as the return code. When STOP RUN is not the last or only statement in a sequence of imperative statements within a sentence, the statements following STOP RUN are not executed.

The STOP RUN statement closes *all* files defined in any of the programs in the run unit.

The STOP RUN statement frees *all* allocated buffers for dynamic-length elementary items in all of the programs in the run unit.

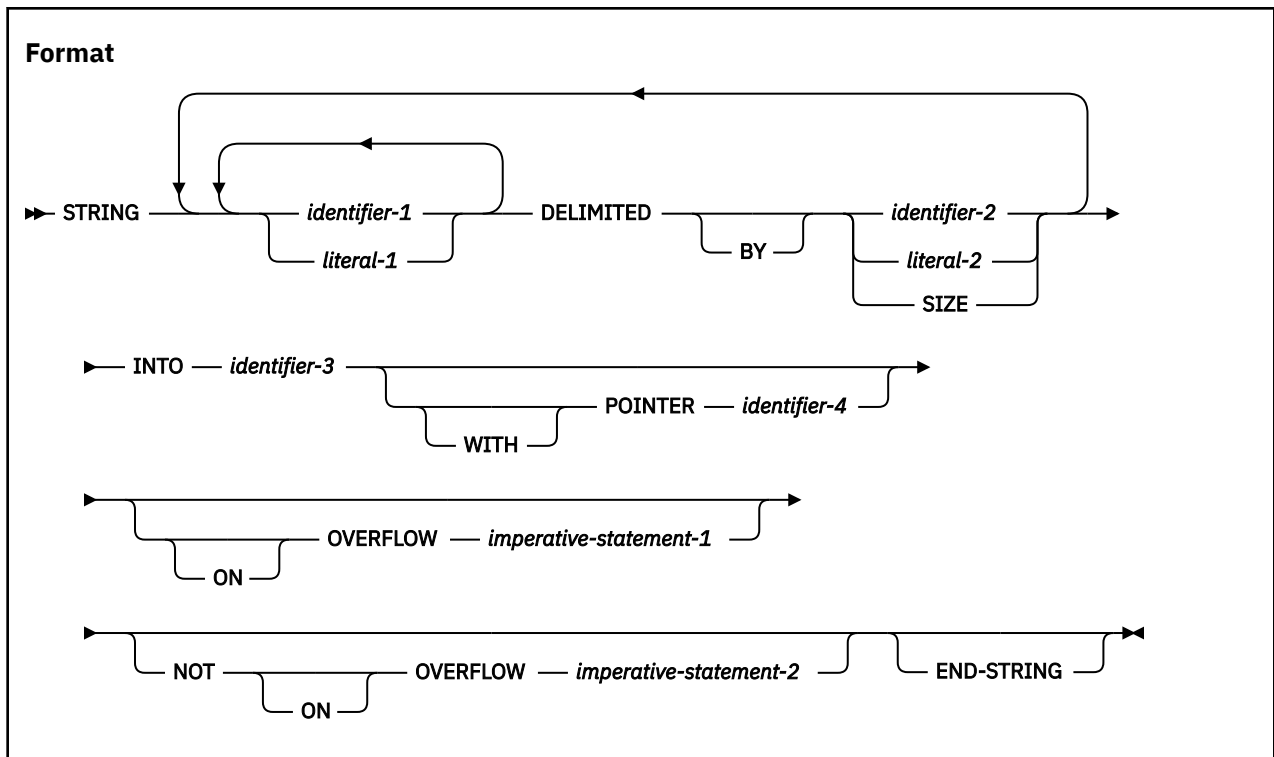
For use of the STOP RUN statement in calling and called programs, see the following table.

Termination statement	Main program	Subprogram
STOP RUN	Returns to the calling program. (Can be the system, which causes the application to end.)	Returns directly to the program that called the main program. (Can be the system, which causes the application to end.)

STRING statement

The STRING statement strings together the partial or complete contents of two or more data items or literals into one single data item.

One STRING statement can be written instead of a series of MOVE statements.



identifier-1, literal-1

Represents the *sending fields*.

DELIMITED BY phrase

Sets the limits of the string.

identifier-2, literal-2

Are delimiters; that is, characters that delimit the data to be transferred.

SIZE

Transfers the complete sending area.

INTO phrase

Identifies the receiving field.

identifier-3

Represents the *receiving field*.

POINTER phrase

Points to a character position in the receiving field. The pointer field indicates a relative alphanumeric character position, DBCS character position, or national character position when the receiving field is of usage DISPLAY, DISPLAY-1, or NATIONAL, respectively.

identifier-4

Represents the *pointer field*. *identifier-4* must be large enough to contain a value equal to the length of the receiving field plus 1. You must initialize *identifier-4* to a nonzero value before execution of the STRING statement begins.

When the POINTER phrase is specified, an explicit pointer field is available to control placement of data in the receiving field. It is required to set the explicit pointer's initial value, which must greater than or equal to 1. For fixed-length data items, the pointer's initial value must less than or equal to the character position count of the receiving field. For dynamic-length elementary items, the pointer's initial value must less than or equal to the specified value of the LIMIT phrase on the item's data description entry, or the default limit if no LIMIT is specified.

When the POINTER phrase is specified and the receiving field is a dynamic-length elementary item, the value of the POINTER field may be greater than the length of the receiver. When the value of the

POINTER field is equal to the length of the receiver plus one, the STRING statement will effectively concatenate the sending fields to the receiver. When the value of the POINTER field is equal to the length of the receiver plus two, or more:

- For alphanumeric (PIC X) dynamic-length elementary item receivers, the intermediate character positions between the end of the receiver and the beginning of the POINTER field will be padded with spaces, and the sending fields will be concatenated starting at the POINTER field position as usual.
- For UTF-8 (PIC U) dynamic-length elementary item receivers, this is not allowed.

The following rules apply:

- All identifiers except *identifier-4* must reference data items described explicitly or implicitly as usage DISPLAY, DISPLAY-1, NATIONAL, or UTF-8.
- All identifiers cannot be dynamic-length group items.
- *literal-1* or *literal-2* must be of category alphanumeric, DBCS, national, or UTF-8 and can be any figurative constant that does not begin with the word ALL (except NULL).
- If *identifier-1* or *identifier-2* references a data item of category numeric, each numeric item must be described as an integer without the symbol 'P' in its PICTURE character-string.
- *identifier-3* must not reference a data item of category numeric-edited, alphanumeric-edited, or national-edited; an external floating-point data item of usage DISPLAY, or an external floating-point data item of usage NATIONAL.
- *identifier-3* must not be described with the JUSTIFIED clause.
- If *identifier-3* is of usage DISPLAY, *identifier-1* and *identifier-2* must be of usage DISPLAY and all literals must be alphanumeric literals. Any figurative constant can be specified except one that begins with the word ALL. Each figurative constant represents a 1-character alphanumeric literal.
- If *identifier-3* is of usage DISPLAY-1, *identifier-1* and *identifier-2* must be of usage DISPLAY-1 and all literals must be DBCS literals. The only figurative constant that can be specified is SPACE, which represents a 1-character DBCS literal. ALL *DBCS-literal* must not be specified.
- If *identifier-3* is of usage NATIONAL, *identifier-1* and *identifier-2* must be of usage NATIONAL and all literals must be national literals. Any figurative constant can be specified except *symbolic-character* and one that begins with the word ALL. Each figurative constant represents a 1-character national literal.
- If *identifier-3* is of usage UTF-8, *identifier-1* and *identifier-2* must be of usage UTF-8 and all literals must be UTF-8 literals. Any figurative constant can be specified except *symbolic-character* and one that begins with the word ALL. Each figurative constant represents a 1-character UTF-8 literal.
- If *identifier-1* or *identifier-2* references an elementary data item of usage DISPLAY that is described as category numeric, numeric-edited, or alphanumeric-edited, the item is treated as if it were redefined as category alphanumeric.
- If *identifier-1* or *identifier-2* references an elementary data item of usage NATIONAL that is described as category numeric, numeric-edited, or national-edited item, the item is treated as if it were redefined as category national.
- *identifier-4* must not be described with the symbol P in its PICTURE character-string.

Evaluation of subscripts, reference modification, variable-lengths, variable locations, and function-identifiers is performed only once, at the beginning of the execution of the STRING statement. Therefore, if *identifier-3* or *identifier-4* is used as a subscript, reference-modifier, or function argument in the STRING statement, or affects the length or location of any of the identifiers in the STRING statement, the values calculated for those subscripts, reference-modifiers, variable lengths, variable locations, and functions are not affected by any results of the STRING statement.

If *identifier-3* and *identifier-4* occupy the same storage area, undefined results will occur, even if the identifiers are defined by the same data description entry.

If *identifier-1* or *identifier-2* occupies the same storage area as *identifier-3* or *identifier-4*, undefined results will occur, even if the identifiers are defined by the same data description entry.

Note: If a UTF-8 receiver is specified, the STRING statement might produce invalid UTF-8 byte sequences if the receiver contained 2-, 3-, or 4-byte encodings prior to STRING statement processing. To avoid this, initialize UTF-8 receivers with a single byte character encoding, such as SPACES.

See [“Data flow” on page 465](#) for details of STRING statement processing.

ON OVERFLOW phrases

imperative-statement-1

Executed when the pointer value (explicit or implicit):

- Is less than 1
- Exceeds a value equal to the length of the receiving field

When either of the above conditions occurs, an overflow condition exists, and no more data is transferred. Then the STRING operation is terminated, the NOT ON OVERFLOW phrase, if specified, is ignored, and control is transferred to the end of the STRING statement or, if the ON OVERFLOW phrase is specified, to *imperative-statement-1*.

If control is transferred to *imperative-statement-1*, execution continues according to the rules for each statement specified in *imperative-statement-1*. If a procedure branching or conditional statement that causes explicit transfer of control is executed, control is transferred according to the rules for that statement; otherwise, upon completion of the execution of *imperative-statement-1*, control is transferred to the end of the STRING statement.

If at the time of execution of a STRING statement, conditions that would cause an overflow condition are not encountered, then after completion of the transfer of data, the ON OVERFLOW phrase, if specified, is ignored. Control is then transferred to the end of the STRING statement, or if the NOT ON OVERFLOW phrase is specified, to *imperative-statement-2*.

If control is transferred to *imperative-statement-2*, execution continues according to the rules for each statement specified in *imperative-statement-2*. If a procedure branching or conditional statement that causes explicit transfer of control is executed, control is transferred according to the rules for that statement. Otherwise, upon completion of the execution of *imperative-statement-2*, control is transferred to the end of the STRING statement.

END-STRING phrase

This explicit scope terminator serves to delimit the scope of the STRING statement. END-STRING permits a conditional STRING statement to be nested in another conditional statement. END-STRING can also be used with an imperative STRING statement.

For more information, see [“Delimited scope statements” on page 293](#).

Data flow

When the STRING statement is executed, characters are transferred from the sending fields to the receiving field. The order in which sending fields are processed is the order in which they are specified.

The following rules apply:

- Characters from the sending fields are transferred to the receiving fields in the following manner:
 - For national sending fields, data is transferred using the rules of the MOVE statement for elementary national-to-national moves, except that no space filling takes place.
 - For UTF-8 sending fields, data is transferred using the rules of the MOVE statement for elementary UTF-8-to-UTF-8 moves, except that no space filling takes place.
 - For DBCS sending fields, data is transferred using the rules of the MOVE statement for elementary DBCS-to-DBCS moves, except that no space filling takes place.

- Otherwise, data is transferred to the receiving fields using the rules of the MOVE statement for elementary alphanumeric-to-alphanumeric moves, except that no space filling takes place (see “MOVE statement” on page 405).
- When DELIMITED BY *identifier-2* or *literal-2* is specified, the contents of each sending item are transferred, character-by-character, beginning with the leftmost character position and continuing until either:
 - A delimiter for this sending field is reached (the delimiter itself is not transferred).
 - The rightmost character of this sending field has been transferred.
- When DELIMITED BY SIZE is specified, each entire sending field is transferred to the receiving field.
- When the receiving field is filled, or when all the sending fields have been processed, the operation is ended.
- When the POINTER phrase is specified, an explicit pointer field is available to the COBOL user to control placement of data in the receiving field. The user must set the explicit pointer's initial value, which must not be less than 1 and not more than the character position count of the receiving field.

Usage note: The pointer field must be defined as a field large enough to contain a value equal to the length of the receiving field plus 1; this precludes arithmetic overflow when the system updates the pointer at the end of the transfer.

- When the POINTER phrase is not specified, no pointer is available to the user. However, a conceptual implicit pointer with an initial value of 1 is used by the system.
- Conceptually, when the STRING statement is executed, the initial pointer value (explicit or implicit) is the first character position within the receiving field into which data is to be transferred. Beginning at that position, data is then positioned, character-by-character, from left to right. After each character is positioned, the explicit or implicit pointer is increased by 1. The value in the pointer field is changed only in this manner. At the end of processing, the pointer value always indicates a value equal to one character position beyond the last character transferred into the receiving field.

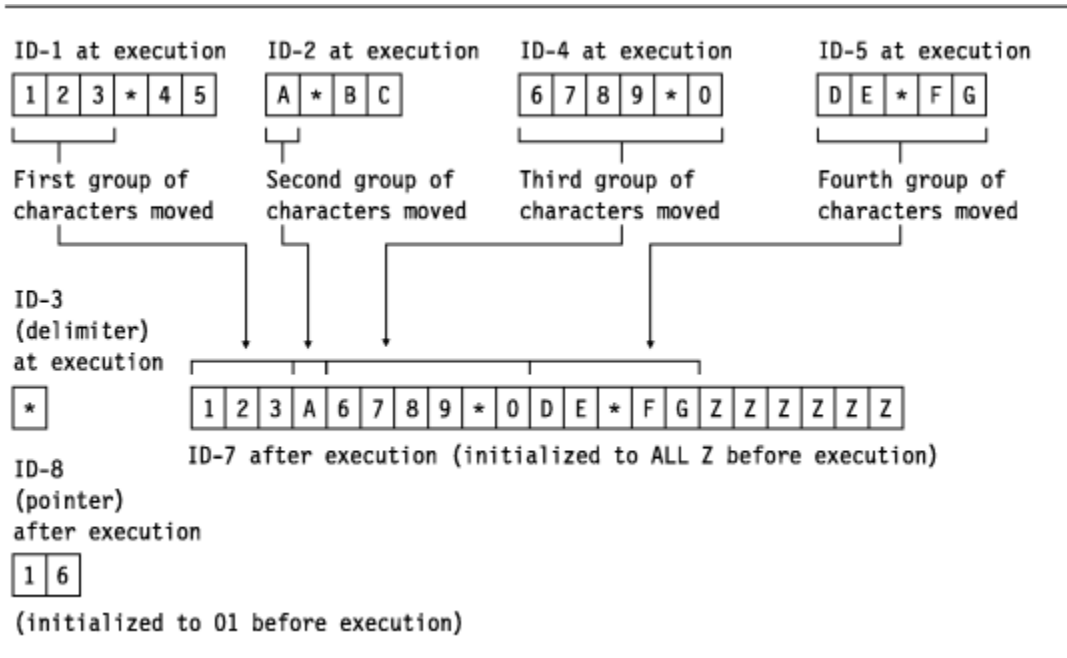
After STRING statement execution is completed, only that part of the receiving field into which data was transferred is changed. The rest of the receiving field contains the data that was present before this execution of the STRING statement.

Example of the STRING statement

This topic lists an example for the STRING statement.

When the following STRING statement is executed, the results obtained will be like those illustrated in the figure after the statement.

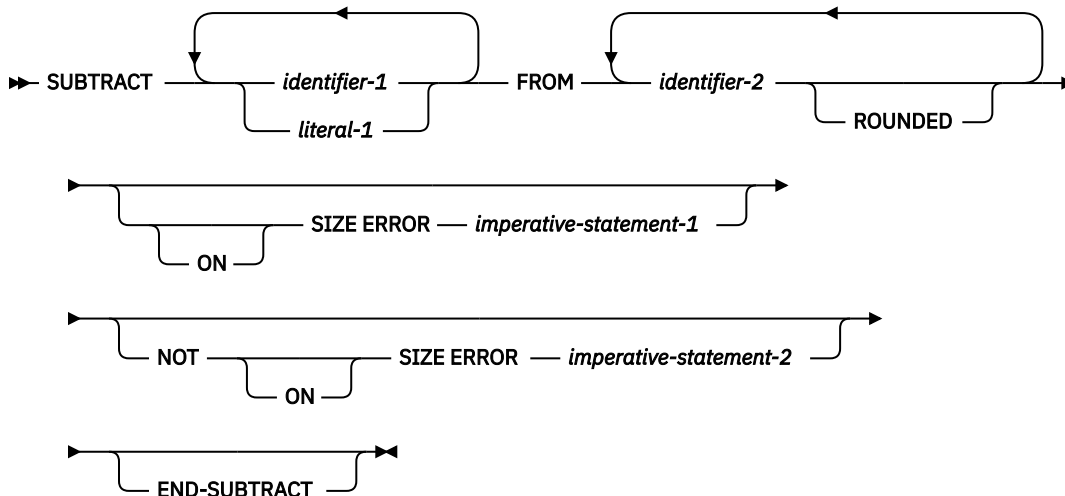
```
STRING ID-1 ID-2 DELIMITED BY ID-3
      ID-4 ID-5 DELIMITED BY SIZE
      INTO ID-7 WITH POINTER ID-8
END-STRING
```



SUBTRACT statement

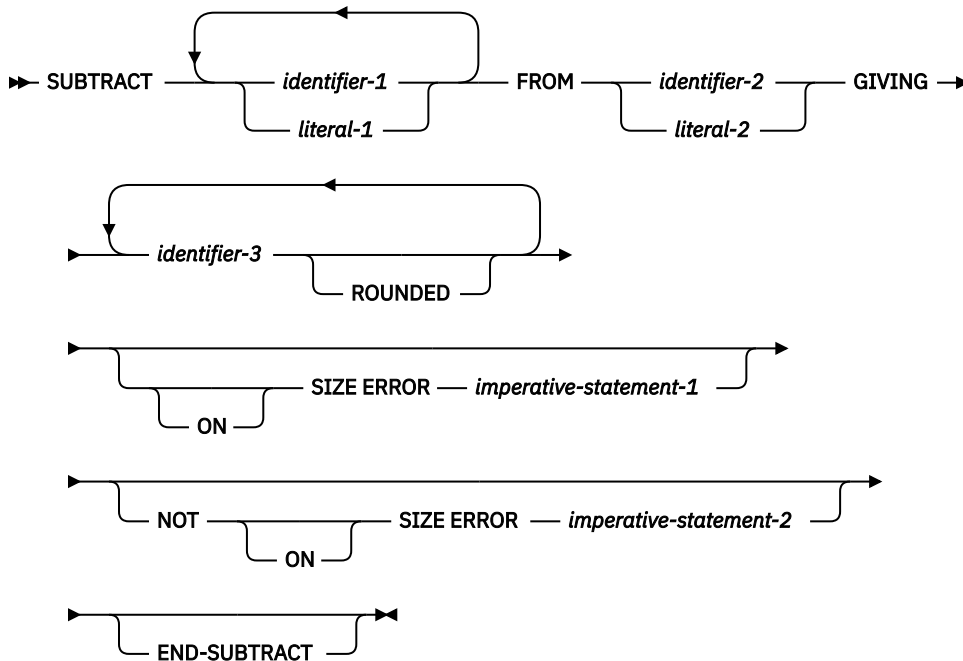
The SUBTRACT statement subtracts one numeric item, or the sum of two or more numeric items, from one or more numeric items, and stores the result.

Format 1: SUBTRACT statement



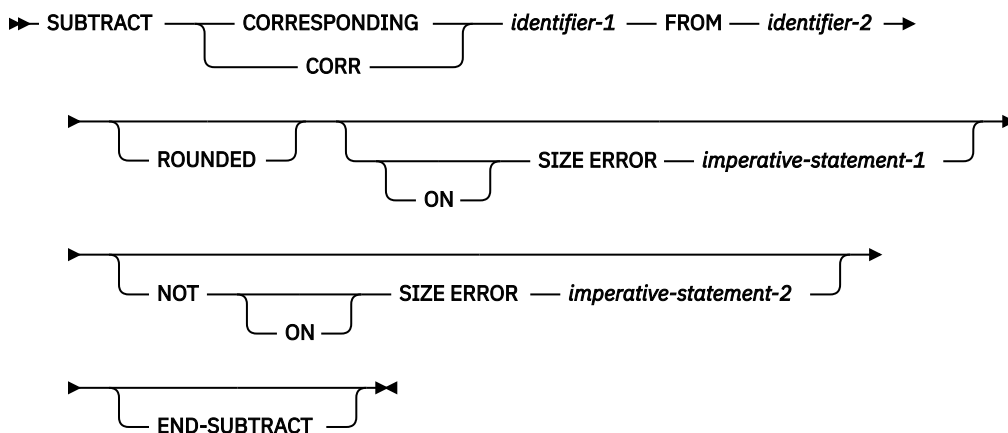
All identifiers or literals preceding the keyword FROM are added together and their sum is subtracted from and stored immediately in *identifier-2*. This process is repeated for each successive occurrence of *identifier-2*, in the left-to-right order in which *identifier-2* is specified.

Format 2: SUBTRACT statement with GIVING phrase



All identifiers or literals preceding the keyword FROM are added together and their sum is subtracted from *identifier-2* or *literal-2*. The result of the subtraction is stored as the new value of each data item referenced by *identifier-3*.

Format 3: SUBTRACT statement with CORRESPONDING phrase



Elementary data items within *identifier-1* are subtracted from, and the results are stored in, the corresponding elementary data items within *identifier-2*.

When the ARITH(COMPAT) compiler option is in effect, the composite of operands can contain a maximum of 30 digits. When the ARITH(EXTEND) compiler option is in effect, the composite of operands can contain a maximum of 31 digits. For more information about arithmetic intermediate results, see *Appendix A. Intermediate results and arithmetic precision* in the *Enterprise COBOL Programming Guide*.

For all formats:

identifier

In format 1, must name an elementary numeric data item.

In format 2, must name an elementary numeric data item, unless the identifier follows the word GIVING. Each identifier following the word GIVING must name a numeric or numeric-edited elementary data item.

In format 3, must name an alphanumeric group item or a national group item.

literal

Must be a numeric literal.

Floating-point data items and literals can be used anywhere numeric data items and literals can be specified.

ROUNDED phrase

For information about the ROUNDED phrase, and for operand considerations, see [“ROUNDED phrase” on page 296](#).

SIZE ERROR phrases

For information about the SIZE ERROR phrases, and for operand considerations, see [“SIZE ERROR phrases” on page 296](#).

CORRESPONDING phrase (format 3)

See [“CORRESPONDING phrase” on page 295](#).

END-SUBTRACT phrase

This explicit scope terminator serves to delimit the scope of the SUBTRACT statement. END-SUBTRACT permits a conditional SUBTRACT statement to be nested in another conditional statement. END-SUBTRACT can also be used with an imperative SUBTRACT statement.

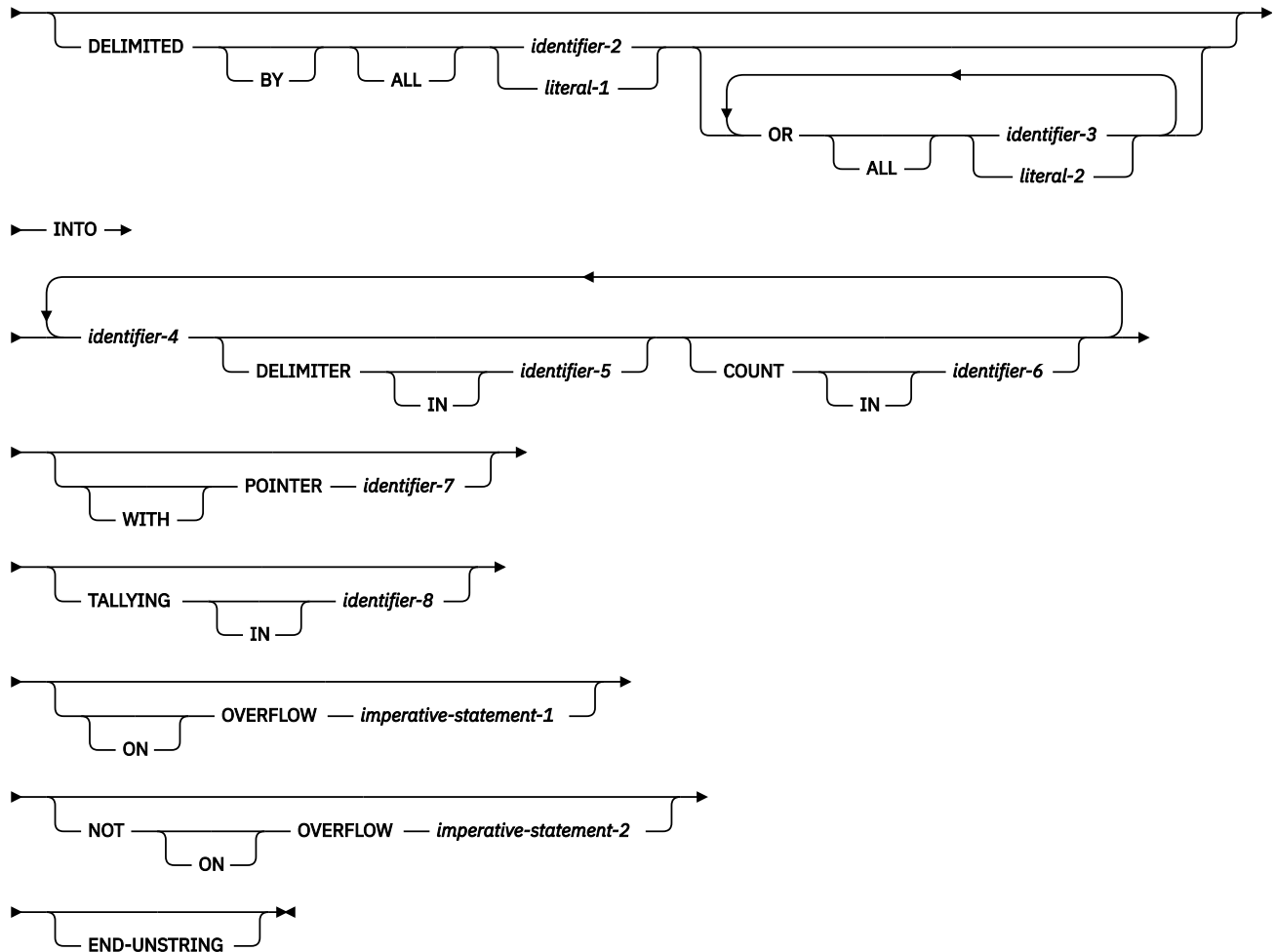
For more information, see [“Delimited scope statements” on page 293](#).

UNSTRING statement

The UNSTRING statement causes contiguous data in a sending field to be separated and placed into multiple receiving fields.

Format

►► UNSTRING — *identifier-1* →



identifier-1

Represents the *sending field*. Data is transferred from this field to the data receiving fields (*identifier-4*).

identifier-1 must reference a data item of category alphabetic, alphanumeric, alphanumeric-edited, DBCS, national, national-edited, or UTF-8.

identifier-2, *literal-1*, *identifier-3*, *literal-2*

Specifies one or more delimiters.

identifier-2 and *identifier-3* must reference data items of category alphabetic, alphanumeric, alphanumeric-edited, DBCS, national, national-edited, or UTF-8.

literal-1 or *literal-2* must be of category alphanumeric, DBCS, national, or UTF-8 and must not be a figurative constant that begins with the word ALL.

identifier-4

Specifies one or more receiving fields.

identifier-4 must reference a data item of category alphabetic, alphanumeric, numeric, DBCS, national, or UTF-8. If the referenced data item is of category numeric, its picture character-string must not contain the picture symbol P, and its usage must be DISPLAY or NATIONAL.

identifier-5

Specifies a field to receive the delimiter associated with *identifier-4*.

identifier-5 must reference a data item of category alphabetic, alphanumeric, DBCS, national, or UTF-8.

identifier-6

Specifies a field to hold the count of characters that are transferred to *identifier-4*.

identifier-6 must be an integer data item defined without the symbol P in its PICTURE character-string.

identifier-7

Specifies a field to hold a relative character position during UNSTRING processing.

identifier-7 must be an integer data item defined without the symbol P in the PICTURE string.

identifier-7 must be described as a data item of sufficient size to contain a value equal to 1 plus the number of character positions in the data item referenced by *identifier-1*.

identifier-8

Specifies a field that is incremented by the number of delimited fields processed.

identifier-8 must be an integer data item defined without the symbol P in its PICTURE character-string.

The following rules apply:

- If *identifier-4* references a data item of usage DISPLAY, *identifier-1*, *identifier-2*, *identifier-3*, and *identifier-5* must also reference data items of usage DISPLAY and all literals must be alphanumeric literals. Any figurative constant can be specified except NULL or one that begins with the word ALL. Each figurative constant represents a 1-character alphanumeric literal.
- If *identifier-4* references a data item of usage DISPLAY-1, *identifier-1*, *identifier-2*, *identifier-3*, and *identifier-5* must also reference data items of usage DISPLAY-1 and all literals must be DBCS literals. Figurative constant SPACE is the only figurative constant that can be specified. Each figurative constant represents a 1-character DBCS literal.
- If *identifier-4* references a data item of usage NATIONAL, *identifier-1*, *identifier-2*, *identifier-3*, and *identifier-5* must also reference data items of usage NATIONAL and all literals must be national literals. Any figurative constant can be specified except NULL or one that begins with the word ALL. Each figurative constant represents a 1-character national literal.
- If *identifier-4* references a data item of usage UTF-8, *identifier-1*, *identifier-2*, *identifier-3*, and *identifier-5* must also reference data items of usage UTF-8 and all literals must be UTF-8 literals. Any figurative constant can be specified except NULL or one that begins with the word ALL. Each figurative constant represents a 1-character UTF-8 literal.

Count fields (*identifier-6*) and pointer fields (*identifier-7*) are incremented by number of character positions (alphanumeric, DBCS, national, or UTF-8), not by number of bytes.

One UNSTRING statement can take the place of a series of MOVE statements, except that evaluation or calculation of certain elements is performed only once, at the beginning of the execution of the UNSTRING statement. For more information, see [“Values at the end of execution of the UNSTRING statement” on page 475](#).

The rules for moving are the same as those for a MOVE statement for an elementary sending item of the category of *identifier-1*, with the appropriate *identifier-4* as the receiving item (see [“MOVE statement” on page 405](#)). For example, rules for moving a DBCS item are used when *identifier-1* is a DBCS item.

DELIMITED BY phrase

This phrase specifies delimiters within the data that control the data transfer.

Each *identifier-2*, *identifier-3*, *literal-1*, or *literal-2* represents one delimiter.

If the DELIMITED BY phrase is *not* specified, the DELIMITER IN and COUNT IN phrases must *not* be specified.

ALL

Multiple contiguous occurrences of any delimiters are treated as if there were only one occurrence; this one occurrence is moved to the delimiter receiving field (*identifier-5*), if specified. The delimiting characters in the sending field are treated as an elementary item of the same usage and category as *identifier-1* and are moved into the current delimiter receiving field according to the rules of the MOVE statement.

When DELIMITED BY ALL is *not* specified, and two or more contiguous occurrences of any delimiter are encountered, the current data receiving field (*identifier-4*) is filled with spaces or zeros, according to the description of the data receiving field.

Delimiter with two or more characters

A delimiter that contains two or more characters is recognized as a delimiter only if the delimiting characters are in both of the following cases:

- Contiguous
- In the sequence specified in the sending field

Two or more delimiters

When two or more delimiters are specified, an OR condition exists, and each nonoverlapping occurrence of any one of the delimiters is recognized in the sending field in the sequence specified.

For example:

```
DELIMITED BY "AB" or "BC"
```

An occurrence of either AB or BC in the sending field is considered a delimiter. An occurrence of ABC is considered an occurrence of AB.

INTO phrase

This phrase specifies the fields where the data is to be moved.

identifier-4 represents the *data receiving fields*.

DELIMITER IN

This phrase specifies the fields where the delimiters are to be moved.

identifier-5 represents the *delimiter receiving fields*.

The DELIMITER IN phrase must *not* be specified if the DELIMITED BY phrase is *not* specified.

COUNT IN

This phrase specifies the field where the count of examined character positions is held.

identifier-6 is the *data count field* for each data transfer. Each field holds the count of examined character positions in the sending field, terminated by the delimiters or the end of the sending field, for the move to this receiving field. The delimiters are not included in this count.

The COUNT IN phrase must *not* be specified if the DELIMITED BY phrase is *not* specified.

POINTER phrase

When the POINTER phrase is specified, the value of the pointer field, *identifier-7*, behaves as if it were increased by 1 for each examined character position in the sending field. When execution of the

UNSTRING statement is completed, the pointer field contains a value equal to its initial value plus the number of character positions examined in the sending field.

When this phrase is specified, the user must initialize the pointer field before execution of the UNSTRING statement begins.

TALLYING IN phrase

When the TALLYING phrase is specified, the area count field, *identifier-8*, contains (at the end of execution of the UNSTRING statement) a value equal to the initial value plus the number of data receiving areas acted upon.

When this phrase is specified, the user must initialize the area count field before execution of the UNSTRING statement begins.

ON OVERFLOW phrases

An overflow condition exists when:

- The pointer value (explicit or implicit) is less than 1.
- The pointer value (explicit or implicit) exceeds a value equal to the length of the sending field.
- All data receiving fields have been acted upon and the sending field still contains unexamined character positions.

When an overflow condition occurs

An overflow condition results in the following actions:

1. No more data is transferred.
2. The UNSTRING operation is terminated.
3. The NOT ON OVERFLOW phrase, if specified, is ignored.
4. Control is transferred to the end of the UNSTRING statement or, if the ON OVERFLOW phrase is specified, to *imperative-statement-1*.

imperative-statement-1

Statement or statements for dealing with an overflow condition.

If control is transferred to *imperative-statement-1*, execution continues according to the rules for each statement specified in *imperative-statement-1*. If a procedure branching or conditional statement that causes explicit transfer of control is executed, control is transferred according to the rules for that statement. Otherwise, upon completion of the execution of *imperative-statement-1*, control is transferred to the end of the UNSTRING statement.

When an overflow condition does not occur

When, during execution of an UNSTRING statement, conditions that would cause an overflow condition are not encountered, then:

1. The transfer of data is completed.
2. The ON OVERFLOW phrase, if specified, is ignored.
3. Control is transferred to the end of the UNSTRING statement or, if the NOT ON OVERFLOW phrase is specified, to *imperative-statement-2*.

imperative-statement-2

Statement or statements for dealing with an overflow condition that does not occur.

If control is transferred to *imperative-statement-2*, execution continues according to the rules for each statement specified in *imperative-statement-2*. If a procedure branching or conditional statement that causes explicit transfer of control is executed, control is transferred according to the rules for

that statement. Otherwise, upon completion of the execution of *imperative-statement-2*, control is transferred to the end of the UNSTRING statement.

The ON OVERFLOW phrase can be used for examining the sending field whereas the NOT ON OVERFLOW phrase is for normal execution when an overflow condition does not occur. You must include NOT ON OVERFLOW if you want to specify procedures to be executed only when an overflow condition does not occur. For example:

```
UNSTRING COLOR-LIST
  ON OVERFLOW
    DISPLAY 'Error: The string is too large'
  NOT ON OVERFLOW *> Execute when the UNSTRING is successful
    PERFORM SORT-COLORS
END-UNSTRING
```

END-UNSTRING phrase

This explicit scope terminator serves to delimit the scope of the UNSTRING statement. END-UNSTRING permits a conditional UNSTRING statement to be nested in another conditional statement. END-UNSTRING can also be used with an imperative UNSTRING statement.

For more information, see [“Delimited scope statements” on page 293](#).

Data flow

The data flow for the UNSTRING statement is based on certain rules.

When the UNSTRING statement is initiated, data is transferred from the sending field to the current data receiving field, according to the following rules:

1. Examine

- If the POINTER phrase is specified, the field is examined, beginning at the relative character position specified by the value in the pointer field.

If the POINTER phrase is *not* specified, the sending field character-string is examined, beginning with the leftmost character position.

- If the DELIMITED BY phrase is specified, the examination proceeds from left to right, examining character positions one-by-one until a delimiter is encountered. If the end of the sending field is reached before a delimiter is found, the examination ends with the last character position in the sending field. If there are more receiving fields, the next one is selected; otherwise, an overflow condition occurs.

If the DELIMITED BY phrase is *not* specified, the number of character positions examined is equal to the size of the current data receiving field, as described in the table below. The size depends on the category treatment of the receiving field, as shown in [Table 40 on page 359](#).

If the DELIMITED BY phrase is *not* specified and the receiver is a dynamic-length elementary item, the number of character positions examined is equal to the length of the sender. All identifiers cannot be dynamic-length group items.

Table 58. <i>Character positions examined when DELIMITED BY is not specified</i>	
If the receiving field is ...	The number of character positions examined is ...
Alphanumeric or alphabetic	Equal to the number of alphanumeric character positions in the current receiving field
DBCS	Equal to the number of DBCS character positions in the current receiving field

<i>Table 58. Character positions examined when DELIMITED BY is not specified (continued)</i>	
If the receiving field is ...	The number of character positions examined is ...
National	Equal to the number of national character positions in the current receiving field
Numeric	Equal to the number of character positions in the integer portion of the current receiving field
UTF-8	Equal to the number of UTF-8 character positions in the current receiving field
Described with the SIGN IS SEPARATE clause	1 less than the size of the current receiving field
Described as a variable-length data item	Determined by the size of the current receiving field at the beginning of the UNSTRING operation

2. Move

- The examined character positions (excluding any delimiter characters) are treated as an elementary data item of the same data category as the sending field except for the cases shown in the table below.

Category of <i>identifier-1</i> (sending-field)	Category of elementary data item
Alphanumeric-edited	Alphanumeric
National-edited	National

That elementary data item is moved to the current data receiving field according to the rules for the MOVE statement for the categories of the sending and receiving fields as described in [“MOVE statement”](#) on page 405.

- If the DELIMITER IN phrase is specified, the delimiting characters in the sending field are treated as an elementary alphanumeric item and are moved to the current delimiter receiving field, according to the rules for the MOVE statement. If the delimiting condition is the end of the sending field, the current delimiter receiving field is filled with spaces.
- If the COUNT IN phrase is specified, a value equal to the number of examined character positions (excluding any delimiters) is moved into the data count field, according to the rules for an elementary move.

3. Successive iterations

- If the DELIMITED BY phrase is specified, the sending field is further examined, beginning with the first character position to the right of the delimiter.

If the DELIMITED BY phrase is *not* specified, the sending field is further examined, beginning with the first character position to the right of the last character position examined.

- For each succeeding data receiving field, this process of examining and moving is repeated until either of the following conditions occurs:
 - All the characters in the sending field have been transferred.
 - There are no more unfilled data receiving fields.

Values at the end of execution of the UNSTRING statement

At the beginning of the execution of the UNSTRING statement, certain operations are performed only once.

The operations are:

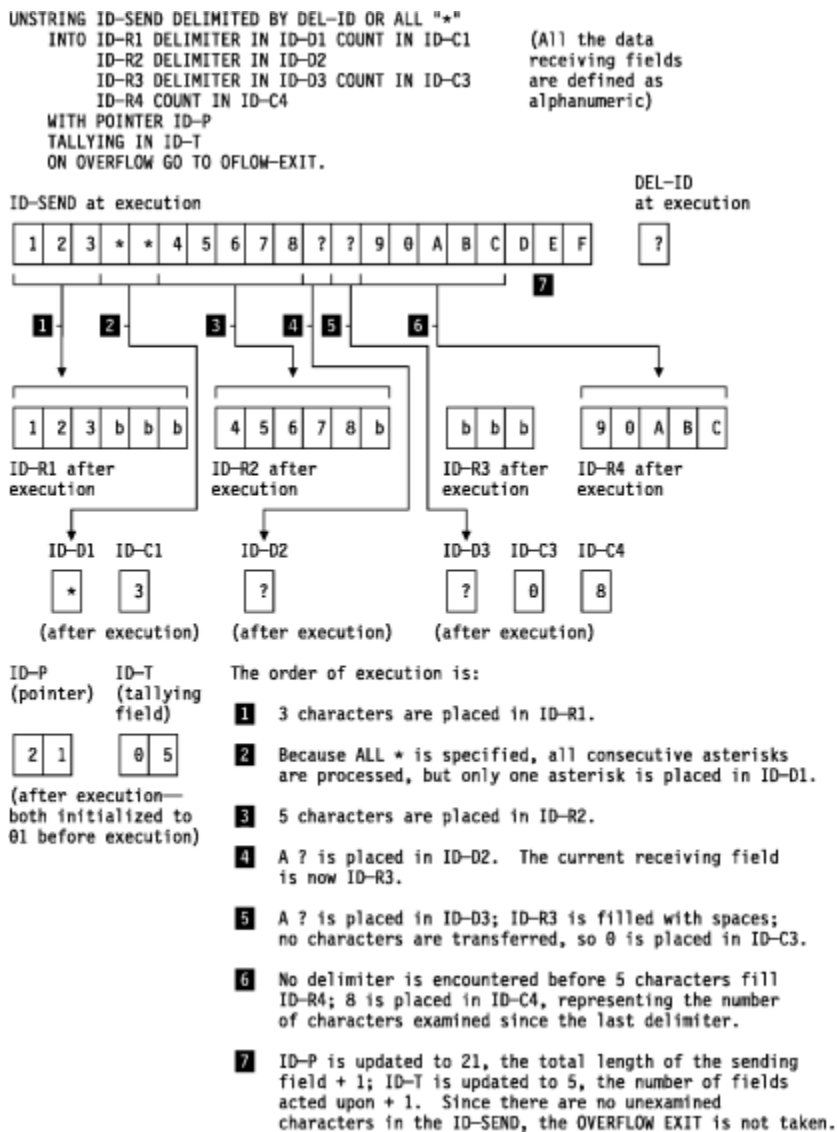
- Calculations of subscripts, reference modifications, variable-lengths, variable locations
- Evaluations of functions

Therefore, if *identifier-4*, *identifier-5*, *identifier-6*, *identifier-7*, or *identifier-8* is used as a subscript, reference-modifier, or function argument in the UNSTRING statement, or affects the length or location of any of the identifiers in the UNSTRING statement, these values are determined at the beginning of the UNSTRING statement, and are *not* affected by any results of the UNSTRING statement.

Example of the UNSTRING statement

This topic lists an example for the UNSTRING statement.

The following figure shows the execution results for an example of the UNSTRING statement.



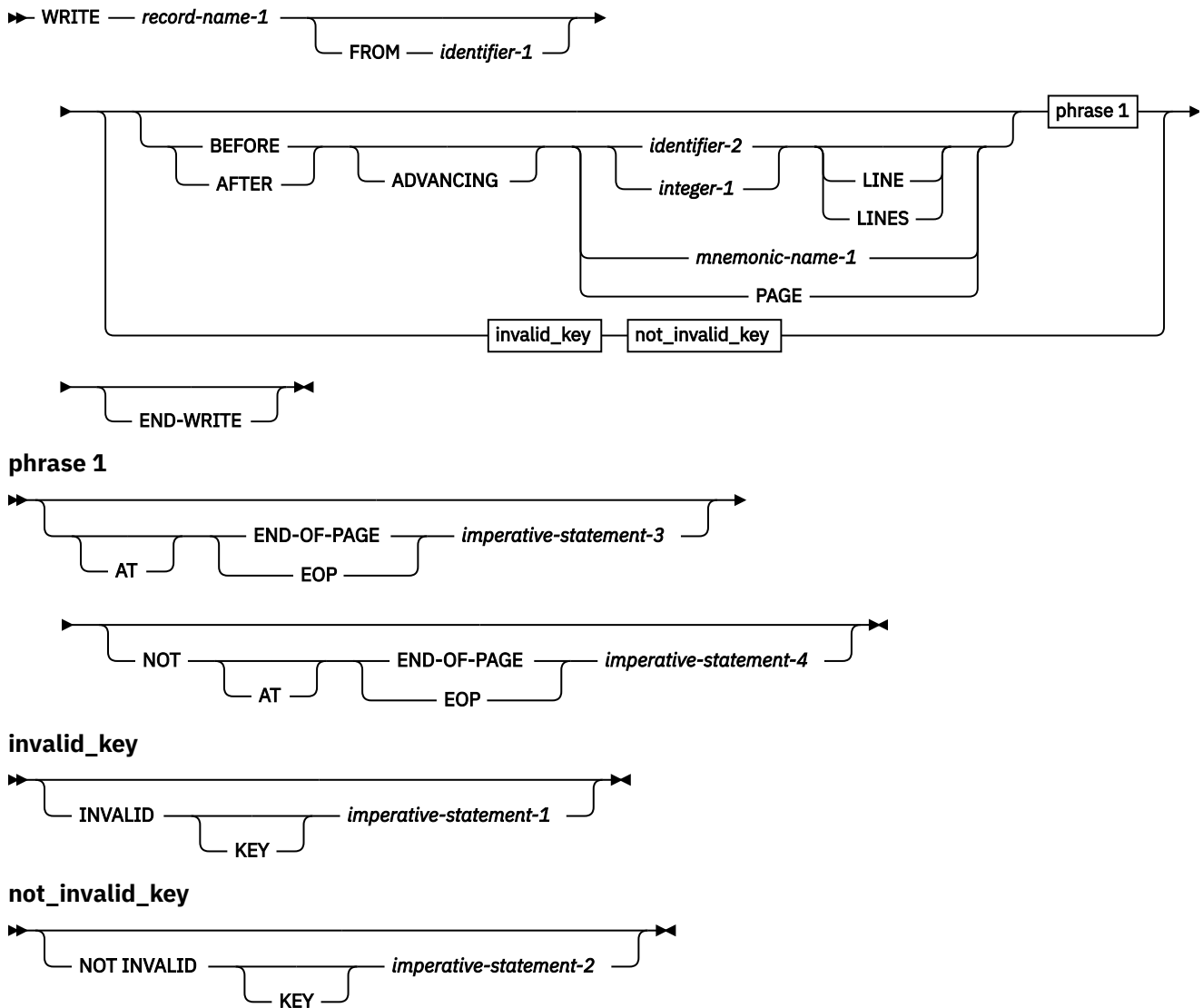
WRITE statement

The WRITE statement releases a logical record to an output or input/output file.

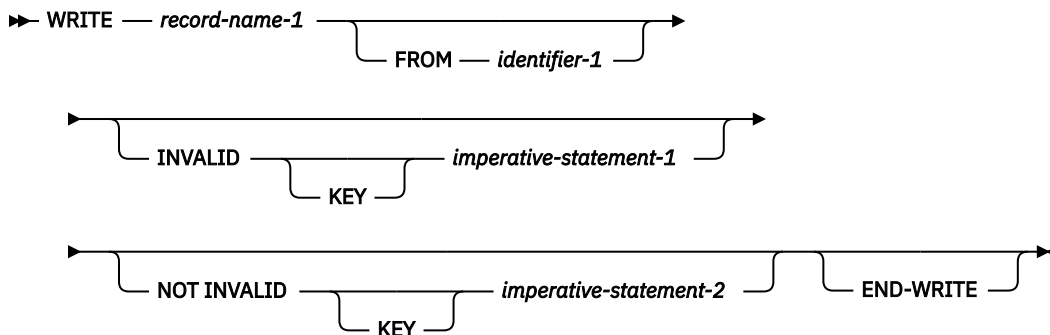
When the WRITE statement is executed:

- The associated sequential file must be open in OUTPUT or EXTEND mode.
- The associated indexed or relative file must be open in OUTPUT, I-O, or EXTEND mode.

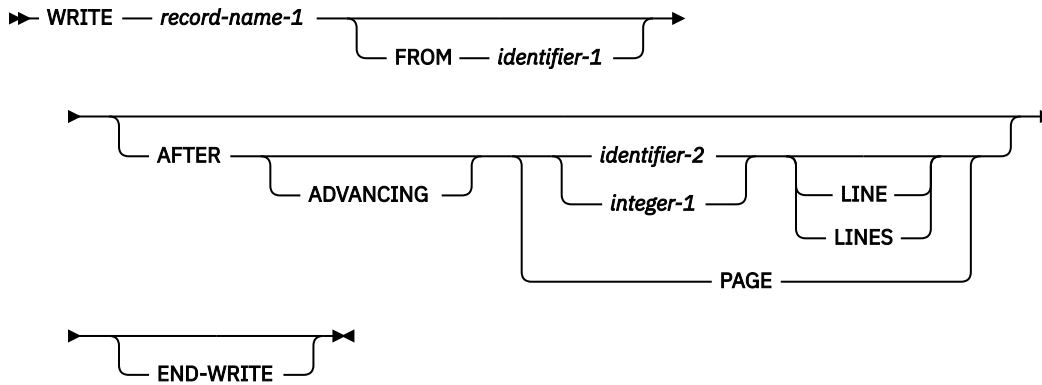
Format 1: WRITE statement for sequential files



Format 2: WRITE statement for indexed and relative files



Format 3: WRITE statement for line-sequential files



record-name-1

Must be defined in a DATA DIVISION FD entry. *record-name-1* can be qualified. It must not be associated with a sort or merge file.

For relative files, the number of character positions in the record being written can be different from the number of character positions in the record being replaced.

FROM phrase

The result of the execution of the WRITE statement with the FROM *identifier-1* phrase is equivalent to the execution of the following statements in the order specified:

```
MOVE identifier-1 TO record-name-1.  
WRITE record-name-1.
```

The MOVE is performed according to the rules for a MOVE statement without the CORRESPONDING phrase.

identifier-1

identifier-1 can reference any of the following items:

- A data item defined in the WORKING-STORAGE SECTION, the LOCAL-STORAGE SECTION, or the LINKAGE SECTION
- A record description for another previously opened file
- An alphanumeric function
- A national function

identifier-1 must be a valid sending item for a MOVE statement with *record-name-1* as the receiving item.

identifier-1 and *record-name-1* must not refer to the same storage area.

After the WRITE statement is executed, the information is still available in *identifier-1*. (See [“INTO and FROM phrases”](#) on page 304 under Common processing facilities.)

identifier-2

Must be an integer data item.

ADVANCING phrase

The ADVANCING phrase controls positioning of the output record on the page.

The BEFORE and AFTER phrases are not supported for VSAM files. QSAM files are sequentially organized. The ADVANCING and END-OF-PAGE phrases control the vertical positioning of each line on a printed page.

You can specify the ADVANCING PAGE and END-OF-PAGE phrases in a single WRITE statement.

If the printed page is held on an intermediate device (a disk, for example), the format can appear different from the expected format when the output is edited or browsed.

ADVANCING phrase rules

When the ADVANCING phrase is specified, the following rules apply:

1. When BEFORE ADVANCING is specified, the line is printed before the page is advanced.
2. When AFTER ADVANCING is specified, the page is advanced before the line is printed.
3. When *identifier-2* is specified, the page is advanced the number of lines equal to the current value in *identifier-2*. *identifier-2* must name an elementary integer data item.
4. When integer is specified, the page is advanced the number of lines equal to the value of integer.
5. Integer or the value in *identifier-2* can be zero.
6. When PAGE is specified, the record is printed on the logical page BEFORE or AFTER (depending on the phrase used) the device is positioned to the next logical page. If PAGE has no meaning for the device used, then BEFORE or AFTER (depending on the phrase specified) ADVANCING 1 LINE is provided.

If the FD entry contains a LINAGE clause, the repositioning is to the first printable line of the next page, as specified in that clause. If the LINAGE clause is omitted, the repositioning is to line 1 of the next succeeding page.
7. When *mnemonic-name* is specified, a skip to channels 1 through 12, or space suppression, takes place. *mnemonic-name* must be equated with *environment-name-1* in the SPECIAL-NAMES paragraph.

The *mnemonic-name* phrase can also be specified for stacker selection with a card punch file. When using stacker selection, WRITE AFTER ADVANCING must be used.

The ADVANCING phrase of the WRITE statement, or the presence of a LINAGE clause on the file, causes a carriage control character to be generated in the record that is written. If the corresponding file is described with the EXTERNAL clause, all file connectors within the run unit must be defined such that carriage control characters will be generated for records that are written. That is, if all the files have a LINAGE clause, some of the programs can use the WRITE statement with the ADVANCING phrase and other programs can use the WRITE statement without the ADVANCING phrase. However, if none of the files has a LINAGE clause, then if any of the programs use the WRITE statement with the ADVANCING phrase, all of the programs in the run unit that have a WRITE statement must use the WRITE statement with the ADVANCING phrase.

When the ADVANCING phrase is omitted, automatic line advancing is provided, as if AFTER ADVANCING 1 LINE had been specified.

When carriage control is required, the WRITE statement with the ADVANCING option is used to write records to the output file. In this case, one character position must be included as the first character of the record (if NOADV is specified). This position is filled with the carriage control character. The type of carriage control character is determined by the compiler as mentioned in the following scenarios:

- When only AFTER is specified for the file, ASA control characters are used (FBA is required).
- When only BEFORE is specified for the file, machine control characters are used (FBM is required).
- When both BEFORE and AFTER are specified for output operations, machine control characters are used (FBM is required).

To indicate that records contain carriage control characters in the output file, specify RECFM=FBA or RECFM=FBM in the DCB parameter of the output file.

If the output file does not match the required RECFM, the COBOL runtime system modifies the specified record format (RECFM) of the output file and changes it to FBA or FBM accordingly.

LINAGE-COUNTER rules

If the LINAGE clause is specified for this file, the associated LINAGE-COUNTER special register is modified during the execution of the WRITE statement, according to the following rules:

1. If ADVANCING PAGE is specified, LINAGE-COUNTER is reset to 1.
2. If ADVANCING *identifier-2* or *integer* is specified, LINAGE-COUNTER is increased by the value in *identifier-2* or *integer*.
3. If the ADVANCING phrase is omitted, LINAGE-COUNTER is increased by 1.
4. When the device is repositioned to the first available line of a new page, LINAGE-COUNTER is reset to 1.

Usage note: If you use the ADV compiler option, the compiler adds 1 byte to the record length in order to allow for the control character. If in your record definition you already reserve the first byte for the control character, you should use the NOADV option. For files defined with the LINAGE clause, the NOADV option has no effect. The compiler processes these files as if the ADV option were specified.

END-OF-PAGE phrases

The AT END-OF-PAGE phrase is not supported for VSAM files.

When END-OF-PAGE is specified, and the logical end of the printed page is reached during execution of the WRITE statement, the END-OF-PAGE imperative-statement is executed. When the END-OF-PAGE phrase is specified, the FD entry for this file must contain a LINAGE clause.

The logical end of the printed page is specified in the associated LINAGE clause.

An END-OF-PAGE condition is reached when execution of a WRITE END-OF-PAGE statement causes printing or spacing within the footing area of a page body. This occurs when execution of such a WRITE statement causes the value in the LINAGE-COUNTER special register to equal or exceed the value specified in the WITH FOOTING phrase of the LINAGE clause. The WRITE statement is executed, and then the END-OF-PAGE imperative-statement is executed.

An automatic page overflow condition is reached whenever the execution of any given WRITE statement (with or without the END-OF-PAGE phrase) cannot be completely executed within the current page body. This occurs when a WRITE statement, if executed, would cause the value in the LINAGE-COUNTER to exceed the number of lines for the page body specified in the LINAGE clause. In this case, the line is printed BEFORE or AFTER (depending on the option specified) the device is repositioned to the first printable line on the next logical page, as specified in the LINAGE clause. If the END-OF-PAGE phrase is specified, the END-OF-PAGE imperative-statement is then executed.

If the WITH FOOTING phrase of the LINAGE clause is not specified, the automatic page overflow condition exists because no end-of-page condition (as distinct from the page overflow condition) can be detected.

If the WITH FOOTING phrase is specified, but the execution of a given WRITE statement would cause the LINAGE-COUNTER to exceed both the footing value and the page body value specified in the LINAGE clause, then both the end-of-page condition and the automatic page overflow condition occur simultaneously.

The keywords END-OF-PAGE and EOP are equivalent.

You can specify both the ADVANCING PAGE phrase and the END-OF-PAGE phrase in a single WRITE statement.

INVALID KEY phrases

The INVALID KEY phrase is not supported for VSAM sequential files.

An invalid key condition is caused by the following cases:

- For sequential files, an attempt is made to write beyond the externally defined boundary of the file.

- For indexed files:
 - An attempt is made to write beyond the externally defined boundary of the file.
 - ACCESS SEQUENTIAL is specified and the file is opened OUTPUT, and the value of the prime record key is not greater than that of the previous record.
 - The file is opened OUTPUT or I-O and the value of the prime record key equals that of an already existing record.
- For relative files:
 - An attempt is made to write beyond the externally defined boundary of the file.
 - When the access mode is random or dynamic and the RELATIVE KEY data item specifies a record that already exists in the file.
 - The number of significant digits in the relative record number is larger than the size of the relative key data item for the file.

When an invalid key condition occurs:

- If the INVALID KEY phrase is specified, *imperative-statement-1* is executed. For details of invalid key processing, see [Invalid key condition](#).
- Otherwise, the WRITE statement is unsuccessful and the contents of *record-name* are unaffected (except for QSAM files) and the following case occurs:
 - For sequential files, the file status key, if specified, is updated and an EXCEPTION/ERROR condition exists.
 If an explicit or implicit EXCEPTION/ERROR procedure is specified for the file, the procedure is executed. If no such procedure is specified, the results are unpredictable.
 - For relative and indexed files, program execution proceeds according to the rules described by [Invalid key condition](#) under Common processing facilities.
 The INVALID KEY conditions that apply to a relative file in OPEN OUTPUT mode also apply to one in OPEN EXTEND mode.
- If the NOT INVALID KEY phrase is specified and a valid key condition exists at the end of the execution of the WRITE statement, control is passed to *imperative-statement-4*.

Both the INVALID KEY phrase and an applicable EXCEPTION/ERROR procedure can be omitted.

END-WRITE phrase

This explicit scope terminator serves to delimit the scope of the WRITE statement. END-WRITE permits a conditional WRITE statement to be nested in another conditional statement. END-WRITE can also be used with an imperative WRITE statement.

For more information, see [“Delimited scope statements” on page 293](#).

WRITE for sequential files

The maximum record dimension for sequential files is established at the time the file is created and cannot subsequently be changed.

After the WRITE statement is executed, the logical record is no longer available in *record-name-1* unless either:

- The associated file is named in a SAME RECORD AREA clause (in which case, the record is also available as a record of the other files named in the SAME RECORD AREA clause)
- The WRITE statement is unsuccessful because of a boundary violation.

In either of these two cases, the logical record is still available in *record-name-1*.

The file position indicator is not affected by execution of the WRITE statement.

The number of character positions required to store the record in a file might or might not be the same as the number of character positions defined by the logical description of that record in the COBOL program. (See “[PICTURE clause editing](#)” on page 219 and “[USAGE clause](#)” on page 237.)

If the FILE STATUS clause is specified in the file-control entry, the associated file status key is updated when the WRITE statement is executed, whether or not execution is successful.

The WRITE statement can only be executed for a sequential file opened in OUTPUT or EXTEND mode for QSAM files.

Punch function files with the IBM 3525

When the punch function is used, the next I-O operation after the READ statement must be a WRITE statement for the punch function file.

If you want to punch additional data into some of the cards and not into others, a dummy WRITE statement must be executed for the null cards, first filling the output area with SPACES.

If stacker selection for the punch function file is required, you can specify the appropriate stacker function-names in the SPECIAL-NAMES paragraph, and then code WRITE ADVANCING statements using the associated mnemonic-names.

Print function files

After the punch function operations (if specified) are completed, you can issue WRITE statements for the print function file.

If you wish to print additional data on some of the data cards and not on others, the WRITE statement for the null cards can be omitted. Any attempt to write beyond the limits of the card results in abnormal termination of the application, thus, the END-OF-PAGE phrase cannot be specified.

Depending on the capabilities of the specific IBM 3525 model in use, the print file can be either a two-line print file or a multiline print file. Up to 64 characters can be printed on each line.

- For a two-line print file, the lines are printed on line 1 (top edge of card) and line 3 (between rows 11 and 12). Line control cannot be specified. Automatic spacing is provided.
- For a multiline print file, up to 25 lines of characters can be printed. Line control can be specified. If line control is not specified, automatic spacing is provided.

Line control is specified by issuing WRITE AFTER ADVANCING statements for the print function file. If line control is used for one such statement, it must be used for all other WRITE statements for the file. The maximum number of printable characters, including any space characters, is 64. Such WRITE statements must not specify space suppression.

Identifier and integer have the same meanings they have for other WRITE AFTER ADVANCING statements. However, such WRITE statements must not increase the line position on the card beyond the card limit, or abnormal termination results.

The mnemonic-name option of the WRITE AFTER ADVANCING statement can also be specified. In the SPECIAL-NAMES paragraph, the environment-names can be associated with the mnemonic-names, as shown in the following table:

<i>Table 59. Meanings of environment-names in SPECIAL NAMES paragraph</i>	
environment-name	Meaning
C02	Line 3
C03	Line 5
C04	Line 7
C05	Line 9

Table 59. *Meanings of environment-names in SPECIAL NAMES paragraph (continued)*

environment-name	Meaning
...	...
C22	Line 21
C12	Line 23

Advanced Function Printing

When you use the WRITE ADVANCING phrase with a mnemonic-name associated with environment-name AFP-5A, a Print Services Facility (PSF) control character is placed in the control character position of the output record. This control character (X'5A') allows Advanced Function Printing (AFP) services to be used. For more information, refer to the documentation for the Print Services Facility product: PSF for OS/390® & z/OS (5655-B17).

WRITE for indexed files

Before the WRITE statement is executed for indexed files, you must set the prime record key (the RECORD KEY data item, as defined in the file-control entry) to the required value. Note that RECORD KEY values must be unique within a file.

If the ALTERNATE RECORD KEY clause is also specified in the file-control entry, each alternate record key must be unique, unless the DUPLICATES phrase is specified. If the DUPLICATES phrase is specified, alternate record key values might not be unique. In this case, the system stores the records so that later sequential access to the records allows retrieval in the same order in which they were stored.

When ACCESS IS SEQUENTIAL is specified in the file-control entry, records must be released in ascending order of RECORD KEY values.

When ACCESS IS RANDOM or ACCESS IS DYNAMIC is specified in the file-control entry, records can be released in any programmer-specified order.

WRITE for relative files

For relative record OUTPUT files, the WRITE statement causes actions as described in the topic.

- If ACCESS IS SEQUENTIAL is specified:

The first record released has relative record number 1, the second record released has relative record number 2, the third number 3, and so on.

If the RELATIVE KEY is specified in the file-control entry, the relative record number of the record just released is placed in the RELATIVE KEY during execution of the WRITE statement.

- If ACCESS IS RANDOM or ACCESS IS DYNAMIC is specified, the RELATIVE KEY must contain the required relative record number for this record before the WRITE statement is executed. When the WRITE statement is executed, this record is placed at the specified relative record number position in the file.

For I-O files, either ACCESS IS RANDOM or ACCESS IS DYNAMIC must be specified; the WRITE statement inserts new records into the file. The RELATIVE KEY must contain the required relative record number for this record before the WRITE statement is executed. When the WRITE statement is executed, this record is placed at the specified relative record number position in the file.

XML GENERATE statement

The XML GENERATE statement converts data to XML format.

Format

► XML GENERATE — *identifier-1* — FROM — *identifier-2* —►

COUNT — *identifier-3* —
IN

ENCODING — *codepage* —
WITH

XML-DECLARATION —
WITH
ATTRIBUTES —
WITH

NAMESPACE — IS — *identifier-4* —
literal-4
NAMESPACE-PREFIX — IS — *identifier-5* —
literal-5

NAME — OF — *identifier-6* — IS — *literal-6* —

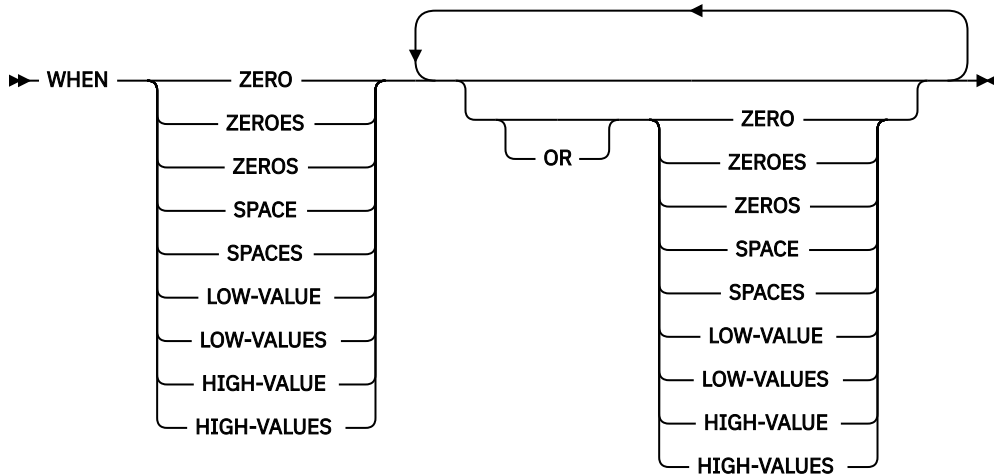
TYPE — OF — *identifier-7* — IS — ATTRIBUTE —
ELEMENT —
CONTENT

SUPPRESS — *identifier-8* —
when-phrase —
generic-suppression-phrase

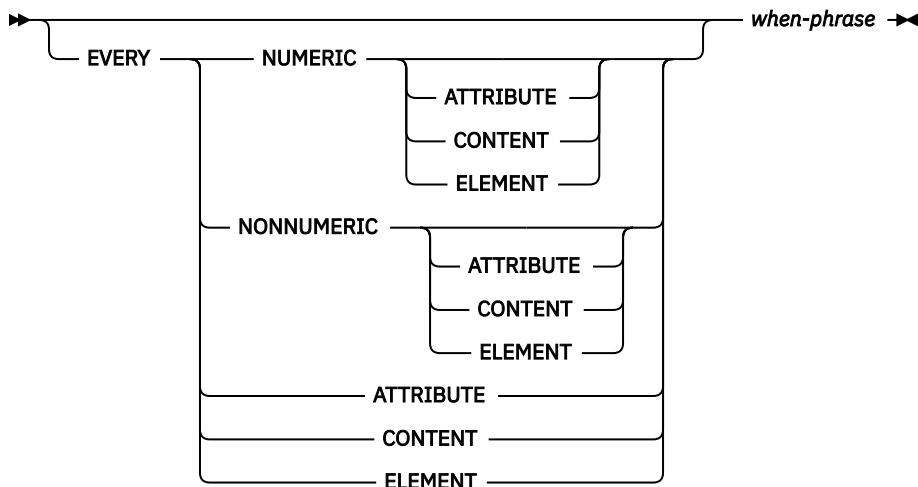
EXCEPTION — *imperative-statement-1* —
ON

NOT — ON — EXCEPTION — *imperative-statement-2* — END-XML —

***when-phrase* Format**



***generic-suppression-phrase* Format**



identifier-1

The receiving area for a generated XML document. *identifier-1* must reference one of the following items:

- An elementary data item of category alphanumeric
- An alphanumeric group item
- An elementary data item of category national
- A national group item

When *identifier-1* references a national group item, *identifier-1* is processed as an elementary data item of category national. When *identifier-1* references an alphanumeric group item, *identifier-1* is treated as though it were an elementary data item of category alphanumeric.

identifier-1 must not be described with the JUSTIFIED clause, and cannot be a function identifier. *identifier-1* can be subscripted or reference modified.

identifier-1 must not overlap *identifier-2*, *identifier-3*, *codepage* (if an identifier), *identifier-4*, or *identifier-5*.

identifier-1 must not be a dynamic-length group item or a dynamic-length elementary item.

The generated XML output is encoded as described in the documentation of the ENCODING phrase.

identifier-1 must reference a data item of category national, or the ENCODING phrase must specify 1208, if any of the following statements are true:

- The CODEPAGE compiler option specifies an EBCDIC DBCS code page.
- *identifier-4* or *identifier-5* references a data item of category national.
- *literal-4*, *literal-5*, or *literal-6* is of category national.
- The generated XML includes data from *identifier-2* for:
 - Any data item of class national or class DBCS
 - Any data item with a DBCS name (that is, a data item whose name consists of DBCS characters)
 - Any data item of class alphanumeric that contains DBCS characters

identifier-1 must be sufficient enough to contain the generated XML document. Typically, it must be from 5 to 10 times the size of *identifier-2*, depending on the length of the data-name or data-names within *identifier-2*. If *identifier-1* is not sufficient enough, an error condition exists at the end of the XML GENERATE statement.

identifier-2

The group or elementary data item to be converted to XML format.

If *identifier-2* references a national group item, *identifier-2* is processed as a group item. When *identifier-2* includes a subordinate national group item, that subordinate item is processed as a group item.

identifier-2 cannot be a function identifier or be reference modified, but it can be subscripted.

identifier-2 must not overlap *identifier-1* or *identifier-3*.

identifier-2 must not be a dynamic-length group item or a dynamic-length elementary item.

The data description entry for *identifier-2* must not contain a RENAME clause.

The following data items that are specified by *identifier-2* are ignored by the XML GENERATE statement:

- Any subordinate unnamed elementary data items or elementary FILLER data items
- Any slack bytes inserted for SYNCHRONIZED items
- Any data item subordinate to *identifier-2* that is described with the REDEFINES clause or that is subordinate to such a redefining item
- Any data item subordinate to *identifier-2* that is described with the RENAME clause
- Any group data item all of whose subordinate data items are ignored

All data items specified by *identifier-2* that are not ignored according to the previous rules must satisfy the following conditions:

- Each elementary data item must either have class alphabetic, alphanumeric, numeric, or national, or be an index data item. (That is, no elementary data item can be described with the USAGE POINTER, USAGE FUNCTION-POINTER, USAGE PROCEDURE-POINTER, or USAGE OBJECT REFERENCE phrase.)
- There must be at least one such elementary data item.
- Each non-FILLER data-name must be unique within any immediately superordinate group data item.
- Any DBCS data-names, when converted to Unicode, must be legal as names in the XML specification, version 1.0. For details about the XML specification, see [XML specification](#).

For example, consider the following data declaration:

```
01 STRUCT.
```

```

02 STAT PIC X(4).
02 IN-AREA PIC X(100).
02 OK-AREA REDEFINES IN-AREA.
   03 FLAGS PIC X.
   03 PIC X(3).
   03 COUNTER USAGE COMP-5 PIC S9(9).
   03 ASFNPTR REDEFINES COUNTER USAGE FUNCTION-POINTER.
   03 UNREFERENCED PIC X(92).
02 NG-AREA1 REDEFINES IN-AREA.
   03 FLAGS PIC X.
   03 PIC X(3).
   03 PTR USAGE POINTER.
   03 ASNUM REDEFINES PTR USAGE COMP-5 PIC S9(9).
   03 PIC X(92).
02 NG-AREA2 REDEFINES IN-AREA.
   03 FN-CODE PIC X.
   03 UNREFERENCED PIC X(3).
   03 QTYONHAND USAGE BINARY PIC 9(5).
   03 DESC USAGE NATIONAL PIC N(40).
   03 UNREFERENCED PIC X(12).

```

The following data items from the previous example can be specified as *identifier-2*:

- STRUCT, of which subordinate data items STAT and IN-AREA would be converted to XML format. (OK-AREA, NG-AREA1, and NG-AREA2 are ignored because they specify the REDEFINES clause.)
- OK-AREA, of which subordinate data items FLAGS, COUNTER, and UNREFERENCED would be converted. (The item whose data description entry specifies 03 PIC X(3) is ignored because it is an elementary FILLER data item. ASFNPTR is ignored because it specifies the REDEFINES clause.)
- Any of the elementary data items that are subordinate to STRUCT except:
 - ASFNPTR or PTR (disallowed usage)
 - UNREFERENCED OF NG-AREA2 (nonunique names for data items that are otherwise eligible)
 - Any FILLER data items

The following data items cannot be specified as *identifier-2*:

- NG-AREA1, because subordinate data item PTR specifies USAGE POINTER but does not specify the REDEFINES clause. (PTR would be ignored if it specified the REDEFINES clause.)
- NG-AREA2, because subordinate elementary data items have the nonunique name UNREFERENCED.

COUNT IN phrase

If the COUNT IN phrase is specified, *identifier-3* contains (after execution of the XML GENERATE statement) the count of generated XML character encoding units. If *identifier-1* (the receiver) has category national, the count is in UTF-16 character encoding units. For all other encodings (including UTF-8), the count is in bytes.

identifier-3

The data count field. Must be an integer data item defined without the symbol P in its picture string.

identifier-3 must not overlap *identifier-1*, *identifier-2*, *codepage* (if an identifier), *identifier-4*, or *identifier-5*.

ENCODING phrase

The ENCODING phrase, if specified, determines the encoding of the generated XML document.

codepage

Must be an unsigned integer data item or unsigned integer literal and must represent a valid coded character set identifier (CCSID). Must identify one of the code pages supported for COBOL XML processing as described in *The encoding of XML documents* in the *Enterprise COBOL Programming Guide*.

If *identifier-1* references a data item of category national, *codepage* must specify 1200, the CCSID for Unicode UTF-16.

If *identifier-1* references a data item of category alphanumeric, *codepage* must specify 1208 or the CCSID of a supported EBCDIC code page as listed in *The encoding of XML documents* in the *Enterprise COBOL Programming Guide*.

If *codepage* is an identifier, it must not overlap *identifier-1* or *identifier-3*.

If the ENCODING phrase is omitted and *identifier-1* is of category national, the document encoding is Unicode UTF-16, CCSID 1200.

If the ENCODING phrase is omitted and *identifier-1* is of category alphanumeric, the XML document is encoded using the code page specified by the CODEPAGE compiler option in effect when the source code was compiled.

If the ENCODING phrase is omitted and *identifier-1* is of category alphanumeric, the XML document is encoded using the code page specified by the EBCDIC_CODEPAGE environment variable in effect when the source code was compiled.

XML-DECLARATION phrase

If the XML-DECLARATION phrase is specified, the generated XML document starts with an XML declaration that includes the XML version information and an encoding declaration.

If *identifier-1* is of category national, the encoding declaration has the value UTF-16 (encoding="UTF-16").

If *identifier-1* is of category alphanumeric, the encoding declaration is derived from the ENCODING phrase, if specified, or from the CODEPAGE compiler option in effect for the program if the ENCODING phrase is not specified. See the description of the ENCODING phrase for further details.

For an example of the effect of coding the XML-DECLARATION phrase, see *Generating XML output* in the *Enterprise COBOL Programming Guide*.

If the XML-DECLARATION phrase is omitted, the generated XML document does not include an XML declaration.

ATTRIBUTES phrase

If the ATTRIBUTES phrase is specified, each eligible item included in the generated XML document is expressed as an attribute of the XML element that corresponds to the data item immediately superordinate to that eligible item, rather than as a child element of the XML element. To be eligible, a data item must be elementary, must have a name other than FILLER, and must not specify an OCCURS clause in its data description entry.

If the TYPE phrase is specified for particular identifiers, the TYPE phrase takes precedence for those identifiers over the WITH ATTRIBUTES phrase.

For an example of the effect of the ATTRIBUTES phrase, see *Generating XML output* in the *Enterprise COBOL Programming Guide*.

NAMESPACE and NAMESPACE-PREFIX phrases

Use the NAMESPACE phrase to identify a *namespace* for the generated XML document. If the NAMESPACE phrase is not specified, or if *identifier-4* has length zero or contains all spaces, the element names of XML documents produced by the XML GENERATE statement are not in any namespace.

Use the NAMESPACE-PREFIX phrase to qualify the start and end tag of each element in the generated XML document with a prefix.

If the NAMESPACE-PREFIX phrase is not specified, or if *identifier-5* is of length zero or contains all spaces, the namespace specified by the NAMESPACE phrase specifies the default namespace for the document. In this case, the namespace declared on the root element applies by default to each element name in the document, including that of the root element. (Default namespace declarations do not apply directly to attribute names.)

If the NAMESPACE-PREFIX phrase is specified, and *identifier-5* is not of length zero and does not contain all spaces, then the start and end tag of each element in the generated document is qualified with the specified prefix. The prefix should therefore preferably be short. When the XML GENERATE

statement is executed, the prefix must be a valid XML name, but without the colon (:), as defined in *Namespaces in XML 1.0*. The prefix can have trailing spaces, which are removed before use.

identifier-4, literal-4; identifier-5, literal-5

identifier-4, literal-4: The namespace identifier, which must be a valid *Uniform Resource Identifier (URI)* as defined in *Uniform Resource Identifier (URI): Generic Syntax*.

identifier-5, literal-5: The namespace prefix, which serves as an alias for the namespace identifier.

identifier-4 and *identifier-5* must reference data items of category alphanumeric or national.

identifier-4 and *identifier-5* must not overlap *identifier-1* or *identifier-3*.

literal-4 and *literal-5* must be of category alphanumeric or national, and must not be figurative constants.

For full details about namespaces, see *Namespaces in XML 1.0*.

For examples that show the use of the NAMESPACE and NAMESPACE-PREFIX phrases, see *Generating XML output* in the *Enterprise COBOL Programming Guide*.

NAME phrase

Allows you to supply element and attribute names.

identifier-6 must reference *identifier-2* or one of its subordinate data items. It cannot be a function identifier and cannot be reference modified or subscripted. It must not specify any data item which is ignored by the XML GENERATE statement. For more information about *identifier-2*, see [the description of identifier-2](#). If *identifier-6* is specified more than once in the NAME phrase, the last specification is used.

literal-6 must be an alphanumeric or national literal containing the attribute or element name to be generated in the XML document corresponding to *identifier-6*. It must be a valid XML local name. If *literal-6* is a national literal, *identifier-1* must reference a data item of category national or the encoding phrase must specify 1208.

TYPE phrase

Allows you to control attribute and element generation.

identifier-7 must reference an elementary data item that is subordinate to *identifier-2*. It cannot be a function identifier and cannot be reference modified or subscripted. It must not specify any data item which is ignored by the XML GENERATE statement. For more information about *identifier-2*, see [the description of identifier-2](#). If *identifier-7* is specified more than once in the TYPE phrase, the last specification is used.

- If the XML GENERATE statement also includes a WITH ATTRIBUTES phrase, the TYPE phrase has precedence for *identifier-7*.
- When ATTRIBUTE is specified, *identifier-7* must be eligible to be an XML attribute. *identifier-7* is expressed in the generated XML as an attribute of the XML element immediately superordinate to *identifier-7* rather than as a child element.
- When ELEMENT is specified, *identifier-7* is expressed in the generated XML as an element. The XML element name is derived from *identifier-7* and the element character content is derived from the converted content of *identifier-7* as described in [“Operation of XML GENERATE” on page 491](#).
- When CONTENT is specified, *identifier-7* is expressed in the generated XML as element character content of the XML element that corresponds to the data item immediately superordinate to *identifier-7*. The value of the element character content is derived from the converted content of *identifier-7* as described in [“Operation of XML GENERATE” on page 491](#). When CONTENT is specified for multiple identifiers all corresponding to the same superordinate identifier, the multiple contributions to the element character content are concatenated.

SUPPRESS phrase

Allows you to identify and unconditionally suppress items that are subordinate to *identifier-2* and selectively generate output for the XML GENERATE statement. If the SUPPRESS phrase is specified, *identifier-1* must be sufficient enough to contain the generated XML document before any suppression.

With the *generic-suppression-phrase*, elementary items subordinate to *identifier-2* that are not otherwise ignored by XML GENERATE operations are identified generically for potential suppression. Either items of class numeric, if the NUMERIC keyword is specified, or items that are not of class numeric, if the NONNUMERIC keyword is specified, or both, might be suppressed. If the ATTRIBUTE keyword is specified, only items that would be expressed in the generated XML document as an XML attribute are identified for potential suppression. If the ELEMENT keyword is specified, only items that would be expressed in the generated XML document as an XML element are identified for potential suppression. If the CONTENT keyword is specified, only items that would be expressed in the generated XML document as element character content of the XML element corresponding to the data item superordinate to the CONTENT data item are identified for potential suppression.

If multiple *generic-suppression-phrase* are specified, the effect is cumulative.

identifier-8 explicitly identifies items for potential suppression. If the WHEN phrase is specified, *identifier-8* must reference an elementary data item that is subordinate to *identifier-2* and that is not otherwise ignored by the XML GENERATE operations. *identifier-8* cannot be a function identifier and cannot be reference modified or subscripted. If the WHEN phrase is omitted, *identifier-8* can reference not only an elementary data item but also a group data item. That group data item and all data items that are subordinate to the group item are suppressed. If *identifier-8* is specified more than once in the SUPPRESS phrase, the last specification is used. The explicit suppression specification for *identifier-8* overrides the suppression specification that is implied by any *generic-suppression-phrase*, if *identifier-8* is also one of the identifiers generically identified.

If *identifier-8* is specified, the following rules apply to it:

- If ZERO, ZEROES, or ZEROS is specified in the WHEN phrase, *identifier-8* must not be of USAGE DISPLAY-1.
- If SPACE or SPACES is specified in the WHEN phrase, *identifier-8* must be of USAGE DISPLAY, DISPLAY-1, or NATIONAL. If *identifier-8* is a zoned or national decimal item, it must be an integer.
- If LOW-VALUE, LOW-VALUES, HIGH-VALUE, or HIGH-VALUES is specified in the WHEN phrase, *identifier-8* must be of USAGE DISPLAY or NATIONAL. If *identifier-8* is a zoned or national decimal item, it must be an integer.

If the *generic-suppression-phrase* is specified, data items are selected for potential suppression according to the following rules:

- If ZERO, ZEROES, or ZEROS is specified in the WHEN phrase, all data items except those that are defined with USAGE DISPLAY-1 are selected.
- If SPACE or SPACES is specified in the WHEN phrase, data items of USAGE DISPLAY, DISPLAY-1, or NATIONAL are selected. For zoned or national decimal items, only integers are selected.
- If LOW-VALUE, LOW-VALUES, HIGH-VALUE, or HIGH-VALUES is specified in the WHEN phrase, data items of USAGE DISPLAY or NATIONAL are selected. For zoned or national decimal items, only integers are selected.

The comparison operation that determines whether an item will be suppressed is a relation condition as shown in the table of Comparisons involving figurative constants. That is, the comparison is a numeric comparison if the value specified is ZERO, ZEROS, or ZEROES, and the item is of class numeric. For all other cases, the comparison operation is an alphanumeric, DBCS, or national comparison, depending on whether the item is of usage DISPLAY, DISPLAY-1 or NATIONAL, respectively.

When the SUPPRESS phrase is specified, a group item subordinate to *identifier-2* is suppressed in the generated XML document if all the eligible items subordinate to the group item are suppressed or if, after suppressing any subordinate items, the XML corresponding to the group item would be an empty element with no attributes. The root element is always generated, even if all the items subordinate to *identifier-2* are suppressed.

ON EXCEPTION phrase

An exception condition exists when an error occurs during generation of the XML document, for example if *identifier-1* is not sufficient enough to contain the generated XML document. In this case, XML generation stops and the content of the receiver, *identifier-1*, is undefined. If the COUNT IN

phrase is specified, *identifier-3* contains the number of character positions that were generated, which can range from 0 to the length of *identifier-1*.

If the ON EXCEPTION phrase is specified, control is transferred to *imperative-statement-1*. If the ON EXCEPTION phrase is not specified, the NOT ON EXCEPTION phrase, if any, is ignored, and control is transferred to the end of the XML GENERATE statement. Special register XML-CODE contains an exception code, as detailed in *Handling XML GENERATE exceptions* in the *Enterprise COBOL Programming Guide*.

NOT ON EXCEPTION phrase

If an exception condition does not occur during generation of the XML document, control is passed to *imperative-statement-2*, if specified, otherwise to the end of the XML GENERATE statement. The ON EXCEPTION phrase, if specified, is ignored. Special register XML-CODE contains zero after execution of the XML GENERATE statement.

END-XML phrase

This explicit scope terminator delimits the scope of XML GENERATE or XML PARSE statements. END-XML permits a conditional XML GENERATE or XML PARSE statement (that is, an XML GENERATE or XML PARSE statement that specifies the ON EXCEPTION or NOT ON EXCEPTION phrase) to be nested in another conditional statement.

The scope of a conditional XML GENERATE or XML PARSE statement can be terminated by:

- An END-XML phrase at the same level of nesting
- A separator period

END-XML can also be used with an XML GENERATE or XML PARSE statement that does not specify either the ON EXCEPTION or the NOT ON EXCEPTION phrase.

For more information about explicit scope terminators, see [“Delimited scope statements” on page 293](#).

Nested XML GENERATE or XML PARSE statements

When a given XML GENERATE or XML PARSE statement appears as *imperative-statement-1* or *imperative-statement-2*, or as part of *imperative-statement-1* or *imperative-statement-2* of another XML GENERATE or XML PARSE statement, that given XML GENERATE or XML PARSE statement is a *nested* XML GENERATE or XML PARSE statement.

Nested XML GENERATE or XML PARSE statements are considered to be matched XML GENERATE and END-XML combinations, or XML PARSE and END-XML combinations, proceeding from left to right. Thus, any END-XML phrase that is encountered is matched with the nearest preceding XML GENERATE or XML PARSE statement that has not been implicitly or explicitly terminated.

Operation of XML GENERATE

The content of each eligible elementary data item within *identifier-2* that has not been suppressed from XML generation according to a SUPPRESS phrase, is converted to character format.

Only the first definition of each storage area is processed. Redefinitions of data items are not included. Data items that are effectively defined by the RENAMES clause are also not included.

For information about the format conversion of elementary data, see [“Format conversion of elementary data” on page 492](#) and [“Trimming of generated XML data” on page 493](#).

If the TYPE OF phrase is specified, the converted content is then processed as element character content or attribute value, according to the specifications on that phrase. If the TYPE OF phrase is not specified, by default the converted content is inserted as element character content, or, if the WITH ATTRIBUTES phrase is specified and the data item is eligible to be expressed as an attribute, as the value of the attribute, in the generated XML document.

The XML element names and attribute names are obtained from the NAME phrase if specified; otherwise by default they are derived from the data-names within *identifier-2* as described in [“XML element name and attribute name formation” on page 493](#). The names of group items that contain the selected

elementary items are retained as parent elements. If the NAMESPACE-PREFIX phrase is specified, the prefix value, minus any trailing spaces, is used to qualify the start and end tag of each element.

No additional white space (new lines, indentation, and so forth) is inserted to make the generated XML more readable. An XML declaration is generated if the XML-DECLARATION phrase is specified.

If the receiving area specified by *identifier-1* is not sufficient enough to contain the resulting XML document, an error condition exists. See the description of the ON EXCEPTION phrase mentioned previously for details.

If *identifier-1* is longer than the generated XML document, only that part of *identifier-1* in which XML is generated is changed. The rest of *identifier-1* contains the data that was present before this execution of the XML GENERATE statement. To avoid referring to that data, either initialize *identifier-1* to spaces before the XML GENERATE statement or specify the COUNT IN phrase.

If the COUNT IN phrase is specified, *identifier-3* contains (after execution of the XML GENERATE statement) the total number of character positions (UTF-16 encoding units or bytes) that were generated. You can use *identifier-3* as a reference modification length field to refer to the part of *identifier-1* that contains the generated XML document.

After execution of the XML GENERATE statement, special register XML-CODE contains either zero, which indicates successful completion, or a nonzero exception code. For details, see *Handling XML GENERATE exceptions* in the *Enterprise COBOL Programming Guide*.

The XML PARSE statement also uses special register XML-CODE. Therefore if you code an XML GENERATE statement in the processing procedure of an XML PARSE statement, save the value of XML-CODE before that XML GENERATE statement executes and restore the saved value after the XML GENERATE statement terminates.

A byte order mark is not generated for XML documents that have Unicode encoding.

Format conversion of elementary data

Elementary data items within *identifier-2* are converted in a sequence of several steps, some of them are optional as described here.

Conversion to character format:

Elementary data items are converted to character format depending on the type of the data item:

- Data items of category alphabetic, alphanumeric, alphanumeric-edited, DBCS, external floating-point, national, national-edited, and numeric-edited are not converted.
- Fixed-point numeric data items other than COMPUTATIONAL-5 (COMP-5) binary data items or binary data items compiled with the TRUNC(BIN) compiler option are converted as if they were moved to a numeric-edited item that has:
 - As many integer positions as the numeric item has, but with at least one integer position
 - An explicit decimal point, if the numeric item has at least one decimal position
 - The same number of decimal positions as the numeric item has
 - A leading '-' picture symbol if the data item is signed (has an S in its PICTURE clause)
- COMPUTATIONAL-5 (COMP-5) binary data items or binary data items compiled with the TRUNC(BIN) compiler option are converted in the same way as the other fixed-point numeric items, except for the number of integer positions. The number of integer positions is computed depending on the number of '9' symbols in the picture character string as follows:
 - 5 minus the number of decimal places, if the data item has 1 to 4 '9' picture symbols
 - 10 minus the number of decimal places, if the data item has 5 to 9 '9' picture symbols
 - 20 minus the number of decimal places, if the data item has 10 to 18 '9' picture symbols
- Internal floating-point data items are converted as if they were moved to a data item as follows:
 - For COMP-1: an external floating-point data item with PICTURE -9.9(8)E+99

- For COMP-2: an external floating-point data item with PICTURE -9.9(17)E+99 (illegal because of the number of digit positions)
- Index data items are converted as if they were declared USAGE COMP-5 PICTURE S9(9).

Trimming:

After any conversion to character format, leading and trailing spaces and leading zeroes are eliminated, as described under [“Trimming of generated XML data”](#) on page 493.

Conversion to the document encoding:

If *identifier-1* is a data item of category national, any nonnational values are converted to national format.

Conversion of special characters to XML references:

Any remaining instances of the five characters & (ampersand), ' (apostrophe), > (greater-than sign), < (less-than sign), and " (quotation mark) are converted into the equivalent XML references '&';', '>', '<', and '"', respectively.

Replacement of out-of-range Unicode characters:

Any remaining Unicode character that has a Unicode scalar value greater than x'FFFF' is replaced by an XML character reference. For example, if the document contains a character with Unicode scalar value x'10813', in UTF-16, that value is represented by the surrogate pair (NX'D802', NX'DC13'), which is replaced by the reference '𐠓'. For a document encoding of UTF-8, the byte sequence that is equivalent to character reference '𐠓' is X'F090A093'.

Trimming of generated XML data

Trimming is performed on data values after their conversion to character format.

For more information about the conversion, see [“Format conversion of elementary data”](#) on page 492.

For values converted from signed numeric values, the leading space is removed if the value is positive.

For values converted from numeric items, leading zeroes (after any initial minus sign) up to but not including the digit immediately before the actual or implied decimal point are eliminated. Trailing zeroes after a decimal point are retained. For example:

- -012.340 becomes -12.340.
- 0000.45 becomes 0.45.
- 0013 becomes 13.
- 0000 becomes 0.

Character values from data items of class alphabetic, alphanumeric, DBCS, and national have either trailing or leading spaces removed, depending on whether the corresponding data items have left (default) or right justification, respectively. That is, trailing spaces are removed from values whose corresponding data items do not specify the JUSTIFIED clause. Leading spaces are removed from values whose data items do specify the JUSTIFIED clause. If a character value consists solely of spaces, one space remains as the value after trimming is finished.

XML element name and attribute name formation

In the XML documents that are generated from *identifier-2*, the XML element names and attribute names are obtained from the NAME phrase if specified; otherwise they are derived from the names of the data item specified by *identifier-2* and from any eligible data-names that are subordinate to *identifier-2*.

The formation of XML element name and attribute name is as follows:

- The exact mixed-case spelling of data-names from the data description entry is retained. The spellings from any references to data items (for example, in an OCCURS DEPENDING ON clause) are not used.
- Data-names that start with a digit are prefixed by an underscore. For example, the data-name '3D' becomes XML tag or attribute name '_3D'.

- Data-names that start with the characters 'xml', in any combination of uppercase and lowercase, are prefixed by an underscore. For example, the data-name 'Xml' becomes XML tag or attribute name '_Xml'.

DBCS data-names, when translated to Unicode, must be legal as names in the XML specification, version 1.0. For details about the XML specification, see [XML specification](#).

XML PARSE statement

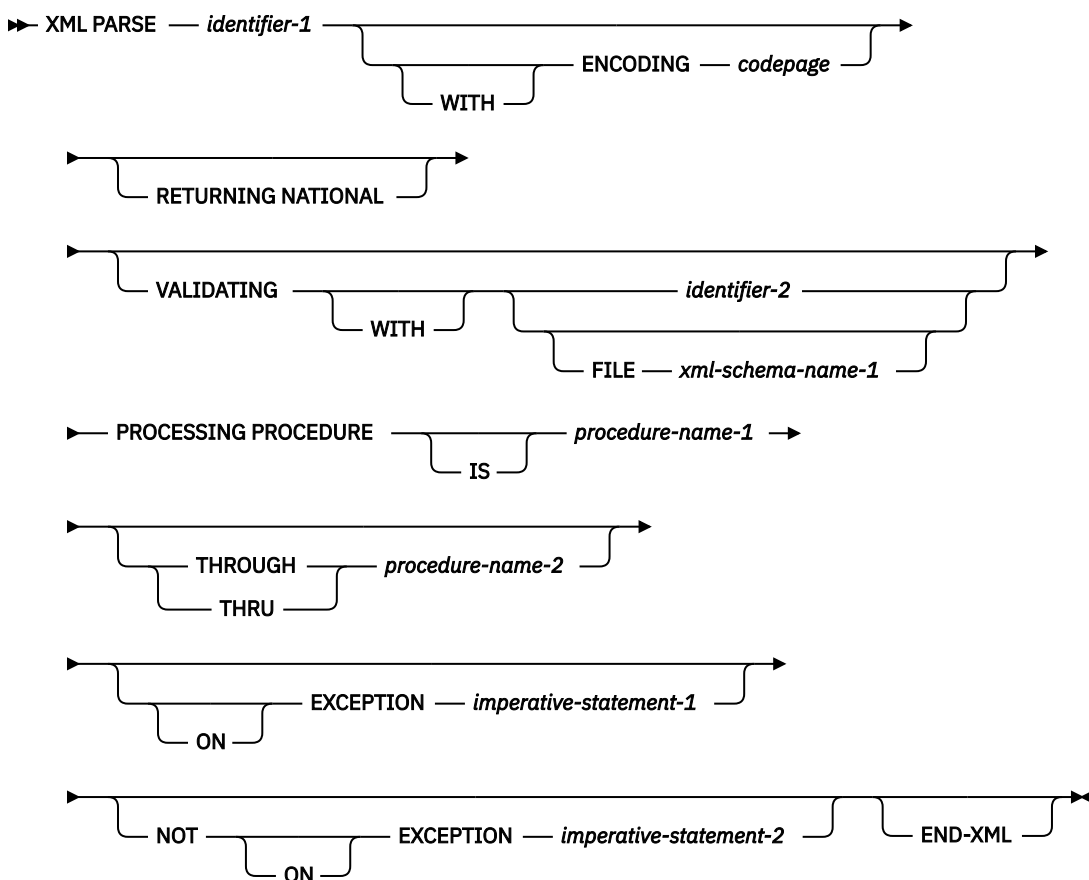
The XML PARSE statement is the COBOL language interface to either of two high-speed XML parsers, depending on the setting of the XMLPARSE compiler option.

The two high-speed XML parsers are:

- The z/OS XML System Services parser, for enhanced parsing capabilities. This parser is selected by the XMLPARSE(XMLSS) compiler option.
- The XML parser that is provided in the COBOL run time, for compatibility with Enterprise COBOL for z/OS 3. The compatible parser is selected by the XMLPARSE(COMPAT) compiler option.

The XML PARSE statement parses an XML document into its individual pieces and passes each piece, one at a time, to a user-written processing procedure.

Format



identifier-1

identifier-1 must be an elementary data item of category national, a national group, an elementary data item of category alphanumeric, or an alphanumeric group item. *identifier-1* cannot be a function-identifier. *identifier-1* contains the XML document character stream.

If *identifier-1* is a national group item, *identifier-1* is processed as an elementary data item of category national.

If *identifier-1* is of category national, its content must be encoded using Unicode UTF-16BE (CCSID 1200). If the XMLPARSE(COMPAT) compiler option is in effect, *identifier-1* must not contain any character entities that are represented using multiple encoding units. Use a character reference to represent any such characters, for example:

- "񧘃" or
- "𐠓"

The letter x must be lowercase.

identifier-1 must not be a dynamic-length group item or a dynamic-length elementary item.

If *identifier-1* is of category alphanumeric, its content must be encoded using one of the character sets listed in *Coded character sets for XML documents* in the *Enterprise COBOL Programming Guide*. If the XMLPARSE(COMPAT) compiler option is in effect, and *identifier-1* is alphanumeric and contains an XML document that does not specify an encoding declaration, the XML document is parsed with the code page specified by the CODEPAGE compiler option.

If the XMLPARSE(XMLSS) compiler option is in effect, the XML document is parsed with the code page specified in the ENCODING phrase; if the ENCODING phrase is not used, the document is parsed with the code page specified by the CODEPAGE compiler option. Any encoding declaration in the XML document is ignored.

RETURNING NATIONAL phrase

The RETURNING NATIONAL phrase can be specified only when the XMLPARSE(XMLSS) compiler option is in effect.

When *identifier-1* references a data item of category alphanumeric and the RETURNING NATIONAL phrase is specified, XML document fragments are automatically converted to Unicode UTF-16 representation and returned to the processing procedure in the national special registers XML-NTEXT, XML-NNAMESPACE, and XML-NNAMESPACE-PREFIX.

When the RETURNING NATIONAL phrase is not specified and *identifier-1* references a data item of category alphanumeric, the XML document fragments are returned to the processing procedure in the alphanumeric special registers XML-TEXT, XML-NAMESPACE, and XML-NAMESPACE-PREFIX except that when XMLPARSE(COMPAT) is in effect, text for the ATTRIBUTE-NATIONAL-CHARACTER and CONTENT-NATIONAL-CHARACTER XML events is always returned in special register XML-NTEXT.

When *identifier-1* references a national data item, XML document fragments are always returned in Unicode UTF-16 representation in the national special registers XML-NTEXT, XML-NNAMESPACE, and XML-NNAMESPACE-PREFIX.

VALIDATING phrase

The VALIDATING phrase specifies that the parser should validate the XML document against an XML schema while parsing it. In Enterprise COBOL, the schema used for XML validation is in a preprocessed format known as *Optimized Schema Representation* or *OSR*. The VALIDATING phrase can be specified only when the XMLPARSE(XMLSS) compiler option is in effect.

See *Parsing XML documents with validation* in the *Enterprise COBOL Programming Guide* for details.

identifier-2 must not be a dynamic-length group item or a dynamic-length elementary item.

If the FILE keyword is not specified, *identifier-2* must reference a data item that contains the optimized XML schema. *identifier-2* must be of category alphanumeric and cannot be a function-identifier.

If the FILE keyword is specified, *xml-schema-name-1* identifies an existing z/OS UNIX file or MVS data set that contains the optimized XML schema. *xml-schema-name-1* must be associated with the external file name of the schema by using the XML-SCHEMA clause. For more information about the XML-SCHEMA clause, see [“SPECIAL-NAMES paragraph” on page 124](#).

Restriction: XML validation using the FILE keyword is not supported under CICS.

During parsing with validation, normal XML events are returned as for nonvalidating parsing until an exception occurs due to a validation error or other error in the document.

When an XML document is not valid, the parser signals an XML exception and passes control to the processing procedure with special register XML-EVENT containing 'EXCEPTION' and special-register XML-CODE containing return code 24 in the high-order halfword and a reason code in the low-order halfword.

For information about the return code and reason code for exceptions that might occur when parsing XML documents with validation, see *XML PARSE exceptions with XMLPARSE(XMLSS) in effect* in the *Enterprise COBOL Programming Guide*.

ENCODING phrase

The ENCODING phrase can be specified only when the XMLPARSE(XMLSS) compiler option is in effect.

The ENCODING phrase specifies an encoding that is assumed for the source XML document in *identifier-1*. *codepage* must be an unsigned integer data item or an unsigned integer literal that represents a valid coded character set identifier (CCSID). The ENCODING phrase specification overrides the encoding specified by the CODEPAGE compiler option. The encoding specified in any XML declaration is always ignored.

If *identifier-1* references a data item of category national, *codepage* must specify CCSID 1200, for Unicode UTF-16.

If *identifier-1* references a data item of category alphanumeric, *codepage* must specify CCSID 1208 for UTF-8 or a CCSID for a supported EBCDIC or ASCII codepage. See *Coded character sets for XML documents* in the *Enterprise COBOL Programming Guide* for details.

PROCESSING PROCEDURE phrase

Specifies the name of a procedure to handle the various events that the XML parser generates.

procedure-name-1, procedure-name-2

Must name a section or paragraph in the PROCEDURE DIVISION. When both *procedure-name-1* and *procedure-name-2* are specified, if either is a procedure name in a declarative procedure, both must be procedure names in the same declarative procedure.

procedure-name-1

Specifies the first (or only) section or paragraph in the processing procedure.

procedure-name-2

Specifies the last section or paragraph in the processing procedure.

For each XML event, the parser transfers control to the first statement of the procedure named *procedure-name-1*. Control is always returned from the processing procedure to the XML parser. The point from which control is returned is determined as follows:

- If *procedure-name-1* is a paragraph name and *procedure-name-2* is not specified, the return is made after the execution of the last statement of the *procedure-name-1* paragraph.
- If *procedure-name-1* is a section name and *procedure-name-2* is not specified, the return is made after the execution of the last statement of the last paragraph in the *procedure-name-1* section.
- If *procedure-name-2* is specified and it is a paragraph name, the return is made after the execution of the last statement of the *procedure-name-2* paragraph.
- If *procedure-name-2* is specified and it is a section name, the return is made after the execution of the last statement of the last paragraph in the *procedure-name-2* section.

The only necessary relationship between *procedure-name-1* and *procedure-name-2* is that they define a consecutive sequence of operations to execute, beginning at the procedure named by *procedure-name-1* and ending with the execution of the procedure named by *procedure-name-2*.

If there are two or more logical paths to the return point, then *procedure-name-2* can name a paragraph that consists of only an EXIT statement; all the paths to the return point must then lead to this paragraph.

The processing procedure consists of all the statements at which XML events are handled. The range of the processing procedure includes all statements executed by CALL, EXIT, GO TO, GOBACK,

INVOKE, MERGE, PERFORM, and SORT statements that are in the range of the processing procedure, as well as all statements in declarative procedures that are executed as a result of the execution of statements in the range of the processing procedure.

The range of the processing procedure must not cause the execution of any GOBACK or EXIT PROGRAM statement, except to return control from a method or program to which control was passed by an INVOKE or CALL statement, respectively, that is executed in the range of the processing procedure.

The range of the processing procedure must not cause the execution of an XML PARSE statement, unless the XML PARSE statement is executed in a method or outermost program to which control was passed by an INVOKE or CALL statement that is executed in the range of the processing procedure.

A program executing on multiple threads can execute the same XML statement or different XML statements simultaneously.

The processing procedure can terminate the run unit with a STOP RUN statement.

For more details about the processing procedure, see [“Control flow” on page 498](#).

ON EXCEPTION

The ON EXCEPTION phrase specifies imperative statements that are executed when the XML PARSE statement raises an exception condition.

An exception condition exists when the XML parser detects an error in processing the XML document. The parser first signals an XML exception by passing control to the processing procedure with special register XML-EVENT containing 'EXCEPTION'. The parser also provides a numeric error code in special register XML-CODE, as detailed in *Handling XML PARSE exceptions* in the *Enterprise COBOL Programming Guide*.

An exception condition also exists if the processing procedure sets XML-CODE to -1 before returning to the parser for any normal XML event. In this case, the parser does not signal an EXCEPTION XML event and parsing is terminated.

If the ON EXCEPTION phrase is specified, the parser transfers control to *imperative-statement-1*. If the ON EXCEPTION phrase is not specified, the NOT ON EXCEPTION phrase, if any, is ignored and control is transferred to the end of the XML PARSE statement.

Special register XML-CODE contains the numeric error code for the XML exception or -1 after execution of the XML PARSE statement.

If the processing procedure handles the XML exception event and sets XML-CODE to zero before returning control to the parser, the exception condition no longer exists. If no other unhandled exceptions occur before termination of the parser, control is transferred to *imperative-statement-2* of the NOT ON EXCEPTION phrase, if specified.

NOT ON EXCEPTION

The NOT ON EXCEPTION phrase specifies imperative statements that are executed when no exception condition exists at the termination of XML PARSE processing.

If an exception condition does not exist at termination of XML PARSE processing, control is transferred to *imperative-statement-2* of the NOT ON EXCEPTION phrase, if specified. If the NOT ON EXCEPTION phrase is not specified, control is transferred to the end of the XML PARSE statement. The ON EXCEPTION phrase, if specified, is ignored.

Special register XML-CODE contains zero after execution of the XML PARSE statement.

END-XML phrase

This explicit scope terminator delimits the scope of XML GENERATE or XML PARSE statements. END-XML permits a conditional XML GENERATE or XML PARSE statement (that is, an XML GENERATE or XML PARSE statement that specifies the ON EXCEPTION or NOT ON EXCEPTION phrase) to be nested in another conditional statement.

The scope of a conditional XML GENERATE or XML PARSE statement can be terminated by:

- An END-XML phrase at the same level of nesting

- A separator period

END-XML can also be used with an XML GENERATE or XML PARSE statement that does not specify either the ON EXCEPTION or NOT ON EXCEPTION phrase.

For more information about explicit scope terminators, see [“Delimited scope statements” on page 293](#).

Nested XML GENERATE or XML PARSE statements

When a given XML GENERATE or XML PARSE statement appears as *imperative-statement-1* or *imperative-statement-2*, or as part of *imperative-statement-1* or *imperative-statement-2* of another XML GENERATE or XML PARSE statement, that given XML GENERATE or XML PARSE statement is a *nested* XML GENERATE or XML PARSE statement.

Nested XML GENERATE or XML PARSE statements are considered to be matched XML GENERATE and END-XML, or XML PARSE and END-XML combinations proceeding from left to right. Thus, any END-XML phrase that is encountered is matched with the nearest preceding XML GENERATE or XML PARSE statement that has not been implicitly or explicitly terminated.

Control flow

When the XML parser receives control from an XML PARSE statement, the parser analyzes the XML document and transfers control at specific points in the process.

The points are:

- The start of the parsing process
- When a document fragment is found
- When the parser detects an error in parsing the XML document
- The end of processing the XML document

Control returns to the XML parser when the end of the processing procedure is reached.

The exchange of control between the parser and the processing procedure continues until either:

- The entire XML document has been parsed, ending with the END-OF-DOCUMENT event.
- The processing procedure terminates parsing deliberately by setting XML-CODE to -1 before returning to the parser.
- When the XMLPARSE(XMLSS) compiler option is in effect: The parser detects an exception of any kind.
- When the XMLPARSE(COMPAT) compiler option is in effect: The parser detects an exception (other than an encoding conflict) and the processing procedure does not reset special register XML-CODE to zero before returning to the parser.
- When the XMLPARSE(COMPAT) compiler option is in effect: The parser detects an encoding conflict exception and the processing procedure does not reset special register XML-CODE to zero or to the CCSID of the document encoding.

In each case, the processing procedure returns control to the parser. Then, the parser terminates and returns control to the XML PARSE statement with the XML-CODE special register containing the most recent value set by the parser or -1 (which might have been set by the parser or by the processing procedure).

For each XML event passed to the processing procedure, the XML-CODE and XML-EVENT special registers contain information about the particular event. Special register XML-EVENT is set to the event name, such as 'START-OF-DOCUMENT'. For most events, the XML-TEXT or XML-NTEXT special register contains document text. Additionally, when the XMLPARSE(XMLSS) compiler option is in effect, the XML-NAMESPACE and XML-NAMESPACE-PREFIX or the XML-NNAMESPACE and XML-NNAMESPACE-PREFIX special registers contain a namespace identifier and namespace prefix when applicable. See [“XML-EVENT” on page 29](#) for details.

The content of the XML-CODE special register is defined during and after execution of an XML PARSE statement. The contents of all other XML special registers are undefined outside the range of the processing procedure.

For normal XML events, special register XML-CODE contains zero when the processing procedure receives control. For XML exception events, XML-CODE contains an XML exception code as described in *XML PARSE exceptions* in the *Enterprise COBOL Programming Guide*.

For more information about the XML special registers, see:

- [“XML-CODE” on page 28](#)
- [“XML-EVENT” on page 29](#)
- [“XML-INFORMATION” on page 34](#)
- [“XML-NAMESPACE” on page 34](#)
- [“XML-NAMESPACE-PREFIX” on page 36](#)
- [“XML-NNAMESPACE” on page 35](#)
- [“XML-NNAMESPACE-PREFIX” on page 36](#)
- [“XML-NTEXT” on page 37](#)
- [“XML-TEXT” on page 37](#)

For an introduction to special registers, see [“Special registers” on page 17](#)

For more information about the EXCEPTION event and exception processing, see *Handling XML PARSE exceptions* in the *Enterprise COBOL Programming Guide*.

Part 7. Intrinsic functions

An *intrinsic function* is a function that performs a mathematical, character, or logical operation. You can use intrinsic functions to make reference to a data item whose value is derived automatically during execution.

Data processing problems often require the use of values that are not directly accessible in the data storage associated with the object program, but instead must be derived through performing operations on other data. An *intrinsic function* is a function that performs a mathematical, character, or logical operation, and thereby allows you to make reference to a data item whose value is derived automatically during execution.

The intrinsic functions can be grouped into six categories, based on the type of service performed:

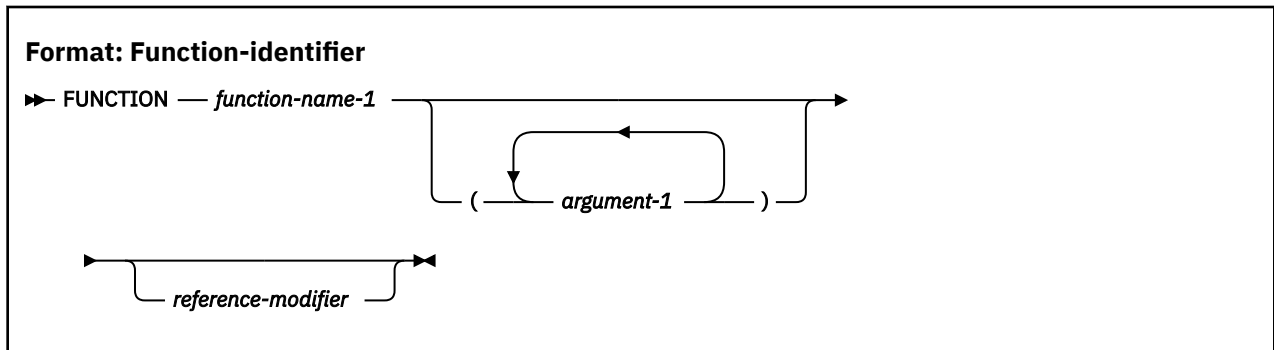
- Mathematical
- Statistical
- Date/time
- Financial
- Character-handling
- General

You can reference a function by specifying its name, along with any required arguments, in a PROCEDURE DIVISION statement.

Functions are elementary data items, and return alphanumeric character, national character, numeric, or integer values. Functions cannot serve as receiving operands.

Chapter 29. Specifying a function

This topic describes the general format of a function-identifier.



Note: The keyword `FUNCTION` may be omitted if you specify the function name in the `REPOSITORY` paragraph of your program.

function-name-1

function-name-1 must be one of the intrinsic function names.

argument-1

argument-1 must be an identifier, a literal (other than a figurative constant), or an arithmetic expression that satisfies the argument requirements for the specified function.

reference-modifier

Can be specified only for functions of type alphanumeric or national.

A function-identifier can be specified wherever a data item of the type of the function is allowed. The argument to a function can be any function or an expression containing a function, including another evaluation of the same function, whose result meets the requirements for the argument.

Within a `PROCEDURE DIVISION` statement, each function-identifier is evaluated at the same time as any reference modification or subscripting associated with an identifier in that same position would be evaluated.

Function definition and evaluation

The class and characteristics of a function, and the number and types of arguments it requires, are determined by its function definition.

These characteristics include:

- For functions of type alphanumeric, national, and UTF-8, the size of the returned value
- For functions of type numeric and integer, the sign of the returned value, and whether the function is integer
- The actual value returned by the function

For some functions, the class and characteristics are determined by the arguments to the function.

The evaluation of any intrinsic function is not affected by the context in which it appears; in other words, function evaluation is not affected by operations or operands outside the function. However, evaluation of a function can be affected by the attributes of its arguments.

Within a `PROCEDURE DIVISION` statement, each function-identifier is evaluated at the same time as any reference modification or subscripting associated with an identifier in that same position would be evaluated.

Types of functions

The topic introduces types of functions in COBOL.

COBOL has the following types of functions:

- Alphanumeric
- National
- UTF-8
- Numeric
- Integer

Alphanumeric functions are of class and category alphanumeric. The value returned has an implicit usage of `DISPLAY`. The number of character positions in the value returned is determined by the function definition.

National functions are of class and category national. The value returned has an implicit usage of `NATIONAL` and is represented in national characters (UTF-16). The number of character positions in the value returned is determined by the function definition.

UTF-8 functions are of class and category UTF-8. The value returned has an implicit usage of UTF-8 and is represented in UTF-8 characters (UTF-8). The number of character positions in the value returned is determined by the function definition.

Numeric functions are of class and category numeric. The returned value is always considered to have an operational sign and is a numeric intermediate result. For more information, see *Using numeric intrinsic functions* in the *Enterprise COBOL Programming Guide*.

Integer functions are of class and category numeric. The returned value is always considered to have an operational sign and is an integer intermediate result. The number of digit positions in the value returned is determined by the function definition. For more information, see *Using numeric intrinsic functions* in the *Enterprise COBOL Programming Guide*.

Rules for usage

The topic describes rules of using different types of functions.

Alphanumeric functions

An alphanumeric function can be specified anywhere in the general formats that a data item of class and category alphanumeric is permitted and where the rules associated with the general formats do not specifically prohibit reference to functions, except as noted in the following list.

An alphanumeric function can be used as an argument for any function that allows an alphanumeric argument.

Reference modification of an alphanumeric function is allowed. If reference modification is specified for a function, the evaluation of the reference modification takes place immediately after the evaluation of the function; that is, the function's returned value is reference-modified.

An alphanumeric function cannot be used:

- As a receiving operand of any statement
- Where the rules associated with the general formats require the data item being referenced to have particular characteristics (such as class and category, usage, size, and permissible values) and the evaluation of the function according to its definition and the particular arguments specified would not have those characteristics

National functions

A national function can be specified anywhere in the general formats that a data item of class and category national is permitted and where the rules associated with the general formats do not specifically prohibit reference to functions, except as noted in the following list.

A national function can be used as an argument for any function that allows a national argument.

Reference modification of a national function is allowed. If reference modification is specified for a function, the evaluation of the reference modification takes place immediately after the evaluation of the function; that is, the function's returned value is reference-modified.

A national function cannot be used:

- As a receiving operand of any statement
- Where the rules associated with the general formats require the data item being referenced to have particular characteristics (such as class and category, usage, size, and permissible values) and the evaluation of the function according to its definition and the particular arguments specified would not have those characteristics

UTF-8 functions

A UTF-8 function can be specified anywhere in the general formats that a data item of class and category UTF-8 is permitted and where the rules associated with the general formats do not specifically prohibit reference to functions, except as noted in the following list.

A UTF-8 function can be used as an argument for any function that allows a UTF-8 argument.

Reference modification of a UTF-8 function is allowed. If reference modification is specified for a function, the evaluation of the reference modification takes place immediately after the evaluation of the function; that is, the function's returned value is reference-modified.

A UTF-8 function cannot be used:

- As a receiving operand of any statement
- Where the rules associated with the general formats require the data item being referenced to have particular characteristics (such as class and category, usage, size, and permissible values) and the evaluation of the function according to its definition and the particular arguments specified would not have those characteristics

Numeric functions

A numeric function can be used only where an arithmetic expression can be specified.

A numeric function can be referenced as an argument for a function that allows a numeric argument.

A numeric function cannot be used where an integer operand is required, even if the particular reference would yield an integer value. The INTEGER or INTEGER-PART functions can be used to force the type of a numeric argument to be an integer.

Integer functions

An integer function can be used only where an arithmetic expression can be specified.

An integer function can be referenced as an argument for a function that allows a numeric argument.

Usage notes:

- A function-identifier cannot be used in the BY REFERENCE phrase of a CALL statement (that is, *identifier-2* of the CALL statement must not be a function-identifier).
- The COPY statement accepts function-identifiers of all types in the REPLACING phrase.

Arguments

The value returned by some functions is determined by the arguments specified in the function-identifier when the functions are evaluated. Some functions require no arguments; others require a fixed number of arguments, and still others accept a variable number of arguments.

An argument must be one of the following items:

- A data item identifier
- An arithmetic expression
- A function-identifier

- A literal other than a figurative constant
- A special-register

See [Chapter 30, “Function definitions,”](#) on page 515 for function-specific argument specifications.

The types of arguments are:

Alphabetic

An elementary data item of the class alphabetic or an alphanumeric literal containing only alphabetic characters. The content of the argument is used to determine the value of the function. The length of the argument can be used to determine the value of the function.

Alphanumeric

A data item of the class alphabetic or alphanumeric or an alphanumeric literal. The content of the argument is used to determine the value of the function. The length of the argument can be used to determine the value of the function.

DBCS

An elementary data item of class DBCS or a DBCS literal. The content of the argument is used to determine the value of the function. The length of the argument can be used to determine the value of the function. (A DBCS data item or literal can be used as an argument only for the NATIONAL-OF function.)

National

A data item of class national (category national, national-edited, or numeric-edited). The content of the argument is used to determine the value of the function. The length of the argument can be used to determine the value of the function.

UTF-8

A data item of class UTF-8 (category UTF-8). The content of the argument is used to determine the value of the function. The length of the argument can be used to determine the value of the function.

Integer

An arithmetic expression that always results in an integer value. The value of the expression, including its sign, is used to determine the value of the function.

Numeric

An arithmetic expression. The expression can include numeric literals and data items of categories numeric, internal floating-point, and external floating-point. The numeric data items can have any usage permitted for the category of the data item (including NATIONAL). The value of the expression, including its sign, is used to determine the value of the function.

Keyword

A keyword shall be specified in accordance with the intrinsic function definition. The TRIM intrinsic function is an example of an intrinsic function with a keyword argument. The keywords LEADING and TRAILING may be specified as the second and optional argument of TRIM.

Some functions place constraints on their arguments, such as the acceptable range of values. If the values assigned as arguments for a function do not comply with specified constraints, the returned value is undefined.

If a nested function is used as an argument, the evaluation of its arguments is not affected by the arguments in the outer function.

Only those arguments at the same function level interact with each other. This interaction occurs in two areas:

- The computation of an arithmetic expression that appears as a function argument is affected by other arguments for that function.
- The evaluation of the function takes into consideration the attributes of all of its arguments.

When a function is evaluated, its arguments are evaluated individually in the order specified in the list of arguments, from left to right. The argument being evaluated can be a function-identifier or an expression that includes function-identifiers.

If an arithmetic expression is specified as an argument and if the first operator in the expression is a unary plus or a unary minus, the expression must be immediately preceded by a left parenthesis.

Floating-point literals are allowed wherever a numeric argument is allowed and in arithmetic expressions used in functions that allow a numeric argument.

Internal floating-point items and external floating-point items (both display floating-point and national floating-point) can be used wherever a numeric argument is allowed and in arithmetic expressions as arguments to a function that allows a numeric argument.

Floating-point items and floating-point literals *cannot* be used where an integer argument is required or where an argument of class alphanumeric or national is required (such as in the LOWER-CASE, REVERSE, UPPER-CASE, NUMVAL, and NUMVAL-C functions).

Where a function allows an alphanumeric, a national, or a UTF-8 argument to be specified, an alphanumeric group, a national or a UTF-8 group, respectively, can also be specified. However, an unbounded group cannot be specified as a function argument, except when it is used in the LENGTH intrinsic function.

Examples

See examples of using different types of intrinsic functions.

The following statement illustrates the use of intrinsic function UPPER-CASE to replace each lowercase letter in an alphanumeric argument with the corresponding uppercase letter.

```
MOVE FUNCTION UPPER-CASE('hello') TO DATA-NAME.
```

This statement moves HELLO into DATA-NAME.

The following statement illustrates the use of intrinsic function LOWER-CASE to replace each uppercase letter in a national argument with the corresponding lowercase letter.

```
MOVE FUNCTION LOWER-CASE(N'HELLO') TO N-DATA-NAME.
```

This statement moves national characters hello into N-DATA-NAME.

The following statement illustrates the use of a numeric intrinsic function:

```
COMPUTE NUM-ITEM = FUNCTION SUM(A B C)
```

This statement uses the numeric function SUM to add the values of A, B, and C and places the result in NUM-ITEM.

ALL subscripting

When a function allows an argument to be repeated a variable number of times, you can refer to a table by specifying the data-name and any qualifiers that identify the table. This can be followed immediately by subscripting where one or more of the subscripts is the word ALL.

Tip: The evaluation of an ALL subscript must result in at least one argument or the value returned by the function will be undefined; however, the situation can be diagnosed at run time by specifying the SSRANGE compiler option.

Specifying ALL as a subscript is equivalent to specifying all table elements possible using every valid subscript in that subscript position.

For a table argument specified as Table-name(ALL), the order of the implicit specification of each table element as an argument is from left to right, where the first (or leftmost) argument is Table-name(1) and ALL has been replaced by 1. The next argument is Table-name(2), where the subscript has been

incremented by 1. This process continues, with the subscript being incremented by 1 to produce an implicit argument, until the ALL subscript has been incremented through its range of values.

For example,

```
FUNCTION MAX(Table(ALL))
```

is equivalent to

```
FUNCTION MAX(Table(1) Table(2) Table(3) ... Table(n))
```

where n is the number of elements in Table.

If there are multiple ALL subscripts, Table-name(ALL, ALL, ALL), the first implicit argument is Table-name(1, 1, 1), where each ALL has been replaced by 1. The next argument is Table-name(1, 1, 2), where the rightmost subscript has been incremented by 1. The subscript represented by the rightmost ALL is incremented through its range of values to produce an implicit argument for each value.

Once a subscript specified as ALL has been incremented through its range of values, the next subscript to the left that is specified as ALL is incremented by 1. Each subscript specified as ALL to the right of the newly incremented subscript is set to 1 to produce an implicit argument. Once again, the subscript represented by the rightmost ALL is incremented through its range of values to produce an implicit argument for each value. This process is repeated until each subscript specified as ALL has been incremented through its range of values.

For example,

```
FUNCTION MAX(Table(ALL, ALL))
```

is equivalent to

```
FUNCTION MAX(Table(1, 1) Table(1, 2) Table(1, 3) ... Table(1, n)
              Table(2, 1) Table(2, 2) Table(2, 3) ... Table(2, n)
              Table(3, 1) Table(3, 2) Table(3, 3) ... Table(3, n)
              ...
              Table(m, 1) Table(m, 2) Table(m, 3) ... Table(m, n))
```

where n is the number of elements in the column dimension of Table, and m is the number of elements in the row dimension of Table.

ALL subscripts can be combined with literal, data-name, or index-name subscripts to reference multidimensional tables.

For example,

```
FUNCTION MAX(Table(ALL, 2))
```

is equivalent to

```
FUNCTION MAX(Table(1, 2)
              Table(2, 2)
              Table(3, 2)
              ...
              Table(m, 2))
```

where m is the number of elements in the row dimension of Table.

If an ALL subscript is specified for an argument and the argument is reference-modified, then the reference-modifier is applied to each of the implicitly specified elements of the table.

If an ALL subscript is specified for an operand that is reference-modified, the reference-modifier is applied to each of the implicitly specified elements of the table.

If the ALL subscript is associated with an OCCURS DEPENDING ON clause, the range of values is determined by the object of the OCCURS DEPENDING ON clause.

For example, given a payroll record definition such as:

```
01 PAYROLL.  
  02 PAYROLL-WEEK    PIC 99.  
  02 PAYROLL-HOURS    PIC 999 OCCURS 1 TO 52  
    DEPENDING ON PAYROLL-WEEK.
```

The following COMPUTE statements could be used to identify total year-to-date hours, the maximum hours worked in any week, and the specific week corresponding to the maximum hours:

```
COMPUTE YTD-HOURS = FUNCTION SUM (PAYROLL-HOURS(ALL))  
COMPUTE MAX-HOURS = FUNCTION MAX (PAYROLL-HOURS(ALL))  
COMPUTE MAX-WEEK  = FUNCTION ORD-MAX (PAYROLL-HOURS(ALL))
```

In these function invocations, the subscript ALL is used to reference all elements of the PAYROLL - HOURS array (depending on the execution time value of the PAYROLL - WEEK field).

Format of arguments and return values for date and time intrinsic functions

The descriptions of intrinsic functions that process dates and times refer to the following different types of date and time related values:

- [Integer date form](#)
- [Standard date form](#)
- [Julian date form](#)
- [UTC offset value](#)
- [Standard numeric time form](#)
- [“Date and time formats” on page 510](#)

A description of each of these values, including a range of permissible values, is given as follows.

Note: The definitions are intended to be consistent with ISO 8601 *Data elements and interchange formats – Information interchange – Representation of dates and times*.

Integer date form

A value in integer date form is a positive integer that represents a number of days succeeding 31 December, 1600 in the Gregorian calendar. It must be greater than zero and less than or equal to the value of FUNCTION INTEGER-OF-DATE (99991231), which is 3,067,671.

Note: The INTDATE compiler option affects the starting date of the integer date functions. For details, see *INTDATE* in the *Enterprise COBOL Programming Guide*.

Standard date form

A value in standard date form is an integer of the form YYYYMMDD, whose value is obtained from the calculation $(YYYY * 10,000) + (MM * 100) + DD$, where:

- YYYY represents the year in the Gregorian calendar. It must be an integer greater than 1600 but not greater than 9999.
- MM represents a month and must be a positive integer less than 13.

- DD represents a day and must be a positive integer less than 32 if it is valid for the specified month and year combination.

Julian date form

A value in Julian date form is an integer of the form YYYYDDD whose value is obtained from the calculation $(YYYY * 1000) + DDD$, where:

- YYYY represents the year in the Gregorian calendar. It must be an integer greater than 1600 but not greater than 9999.
- DDD represents the day of the year. It must be a positive integer less than 367 if it is valid for the year specified.

UTC offset value

A UTC offset value is an integer representation of offset from UTC (Coordinated Universal Time) expressed in minutes. The value must be greater than or equal to -1439 and less than or equal to 1439.

Note: The offset value 1439 represents 23 hours 59 minutes, which is one minute less than a day.

Standard numeric time form

A value in standard numeric time form is a numeric value representing seconds past midnight. The value must be greater than or equal to zero and less than 86,400.

Date and time formats

For functions FORMATTED-CURRENT-DATE, FORMATTED-DATE, FORMATTED-TIME, and FORMATTED-DATETIME, the format literal argument indicates the format of the date or time value that is the result of the function. If the specified format literal is alphanumeric, the result of the function will be an alphanumeric value; if the specified format literal is national, the result of the function will be a national value; and if the format literal is UTF-8, the result of the function will be a UTF-8 value.

For functions INTEGER-OF-FORMATTED-DATE, SECONDS-FROM-FORMATTED-TIME, and TEST-FORMATTED-DATETIME, the format literal indicates the format of the date or time value specified as the second argument of the function.

The permissible format strings are listed as follows. For a full description of each subfield in the format literals, including a range of permissible values in data associated with the formats, see the [Meaning of date and time format subfields and permissible values of associated data](#) section.

<i>Table 60. The permissible format strings for date</i>	
Date formats	Format literals
Basic calendar date	YYYYMMDD
Extended calendar date	YYYY-MM-DD
Basic ordinal date	YYYYDDD
Extended ordinal date	YYYY-DDD
Basic week date	YYYYWwwD
Extended week date	YYYY-Www-D

<i>Table 61. The permissible format strings for integer-seconds time</i>	
Integer-seconds time formats	Format literals
Basic local time	hhmmss

<i>Table 61. The permissible format strings for integer-seconds time (continued)</i>	
Integer-seconds time formats	Format literals
Extended local time	hh:mm:ss
Basic Coordinated Universal Time (UTC)	hhmmssZ
Extended UTC time	hh:mm:ssZ
Basic offset time	hhmmss+hhmm
Extended offset time	hh:mm:ss+hh:mm

<i>Table 62. The permissible format strings for fractional-seconds time</i>	
Fractional-seconds time formats	Format literals
Basic local time	hhmmss.ssss
Extended local time	hh:mm:ss.ssss
Basic UTC time	hhmmss.ssssZ
Extended UTC time	hh:mm:ss.ssssZ
Basic offset time	hhmmss.ssss+hhmm
Extended offset time	hh:mm:ss.ssss+hh:mm

Note: In [Table 62 on page 511](#), the period is used as the decimal separator, and four "s" characters after the period are used for illustrative purposes. The number of "s" characters that might be specified after the decimal separator in these formats might range from 1 to 9.

Combined date and time formats

A basic combined date and time format consists of a basic date format followed by an uppercase "T" character and a basic time format. For example, "YYYYMMDDThhmmss.sss+hhmm".

An extended combined date and time format consists of an extended date format, followed by an uppercase "T" character and an extended time format. For example, "YYYY-MM-DDThh:mm:ss.ssss+hh:mm".

Combinations of basic date formats with extended time formats, or of extended date formats with basic time formats, are not allowed.

The uppercase "T" character that occurs in both basic and extended combined and time formats, which separates the date format portion from the time format portion, appears in the data associated with the format.

Meaning of date and time format subfields and permissible values of associated data

For date formats:

- Characters "YYYY" represent the year subfield. Data associated with this subfield must have a value greater than 1600 and less than or equal to 9999.
- Characters "MM" represent the month-of-year subfield. Data associated with this subfield must have a value between 01 and 12.
- Characters "DD" in calendar date formats represent the day-of-month subfield. Data associated with this subfield must have a value between 01 and 28, 29, 30, or 31, depending on the month subfield and year subfield of the data.
- Characters "DDD" in ordinal date formats represent the day-of-year subfield. For a leap year, the data associated with this subfield must have a value between 001 and 366, and between 001 and 365 otherwise.

- Characters “ww” in week date formats represent the week-of-year subfield. Data associated with this subfield must have a value between 01 and 52.
- Character “D” in week date formats represents the day-of-week subfield. Data associated with this subfield must have a value between 1 and 7.
- Character “W” in week date formats appears in associated data values as is at the same location indicated in the corresponding format literal.
- Character “-” in all extended date formats appears in associated data values as is at the same location indicated in the corresponding format literal.

For time formats:

- Characters “hh” represent the hours subfield. Data associated with this subfield must have a value between 00 and 23.
- Characters “mm” represent the minutes subfield. Data associated with this subfield must have a value between 00 and 59.
- Characters “ss” represent the seconds subfield. Data associated with this subfield must have a value between 00 and 59.
- In fractional seconds time formats, the fractional seconds subfield is introduced with a "." character and the rest of the subfield consists of a minimum of 1 and a maximum of 9 “s” characters. In data associated with this subfield, the "." character appears as is, and corresponding to each "s" character is a number between 0 and 9.
- In offset time formats, the "+" character introduces the "offset from UTC" subformat, which consists of characters "hh", representing the offset-hours subfield, followed by character ":" (extended time formats only), followed by characters "mm", representing the offset-minutes subfield.
- In data associated with the "offset from UTC" subformat:
 - A "+" character appears at the same location as the "+" character in the format literal when the time portion of the data is adjusted downward by the offset values to represent UTC.
 - A "-" character appears at the same location as the "+" character in the format literal when the time portion of the data is adjusted upward by the offset values to represent UTC.
 - A "0" character appears at the same location as the "+" character in the format literal when offset from UTC is not available on the system.
 - Data for the offset-hours subfield must have a value between 00 and 23 when offset from UTC is available on the system, and must have a value of 00 otherwise.
 - Data for the offset-minutes subfield must have a value between 00 and 59 when offset from UTC is available on the system, and must have a value of 00 otherwise.
- In UTC time formats, the last character is "Z", which indicates a UTC time. In data associated with the UTC time formats, the "Z" character appears as is at the end of the data value.
- Character ":" in all extended time formats appears in associated data values as is at the same location indicated in the corresponding format literal.

The following are the examples:

Example 1

The following example demonstrates the correspondence between the value returned by the FORMATTED-CURRENT-DATE intrinsic function and the format literal specified for its argument.

```
DISPLAY FUNCTION FORMATTED-CURRENT-DATE ('YYYY-MM-DDThh:mm:ss.ss+hh:mm')
```

The output is:

```
2020-10-28T01:11:36.13-04:00
```

Example 2

The following example involving a call to the SECONDS-FROM-FORMATTED-TIME intrinsic function demonstrates a combined date and time format literal and corresponding value that is in the format indicated by the literal.

```
01 SEC COMP-2.  
:  
COMPUTE SEC = FUNCTION SECONDS-FROM-FORMATTED-TIME(  
                                'YYYY-MM-DDThh:mm:ss.ss',  
                                '1987-12-26T00:45:23.06')  
DISPLAY SEC
```

The output is:

```
.27230599999999999E 04
```


Chapter 30. Function definitions

This section provides an overview of the argument type, function type, and value returned for each of the intrinsic functions.

For more information about the intrinsic functions, see [Table 63 on page 515](#).

Argument types and function types are abbreviated as follows:

Abbreviation	Meaning
A	Alphabetic
D	DBCS
I	Integer
K	Keyword
N	Numeric
O	Other, as specified in the function definition (pointer, function-pointer, procedure-pointer, or object reference)
U	National
X	Alphanumeric
UT	UTF-8

Each intrinsic function is described in detail in the topics in the following sections.

<i>Table 63. Table of functions</i>			
Function name	Arguments	Function type	Value returned
ABS	N1	I or N	Absolute value of N1
ACOS	N1	N	Arccosine of N1
ANNUITY	N1, I2	N	Ratio of annuity paid for I2 periods at interest of N1 to initial investment of one
ASIN	N1	N	Arcsine of N1
ATAN	N1	N	Arctangent of N1
BIT-OF	A1, D1, I1, N1, X1, U1, UT1, or O1	X	Alphanumeric character string consisting of characters "1" and "0" that correspond to the binary value of each byte in the argument
BIT-TO-CHAR	X1	X	Character string consisting of bytes that correspond to the bit pattern indicated by the sequence of "0" and "1" characters in the argument
BYTE-LENGTH	A1, D1, N1, X1, U1, UT1, or O1	I	Integer that is equal to the length of the argument in bytes
CHAR	I1	X	Character in position I1 of program collating sequence

Table 63. <i>Table of functions</i> (continued)			
Function name	Arguments	Function type	Value returned
COMBINED-DATETIME	I1, N2	N	Numeric representation of combined integer date and standard numeric time
CONTENT-OF	A1, I1, N1, U1, X1, or UT1	I, N, U, X, or UT	Content of A1, I1, N1, U1, X1, or UT1
COS	N1	N	Cosine of N1
CURRENT-DATE	None	X	Current date and time and difference from Greenwich mean time
DATE-OF-INTEGER	I1	I	Standard date equivalent (YYYYMMDD) of integer date
DATE-TO-YYYYMMDD	I1, I2	I	Standard date equivalent (YYYYMMDD) of I1 (standard date with a windowed year, YYYYMMDD), according to the 100-year interval whose ending year is specified by the sum of I2 and the year at execution time
DAY-OF-INTEGER	I1	I	Julian date equivalent (YYYYDDD) of integer date
DAY-TO-YYYYDDD	I1, I2	I	Julian date equivalent (YYYYDDD) of I1 (Julian date with a windowed year, YYYYDDD), according to the 100-year interval whose ending year is specified by the sum of I2 and the year at execution time
DISPLAY-OF	U1, UT1, I2	X	Each character in U1 or UT1 converted to a corresponding character representation using a code page identified by I2, if specified, or a default code page selected at compile time if I2 is unspecified
E	None	N	Approximation of <i>e</i> , the base of natural logarithms
EXP	N1	N	Approximation of the value of <i>e</i> raised to the power of N1
EXP10	N1	N	Approximation of the value of 10 raised to the power of N1
FACTORIAL	I1	I	Factorial of I1
FORMATTED-CURRENT-DATE	U1, X1, or UT1	U, X, or UT	Formatted date equivalent of current date and time in the format specified in <i>argument-1</i>
FORMATTED-DATE	U1, X1, or UT1, I2	U, X, or UT	Formatted date equivalent of integer date contained in <i>argument-2</i> in the format specified in <i>argument-1</i>
FORMATTED-DATETIME	U1, X1, or UT1, I2, N3, I4	U, X, or UT	Formatted date (from integer date in <i>argument-2</i>) and time (from standard numeric time in <i>argument-3</i>) in the format specified by <i>argument-1</i> . Offset from UTC, if the format requires it, is supplied by <i>argument-4</i>

Table 63. <i>Table of functions</i> (continued)			
Function name	Arguments	Function type	Value returned
FORMATTED-TIME	U1, X1, or UT1, N2, I3	U, X, or UT	Formatted time equivalent of standard numeric time contained in <i>argument-2</i> in the format specified in <i>argument-1</i> . Offset from UTC, if the format requires it, is supplied by <i>argument-3</i>
HEX-OF	A1, D1, I1, N1, X1, U1, UT1, or O1	X	Alphanumeric character string consisting of the bytes of the argument converted to a hexadecimal representation
HEX-TO-CHAR	X1	X	Character string consisting of bytes that correspond to the hexadecimal digit characters in the argument
INTEGER	N1	I	The greatest integer not greater than N1
INTEGER-OF-DATE	I1	I	Integer date equivalent of standard date (YYYYMMDD)
INTEGER-OF-DAY	I1	I	Integer date equivalent of Julian date (YYYYDDD)
INTEGER-OF-FORMATTED-DATE	U1, UT1, or X1, U2, UT2, or X2	I	Integer date equivalent of date contained in <i>argument-2</i> whose format is described by <i>argument-1</i>
INTEGER-PART	N1	I	Integer part of N1
LENGTH	A1, N1, O1, X1, U1, or UT1	I	Length of argument in national character positions or in alphanumeric character positions or bytes, depending on the argument type
LOG	N1	N	Natural logarithm of N1
LOG10	N1	N	Logarithm to base 10 of N1
LOWER-CASE	A1 or X1	X	All letters in the argument set to lowercase
	U1	U	All letters in the argument set to lowercase
	UT1	UT	All letters in the argument set to lowercase
MAX	A1...	X	Value of maximum argument; note that the type of function depends on the arguments
	I1...	I	Value of maximum argument; note that the type of function depends on the arguments
	N1...	N	Value of maximum argument; note that the type of function depends on the arguments
	X1...	X	Value of maximum argument; note that the type of function depends on the arguments
	U1...	U	Value of maximum argument; note that the type of function depends on the arguments
MEAN	N1...	N	Arithmetic mean of arguments
MEDIAN	N1...	N	Median of arguments

Table 63. <i>Table of functions</i> (continued)			
Function name	Arguments	Function type	Value returned
MIDRANGE	N1...	N	Mean of minimum and maximum arguments
MIN	A1...	X	Value of minimum argument; note that the type of function depends on the arguments
	I1...	I	Value of minimum argument; note that the type of function depends on the arguments
	N1...	N	Value of minimum argument; note that the type of function depends on the arguments
	X1...	X	Value of minimum argument; note that the type of function depends on the arguments
	U1...	U	Value of minimum argument; note that the type of function depends on the arguments
MOD	I1, I2	I	I1 modulo I2
NATIONAL-OF	A1, X1, D1, or UT1	U	The characters in the argument converted to national characters, using the code page identified by I2, if specified, or a default code page selected at compile time if I2 is unspecified
	A1, X1, D1, or UT1; I2	U	The characters in the argument converted to national characters, using the code page identified by I2, if specified, or a default code page selected at compile time if I2 is unspecified
NUMVAL	X1 or U1	N	Numeric value of simple numeric string
NUMVAL-C	X1 or U1; X1, X2; U1, U2	N	Numeric value of numeric string with optional commas and currency sign
NUMVAL-F	X1 or U1	N	Numeric value or approximation of the numeric value represented by the alphanumeric character string or national character string specified as the argument
ORD	A1 or X1	I	Ordinal position of the argument in collating sequence
ORD-MAX	A1..., N1..., X1..., or U1...	I	Ordinal position of maximum argument
ORD-MIN	A1..., N1..., X1..., or U1...	I	Ordinal position of minimum argument
PI	None	N	Value that is an approximation of π , the ratio of the circumference of a circle to its diameter.
PRESENT-VALUE	N1, N2...	N	Present value of a series of future period-end amounts, N2, at a discount rate of N1
RANDOM	I1, none	N	Random number

Table 63. <i>Table of functions</i> (continued)			
Function name	Arguments	Function type	Value returned
RANGE	I1...	I	Value of maximum argument minus value of minimum argument; note that the type of function depends on the arguments.
	N1...	N	Value of maximum argument minus value of minimum argument; note that the type of function depends on the arguments.
REM	N1, N2	N	Remainder of N1/N2
REVERSE	A1 or X1	X	Reverse order of the characters of the argument
	U1	U	Reverse order of the characters of the argument
SECONDS-FROM-FORMATTED-TIME	U1, UT1, or X1, U2, UT2, or X2	N	Standard numeric time equivalent of the data contained in <i>argument-2</i> as described by the format specified in <i>argument-1</i>
SECONDS-PAST-MIDNIGHT		N	Seconds past midnight as provided by the system
SIGN	N1	I	+1, 0, or -1 depending on the sign of the argument
SIN	N1	N	Sine of N1
SQRT	N1	N	Square root of N1
STANDARD-DEVIATION	N1...	N	Standard deviation of arguments
SUM	I1...	I	Sum of arguments; note that the type of function depends on the arguments.
	N1...	N	Sum of arguments; note that the type of function depends on the arguments.
TAN	N1	N	Tangent of N1
TEST-DATE-YYYYMMDD	I1	I	0 if <i>argument-1</i> is a valid standard date; otherwise identifies the sub-field in error
TEST-DAY-YYYYDDD	I1	I	0 if <i>argument-1</i> is a valid Julian date; otherwise identifies the sub-field in error
TEST-FORMATTED-DATETIME	U1, UT1, or X1, U2, UT2, or X2	I	0 if <i>argument-2</i> conforms in form to the format specified in <i>argument-1</i> and represents a valid date, time or combined representation according to that description; otherwise, identifies the character in error

Table 63. **Table of functions** (continued)

Function name	Arguments	Function type	Value returned
TEST-NUMVAL	X1 or U1	I	- If the content of <i>argument-1</i> conforms to the argument rules for the NUMVAL function, the returned value is 0. - If one or more characters are in error, the returned value is the position of the first character in error. - Otherwise, the returned value is (FUNCTION LENGTH (<i>argument-1</i>) + 1).
TEST-NUMVAL-C	X1 or U1; X1, X2; U1, U2	I	- If the content of <i>argument-1</i> conforms to the argument rules for the NUMVAL-C function, the returned value is 0. - If one or more characters are in error, the returned value is the position of the first character in error. - Otherwise, the returned value is (FUNCTION LENGTH (<i>argument-1</i>) + 1).
TEST-NUMVAL-F	X1 or U1	I	- If the content of <i>argument-1</i> conforms to the argument rules for the NUMVAL-F function, the returned value is 0. - If one or more characters are in error, the returned value is the position of the first character in error. - Otherwise, the returned value is (FUNCTION LENGTH (<i>argument-1</i>) + 1).
TRIM	A1, X1, U1, or UT1; K1	X, U, or UT	Character string that contains the characters in A1, X1, or U1 with leading spaces or trailing spaces deleted (if K1 is specified)
	A1, X1, U1, or UT1	X, U, or UT	Character string that contains the characters in A1, X1, or U1 with both leading spaces and trailing spaces deleted (if K1 is unspecified)
ULENGTH	A1, X1, U1, or UT1	I	Length of A1, X1, U1, or UT1 in UTF-8 or UTF-16 characters
UPOS	A1, X1, U1, or UT1, I2	I	Index of the I2th UTF-8 or UTF-16 character of A1, X1, U1, or UT1
UPPER-CASE	A1 or X1	X	All letters in the argument set to uppercase
	U1	U	All letters in the argument set to uppercase
	UT1	UT	All letters in the argument set to uppercase
USUBSTR	A1, X1, U1, or UT1, I2, I3	X, U, or UT	Alphanumeric character string that contains the I3 UTF-8 or UTF-16 characters of A1, X1, U1, or UT1, starting at the I2th character position
USUPPLEMENTARY	A1, X1, U1, or UT1	I	Index of the first Unicode supplementary character of A1, X1, U1, or UT1

<i>Table 63. Table of functions (continued)</i>			
Function name	Arguments	Function type	Value returned
UUID4	None	X	36-character alphanumeric string that is a version 4 universally unique identifier (UUID)
UVALID	A1, X1, U1, or UT1	I	0 if A1, X1, U1, or UT1 contains valid Unicode UTF-8 or UTF-16 data, or the index of the first invalid element of A1, X1, U1, or UT1
UWIDTH	A1, X1, U1, or UT1, I2	I	Width in bytes of the I2th UTF-8 or UTF-16 character of A1, X1, U1, or UT1
VARIANCE	N1...	N	Variance of arguments
WHEN-COMPILED	None	X	Date and time when program was compiled
YEAR-TO-YYYY	I1, I2	I	Expanded year equivalent (YYYY) of I1 (windowed year, YY), according to the 100-year interval whose ending year is specified by the sum of I2 and the year at execution time

Chapter 31. ABS

The ABS function returns the absolute value of the argument.

The function type depends on the argument type as follows:

Argument type	Function type
Integer	Integer
Numeric	Numeric

Format

➤ FUNCTION ABS — (— *argument-1* —) ➤

argument-1

Must be of class numeric.

The equivalent arithmetic expression is as follows:

- When the value of *argument-1* is zero or positive, (*argument-1*) is returned.
- When the value of *argument-1* is negative, (– (*argument-1*)) is returned.

Chapter 32. ACOS

The ACOS function returns a numeric value in radians that approximates the arccosine of the argument specified.

The function type is numeric.

Format

►► FUNCTION ACOS — (— *argument-1* —) ►◄

argument-1

Must be class numeric. The value of *argument-1* must be greater than or equal to -1 and less than or equal to +1.

The returned value is the approximation of the arccosine of the argument and is greater than or equal to zero and less than or equal to Pi.

Chapter 33. ANNUITY

The ANNUITY function returns a numeric value that approximates the ratio of an annuity paid at the end of each period, for a given number of periods, at a given interest rate, to an initial value of one.

The number of periods is specified by *argument-2*; the rate of interest is specified by *argument-1*. For example, if *argument-1* is zero and *argument-2* is four, the value returned is the approximation of the ratio 1 / 4.

The function type is numeric.

Format

►► FUNCTION ANNUITY — (— *argument-1* — *argument-2* —) ►◄

argument-1

Must be class numeric. The value of *argument-1* must be greater than or equal to zero.

argument-2

Must be a positive integer.

When the value of *argument-1* is zero, the value returned by the function is the approximation of:

$1 / \textit{argument-2}$

When the value of *argument-1* is not zero, the value of the function is the approximation of:

$\textit{argument-1} / (1 - (1 + \textit{argument-1})^{(- \textit{argument-2})})$

Chapter 34. ASIN

The ASIN function returns a numeric value in radians that approximates the arcsine of the argument specified.

The function type is numeric.

Format

►► FUNCTION ASIN — (— *argument-1* —) ►◄

argument-1

Must be class numeric. The value of *argument-1* must be greater than or equal to -1 and less than or equal to +1.

The returned value is the approximation of the arcsine of *argument-1* and is greater than or equal to -Pi/2 and less than or equal to +Pi/2.

Chapter 35. ATAN

The ATAN function returns a numeric value in radians that approximates the arctangent of the argument specified.

The function type is numeric.

Format

►► FUNCTION ATAN — (— *argument-1* —) ►◄

argument-1

Must be class numeric.

The returned value is the approximation of the arctangent of *argument-1* and is greater than $-\pi/2$ and less than $+\pi/2$.

Chapter 36. BIT-OF

The BIT-OF function returns an alphanumeric character string consisting of characters "1" and "0" that correspond to the binary value of each byte in the input argument.

The type of the function is alphanumeric.

Format

➡ **FUNCTION BIT-OF** — (— *argument-1* —) ➡

argument-1

Can be a data item, literal, or intrinsic function result of any data class. *argument-1* identifies the source character string for the conversion.

The returned value is an alphanumeric character string consisting of the bytes of *argument-1* converted to the bit pattern corresponding to the binary value of each byte in *argument-1*. The length of the output character string in bytes is eight times the length of *argument-1* in bytes.

Note: If *argument-1* is invalid, the behavior is undefined.

Examples

- FUNCTION BIT-OF('Hello, world!') returns
'110010001000010110010011100100111001011001101011010000001010011010010110
10011001100100111000010001011010'

- 01 BIN PIC 9(9) BINARY VALUE 12.
-
-

[illegible]

- 01 PAC PIC 9(5) COMP-3 VALUE 12345.
.
.

FUNCTION BIT-OF(PAC) returns '000100100011010001011111'

- 01 ZON PIC 9(5) VALUE 12345.
.
.

FUNCTION BIT-OF(ZON) returns '1111000111110010111100111111010011110101'

- FUNCTION BIT-OF(NATIONAL-OF(' ')) returns '0000000000100000'

Chapter 37. BIT-TO-CHAR

The BIT-TO-CHAR function returns a character string consisting of bytes that correspond to the bit pattern indicated by the sequence of "0" and "1" characters in the input argument.

The function type is alphanumeric.

Format

► FUNCTION BIT-TO-CHAR — (— *argument-1* —) ►

argument-1

Must be an alphanumeric literal, alphanumeric data item, or alphanumeric group item. *argument-1* must consist only of the characters "0" and "1". The length of *argument-1* must be a multiple of 8 bytes.

The returned value is a character string consisting of bytes that correspond to the bit pattern indicated by the sequence of "0" and "1" characters in *argument-1*. The length of the result string is equal to the length of the input string divided by 8.

Example

The following example shows how to use the BIT-TO-CHAR function to construct a two-byte header field, WS-HEADER-BYTES, from a string of 16 ones and zeros in WS-HEADER-BITS. The program then reverses the operation by taking WS-HEADER-BYTES and reconstructs the original string of ones and zeros in WS-BITS-ROUNDTrip by using the BIT-OF intrinsic function.

```
IDENTIFICATION DIVISION.
PROGRAM-ID. B2CDEMO.
ENVIRONMENT DIVISION.
DATA DIVISION.
WORKING-STORAGE SECTION.
* Bit string representing a 2-byte header:
* Byte 1: flags (A3), Byte 2: length (2A)
  01 WS-HEADER-BITS PIC X(16) VALUE '1010001100101010'.
  01 WS-HEADER-BYTES PIC X(2).
  01 WS-HEADER-HEX PIC X(4).
* For round-trip verification:
  01 WS-BITS-ROUNDTrip PIC X(16).
PROCEDURE DIVISION.
* Convert the bit string to its corresponding bytes
  MOVE FUNCTION BIT-TO-CHAR(WS-HEADER-BITS)
    TO WS-HEADER-BYTES
* Show the result in hexadecimal (e.g., A32A)
  MOVE FUNCTION HEX-OF(WS-HEADER-BYTES)
    TO WS-HEADER-HEX
  DISPLAY 'Header (hex): ' WS-HEADER-HEX
* Optional: verify round-trip using CHAR-TO-BIT
  MOVE FUNCTION BIT-OF(WS-HEADER-BYTES)
    TO WS-BITS-ROUNDTrip
  DISPLAY 'Bits (rt ): ' WS-BITS-ROUNDTrip
  STOP RUN.
```

The output of the program is:

```
Header (hex): A32A
Bits (rt ): 1010001100101010
```

FUNCTION BIT-TO-CHAR(MY-BIT-DATA) returns a character string with value x'FC88'.

Chapter 38. BYTE-LENGTH

The BYTE-LENGTH function returns an integer that is equal to the length of the argument in bytes.

The function type is integer.

Format

►► FUNCTION BYTE-LENGTH — (— *argument-1* —) ►◄

argument-1

Can be:

- An alphanumeric, national, UTF-8, or DBCS literal
- A group item (including unbounded groups) or an elementary data item of any class, including DBCS
- A data item described with USAGE POINTER, PROCEDURE-POINTER, FUNCTION-POINTER, or OBJECT REFERENCE
- The ADDRESS OF special register
- The LENGTH OF special register
- The XML-NTEXT special register
- The XML-TEXT special register

The returned value is a nine-digit integer determined as follows:

- The returned value is an integer that is the length of *argument-1* in number of bytes.
- If *argument-1* is an alphanumeric, national or UTF-8 group item, the value returned is equal to the length of *argument-1* in bytes. If any data item subordinate to *argument-1* is described with the DEPENDING phrase of the OCCURS clause, the length of *argument-1* is determined using the contents of the data item specified in the DEPENDING phrase. This evaluation is accomplished according to the rules of the OCCURS clause for a sending data item. For more information, see the discussions of the [“OCCURS clause” on page 200](#) and the [“USAGE clause” on page 237](#).

The returned value includes implicit FILLER positions, if any.

The only difference between the BYTE-LENGTH and LENGTH functions is that BYTE-LENGTH always returns the byte length of *argument-1*, even when *argument-1* is of class national or UTF-8. The BYTE-LENGTH function also accepts DBCS arguments.

Function BYTE-LENGTH is similar to the LENGTH OF special register, which also always returns the byte length of its argument, but the LENGTH OF special register can be used in more contexts. For more information, see *Finding the length of data items* in the *Enterprise COBOL Programming Guide*.

Related references

Chapter 65, “LENGTH,” on page 591

[“LENGTH OF” on page 22](#)

Chapter 39. CHAR

The CHAR function returns a one-character alphanumeric value that is a character in the program collating sequence having the ordinal position equal to the value of the argument specified.

The function type is alphanumeric.

Format

► FUNCTION CHAR — (— *argument-1* —) ►

argument-1

Must be an integer. The value must be greater than zero and less than or equal to the number of positions in the collating sequence associated with alphanumeric data items (a maximum of 256).

If more than one character has the same position in the program collating sequence, the character returned as the function value is that of the first literal specified for that character position in the ALPHABET clause.

If the current program collating sequence was not specified by an ALPHABET clause, the single-byte EBCDIC collating sequence is used. (See [“Conditional expressions” on page 268.](#))

Chapter 40. COMBINED-DATETIME

The COMBINED-DATETIME function combines a date in integer format with a time in standard numeric format into a single numeric value. From this value, you can derive both the date and time components. The return value is an approximation of the mathematical calculation described in the following explanation.

The function type is numeric.

Format

► FUNCTION COMBINED-DATETIME — (— *argument-1* — *argument-2* —) ►

argument-1

Must be in integer date form. For details, see [“Integer date form”](#) on page 509.

A value in integer date form is a positive integer that represents a number of days succeeding 31 December 1600, in the Gregorian calendar. It is based on a starting date of Monday, 1 January 1601 and integer date 1 represents Monday, 1 January 1601.

The INTDATE compiler option affects the starting date for the integer date functions. For details, see in the *Enterprise COBOL Programming Guide*.

argument-2

Must be in standard numeric time form. For details, see [“Standard numeric time form”](#) on page 510.

A value in standard numeric time form is a numeric value representing seconds past midnight.

Return value

The function determines the returned value using the arithmetic expression *argument-1* + (*argument-2*/100000). The integer part of the result represents the date, while the fractional part represents the time. The return value is a long-precision (64-bit) floating-point approximation of the calculation.

The INTDATE compiler option affects the starting date for the integer date functions. For details, see [INTDATE](#) in the *Enterprise COBOL Programming Guide*.

Examples

Given the integer date form value "143951", which represents the date 15 February 1995, and the standard numeric time form value "18867.812479168304", which represents the time "05:14:27.812479168304", the returned value would be the approximation of "143951.18867812479168304" calculated from : 143951 + (18867.812479168304 / 100000). See code examples for details.

The following example shows how to use the ARTHI (COMP) function:

```
01 MYRESULT PIC 9(8).9(10).
01 MYDATE PIC 9(18).
01 MYTIME PIC 9(06)V9(12).
PROCEDURE DIVISION.
    COMPUTE MYRESULT = 0.
    COMPUTE MYDATE = 143951.
    COMPUTE MYTIME = 18867.812479168304.
    COMPUTE MYRESULT =
```

```

        FUNCTION COMBINED-DATETIME (MYDATE, MYTIME).
        DISPLAY 'COMBINED-DATE-TIME: ' MYRESULT.
OUTPUT
COMBINED-DATE-TIME: 00143951.1886781248

```

The following example shows how to use the ARTHI (EXTEND) function:

```

WORKING-STORAGE SECTION.
01 MYRESULT PIC 9(9).9(17).
01 MYDATE PIC 9(31).
01 MYTIME PIC 9(19)V9(12).
PROCEDURE DIVISION.
    COMPUTE MYRESULT = 0.
    COMPUTE MYDATE = 143951.
    COMPUTE MYTIME = 18867.812479168304
    COMPUTE MYRESULT =
        FUNCTION COMBINED-DATETIME (MYDATE, MYTIME).
    DISPLAY 'COMBINED-DATE-TIME: ' MYRESULT.
OUTPUT
COMBINED-DATE-TIME: 000143951.18867812478856649

```

Chapter 41. CONTENT-OF

The CONTENT-OF intrinsic function returns the content of the argument.

The type of this function depends on the type of *argument-1* as follows:

Table 64. CONTENT-OF function type depending on the argument-1 types	
<i>argument-1</i> type	Function type
Alphabetic	Alphanumeric
Alphanumeric	Alphanumeric
National	National
UTF-8	UTF-8
Integer	Integer
Numeric	Numeric

Format

➤ FUNCTION CONTENT-OF — (— *argument-1* —) ➤

argument-1

Must be class alphabetic, alphanumeric, national, UTF-8, integer, or numeric.

The returned value is the content of *argument-1*. The size of the returned value is the size of *argument-1*.

The CONTENT-OF intrinsic function is useful when you want to pass an argument to a user-defined function that is effectively BY CONTENT. To do this, the formal parameter of the user-defined function must be passed BY REFERENCE on the USING phrase in the function definition, and then on the function invocation, the argument would be specified using the CONTENT-OF intrinsic function as a wrapper around the argument. For an example, see *Passing arguments BY CONTENT to user-defined functions* in the *Enterprise COBOL Programming Guide*.

Chapter 42. CURRENT-DATE

The CURRENT-DATE function returns a 21-character alphanumeric value that represents the calendar date, time of day, and time differential from Greenwich mean time provided by the system on which the function is evaluated.

The function type is alphanumeric.

Format ➤ FUNCTION CURRENT-DATE ➤
--

Reading from left to right, the 21 character positions of the returned value are as follows:

Character positions	Contents
1-4	Four numeric digits of the year in the Gregorian calendar
5-6	Two numeric digits of the month of the year, in the range 01 through 12
7-8	Two numeric digits of the day of the month, in the range 01 through 31
9-10	Two numeric digits of the hours past midnight, in the range 00 through 23
11-12	Two numeric digits of the minutes past the hour, in the range 00 through 59
13-14	Two numeric digits of the seconds past the minute, in the range 00 through 59
15-16	Two numeric digits of the hundredths of a second past the second, in the range 00 through 99. The value 00 is returned if the system on which the function is evaluated does not have the facility to provide the fractional part of a second.
17	Either the character '-' or the character '+'. The character '-' is returned if the local time indicated in the previous character positions is behind Greenwich mean time. The character '+' is returned if the local time indicated is the same as or ahead of Greenwich mean time. The character '0' is returned if the system on which this function is evaluated does not have the facility to provide the local time differential factor.
18-19	If character position 17 is '-', two numeric digits are returned in the range 00 through 12 indicating the number of hours that the reported time is behind Greenwich mean time. If character position 17 is '+', two numeric digits are returned in the range 00 through 13 indicating the number of hours that the reported time is ahead of Greenwich mean time. If character position 17 is '0', the value 00 is returned.
20-21	Two numeric digits are returned in the range 00 through 59 indicating the number of additional minutes that the reported time is ahead of or behind Greenwich mean time, depending on whether character position 17 is '+' or '-', respectively. If character position 17 is '0', the value 00 is returned.

Example

Given the current z/OS operating environment timestamp is "1995-02-15 05:14:27.812479168304" Eastern Standard Time, FUNCTION CURRENT-DATE returns a 21-character alphanumeric field that can be used as follows:

```
WORKING-STORAGE SECTION.  
01 WS-CURRENT-DATE-FIELDS.  
   05 WS-CURRENT-DATE.  
       10 WS-CURRENT-YEAR      PIC 9(4).  
       10 WS-CURRENT-MONTH    PIC 9(2).  
       10 WS-CURRENT-DAY      PIC 9(2).  
   05 WS-CURRENT-TIME.  
       10 WS-CURRENT-HOUR     PIC 9(2).  
       10 WS-CURRENT-MINUTE   PIC 9(2).  
       10 WS-CURRENT-SECOND   PIC 9(2).  
       10 WS-CURRENT-MS       PIC 9(2).  
   05 WS-DIFF-FROM-GMT        PIC S9(4) SIGN IS LEADING SEPARATE.  
PROCEDURE DIVISION.  
    MOVE FUNCTION CURRENT-DATE TO WS-CURRENT-DATE-FIELDS
```

WS-CURRENT-DATE-FIELDS contains "1995021505142781-0500".

For more information, see *Examples: numeric intrinsic functions* in the *Enterprise COBOL Programming Guide*.

Chapter 43. COS

The COS function returns a numeric value that approximates the cosine of the angle or arc specified by the argument in radians.

The function type is numeric.

Format

► FUNCTION COS — (— *argument-1* —) ►

argument-1

Must be class numeric.

The returned value is the approximation of the cosine of the argument and is greater than or equal to -1 and less than or equal to +1.

Chapter 44. CURRENT-DATE

The CURRENT-DATE function returns a 21-character alphanumeric value that represents the calendar date, time of day, and time differential from Greenwich mean time provided by the system on which the function is evaluated.

The function type is alphanumeric.

Format ➤ FUNCTION CURRENT-DATE ➤
--

Reading from left to right, the 21 character positions of the returned value are as follows:

Character positions	Contents
1-4	Four numeric digits of the year in the Gregorian calendar
5-6	Two numeric digits of the month of the year, in the range 01 through 12
7-8	Two numeric digits of the day of the month, in the range 01 through 31
9-10	Two numeric digits of the hours past midnight, in the range 00 through 23
11-12	Two numeric digits of the minutes past the hour, in the range 00 through 59
13-14	Two numeric digits of the seconds past the minute, in the range 00 through 59
15-16	Two numeric digits of the hundredths of a second past the second, in the range 00 through 99. The value 00 is returned if the system on which the function is evaluated does not have the facility to provide the fractional part of a second.
17	Either the character '-' or the character '+'. The character '-' is returned if the local time indicated in the previous character positions is behind Greenwich mean time. The character '+' is returned if the local time indicated is the same as or ahead of Greenwich mean time. The character '0' is returned if the system on which this function is evaluated does not have the facility to provide the local time differential factor.
18-19	If character position 17 is '-', two numeric digits are returned in the range 00 through 12 indicating the number of hours that the reported time is behind Greenwich mean time. If character position 17 is '+', two numeric digits are returned in the range 00 through 13 indicating the number of hours that the reported time is ahead of Greenwich mean time. If character position 17 is '0', the value 00 is returned.
20-21	Two numeric digits are returned in the range 00 through 59 indicating the number of additional minutes that the reported time is ahead of or behind Greenwich mean time, depending on whether character position 17 is '+' or '-', respectively. If character position 17 is '0', the value 00 is returned.

Example

Given the current z/OS operating environment timestamp is "1995-02-15 05:14:27.812479168304" Eastern Standard Time, FUNCTION CURRENT-DATE returns a 21-character alphanumeric field that can be used as follows:

```
WORKING-STORAGE SECTION.  
01 WS-CURRENT-DATE-FIELDS.  
   05 WS-CURRENT-DATE.  
       10 WS-CURRENT-YEAR      PIC 9(4).  
       10 WS-CURRENT-MONTH    PIC 9(2).  
       10 WS-CURRENT-DAY      PIC 9(2).  
   05 WS-CURRENT-TIME.  
       10 WS-CURRENT-HOUR     PIC 9(2).  
       10 WS-CURRENT-MINUTE   PIC 9(2).  
       10 WS-CURRENT-SECOND   PIC 9(2).  
       10 WS-CURRENT-MS       PIC 9(2).  
   05 WS-DIFF-FROM-GMT        PIC S9(4) SIGN IS LEADING SEPARATE.  
PROCEDURE DIVISION.  
    MOVE FUNCTION CURRENT-DATE TO WS-CURRENT-DATE-FIELDS
```

WS-CURRENT-DATE-FIELDS contains "1995021505142781-0500".

For more information, see *Examples: numeric intrinsic functions* in the *Enterprise COBOL Programming Guide*.

Chapter 45. DATE-OF-INTEGER

The DATE-OF-INTEGER function converts a date in the Gregorian calendar from integer date form to standard date form (YYYYMMDD).

The function type is integer.

The function result is an eight-digit integer.

Format

►► FUNCTION DATE-OF-INTEGER — (— *argument-1* —) ►◄

argument-1

A positive integer that represents a number of days succeeding December 31, 1600, in the Gregorian calendar. The valid range is 1 to 3,067,671, which corresponds to dates ranging from January 1, 1601 thru December 31, 9999.

The INTDATE compiler option affects the starting date for the integer date functions. For details, see *INTDATE* in the *Enterprise COBOL Programming Guide*.

The returned value represents the International Standards Organization (ISO) standard date equivalent to the integer specified as *argument-1*.

The returned value is an integer of the form YYYYMMDD where YYYY represents a year in the Gregorian calendar; MM represents the month of that year; and DD represents the day of that month.

Chapter 46. DATE-TO-YYYYMMDD

The DATE-TO-YYYYMMDD function converts *argument-1* from a date with a two-digit year (YYnnnn) to a date with a four-digit year (YYYYnnnn). *argument-2*, when added to the year at the time of execution, defines the ending year of a 100-year interval, or sliding century window, into which the year of *argument-1* falls.

The function type is integer.

Format

➡ FUNCTION DATE-TO-YYYYMMDD — (— *argument-1* — *argument-2*) ➡

argument-1

Must be zero or a positive integer less than 991232.

Note: The COBOL run time does not verify that the value is a valid date.

argument-2

Must be an integer. If *argument-2* is omitted, the function is evaluated assuming the value 50 was specified.

The sum of the year at the time of execution and the value of *argument-2* must be less than 10,000 and greater than 1,699.

See the following examples with returned values from the DATE-TO-YYYYMMDD function:

Current year	<i>argument-1</i> value	<i>argument-2</i> value	Returned value
2002	851003	120	20851003
2002	851003	-20	18851003
2002	851003	10	19851003
1994	981002	-10	18981002

Chapter 47. DAY-OF-INTEGER

The DAY-OF-INTEGER function converts a date in the Gregorian calendar from integer date form to Julian date form (YYYYDDD).

The function type is integer.

The function result is a seven-digit integer.

Format

►► FUNCTION DAY-OF-INTEGER — (— *argument-1* —) -◄◄

argument-1

A positive integer that represents a number of days succeeding December 31, 1600, in the Gregorian calendar. The valid range is 1 to 3,067,671, which corresponds to dates ranging from January 1, 1601 thru December 31, 9999.

The INTDATE compiler option affects the starting date for the integer date functions. For details, see *INTDATE* in the *Enterprise COBOL Programming Guide*.

The returned value represents the Julian equivalent of the integer specified as *argument-1*. The returned value is an integer of the form YYYYDDD where YYYY represents a year in the Gregorian calendar and DDD represents the day of that year.

Chapter 48. DAY-TO-YYYYDDD

The DAY-TO-YYYYDDD function converts *argument-1* from a date with a two-digit year (YYnnn) to a date with a four-digit year (YYYYnnn). *argument-2*, when added to the year at the time of execution, defines the ending year of a 100-year interval, or sliding century window, into which the year of *argument-1* falls.

The function type is integer.

Format

➤ FUNCTION DAY-TO-YYYYDDD — (— *argument-1* — *argument-2*) ➤

argument-1

Must be zero or a positive integer less than 99367.

The COBOL run time does not verify that the value is a valid date.

argument-2

Must be an integer. If *argument-2* is omitted, the function is evaluated assuming the value 50 was specified.

The sum of the year at the time of execution and the value of *argument-2* must be less than 10,000 and greater than 1,699.

Some examples of returned values from the DAY-TO-YYYYDDD function follow:

Current year	<i>argument-1</i> value	<i>argument-2</i> value	Returned value
2002	10004	-20	1910004
2002	10004	-120	1810004
2002	10004	20	2010004
2013	95005	-10	1995005

Chapter 49. DISPLAY-OF

The DISPLAY-OF function returns an alphanumeric character string consisting of the content of *argument-1* converted to a specific code page representation.

The type of the function is alphanumeric.

Format

► FUNCTION DISPLAY-OF — (— *argument-1* — *argument-2*) ◄

argument-1

Must be of class national (categories national, national-edited, and numeric-edited described with usage NATIONAL) or class UTF-8. *argument-1* identifies the source string for the conversion.

argument-2

Must be an integer. *argument-2* identifies the output code page for the conversion.

argument-2 must be a valid CCSID number and must identify an EBCDIC, ASCII, UTF-8, or EUC code page. An EBCDIC or ASCII code page can contain both single-byte and double-byte characters.

If *argument-2* is omitted, the output code page is the one that was in effect for the CODEPAGE compiler option when the source code was compiled.

The returned value is an alphanumeric character string consisting of the characters of *argument-1* converted to the output code page representation. When a source character cannot be converted to a character in the output code page, the source character is replaced with a substitution character. The following table shows substitution characters for some widely-used code pages:

Output code page	Substitution character
SBCS ASCII PC Windows SBCS	X'7F'
EBCDIC SBCS	X'3F'
ASCII DBCS	X'FCFC'
EBCDIC DBCS (except for Thai)	X'FEFE'
EBCDIC DBCS (Thai)	X'41B8'
PC DBCS (Japanese or Chinese)	X'FCFC'
PC DBCS (Korean)	X'BFFC'
EUC (Korean)	X'AFFE'
EUC (Japanese)	X'747E'
UTF-8	From SBCS: X'1A' From MBCS: X'EFBFD'
UTF-16	From SBCS: X'001A' From MBCS: X'FFFD'

No exception condition is raised.

The length of the returned value depends on the content of *argument-1* and the characteristics of the output code page.

Usage notes

- The CCSID for UTF-8 is 1208.
- If the output code page includes DBCS characters, the returned value can be a mixed SBCS and DBCS string.
- The DISPLAY-OF function, with *argument-2* specified, can be used to generate character data represented in a code page that differs from that specified in the CODEPAGE compiler option. Subsequent COBOL operations on that data can involve implicit conversions that assume the data is represented in the EBCDIC code page specified in the CODEPAGE compiler option. See *Converting to or from national (Unicode) representation* in the *Enterprise COBOL Programming Guide* for examples and programming techniques for processing data represented using more than one code page within a single program.

Exception

If the conversion fails, a severe runtime error occurs. Verify that the z/OS Unicode conversion services are installed and are configured to include the table for converting from CCSID 1200 to the output code page. See the *Customization Guide* for installation requirements to support the conversion.

Chapter 50. E

The E function returns an approximation of e , the base of natural logarithms.

The function type is numeric.

Format

►► FUNCTION E ◄◄

When ARITH(COMPAT) is in effect, FUNCTION E returns the long precision (64-bit) floating-point approximation of 2.718281828459045235360287471352662.

When ARITH(EXTEND) is in effect, FUNCTION E returns the extended precision (128-bit) floating-point approximation of 2.718281828459045235360287471352662.

Chapter 51. EXP

The EXP function returns an approximation of the value of *e* raised to the power of the argument.

The function type is numeric.

Format

➤ FUNCTION EXP — (— *argument-1* —) ➤

argument-1

Must be of class numeric.

When LP(32) is in effect, the EXP function produces return values that are identical to the Language Environment callable service CEESDEXP when ARITH(COMPAT) is in effect and CEESQEXP when ARITH(EXTEND) is in effect. The COBOL expression, (FUNCTION E ** (*argument-1*)), is an approximation of this value.

When LP(64) is in effect, the EXP function produces return values that are identical to the C runtime library function `exp(argument-1)` when ARITH(COMPAT) is in effect and `expl(argument-1)` when ARITH(EXTEND) is in effect. The COBOL expression, (FUNCTION E ** (*argument-1*)), is an approximation of this value.

Chapter 52. EXP10

The EXP10 function returns an approximation of the value of 10 raised to the power of the argument.
The function type is numeric.

Format

►► FUNCTION EXP10 — (— *argument-1* —) ►◄

argument-1

Must be of class numeric.

When LP(32) is in effect, the EXP10 function produces return values that are identical to the Language Environment callable service CEESDXPD with *parm1* set to 10.0 when ARITH(COMPAT) is in effect and CEESQXPQ with *parm1* set to 10.0 when ARITH(EXTEND) is in effect. The COBOL expression, $(10^{**}(\textit{argument-1}))$, is an approximation of this value.

When LP(64) is in effect, the EXP10 function produces return values that are identical to the C runtime library function `pow(10,argument-1)` when ARITH(COMPAT) is in effect and `powl(10,argument-1)` when ARITH(EXTEND) is in effect. The COBOL expression, $(10^{**}(\textit{argument-1}))$, is an approximation of this value.

Chapter 53. FACTORIAL

The FACTORIAL function returns an integer that is the factorial of the argument specified.

The function type is integer.

Format

►► FUNCTION FACTORIAL — (— *argument-1* —) ◄◄

argument-1

If the ARITH(COMPAT) compiler option is in effect, *argument-1* must be an integer greater than or equal to zero and less than or equal to 28. If the ARITH(EXTEND) compiler option is in effect, *argument-1* must be an integer greater than or equal to zero and less than or equal to 29.

If the value of *argument-1* is zero, the value 1 is returned; otherwise, the factorial of *argument-1* is returned.

Chapter 54. FORMATTED-CURRENT-DATE

The FORMATTED-CURRENT-DATE function returns a character string that represents the current date and time provided by the system on which the function is evaluated. The content of the returned value is formatted according to the format in the argument.

The function type depends on the argument type, as follows:

Table 65. FORMATTED-CURRENT-DATE function type depending on the argument types	
Argument type	Function type
Alphanumeric	Alphanumeric
National	National
UTF-8	UTF-8

Format

➤ FUNCTION FORMATTED-CURRENT-DATE — (— *argument-1* —) ➤

argument-1

Must be a national, a UTF-8, or an alphanumeric literal.

The content of *argument-1* must be a combined date and time format. For details, see [“Date and time formats” on page 510](#).

The returned value is a representation of the current date and time provided by the system on which the function is evaluated. The returned value is formatted according to the format in *argument-1*. The accuracy of the portion of the returned value that corresponds to the time format is determined by the z/OS operating environment, up to and including millionth (1/1000000) of a second.

Example

Given the format "YYYYMMDDThhmmss.ss+hhmm" and a current z/OS operating environment timestamp of "1995-02-15 05:14:27.812479168304" Eastern Standard Time, the returned value will be "19950215T05142781-0500".

Note: Given this particular format, "YYYYMMDDThhmmss.ss+hhmm", the only difference between the returned value for CURRENT-DATE and that for FORMATTED-CURRENT-DATE is the presence of the character "T" that separates the date portion from the time portion in the returned value of the latter function.

Chapter 55. FORMATTED-DATE

The FORMATTED-DATE function converts a date from its integer date form to the requested format.

The function type depends on the type of *argument-1* as follows:

Table 66. FORMATTED-DATE function type depending on the argument-1 types	
argument-1 type	Function type
Alphanumeric	Alphanumeric
National	National
UTF-8	UTF-8

Format

►► FUNCTION FORMATTED-DATE — (— *argument-1* — *argument-2* —) ►►

argument-1

Must be a national, a UTF-8 or an alphanumeric literal.

The content of argument-1 must be a date format. For details, see [“Date and time formats” on page 510.](#)

argument-2

It must be in integer date form. For details, see [“Integer date form” on page 509.](#)

A value in integer date form is a positive integer that represents a number of days succeeding 31 December 1600, in the Gregorian calendar. It is based on a starting date of Monday, 1 January 1601 and integer date 1 represents Monday, 1 January 1601.

The INTDATE compiler option affects the starting date for the integer date functions. For details, see *INTDATE* in the *Enterprise COBOL Programming Guide*.

The returned value is a representation of the date contained in *argument-2* according to the format in *argument-1*.

Example

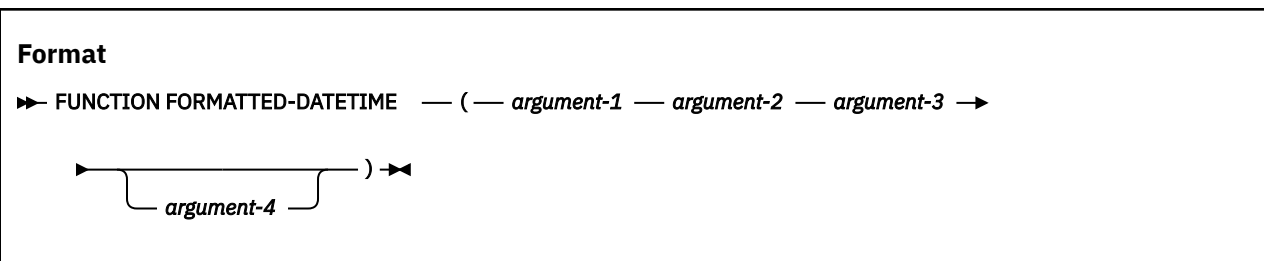
Given the date format "YYYYMMDD" and the value "143951", which represents the date 15 February 1995, the returned value would be "19950215".

Chapter 56. FORMATTED-DATETIME

The FORMATTED-DATETIME function uses a combined date and time format to convert and combine a date in the integer date form and a numeric time expressed as seconds past midnight to a formatted date and time representation according to that combined date and time format.

The type of this function depends on the type of *argument-1* as follows:

Table 67. FORMATTED-DATETIME function type depending on the argument-1 types	
argument-1 type	Function type
Alphanumeric	Alphanumeric
National	National
UTF-8	UTF-8



argument-1

Must be a national, a UTF-8, or an alphanumeric literal.

The content of *argument-1* must be a date format. For details, see [“Date and time formats” on page 510](#).

argument-2

Must be in integer date form. For details, see [“Integer date form” on page 509](#).

A value in integer date form is a positive integer that represents a number of days succeeding 31 December 1600, in the Gregorian calendar. It is based on a starting date of Monday, 1 January 1601 and integer date 1 represents Monday, 1 January 1601.

The INTDATE compiler option affects the starting date for the integer date functions. For details, see INTDATE in the *Enterprise COBOL Programming Guide*.

argument-3

Must be a numeric value in standard numeric time form. For details, see [“Standard numeric time form” on page 510](#).

A value in standard numeric time form is a numeric value that represents seconds past midnight.

argument-4

Must be an integer specifying the offset from Coordinated Universal Time (UTC) expressed in minutes. If *argument-4* is specified, the magnitude of the value must be less than or equal to 1439. For details, see [“UTC offset value” on page 510](#).

Note:

- An offset time format is a time format with the offset appended at the end, for example, hhmmss+hhmm, hh:mm:ss+hh:mm, hhmmss.ssss+hhmm, and hh:mm:ss.ssss+hh:mm.
- A UTC time format is a time format in the UTC timezone, for example, hhmmssZ, hh:mm:ssZ, hhmmss.ssssZ, or hh:mm:ss.ssssZ.

If *argument-4* is omitted and the time portion of the format in *argument-1* is a UTC format or an offset format, the function will be evaluated as though 0 was specified for *argument-4*.

Note: The offset value "1439" represents 23 hours and 59 minutes, which is one minute less than a day.

Returned values

- The returned value is a representation of the date contained in *argument-2* combined with the time contained in *argument-3* according to the format in *argument-1*.
- If the format in *argument-1* indicates that the returned value is expressed in UTC, the time portion of the returned value reflects the adjustment of the value in *argument-3* by the offset in *argument-4*.
- If the format in *argument-1* indicates that the time is to be returned as an offset from UTC, the value in *argument-3* is reflected directly in the time portion of the returned value, and the offset in *argument-4* is reflected directly in the offset portion of the returned value.

Example

If the first argument has the format "YYMMDDThhmmss.ss+hhmm", the second argument the value "143951", the third argument the value "18867.812479168304", and the fourth argument the value "+300", the returned value would be "19950215T05142781+0500".

Chapter 57. FORMATTED-TIME

The FORMATTED-TIME function uses a format to convert a value that represents seconds past midnight to a formatted time of day in the requested format.

The type of this function depends on the type of *argument-1* as follows:

Table 68. FORMATTED-TIME function type depending on the argument-1 types	
argument-1 type	Function type
Alphanumeric	Alphanumeric
National	National
UTF-8	UTF-8

Format ➤ FUNCTION FORMATTED-TIME — (— <i>argument-1</i> — <i>argument-2</i> — <i>argument-3</i>) ➤
--

argument-1

Must be a national, a UTF-8, or an alphanumeric literal.

The content of *argument-1* must be a time format. For details, see [“Date and time formats” on page 510](#).

argument-2

Must be a numeric value in standard numeric time form. For details, see [“Standard numeric time form” on page 510](#).

A value in standard numeric time form is a numeric value that represents seconds past midnight.

argument-3

Argument-3 is an integer representation of the offset from Coordinated Universal Time (UTC) expressed in minutes. If *argument-3* is specified, the magnitude of the value must be less than or equal to 1439. For details, see [“UTC offset value” on page 510](#).

Note: The offset value 1439 represents 23 hours and 59 minutes, which is one minute less than a day.

Argument-3 must not be specified if the time portion of the format in *argument-1* is neither a UTC format nor an offset format.

If *argument-3* is omitted and the time portion of the format in *argument-1* is a UTC format or an offset format, the function will be evaluated as though 0 was specified for *argument-3*.

Returned values

- The returned value is a representation of the standard numeric time contained in *argument-2* according to the format in *argument-1*.
- If the format in *argument-1* indicates that the returned value is expressed in UTC, the time portion of the returned value reflects the adjustment of the value in *argument-2* by the offset in *argument-3*.
- If the format in *argument-1* indicates that the time is returned as an offset from UTC, the value in *argument-2* is reflected directly in the time portion of the returned value, and the offset in *argument-3* is reflected directly in the offset portion of the returned value.

Example

If the first argument has the format "hhmmss.ss+hhmm", the second argument the value "18867.812479168304" which represents the local time, and the third argument the value "-300", which represents the five hours that Eastern Standard Time (EST) differs from UTC, the returned value would be "05142781-0500".

Chapter 58. HEX-OF

The HEX-OF function returns an alphanumeric character string consisting of the bytes of the input argument converted to a hexadecimal representation.

The type of the function is alphanumeric.

Format

► FUNCTION HEX-OF — (— *argument-1* —) ◄

argument-1

Can be a data item, literal, or intrinsic function result of any data class. *argument-1* identifies the source character string for the conversion.

The returned value is an alphanumeric character string consisting of the bytes of *argument-1* converted to a hexadecimal representation. The length of the output character string in bytes is two times the length of *argument-1* in bytes.

Note: If *argument-1* is invalid, the behavior is undefined.

Examples

- FUNCTION HEX-OF('Hello, world!') returns 'C8859393966B40A6969993845A'

- 01 BIN PIC 9(9) BINARY VALUE 12.
.
.

FUNCTION HEX-OF(BIN) returns '0000000C'

- 01 PAC PIC 9(5) COMP-3 VALUE 12345.
.
.

FUNCTION HEX-OF(PAC) returns '12345F'

- 01 ZON PIC 9(5) VALUE 12345.
.
.

FUNCTION HEX-OF(ZON) returns 'F1F2F3F4F5'

- FUNCTION HEX-OF(FUNCTION NATIONAL-OF(' ')) returns '0020'

Chapter 59. HEX-TO-CHAR

The HEX-TO-CHAR function returns a character string consisting of bytes that correspond to the hexadecimal digit characters in the input argument.

The function type is alphanumeric.

Format

► FUNCTION HEX-TO-CHAR — (— *argument-1* —) ◄

argument-1

Must be an alphanumeric literal, alphanumeric data item, or alphanumeric group item. *argument-1* must consist only of the characters "0" through "9", "A" through "F", and "a" through "f". The length of *argument-1* must be a multiple of 2 bytes.

The returned value is a character string consisting of bytes that correspond to the hexadecimal digit characters in *argument-1*. The length of the result string is equal to the length of the input string divided by 2.

Example

```
MOVE 'FFAABB' TO MY-HEX-DATA
```

FUNCTION HEX-TO-CHAR(MY-HEX-DATA) returns a character string with value x'FFAABB'.

Chapter 60. INTEGER

The INTEGER function returns the greatest integer value that is less than or equal to the argument specified.

The function type is integer.

Format

► FUNCTION INTEGER — (— *argument-1* —) ◄

argument-1

Must be class numeric.

The returned value is the greatest integer less than or equal to the value of *argument-1*. For example, FUNCTION INTEGER (2.5) returns a value of 2 and FUNCTION INTEGER (-2.5) returns a value of -3.

Chapter 61. INTEGER-OF-DATE

The INTEGER-OF-DATE function converts a date in the Gregorian calendar from standard date form (YYYYMMDD) to integer date form.

The function type is integer.

The function result is a seven-digit integer with a range from 1 to 3,067,671.

Format

►► FUNCTION INTEGER-OF-DATE — (— *argument-1* —) ►►

argument-1

Must be an integer of the form YYYYMMDD, whose value is obtained from the calculation $(YYYY * 10,000) + (MM * 100) + DD$, where:

- YYYY represents the year in the Gregorian calendar. It must be an integer greater than 1600, but not greater than 9999.
- MM represents a month and must be a positive integer less than 13.
- DD represents a day and must be a positive integer less than 32, provided that it is valid for the specified month and year combination.

The returned value is an integer that is the number of days that the date represented by *argument-1* succeeds December 31, 1600 in the Gregorian calendar.

The INTDATE compiler option affects the starting date for the integer date functions. For details, see *INTDATE* in the *Enterprise COBOL Programming Guide*.

Chapter 62. INTEGER-OF-DAY

The INTEGER-OF-DAY function converts a date in the Gregorian calendar from Julian date form (YYYYDDD) to integer date form.

The function type is integer.

The function result is a seven-digit integer.

Format

► FUNCTION INTEGER-OF-DAY — (— *argument-1* —) ►

argument-1

Must be an integer of the form YYYYDDD whose value is obtained from the calculation (YYYY * 1000) + DDD, where:

- YYYY represents the year in the Gregorian calendar. It must be an integer greater than 1600, but not greater than 9999.
- DDD represents the day of the year. It must be a positive integer less than 367, provided that it is valid for the year specified.

The returned value is an integer that is the number of days that the date represented by *argument-1* succeeds December 31, 1600 in the Gregorian calendar.

The INTDATE compiler option affects the starting date for the integer date functions. For details, see *INTDATE* in the *Enterprise COBOL Programming Guide*.

Chapter 63. INTEGER-OF-FORMATTED-DATE

The INTEGER-OF-FORMATTED-DATE function converts a date that is in a specified format to an integer date form.

The function type is integer.

Format

► FUNCTION INTEGER-OF-FORMATTED-DATE — (— *argument-1* — *argument-2* —) ◄

argument-1

Must be a national, UTF-8, or an alphanumeric literal in either a date format or a combined date and time format. For details, see [“Date and time formats” on page 510](#).

argument-2

Must be a data item of the same class as *argument-1*.

If *argument-1* is a date format, the content of *argument-2* should be a valid date in that format.

If *argument-1* is a combined date and time format, the content of *argument-2* should be a valid combined date and time in that format.

Returned values

The returned value is in the integer date form equivalent of the date represented by *argument-2* when analyzed according to *argument-1*. A value in integer date form is a positive integer that represents a number of days succeeding 31 December, 1600 in the Gregorian calendar. It is based on a starting date of Monday, 1 January, 1601 and integer date 1 represents Monday, 1 January, 1601.

The INTDATE compiler option affects the starting date for the integer date functions. For details, see in the *Enterprise COBOL Programming Guide*.

Note: If *argument-1* contains a combined date and time format, the time portion of *argument-2* is validated against the format in *argument-1*, but the returned value will not be impacted by the validated result.

Example

If the format of the first argument is "YYYYMMDD" and the value for the second argument is "19950215", the returned value would be 143951. The same value would be returned if the format of the first argument is "YYYYMMDDThhmmss.ss+hhmm" and the value for the second argument is "19950215T05142781+0500".

Chapter 64. INTEGER-PART

The INTEGER-PART function returns an integer that is the integer portion of the argument specified.

The function type is integer.

Format

► FUNCTION INTEGER-PART — (— *argument-1* —) ◄

argument-1

Must be class numeric.

If the value of *argument-1* is zero, the returned value is zero. If the value of *argument-1* is positive, the returned value is the greatest integer less than or equal to the value of *argument-1*. If the value of *argument-1* is negative, the returned value is the least integer greater than or equal to the value of *argument-1*.

Chapter 65. LENGTH

The LENGTH function returns an integer equal to the length of the argument in national character positions for arguments of usage NATIONAL and in alphanumeric character positions or bytes for all other arguments. An alphanumeric character position and a byte are equivalent.

The type of the function is integer.

Format

►► FUNCTION LENGTH — (— *argument-1* —) ►►

argument-1

Can be:

- An alphanumeric literal, a national literal, or a UTF-8 literal
- A group item (including unbounded groups) or an elementary data item of any class except DBCS
- A data item described with usage POINTER, PROCEDURE-POINTER, FUNCTION-POINTER, or OBJECT REFERENCE
- The ADDRESS OF special register
- The LENGTH OF special register
- The XML-NTEXT special register
- The XML-TEXT special register

The returned value is a 9-digit integer if LP(32) is in effect or an 18-digit integer if LP(64) is in effect and is determined as follows:

- If *argument-1* is an alphanumeric literal or an elementary data item of class alphabetic or alphanumeric, the value returned is equal to the number of alphanumeric character positions in the argument.

If *argument-1* is a null-terminated alphanumeric literal, the returned value is equal to the number of alphanumeric character positions in the literal excluding the null character at the end of the literal.

The length of an alphanumeric data item or literal containing a mix of single-byte and double-byte characters is counted as though each byte were a single-byte character.

- If *argument-1* is an alphanumeric group item, the value returned is equal to the length of *argument-1* in alphanumeric character positions regardless of the content of the group. If any data item subordinate to *argument-1* is described with the DEPENDING phrase of the OCCURS clause, the length of *argument-1* is determined using the contents of the data item specified in the DEPENDING phrase. This evaluation is accomplished according to the rules of the OCCURS clause for a sending data item. For more information, see the discussions of the [“OCCURS clause” on page 200](#) and the [“USAGE clause” on page 237](#).

The returned value includes implicit FILLER positions, if any.

- If *argument-1* is a national literal or an elementary data item described with usage NATIONAL, the value returned is equal to the length of *argument-1* in national character positions.

For example, if *argument-1* is defined as PIC 9(3) with usage NATIONAL, the returned value is 3, although the storage size of the argument is 6 bytes.

- If *argument-1* is a national group item, the value returned is equal to the length of *argument-1* in national character positions. If any data item subordinate to *argument-1* is described with the DEPENDING phrase of the OCCURS clause, the length of *argument-1* is determined using the contents of the data item specified in the DEPENDING phrase. This evaluation is accomplished according to the rules of the OCCURS clause for a sending data item. For more information, see the discussions of the [“OCCURS clause” on page 200](#) and the [“USAGE clause” on page 237](#).

The returned value includes implicit FILLER positions, if any.

- If *argument-1* is a UTF-8 literal or an elementary data item described with usage UTF-8, the value returned is equal to the length of *argument-1* in UTF-8 character positions.

For example, if *argument-1* is defined as PIC U(*n*), the returned value is always *n*, even though the storage size of the argument is 4**n* bytes and the actual number of bytes used by the item varies depending on the actual data in the item. This character length is known at compile time as this is a fixed character-length UTF-8 item.

- If *argument-1* is defined with the BYTE-LENGTH phrase of the PICTURE clause or the DYNAMIC LENGTH clause, then the character length is computed at runtime by examining all bytes occupied by the data in *argument-1* and counting the number of characters represented.
- If *argument-1* is a UTF-8 group item, the value returned is equal to the length of *argument-1* in UTF-8 character positions. In this case, like in the BYTE-LENGTH and DYNAMIC LENGTH case of UTF-8 data items, the character length is computed at run time by examining the data in the group item. If any data item subordinate to *argument-1* is described with the DEPENDING phrase of the OCCURS clause, the length of *argument-1* is determined using the contents of the data item specified in the DEPENDING phrase. This evaluation is accomplished according to the rules of the OCCURS clause for a sending data item. For more information, see the discussions of the [“OCCURS clause” on page 200](#) and the [“USAGE clause” on page 237](#).

The returned value includes implicit FILLER positions, if any.

- Otherwise, the returned value is the number of bytes of storage occupied by *argument-1*.

Chapter 66. LOG

The LOG function returns a numeric value that approximates the logarithm to the base e (natural log) of the argument specified.

The function type is numeric.

Format

► FUNCTION LOG — (— *argument-1* —) ►

argument-1

Must be class numeric. The value of *argument-1* must be greater than zero.

The returned value is the approximation of the logarithm to the base e of *argument-1*.

Chapter 67. LOG10

The LOG10 function returns a numeric value that approximates the logarithm to the base 10 of the argument specified.

The function type is numeric.

Format

►► FUNCTION LOG10 — (— *argument-1* —) ►◄

argument-1

Must be class numeric. The value of *argument-1* must be greater than zero.

The returned value is the approximation of the logarithm to the base 10 of *argument-1*.

Chapter 68. LOWER-CASE

The LOWER-CASE function returns a character string that contains the characters in the argument with each uppercase letter replaced by the corresponding lowercase letter.

The function type depends on the type of the argument, as follows:

Argument type	Function type
Alphabetic	Alphanumeric
Alphanumeric	Alphanumeric
National	National
UTF-8	UTF-8

Format ➤ FUNCTION LOWER-CASE — (— <i>argument-1</i> —) ➤
--

argument-1

Must be class alphabetic, alphanumeric, national, or UTF-8 and must be at least one character position in length.

Note: If *argument-1* is of the alphanumeric class, it must not contain UTF-8 encoded data.

The same character string as *argument-1* is returned, except that each uppercase letter is replaced by the corresponding lowercase letter.

If *argument-1* is of class alphabetic or alphanumeric, the uppercase letters 'A' through 'Z' are replaced by the corresponding lowercase letters 'a' through 'z', where the range of 'A' through 'Z' and the range of 'a' through 'z' are as shown in [“EBCDIC collating sequence”](#) on page 763, regardless of the code page in effect.

If *argument-1* is of class national or UTF-8, each uppercase letter is replaced by its corresponding lowercase letter based on the specification given in the Unicode database UnicodeData.txt, available from the Unicode Consortium at <http://www.unicode.org/>.

If *argument-1* is not of class UTF-8, the character string returned has the same length as *argument-1*. For UTF-8 arguments, the returned string may have a different byte length than the byte length of *argument-1*.

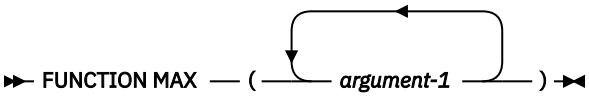
The character string returned has the same length as *argument-1*.

Chapter 69. MAX

The MAX function returns the content of the argument that contains the maximum value.

The function type depends on the argument type, as follows:

Argument type	Function type
Alphabetic	Alphanumeric
Alphanumeric	Alphanumeric
National	National
All arguments integer (includes integer arguments of usage NATIONAL)	Integer
Numeric (some arguments can be integer) (includes numeric arguments of usage NATIONAL)	Numeric

Format ➤ FUNCTION MAX — (— <i>argument-1</i> —) ➤ 

argument-1

Must be class alphabetic, alphanumeric, national, or numeric.

All arguments must be of the same class, except that a combination of alphabetic and alphanumeric arguments is allowed.

The returned value is the content of *argument-1* having the greatest value. The comparisons used to determine the greatest value are made according to the rules for simple conditions. For more information, see [“Conditional expressions”](#) on page 268.

If more than one *argument-1* has the same greatest value, the leftmost *argument-1* having that value is returned.

If the type of the function is alphanumeric or national, the size of the returned value is the size of the selected *argument-1*.

Chapter 70. MEAN

The MEAN function returns a numeric value that approximates the arithmetic average of its arguments.

The function type is numeric.

Format

► FUNCTION MEAN — (— *argument-1* —) ◄

argument-1

Must be class numeric.

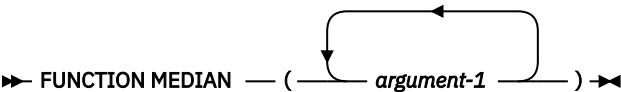
The returned value is the arithmetic mean of the *argument-1* series. The returned value is defined as the sum of the *argument-1* series divided by the number of occurrences referenced by *argument-1*.

Chapter 71. MEDIAN

The MEDIAN function returns the content of the argument whose value is the middle value in the list formed by arranging the arguments in sorted order.

The function type is numeric.

Format



argument-1

Must be class numeric.

The returned value is the content of *argument-1* having the middle value in the list formed by arranging all *argument-1* values in sorted order.

If the number of occurrences referenced by *argument-1* is odd, the returned value is such that at least half of the occurrences referenced by *argument-1* are greater than or equal to the returned value and at least half are less than or equal. If the number of occurrences referenced by *argument-1* is even, the returned value is the arithmetic mean of the values referenced by the two middle occurrences.

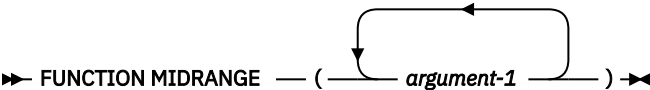
The comparisons used to arrange the argument values in sorted order are made according to the rules for simple conditions. For more information, see [“Conditional expressions” on page 268](#).

Chapter 72. MIDRANGE

The MIDRANGE function returns a numeric value that approximates the arithmetic average of the values of the minimum argument and the maximum argument.

The function type is numeric.

Format



argument-1

Must be class numeric.

The returned value is the arithmetic mean of the value of the greatest *argument-1* and the value of the least *argument-1*. The comparisons used to determine the greatest and least values are made according to the rules for simple conditions. For more information, see [“Conditional expressions” on page 268](#).

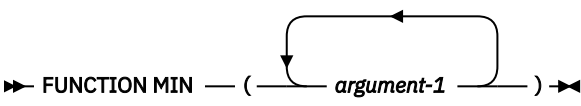
Chapter 73. MIN

The MIN function returns the content of the argument that contains the minimum value.

The function type depends on the argument type, as follows:

Argument type	Function type
Alphabetic	Alphanumeric
Alphanumeric	Alphanumeric
National	National
All arguments integer (includes integer arguments of usage NATIONAL)	Integer
Numeric (some arguments can be integer) (includes numeric arguments of usage NATIONAL)	Numeric

Format



➤ FUNCTION MIN — (*argument-1*) ➤

argument-1

Must be class alphabetic, alphanumeric, national, or numeric.

All arguments must be of the same class, except that a combination of alphabetic and alphanumeric arguments is allowed.

The returned value is the content of *argument-1* having the least value. The comparisons used to determine the least value are made according to the rules for simple conditions. For more information, see [“Conditional expressions”](#) on page 268.

If more than one *argument-1* has the same least value, the leftmost *argument-1* having that value is returned.

If the type of the function is alphanumeric or national, the size of the returned value is the size of the selected *argument-1*.

Chapter 74. MOD

The MOD function returns an integer value that is *argument-1* modulo *argument-2*.

The function type is integer.

The function result is an integer with as many digits as the shorter of *argument-1* and *argument-2*.

Format

➤ FUNCTION MOD — (— *argument-1* — *argument-2* —) ➤

argument-1

Must be an integer.

argument-2

Must be an integer. Must not be zero.

The returned value is *argument-1* modulo *argument-2*. The returned value is defined as:

$argument-1 - (argument-2 * FUNCTION\ INTEGER(argument-1 / argument-2))$

The following table lists expected results for some values of *argument-1* and *argument-2*.

<i>argument-1</i>	<i>argument-2</i>	Returned value
11	5	1
-11	5	4
11	-5	-4
-11	-5	-1

Chapter 75. NATIONAL-OF

The NATIONAL-OF function returns a national character string consisting of the national character representation of the characters in *argument-1*.

The type of the function is national.

Format

►► FUNCTION NATIONAL-OF — (— *argument-1* — *argument-2*) ►►

argument-1

Must be of class alphabetic, alphanumeric, UTF-8, or DBCS. *argument-1* specifies the source string for the conversion.

argument-2

Must be an integer. *argument-2* identifies the source code page for the conversion.

argument-2 must be a valid CCSID number and must identify an EBCDIC, ASCII, UTF-8, or EUC code page. An EBCDIC or ASCII code page can contain both single-byte and double-byte characters.

If *argument-2* is omitted and *argument-1* is of class UTF-8, then the source code page is 1208.

Otherwise, the source code page is the one that was in effect for the CODEPAGE compiler option when the source code was compiled.

The returned value is a national character string consisting of the characters of *argument-1* converted to national character representation. When a source character cannot be converted to a national character, the source character is converted to a substitution character. The substitution character is:

- X'001A' if converting a single-byte character
- X'FFFD' if converting a multi-byte character

No exception condition is raised.

The length of the returned value depends on the content of *argument-1* and the characteristics of the source code page.

Usage note: The CCSID for UTF-8 is 1208.

Exception: If the conversion fails, a severe runtime error occurs. Verify that the z/OS Unicode conversion services are installed and are configured to include the table for converting from the source code page to CCSID 1200. See the *Customization Guide* for installation requirements to support the conversion.

Chapter 76. NUMVAL

The NUMVAL function returns the numeric value represented by the alphanumeric character string or national character string specified as the argument. The function removes any leading or trailing spaces in the string to produce a numeric value.

The function type is numeric.

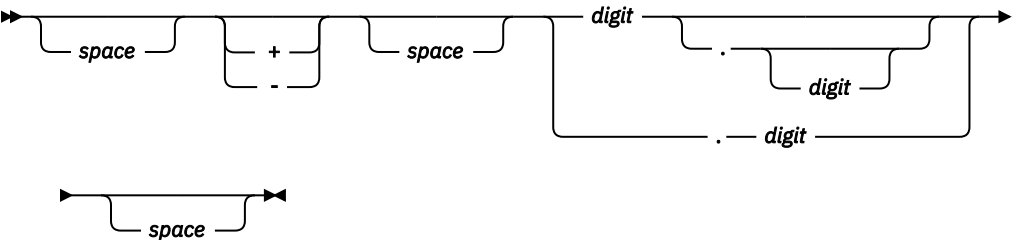
Format

➤ FUNCTION NUMVAL — (— *argument-1* —) ➤

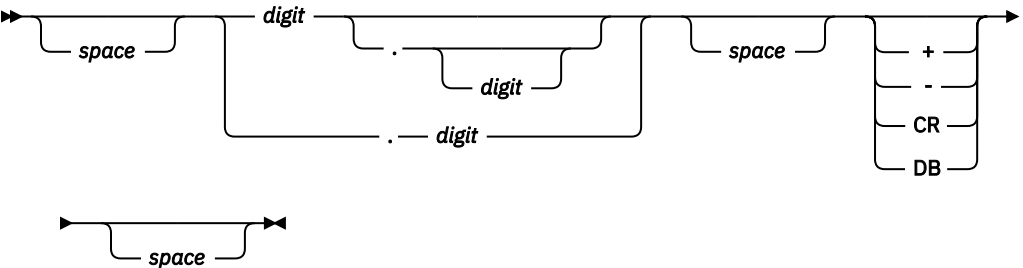
argument-1

Must be an alphanumeric literal, a national literal, or a data item of class national or class alphanumeric that contains a character string in either of the following formats:

Format 1: *argument-1*



Format 2: *argument-1*, monetary format



space

A string of one or more spaces.

digit

A string of one or more digits. If the ARITH(COMPAT) compiler option is in effect, the total number of digits must not exceed 18. If the ARITH(EXTEND) compiler option is in effect, the total number of digits must not exceed 31.

If the DECIMAL-POINT IS COMMA clause is specified in the SPECIAL-NAMES paragraph, a comma must be used in *argument-1* rather than a decimal point.

The returned value is a floating-point approximation of the numeric value represented by *argument-1*. The precision of the returned value depends on the setting of the ARITH compiler option. For details, see *Converting to numbers (NUMVAL, NUMVAL-C, NUMVAL-F)* in the *Enterprise COBOL Programming Guide*.

Chapter 77. NUMVAL-C

The NUMVAL-C function returns the numeric value represented by the alphanumeric character string or national character string specified as *argument-1*. The function removes the currency string, if any, and any grouping separators (commas or periods) to produce a numeric value.

The function type is numeric.

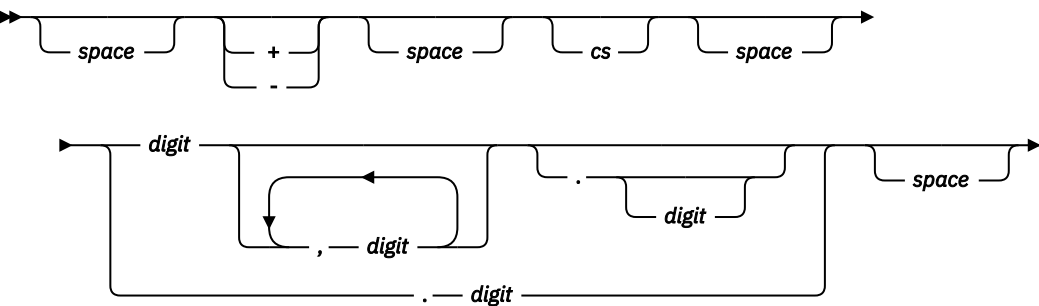
Format

► FUNCTION NUMVAL-C — (— *argument-1* — *argument-2*) ►

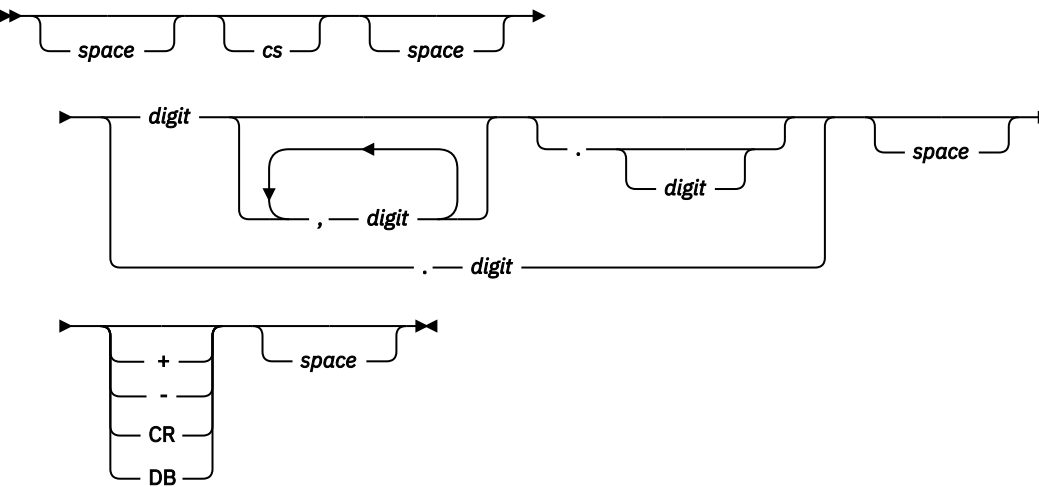
argument-1

Must be an alphanumeric literal, a national literal, or a data item of class alphanumeric or class national that contains a character string in either of the following formats:

Format 1: *argument-1*



Format 2: *argument-1*, monetary format



space

A string of one or more spaces.

cs

The string of one or more characters that form the currency sign. At most one copy of the characters specified by *cs* can occur in *argument-1*.

digit

A string of one or more digits. If the ARITH(COMPAT) compiler option is in effect, the total number of digits must not exceed 18. If the ARITH(EXTEND) compiler option is in effect, the total number of digits must not exceed 31.

If the DECIMAL-POINT IS COMMA clause is specified in the SPECIAL-NAMES paragraph, the functions of the comma and decimal point in *argument-1* are reversed.

argument-2

Specifies the currency string value.

The following rules apply:

- *argument-2* must be specified if the program contains more than one CURRENCY SIGN clause.
- *argument-2*, if specified, must be of the same class as *argument-1*.
- *argument-2* must not contain any of the digits 0 through 9, any leading or trailing spaces, or any of the special characters '+', '-', ',', or '.'.
- *argument-2* can be of any length valid for an elementary or group data item of the class of *argument-2*, including zero.
- Matching of *argument-2* is case sensitive. For example, if you specify *argument-2* as 'CHF', it will not match 'ChF', 'chf' or 'chF'.

If *argument-2* is not specified, the character used for *cs* is the currency symbol specified for the program.

The returned value is a floating-point approximation of the numeric value represented by *argument-1*. The precision of the returned value depends on the setting of the ARITH compiler option. For details, see *Converting to numbers (NUMVAL, NUMVAL-C, NUMVAL-F)* in the *Enterprise COBOL Programming Guide*.

Chapter 78. NUMVAL-F

The NUMVAL-F function returns the numeric value represented by the alphanumeric character string or national character string specified as the argument. The function removes any leading or trailing spaces in the string to produce a numeric value.

The function type is numeric.

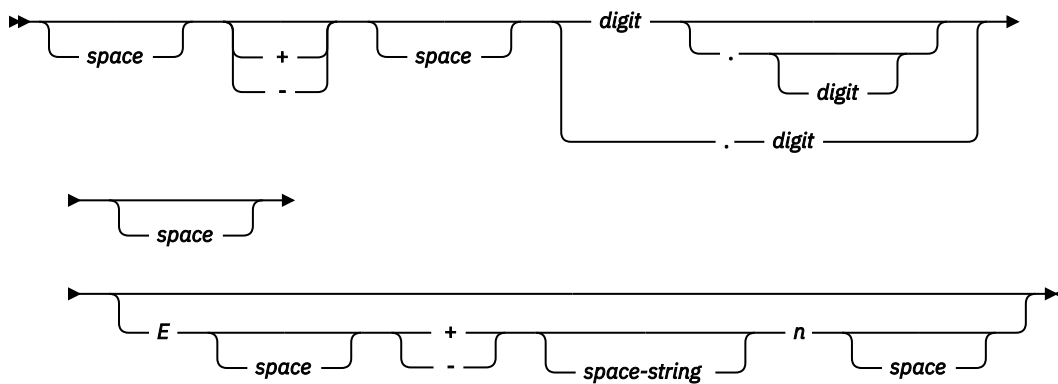
Format

► FUNCTION NUMVAL-F — (— *argument-1* —) ►

argument-1

Must be an alphanumeric literal, a national literal, or a data item of class national or class alphanumeric that contains a character string in the following format:

Format: *argument-1*



space

A string of one or more spaces.

digit

A string of one or more digits.

If the ARITH(COMPAT) compiler option is in effect, the total number of digits must not exceed 18.

If the ARITH(EXTEND) compiler option is in effect, the total number of digits must not exceed 31.

If the exponent clause is specified, the mantissa must not exceed 16 digits.

n

A string of one to four digits representing the exponent value.

E

If *argument-1* is alphanumeric, E must be either an uppercase or lowercase E character.

If *argument-1* is national, E must be either an uppercase or lowercase E national character.

If the DECIMAL-POINT IS COMMA clause is specified in the SPECIAL-NAMES paragraph, a comma must be used in *argument-1* rather than a decimal point.

The returned value is a floating-point approximation of the numeric value represented by *argument-1*. The precision of the returned value depends on the setting of the ARITH compiler option. For details, see *Converting to numbers (NUMVAL, NUMVAL-C, NUMVAL-F)* in the *Enterprise COBOL Programming Guide*.

Chapter 79. ORD

The ORD function returns an integer value that is the ordinal position of its argument in the collating sequence for the program. The lowest ordinal position is 1.

The function type is integer.

The function result is a three-digit integer.

Format

► FUNCTION ORD — (— *argument-1* —) ►

argument-1

Must be one character in length and must be class alphabetic or alphanumeric.

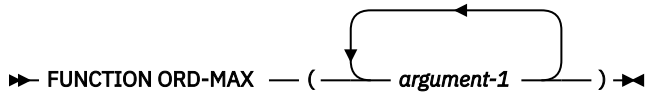
The returned value is the ordinal position of *argument-1* in the collating sequence for the program; it ranges from 1 to 256 depending on the collating sequence.

Chapter 80. ORD-MAX

The ORD-MAX function returns a value that is the ordinal position in the argument list of the argument that contains the maximum value.

The function type is integer.

Format



argument-1

Must be class alphabetic, alphanumeric, national, or numeric.

All arguments must be of the same class, except that a combination of alphabetic and alphanumeric arguments is allowed.

The returned value is the ordinal number that corresponds to the position of the *argument-1* having the greatest value in the *argument-1* series.

The comparisons used to determine the greatest-valued *argument-1* are made according to the rules for simple conditions. For more information, see [“Conditional expressions”](#) on page 268.

If more than one *argument-1* has the same greatest value, the number returned corresponds to the position of the leftmost *argument-1* having that value.

Chapter 81. ORD-MIN

The ORD-MIN function returns a value that is the ordinal position in the argument list of the argument that contains the minimum value.

The function type is integer.

Format

►► FUNCTION ORD-MIN — ( *argument-1*) ◄◄

argument-1

Must be class alphabetic, alphanumeric, national, or numeric.

All arguments must be of the same class, except that a combination of alphabetic and alphanumeric arguments is allowed.

The returned value is the ordinal number that corresponds to the position of the *argument-1* having the least value in the *argument-1* series.

The comparisons used to determine the least-valued *argument-1* are made according to the rules for simple conditions. For more information, see [“Conditional expressions”](#) on page 268.

If more than one *argument-1* has the same least value, the number returned corresponds to the position of the leftmost *argument-1* having that value.

Chapter 82. PI

The PI function returns a value that is an approximation of π , the ratio of the circumference of a circle to its diameter.

The function type is numeric.

Format

► FUNCTION PI ◄

When ARITH(COMPAT) is in effect, FUNCTION PI returns the long precision (64-bit) floating-point approximation of 3.141592653589793238462643383279503.

When ARITH(EXTEND) is in effect, FUNCTION PI returns the extended precision (128-bit) floating-point approximation of 3.141592653589793238462643383279503.

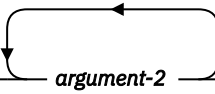
Chapter 83. PRESENT-VALUE

The PRESENT-VALUE function returns a value that approximates the present value of a series of future period-end amounts specified by *argument-2* at a discount rate specified by *argument-1*.

The function type is numeric.

Format

► FUNCTION PRESENT-VALUE — (— *argument-1* — *argument-2* —) ►



argument-1

Must be class numeric. Must be greater than -1.

argument-2

Must be class numeric.

The returned value is an approximation of the summation of a series of calculations with each term in the following form:

$$\text{argument-2} / (1 + \text{argument-1})^{** n}$$

There is one term for each occurrence of *argument-2*. The exponent *n* is incremented from 1 by 1 for each term in the series.

Chapter 84. RANDOM

The RANDOM function returns a numeric value that is a pseudorandom number from a rectangular distribution.

The function type is numeric.

Format

►► FUNCTION RANDOM ———— (— *argument-1* —)

argument-1

If *argument-1* is specified, it must be zero or a positive integer. However, only values in the range from zero up to and including 2,147,483,645 yield a distinct sequence of pseudorandom numbers.

If a subsequent reference specifies *argument-1*, a new sequence of pseudorandom numbers is started.

If the first reference to this function in the run unit does not specify *argument-1*, the seed value used will be zero.

In each case, subsequent references without specifying *argument-1* return the next number in the current sequence.

The returned value is exclusively between zero and one.

For a given seed value, the sequence of pseudorandom numbers is always the same.

The RANDOM function can be used in threaded programs. For an initial seed, a single sequence of pseudorandom numbers is returned, regardless of the thread that is running when RANDOM is invoked.

Chapter 85. RANGE

The RANGE function returns a value that is equal to the value of the maximum argument minus the value of the minimum argument.

The function type depends on the argument types, as follows:

Argument type	Function type
All arguments integer	Integer
Numeric (some arguments can be integer)	Numeric

Format

➡ FUNCTION RANGE

(

argument-1

)

➡

argument-1
Must be class numeric.

The returned value is equal to *argument-1* with the greatest value minus the *argument-1* with the least value. The comparisons used to determine the greatest and least values are made according to the rules for simple conditions. For more information, see [“Conditional expressions” on page 268](#).

Chapter 86. REM

The REM function returns a numeric value that is the remainder of *argument-1* divided by *argument-2*.

The function type is numeric.

Format

► FUNCTION REM — (— *argument-1* — *argument-2* —) ◄

argument-1

Must be class numeric.

argument-2

Must be class numeric. Must not be zero.

The returned value is the remainder of *argument-1* divided by *argument-2*. It is defined as the expression:

$argument-1 - (argument-2 * FUNCTION\ INTEGER-PART (argument-1 / argument-2))$

Chapter 87. REVERSE

The REVERSE function returns a character value of the same length as the argument, whose characters are the same as those specified in the argument except that they are in reverse order. For arguments of type national, character positions are reversed; UTF-16 characters that are surrogate pairs are treated as one character and UTF-16 characters that are not surrogate pairs are treated as one character.

The function type depends on the type of the argument, as follows:

Argument type	Function type
Alphabetic	Alphanumeric
Alphanumeric	Alphanumeric
National	National

Format ►► FUNCTION REVERSE — (— <i>argument-1</i> —) ►►

argument-1

Must be class alphabetic, alphanumeric, or national and must be at least one character in length. *argument-1* must contain valid UTF-8 or UTF-16 encoded characters:

- If *argument-1* is of class alphabetic or alphanumeric, it must contain valid UTF-8 data.
- If *argument-1* is of class national, it must contain valid UTF-16 data.

The returned value is a character string of the same length as *argument-1*, with the characters of *argument-1* in reversed order. For example, if *argument-1* contains ABC, the returned value is CBA.

Example 1

If *argument-1* is an alphanumeric data item that contains the UTF-8 value x'4BC3A4666572' ('Käfer'), the returned value is x'726566C3A44B' ('refäK').

Example 2

If *argument-1* is a national data item that contains the UTF-16 value x'0054 00F6 D847DDF3 0062 0075 0072 D858DC6B 0073' ('Tōber 繁'), the returned value is x'0073 D858DC6B 0072 0075 0062 D847DDF3 00F6 0054' ('s 繁ebōT').

Example 3

If *argument-1* is a UTF-8 encoded item and the UTF-8 argument contains composed characters, the combining characters are counted individually. For example, when encoded in UTF-8, the Unicode character ä can be x'C3A4' or x'61CC88'. With either of the UTF-8 characters in *argument-1*, the returned values of the REVERSE function are different. See the following table for details.

Table 69. REVERSE function of character Kä			
Character	Unicode encoding	UTF-8 encoding	Returned values of the REVERSE function
Kä	U+004B + U+00E4 (precomposed form, latin capital letter <i>K</i> + latin small letter <i>a</i> with diaeresis)	x'4BC3A4' (<i>Kä</i>)	x'C3A44B' (<i>äK</i>)
	U+004B + U+0061 + U+0308 (canonical decomposition, latin capital letter <i>K</i> + latin small letter <i>a</i> + combining diaeresis)	x'4B61CC88' (<i>Kä</i>)	x'CC88614B' (<i>äK</i>)

Chapter 88. SECONDS-FROM-FORMATTED-TIME

The SECONDS-FROM-FORMATTED-TIME function converts a time that is in a specified format to a numeric value that represents the number of seconds after midnight.

The function type is numeric.

Format

► FUNCTION SECONDS-FROM-FORMATTED-TIME — (— *argument-1* — *argument-2* —) ►

argument-1

Must be a national, UTF-8, or an alphanumeric literal in either a time format or a combined date and time format. For details, see [“Date and time formats” on page 510](#).

argument-2

Must be of the same type as *argument-1*.

If *argument-1* is a time format, the content of *argument-2* should be a time in that format.

If *argument-1* is a combined date and time format, the content of *argument-2* should be a valid date and time in that format.

Returned values

The equivalent arithmetic expression is $((H * 3600) + (M * 60) + S)$, where H is the portion of *argument-2* corresponding to the hours subfield of *argument-1*, M is the portion of *argument-2* corresponding to the minutes subfield of *argument-1*, and S is the portion of *argument-2* corresponding to the seconds subfield of *argument-1*.

Example

If the format for the first argument is "hhmmss.ss+hhmm" and the value for the second argument is "05142781+0500", the returned value would be 18867.81. The same value would be returned if the format for the first argument is "YYYYMMDDThhmmss.ss+hhmm" and the value for the second argument is "19950215T051427.81+0500".

Chapter 89. SECONDS-PAST-MIDNIGHT

The SECONDS-PAST-MIDNIGHT function returns a value in standard numeric time form that represents the current local time of day provided by the system on which the function is evaluated.

The function type is numeric.

Format

►► FUNCTION SECONDS-PAST-MIDNIGHT ◄◄

The SECONDS-PAST-MIDNIGHT function has no parameters.

Returned values

The returned value is in standard numeric time form that represents the current local time of the day provided by the system on which the function is evaluated, expressed in seconds past midnight. The precision is indicated by the z/OS operating environment and the precision of the user's data type.

Example

If the current time in the z/OS operating environment is 05:14:27.812479168304, the returned value would be as close to 18867.812479168304 as the z/OS operating environment is capable of providing.

Chapter 90. SIGN

The SIGN function returns +1, 0, or -1 depending on the sign of the argument.

The function type is integer.

Format

►► FUNCTION SIGN — (— *argument-1* —) ◄◄

argument-1

Must be of class numeric.

The returned value is as follows:

- If the value of *argument-1* is greater than zero, the returned value is 1.
- If the value of *argument-1* is zero, the returned value is 0.
- If the value of *argument-1* is less than zero, the returned value is -1.

Chapter 91. SIN

The SIN function returns a numeric value that approximates the sine of the angle or arc specified by the argument in radians.

The function type is numeric.

Format

►► FUNCTION SIN — (— *argument-1* —) ◄◄

argument-1

Must be class numeric.

The returned value is the approximation of the sine of *argument-1* and is greater than or equal to -1 and less than or equal to +1.

Chapter 92. SQRT

The SQRT function returns a numeric value that approximates the square root of the argument specified.

The function type is numeric.

Format

►► FUNCTION SQRT — (— *argument-1* —) ►◄

argument-1

Must be class numeric. The value of *argument-1* must be zero or positive.

The returned value is the absolute value of the approximation of the square root of *argument-1*.

Chapter 93. STANDARD-DEVIATION

The STANDARD-DEVIATION function returns a numeric value that approximates the standard deviation of its arguments.

The function type is numeric.

Format

➤

FUNCTION STANDARD-DEVIATION

—

(

↖

argument-1

↗

)

➤

argument-1

Must be class numeric.

The returned value is the approximation of the standard deviation of the *argument-1* series. The returned value is calculated as follows:

1. The difference between each *argument-1* and the arithmetic mean of the *argument-1* series is calculated and squared.
2. The values obtained are then added together. This quantity is divided by the number of values in the *argument-1* series.
3. The square root of the quotient obtained is then calculated. The returned value is the absolute value of this square root.

If the *argument-1* series consists of only one value, or if the *argument-1* series consists of all variable-occurrence data items and the total number of occurrences for all of them is one, the returned value is zero.

Chapter 94. SUM

The SUM function returns a value that is the sum of the arguments.

The function type depends on the argument types, as follows:

Argument type	Function type
All arguments integer	Integer
Numeric (some arguments can be integer)	Numeric

Format

» FUNCTION SUM

(

argument-1

)

«

argument-1

Must be class numeric.

The returned value is the sum of the arguments. If the *argument-1* series are all integers, the value returned is an integer. If the *argument-1* series are not all integers, a numeric value is returned.

Chapter 95. TAN

The TAN function returns a numeric value that approximates the tangent of the angle or arc that is specified by the argument in radians.

The function type is numeric.

Format

► FUNCTION TAN — (— *argument-1* —) ►

argument-1

Must be class numeric.

The returned value is the approximation of the tangent of *argument-1*.

Chapter 96. TEST-DATE-YYYYMMDD

The TEST-DATE-YYYYMMDD function tests whether a date in standard date form (YYYYMMDD) is a valid date in the Gregorian calendar. *Argument-1* of the INTEGER-OF-DATE function must be in standard date form.

The function type is integer.

Format

►► FUNCTION TEST-DATE-YYYYMMDD — (— *argument-1* —) ►►

argument-1

Must be an integer.

Returned values

- If the value of *argument-1* is less than 16010000 or greater than 99999999, the returned value is 1, which means the year is not within the range of 1601 to 9999.
- If the value of FUNCTION MOD (*argument-1* 10000) is less than 100 or greater than 1299, the returned value is 2, which means the month is not within the range of 1 to 12.
- If the value of FUNCTION MOD (*argument-1* 100) is less than 1 or greater than the number of days in the month determined by FUNCTION INTEGER (FUNCTION MOD (*argument-1* 10000)/100) of the year determined by FUNCTION INTEGER (*argument-1*/10000), the returned value is 3, which means the day is not valid for the given year and month.
- Otherwise, the returned value is 0 (zero) , which means the date is valid.

Examples

FUNCTION TEST-DATE-YYYYMMDD (19950215) returns 0 because the date is valid.

FUNCTION TEST-DATE-YYYYMMDD (12950215) returns 1 because the year is invalid and the value of *argument-1* is less than 16010000.

FUNCTION TEST-DATE-YYYYMMDD (912950215) returns 1 because the year is invalid and the value of *argument-1* is greater than 99999999.

FUNCTION TEST-DATE-YYYYMMDD (19921415) returns 2 because the month is not within the range of 1 to 12 and the value of FUNCTION MOD (*argument-1* 10000) is less than 100 or greater than 1299.

FUNCTION TEST-DATE-YYYYMMDD (19950240) returns 3 because the day is invalid for the given year and given month.

Related references

“Format of arguments and return values
for date and time intrinsic functions” on page 509
[Chapter 61, “INTEGER-OF-DATE,” on page 583](#)

Chapter 97. TEST-DAY-YYYYDDD

The TEST-DAY-YYYYDDD function tests whether a date in Julian date form (YYYYDDD) is a valid date in the Gregorian calendar. *Argument-1* of the INTEGER-OF-DAY function must be in Julian date form.

The function type is integer.

Format

► FUNCTION TEST-DAY-YYYYDDD — (— *argument-1* —) ◄

argument-1

Must be an integer.

Returned values

- If the value of *argument-1* is less than 1601000 or greater than 9999999, the returned value is 1, which means the year is not within the range of 1601 to 9999.
- If the value of FUNCTION MOD (*argument-1* 1000) is less than 1 or greater than the number of days in the year determined by FUNCTION INTEGER (*argument-1*/1000), the returned value is 2, which means the day is not valid in the given year.
- Otherwise, the returned value is 0 (zero), which means the date is valid.

Examples

FUNCTION TEST-DAY-YYYYDDD (1995146) returns 0 because the date is valid.

FUNCTION TEST-DAY-YYYYDDD (1295146) returns 1 because the year is invalid and the value of *argument-1* is less than 1601000.

FUNCTION TEST-DAY-YYYYDDD (1995446) returns 2 because the day is not valid in the given year and the value of FUNCTION MOD (*argument-1* 1000) is less than 1 or greater than the number of days in the year determined by FUNCTION INTEGER (*argument-1*/1000).

Related references

[“Format of arguments and return values for date and time intrinsic functions” on page 509](#)
[Chapter 62, “INTEGER-OF-DAY,” on page 585](#)

Chapter 98. TEST-FORMATTED-DATETIME

The TEST-FORMATTED-DATETIME function tests whether a data item that represents a date, a time, or a combined date and time is valid according to the specified format.

The function type is integer.

Format

►► FUNCTION TEST-FORMATTED-DATETIME — (— *argument-1* — *argument-2* —) ►►

argument-1

Must be a national, UTF-8, or an alphanumeric literal in a date format, a time format, or a combined date and time format. For details, see [“Date and time formats”](#) on page 510.

Note: The permitted values associated with date and time formats are specified in the [Date and time formats](#) of “Format of arguments and return values for date and time intrinsic functions” on page 509.

argument-2

Must be of the same type as *argument-1*.

Returned values

If no format problems or range problems occur during the evaluation of *argument-2* according to the format in *argument-1*, the returned value is zero. Otherwise, the returned value is the ordinal character position where the first error in *argument-2* is detected. In FUNCTION TEST-FORMATTED-DATETIME ("YYYYMMDD", A-DATE), where A-DATE is an 8-character data item, if A-DATE contains the value 20051314, the returned value will be 6, which indicates that the character 3 is in error because the month portion of A-DATE (character positions 5 and 6) contains 13. If A-DATE contains the value 15990316, the returned value will be 2, which indicates that the second character 5 is in error. The character 5 occupies the first position in which it can be determined that the year is less than 1601.

Examples

FUNCTION TEST-FORMATTED-DATETIME ("YYYYMMDD", "19950215") returns 0 because the data item in *argument-2* representing a date is valid according to the specified format in *argument-1*.

FUNCTION TEST-FORMATTED-DATETIME ("YYYYMMDD", "19959215") returns 5 because the data item in *argument-2* has an incorrect character at position 5 for the first digit of the month in YYYYMMDD.

FUNCTION TEST-FORMATTED-DATETIME ("YYYYMMDDThhmmss", "19950215T0514:27") returns 14 because the data item in *argument-2* has an incorrect colon ":" character at position 14 instead of the first character of the number of seconds according to the specified format in *argument-1*.

Chapter 99. TEST-NUMVAL

The TEST-NUMVAL function verifies that the contents of *argument-1* conform to the specification for *argument-1* of the NUMVAL function.

The function type is integer.

Format

►► FUNCTION TEST-NUMVAL — (— *argument-1* —) ►►

argument-1

Must be an alphanumeric literal, a national literal, or a data item of class alphanumeric or class national.

The returned value is as follows:

- If the content of *argument-1* conforms to the argument rules for the NUMVAL function, the returned value is 0.
- If one or more characters are in error, the returned value is the position of the first character in error.

Notes:

- If one or more spaces are embedded within a string of numeric characters, the returned value is the position of the first non-space character following the spaces, because one or more spaces following one or more digits is valid. For example, if *argument-1* is '0 1', the returned value will be 3.
- If the ARITH(COMPAT) compiler option is in effect, the returned value is the position of the 19th digit if no prior error is found, because the character in error for an argument that is greater than 18 digits is the 19th digit.
- If the ARITH(EXTEND) compiler option is in effect, the returned value is the position of the 32nd digit if no prior error is found, because the character in error for an argument that is greater than 31 digits is the 32nd digit.
- Otherwise, the returned value is (FUNCTION LENGTH (*argument-1*) + 1).

These errors include, but are not limited to:

- *argument-1* is zero-length.
- *argument-1* contains only spaces.
- *argument-1* contains valid characters but is incomplete, such as the string '+.'.

Chapter 100. TEST-NUMVAL-C

The TEST-NUMVAL-C function verifies that the contents of *argument-1* conform to the specification for *argument-1* of the NUMVAL-C function.

The function type is integer.

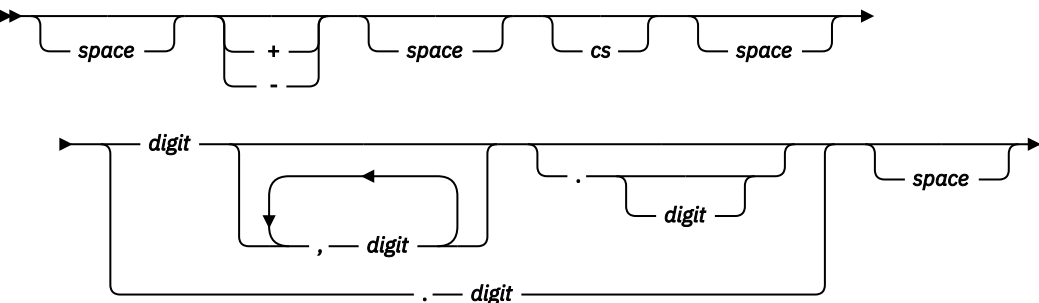
Format

► FUNCTION TEST-NUMVAL-C — (— *argument-1* — *argument-2*) ►

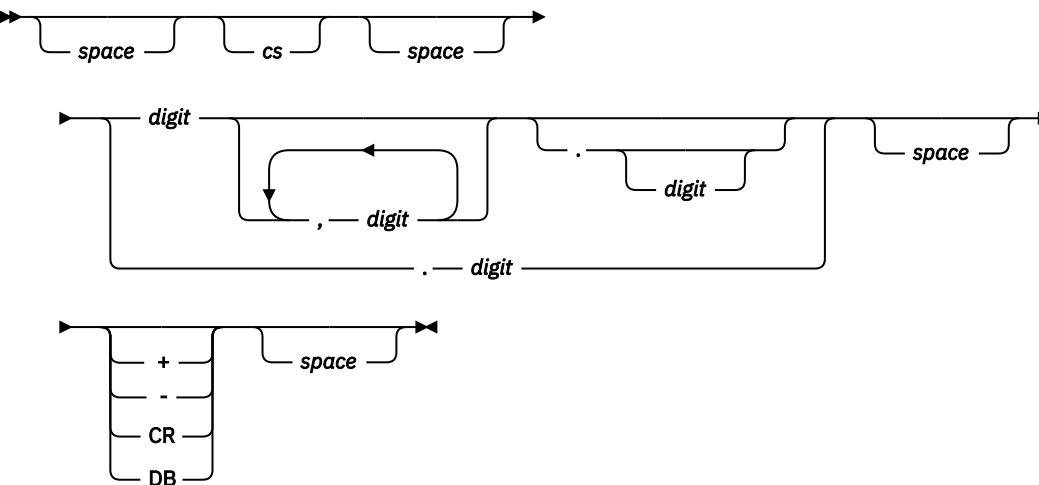
argument-1

Must be an alphanumeric literal, a national literal, or a data item of class alphanumeric or class national that contains a character string in either of the following formats:

Format 1: *argument-1*



Format 2: *argument-1*, monetary format



space

A string of one or more spaces.

cs

The string of one or more characters that form the currency sign. At most one copy of the characters specified by *cs* can occur in *argument-1*.

digit

A string of one or more digits.

If the ARITH(COMPAT) compiler option is in effect, the total number of digits must not exceed 18.

If the ARITH(EXTEND) compiler option is in effect, the total number of digits must not exceed 31.

If the DECIMAL-POINT IS COMMA clause is specified in the SPECIAL-NAMES paragraph, the functions of the comma and decimal point in *argument-1* are reversed.

argument-2

Specifies the currency string value.

The following rules apply:

- *argument-2* must be specified if the program contains more than one CURRENCY SIGN clause.
- *argument-2*, if specified, must be of the same class as *argument-1*.
- *argument-2* must not contain any of the digits 0 through 9, any leading or trailing spaces, or any of the special characters '+', '-', '.', or ','.
- *argument-2* can be of any length valid for an elementary or group data item of the class of *argument-2*, including zero.
- Matching of *argument-2* is case sensitive. For example, if you specify *argument-2* as 'CHF', it will not match 'ChF', 'chf' or 'chF'.

If *argument-2* is not specified, the character used for *cs* is the currency symbol specified for the program.

The returned value is as follows:

- If the content of *argument-1* conforms to the argument rules for the NUMVAL-C function, the returned value is 0.
- If one or more characters are in error, the returned value is the position of the first character in error.

Notes:

- If one or more spaces are embedded within a string of numeric characters, the returned value is the position of the first non-space character following the spaces, because one or more spaces following one or more digits is valid. For example, if *argument-1* is '0 1', the returned value will be 3.
- If the ARITH(COMPAT) compiler option is in effect, the returned value is the position of the 19th digit if no prior error is found, because the character in error for an argument that is greater than 18 digits is the 19th digit.
- If the ARITH(EXTEND) compiler option is in effect, the returned value is the position of the 32nd digit if no prior error is found, because the character in error for an argument that is greater than 31 digits is the 32nd digit.
- Otherwise, the returned value is (FUNCTION LENGTH (*argument-1*) + 1).

These errors include, but are not limited to:

- *argument-1* is zero-length.
- *argument-1* contains only spaces.
- *argument-1* contains valid characters but is incomplete, such as the string '+.'.

Chapter 101. TEST-NUMVAL-F

The TEST-NUMVAL-F function verifies that the contents of *argument-1* conform to the specification for *argument-1* of the NUMVAL-F function.

The function type is integer.

Format

► FUNCTION TEST-NUMVAL-F — (— *argument-1* —) ►◄

argument-1

Must be an alphanumeric literal, a national literal, or a data item of class alphanumeric or class national.

The returned value is as follows:

- If the content of *argument-1* conforms to the argument rules for the NUMVAL-F function, the returned value is 0.
- If one or more characters are in error, the returned value is the position of the first character in error.

Notes:

- If one or more spaces are embedded within a string of numeric characters, the returned value is the position of the first non-space character following the spaces, because one or more spaces following one or more digits is valid. For example, if *argument-1* is '0 1', the returned value will be 3.
- If the ARITH(COMPAT) compiler option is in effect, the returned value is the position of the 19th digit if no prior error is found, because the character in error for an argument that is greater than 18 digits is the 19th digit.
- If the ARITH(EXTEND) compiler option is in effect, the returned value is the position of the 32nd digit if no prior error is found, because the character in error for an argument that is greater than 31 digits is the 32nd digit.
- If the exponent value in the argument contains more than four significant digits, the returned value is the position of the fifth digit of the exponent.
- Otherwise, the returned value is (FUNCTION LENGTH (*argument-1*) + 1).

These errors include, but are not limited to:

- *argument-1* is zero-length.
- *argument-1* contains only spaces.
- *argument-1* contains valid characters but is incomplete, such as the string ' +.!

Chapter 102. TRIM

The TRIM function returns a character string that contains the characters in the argument with leading spaces, trailing spaces, or both, removed.

The function type depends on the argument type as follows:

Table 70. TRIM function types depending on the argument types	
Argument type	Function type
Alphabetic	Alphanumeric
Alphanumeric	Alphanumeric
National	National
UTF-8	UTF-8

Format

argument-1

Must be a data item of class alphabetic, alphanumeric, national, or UTF-8.

The returned value is:

- If LEADING is specified, the returned value is a character string that consists of the characters in *argument-1* beginning from the leftmost character position that does not contain a space character through the rightmost character position.
- If TRAILING is specified, the returned value is a character string that consists of the characters in *argument-1* beginning from the leftmost character position through the rightmost character position that does not contain a space character.
- If neither LEADING nor TRAILING is specified, the returned value is a character string that consists of the characters in *argument-1* beginning from the leftmost character position that does not contain a space character through the rightmost character position that does not contain a space character.
- If *argument-1* contains all spaces or *argument-1* is of length zero, the returned value is of length zero.

Examples

- FUNCTION TRIM(" Hello, world! ", LEADING) returns "Hello, world! "
- FUNCTION TRIM(" Hello, world! ", TRAILING) returns " Hello, world!"
- FUNCTION TRIM(" Hello, world! ") returns "Hello, world!"
- FUNCTION TRIM(" ") returns ""
- FUNCTION TRIM("") returns ""

Chapter 103. ULENGTH

The ULENGTH function returns an integer value that is equal to the number of UTF-8 or UTF-16 characters in a character data item argument that contains UTF-8 or UTF-16 data.

The function type is integer.

Format

➤ FUNCTION ULENGTH — (— *argument-1* —) ➤

argument-1

Must be of class alphabetic, alphanumeric, national or UTF-8. *argument-1* must contain valid UTF-8 or UTF-16 encoded characters:

- If *argument-1* is of class alphabetic, alphanumeric or UTF-8, it must contain valid UTF-8 data.
- If *argument-1* is of class national, it must contain valid UTF-16 data.

The returned value is the number of UTF-8 or UTF-16 characters in *argument-1*. If LP(32) is in effect, the returned value is a 9-digit integer; if LP(64) is in effect, the returned value is an 18-digit integer.

Example 1

If *argument-1* is a UTF-8 encoded item and the UTF-8 argument contains composed characters, the combining characters are counted individually in determining the length. For example, when encoded in UTF-8, the Unicode character *ä* can be x'C3A4' or x'61CC88'. With either of the UTF-8 characters as *argument-1*, the returned values of the ULENGTH function are different. See the following table for details.

Table 71. ULENGTH function of character *ä*

Character	Unicode encoding	UTF-8 encoding	Returned value of the ULENGTH function
<i>ä</i>	U+00E4 (precomposed form, latin small letter <i>a</i> with diaeresis)	x'C3A4'	1
	U+0061 + U+0308 (canonical decomposition, latin small letter <i>a</i> + combining diaeresis)	x'61CC88'	2

Example 2

If *argument-1* is a national data item that contains UTF-16 data and *argument-1* contains surrogate pairs, each pair of low and high surrogates will be counted as one UTF-16 character. For example, if B is a national item that contains the UTF-16 value nx'005400F6006200750072D858DC6B0073' ('Töber'  = X'D858DC6B' is counted as one UTF-16 character.

Chapter 104. UPOS

The UPOS function returns an integer value that is equal to the index of the *n*th UTF-8 or UTF-16 character in a character data item argument that contains UTF-8 or UTF-16.

The function type is integer.

Format

► FUNCTION UPOS — (— *argument-1* — *argument-2* —) ►

argument-1

Must be of class alphabetic, alphanumeric, national or UTF-8. *argument-1* must contain valid UTF-8 or UTF-16 encoded characters:

- If *argument-1* is of class alphabetic, alphanumeric or UTF-8, it must contain valid UTF-8 data.
- If *argument-1* is of class national, it must contain valid UTF-16 data.

argument-2

Must be an integer.

Suppose *argument-1* is alphabetic or alphanumeric and *argument-2*=*n*, the returned value is the byte position of the *n*th UTF-8 character in *argument-1*. Suppose *argument-1* is a national data item and *argument-2*=*n*, the returned value is the byte position of the *n*th UTF-16 character in *argument-1*.

If *argument-2* is not positive or if *argument-2* is larger than ULENGTH(*argument-1*), zero is returned. Otherwise, if *argument-2*=*n*, the returned value is the byte position in *argument-1* where the *n*th UTF-8 or UTF-16 character starts.

The returned value of UPOS is a 9-digit integer if LP(32) is in effect or an 18-digit integer if LP(64) is in effect.

Example 1

If A is an alphanumeric item that contains the UTF-8 value x'4BC3A4666572' ('Käfer'), the returned values are as follows:

- UPOS(A 1) returns 1
- UPOS(A 2) returns 2
- UPOS(A 3) returns 4
- UPOS(A 4) returns 5
- UPOS(A 5) returns 6

Example 2

If B is a national item that contains the UTF-16 value nx'005400F6006200750072D858DC6B0073' ('Töber s'), the returned values are as follows:

- UPOS (B 1) returns 1
- UPOS (B 2) returns 3
- UPOS (B 3) returns 5
- UPOS (B 4) returns 7
- UPOS (B 5) returns 9
- UPOS (B 6) returns 11
- UPOS (B 7) returns 15

Example 3

If *argument-1* is a UTF-8 encoded item and the UTF-8 argument contains composed characters, the combining characters are counted individually. For example, when encoded in UTF-8, the Unicode character ä can be x'C3A4' or x'61CC88'. With either of the UTF-8 characters in *argument-1*, the returned values of the UPOS function are different. See the following table for details.

Table 72. Returned values of the UPOS function			
argument-1	Unicode encoding	UTF-8 encoding	Returned values of the UPOS function
C = äK	U+00E4 + U+004B (precomposed form, latin small letter <i>a</i> with diaeresis + latin capital letter <i>K</i>)	x'C3A44B' (äK)	UPOS(C 1) returns 1 UPOS(C 2) returns 3 UPOS(C 3) returns 0
	U+0061 + U+0308 + U+004B (canonical decomposition, latin small letter <i>a</i> + combining diaeresis + latin capital letter <i>K</i>)	x'61CC884B' (äK)	UPOS(C 1) returns 1 UPOS(C 2) returns 2 UPOS(C 3) returns 4

Chapter 105. UPPER-CASE

The UPPER-CASE function returns a character string that contains the characters in the argument with each lowercase letter replaced by the corresponding uppercase letter.

The function type depends on the type of the argument, as follows:

Argument type	Function type
Alphabetic	Alphanumeric
Alphanumeric	Alphanumeric
National	National
UTF-8	UTF-8

Format ➤ FUNCTION UPPER-CASE — (— <i>argument-1</i> —) ->

argument-1

Must be of class alphabetic, alphanumeric, national, or UTF-8 and must be at least one character position in length.

Note: If *argument-1* is of the alphanumeric class, it must not contain UTF-8 encoded data.

The same character string as *argument-1* is returned, except that each lowercase letter is replaced by the corresponding uppercase letter.

If *argument-1* is alphabetic or alphanumeric, the lowercase letters 'a' through 'z' are replaced by the corresponding uppercase letters 'A' through 'Z', where the range of 'a' through 'z' and the range of 'A' through 'Z' are as shown in [“EBCDIC collating sequence” on page 763](#), regardless of the code page in effect.

If *argument-1* is national or UTF-8, each lowercase letter is replaced by its corresponding uppercase letter based on the specification given in the Unicode database UnicodeData.txt, available from the Unicode Consortium at www.unicode.org/.

If *argument-1* is not of class UTF-8, the character string returned has the same length as *argument-1*. For UTF-8 arguments, the returned string may have a different byte length than the byte length of *argument-1*.

Chapter 106. USUBSTR

The USUBSTR function returns a substring of the data in a character data item argument that contains UTF-8 or UTF-16 data.

The function type is alphanumeric, national, or UTF-8, depending on the class of *argument-1*.

Format

► FUNCTION USUBSTR — (— *argument-1* — *argument-2* — *argument-3* —) ►

argument-1

Must be of class alphabetic, alphanumeric, national or UTF-8. *argument-1* must contain valid UTF-8 or UTF-16 encoded characters:

- If *argument-1* is of class alphabetic, alphanumeric or UTF-8, it must contain valid UTF-8 data.
- If *argument-1* is of class national, it must contain valid UTF-16 data.

argument-2

Must be an integer that is greater than zero. It represents the starting position of a substring in *argument-1*.

argument-3

Must be an integer that is greater than or equal to zero. It represents the length of a substring in *argument-1*.

Note: The sum of *argument-2* and *argument-3* minus one must be less than or equal to ULENGTH(*argument-1*).

Suppose *argument-1* is alphabetic or alphanumeric, *argument-2* = *n* and *argument-3* = *m*, the returned value is an alphanumeric item that contains *m* UTF-8 characters from *argument-1*, starting with the *n*th UTF-8 character. Suppose *argument-1* is a national data item, *argument-2* = *n* and *argument-3* = *m*, the returned value is a national item that contains *m* UTF-16 characters from *argument-1*, starting with the *n*th UTF-16 character.

Example 1

If A is an alphanumeric item that contains the UTF-8 value x'4BC3A4666572' ('Käfer'), the returned values are as follows:

- USUBSTR(A 1 2) returns x'4BC3A4' ('Kä')
- USUBSTR(A 2 1) returns x'C3A4' ('ä')
- USUBSTR(A 2 2) returns x'C3A466' ('äf')
- USUBSTR(A 3 2) returns x'6665' ('fe')

Example 2

If B is a national item that contains the UTF-16 value nx'005400F6006200750072D858DC6B0073' ('Töber 森林'), the returned values are as follows:

- USUBSTR(B 1 2) returns x'005400F6' ('Tö')
- USUBSTR(B 2 1) returns x'00F6' ('ö')
- USUBSTR(B 2 2) returns x'00F60062' ('öb')
- USUBSTR(B 3 2) returns x'00620075' ('be')
- USUBSTR(B 5 2) returns x'0072D858DC6B' ('r 森')

- USUBSTR(B 6 2) returns x'D858DC6B0073' ('äKs')

Example 3

If *argument-1* is a UTF-8 encoded item and the UTF-8 argument contains composed characters, the combining characters are counted individually. For example, when encoded in UTF-8, the Unicode character ä can be x'C3A4' or x'61CC88'. With either of the UTF-8 characters in *argument-1*, the returned values of the USUBSTR function are different. See the following table for details.

Table 73. Returned values of the USUBSTR function			
argument-1	Unicode encoding	UTF-8 encoding	Returned values of the USUBSTR function
C = äK	U+00E4 + U+004B (precomposed form, latin small letter <i>a</i> with diaeresis + latin capital letter <i>K</i>)	x'C3A44B' (äK)	USUBSTR (C 1 1) returns x'C3A4' (ä) USUBSTR (C 2 1) returns x'4B' (K) USUBSTR (C 1 2) returns x'C3A44B' (äK)
	U+0061 + U+0308 + U+004B (canonical decomposition, latin small letter <i>a</i> + combining diaeresis + latin capital letter <i>K</i>)	x'61CC884B' (äK)	USUBSTR (C 1 1) returns x'61' (a) USUBSTR (C 2 1) returns x'CC88' (") USUBSTR (C 1 2) returns x'61CC88' (ä) USUBSTR (C 1 3) returns x'61CC884B' (äK)

Chapter 107. USUPPLEMENTARY

The USUPPLEMENTARY function returns an integer value that is equal to the index of the first Unicode supplementary character in a character data item argument that is encoded in UTF-8 or UTF-16.

A Unicode supplementary character is a character above U+FFFF, that is, a character outside of the Basic Multilingual Plane (BMP). These characters are encoded in UTF-16 with a surrogate pair (two 16-bit code units), or are encoded in UTF-8 with a 4-byte representation.

The function type is integer.

Format

►► FUNCTION USUPPLEMENTARY — (— *argument-1* —) ►►

argument-1

Must be of class alphabetic, alphanumeric, or national. *argument-1* must contain valid UTF-8 or UTF-16 data based on its class:

- If *argument-1* is of class alphabetic or alphanumeric, it must contain valid UTF-8 data.
- If *argument-1* is of class national, it must contain valid UTF-16 data.

The returned value is an integer, which differs based on the *argument-1* value, and is 9-digit if LP(32) is in effect or 18-digit if LP(64) is in effect:

- If the contents of *argument-1* are not valid Unicode (UTF-8 or UTF-16, depending on class), the returned result is unpredictable.
- If *argument-1* contains no supplementary characters, the returned value is zero.
- If *argument-1* is of class alphabetic or alphanumeric, the returned value is the byte position of the first UTF-8 supplementary character in *argument-1*.
- If *argument-1* is of class national, the returned value is the index, in UTF-16 encoding units, of the first UTF-16 supplementary character in *argument-1*.

Example 1

For example, the musical G-clef symbol is represented in UTF-16 Unicode by the surrogate pair nx'D834DD1E', or in UTF-8 Unicode by x'F09D849E'. Thus, for the following COBOL program fragment, the output of both DISPLAY statements is value 3.

```
01 N pic N(4) value nx'00200020D834DD1E'.
01 X pic X(6) value x'2020F09D849E'.
01 I pic 9.
...
Compute I = function Usupplementary(N)
Display I
Compute I = function Usupplementary(X)
Display I
```

Example 2

If *argument-1* is a UTF-8 encoded item and the UTF-8 argument contains composed characters, the combining characters are counted individually. For example, when encoded in UTF-8, the Unicode character â can be x'C3A4' or x'61CC88'. With either of the UTF-8 characters in *argument-1*, the returned values of the USUPPLEMENTARY function are different. See the following table for details.

Table 74. Returned values of the USUPPLEMENTARY function

argument-1	Unicode encoding	UTF-8 encoding	Returned values of the USUPPLEMENTARY function
<i>B</i> = äK	U+00E4 + U+004B (precomposed form, latin small letter <i>a</i> with diaeresis + latin capital letter <i>K</i>)	x'C3A4F0A1B7A44B' (äK)	USUPPLEMENTARY (B) returns 3
	U+0061 + U+0308 + U+004B (canonical decomposition, latin small letter <i>a</i> + combining diaeresis + latin capital letter <i>K</i>)	x'61CC88F0A1B7A44B' (äK)	USUPPLEMENTARY (B) returns 4

Chapter 108. UUID4

The UUID4 function returns a 36-character alphanumeric string that is a version 4 universally unique identifier(UUID).

The function type is alphanumeric.

Format

►► Function UUID4 ◄◄

Version 4 UUID is generated based on random numbers. It produces unique identifiers from separate applications without the coordination of a centralized agent or process. To maintain a high degree of randomness, the generation might take up processing time. The performance can be improved¹ by leveraging the random number generation facility in the IBM Z[®] hardware when it is available.

Note:

1. This performance improvement requires Message-Security-Assist Extension (MSA 7) that is supported in IBM z14[®] and higher architectures. The intrinsic function will automatically detect the availability of the facility and use it.

Example

The following COBOL program fragment produces output like 'UUID4: 4ed161b5-0d3c-4f06-8381-5f14678e13da', with subsequent executions producing different UUID values.

```
DISPLAY "UUID4: " FUNCTION UUID4
```


Chapter 109. UVALID

If a character data item contains valid UTF-8 or UTF-16 data, the UVALID function returns the value zero. If a character data item contains invalid UTF-8 or UTF-16 data, the UVALID function returns the index of the first invalid element.

The function type is integer.

Format

➤ FUNCTION UVALID — (— *argument-1* —) ➤

argument-1

Must be of class alphabetic, alphanumeric, national or UTF-8.

Must be of class alphabetic, alphanumeric, or national.

The returned value is an integer, which differs based on *argument-1*, and is 9-digit if LP(32) is in effect or 18-digit if LP(64) is in effect:

- If *argument-1* is of class alphabetic, alphanumeric or UTF-8, and it consists of valid UTF-8 encoded Unicode data, the returned value is zero.
- If *argument-1* is of class alphabetic, or alphanumeric, and it consists of valid UTF-8 encoded Unicode data, the returned value is zero.
- If *argument-1* is of class alphabetic, alphanumeric or UTF-8, and it contains invalid UTF-8 encoded Unicode data, the returned value is the position of the first byte where the invalid UTF-8 data starts.
- If *argument-1* is of class alphabetic, or alphanumeric, and it contains invalid UTF-8 encoded Unicode data, the returned value is the position of the first byte where the invalid UTF-8 data starts.
- If *argument-1* is of class national, and it consists of valid UTF-16 encoded Unicode data, the returned value is zero.
- If *argument-1* is of class national, and it contains invalid UTF-16 encoded Unicode data, the returned value is the position of the first UTF-16 encoding unit where the invalid UTF-16 data starts. This position is one plus the number of well-formed UTF-16 encoding units that precede the invalid data.

Note: The UVALID function indicates whether the character string contains well-formed Unicode UTF-8 or UTF-16 data. It does not indicate whether any or all of the Unicode code points represented by the character string are assigned to characters.

For UTF-8 data, the validity of a byte varies according to its range as listed in the table:

Table 75. Byte validity for UTF-8 data		
Value Range	Dependency	Validity
x'00' - x'7F'	None	Valid
x'80' - x'C1'	None	Invalid
x'C2' - x'DF'	Followed by another byte that is in the range x'80' to x'BF'	Valid

Table 75. Byte validity for UTF-8 data (continued)		
Value Range	Dependency	Validity
x'E0' - x'EF'	If the first byte is x'E0', followed by two more bytes that meet the following requirements: - The second byte is in the range x'A0' to x'BF' - The third byte is in the range x'80' to x'BF'	Valid
	If the first byte is in the range x'E1' to x'EC', both the second and third bytes are in the range x'80' to x'BF'	Valid
	If the first byte is x'ED', followed by two more bytes that meet the following requirements: - The second byte is in the range x'80' to x'9F' - The third byte is in the range x'80' to x'BF'	Valid
	If the first byte is in the range x'EE' to x'EF', both the second and third bytes are in the range x'80' to x'BF'	Valid
x'F0' - x'F4'	If the first byte is x'F0', followed by three more bytes that meet the following requirements: - The second byte is in the range x'90' to x'BF' - The third byte is in the range x'80' to x'BF' - The fourth byte is in the range x'80' to x'BF'	Valid
	If the first byte is in the range x'F1' to x'F3', all the second, third, and fourth bytes are in the range x'80' to x'BF'	Valid
	If the first byte is x'F4', followed by three more bytes that meet the following requirements: - The second byte is in the range x'80' to x'8f' - The third byte is in the range x'80' to x'BF' - The fourth byte is in the range x'80' to x'BF'	Valid
x'F5' - x'FF'	None	Invalid

For UTF-16 data, the validity of an encoding unit varies according to its range as listed in the table:

Table 76. Encoding unit validity for UTF-16 data			
Value Range	Dependency	Validity	Number of bytes if converted to UTF-8
nx'0000' - nx'007F'	None	Valid	1
nx'0080' - nx'07FF'	None	Valid	2
nx'0800' - nx'D7FF'	None	Valid	3
nx'D800' - nx'DBFF'	Must be followed by a second encoding unit with a value in the range nx'DC00' to nx'DFFF'	Valid	4 (A Unicode surrogate pair)
	Other cases	Invalid	Not applicable
nx'E000' - nx'FFFF'	None	Valid	3

Example 1

If A is an alphabetic or alphanumeric data item that contains value x'4BC3A4666572' ('Käfer') in UTF-8 encoding, the returned value from UVALID(A) is 0.

Example 2

If B is a national data item that contains value nx'005400F6006200750072D858DC6B0073' ('Töber 's') in UTF-16 encoding, the returned value from UVALID(B) is 0.

Example 3

If C is a national data item that contains value nx'0054D9C3006200750072D858DC6B0073' in UTF-16 encoding, the returned value from UVALID(C) is 2 because x'D9C3' does not have a low surrogate pair.

Example 4

If D is a national data item that contains value nx'005400F60062DC010072D858DC6B0073' in UTF-16 encoding, the returned value from UVALID(D) is 4 because x'DC01' does not have a corresponding high surrogate pair.

Chapter 110. UWIDTH

The UWIDTH function returns an integer value that is equal to the width in bytes of the *n*th UTF-8 or UTF-16 character in a character data item argument that is encoded in UTF-8 or UTF-16.

The function type is integer.

Format

► FUNCTION UWIDTH — (— *argument-1* — *argument-2* —) ►

argument-1

Must be of class alphabetic, alphanumeric, national , or UTF-8. *argument-1* must contain valid UTF-8 or UTF-16 encoded characters:

- If *argument-1* is of class alphabetic, alphanumeric , or UTF-8, it must contain valid UTF-8 data.
- If *argument-1* is of class national, it must contain valid UTF-16 data.

argument-2

Must be an integer.

The returned value is an integer.

If *argument-2* is not positive or if *argument-2* is larger than ULENGTH(*argument-1*), zero is returned. Otherwise, if *argument-2*=*n*, the returned value is the width in bytes of the *n*th UTF-8 or UTF-16 character in *argument-1*.

Example 1

If A is an alphanumeric item that contains the UTF-8 value x'4BC3A4666572' ('Käfer'), the returned values are as follows:

- UWIDTH(A 1) returns 1
- UWIDTH(A 2) returns 2
- UWIDTH(A 3) returns 1
- UWIDTH(A 4) returns 1
- UWIDTH(A 5) returns 1

Example 2

If B is a national item that contains the UTF-16 value nx'005400F6006200750072D858DC6B0073' ('Töber s'), the returned values are as follows:

- UWIDTH (B 1) returns 2
- UWIDTH (B 2) returns 2
- UWIDTH (B 3) returns 2
- UWIDTH (B 4) returns 2
- UWIDTH (B 5) returns 2
- UWIDTH (B 6) returns 4
- UWIDTH (B 7) returns 2

Example 3

If *argument-1* is a UTF-8 encoded item and the UTF-8 argument contains composed characters, the combining characters are counted individually. For example, when encoded in UTF-8, the Unicode character ä can be x'C3A4' or x'61CC88'. With either of the UTF-8 characters in *argument-1*, the returned values of the UWIDTH function are different. See the following table for details.

Table 77. Returned values of the UWIDTH function			
argument-1	Unicode encoding	UTF-8 encoding	Returned values of the UWIDTH function
C = äK	U+00E4 + U+004B (precomposed form, latin small letter <i>a</i> with diaeresis + latin capital letter <i>K</i>)	x'C3A44B' (äK)	UWIDTH (C 1) returns 2 UWIDTH (C 2) returns 1 UWIDTH (C 3) returns 0
	U+0061 + U+0308 + U+004B (canonical decomposition, latin small letter <i>a</i> + combining diaeresis + latin capital letter <i>K</i>)	x'61CC884B' (äK)	UWIDTH (C 1) returns 1 UWIDTH (C 2) returns 2 UWIDTH (C 3) returns 1

Chapter 111. VARIANCE

The VARIANCE function returns a numeric value that approximates the variance of its arguments.
The function type is numeric.

Format

➤

FUNCTION VARIANCE

—

argument-1

)

➤

argument-1
Must be class numeric.

The returned value is the approximation of the variance of the *argument-1* series.

The returned value is defined as the square of the standard deviation of the *argument-1* series. This value is calculated as follows:

1. The difference between each *argument-1* value and the arithmetic mean of the *argument-1* series is calculated and squared.
2. The values obtained are then added together. This quantity is divided by the number of values in the argument series.

If the *argument-1* series consists of only one value, or if the *argument-1* series consists of all variable-occurrence data items and the total number of occurrences for all of them is one, the returned value is zero.

Chapter 112. WHEN-COMPILED

The WHEN-COMPILED function returns the date and time that the program was compiled as provided by the system on which the program was compiled.

The function type is alphanumeric.

Format ► FUNCTION WHEN-COMPILED ◄

Reading from left to right, the 21 character positions of the returned value are as follows:

Character positions	Contents
1-4	Four numeric digits of the year in the Gregorian calendar
5-6	Two numeric digits of the month of the year, in the range 01 through 12
7-8	Two numeric digits of the day of the month, in the range 01 through 31
9-10	Two numeric digits of the hours past midnight, in the range 00 through 23
11-12	Two numeric digits of the minutes past the hour, in the range 00 through 59
13-14	Two numeric digits of the seconds past the minute, in the range 00 through 59
15-16	Two numeric digits of the hundredths of a second past the second, in the range 00 through 99. The value 00 is returned if the system on which the function is evaluated does not have the facility to provide the fractional part of a second.
17	Either the character '-' or the character '+'. The character '-' is returned if the local time indicated in the previous character positions is behind Greenwich mean time. The character '+' is returned if the local time indicated is the same as or ahead of Greenwich mean time. The character '0' is returned if the system on which this function is evaluated does not have the facility to provide the local time differential factor.
18-19	If character position 17 is '-', two numeric digits are returned in the range 00 through 12 indicating the number of hours that the reported time is behind Greenwich mean time. If character position 17 is '+', two numeric digits are returned in the range 00 through 13 indicating the number of hours that the reported time is ahead of Greenwich mean time. If character position 17 is '0', the value 00 is returned.
20-21	Two numeric digits are returned in the range 00 through 59 indicating the number of additional minutes that the reported time is ahead of or behind Greenwich mean time, depending on whether character position 17 is '+' or '-', respectively. If character position 17 is '0', the value 00 is returned.

The returned value is the date and time of compilation of the source unit that contains this function. If a program is a contained program, the returned value is the compilation date and time associated with the containing program.

Chapter 113. YEAR-TO-YYYY

The YEAR-TO-YYYY function converts *argument-1*, a two-digit year, to a four-digit year. *argument-2*, when added to the year at the time of execution, defines the ending year of a 100-year interval, or sliding century window, into which the year of *argument-1* falls.

The function type is integer.

Format

➤ FUNCTION YEAR-TO-YYYY — (— *argument-1* — *argument-2*) ➤

argument-1

Must be a non-negative integer that is less than 100.

argument-2

Must be an integer. If *argument-2* is omitted, the function is evaluated assuming the value 50 was specified.

The sum of the year at the time of execution and the value of *argument-2* must be less than 10,000 and greater than 1,699.

Examples of return values from the YEAR-TO-YYYY function are shown in the following table.

Current year	<i>argument-1</i> value	<i>argument-2</i> value	Returned value
1995	4	23	2004
1995	4	-15	1904
2008	98	23	1998
2008	98	-15	1898

Part 8. Compiler-directing statements and compiler directives

Chapter 114. Compiler-directing statements

A *compiler-directing statement* is a statement that causes the compiler to take a specific action during compilation.

You can use compiler-directing statements for the following purposes:

- Extended source library control (BASIS, DELETE, and INSERT statements)
- Source text manipulation (COPY and REPLACE statements)
- Exception handling (USE statement)
- Controlling compiler listings (*CONTROL, *CBL, EJECT, TITLE, SKIP1, SKIP2, and SKIP3 statements)
- Specifying compiler options (CBL and PROCESS statements)
- Specifying COBOL exception handling procedures (USE statements)

The SERVICE LABEL statement is used with Language Environment condition handling. It is also generated by the CICS integrated translator (and the separate CICS translator).

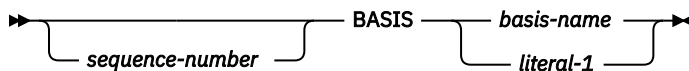
The following compiler directing statements have no effect: ENTER, READY or RESET TRACE, and SERVICE RELOAD.

BASIS statement

The BASIS statement is an extended source text library statement. It provides a complete COBOL program as the source for a compilation.

A complete program can be stored as an entry in a user-defined library and can be used as the source for a compilation. Compiler input is a BASIS statement, optionally followed by any number of INSERT and DELETE statements.

Format



sequence-number

Can optionally appear in columns 1 through 6, followed by a space. The content of this field is ignored.

BASIS

Can appear anywhere in columns 1 through 72, followed by *basis-name*. There must be no other text in the statement.

basis-name, literal-1

Is the name by which the library entry is known to the system environment.

For rules of formation and processing rules, see the description under *literal-1* and *text-name* of the “COPY statement” on page 696.

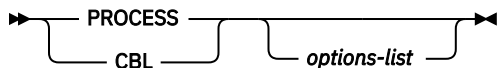
The source file remains unchanged after execution of the BASIS statement.

Usage note: If INSERT or DELETE statements are used to modify the COBOL source text provided by a BASIS statement, the sequence field of the COBOL source text must contain numeric sequence numbers in ascending order.

PROCESS(CBL) statement

With the PROCESS(CBL) statement, you can specify compiler options to be used in the compilation of the program. The PROCESS(CBL) statement is placed before the IDENTIFICATION DIVISION header of an outermost program.

Format



options-list

A series of one or more compiler options, each one separated by a comma or a space.

For more information about compiler options, see *Compiler options* in the *Enterprise COBOL Programming Guide*.

The PROCESS(CBL) statement can be preceded by a sequence number in columns 1 through 6. The first character of the sequence number must be numeric, and PROCESS or CBL can begin in column 8 or after; if a sequence number is not specified, PROCESS or CBL can begin in column 1 or after.

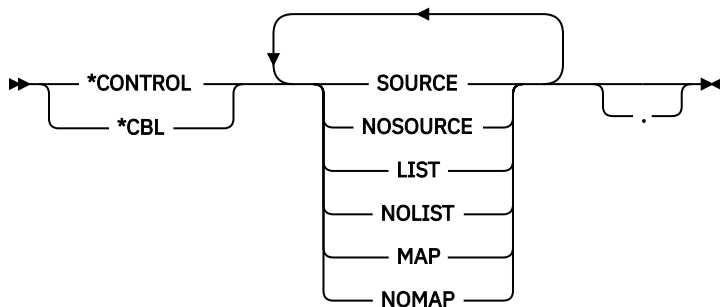
The PROCESS(CBL) statement must end before or at column 72, and options cannot be continued across multiple PROCESS(CBL) statements. However, you can use more than one PROCESS(CBL) statement. Multiple PROCESS(CBL) statements must follow one another with no intervening statements of any other type.

The PROCESS(CBL) statement must be placed before any comment lines or other compiler-directing statements.

*CONTROL (*CBL) statement

With the *CONTROL (or *CBL) statement, you can selectively display or suppress the listing of source code, object code, and storage maps throughout the source text.

Format



For a complete discussion of the output produced by these options, see *Getting listings* in the *Enterprise COBOL Programming Guide*.

The *CONTROL and *CBL statements are synonymous. *CONTROL is accepted anywhere that *CBL is accepted.

The characters *CONTROL or *CBL can start in any column beginning with column 7, followed by at least one space or comma and one or more option keywords. The option keywords must be separated by one or more spaces or commas. This statement must be the only statement on the line, and continuation is not allowed. The statement can be terminated with a period.

The *CONTROL and *CBL statements must be embedded in a program source. For example, in the case of batch applications, the *CONTROL and *CBL statements must be placed between the PROCESS (CBL) statement and the end of the program (or END PROGRAM marker, if specified).

The source line containing the *CONTROL (*CBL) statement will not appear in the source listing.

If an option is defined at installation as a fixed option, that fixed option takes precedence over all of the following parameter and statements:

- PARM (if available)
- CBL statement
- *CONTROL (*CBL) statement

The requested options are handled in the following manner:

1. If an option or its negation appears more than once in a *CONTROL statement, the last occurrence of the option word is used.
2. If the corresponding option has been requested as a parameter to the compiler, then a *CONTROL statement with the negation of the option word must precede the portions of the source text for which listing output is to be inhibited. Listing output then resumes when a *CONTROL statement with the affirmative option word is encountered.
3. If the negation of the corresponding option has been requested as a parameter to the compiler, then that listing is *always* inhibited.
4. The *CONTROL statement is in effect only within the source program in which it is written, including any contained programs. It does not remain in effect across batch compiles of two or more COBOL source programs.

Source code listing

The topic lists statements that control the listing of the input source text lines.

The statement can be any of the following one:

```
*CONTROL SOURCE      [*CBL SOURCE]
*CONTROL NOSOURCE    [*CBL NOSOURCE]
```

If a *CONTROL NOSOURCE statement is encountered and SOURCE has been requested as a compilation option, printing of the source listing is suppressed from this point on. An informational (I-level) message is issued stating that printing of the source has been suppressed.

Object code listing

The topic lists statements that control the listing of generated object code in the PROCEDURE DIVISION.

The statement can be any of the following one:

```
*CONTROL LIST        [*CBL LIST]
*CONTROL NOLIST      [*CBL NOLIST]
```

If a *CONTROL NOLIST statement is encountered, and LIST has been requested as a compilation option, listing of generated object code is suppressed from this point on.

Storage map listing

The topic lists statements that control the listing of storage map entries occurring in the DATA DIVISION.

The statement can be any of the following one:

```
*CONTROL MAP         [*CBL MAP]
*CONTROL NOMAP       [*CBL NOMAP]
```

If a `*CONTROL NOMAP` statement is encountered, and MAP has been requested as a compilation option, listing of storage map entries is suppressed from this point on.

For example, either of the following sets of statements produces a storage map listing in which A and B will not appear:

```
*CONTROL NOMAP      *CBL NOMAP
  01 A               01 A
  02 B               02 B
*CONTROL MAP         *CBL MAP
```

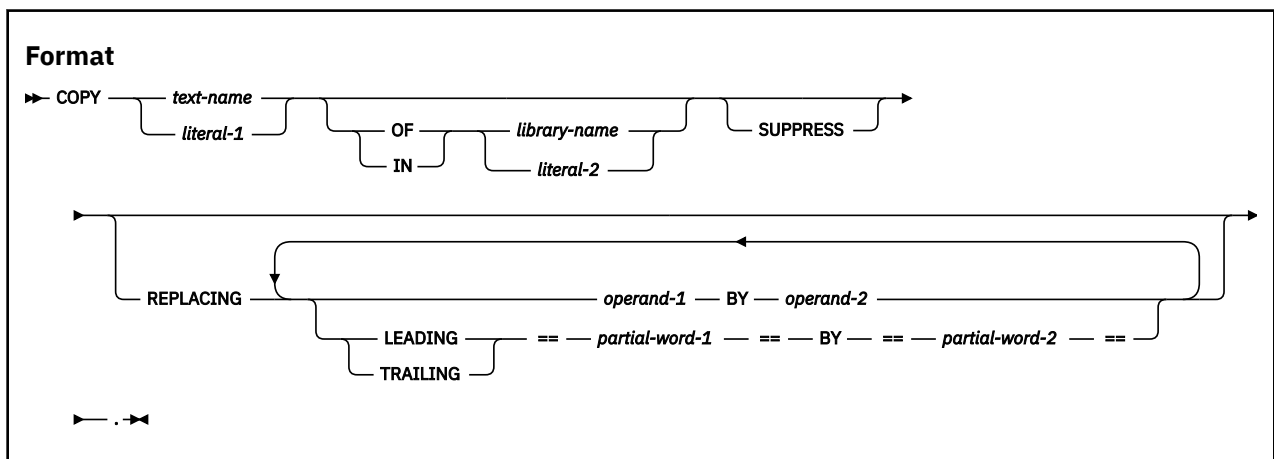
COPY statement

The COPY statement is a library statement that places prewritten text in a COBOL compilation unit.

Prewritten source code entries can be included in a compilation unit at compile time. Thus, an installation can use standard file descriptions, record descriptions, or procedures without recoding them. These entries and procedures can then be saved in user-created libraries; they can then be included in programs and class definitions by means of the COPY statement.

Compilation of the source code containing COPY statements is logically equivalent to processing all COPY statements before processing the resulting source text.

The effect of processing a COPY statement is that the library text associated with *text-name* is copied into the compilation unit, logically replacing the entire COPY statement, beginning with the word COPY and ending with the period, inclusive. When the REPLACING phrase is not specified, the library text is copied unchanged.



text-name, *library-name*

text-name identifies the copy text. *library-name* identifies where the copy text exists.

- Can be from 1-30 characters in length
- Can contain the following characters: Latin uppercase letters A-Z, Latin lowercase letters a-z, digits 0-9, and hyphen
- The first or last character must not be a hyphen
- Cannot contain an underscore

Neither *text-name* nor *library-name* need to be unique within a program. They can be identical to other user-defined words in the program.

text-name need not be qualified. If *text-name* is not qualified, a library-name of SYSLIB is assumed.

When the compiler searches for COPY members in PDS or PDSE datasets, including those specified in the COPYLOC option with the DSN argument, only the first eight characters of *text-name* are used

as the identifying name. When the compiler searches for COPY text in z/OS Unix directories, such as those directories specified via the -I option of the cob2 command or those directories specified via the SYSLIB environment variable (when the compiler is invoked from cob2) or those directories specified via the COPYLOC option with the PATH argument, all characters are significant.

If you have specified only z/OS UNIX search locations for COPY members, the restrictions mentioned previously on the legal characters in *text-name* do not apply and any valid COBOL character can be used in the name.

For details, see [“Copy member search order”](#) on page 705.

literal-1*, *literal-2

Must be alphanumeric literals. *literal-1* identifies the copy text. *literal-2* identifies where the copy text exists.

When compiling from JCL or TSO:

- Literals can be from 1-30 characters in length.
- Literals can contain characters: A-Z, a-z, 0-9, hyphen, @, #, or \$.
- The first or last character must not be a hyphen.
- Literals cannot contain an underscore.
- Only the first eight characters are used as the identifying name.

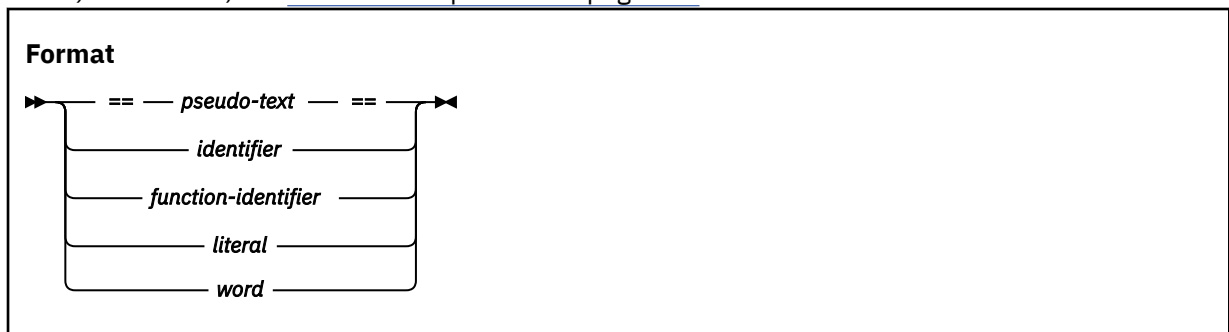
When compiling with the cob2 command and processing COPY text residing in the z/OS UNIX file system, the literal can be from 1 to 160 characters in length and the restrictions on characters in the literal mentioned previously do not apply and any character that is legal in a z/OS UNIX file name can appear in the literal.

The uniqueness of *text-name* and *library-name* is determined after the formation and conversion rules for a system-dependent name have been applied.

For information about the mapping of characters in the *text-name*, *library-name*, and literals, see *Compiler-directing statements* in the *Enterprise COBOL Programming Guide*.

operand-1*, *operand-2

Can be pseudo-text, an identifier, a function-identifier, a literal, or a COBOL word (except the word COPY). For details, see [“REPLACING phrase”](#) on page 698.



partial-word-1*, *partial-word-2

Can be a partial-word. For details, see [“REPLACING phrase”](#) on page 698.

Each COPY statement must be preceded by a space and ended with a separator period.

A COPY statement can appear in the source text anywhere a character string or a separator can appear.

COPY statements can be nested, and any COPY statement in a chain of nested COPY statements can have the REPLACING phrase, **as long as there is only one COPY statement in the chain that includes the REPLACING phrase**. When the REPLACING phrase is specified for a COPY statement that appears in a chain of nested COPY statements, the REPLACING phrase applies to all library text that is included by COPY statements nested under the COPY statement that has the REPLACING phrase.

A nested COPY statement cannot cause recursion. That is, a COPY member can be named only once in a set of nested COPY statements until the end-of-file for that COPY member is reached. For example,

assume that the source text contains the statement: COPY X. and library text X contains the statement: COPY Y..

In this case, library text contained in Y must not have a COPY X or a COPY Y statement.

For details, see [“Comparison and replacement rules” on page 699](#).

Library text copied from the library is placed into the same area of the resultant program as it is in the library. Library text must conform to the rules for the 85 COBOL Standard format. Library text can consist of or include any words, identifiers, or literals that can be written in the source text. This includes DBCS user-defined words, DBCS literals, and national literals.

Note: Characters outside those defined for COBOL words and separators must not appear in library text or pseudo-text except in comment lines, inline comments, comment-entries, alphanumeric literals, DBCS literals, or national literals.

SUPPRESS phrase

The SUPPRESS phrase specifies that the library text is not to be printed on the source listing.

REPLACING phrase

When the REPLACING phrase is specified, the library text is copied, and each properly matched occurrence of *operand-1* or *partial-word-1* within the library text is replaced by the associated *operand-2* or *partial-word-2*.

In the discussion that follows, when the LEADING or TRAILING keyword of the REPLACING phrase is specified, each operand of the REPLACING phrase must be a partial-word. Otherwise, each *operand* can consist of one of the following items:

- Pseudo-text
- An identifier
- A literal
- A COBOL word (except the word COPY)
- A function-identifier

pseudo-text

A sequence of text words that are bounded by, but not including, pseudo-text delimiters (==). Both characters of each pseudo-text delimiter must appear on one line.

Individual text words within pseudo-text can be up to 322 characters long. They can be continued subject to the normal continuation rules for source code format.

A text word must be delimited by separators. For more information, see [Chapter 1, “Characters,” on page 3](#).

pseudo-text-1 refers to pseudo-text when used for *operand-1*, and *pseudo-text-2* refers to pseudo-text when used for *operand-2*.

pseudo-text-1 can be one or more text words. It can consist solely of the separator comma or separator semicolon. *pseudo-text-2* can be zero or more text words. It can consist solely of space characters, comment lines, or inline comments.

Each text word in *pseudo-text-2* that is to be copied into the program is placed in the same area of the resultant program as the area in which it appears in *pseudo-text-2*.

Pseudo-text can consist of or include any words (except COPY), identifiers, or literals that can be written in the source text. This includes DBCS user-defined words, DBCS literals, and national literals.

DBCS user-defined words must be wholly formed; that is, there is no partial-word replacement for DBCS words.

Words or literals containing DBCS characters cannot be continued across lines.

Use pseudo-text when you replace a PICTURE character-string. To avoid ambiguities, the entire PICTURE clause, including the keyword PICTURE or PIC, should be specified in *pseudo-text-1*.

identifier

Can be defined in any section of the DATA DIVISION.

literal

Can be numeric, alphanumeric, DBCS, or national.

word

Can be any single COBOL word (except COPY), including DBCS user-defined words. DBCS user-defined words must be wholly formed. You cannot replace part of a DBCS word.

You can include the nonseparator COBOL characters (for example, + * / \$ < > =) as part of a COBOL word when used as REPLACING operands. In addition, a hyphen or underscore can be at the beginning of the word or a hyphen can be at the end of the word.

function-identifier

A sequence of character strings and separators that uniquely references the data item that results from the evaluation of a function. For more information, see [“Function-identifier” on page 77](#).

partial-word

A single text word that is bounded by, but not including, pseudo-text delimiters (==). Both characters of each pseudo-text delimiter must appear on one line. However, the text word within a partial-word can be continued.

The following rules apply to *partial-word-1* and *partial-word-2*:

- *partial-word-1* consists of one text word.
- *partial-word-2* consists of zero or one text word.
- *partial-word-1* and *partial-word-2* cannot be an alphanumeric literal, national literal, DBCS literal, or DBCS word.

For purposes of matching, each identifier, literal, word, or function-identifier is treated as pseudo-text that contains only that identifier, literal, word, or function-identifier, respectively.

Comparison and replacement rules

This topic introduces detailed rules for comparison and replacement.

- Only one COPY REPLACING statement is allowed in a nested COPY chain. The addition of a second COPY REPLACING statement in the same nested COPY chain will result in a severe-level compiler error (RC=12), and the issuing of the following error message:

```
IGYLI0078-S   A second "COPY" statement with "REPLACING" phrase was found within a nested
"COPY".
               The second "COPY" statement was discarded.
```

- Arithmetic and logical operators are considered text words and can be replaced only through a pseudo-text operand.
- Beginning and ending blanks are not included in the text comparison process. Embedded blanks are used in the text comparison process to separate multiple text words.
- When *operand-1* is a figurative constant, *operand-1* matches only the same exact figurative constant. For example, if ALL "AB" is specified in *operand-1*, "ABAB" in the library text is not considered a match; only ALL "AB" is considered a match.
- Any separator comma, semicolon, or space that precedes the leftmost word in the library text is copied into the source text. Beginning with the leftmost library text word and the first *operand-1* or *partial-word-1* specified in the REPLACING phrase, the entire REPLACING operand that precedes the keyword BY is compared to an equivalent number of contiguous library text words.

- *operand-1* matches the library text only if the ordered sequence of text words that forms *operand-1* is equal, character for character, to the ordered sequence of library words. For national characters, the sequence of national characters must be equal, national character for national character, to the ordered sequence of library words.

- When the LEADING phrase is specified, *partial-word-1* matches the library text only if the contiguous sequence of characters that forms *partial-word-1* is equal, character for character, to an equal number of contiguous characters that start with the leftmost character position of a library text word.

When the TRAILING phrase is specified, *partial-word-1* matches the library text only if the contiguous sequence of characters that forms *partial-word-1* is equal, character for character, to an equal number of contiguous characters that end with the rightmost character position of a library text word.

- For matching purposes, each occurrence of a separator comma, a separator semicolon, or a sequence of one or more separator spaces is considered to be a single space.

However, when *operand-1* or *partial-word-1* consists solely of a separator comma or separator semicolon, the *operand-1* or *partial-word-1* participates in the match as a text word. In this case, the space that follows the comma or semicolon separator can be omitted.

When the library text contains a closing quotation mark that is not immediately followed by a separator space, a separator comma, a separator semicolon, or a separator period, the closing quotation mark is considered a separator quotation mark.

- If no match occurs, the comparison is repeated with each successive *operand-1* or *partial-word-1* if specified, until either a match is found or no further REPLACING operands exist.
- Whenever a match occurs between *operand-1* and the library text, the corresponding *operand-2* is copied into the source text. Whenever a match occurs between *partial-word-1* and the library text word, the matched characters of that library text word are either replaced by *partial-word-2* or deleted when *partial-word-2* consists of zero text words.
- When all operands are compared and no match occurs, the leftmost library text word is copied into the source text.
- Once a library text word is finished being processed and is either copied into the source text unchanged or replaced because of a match, the next successive uncopied library text word is then considered to be the leftmost text word, and the comparison cycle starts again, beginning with the first occurrence of *operand-1* or *partial-word-1*. The process continues until the rightmost library text word is compared.
- The sequence of text words in the library text, *pseudo-text-1*, and *partial-word-1* is determined by the rules for reference format. For more information, see [Chapter 6, "Reference format," on page 55](#).
- When text words are placed in the source text, additional spaces are introduced only between text words where there already exists a space, including the assumed space between source lines.
- The following rules apply to comment lines, inline comments, and blank lines:
 - Comment lines, inline comments, or blank lines in the library text, *pseudo-text-1*, or *partial-word-1* are ignored for purposes of matching.
 - Comment lines, inline comments, or blank lines in the library text are copied into the resultant source text unchanged with the following exception: a comment line, an inline comment, or a blank line in the library text is not copied if that comment line, inline comment, or blank line appears in the sequence of text words that match *pseudo-text-1* or *partial-word-1*.
 - Comment lines, inline comments, or blank lines in *pseudo-text-2* or *partial-word-2* are copied into the resultant program unchanged whenever *pseudo-text-2* or *partial-word-2* is placed into the source text as a result of text replacement.
 - If the word COPY appears in a comment-entry, or in the place where a comment-entry can appear, it is considered part of the comment-entry.
- COPY REPLACING does not affect the EJECT, SKIP1, SKIP2, SKIP3, or TITLE compiler-directing statements.
- Lines that contain *CONTROL (*CBL), EJECT, SKIP1, SKIP2, SKIP3, or TITLE statements can appear in the library text. Such lines are copied into the resultant source text unchanged.
- The following rules apply to debugging lines:

- Debugging lines are permitted in library text and pseudo-text. Text words within a debugging line participate in the matching rules as if the letter "D" did not appear in the indicator area. A debugging line is specified within pseudo-text if the debugging line begins in the source text after the opening pseudo-text delimiter but before the matching closing pseudo-text delimiter.
- If more lines are introduced into the source text as a result of a COPY statement, each text word that is introduced appears on a debugging line if the COPY statement begins on a debugging line or if the text word that is introduced appears on a debugging line in library text. When a text word specified in the BY phrase is introduced, it appears on a debugging line if the first library text word that is being replaced is specified on a debugging line.
- When a COPY statement is specified on a debugging line, the copied text is treated as though it appeared on a debugging line, except that comment lines in the text appear as comment lines in the resulting source text.
- After all COPY and REPLACE statements are processed, a debugging line will be considered to have all the characteristics of a comment line, if the WITH DEBUGGING MODE clause is not specified in the SOURCE-COMPUTER paragraph.
- The syntactic correctness of the entire COBOL source text cannot be determined until all COPY and REPLACE statements have been completely processed, because the syntactic correctness of the library text cannot be independently determined.
- (This rule applies to pseudo-text only.) If the source text has occurrences of a dummy operand :TAG: that is delimited by colons in the program text, the compiler replaces the dummy operand with the required text. "Example 3" on page 702 shows how it is used with the dummy operand :TAG:. The colons serve as separators and make TAG a stand-alone operand. Note that parenthesis can also be used as delimiters, as in (TAG).
- The COPY statement with REPLACING phrase can be used to replace parts of words, either by using the LEADING|TRAILING *partial-word-1* BY *partial-word-2* phrase, or by using the pseudo-text :TAG: or (TAG) method.
- After replacement, text words are placed in the source text according to the 85 COBOL Standard format rules.

Comparison and replacement examples

This topic shows examples of comparison and replacement.

Sequences of code (such as file and data descriptions, error, and exception routines) that are common to a number of programs can be saved in a library, and then used with the COPY statement. If naming conventions are established for such common code, the REPLACING phrase need not be specified. If the names change from one program to another, the REPLACING phrase can be used to supply meaningful names for this program.

Example 1

In this example, the library text PAYLIB consists of the following DATA DIVISION entries:

```
01  A.
   02  B      PIC S99.
   02  C      PIC S9(5)V99.
   02  D      PIC S9999 OCCURS 1 TO 52 TIMES
           DEPENDING ON B OF A.
```

You can use the COPY statement in the DATA DIVISION of a program as follows:

```
COPY PAYLIB.
```

In this program, the library text is copied. The resulting text is treated as if it were written as follows:

```
01  A.
02  B      PIC S99.
02  C      PIC S9(5)V99.
02  D      PIC S9999 OCCURS 1 TO 52 TIMES
          DEPENDING ON B OF A.
```

Example 2

To change some or all of the names within the library text, you can use the REPLACING phrase:

```
COPY PAYLIB REPLACING A BY PAYROLL
                      B BY PAY-CODE
                      C BY GROSS-PAY
                      D BY HOURS.
```

In this program, the library text is copied. The resulting text is treated as if it were written as follows:

```
01  PAYROLL.
02  PAY-CODE      PIC S99.
02  GROSS-PAY     PIC S9(5)V99.
02  HOURS         PIC S9999 OCCURS 1 TO 52 TIMES
          DEPENDING ON PAY-CODE OF PAYROLL.
```

The changes that are shown are made only for this program. The text remains unchanged as it appears in the library.

Example 3

If the following conventions are followed in the library text, parts of names (for example, the prefix portion of data names) can be changed with the REPLACING phrase.

In this example, the library text PAYLIB consists of the following DATA DIVISION entries:

```
01  :TAG:.
02  :TAG:-WEEK      PIC S99.
02  :TAG:-GROSS-PAY PIC S9(5)V99.
02  :TAG:-HOURS     PIC S9999 OCCURS 1 TO 52 TIMES
          DEPENDING ON :TAG:-WEEK OF :TAG:.
```

You can use the COPY statement in the DATA DIVISION of a program as follows:

```
COPY PAYLIB REPLACING ==:TAG:== BY ==Payroll==.
```

Usage Note: In this example, the use of either colons or parentheses as delimiters is required in the library text. (Colons and parenthesis are the only valid delimiters.) Colons are recommended for clarity because parentheses can be used for a subscript, for instance in referencing a table element.

In this program, the library text is copied. The resulting text is treated as if it were written as follows:

```
01  PAYROLL.
02  PAYROLL-WEEK      PIC S99.
02  PAYROLL-GROSS-PAY PIC S9(5)V99.
02  PAYROLL-HOURS     PIC S9999 OCCURS 1 TO 52 TIMES
          DEPENDING ON PAYROLL-WEEK OF PAYROLL.
```

The changes that are shown are made only for this program. The text remains unchanged as it appears in the library.

Example 4

This example shows how to selectively replace level numbers without replacing the numbers in the PICTURE clause:

```
COPY xxx REPLACING ==(01)== BY ==(01)==  
                == 01 == BY == 05 ==.
```

Example 5

This example demonstrates use of the LEADING keyword of the REPLACING phrase in the COPY statement. The library text PAYLIB consists of the following DATA DIVISION entries:

```
01 DEPT.  
  02 DEPT-WEEK          PIC S99.  
  02 DEPT-GROSS-PAY     PIC S9(5)V99.  
  02 DEPT-HOURS         PIC S999 OCCURS 1 TO 52 TIMES  
    DEPENDING ON DEPT-WEEK OF DEPT.
```

You can use the COPY statement in the DATA DIVISION of a program as follows:

```
COPY PAYLIB REPLACING LEADING == DEPT == BY == PAYROLL ==.
```

In this program, the library text is copied. The resulting text is treated as if it were written as follows:

```
01 PAYROLL.  
  02 PAYROLL-WEEK       PIC S99.  
  02 PAYROLL-GROSS-PAY  PIC S9(5)V99.  
  02 PAYROLL-HOURS      PIC S999 OCCURS 1 TO 52 TIMES  
    DEPENDING ON PAYROLL-WEEK OF PAYROLL.
```

The changes that are shown are made only for this program. The text remains unchanged as it appears in the library.

Example 6

This example demonstrates use of the TRAILING keyword of the REPLACING phrase in the COPY statement. The library text PAYLIB consists of the following DATA DIVISION entries:

```
01 PAYROLL.  
  02 PAYROLL-WEEK       PIC S99.  
  02 PAYROLL-GROSS-PAY  PIC S9(5)V99.  
  02 PAYROLL-HOURS      PIC S999 OCCURS 1 TO 52 TIMES  
    DEPENDING ON PAYROLL-WEEK OF PAYROLL.
```

You can use the COPY statement in the DATA DIVISION of a program as follows:

```
COPY PAYLIB REPLACING TRAILING == GROSS-PAY == BY == NET-PAY ==.
```

In this program, the library text is copied. The resulting text is treated as if it were written as follows:

```
01 PAYROLL.  
  02 PAYROLL-WEEK       PIC S99.  
  02 PAYROLL-NET-PAY    PIC S9(5)V99.  
  02 PAYROLL-HOURS      PIC S999 OCCURS 1 TO 52 TIMES  
    DEPENDING ON PAYROLL-WEEK OF PAYROLL.
```

The changes that are shown are made only for this program. The text remains unchanged as it appears in the library.

Example 7

This example demonstrates a scenario where two types of partial-word replacement are specified in a single REPLACING phrase. The library text PAYLIB consists of the following DATA DIVISION entries:

```
01 PAYROLL.  
  02 PAYROLL-WEEK      PIC S99.  
  02 :TAG:-GROSS-PAY   PIC S9(5)V99.  
  02 PAYROLL-HOURS     PIC S999 OCCURS 1 TO 52 TIMES  
      DEPENDING ON PAYROLL-WEEK OF PAYROLL.
```

You can use the COPY statement in the DATA DIVISION of a program as follows:

```
COPY PAYLIB REPLACING == :TAG: == BY == PAYROLL ==  
      TRAILING == GROSS-PAY == BY == NET-PAY ==.
```

In this program, the library text is copied. The resulting text is treated as if it were written as follows:

```
01 PAYROLL.  
  02 PAYROLL-WEEK      PIC S99.  
  02 PAYROLL-GROSS-PAY PIC S9(5)V99.  
  02 PAYROLL-HOURS     PIC S999 OCCURS 1 TO 52 TIMES  
      DEPENDING ON PAYROLL-WEEK OF PAYROLL.
```

Two types of partial-word replacement are specified in the same REPLACING operation, but as usual, one replacement is done on a single library text word. Therefore, even though :TAG: -GROSS-PAY is considered a match with the first operand of both replacement operations, after the first match and replacement is performed, no more replacement is performed on that word.

The changes that are shown are made only for this program. The text remains unchanged as it appears in the library.

Example 8

This example demonstrates support for the REPLACING phrase in a chain of nested COPY statements. In this example, it is the outermost COPY statement that has the REPLACING phrase, but note that the REPLACING phrase can be specified on any of the nested COPY statements in the chain, **provided it is specified on only one of them**. The library text PAYLIB consists of the following DATA DIVISION entries:

```
01 PAYROLL.  
  02 PAYROLL-WEEK      PIC S99.  
  02 :TAG:-GROSS-PAY   PIC S9(5)V99.  
  02 PAYROLL-HOURS     PIC S999 OCCURS 1 TO 52 TIMES  
      DEPENDING ON PAYROLL-WEEK OF PAYROLL.  
  COPY PAYLIB2
```

The library text PAYLIB2 consists of the following DATA DIVISION entries:

```
01 PAYROLL2.  
  02 PAYROLL2-WEEK     PIC S99.  
  02 :TAG:2-GROSS-PAY  PIC S9(5)V99.  
  02 PAYROLL2-HOURS    PIC S999 OCCURS 1 TO 52 TIMES  
      DEPENDING ON PAYROLL2-WEEK OF PAYROLL2.
```

You can use the COPY statement in the DATA DIVISION of a program as follows:

```
COPY PAYLIB REPLACING == :TAG: == BY == PAYROLL ==  
      TRAILING == GROSS-PAY == BY == NET-PAY ==.
```

In this program, the library text is copied. The resulting text is treated as if it were written as follows:

```
01 PAYROLL.
02 PAYROLL-WEEK      PIC S99.
02 PAYROLL-NET-PAY   PIC S9(5)V99.
02 PAYROLL-HOURS     PIC S999 OCCURS 1 TO 52 TIMES
    DEPENDING ON PAYROLL-WEEK OF PAYROLL.

01 PAYROLL2.
02 PAYROLL2-WEEK     PIC S99.
02 PAYROLL2-NET-PAY  PIC S9(5)V99.
02 PAYROLL2-HOURS    PIC S999 OCCURS 1 TO 52 TIMES
    DEPENDING ON PAYROLL2-WEEK OF PAYROLL2.
```

The REPLACING phrase in the outermost COPY statement applies not only to the library text in PAYLIB but also to the text in PAYLIB2.

The changes that are shown are made only for this program. The text remains unchanged as it appears in the library.

Copy member search order

This topic discusses the search order of copy members when a COPY statement is processed.

When the compiler is invoked from JCL or any method other than from z/OS UNIX, the search for a copy member is performed according to the following rules:

- If an explicit library name is specified in the COPY statement, the name is assumed to correspond to a ddname that has been allocated to a concatenation of datasets where the compiler will search for the copy member. If no library name is specified in the COPY statement, the concatenation of datasets allocated to the SYSLIB ddname is searched.
- If the copy member was not found during the above search, then all locations specified via the COPYLOC option will be searched in the order that they were specified. These locations can include a mix of z/OS UNIX directories and PDS or PDSE datasets.

When the compiler is invoked in z/OS UNIX using the cob2 command, the search for a copy member is performed according to the following rules:

- If an explicit library name is specified in the COPY statement and the name is a literal, the name will be interpreted as a z/OS UNIX directory and that directory will be searched.
- If an explicit library name is specified in the COPY statement and it is not a literal, the name is interpreted to be an environment variable that specifies a colon-separated list of z/OS UNIX directories. If the environment variable exists, those z/OS UNIX directories will be searched in the order they appear in the list. If the environment variable does not exist, the current directory will be searched.
- If an explicit library name was not specified in the COPY statement, the compiler searches for copy members in this order:
 - The current z/OS UNIX directory is searched for the copy member.
 - Any z/OS UNIX directories specified via the -I option of cob2 are searched.
 - Any z/OS UNIX directories specified via the SYSLIB environment variable are searched.
- If the copy member was not found during the above search; then all locations specified via the COPYLOC option are searched in the order that they were specified. These locations can include a mix of z/OS UNIX directories and PDS or PDSE datasets.

Note: When the compiler searches for a copy member in a z/OS UNIX directory and the copy member name in the corresponding COPY statement is not a literal and is not found to be referring to an environment variable, then the compiler searches for multiple different versions of the provided name with the following extensions: .cpy, .CPY, .cb1, .CBL, .cob, and .COB. Each extension is tested with an uppercase version and a lowercase version of the copy member name.

DELETE statement

The DELETE statement is an extended source library statement. It removes COBOL statements from a source program that was included by a BASIS statement.

Format

→ sequence-number DELETE — *sequence-number-field* →

sequence-number

Can optionally appear in columns 1 through 6, followed by a space. The content of this field is ignored.

DELETE

Can appear anywhere within columns 1 through 72. The keyword DELETE must be followed by a space and the *sequence-number-field*. There must be no other text in the statement.

sequence-number-field

Each number must be equal to a *sequence-number* in the BASIS source program. This *sequence-number* is the six-digit number the programmer assigns in columns 1 through 6 of the COBOL coding form. The numbers referenced in the *sequence-number-field* of INSERT or DELETE statements must always be specified in ascending numeric order.

The *sequence-number-field* must be one of the following options:

- A single number
- A series of single numbers
- A range of numbers (indicated by separating the two bounding numbers of the range by a hyphen)
- A series of ranges of numbers
- Any combination of one or more single numbers and one or more ranges of numbers

Each entry in the *sequence-number-field* must be separated from the preceding entry by a comma followed by a space. For example:

```
000250 DELETE 000010-000050, 000400, 000450
```

Source program statements can follow a DELETE statement. These source program statements are then inserted into the BASIS source program before the statement following the last statement deleted (that is, in the example mentioned previously, before the next statement following deleted statement 000450).

If a DELETE statement begins in column 12 or higher and a valid *sequence-number-field* does not follow the keyword DELETE, the compiler assumes that this DELETE statement is a COBOL DELETE statement.

Usage note: If INSERT or DELETE statements are used to modify the COBOL source program provided by a BASIS statement, the sequence field of the COBOL source program must contain numeric sequence numbers in ascending order. The source file remains unchanged. Any INSERT or DELETE statements referring to these sequence numbers must occur in ascending order.

EJECT statement

The EJECT statement specifies that the next source statement is to be printed at the beginning of the next page.

Format

→ EJECT →

The EJECT statement must be the only statement on the line. It can be written in either Area A or Area B, and can be terminated with a separator period.

The EJECT statement must be embedded in a program source. For example, in the case of batch applications, the EJECT statement must be placed between the CBL (PROCESS) statement and the end of the program (or the END PROGRAM marker, if specified).

The EJECT statement has no effect on the compilation of the source unit itself.

ENTER statement

The ENTER statement is designed to facilitate the use of more than one source language in the same source program. However, only COBOL is allowed in the source program.

The ENTER statement is syntax checked but has no effect on the execution of the program.

Format

➤ ENTER — *language-name-1* — *routine-name-1* . ➤

language-name-1

A system name that has no defined meaning. It must be either a correctly formed user-defined word or the word "COBOL." At least one character must be alphabetic.

routine-name-1

Must follow the rules for formation of a user-defined word. At least one character must be alphabetic.

INSERT statement

The INSERT statement is a library statement that adds COBOL statements to a source program that was included by a BASIS statement.

Format

➤ *sequence-number* — INSERT — *sequence-number-field* ➤

sequence-number

Can optionally appear in columns 1 through 6, followed by a space. The content of this field is ignored.

INSERT

Can appear anywhere within columns 1 through 72, followed by a space and the *sequence-number-field*. There must be no other text in the statement.

sequence-number-field

A number that must be equal to a *sequence-number* in the BASIS source program. This *sequence-number* is a six-digit number that the programmer assigns in columns 1 through 6 of the COBOL source line.

The numbers referenced in the *sequence-number-field* of INSERT or DELETE statements must always be specified in ascending numeric order.

The *sequence-number-field* must be a single number (for example, 000130). At least one new source program statement must follow the INSERT statement for insertion after the statement number specified by the *sequence-number-field*.

New source program statements following the INSERT statement can include any COBOL syntax.

Usage note: If INSERT or DELETE statements are used to modify the COBOL source program provided by a BASIS statement, the sequence field of the COBOL source program must contain numeric sequence

numbers in ascending order. The source file remains unchanged. Any INSERT or DELETE statements referring to these sequence numbers must occur in ascending order.

READY or RESET TRACE statement

The READY or RESET TRACE statement was designed to trace the execution of procedures. The READY or RESET TRACE statement can appear only in the PROCEDURE DIVISION, but has no effect on your program.

Format

➤➤ READY TRACE — . ➤➤
RESET

You can trace the execution of procedures by using the USE FOR DEBUGGING declarative as described in *Example: USE FOR DEBUGGING* in the *Enterprise COBOL Programming Guide*.

REPLACE statement

The REPLACE statement is used to replace source text.

A REPLACE statement can occur anywhere in the source text that a character-string can occur. It must be preceded by a separator period except when it is the first statement in a separately compiled program. It must end with a separator period.

The REPLACE statement provides a means of applying a change to an entire COBOL compilation group, or part of a compilation group, without manually having to find and modify all places that need to be changed. It is an easy method of doing simple string substitutions. It is similar in action to the REPLACING phrase of the COPY statement, except that it acts on the entire source text, not just on the text in COPY libraries.

If the word REPLACE appears in a comment-entry or in the place where a comment-entry can appear, it is considered part of the comment-entry.

Format 1

➤➤ REPLACE ➤➤

➤➤ — *pseudo-text-1* — BY — *pseudo-text-2* — ➤➤
LEADING — *partial-word-1* — BY — *partial-word-2* —
TRAILING —
➤➤ . ➤➤

Each matched occurrence of *pseudo-text-1* in the source text is replaced by the corresponding *pseudo-text-2*.

Format 2

➤➤ REPLACE OFF. ➤➤

Any text replacement currently in effect is discontinued with the format-2 form of REPLACE. If format 2 is not specified, a specific occurrence of the REPLACE statement is in effect from the point at which

it is specified until the next occurrence of a REPLACE statement or the end of the separately compiled program.

pseudo-text-1, pseudo-text-2

A sequence of text words that are bounded by, but not including, pseudo-text delimiters (==). Both characters of each pseudo-text delimiter must appear on one line.

Individual text words within pseudo-text can be up to 322 characters long. They can be continued subject to the normal continuation rules for source code format.

A text word must be delimited by separators. For more information, see [Chapter 1, “Characters,” on page 3](#).

pseudo-text-1 can be one or more text words. It can consist solely of the separator comma or separator semicolon. *pseudo-text-2* can be zero or more text words. It can consist solely of space characters, comment lines, or inline comments.

Each text word in *pseudo-text-2* that is to be copied into the program is placed in the same area of the resultant program as the area in which it appears in *pseudo-text-2*.

Pseudo-text can consist of or include any words (except COPY), identifiers, or literals that can be written in the source text. This includes DBCS user-defined words, DBCS literals, and national literals.

DBCS user-defined words must be wholly formed; that is, there is no partial-word replacement for DBCS words.

Words or literals containing DBCS characters cannot be continued across lines.

partial-word-1, partial-word-2

A single text word that is bounded by, but not including, pseudo-text delimiters (==). Both characters of each pseudo-text delimiter must appear on one line. However, the text word within a partial-word can be continued.

The following rules apply to *partial-word-1* and *partial-word-2*:

- *partial-word-1* consists of one text word.
- *partial-word-2* consists of zero or one text word.
- *partial-word-1* and *partial-word-2* cannot be an alphanumeric literal, national literal, DBCS literal, or DBCS word.

The compiler processes REPLACE statements in source text after the processing of any COPY statements. COPY must be processed first to assemble complete source text. Then, REPLACE can be used to modify that source text, performing simple string substitution. REPLACE statements cannot themselves contain COPY statements.

The text that is produced as a result of the processing of a REPLACE statement must not contain a REPLACE statement.

Continuation rules for pseudo-text and partial-word

The character-strings and separators that comprise pseudo-text and partial-words can start in either Area A or Area B. However, if a hyphen is in the indicator area of a line, and that hyphen follows the opening pseudo-text or partial-word delimiter, Area A of the line must be blank, and the normal rules for continuation of lines apply to the formation of text words. See [“Continuation lines” on page 58](#).

Example

The following example shows the use of REPLACE statement to replace source text.

```
IDENTIFICATION DIVISION.  
PROGRAM-ID. REPLEXMP.  
DATA DIVISION.  
WORKING-STORAGE SECTION.  
    REPLACE =="(Hello, World!)"== BY =="(Hello, Mom!)"==.  
01 WS-STRING1 PIC X(30) VALUE "(Hello, World!)".
```

```

01 WS-STRING2 PIC X(30) VALUE "Hello, COBOL!".
PROCEDURE DIVISION.
  DISPLAY "Original: " WS-STRING1
  REPLACE LEADING ==XX== BY ====
              ==:TAG:== BY ==STRING==
              TRAILING ==1== BY ==2==.
  DISPLAY "Modified: " XX-WS-:TAG:1
  REPLACE OFF.
  GOBACK.

```

The output of this program is:

```

Original: (Hello, Mom!)
Modified: Hello, COBOL!

```

Comparison rules

The comparison operation that determines text replacement starts with the leftmost source text word that follows the REPLACE statement, and with the first word of *pseudo-text-1* or *partial-word-1*.

- *pseudo-text-1* is compared to an equivalent number of contiguous source text words. *pseudo-text-1* matches the source text only if the ordered sequence of text words that forms *pseudo-text-1* is equal, character for character, to the ordered sequence of source text words. For national characters, the sequence of national characters must be equal, national character for national character, to the ordered sequence of library words.
- When the LEADING phrase is specified, *partial-word-1* matches the source text only if the contiguous sequence of characters that forms *partial-word-1* is equal, character for character, to an equal number of contiguous characters that start with the leftmost character position of a source text word.

When the TRAILING phrase is specified, *partial-word-1* matches the source text only if the contiguous sequence of characters that forms *partial-word-1* is equal, character for character, to an equal number of contiguous characters that end with the rightmost character position of a source text word.

- For matching purposes, each occurrence of a separator comma, a separator semicolon, or a sequence of one or more separator spaces is considered to be a single space.

However, when *pseudo-text-1* or *partial-word-1* consists solely of a separator comma or separator semicolon, the comma or semicolon participates in the match as a text word. In this case, the space that follows the comma or semicolon separator can be omitted.

When the source text contains a closing quotation mark that is not immediately followed by a separator space, a separator comma, a separator semicolon, or a separator period, the closing quotation mark is considered a separator quotation mark.

- If no match occurs, the comparison is repeated with each successive occurrence of *pseudo-text-1* or *partial-word-1* if specified, until either a match is found or no further REPLACING operands exist.
- When all occurrences of *pseudo-text-1* or *partial-word-1* are compared and no match occurs, the next successive source text word is then considered to be the leftmost source text word, and the comparison cycle starts again, beginning with the first occurrence of *pseudo-text-1* or *partial-word-1*.
- Whenever a match occurs between *pseudo-text-1* and the source text, the corresponding *pseudo-text-2* replaces the matched text in the source text. Whenever a match occurs between *partial-word-1* and the source text word, the matched characters of that source text word are either replaced by *partial-word-2* or deleted if *partial-word-2* consists of zero text words. The source text word that immediately follows the rightmost text word that participated in the match is then considered as the leftmost source text word. The comparison cycle starts again, beginning with the first occurrence of *pseudo-text-1* or *partial-word-1*.
- The comparison operation continues until the rightmost text word in the source text that is within the scope of the REPLACE statement has either participated in a match, or been considered as a leftmost source text word and participated in a complete comparison cycle.

Replacement rules

This topic introduces detailed rules for replacement.

- The sequence of text words in the source text, *pseudo-text-1*, and *partial-word-1* is determined by the rules for reference format. For more information, see [Chapter 6, “Reference format,” on page 55](#).
- Text words that are inserted into the source text as a result of processing a REPLACE statement are placed in the source text according to the rules for reference format. When inserting text words of *pseudo-text-2* or *partial-word-2* into the source text, additional spaces are introduced only between text words where there already exists a space, including the assumed space between source lines.
- If more lines are introduced into the source text as a result of the processing of REPLACE statements, the indicator area of the introduced lines contains the same character as the line on which the text being replaced begins, unless that line contains a hyphen, in which case the introduced line contains a space.
- If any literal within *pseudo-text-2* or *partial-word-2* is of a length too great to be accommodated on a single line without continuation to another line in the resultant program, and the literal is not being placed on a debugging line, more continuation lines are introduced that contain the remainder of the literal. If replacement requires the continued literal to be continued on a debugging line, the program is in error.
- Each word in *pseudo-text-2* or *partial-word-2* that is to be placed into the resultant program begins in the same area of the resultant program as it appears in *pseudo-text-2* or *partial-word-2*.
- The following rules apply to comment lines, inline comments, and blank lines:
 - Comment lines, inline comments, or blank lines in the source text, *pseudo-text-1*, or *partial-word-1* are ignored for purposes of matching.
 - Comment lines, inline comments, or blank lines in the source text are copied into the resultant source text unchanged with the following exception: a comment line, an inline comment, or a blank line in the source text is not copied if that comment line, inline comment, or blank line appears in the sequence of text words that match *pseudo-text-1* or *partial-word-1*.
 - Comment lines, inline comments, or blank lines in *pseudo-text-2* or *partial-word-2* are copied into the resultant program unchanged whenever *pseudo-text-2* or *partial-word-2* is placed into the source text as a result of text replacement.
- Lines that contain *CONTROL (*CBL), EJECT, SKIP1, SKIP2, SKIP3, or TITLE statements can appear in the source text. Such lines are copied into the resultant source text unchanged.
- The following rules apply to debugging lines:
 - Debugging lines are permitted in pseudo-text or partial-words. Text words within a debugging line participate in the matching rules as if the letter "D" did not appear in the indicator area.
 - When a REPLACE statement is specified on a debugging line, the statement is treated as if the letter "D" did not appear in the indicator area.
 - After all COPY and REPLACE statements are processed, a debugging line is considered to have all the characteristics of a comment line if the WITH DEBUGGING MODE clause is not specified in the SOURCE-COMPUTER paragraph.
- Except for COPY and REPLACE statements, the syntactic correctness of the source text cannot be determined until all COPY and REPLACE statements are completely processed.
- (This rule only applies to pseudo-text.) If the source text has occurrences of a dummy operand :TAG: that is delimited by colons in the program text, the compiler replaces the dummy operand with the required text. The colons serve as separators and make TAG a stand-alone operand.

For example, you can use the REPLACE statement in the DATA DIVISION of a program as follows:

```
REPLACE ==:TAG:== BY ==Payroll==.

01  :TAG:.
02  :TAG:-WEEK      PIC S99.
02  :TAG:-GROSS-PAY PIC S9(5)V99.
```

```
02 :TAG:-HOURS          PIC S999 OCCURS 1 TO 52 TIMES  
   DEPENDING ON :TAG:-WEEK OF :TAG:.
```

- The REPLACE statement can be used to replace parts of words, either by using the LEADING | TRAILING *partial-word-1* BY *partial-word-2* phrase, or by using the pseudo-text :TAG: method.

SERVICE LABEL statement

This statement is generated by the CICS integrated language translator (and the separate CICS translator) to indicate control flow. It is also used after calls to CEE3SRP when using Language Environment condition handling. For more information about CEE3SRP, see the *Language Environment Programming Guide*.

Format

►► SERVICE LABEL ◄◄

The SERVICE LABEL statement can appear only in the PROCEDURE DIVISION, but not in the declaratives section.

SERVICE RELOAD statement

The SERVICE RELOAD statement is syntax checked, but has no effect on the execution of the program.

Format

►► SERVICE RELOAD — *identifier-1* ◄◄

SKIP statements

The SKIP1, SKIP2, and SKIP3 statements specify blank lines that the compiler should add when printing the source listing. SKIP statements have no effect on the compilation of the source text itself.

Format

►► SKIP1 — . — ◄◄
 |
 | SKIP2
 |
 | SKIP3

SKIP1

Specifies a single blank line to be inserted in the source listing.

SKIP2

Specifies two blank lines to be inserted in the source listing.

SKIP3

Specifies three blank lines to be inserted in the source listing.

SKIP1, SKIP2, or SKIP3 can be written anywhere in either Area A or Area B, and can be terminated with a separator period. It must be the only statement on the line.

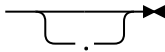
The SKIP statements must be embedded in a program source. For example, in the case of batch applications, a SKIP1, SKIP2, or SKIP3 statement must be placed between the CBL (PROCESS) statement and the end of the program or class (or the END CLASS marker or END PROGRAM marker, if specified).

TITLE statement

The TITLE statement specifies a title to be printed at the beginning of each page of the source listing produced during compilation.

If no TITLE statement is found, a title containing the identification of the compiler and the current release level is generated. The title is left-justified on the title line.

Format

►► TITLE — *literal* 

literal

Must be an alphanumeric literal, DBCS literal, or national literal and can be followed by a separator period.

Must not be a figurative constant.

In addition to the default or chosen title, the right side of the title line contains the following items:

- For programs, the name of the program from the PROGRAM-ID paragraph for the outermost program. (This space is blank on pages preceding the PROGRAM-ID paragraph for the outermost program.)
- For classes, the name of the class from the CLASS-ID paragraph.
- Current page number.
- Date and time of compilation.

The TITLE statement:

- Forces a new page immediately, if the SOURCE compiler option is in effect
- Is not itself printed on the source listing
- Has no other effect on compilation
- Has no effect on program execution
- Cannot be continued on another line
- Can appear anywhere in any of the divisions

A title line is produced for each page in the listing produced by the LIST option. This title line uses the last TITLE statement found in the source statements or the default.

The word TITLE can begin in either Area A or Area B.

The TITLE statement must be embedded in a class or program source. For example, in the case of batch applications, the TITLE statement must be placed between the CBL (PROCESS) statement and the end of the class or program (or the END CLASS marker or END PROGRAM marker, if specified).

No other statement can appear on the same line as the TITLE statement.

USE statement

The USE statement defines the conditions under which the procedures that follow the statement will be executed.

The formats for the USE statement are:

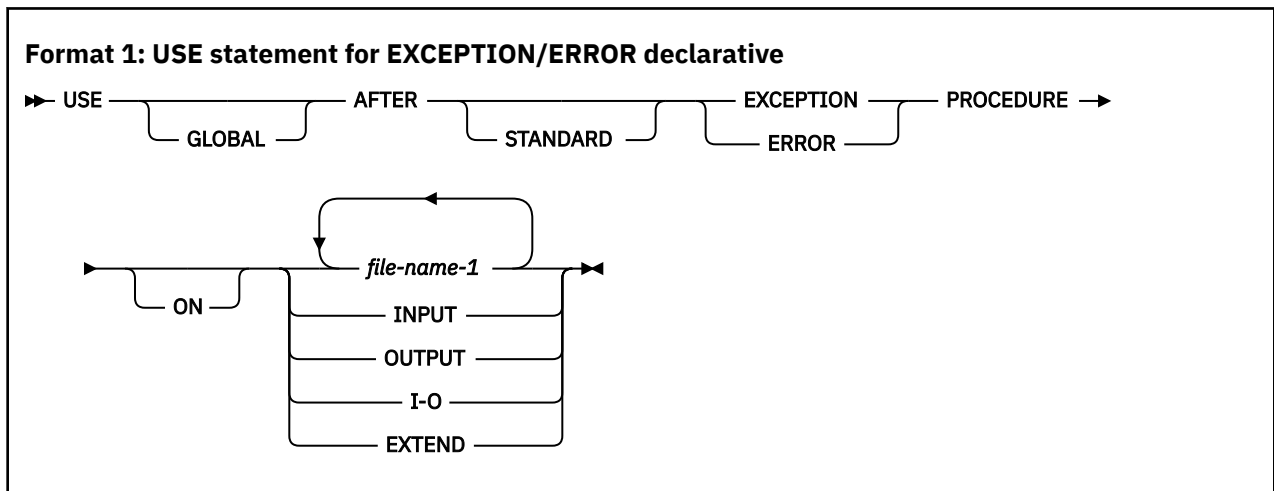
- EXCEPTION/ERROR declarative
- DEBUGGING declarative

For general information about declaratives, see [“Declaratives” on page 264](#).

EXCEPTION/ERROR declarative

The EXCEPTION/ERROR declarative specifies procedures for input/output exception or error handling that are to be executed in addition to the standard system procedures.

The words EXCEPTION and ERROR are synonymous and can be used interchangeably.



file-name-1

Valid for all files. When this option is specified, the procedure is executed only for the files named. No file-name can refer to a sort or merge file. For any given file, only one EXCEPTION/ERROR procedure can be specified; thus, file-name specification must not cause simultaneous requests for execution of more than one EXCEPTION/ERROR procedure.

A USE AFTER EXCEPTION/ERROR declarative statement specifying the name of a file takes precedence over a declarative statement specifying the open mode of the file.

INPUT

Valid for all files. When this option is specified, the procedure is executed for all files opened in INPUT mode or in the process of being opened in INPUT mode that get an error.

OUTPUT

Valid for all files. When this option is specified, the procedure is executed for all files opened in OUTPUT mode or in the process of being opened in OUTPUT mode that get an error.

I-O

Valid for all direct-access files. When this option is specified, the procedure is executed for all files opened in I-O mode or in the process of being opened in I-O mode that get an error.

EXTEND

Valid for all files. When this option is specified, the procedure is executed for all files opened in EXTEND mode or in the process of being opened in EXTEND mode that get an error.

The EXCEPTION/ERROR procedure is executed:

- Either after completing the system-defined input/output error routine, or
- Upon recognition of an INVALID KEY or AT END condition when an INVALID KEY or AT END phrase has not been specified in the input/output statement, or
- Upon recognition of an IBM-defined condition that causes file status key 1 to be set to 9. (See [“File status key”](#) on page 299.)

After execution of the EXCEPTION/ERROR procedure, control is returned to the invoking routine in the input/output control system. If the input/output status value does not indicate a critical input/output error, the input/output control system returns control to the next executable statement following the input/output statement whose execution caused the exception.

An applicable EXCEPTION/ERROR procedure is activated when an input/output error occurs during execution of a READ, WRITE, REWRITE, START, OPEN, CLOSE, or DELETE statement. To determine what conditions are errors, see [“Common processing facilities” on page 299](#).

The following rules apply to declarative procedures:

- A declarative procedure can be performed from a nondeclarative procedure.
- A nondeclarative procedure can be performed from a declarative procedure.
- A declarative procedure can be referenced in a GO TO statement in a declarative procedure.
- A nondeclarative procedure can be referenced in a GO TO statement in a declarative procedure.

You can include a statement that executes a previously called USE procedure that is still in control. However, to avoid an infinite loop, you must be sure that there is an eventual exit at the end.

You cannot use a GOBACK statement or a STOP RUN statement when an EXCEPTION/ERROR declarative is active due to a QSAM abend for a READ, WRITE, or REWRITE statement. You cannot use an EXIT PROGRAM statement in a non-nested subprogram when an EXCEPTION/ERROR declarative is active due to a QSAM abend for a READ, WRITE, or REWRITE statement. When a QSAM abend occurs during a READ, WRITE, or REWRITE statement, the file status code can be "34" or "90".

You cannot use a GOBACK statement or an EXIT PROGRAM statement while a declarative is active in a nested program. You cannot use a GOBACK statement or an EXIT METHOD statement while a declarative is active in a method.

EXCEPTION/ERROR procedures can be used to check the file status key values whenever an input/output error occurs.

Precedence rules for nested programs

Special precedence rules are followed when programs are contained within other programs.

In applying these rules, only the first qualifying declarative is selected for execution. The order of precedence for selecting a declarative is:

1. A file-specific declarative (that is, a declarative of the form USE AFTER ERROR ON *file-name-1*) within the program that contains the statement that caused the qualifying condition.
2. A mode-specific declarative (that is, a declarative of the form USE AFTER ERROR ON INPUT) within the program that contains the statement that caused the qualifying condition.
3. A file-specific declarative that specifies the GLOBAL phrase and is within the program directly containing the program that was last examined for a qualifying declarative.
4. A mode-specific declarative that specifies the GLOBAL phrase and is within the program directly containing the program that was last examined for a qualifying condition.

Steps 3 and 4 are repeated until the last examined program is the outermost program, or until a qualifying declarative has been found.

DEBUGGING declarative

Debugging sections are permitted only in the outermost program; they are not valid in nested programs. Debugging sections are never triggered by procedures contained in nested programs.

Debugging sections are not permitted in:

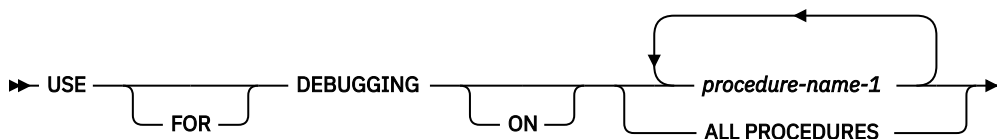
- A method
- A program defined with the recursive attribute
- A program compiled with the THREAD compiler option

The WITH DEBUGGING MODE clause of the SOURCE-COMPUTER paragraph activates all debugging sections and lines that have been compiled into the object code. See [Appendix F, “Source language debugging,” on page 771](#) for additional details.

When the debugging mode is suppressed by not specifying the WITH DEBUGGING MODE clause, all USE FOR DEBUGGING declarative procedures and all debugging lines are inhibited.

Automatic execution of a debugging section is not caused by a statement that appears in a debugging section.

Format 2: USE statement for DEBUGGING declarative



USE FOR DEBUGGING

All debugging statements must be written together in a section immediately after the DECLARATIVES header.

Except for the USE FOR DEBUGGING sentence itself, within the debugging procedure there must be no reference to any nondeclarative procedures.

procedure-name-1

Must not be defined in a debugging session.

Table 78 on page 716 shows, for each valid option, the points during execution when the USE FOR DEBUGGING procedures are executed.

Any given procedure-name can appear in only one USE FOR DEBUGGING sentence, and only once in that sentence. All procedures must appear in the outermost program.

ALL PROCEDURES

procedure-name-1 must not be specified in any USE FOR DEBUGGING sentences. The ALL PROCEDURES phrase can be specified only once in a program. Only the procedures contained in the outermost program will trigger execution of the debugging section.

Table 78. Execution of debugging declaratives	
USE FOR DEBUGGING operand	Upon execution of the following, the USE FOR DEBUGGING procedures are executed immediately
<i>procedure-name-1</i>	Before each execution of the named procedure After the execution of an ALTER statement referring to the named procedure
ALL PROCEDURES	Before each execution of each nondebugging procedure in the outermost program After the execution of each ALTER statement in the outermost program (except ALTER statements in declarative procedures)

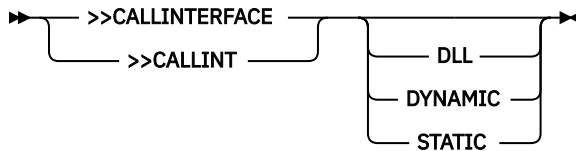
Chapter 115. Compiler directives

A *compiler directive* is a statement that causes the compiler to take a specific action during compilation.

CALLINTERFACE

The CALLINTERFACE directive specifies the interface convention for CALL and SET statements. The convention specified stays in effect until another CALLINTERFACE directive is encountered in the source.

Format



DLL

Specifies that the interface convention for subsequent CALL statements is a call to a DLL, and that subsequent SET function-pointer and procedure-pointer statements are treated as if the DLL compiler option was in effect.

DYNAMIC

Specifies that the interface convention for subsequent CALL literal statements and subsequent SET function-pointer and procedure-pointer statements is dynamic (as if the DYNAM compiler option was in effect).

STATIC

Specifies that the interface convention for subsequent CALL statements and subsequent SET function pointer and procedure pointer statements is static (as if the NODLL and NODYNAM compiler options were in effect).

If the >>CALLINTERFACE directive has no suboptions, the interface convention for subsequent CALL and SET statements is determined by the setting of the DLL and DYNAM compiler options.

The >>CALLINTERFACE directive has no effect on the CANCEL statement.

The >>CALLINTERFACE directive can only appear in the procedure division.

The positions of CALL statements relative to the CALLINTERFACE directive are determined following any processing of COPY and REPLACE statements. For example, CALL statements and CALLINTERFACE directives in COPY text are processed by the rules specified for the CALLINTERFACE directive.

Syntax and general rules

You must specify >>CALLINTERFACE on a line by itself, in either Area A or Area B.

You cannot specify >>CALLINTERFACE in the following cases:

- Within a COPY or REPLACE statement
- Between the lines of a continued character string
- In the middle of a COBOL statement

The >>CALLINTERFACE specification is limited to the current compilation unit.

The REPLACE statement and REPLACING phrase of the COPY statement do not affect the CALLINTERFACE directive.

Precedence of CALLINTERFACE directive versus DLL and DYNAM compiler options

If you specify both the CALLINTERFACE directive (with suboptions) and the DLL or the DYNAM compiler option, the directive overrides the compiler option in effect and determines the interface to be used for subsequent CALL and SET statements.

If you specify the CALLINTERFACE directive without any suboptions, the DLL or the DYNAM compiler option will determine the interface to use for subsequent CALL and SET statements.

DATA

The DATA directive is supported when the compiler option LP (64) is in effect. It specifies the storage location of identifiers in WORKING-STORAGE section following the directive, and applies to level 01 and 77 data items. The storage location specified stays in effect until another DATA directive is encountered in the source. The directive must not be placed in the middle of a group data item. The directive also applies to external data items.

Format

► ► >>DATA — *integer-1* ◄◄

integer-1

Must be an unsigned integer with a value of 64 or 31. The default for LP (64) is 64. The directive is not supported in LP(32).

Location of data items in WORKING-STORAGE Section

Data items in WORKING-STORAGE, which is also called storage location or data location, can reside beyond the bar, or under the bar, which informally are referred as *64-bit data items*, and *31-bit data items* respectively. This applies to programs compiled with the LP(64) option. By default, data items in WORKING-STORAGE are allocated beyond the bar when the LP (64) compiler option is in effect. The DATA compiler directive can be used to change this default to a different storage location.

Note: All subordinate items belonging to the same group item must be allocated in the same storage location.

>>DATA 31 specifies that storage should reside under the 2GB bar, accessible by AMODE 31 or AMODE 64 programs.

>>DATA 64 specifies that storage could reside beyond the 2GB bar, accessible by AMODE 64 programs only.

Related topics

Dynamic call between AMODE 31 and AMODE 64 programs (*Enterprise COBOL Programming Guide*)
[“WORKING-STORAGE SECTION” on page 163](#)

INLINE

The INLINE directive lets you selectively prevent the compiler from considering procedures eligible for inlining. Specifying >>INLINE OFF prevents the compiler from inlining procedures referenced by PERFORM statements. If the NOINLINE compiler option is in effect, all INLINE directives are ignored. The relative location of an INLINE directive to the definition of a procedure name is important for determining the inlining behavior for that procedure.

Format

► ► >>INLINE — ON — ◄◄
 OFF —

ON

Specifies that the compiler determines if the procedures within the scope of the directive are inlined in a specific PERFORM statement when OPTIMIZE(1) or OPTIMIZE(2) is in effect.

OFF

Specifies that procedures within the scope of the directive will not be inlined when referenced by PERFORM statements, no matter which optimization level setting is in effect.

Note: The word *inlining* here implies that the compiler might choose to replace the PERFORM of a procedure (paragraph or section) that is performed more than once with a copy of that procedure's code. By inserting the procedure code at the location of the PERFORM, the compiler saves the overhead of branching logic to and from the procedure. Note that if a procedure is only performed once, the compiler might move the code for that procedure to the PERFORM site, even with >>INLINE OFF.

Syntax and general rules

You must specify >>INLINE ON or >>INLINE OFF on a line by itself, in either Area A or Area B.

You cannot specify >>INLINE ON or >>INLINE OFF in the following cases:

- Within a COPY or REPLACE statement
- Between the lines of a continued character string
- In the middle of a COBOL statement

The >>INLINE ON or >>INLINE OFF specification is limited to the current compilation unit.

Note: The INLINE directive can appear multiple times in the program.

Inlining eligibility of procedures for PERFORM statements

A procedure is in the scope of a particular INLINE directive when:

- This INLINE directive appears before the definition of the procedure name

and

- There are no other intervening INLINE directives between the definition of the procedure name and this particular INLINE directive

A procedure is eligible for inlining for statements of the form "PERFORM *procedure-name-1* [THROUGH *procedure-name-2*]", if and only if:

- The procedure is in the scope of an INLINE ON directive

or

- The INLINE compiler option is in effect and the procedure is not in the scope of an INLINE OFF directive

For a section that contains paragraphs, it is possible that this section is under the scope of one INLINE directive while some of its paragraphs are under the scope of another INLINE directive.

Example

Following is an example of the effect of the inlining directive on a section that is comprised of one or more paragraphs:

```
>>INLINE ON
MY-SUBROUTINE SECTION.
MY-PARAGRAPH-ONE.
.
.
>>INLINE OFF
MY-PARAGRAPH-TWO.
.
```

```
.  
MY-PARAGRAPH-THREE .  
.  
.  
EXIT .
```

Notes:

1. Procedure MY-SUBROUTINE will be eligible for inlining in any statements that PERFORM it.
2. Procedure MY-PARAGRAPH-ONE will be eligible for inlining in any statements that PERFORM it.
3. Procedure MY-PARAGRAPH-TWO will not be eligible for inlining in any statements that PERFORM it.
4. Procedure MY-PARAGRAPH-THREE will not be eligible for inlining in any statements that PERFORM it.
5. Procedures MY-PARAGRAPH-ONE through MY-PARAGRAPH-THREE will be eligible for inlining in any statements that PERFORM "MY-PARAGRAPH-ONE THRU MY-PARAGRAPH-THREE", because MY-PARAGRAPH-ONE is eligible for inlining in a PERFORM statement.

Related references

INLINE compiler option (*Enterprise COBOL Programming Guide*)

Conditional compilation

Conditional compilation provides a way of including or omitting selected lines of source code depending on the values of literals specified by the DEFINE directive. In this way, you can create multiple variants of the same program without the need to maintain separate source streams.

The compiler directives that are used for conditional compilation are the DEFINE directive, the EVALUATE directive, and the IF directive. The DEFINE directive is used to define compilation variables that are referenced in the EVALUATE and IF directives to select lines of source code that are to be included or omitted in a compilation group.

Conditional compilation directives are processed according to the following rules:

- Within a source file, if a conditional compilation directive appears before a COPY or REPLACE statement, it is processed before the COPY or REPLACE statement is processed. This means that conditional compilation directives may be used to exclude COPY and REPLACE statements from a program.
- Conditional compilation directives are not affected by substitutions made as the result of REPLACE statements or the REPLACING phrase of COPY statements.
- Conditional compilation directives may appear in copybooks.

Notes:

- Conditional compilation directives can be included in a file that contains the BASIS statement, but in that file, conditional compilation directives do not control the inclusion or exclusion of source from that file. Instead, the conditional compilation directives will be processed like any other source lines in the BASIS file that are not INSERT or DELETE statements, and those directives will be passed through to the source that is being assembled to be processed later during the library phase.
- Conditional compilation directives along with their included and excluded source text can be viewed in the listing. Excluded source text can be omitted from the listing by using the CONDCOMP (SKIPSR) option.

Related references

[“DEFINE” on page 721](#)

[“EVALUATE” on page 722](#)

[“IF” on page 724](#)

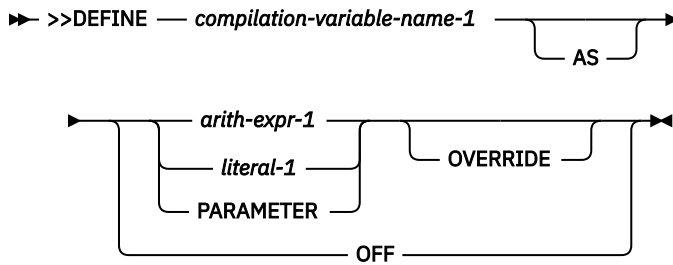
CONDCOMP (*Enterprise COBOL Programming Guide*)

Example: conditional compilation output (*Enterprise COBOL Programming Guide*)

DEFINE

The DEFINE directive defines or undefines a compilation variable. The compilation variables can be used within any of the conditional compilation directives (DEFINE, EVALUATE, and IF). The compilation variable is treated as a symbolic reference to the literal value it currently represents.

Format



>>DEFINE

Must begin on a new line in area A or B and must be specified entirely on that line.

compilation-variable-name-1

Must not be the same as a conditional compiler directive keyword and must not be one of the predefined compilation variable names.

If a DEFINE directive does not specify the OFF or the OVERRIDE phrase, then one of the following conditions must be true:

- *compilation-variable-name-1* was not declared previously within the same compilation group.
- The previous DEFINE directive referring to *compilation-variable-name-1* must have been specified with the OFF phrase.
- The previous DEFINE directive referring to *compilation-variable-name-1* must have specified the same value.

literal-1

Must be one of the following items:

- An alphanumeric literal, which can be specified as a regular alphanumeric literal ('abcd') or as a hex literal (x'F1F2F3'). National literals, DBCS literals, and null-terminated alphanumeric literals (Z literals) are not supported.
- An integer literal.
- A boolean literal (only B'0' and B'1' are supported).

arith-expr-1

Must be formed in accordance with the arithmetic expression rules as described in [“Compile-time arithmetic expressions”](#) on page 727.

General rules

- DEFINE directives that appear in code that is omitted as the result of other conditional compilation directives are not processed.
- In the text that follows a DEFINE directive that defines *compilation-variable-name-1* and does not include the OFF phrase, *compilation-variable-name-1* can be used in a conditional compilation directive wherever a literal of the category associated with the name is permitted, including in a [defined condition](#) and a [boolean condition](#).
- In the text that follows a DEFINE directive specifying *compilation-variable-name-1* with the OFF phrase, *compilation-variable-name-1* can be used only in a [defined condition](#), unless *compilation-variable-name-1* is redefined in a subsequent DEFINE directive without the OFF phrase.

- If the OVERRIDE phrase is specified, *compilation-variable-name-1* is unconditionally set to reference the value of the specified operand.
- If the AS PARAMETER phrase is specified, the value referenced by *compilation-variable-name-1* is obtained from a DEFINE option for *compilation-variable-name-1*, if such an option exists. If there is no DEFINE option for *compilation-variable-name-1*, *compilation-variable-name-1* is not defined.
- If *arith-expr-1* is specified, *arith-expr-1* is evaluated according to “Compile-time arithmetic expressions” on page 727, and *compilation-variable-name-1* references the resultant value.
- If *literal-1* is specified, *compilation-variable-name-1* references *literal-1*.

Related references

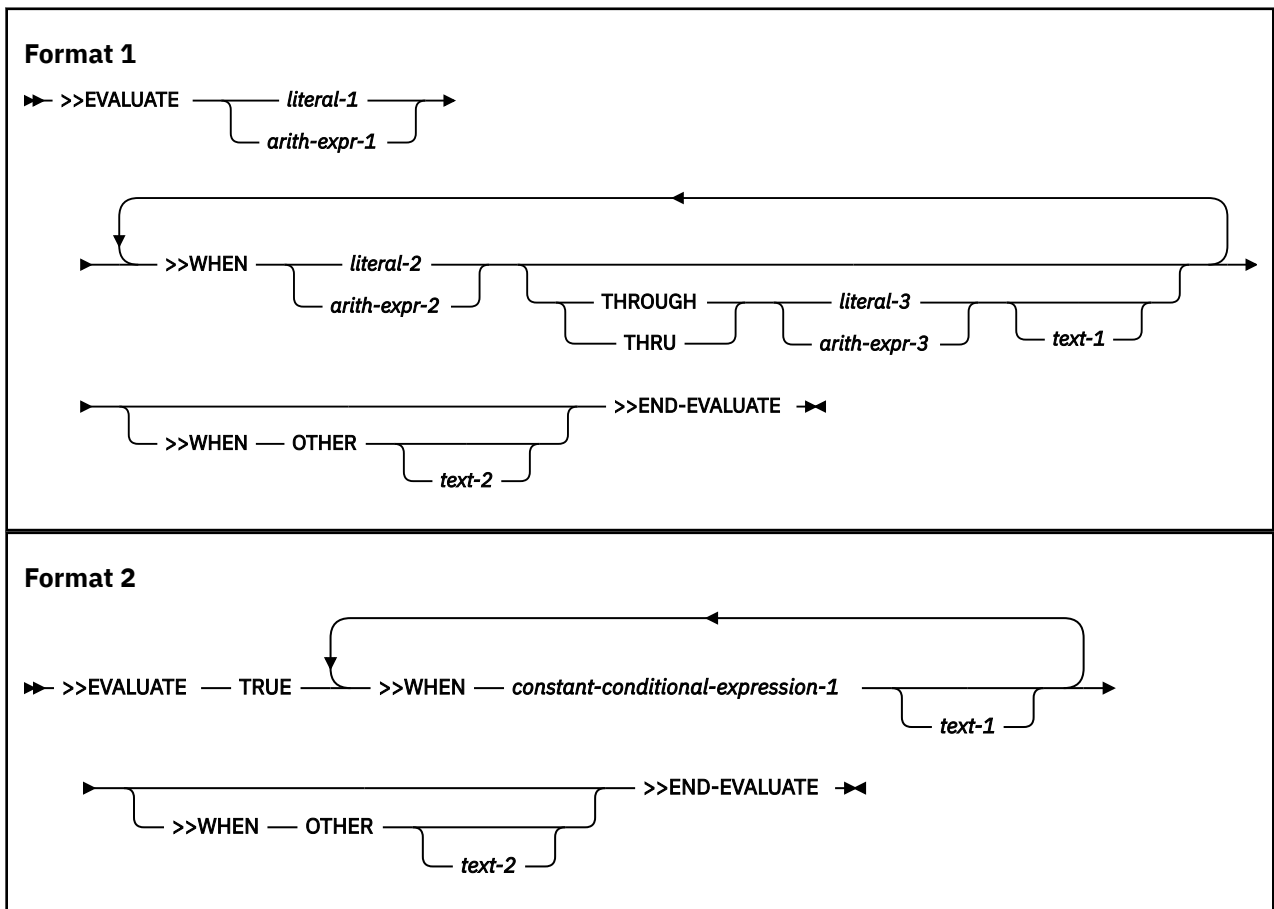
“Defined conditions” on page 727

“Predefined compilation variables” on page 728

DEFINE (*Enterprise COBOL Programming Guide*)

EVALUATE

The EVALUATE directive provides a multi-branch method of choosing the source lines to include in a compilation group.



For descriptive purposes, in this topic:

- *operand-1* refers to *literal-1* or *arith-expr-1* in format 1, and to the TRUE keyword in format 2.
- *operand-2* refers to *literal-2* or *arith-expr-2* in format 1, and to *constant-conditional-expression-1* in format 2.
- *operand-3* refers to *literal-3* or *arith-expr-3* in format 1.

All formats:

>>EVALUATE, >>WHEN, >>WHEN OTHER, >>END-EVALUATE

Must begin on a new line in area A or B and must be specified entirely on that line.

text-1, text-2

Must begin on a new line and may consist of multiple lines.

May be any kind of source lines, including compiler directives. *text-1* or *text-2* may also include COPY statements.

The phrases of a given EVALUATE directive must be specified all in the same library text or all in source text. For purposes of this rule, *text-1* and *text-2* are not considered phrases of the EVALUATE directive. A nested EVALUATE directive specified in *text-1* or *text-2* is considered a new EVALUATE directive.

Format 1:

>>EVALUATE

All operands of one EVALUATE directive must be of the same category. For this rule, an arithmetic expression is of category numeric.

literal-1, arith-expr-1

Selection subjects.

literal-2, literal-3, arith-expr-2, arith-expr-3

Selection objects.

THROUGH, THRU

The words THROUGH and THRU are equivalent. If the THROUGH phrase is specified, all selection subjects and selection objects must be of category numeric.

arith-expr-1, arith-expr-2, arith-expr-3

Must be formed in accordance with the arithmetic expression rules as described in [“Compile-time arithmetic expressions”](#) on page 727.

Format 2:

constant-conditional-expression-1

Must be formed in accordance with the constant conditional expression rules as described in [“Constant conditional expressions”](#) on page 726.

General rules

All Formats:

- *text-1* and *text-2* are not part of the EVALUATE compiler directive line. *text-1* and *text-2* that are in the first true branch of the EVALUATE statement are subject to the matching and replacing rules of the COPY statement and REPLACE statement.
- If the END-EVALUATE phrase is reached without any WHEN phrase evaluating to TRUE, or without encountering a WHEN OTHER phrase, all lines of *text-1* associated with all WHEN phrases are omitted from the resultant text.

Format 1:

- The selection subject is compared against the values specified in each WHEN phrase in turn as follows:
 - If the THROUGH phrase is not specified, a TRUE result is returned if the selection subject is equal to *operand-2*.
 - If the THROUGH phrase is specified, a TRUE result is returned if the selection subject is greater than or equal to *operand-2* and less than or equal to *operand-3*.
- If a WHEN phrase evaluates to TRUE, all the lines of *text-1* associated with that WHEN phrase are included in the resultant text. All lines of *text-1* associated with other WHEN phrases in that EVALUATE directive and all lines of *text-2* associated with a WHEN OTHER phrase are omitted from the resultant text.

- Format 2:

- ## Related references

“REPLACE statement” on page 708

The IF directive provides for a one-way or two-way conditional compilation.



constant-conditional-expression-1

text-1, text-2

The phrases of a given IF directive must be specified all in the same library text or all in source text. For purposes of this rule, *text-1* and *text-2* are not considered phrases of the IF directive. A nested IF directive specified in *text-1* or in *text-2* is considered a new IF directive.

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- If *constant-conditional-expression-1* evaluates to FALSE, all lines of *text-2* are included in the resultant text and all lines of *text-1* are omitted from the resultant text.

Related references

[“COPY statement” on page 696](#)

[“REPLACE statement” on page 708](#)

Examples of conditional compilation

Example 1: use predefined compilation variables to enable a program to be used in both CICS and BATCH:

Suppose that the CICS compiler option is in effect and the compiler works with the integrated CICS translator:

```
>>IF IGY-CICS
    EXEC CICS READ FILE-1... *> Read a record in CICS
>>ELSE
    READ FILE-2              *> Read a record in BATCH
>>END-IF
```

Example 2: import numeric variable value from outside the source and test it

Suppose that DEFINE(VAR1=10) is in effect:

```
>>DEFINE VAR1 AS PARAMETER
>>DEFINE VAR2 AS VAR1 + 2
>>IF VAR2 < 12
    compute x = x + 1 *> this code should NOT be included
>>ELSE
    compute x = x - 1 *> this code should be included
>>END-IF
```

Example 3: use the format 1 EVALUATE directive with numeric compilation variables

```
>>DEFINE VAR1 AS 6
>>DEFINE VAR2 AS 1
>>EVALUATE VAR1
>>WHEN VAR2 + 2
    compute x = x + 1 *> this code should NOT be included
>>WHEN 4 THRU 8
    compute x = x - 1 *> this code should be included
>>WHEN OTHER
    compute x = x * 2 *> this code should NOT be included
>>END-EVALUATE
```

Example 4: use the format 2 EVALUATE directive with alphanumeric compilation variables

```
>>DEFINE VAR1 AS 'MOO'
>>EVALUATE TRUE
>>WHEN VAR2 IS DEFINED
    compute x = x + 1 *> this code should NOT be included
>>WHEN VAR1 IS EQUAL TO 'GOO' OR VAR1 IS EQUAL TO 'MOO'
    compute x = x - 1 *> this code should be included
>>END-EVALUATE
```

Example 5: use **OVERWRITE** and **OFF** in the **DEFINE** directive

```
>>DEFINE VAR AS 12
.
.
>>DEFINE VAR OFF
.
.
>>IF VAR IS DEFINED
    compute x = x + 1 *> this code should NOT be included
>>ELSE
    compute x = x - 1 *> this code should be included
>>END-IF
.
.
>>DEFINE VAR AS 16
.
.
>>DEFINE VAR AS VAR - 2 OVERWRITE
.
.
>>IF VAR IS EQUAL TO 16
    compute x = x + 1 *> this code should NOT be included
>>ELSE
    compute x = x - 1 *> this code should be included
>>END-IF
```

Example 6: general use of boolean literals and compilation variables

```
>>DEFINE B1 B'1' *> B1 is category boolean
>>DEFINE B2 B'0' *> B2 is category boolean
.
.
>>IF B1 AND B2
    display "Both B1 and B2 are true" *> not included
>>ELSE
    >>IF B1
        display "Only B1 is true" *> included
    >>ELSE
        >>IF B2
            display "Only B2 is true" *> not included
        >>ELSE
            display "Neither B1 nor B2 is true" *> not included
        >>END-IF
    >>END-IF
>>END-IF
```

Constant conditional expressions

A constant conditional expression is an expression that is specified in conditional compilation directives and evaluated during the processing of those directives to determine the text that is included in the resultant program.

Note: In this topic, "literals" also include compilation variables, which means that you can use compilation variables in constant conditional expressions.

A constant conditional expression shall be one of the following items:

- A relation condition in which both operands are literals or arithmetic expressions that contain only literal terms. The condition shall follow the rules for relation conditions, with the following additions:
 - The operands shall be of the same category. An arithmetic expression is of the category numeric.
 - If literals are specified and they are not numeric literals, the relational operator shall be "IS EQUAL TO", "IS NOT EQUAL TO", "IS =", or "IS NOT =".
- A defined condition.
- A boolean condition.
- A complex condition formed by combining the above forms of simple conditions into complex conditions by using AND, OR, and NOT. Abbreviated combined relation conditions shall not be specified.

Related references

["Relation conditions" on page 272](#)

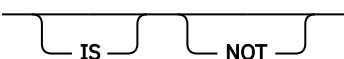
["Defined conditions" on page 727](#)

["Abbreviated combined relation conditions" on page 287](#)

Defined conditions

A defined condition expression tests whether a compilation variable is defined.

Format

► *compilation-variable-name-1*  DEFINED ►

compilation-variable-name-1

Must not be the same as a conditional compiler directive keyword.

IS DEFINED

A defined condition that uses the IS DEFINED syntax evaluates to TRUE if the *compilation-variable-name-1* is defined.

If a defined condition references a compilation variable that was defined via a DEFINE compiler option, but preceding the defined condition in the program there is neither a corresponding DEFINE directive with the AS PARAMETER phrase nor a DEFINE directive without the OFF phrase for the compilation variable, then the defined condition for the compilation variable evaluates to FALSE.

IS NOT DEFINED

A defined condition that uses the IS NOT DEFINED syntax evaluates to TRUE if the *compilation-variable-name-1* is not defined.

A compilation variable whose most recent definition is via a DEFINE directive with the OFF phrase is considered to be not defined.

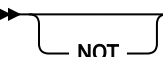
Related references

“Predefined compilation variables” on page 728
DEFINE (*Enterprise COBOL Programming Guide*)

Boolean conditions

A boolean condition determines whether a boolean literal is true or false.

Format

►  *boolean-literal-1* ►

boolean-literal-1

Evaluates to true if it is B'1', and evaluates to false if it is B'0'.

The condition NOT *boolean-literal-1* evaluates to the reverse truth-value of *boolean-literal-1*.

Related references

DEFINE (*Enterprise COBOL Programming Guide*)

Compile-time arithmetic expressions

You can specify a compile-time arithmetic expression in the DEFINE and EVALUATE directives and as part of a constant conditional expression, such as those found in IF directives or WHEN phrases of the EVALUATE directives.

Note: In this topic, "literals" also include compilation variables, which means that you can use compilation variables in compile-time arithmetic expressions.

A compile-time arithmetic expression follows the usual arithmetic expression rules, with the following exceptions:

- The exponentiation operator shall not be specified.

- All operands shall be integer numeric literals or arithmetic expressions in which all operands are integer numeric literals.
- The expression shall be specified in such a way that a division by zero does not occur.
- Intermediate results are computed according to the rules described in *Fixed-point data and intermediate results* in the *Enterprise COBOL Programming Guide*. For that purpose, the integer operands of compile-time arithmetic expressions can be considered fixed-point numbers with 0 decimal digits. The ARITH(COMPAT|EXTEND) option setting is taken into account when deciding how many digits of precision to maintain for intermediate results.

Related references

[“Arithmetic expressions” on page 266](#)

ARITH (*Enterprise COBOL Programming Guide*)

Predefined compilation variables

There are compilation variables that are defined automatically by the compiler. These compilation variables listed in this topic can be referenced in conditional compilation directives wherever a compilation variable is allowed.

Table 79. Predefined compilation variables		
Predefined compilation variable name ¹	Description	Value
IGY-ARCH	Indicates the target architecture for which the source code is being compiled.	The value of the ARCH option that was used to compile the program: 10, 11, 12, 13, or 14.
IGY-CICS	Indicates whether embedded CICS statements are accepted.	B'1' if the CICS compiler option is in effect; B'0' otherwise.
IGY-COMPILER-VRM	Indicates the version of the compiler.	An integer in the format VVRRMM, where: • VV represents the version number. • RR represents the release number. • MM represents the modification number. For example, compiler version 6.4.0 has an IGY-COMPILER-VRM value of 060400.
IGY-DLL	Indicates whether the program is compiled as DLL code.	B'1' if the DLL compiler option is in effect; B'0' otherwise.
IGY-DYNAM	Indicates whether programs invoked through the CALL literal statement will be loaded or deleted dynamically at run time.	B'1' if the DYNAM compiler option is in effect; B'0' otherwise.
IGY-LP	Indicates whether an AMODE 31 or AMODE 64 program should be generated with the related language features enabled.	The value of the LP option that is used to compile the program: 32 or 64.
IGY-OPTIMIZE	Indicates the optimization level.	The optimization level that was used to compile the program: 0, 1 or 2.

Table 79. Predefined compilation variables (continued)

Predefined compilation variable name ¹	Description	Value
IGY-SQL	Indicates whether processing of embedded SQL statements is enabled.	B'1' if the SQL compiler option is in effect; B'0' otherwise.
IGY-SQLIMS	Indicates whether processing of embedded SQLIMS statements is enabled.	B'1' if the SQLIMS compiler option is in effect; B'0' otherwise.
IGY-THREAD	Indicates whether the program is compiled with multithread support enabled.	B'1' if the THREAD compiler option is in effect; B'0' otherwise.
1. The older predefined compilation variables without the <i>IGY-</i> prefix are tolerated. It is suggested that the <i>IGY-</i> prefixed names be used. Do not use the <i>IGY-</i> prefix when you define your own compilation variables.		

Related references

[“DEFINE” on page 721](#)

DEFINE (*Enterprise COBOL Programming Guide*)

COBOL/Java interoperability

Enterprise COBOL provides two directives, JAVA-CALLABLE and JAVA-SHAREABLE, that facilitate interoperability between COBOL programs and Java programs.

JAVA-CALLABLE

The JAVA-CALLABLE directive instructs the compiler to make the COBOL program automatically callable from Java. When a COBOL program that is callable from Java is compiled, a program referred to as a *native method call stub program* is automatically generated by the compiler. This program is an interface between the Java caller and the user COBOL program. When the call stub program receives control from Java, it performs conversion of incoming argument values to their corresponding COBOL format and then dynamically calls the user COBOL program, which must be located by the application's STEPLIB. If the user program returns a value, then that value is converted to its corresponding Java format before the call stub returns.

Format

►► >>JAVA-CALLABLE ◀◀

General rules

The JAVA-CALLABLE directive must appear before the PROCEDURE DIVISION header for the outermost program of a compilation unit, and in a file that contains multiple compilation units, the JAVA-CALLABLE directive can only be used in the first function or program compilation unit in the file..

Since a Java-callable program is invoked dynamically by the call stub, the program name must be 8 characters or less.

The generated native method call stub program is written to a z/OS UNIX directory that is specified by the JAVAIO(OUTPATH(zos-unix-directory)) compiler option. If that option is not in effect, the default output location is the current directory if the compiler is being run from the cob2 utility; otherwise, the output location is the home directory of the userid under which the compiler is running.

The name of the COBOL native method call stub program has the following format:

```
<cobol-program-name>_java_native.cbl
```

where `cobol-program-name` is the name of the COBOL program that is being compiled as specified in the program's IDENTIFICATION DIVISION.

Handling parameters and returned values

- If the COBOL program containing the `JAVA-CALLABLE` directive specifies parameters through the `USING` phrase of its `PROCEDURE DIVISION` header, the native method call stub program that is generated by the compiler automatically handles conversions between incoming Java arguments and the parameter types of the user COBOL program before calling that program. Using the data definition of the COBOL program's parameters, the compiler assumes the incoming Java argument values are in a format that adheres to the COBOL/Java type mapping and code to do the conversion is generated accordingly. For mapping details, see [Legal COBOL types for Java interoperability and corresponding Java types](#).

Note: For parameters that are fixed length tables, if the numbers of elements in the incoming Java array argument does not match the number of elements indicated in the `OCCURS` clause of the table definition in the corresponding COBOL parameter, a runtime error will occur. For mapping details, see [Legal COBOL types for Java interoperability and corresponding Java types](#).

- If the COBOL program containing the `JAVA-CALLABLE` directive also contains a `RETURNING` phrase in its `PROCEDURE DIVISION` header, then after the user program is called, the native method call stub program will automatically convert the returned COBOL value to its corresponding Java value that is then returned to the Java caller. For mapping details, see [Legal COBOL types for Java interoperability and corresponding Java types](#).

Note: When a call is made from a Java program to a COBOL program compiled with the `JAVA-CALLABLE` directive, the Java program will use a signature for the COBOL native method that is generated by the COBOL compiler itself and is based on the definition of the COBOL program and its parameters and possible returned value. This allows the Java program to perform compile-time checking to ensure that only arguments of the appropriate Java type can be passed to the COBOL program.

Data access properties

Some data types are treated as read-only data items and some are treated as read/write data items when passed as parameters in the following cases:

- A COBOL parameter that corresponds to a primitive Java type (that is, `byte`, `short`, `int`, `long`, `float`, `double` or `boolean`) is treated as a read-only data item in a COBOL/Java interoperable application, regardless of whether the parameter is defined as `BY REFERENCE`, `BY VALUE`, or `BY CONTENT`. If a COBOL program modifies such a parameter value, the change is not reflected in the Java caller. This also applies to COBOL parameters that correspond to the immutable Java classes `java.lang.String` and `java.math.BigDecimal`.
- A COBOL parameter that corresponds to a single-dimension array of a primitive Java type (that is, `byte[]`, `short[]`, `int[]`, `long[]`, `float[]`, `double[]`, or `boolean[]`). If the COBOL program modifies such a parameter value, the change is reflected in the Java caller. This behavior is consistent with the semantics of arrays of primitive types when they are used as parameters in a pure Java application.
- Changes to a COBOL parameter that is an alphanumeric group item will be reflected in the Java caller.

Building COBOL native method call stub programs

The COBOL native method call stub program, along with any other COBOL stub programs that are generated by the compiler for the application, must be compiled into a Java native method DLL using the `cjbuild` tool. See *Building and running non-OO applications that interoperate with Java in Enterprise COBOL Programming Guide* for more details. This native method DLL will be loaded into the JVM at run time when a call is made to a COBOL program from Java. If the output location of the DLL is specified as

a data set to `cjbuild`, then that data set must be in your STEPLIB at run time. If the output location of the DLL is specified as a z/OS UNIX directory, then JVM property `java.library.path` should be set to indicate that directory, which can be done by adding `-Djava.library.path=<path>` to the "java" command used to execute the Java application.

Invoking COBOL programs from Java

COBOL programs containing the `JAVA-CALLABLE` directive are considered to be native methods in a class called "progs". The class is part of a package whose name is provided to the `cjbuild` utility. If no package name is specified when using `cjbuild`, the name of the package defaults to `enterprise.COBOL`.

For example, if a COBOL program named COBPROG1 contains the JAVA-CALLABLE directive and the native method DLL for the associated application is compiled with the cjbuild utility using the default package name, then COBPROG1 can be invoked in Java as a COBOL native method as follows:

```
import enterprise.COBOL.*;

:
enterprise.COBOL.progs.COBPROG1(...);
```

If the name of a Java-callable COBOL program is not specified as a literal in the IDENTIFICATION DIVISION, then the name of the corresponding COBOL native method in Java will be the upper-cased version of the program name. If the name of the Java-callable program is specified as a literal in the IDENTIFICATION DIVISION, then the case of the letters in the corresponding COBOL native method name in Java will match the case of the letters in the literal exactly.

Related references

COBOL/Java interoperability outside of the object-oriented (OO) COBOL framework (*Enterprise COBOL Programming Guide*)

Compiling COBOL applications that interoperate with Java (*Enterprise COBOL Programming Guide*)

JAVAIOP (*Enterprise COBOL Programming Guide*)

“Mapping between COBOL and Java data types for non-OO COBOL/Java interoperability” on page 733

JAVA-SHAREABLE

Use the JAVA-SHAREABLE ON and JAVA-SHAREABLE OFF directives to bracket one or more WORKING-STORAGE data items to indicate that they are to be made read/write accessible from Java applications interoperating with this COBOL program.

Note: In order for a Java method to access a COBOL program's WORKING-STORAGE, the COBOL program must be part of the run unit associated with the COBOL/Java application, and the program must also have been invoked at least once, either directly by a Java method or indirectly by another COBOL program that is callable from Java, so that its WORKING-STORAGE is initialized before the Java method tries to access it.

Format

► ►JAVA-SHAREABLE 

General rules

The JAVA-SHAREABLE directive must be specified in the DATA DIVISION and is only relevant for data items defined in the WORKING-STORAGE SECTION and only for data items that are Java-compatible. See [“Mapping between COBOL and Java data types for non-OO COBOL/Java interoperability”](#) on page 733 for details. Otherwise, the data item is ignored.

If JAVA-SHAREABLE ON/OFF appears in the middle of a group definition, it will take effect starting with the next 01/77 level data item and will apply to that item and its subordinates if any, and will also apply to any subsequently defined data items until the end of the data division is encountered.

Because the mechanism for sharing data between COBOL and Java involves external data items in COBOL, data items that are Java-shareable should have a unique name across the entire COBOL part of the application; otherwise, there's a possibility for a name collision. The name of Java-shareable data items must also be 26 characters or less.

The generated stub programs and Java interface classes are written to a z/OS UNIX directory that is specified by the JAVAIO(OUTPATH(zos-unix-directory)) compiler option. If that option is not in effect, the default output location is the current directory if the compiler is being run from the cob2 utility; otherwise, the output location is the home directory of the userid under which the compiler is running.

Building COBOL stub programs and Java interface classes

- The WORKING-STORAGE initialization stub program, along with any other COBOL stub programs generated by the compiler for the application, must be compiled into an application DLL using the cjbuild tool, which will also build any generated Java interface classes. For details, see Building and running non-OO applications that interoperate with Java in *Enterprise COBOL Programming Guide*. This application DLL will be loaded into the JVM at run time when Java makes a call to COBOL. If the output location of the DLL is specified as a data set to cjbuild, then that data set must be in your STEPLIB at run time. If the output location of the DLL is specified as a z/OS UNIX directory, then JVM property java.library.path should be set to that directory, which can be done by adding -Djava.library.path=<path> to the "java" command used to execute the Java application.

Accessing COBOL WORKING-STORAGE items from Java

The Java strg class used for accessing a program's WORKING-STORAGE items from Java is considered to be part of a package. The package name can be specified using the --pkgname option of the cjbuild utility when it is invoked to build the native method DLL for an interoperable application. If the name is not specified, it defaults to enterprise.COBOL.

The strg class consists of a 2-level hierarchy of classes and objects that can be used to access COBOL WORKING-STORAGE items. In particular, strg contains one nested class per each COBOL program that is part of your interoperable application, and for each such class, there is one object per 01/77-level WORKING-STORAGE item that is accessible from Java.

The name of the Java object corresponding to the COBOL data item contains only uppercase letters. This is true even if the data item is defined in your COBOL program with some lowercase letters. This is due to the fact that COBOL is a case insensitive language.

For example, if you have two COBOL programs in your application, PROG1 and PROG2, and PROG1 has shareable WORKING-STORAGE item DATA1, and PROG2 has shareable WORKING-STORAGE item DATA2, where DATA1 and DATA2 are both defined with picture string PIC S9(9) COMP-5, then DATA1 and DATA2 can be accessed from your Java application as follows:

```
import enterprise.COBOL.*;
:
int data1 = enterprise.COBOL.strg.PROG1.DATA1.get();
enterprise.COBOL.strg.PROG1.DATA1.put(24);

int data2 = enterprise.COBOL.strg.PROG2.DATA2.get();
enterprise.COBOL.strg.PROG2.DATA2.put(12);
```

• Accessing groups

If a Java-shareable WORKING-STORAGE item is a group, then the corresponding Java accessor class has a structure that mimics the structure of the COBOL group.

For example, consider a COBOL program PROG1 with the following COBOL group G1 in WORKING-STORAGE:


```
>>JAVA-SHAREABLE ON
01 G1.
   03 N1 PIC S9(9) COMP-5.
   03 G1SUB.
       05 S1 PIC X(20).
>>JAVA-SHAREABLE OFF
```

A Java program can access the items in group G1 as follows:

```
import enterprise.COBOL.*;
:
int n1 = enterprise.COBOL.strg.PROG1.G1.N1.get();
String s2 = enterprise.COBOL.strg.PROG1.G1.G1SUB.S1.get();
```

• Accessing tables

To access an element of a table, the corresponding Java array variable must be indexed.

For example, consider the following COBOL program PROG1 with the following COBOL table:

```
>>JAVA-SHAREABLE ON
01 LIST.
   03 NUM PIC S9(9) COMP-5 OCCURS 10 TIMES.
>>JAVA-SHAREABLE OFF
```

Java code can access elements of table NUM as follows:

```
import enterprise.COBOL.*;
int n1 = enterprise.COBOL.PROG1.LIST.NUM[0].get();
// get value of NUM(1)
enterprise.COBOL.strg.PROG1.LIST.NUM[0].put(12);
// Set NUM(1) to 12
```

Related references

COBOL/Java interoperability outside of the object-oriented (OO) COBOL framework (*Enterprise COBOL Programming Guide*)

Compiling COBOL applications that interoperate with Java (*Enterprise COBOL Programming Guide*)

JAVAIOP (*Enterprise COBOL Programming Guide*)

[“Mapping between COBOL and Java data types for non-OO COBOL/Java interoperability” on page 733](#)

Mapping between COBOL and Java data types for non-OO COBOL/Java interoperability

This section describes the COBOL data types and related Java data types in non-OO COBOL programs that interoperate with Java through use of the JAVA-CALLABLE or JAVA-SHAREABLE directives or through use of the CALL statement to call a static Java method.

A COBOL program and a Java method need to pass data between one another under the following circumstances:

- A Java method passes arguments to a COBOL native method
- A Java method receives a returned value from a COBOL native method
- A COBOL program passes arguments to a static Java method
- A COBOL program receives a returned value from a static Java method
- A Java method directly accesses a data item in a native COBOL method's working-storage

To deal with these scenarios, Enterprise COBOL defines the following legal COBOL types for Java interoperability and their corresponding Java types:

Table 80. Java-compatible elementary COBOL types	
Elementary COBOL Data Types	Java Data Types
PIC X	byte

Table 80. Java-compatible elementary COBOL types (continued)

Elementary COBOL Data Types	Java Data Types
PIC X(m) DISPLAY, $m > 1$ PIC N(m) NATIONAL, $m > 0$ PIC U(m) UTF-8, $m > 0$ PIC U BYTE-LENGTH m, $m > 0$ PIC X DYNAMIC [LIMIT n] (not supported by the JAVA-SHAREABLE directive and will be ignored) PIC U DYNAMIC [LIMIT n] (not supported by the JAVA-SHAREABLE directive and will be ignored)	String ¹
PIC S9(n) COMP-5, $1 \leq n \leq 4$	short
PIC S9(n) COMP-5, $5 \leq n \leq 9$	int
PIC S9(n) COMP-5, $10 \leq n \leq 18$	long
COMP-1 (4-byte IBM hex float)	float (4-byte IEEE float)
COMP-2 (8-byte IBM hex float)	double (8-byte IEEE float)
usage COMP-3/PACKED-DECIMAL numeric	java.math.BigDecimal ¹
usage DISPLAY numeric (zoned)	java.math.BigDecimal ¹
<pre> 01 flag pic x. 88 flag-false value x'00'. 88 flag-true value x'01'. </pre>	boolean

Table 81. Java-compatible array COBOL types

Array COBOL Data Types	Java Data Types
<pre> 01 byte-list. 03 b1 pic x occurs m times. </pre>	byte[m]
<pre> 01 short-list. 03 s1 pic s9(n) comp-5 occurs m times. </pre> where $1 \leq n \leq 4$.	short[m]
<pre> 01 int-list. 03 i1 pic s9(n) comp-5 occurs m times. </pre> where $5 \leq n \leq 9$.	int[m]
<pre> 01 long-list. 03 l1 pic s9(n) comp-5 occurs m times. </pre> where $10 \leq n \leq 18$.	long[m]

Table 81. Java-compatible array COBOL types (continued)	
Array COBOL Data Types	Java Data Types
01 float-list. 03 f1 pic comp-1 occurs m times.	float[m]
01 double-list. 03 d1 pic comp-2 occurs m times.	double[m]
01 bool-list. 03 b1 pic x occurs m times. 88 flag-false value x'00'. 88 flag-true value x'01'.	boolean[m]
01 dec-list. 03 n1 pic s9(m)[v9(n))] display/comp-3 occurs j times.	BigDecimal[j] ¹
01 alpha-list. 03 s1 pic x(m) occurs n times. where $m > 1$.	String[n] ¹
01 nat-list. 03 n1 pic n(m) occurs n times.	String[n] ¹
01 utf8-list. 03 u1 pic u(m) occurs m times.	String[m] ¹

Table 82. Java-compatible alphanumeric group COBOL types	
Alphanumeric Group COBOL Data Types	Java Data Types
Any alphanumeric group item that does not satisfy the definition of a Java-compatible array type is considered to map to a Java byte array (byte[]) and it is the Java application's responsibility to both read and set the byte array appropriately based on the corresponding COBOL group definition.	byte[]

Note: ¹When these objects are used as arguments, any changes to the corresponding data in COBOL will not be reflected on the Java side of the application.

Alphanumeric groups

An alphanumeric group item in COBOL that does not satisfy the definition of one of the Java-compatible array types can still be used as an argument in a call to a static Java method and as a parameter in a COBOL program that is callable from Java.

When the group is used as a parameter in a Java callable program, it is assumed that the data being passed from Java to the parameter is a byte array object (byte[]), and it is the responsibility of the calling Java method to set up the data in the corresponding byte array to match the byte layout of the receiving

COBOL group item. Conversely, when the group is used as an argument in a call to a static Java method, the called Java method must assume that it is receiving a Java byte array object (byte[]), and it is the responsibility of the Java method to wrap the incoming byte array with a ByteBuffer in order to extract data out of the byte array in a meaningful way based on the layout of the bytes in the corresponding COBOL group.

Using general groups as parameters and arguments offers added flexibility to application developers, allowing them to achieve interoperability between Java and COBOL with complex group items for which there is no clear mapping to a Java type.

Note: When an alphanumeric group item is automatically treated as a Java byte array object because it doesn't satisfy the definition of a Java-compatible array type, the compiler produces informational message IGYPA3415I for the item.

Further considerations

- Alphanumeric group items are only compatible with Java when used as COBOL parameters or arguments. That is, alphanumeric group items cannot be the returned value of a java-callable program or the returned value from a call to a Java static method.
- Single-dimension arrays of each supported Java type are supported. To have the compiler treat an incoming Java argument as a single dimension array, or to treat an outgoing COBOL argument as being intended for a single dimension Java array, the COBOL parameter or argument should be defined using the following format:

```
01 identifier-1 occurs n times.  
   03 identifier-2 <java-compatible-cobol-data-definition>.
```

For example, the following COBOL definition corresponds to a Java `int[]` type:

```
01 list-data occurs 10 times.  
   03 num pic s9(9) comp-5.
```

Note: If the number of elements in an incoming Java array argument does not match the number of elements indicated in the OCCURS clause of a fixed length table definition in the corresponding COBOL parameter, a runtime error will occur.

- For conversions between COBOL COMP-1/COMP-2 items and Java float/double items, a loss of precision may occur due to the difference in the underlying representation. That is, COMP-1/COMP-2 are stored in IBM hex float format, and float/double are stored in IEEE binary float format:
 - Conversions between IBM hexadecimal floating point format and IEEE floating point format are done via Language Environment special purpose C/C++ interface `__fp_htob()`, using rounding mode `_FP_HB_BRN` (biased round to nearest).
 - Conversions between IEEE floating point format and IBM hexadecimal floating point format are done via Language Environment special purpose C/C++ interface `__fp_btob()`, using rounding mode `_FP_HB_BRN` (biased round to nearest).
- A BigDecimal value can be received into either a zoned or packed decimal item in COBOL. Similarly, zoned and packed decimal items are both legal senders for BigDecimal receivers in Java. The conversion between zoned/packed decimal and BigDecimal (and vice versa) is handled automatically by routines in the IBM JZOS toolkit, which must be installed in the environment in which the COBOL / Java interoperable application is running.
- Data items defined as `PIC X(m) / U(m) / N(m)` will be padded with an appropriate space character to a length of m characters when the length of the corresponding Java String object is smaller than m characters. Data items defined as `PIC U BYTE-LENGTH m` will be padded with an appropriate space character to a maximum length of m bytes when the length of the corresponding Java String object is smaller than m bytes. Data items defined as `PIC X/U DYNAMIC [LIMIT n]` will not be padded with spaces and will have the same byte length as the corresponding Java String object, up to a maximum of n bytes if the LIMIT phrase of the DYNAMIC clause is specified.

Appendix A. IBM extensions and COBOL standards

IBM extensions are features, syntax rules, or behaviors defined by IBM rather than by the COBOL standards. IBM extensions generally add features, syntax, or rules that are not specified in the ANSI and ISO COBOL standards. The term 85 COBOL Standard refers to those standards.

IBM extensions

The COBOL standards are listed in [Appendix J, “Industry specifications,”](#) on page 797.

[Table 83 on page 737](#) lists IBM extensions with a brief description. Standard behavior is shown in brackets, [], when the standard behavior is not obvious. Extensions are described in more detail throughout this document, but they are not further identified as extensions.

Many IBM extensions are distinguished from standard language by their syntax. For others, you use compiler options to choose between standard and extension behavior. Generally, the related compiler options are noted in the detailed rules. For information about compiler options, see *Option settings for 85 COBOL Standard conformance* in the *Enterprise COBOL Programming Guide*.

If an item is listed as an extension, all related rules are also extensions. For example, USAGE DISPLAY-1 for DBCS characters is listed as an extension; its many uses in statements and clauses are also extensions, but are not listed separately.

Table 83. IBM extension language elements	
Language area	Extension elements
COBOL words	User-defined words written in DBCS characters Computer-name written in DBCS characters Class-names (for object orientation) Method-names User-defined words can include an underscore, but not as the first character.
National character support (Unicode support)	Support for UTF-16 with USAGE NATIONAL Allowance of UTF-8 with USAGE DISPLAY Usage NATIONAL for data categories national, national-edited, numeric, numeric-edited, external decimal, and external floating-point GROUP-USAGE clause with the NATIONAL phrase National literals (basic and hexadecimal) National character value for figurative constants SPACE, ZERO, QUOTE, HIGH-VALUES, LOW-VALUES, ALL literal Intrinsic functions for data conversion: • DISPLAY-OF • NATIONAL-OF Extended case mapping with UPPER-CASE and LOWER-CASE functions

Table 83. **IBM extension language elements** (continued)

Language area	Extension elements
Implicit items	Special object references: • SELF • SUPER Special registers: • ADDRESS OF • JNIEVPTR • JSON-CODE • JSON-STATUS • LENGTH OF • RETURN-CODE • SHIFT-IN • SHIFT-OUT • SORT-CONTROL • SORT-CORE-SIZE • SORT-FILE-SIZE • SORT-MESSAGE • SORT-MODE-SIZE • SORT-RETURN • TALLY • WHEN-COMPILED • XML-CODE • XML-EVENT • XML-INFORMATION • XML-NAMESPACE • XML-NAMESPACE-PREFIX • XML-NNAMESPACE • XML-NNAMESPACE-PREFIX • XML-NTEXT • XML-TEXT
<u>Figurative constants</u>	Selection of apostrophe (') as the value of the figurative constant QUOTE NULL and NULLS for pointers and object references

Table 83. **IBM extension language elements** (continued)

Language area	Extension elements
Literals	Use of apostrophe (') as an alternative to the quotation mark (") in opening and closing delimiters Mixed single-byte and double-byte characters in alphanumeric literals (mixed literals) <u>Hexadecimal notation for alphanumeric literals</u>, defined by opening delimiters X " and X ' <u>Null-terminated alphanumeric literals</u>, defined by opening delimiters Z " and Z ' <u>DBCS literals</u>, defined by opening delimiters N " , N ' , G " , and G ' . N " and N ' are defined as DBCS when the NSYMBOL(DBCS) compiler option is in effect. Consecutive alphanumeric literals (coding two consecutive alphanumeric literals by ending the first literal in column 72 of a continued line and starting the next literal with a quotation mark in the continuation line) <u>National literals</u> N " , N ' , NX " , NX ' for storing literal content as national characters. N " and N ' are defined as national when the NSYMBOL(NATIONAL) compiler option is in effect. 19- to 31-digit fixed-point numeric literals. [The 85 COBOL Standard specifies a maximum of 18 digits.] Floating-point numeric literals
Comments	Comment lines before the IDENTIFICATION DIVISION header <u>Comment lines and comment entries containing multibyte characters</u> <u>Inline comments</u>
End markers	The following <u>end markers</u>: • END CLASS • END FACTORY • END METHOD • END OBJECT
Indexing and subscripting	<u>Referencing a table with an index-name defined for a different table</u> Specifying a positive signed integer literal following the operator + or - in relative subscripting

Table 83. **IBM extension language elements** (continued)

Language area	Extension elements
IDENTIFICATION DIVISION for programs	Abbreviation ID for IDENTIFICATION <u>RECURSIVE clause</u> An optional separator period following PROGRAM-ID, AUTHOR, INSTALLATION, DATE-WRITTEN, and SECURITY paragraph headers. [The 85 COBOL Standard requires a period following each of these paragraph headers.] An optional separator period following program-name in the PROGRAM-ID paragraph. [The 85 COBOL Standard requires a period following program-name.] An alphanumeric literal for program-name in the PROGRAM-ID paragraph; characters \$, #, and @ in the name of the outermost program, and the underscore can be the first character; program-name up to 160 characters in length. [The 85 COBOL Standard requires that program-name be specified as a user-defined word.]
End markers	<u>Program-name in a literal</u>. [The 85 COBOL Standard requires that program-name be specified as a user-defined word.]
Object-oriented structure	In a class definition: • <u>CLASS-ID paragraph</u> • <u>INHERITS clause</u> • <u>END CLASS marker</u> In a method definition: • <u>METHOD-ID paragraph</u> • <u>EXIT METHOD statement</u> • <u>END METHOD marker</u>
Configuration section	<u>Repository paragraph</u>

Table 83. **IBM extension language elements** (continued)

Language area	Extension elements
<u>SPECIAL-NAMES paragraph</u>	The optional order of clauses. [The 85 COBOL Standard requires that the clauses be coded in the order presented in the syntax diagram.] Optionality of a period after the last clause when no clauses are coded. [The 85 COBOL Standard requires a period, even when no clauses are coded.] <u>Multiple CURRENCY SIGN clauses.</u> [The 85 COBOL Standard allows a single CURRENCY SIGN clause.] WITH PICTURE SYMBOL phrase in the CURRENCY SIGN clause Multiple-character and mixed-case currency signs in the CURRENCY SIGN clause (when the WITH PICTURE SYMBOL phrase is specified). [The 85 COBOL Standard allows only one character, and it is both the currency sign and the currency picture symbol. The standard currency sign must not be: • The same character as any standard picture symbol • A digit 0-9 • One of the special characters * + - , ; () " = / • A space] Use of lower-case alphabetic characters as a currency sign. [The 85 COBOL Standard allows only uppercase characters.]

Table 83. **IBM extension language elements** (continued)

Language area	Extension elements
<u>INPUT-OUTPUT SECTION, FILE-CONTROL paragraph</u>	Optionality of "FILE-CONTROL." when the INPUT-OUTPUT SECTION is specified, no file-control-paragraph is specified, and there are no files defined in the compilation unit. [The 85 COBOL Standard requires that "FILE-CONTROL." be coded if "INPUT-OUTPUT SECTION." is coded.] Optionality of the file-control-paragraph when the "FILE CONTROL." syntax is specified and there are no files defined in the compilation unit. [The 85 COBOL Standard requires that a file-control-paragraph be coded if "INPUT-OUTPUT SECTION." is coded.] <u>PASSWORD clause</u> The second <i>data-name</i> in the <u>FILE STATUS clause</u> Optionality of RECORD in the <u>ALTERNATE RECORD KEY clause</u>. [The 85 COBOL Standard requires the word RECORD.] A numeric, numeric-edited, alphanumeric-edited, alphabetic, internal floating-point, external floating-point, national, national-edited, or DBCS primary or alternate record key data item. [The 85 COBOL Standard requires that the key be alphanumeric.] A primary or alternate record key defined outside the minimum record size for indexed files containing variable-length records. [The 85 COBOL Standard requires that the primary and alternate record keys be within the minimum record size.] A numeric data item of usage DISPLAY or NATIONAL in the <u>FILE STATUS clause</u>. [The 85 COBOL Standard requires an alphanumeric file status data item.] The <u>ORGANIZATION IS LINE SEQUENTIAL clause</u> and line-sequential file control format National literal in the <u>PADDING CHARACTER clause</u>
<u>INPUT-OUTPUT SECTION, I-O-CONTROL paragraph</u>	<u>APPLY WRITE-ONLY clause</u> Specifying only one file-name in the SAME clause in the sequential, indexed, and sort-merge formats of the I-O-control entry. [The 85 COBOL Standard requires at least two file-names.] Optionality of the keyword ON in the RERUN clause. [The 85 COBOL Standard requires that ON be coded.] The line-sequential format I-O-control entry The RERUN clause in the sort-merge I-O-control entry

Table 83. <i>IBM extension language elements</i> (continued)	
Language area	Extension elements
DATA DIVISION	<u>LOCAL-STORAGE SECTION</u> The <u>GLOBAL</u> clause in the <u>LINKAGE SECTION</u> Specifying level numbers that are lower than other level numbers at the same hierarchical level in a data description entry. [The 85 COBOL Standard requires that all elementary or group items at the same level in the hierarchy be assigned identical level numbers.] Data categories internal floating-point, external floating-point, DBCS, national, and national-edited. Data category numeric with usage NATIONAL. Data category numeric-edited with usage NATIONAL.
FILE SECTION	<i>data-name</i> in the <u>LABEL RECORDS</u> clause, for specifying user labels <u>RECORDING MODE</u> clause Line-sequential format file description entry
Sort/merge file description entry	The following clauses: • <u>BLOCK CONTAINS</u> • <u>LABEL RECORDS</u> • <u>VALUE OF</u> • <u>LINAGE</u> • <u>CODE-SET</u> • <u>WITH FOOTING</u> • <u>LINES AT</u>
<u>BLOCK CONTAINS</u> clause	BLOCK CONTAINS 0 for QSAM files. [The 85 COBOL Standard requires that at least 1 CHARACTER or RECORD be specified in the BLOCK CONTAINS clause.]
<u>VALUE OF</u> clause	The lack of VALUE clause effect on execution when specified under an SD
<u>DATA RECORDS</u> clause	Optionality of an 01 record description entry for a specified <i>data-name</i> . [The 85 COBOL Standard requires that an 01 record with the same <i>data-name</i> be specified.]
<u>LINAGE</u> clause	Specifying LINAGE for files opened in EXTEND mode
<u>BLANK WHEN ZERO</u> clause	Alternative spellings ZEROS and ZEROES for ZERO
<u>GLOBAL</u> clause	Specifying GLOBAL in the LINKAGE SECTION
<u>INDEXED BY</u> phrase	Nonunique unreferenced index names

Table 83. **IBM extension language elements** (continued)

Language area	Extension elements
<u>OCCURS clause</u>	Omission of "<i>integer-1 TO</i>" for variable-length tables Complex OCCURS DEPENDING ON. [The 85 COBOL Standard requires that an entry containing OCCURS DEPENDING ON be followed only by subordinate entries, and that no entry containing OCCURS DEPENDING ON be subordinate to an entry containing OCCURS DEPENDING ON.] Implicit qualification of a key specified without qualifiers when the key name is not unique Reference to a table through indexing when no INDEXED BY phrase is specified Keys of usages COMPUTATIONAL-1, COMPUTATIONAL-2, COMPUTATIONAL-3, COMPUTATIONAL-4, and COMPUTATIONAL-5 in the ASCENDING/DESCENDING KEY phrase Acceptance of nonunique index-names that are not referenced <u>Unbounded tables and groups</u>
<u>PICTURE clause</u>	A picture character-string containing 31 to 50 characters. [The 85 COBOL Standard allows a maximum of 30 characters.] Picture symbols G and N Picture symbol E and the external floating-point picture format Coding a trailing comma insertion character or trailing period insertion character immediately followed by a separator comma or separator semicolon in a PICTURE clause that is not the last clause of a data description entry. [The 85 COBOL Standard requires that a PICTURE clause containing a picture ending with a comma or period be the last clause in the entry and that it be followed immediately by a separator period.] Selecting a currency sign and currency symbol with the CURRENCY compiler option Case-sensitive currency symbols The maximum of 31 digits for numeric items of usages DISPLAY and PACKED-DECIMAL and for numeric-edited items of USAGE DISPLAY The effect of the TRUNC compiler option on the value of data items described with a usage of BINARY, COMPUTATIONAL, or COMPUTATIONAL-4
<u>REDEFINES clause</u>	Specifying REDEFINES of a redefined data item At a subordinate level, specifying a redefining data item that has a size greater than the size of the redefined data item
<u>SYNCHRONIZED clause</u>	Specifying SYNCHRONIZED for a level 01 entry

Table 83. **IBM extension language elements** (continued)

Language area	Extension elements
<u>USAGE</u> clause	The following phrases: • NATIVE • COMP-1 and COMPUTATIONAL-1 • COMP-2 and COMPUTATIONAL-2 • COMP-3 and COMPUTATIONAL-3 • COMP-4 and COMPUTATIONAL-4 • COMP-5 and COMPUTATIONAL-5 • DISPLAY-1 • OBJECT REFERENCE • NATIONAL • POINTER • PROCEDURE-POINTER • FUNCTION-POINTER Use of the SYNCHRONIZED clause for items of usage INDEX
<u>VALUE</u> clause for condition-name entries	A VALUE clause in file and LINKAGE SECTION other than in condition-name entries A VALUE clause for a condition-name entry on a group that has usages other than DISPLAY VALUE IS NULL and VALUE IS NULLS
<u>VOLATILE</u> clause	A VOLATILE clause in a format 1 data description entry
PROCEDURE DIVISION	Omission of a section-name Omission of a paragraph-name when a section-name is omitted A method, factory, or object procedure division Referencing data items in the LINKAGE SECTION without a USING phrase in the PROCEDURE DIVISION header (when those data-names are the operand of an ADDRESS OF phrase or ADDRESS OF special register) The following statements: • <u>ENTRY</u> • <u>EXIT METHOD</u> • <u>GOBACK</u> • <u>INVOKE</u> • <u>JSON PARSE</u> • <u>JSON GENERATE</u> • <u>XML PARSE</u> • <u>XML GENERATE</u>

Table 83. **IBM extension language elements** (continued)

Language area	Extension elements
PROCEDURE DIVISION header	The BY VALUE phrase The RETURNING phrase Specifying a data item in the <u>USING phrase</u> when the data item has a REDEFINES clause in its description Specifying multiple instances of a given data item in the <u>USING phrase</u> The formats for method, factory, and object definitions
Declarative Procedures	Performing a nondeclarative procedure from a declarative procedure Referencing a declarative procedure or nondeclarative procedure in a GO TO statement from a declarative procedure. [The 85 COBOL Standard specifies that a declarative procedure must not reference a nondeclarative procedure. A reference to a declarative procedure from either another declarative procedure or a nondeclarative procedure is allowed only with a PERFORM statement.] Executing an active declarative
Procedures	Specifying <i>priority-number</i> as a positive signed numeric literal. [The 85 COBOL Standard requires an unsigned integer.] Omitting the section-header after the declaratives or when there are no declaratives. [The 85 COBOL Standard requires a section-header following the "DECLARATIVES." syntax and following the "END DECLARATIVES." syntax.] Omitting an initial <i>paragraph-name</i> if there are no declaratives. [The 85 COBOL Standard requires a paragraph-name in the following circumstances: • After the USE statement if there are statements in the declarative procedure • Following a section header outside declarative procedures • Before any procedural statement if there are no declaratives and the 85 COBOL Standard requires that procedural statements be within a paragraph.] Specifying paragraphs that are not contained within a section, even if some paragraphs are so contained. [The 85 COBOL Standard requires that paragraphs be within a section except when there are no declaratives. The 85 COBOL Standard requires that either all paragraphs be in sections or that none be.]
<u>Conditional expressions</u>	DBCS and KANJI class conditions Specifying data items of usage COMPUTATIONAL-3 or usage PACKED-DECIMAL in a NUMERIC class test

Table 83. **IBM extension language elements** (continued)

Language area	Extension elements
<u>Relation condition</u>	Enclosing an alphanumeric, DBCS, or national literal in parentheses The data-pointer format, the procedure-pointer and function-pointer format, and the object-reference format Comparison of an index-name with an arithmetic expression Use of parentheses within abbreviated combined relation conditions Note: Enterprise COBOL supports most parenthesis usage as IBM extensions with the following exceptions: • Within the scope of an abbreviated combined relation condition, Enterprise COBOL does not support relational operators inside parentheses. For example: <pre>A = B AND (< C OR D)</pre> • Incorrect usages of parentheses in relation conditions are not accepted by Enterprise COBOL. For example: <pre>(A = 0 AND B) = 0</pre>
<u>CORRESPONDING phrase</u>	Specifying an identifier that is subordinate to a filler item
<u>INVALID KEY phrase</u>	Omission of both the <u>INVALID KEY phrase</u> and an applicable EXCEPTION/ERROR procedure. [The 85 COBOL Standard requires at least one of them.]
<u>ACCEPT statement</u>	The <i>environment-name</i> operand of the FROM phrase The DATE YYYYMMDD phrase The DAY YYYYDDD phrase
<u>ADD statement</u>	A composite of operands greater than 18 digits
<u>ALLOCATE statement</u>	The LOC phrase
<u>CALL statement</u>	The procedure-pointer and function-pointer operands for identifying the program to be called The following phrases and parameters: • ADDRESS OF • LENGTH OF • OMITTED • BY VALUE • RETURNING Specifying a <i>file-name</i> as an argument Specifying the called program-name in an alphabetic or zoned-decimal data item Specifying an argument defined as a subordinate group item. [The 85 COBOL Standard requires that arguments be an elementary data item or a group item defined with level 01.]

Table 83. <i>IBM extension language elements</i> (continued)	
Language area	Extension elements
<u>CANCEL statement</u>	Specifying the name of the program to be canceled in an alphabetic or zoned-decimal data item The effect of the PGMNAME compiler option on the name of the program to be canceled
<u>CLOSE statement</u>	WITH NO REWIND phrase The line-sequential format
<u>COMPUTE statement</u>	The use of the word EQUAL in place of the equal sign (=)
<u>DISPLAY statement</u>	The <i>environment-name</i> operand of the UPON phrase Displaying signed numeric literals and noninteger numeric literals
<u>DIVIDE statement</u>	A composite of operands greater than 18 digits
<u>EXIT statement</u>	Specifying the EXIT statement in a sentence that has statements before or after the EXIT statement or in a paragraph that has other sentences. [The 85 COBOL Standard requires that the EXIT statement be specified in a sentence by itself and that the sentence be the only sentence in the paragraph.]
<u>EXIT PROGRAM statement</u>	Specifying EXIT PROGRAM before the last statement in a sequence of imperative statements. [The 85 COBOL Standard requires that the EXIT PROGRAM statement be specified as the last statement in a sequence of imperative statements.]
<u>GO TO statement</u>	Coding the unconditional format before the last statement in a sequence of imperative statements. [The 85 COBOL Standard requires that an unconditional GO TO be coded: • Only in a single-statement paragraph if no procedure-name is specified • Otherwise, as the last statement of a sentence.]
<u>IF statement</u>	The use of END-IF with the NEXT SENTENCE phrase. [The 85 COBOL Standard disallows use of END-IF with NEXT SENTENCE.]
<u>INITIALIZE statement</u>	DBCS, EGCS, NATIONAL, and NATIONAL-EDITED in the REPLACING phrase Initializing a data item that contains the DEPENDING phrase of the OCCURS clause
<u>MERGE statement</u>	Specifying file-names in a SAME clause
<u>MULTIPLY statement</u>	A composite of operands greater than 18 digits
<u>OPEN statement</u>	The line-sequential format Specifying the EXTEND phrase for files that have a LINAGE clause
<u>PERFORM statement</u>	An empty in-line PERFORM statement A common exit for two or more active PERFORMS

<i>Table 83. IBM extension language elements (continued)</i>	
Language area	Extension elements
<u>READ statement</u>	Omission of both the AT END phrase and an applicable declarative procedure Omission of both the INVALID KEY phrase and an applicable declarative procedure Read into an item that is neither an alphanumeric group item nor an elementary alphanumeric item
<u>RETURN statement</u>	Return into an item that is neither an alphanumeric group item nor an elementary alphanumeric item
<u>REWRITE statement</u>	Omission of both the INVALID KEY phrase and an applicable declarative procedure Rewriting a record with a different number of character positions than the number of character positions in the record being rewritten
<u>SEARCH statement</u>	Specifying END SEARCH with NEXT SENTENCE Omission of both the NEXT SENTENCE phrase and imperative statements in the binary search format
<u>SET statement</u>	The data-pointer format The procedure-pointer and function-pointer format The object reference format
<u>SORT statement</u>	Specifying GIVING file-names in the SAME clause
<u>START statement</u>	Omission of both the INVALID KEY phrase and an applicable exception procedure Use of a key of a category other than alphanumeric
<u>STOP statement</u>	Specifying a noninteger fixed-point literal or a signed numeric integer or noninteger fixed-point literal Coding STOP as other than the last statement in a sentence
<u>STRING statement</u>	Reference modification of the data item specified in the INTO phrase
<u>SUBTRACT statement</u>	A composite of operands greater than 18 digits
<u>UNSTRING statement</u>	Reference modification of the sending field
<u>WRITE statement</u>	INVALID KEY and NOT ON INVALID KEY phrases The line-sequential format For a relative file, writing a different number of character positions than the number of character positions in the record being replaced Specifying both the ADVANCING PAGE and END-OF-PAGE phrases in a single WRITE statement The effect of the ADV compiler option on the length of the record written to a file Using WRITE ADVANCING with stacker selection for a card punch file For a relative or indexed file, omission of both the INVALID KEY phrase and an applicable exception procedure

Table 83. **IBM extension language elements** (continued)

Language area	Extension elements
Intrinsic functions	The effect of the INTDATE compiler options on the INTEGER-OF-DATE and INTEGER-OF-DAY functions The following functions: • Chapter 36, “BIT-OF,” on page 533 • Chapter 37, “BIT-TO-CHAR,” on page 535 • Chapter 46, “DATE-TO-YYYYMMDD,” on page 553 • Chapter 48, “DAY-TO-YYYYDDD,” on page 557 • Chapter 49, “DISPLAY-OF,” on page 559 • Chapter 58, “HEX-OF,” on page 577 • Chapter 59, “HEX-TO-CHAR,” on page 579 • Chapter 75, “NATIONAL-OF,” on page 611 • ULENGTH • UPOS • USUBSTR • USUPPLEMENTARY • Chapter 108, “UUID4,” on page 677 • UVALID • UWIDTH • Chapter 113, “YEAR-TO-YYYY,” on page 689
FACTORIAL function	The effect of the ARITH(EXTEND) compiler option on the range of values permitted in the argument
LENGTH function	Specifying a pointer, the ADDRESS OF special register, or the LENGTH OF special register as an argument to the function
NUMVAL function	The effect of the ARITH(EXTEND) compiler option on the maximum number of digits allowed in the argument
NUMVAL-C function	The effect of the ARITH(EXTEND) compiler option on the maximum number of digits allowed in the argument
NUMVAL-F function	The effect of the ARITH(EXTEND) compiler option on the maximum number of digits allowed in the argument
TEST-NUMVAL function	The effect of the ARITH(EXTEND) compiler option on the maximum number of digits allowed in the argument
TEST-NUMVAL-C function	The effect of the ARITH(EXTEND) compiler option on the maximum number of digits allowed in the argument
TEST-NUMVAL-F function	The effect of the ARITH(EXTEND) compiler option on the maximum number of digits allowed in the argument

Table 83. IBM extension language elements (continued)	
Language area	Extension elements
<u>Compiler-directing statements</u>	The following statements: • <u>BASIS</u> • <u>CBL(PROCESS)</u> • <u>*CONTROL and *CBL</u> • <u>DELETE</u> • <u>EJECT</u> • <u>INSERT</u> • <u>READY or RESET TRACE</u> • <u>SERVICE LABEL</u> • <u>SERVICE RELOAD</u> • <u>SKIP1, SKIP2, and SKIP3</u> • <u>TITLE</u>
<u>Compiler directives</u>	<u>CALLINTERFACE directive</u>
<u>COPY statement</u>	The optionality of the syntax "OF <i>library-name</i>" for specifying a text-name qualifier Literals for specifying <i>text-name</i> and <i>library-name</i> SUPPRESS phrase Nested COPY statements Hyphen as the first or last character in the <i>word</i> form of REPLACING operands The use of any character (other than a COBOL separator) in the <i>word</i> form of REPLACING operands. [The 85 COBOL Standard accepts only the characters used in formation of user-defined words.]

COBOL standards

The COBOL standards, or 85 COBOL Standards are listed in [Appendix J, "Industry specifications,"](#) on [page 797](#). Extensions range from minor relaxation of rules to major capabilities, such as XML support, Unicode support, object-oriented COBOL for Java interoperability, and DBCS character handling.

Appendix B. Obsolete language elements

Obsolete language elements are elements that are categorized as *obsolete* in the 85 COBOL Standard. Those elements are not part of the 2002 COBOL Standard.

This does *not* imply that IBM will remove the 85 COBOL Standard obsolete elements from a future release of Enterprise COBOL.

The following language elements are categorized as obsolete by the 85 COBOL Standard:

- ALTER statement
- AUTHOR paragraph
- Comment entry
- DATA RECORDS clause
- DATE-COMPILED paragraph
- DATE-WRITTEN paragraph
- DEBUG-ITEM special register
- Debugging sections
- ENTER statement
- GO TO without a specified procedure-name
- INSTALLATION paragraph
- LABEL RECORDS clause
- MEMORY SIZE clause
- MULTIPLE FILE TAPE clause
- RERUN clause
- REVERSED phrase
- SECURITY paragraph
- Segmentation module
- STOP *literal* format of the STOP statement
- USE FOR DEBUGGING declarative
- VALUE OF clause
- The figurative constant ALL *literal* with a length greater than one, when the figurative constant is associated with a numeric or numeric-edited item

Appendix C. DBCS notation

Double-Byte Character Set (DBCS) strings in literals, comments, and user-defined words are delimited by shift-out and shift-in characters.

In this document, the shift-out delimiter is represented pictorially by the < character, and the shift-in character is represented pictorially by the > character. The single-byte EBCDIC codes for the shift-out and shift-in delimiters are X'0E' and X'0F', respectively.

The <> symbol denotes contiguous shift-out and shift-in characters. The >< symbol denotes contiguous shift-in and shift-out characters.

DBCS characters are shown in this form: D1D2D3. Latin alphabet characters in DBCS representation are shown in this form: .A.B.C. The dots that precede the letters represent the hexadecimal value X'42'.

Notes:

- In EBCDIC DBCS data containing mixed single-byte and double-byte characters, double-byte character strings are delimited by shift-out and shift-in characters.
- In ASCII DBCS data containing mixed single-byte and double-byte characters, double-byte character strings are not delimited by shift-out and shift-in characters.

Appendix D. Compiler limits

Enterprise COBOL has the following limits for programs and class definitions.

Although the COBOL compiler supports addressing various memory areas in a compile unit up to the limits described in this appendix, a complete application, typically consisting of multiple compile units, is still restricted by the amount of private storage available in the address space in which it runs. In other words, an application might run out of storage before reaching the described limits.

For details about how to calculate the length of data items, see *Finding the length of data items* in the *Enterprise COBOL Programming Guide*.

You can sum the length fields in the MAP (also requires either OFFSET or LIST) option output's PPA4 section for the static memory footprint of a compile unit. For automatic (local-storage) requirements, see *Example: DSA memory map (stack storage map)* in the *Enterprise COBOL Programming Guide*.

In general, the maximum size of tables and of elementary alphanumeric data items in LP(64) is 2,147,483,646 bytes. See the table below for more details.

The LOCAL-STORAGE SECTION is allocated on the Language Environment stack. Its total size is limited by the settings in the Language Environment as well as the requirements of internal variables used by the compiler. The actual limit available for COBOL programs is less than 2,147,483,646 bytes.

The Language Environment STACK64 runtime option controls the allocation of stack storage for AMODE 64 applications. The default value is STACK64(1M,1M,128M). Use this runtime option to specify the maximum size of the stack required by your application.

The WORKING-STORAGE SECTION is allocated on the Language Environment heap. Its total size is limited by the 64-bit storage capacity of the machine.

Table 84. Compiler limits	
Language element	Compiler limit
Maximum length of alphanumeric literals	160 bytes
Maximum length of user-defined words (for example, <i>data-name</i> , <i>file-name</i> , <i>class-name</i>)	30 bytes
Size of program	999,999 lines With SOURCE(DEC): 999,999 lines With SOURCE(HEX): 16,777,215 lines
Size of the following data types: USAGE IS INDEX USAGE IS POINTER USAGE IS FUNCTION-POINTER USAGE IS OBJECT-REFERENCE	With LP(32): 4 bytes With LP(64): 8 bytes
Size of the following data types: USAGE IS PROCEDURE-POINTER	With LP(32): 8 bytes With LP(64): 8 bytes
LENGTH OF special register	With LP(32): USAGE IS BINARY PICTURE 9(9) With LP(64): USAGE IS BINARY PICTURE 9(18)
ADDRESS OF special register	With LP(32): 4 bytes With LP(64): 8 bytes
Number of literals	4,194,303 ^(Note 1)
Total length of literals	4,194,303 bytes ^(Note 1)

<i>Table 84. Compiler limits (continued)</i>	
Language element	Compiler limit
Reserved word table entries	1536
COPY REPLACING ... BY ... (items per COPY statement)	No limit
Number of COPY libraries	No limit
Block size of COPY library	32,760 bytes
IDENTIFICATION DIVISION	
ENVIRONMENT DIVISION	
Configuration section	
Special-names paragraph	
<i>mnemonic-name</i> IS	18
UPSI- <i>n</i> ... (switches)	0-7
<i>alphabet-name</i> IS ...	No limit
Literal THRU ... or ALSO ...	256
Input-Output section	
File-control paragraph	
SELECT <i>file-name</i> ...	A maximum of 65,535 file names can be assigned external names
ASSIGN <i>system-name</i> ...	No limit
ALTERNATE RECORD KEY <i>data-name</i> ...	253
RECORD KEY length	No limit ^(Note 3)
RESERVE <i>integer</i> (buffers)	255 ^(Note 4)
I-O-control paragraph	
RERUN ON <i>system-name</i> ...	32,767
RERUN <i>integer</i> RECORDS	16,777,215
SAME RECORD AREA	255
SAME RECORD AREA FOR <i>file-name</i> ...	255
SAME SORT/MERGE AREA	No limit ^(Note 2)
MULTIPLE FILE <i>file-name</i> ...	No limit ^(Note 2)
DATA DIVISION	
77 data item size	With LP(32): 999,999,999 bytes With LP(64): 2,147,483,646 bytes
01-49 data item size	With LP(32): 999,999,999 bytes With LP(64): 2,147,483,646 bytes
Total 01 + 77 (data items)	No limit
88 condition-names ...	No limit
88 level VALUE clause ...	No limit

Table 84. Compiler limits (continued)	
Language element	Compiler limit
66 RENAMES . . .	No limit
PICTURE clause, number of characters in <i>character-string</i>	50
PICTURE clause, numeric item digit positions	With ARITH(COMPAT): 18 With ARITH(EXTEND): 31
PICTURE clause, numeric-edited character positions	249
Picture symbol replication ()	With LP(32): 999,999,999 bytes With LP(64): 2,147,483,646 bytes
Picture symbol replication (editing)	32,767
Picture symbol replication (), class DBCS items	With LP(32): 499,999,999 bytes With LP(64): 1,073,741,823 bytes
Picture symbol replication (), class national items	With LP(32): 499,999,999 bytes With LP(64): 1,073,741,823 bytes
Elementary item size	With LP(32): 999,999,999 bytes With LP(64): 2,147,483,646 bytes
OCCURS integer	With LP(32): 999,999,999 bytes With LP(64): 2,147,483,646 bytes
Total number of ODOs	4,194,303 ^(Note 1)
Table size	With LP(32): 999,999,999 bytes With LP(64): 2,147,483,646 bytes
Table element size	With LP(32): 999,999,999 bytes With LP(64): 2,147,483,646 bytes
ASCENDING or DESCENDING KEY . . . (per OCCURS clause)	12 KEYS
Total length of keys (per OCCURS clause)	256 bytes
INDEXED BY . . . (index names per OCCURS clause)	12
Total number of indexes (index names) per class or program	65,535
Size of relative index	32,765
FILE SECTION	
FD record description entry	1,048,575 bytes
FD <i>file-name</i> . . .	65,535
LABEL <i>data-name</i> . . . (if no optional clauses)	255
Label record length	80 bytes
BLOCK CONTAINS <i>integer</i>	2,147,483,647 ^(Note 8)
RECORD CONTAINS <i>integer</i>	1,048,575 ^(Note 5)
LINAGE clause values	99,999,999
SD <i>file-name</i> . . .	65,535

Table 84. Compiler limits (continued)	
Language element	Compiler limit
DATA RECORD <i>data-name</i> . . .	No limit ^(Note 2)
LINKAGE SECTION	
Total size	With LP(32): 2,147,483,646 bytes With LP(64): Unlimited, up to the available 64-bit addressing capacity of the machine.
LOCAL-STORAGE SECTION	
Total size	With LP(32): 2,147,483,646 bytes With LP(64): 2,147,483,646 bytes
WORKING-STORAGE SECTION	
Total size of items without the external attribute	With LP(32): 2,147,483,646 bytes With LP(64): Unlimited, up to the available 64-bit addressing capacity of the machine.
Total size of items with the external attribute	With LP(32): 2,147,483,646 bytes With LP(64): Unlimited, up to the available 64-bit addressing capacity of the machine.
PROCEDURE DIVISION	
Procedure and constant area	4,194,303 bytes ^(Note 1)
PROCEDURE DIVISION USING <i>identifier</i> . . .	32,767
Procedure-names	1,048,575 ^(Note 1)
Subscripted data-names per statement	32,767
Statements per line (TEST)	7
ACCEPT statement, record length on input device	32,760
ADD <i>identifier</i> . . .	No limit
ALTER <i>procedure-name-1</i> TO <i>procedure-name-2</i> . . .	4,194,303 ^(Note 1)
CALL . . . BY CONTENT <i>identifier</i>	2,147,483,647 bytes
CALL <i>identifier</i> or <i>literal</i> USING <i>identifier</i> or <i>literal</i> . . .	16,380
CALL <i>literal</i> . . .	4,194,303 ^(Note 1)
Active programs in a run unit	32,767
Number of names called (DYN option)	No limit
CANCEL <i>identifier</i> or <i>literal</i> . . .	No limit
CLOSE file-name . . .	No limit
COMPUTE <i>identifier</i> . . .	No limit
DISPLAY <i>identifier</i> or <i>literal</i> . . .	No limit
DIVIDE <i>identifier</i> . . .	No limit
ENTRY USING <i>identifier</i> or <i>literal</i> . . .	No limit
EVALUATE . . . subjects	64
EVALUATE . . . WHEN clauses	256

Table 84. Compiler limits (continued)	
Language element	Compiler limit
GO <i>procedure-name</i> . . . DEPENDING	255
INSPECT TALLYING and REPLACING clauses	No limit
MERGE <i>file-name</i> ASC or DES KEY . . .	No limit
Total merge key length	4,092 bytes ^(Note 6)
MERGE USING <i>file-name</i> . . .	16 ^(Note 7)
MOVE <i>identifier</i> or <i>literal</i> TO <i>identifier</i> . . .	No limit
MULTIPLY <i>identifier</i> . . .	No limit
OPEN <i>file-name</i> . . .	No limit
PERFORM	4,194,303
PERFORM . . . TIMES <i>identifier</i> or <i>literal</i>	999,999,999
SEARCH ALL . . . maximum key length	No limit
SEARCH ALL . . . total length of keys	No limit
SEARCH . . . WHEN . . .	No limit
SET <i>index</i> or <i>identifier</i> . . . TO	No limit
SET <i>index</i> . . . UP/DOWN	No limit
SORT <i>file-name</i> ASC or DES KEY	No limit
Total sort key length	4,092 bytes ^(Note 6)
SORT USING <i>file-name</i> . . .	16 ^(Note 7)
STRING <i>identifier</i> . . .	No limit
STRING DELIMITED <i>identifier</i> or <i>literal</i> . . .	No limit
UNSTRING DELIMITED <i>identifier</i> or <i>literal</i> . . .	No limit
UNSTRING INTO <i>identifier</i> or <i>literal</i> . . .	No limit
USE . . . ON <i>file-name</i> . . .	No limit
XML PARSE statement, maximum size of <i>identifier</i>	999,999,999 bytes
Intrinsic Function	
LENGTH UPOS UVALID ULENGTH USUPPLEMENTARY	With LP(32): Return maximum 9-digit integer With LP(64): Return maximum 18-digit integer

Table 84. **Compiler limits** (continued)

Language element	Compiler limit
Notes: 1. Items included in 4,194,303 byte limit for procedure plus constant area. 2. Syntax checked, but has no effect on the execution of the program; there is no limit. 3. No compiler limit, but VSAM limits it to 255 bytes. 4. QSAM. 5. Compiler limit shown, but QSAM limits it to 32,760 bytes. 6. For QSAM and VSAM, the limit is 4088 bytes if EQUALS is coded on the OPTION control statement. 7. SORT limit for QSAM and VSAM. 8. Requires large block interface (LBI) support provided by OS/390 DFSMS 2.10.0 or later. On OS/390 systems with earlier releases of DFSMS, the limit is 32,760 bytes. For more information about using large block sizes, see <i>Setting block sizes</i> in the <i>Enterprise COBOL Programming Guide</i>.	

Appendix E. EBCDIC and ASCII collating sequences

A collating sequence defines the order of characters within a coded character set or COBOL alphabet for purposes of sorting, merging, and comparing data and for processing files with indexed organization.

The ascending collating sequences for both the single-byte EBCDIC (Extended Binary Coded Decimal Interchange Code) and single-byte ASCII (American National Standard Code for Information Interchange) character sets are shown in this appendix. The collating sequence is defined by the ordinal number of characters in the character set, relative to 1.

The symbols and associated meanings shown for the EBCDIC collating sequence are those defined in the EBCDIC code page defined with CCSID 1140. Symbols and meanings can vary for other EBCDIC code pages, but the collating sequence is unchanged.

EBCDIC collating sequence

The EBCDIC collating sequence is the order in which characters are defined in EBCDIC.

The following table presents the collating sequence for single-byte EBCDIC code page 1140.

Ellipsis (...) indicates omission of a range of ordinal numbers between predecessor and successor ordinal numbers.

Table 85. <i>EBCDIC collating sequence</i>				
Ordinal number	Symbol	Meaning	Decimal representation	Hex representation
...				
65		Space	64	40
...				
75	¢	Cent sign	74	4A
76	.	Period, decimal point	75	4B
77	<	Less than sign	76	4C
78	(	Left parenthesis	77	4D
79	+	Plus sign	78	4E
80		Vertical bar, logical OR	79	4F
81	&	Ampersand	80	50
...				
91	!	Exclamation point	90	5A
92	\$	Dollar sign	91	5B
93	*	Asterisk	92	5C
94	)	Right parenthesis	93	5D
95	;	Semicolon	94	5E
96	¬	Logical NOT	95	5F
97	-	Minus, hyphen	96	60
98	/	Slash	97	61

Table 85. **EBCDIC collating sequence** (continued)

Ordinal number	Symbol	Meaning	Decimal representation	Hex representation
...				
108	,	Comma	107	6B
109	%	Percent sign	108	6C
110	_	Underscore	109	6D
111	>	Greater than sign	110	6E
112	?	Question mark	111	6F
...				
122	`	Grave accent	121	79
123	:	Colon	122	7A
124	#	Number sign, pound sign	123	7B
125	@	At sign	124	7C
126	'	Apostrophe, prime sign	125	7D
127	=	Equal sign	126	7E
128	"	Quotation marks	127	7F
...				
130	a		129	81
131	b		130	82
132	c		131	83
133	d		132	84
134	e		133	85
135	f		134	86
136	g		135	87
137	h		136	88
138	i		137	89
...				
146	j		145	91
147	k		146	92
148	l		147	93
149	m		148	94
150	n		149	95
151	o		150	96
152	p		151	97
153	q		152	98
154	r		153	99

<i>Table 85. EBCDIC collating sequence (continued)</i>				
Ordinal number	Symbol	Meaning	Decimal representation	Hex representation
...				
160	€	Euro currency sign	159	9F
...				
162	~	Tilde	161	A1
163	s		162	A2
164	t		163	A3
165	u		164	A4
166	v		165	A5
167	w		166	A6
168	x		167	A7
169	y		168	A8
170	z		169	A9
...				
177	^	Caret	176	B0
...				
188	[	Opening square bracket	187	BA
189	]	Closing square bracket	188	BB
...				
193	{	Opening brace	192	C0
194	A		193	C1
195	B		194	C2
196	C		195	C3 [®]
197	D		196	C4
198	E		197	C5
199	F		198	C6
200	G		199	C7
201	H		200	C8
202	I		201	C9
...				
209	}	Closing brace	208	D0
210	J		209	D1
211	K		210	D2
212	L		211	D3
213	M		212	D4

Table 85. **EBCDIC collating sequence** (continued)

Ordinal number	Symbol	Meaning	Decimal representation	Hex representation
214	N		213	D5
215	O		214	D6
216	P		215	D7
217	Q		216	D8
218	R		217	D9
...				
225	\	Backslash	224	E0
...				
227	S		226	E2
228	T		227	E3
229	U		228	E4
230	V		229	E5
231	W		230	E6
232	X		231	E7
233	Y		232	E8
234	Z		233	E9
...				
241	0		240	F0
242	1		241	F1
243	2		242	F2
244	3		243	F3
245	4		244	F4
246	5		245	F5
247	6		246	F6
248	7		247	F7
249	8		248	F8
250	9		249	F9
...				

US English ASCII code page

The ASCII collating sequence is the order in which characters are defined in ASCII.

The following table presents the collating sequence for the US English ASCII code page. The collating sequence is the order in which characters are defined in ANSI INCITS 4, the 7-Bit American National Standard Code for Information Interchange (7-Bit ASCII), and in the International Reference Version of *ISO/IEC 646, 7-Bit Coded Character Set for Information Interchange*.

Ellipsis (. . .) indicates omission of a range of ordinal numbers between predecessor and successor ordinal numbers.

<i>Table 86. ASCII collating sequence</i>				
Ordinal number	Symbol	Meaning	Decimal representation	Hex representation
1		Null	0	0
...				
33		Space	32	20
34	!	Exclamation point	33	21
35	"	Quotation mark	34	22
36	#	Number sign	35	23
37	\$	Dollar sign	36	24
38	%	Percent sign	37	25
39	&	Ampersand	38	26
40	'	Apostrophe, prime sign	39	27
41	(	Opening parenthesis	40	28
42	)	Closing parenthesis	41	29
43	*	Asterisk	42	2A
44	+	Plus sign	43	2B
45	,	Comma	44	2C
46	-	Hyphen, minus	45	2D
47	.	Period, decimal point	46	2E
48	/	Slash, solidus	47	2F
49	0		48	30
50	1		49	31
51	2		50	32
52	3		51	33
53	4		52	34
54	5		53	35
55	6		54	36
56	7		55	37
57	8		56	38
58	9		57	39
59	:	Colon	58	3A
60	;	Semicolon	59	3B
61	<	Less than sign	60	3C
62	=	Equal sign	61	3D

Table 86. **ASCII collating sequence** (continued)

Ordinal number	Symbol	Meaning	Decimal representation	Hex representation
63	>	Greater than sign	62	3E
64	?	Question mark	63	3F
65	@	Commercial At sign	64	40
66	A		65	41
67	B		66	42
68	C		67	43
69	D		68	44
70	E		69	45
71	F		70	46
72	G		71	47
73	H		72	48
74	I		73	49
75	J		74	4A
76	K		75	4B
77	L		76	4C
78	M		77	4D
79	N		78	4E
80	O		79	4F
81	P		80	50
82	Q		81	51
83	R		82	52
84	S		83	53
85	T		84	54
86	U		85	55
87	V		86	56
88	W		87	57
89	X		88	58
90	Y		89	59
91	Z		90	5A
92	[	Opening bracket	91	5B
93	\	Backslash, reverse solidus	92	5C
94	]	Closing bracket	93	5D
95	^	Caret	94	5E
96	_	Underscore	95	5F

<i>Table 86. ASCII collating sequence (continued)</i>				
Ordinal number	Symbol	Meaning	Decimal representation	Hex representation
97	`	Grave accent	96	60
98	a		97	61
99	b		98	62
100	c		99	63
101	d		100	64
102	e		101	65
103	f		102	66
104	g		103	67
105	h		104	68
106	i		105	69
107	j		106	6A
108	k		107	6B
109	l		108	6C
110	m		109	6D
111	n		110	6E
112	o		111	6F
113	p		112	70
114	q		113	71
115	r		114	72
116	s		115	73
117	t		116	74
118	u		117	75
119	v		118	76
120	w		119	77
121	x		120	78
122	y		121	79
123	z		122	7A
124	{	Opening brace	123	7B
125		Vertical bar	124	7C
126	}	Closing brace	125	7D
127	~	Tilde	126	7E

Appendix F. Source language debugging

Several COBOL language elements implement the debugging feature.

COBOL language elements are:

- Debugging lines
- Debugging sections
- DEBUG-ITEM special register
- Compile-time switch (WITH DEBUGGING MODE clause)
- Object-time switch

Debugging lines

A *debugging line* is a statement that is compiled only when the compile-time switch is activated. Debugging lines allow you, for example, to check the value of a data item at certain points in a procedure.

To specify a debugging line in your program, code a D in column 7 (the indicator area). You can include successive debugging lines, but each must have a D in column 7. You cannot break character-strings across two lines.

All your debugging lines must be written so that the program is syntactically correct, whether the debugging lines are compiled or treated as comments.

You can code debugging lines anywhere in your program after the OBJECT-COMPUTER paragraph.

A debugging line that contains only spaces in Area A and in Area B is treated as a blank line.

Debugging sections

Debugging sections are permitted only in the outermost program; they are not valid in nested programs. Debugging sections are never triggered by procedures contained in nested programs.

Debugging sections are declarative procedures. Declarative procedures are described under [“USE statement”](#) on page 713. A debugging section can be called, for example, by a PERFORM statement that causes repeated execution of a procedure. Any associated *procedure-name* debugging declarative section is executed once for each repetition.

A debugging section executes *only* if both the compile-time switch and the object-time switch are activated.

The debug feature recognizes each separate occurrence of an imperative statement *within* an imperative statement as the beginning of a separate statement.

You cannot refer to a procedure defined within a debugging section from a statement outside of the debugging section.

References to the DEBUG-ITEM special register can be made only from within a debugging declarative procedure.

DEBUG-ITEM special register

The DEBUG-ITEM special register provides information for a debugging declarative procedure about the conditions that cause debugging section execution.

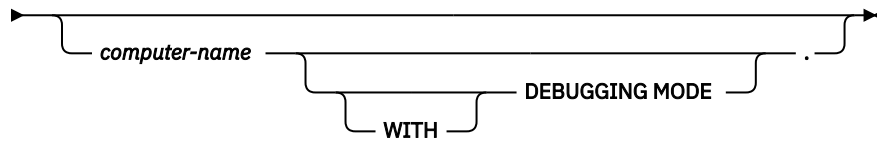
For details of the DEBUG-ITEM special register, see [“DEBUG-ITEM”](#) on page 19.

Activate compile-time switch

The compile-time switch activates the debugging lines and sections. To place the compile-time switch in effect, specify WITH DEBUGGING MODE in the SOURCE-COMPUTER paragraph of the configuration section.

Format

►► SOURCE-COMPUTER. →



WITH DEBUGGING MODE

When WITH DEBUGGING MODE is specified, all debugging sections and debugging lines are compiled.

When WITH DEBUGGING MODE is omitted, all debugging sections and debugging lines are treated as comments.

Usage note: If you include a COPY statement as a debugging line, the letter "D" must appear on the first line of the COPY statement. The compiler treats the copied text as the debugging line or lines. The COPY statement is executed, regardless of whether WITH DEBUGGING MODE is specified or not.

Activate object-time switch

The object-time switch is set when the runtime option DEBUG or NODEBUG is specified. (NODEBUG is the default supplied by IBM.)

For details on the format, see the *Language Environment Programming Guide*.

The USE FOR DEBUGGING declarative procedures are activated when DEBUG is in effect and inhibited when NODEBUG is in effect.

The debugging lines (lines with "D" or "d" in column 7) are not affected by the DEBUG or NODEBUG option; they are always active if they have been compiled.

When WITH DEBUGGING MODE is *not* specified in the SOURCE-COMPUTER paragraph, the object-time switch has no effect on execution of the object program.

You do not have to recompile the source unit to activate or deactivate the object-time switch.

Appendix G. Reserved words

A *reserved word* is a character-string with a predefined meaning in a COBOL source unit.

The following table identifies words that are reserved in Enterprise COBOL and words that you should avoid because they might be reserved in a future release of Enterprise COBOL.

- Words marked *X* under *Reserved* are reserved for function implemented in Enterprise COBOL. If used as user-defined names, these words are flagged with an S-level message.
- Words marked *X* under *Standard only* are 85 COBOL Standard reserved words for function not implemented in Enterprise COBOL. (Some of the function is implemented in the Report Writer Precompiler.) Use of these words as user-defined names is flagged with an S-level message.
- Words marked *X* under *Potential reserved words* are words that might be reserved in a future release of Enterprise COBOL. IBM recommends that you not use these words as user-defined names. Use of these words as user-defined names is flagged with an I-level message.

This column includes words reserved in the 2002 COBOL Standard.

The default reserved words are shown in the following table. You can select a different reserved word table by using the WORD compiler option. For details, see *WORD* in the *Enterprise COBOL Programming Guide*.

Table 87. Reserved words			
Word	Reserved	Standard only	Potential reserved words
+ Arithmetic operator - unary plus or addition	X		
- Arithmetic operator - unary minus or subtraction	X		
* Arithmetic operator - multiplication	X		
/ Arithmetic operator - division	X		
** Arithmetic operator - exponentiation	X		
> Relational operator - greater than	X		
< Relational operator - less than	X		
= Relational operator - equal and assignment operator in COMPUTE	X		
== Pseudo-text delimiter in COPY and REPLACE statements	X		
>= Relational operator - greater than or equal	X		
<= Relational operator - less than or equal	X		
<> Relational operator - not equal		X	
*> Comment indicator	X		
>> Compiler directive indicator		X	
ACCEPT	X		
ACCESS	X		
ACTIVE-CLASS			X

Table 87. **Reserved words** (continued)

Word	Reserved	Standard only	Potential reserved words
ADD	X		
ADDRESS	X		
ADVANCING	X		
AFTER	X		
ALIGNED			X
ALL	X		
ALLOCATE	X		
ALPHABET	X		
ALPHABETIC	X		
ALPHABETIC-LOWER	X		
ALPHABETIC-UPPER	X		
ALPHANUMERIC	X		
ALPHANUMERIC-EDITED	X		
ALSO	X		
ALTER	X		
ALTERNATE	X		
AND	X		
ANY	X		
ANYCASE			X
APPLY	X		
ARE	X		
AREA	X		
AREAS	X		
ASCENDING	X		
ASSIGN	X		
AT	X		
AUTHOR	X		
B-AND			X
B-NOT			X
B-OR			X
B-XOR			X
BASED			X
BASIS	X		
BEFORE	X		

Table 87. **Reserved words** (continued)

Word	Reserved	Standard only	Potential reserved words
BEGINNING	X		
BINARY	X		
BINARY-CHAR			X
BINARY-DOUBLE			X
BINARY-LONG			X
BINARY-SHORT			X
BIT			X
BLANK	X		
BLOCK	X		
BOOLEAN			X
BOTTOM	X		
BY	X		
BYTE-LENGTH	X		
CALL	X		
CANCEL	X		
CBL	X		
CD		X	
CF		X	
CH		X	
CHARACTER	X		
CHARACTERS	X		
CLASS	X		
CLASS-ID	X		
CLOCK-UNITS		X	
CLOSE	X		
COBOL	X		
CODE	X		
CODE-SET	X		
COL			X
COLLATING	X		
COLS			X
COLUMN		X	
COLUMNS			X
COM-REG	X		

Table 87. **Reserved words** (continued)

Word	Reserved	Standard only	Potential reserved words
COMMA	X		
COMMON	X		
COMMUNICATION		X	
COMP	X		
COMP-1	X		
COMP-2	X		
COMP-3	X		
COMP-4	X		
COMP-5	X		
COMPUTATIONAL	X		
COMPUTATIONAL-1	X		
COMPUTATIONAL-2	X		
COMPUTATIONAL-3	X		
COMPUTATIONAL-4	X		
COMPUTATIONAL-5	X		
COMPUTE	X		
CONDITION			X
CONFIGURATION	X		
CONSTANT			X
CONTAINS	X		
CONTENT	X		
CONTINUE	X		
CONTROL		X	
CONTROLS		X	
CONVERTING	X		
COPY	X		
CORR	X		
CORRESPONDING	X		
COUNT	X		
CRT			X
CURRENCY	X		
CURSOR			X
DATA	X		
DATA-POINTER			X

Table 87. **Reserved words** (continued)

Word	Reserved	Standard only	Potential reserved words
DATE	X		
DATE-COMPILED	X		
DATE-WRITTEN	X		
DAY	X		
DAY-OF-WEEK	X		
DBCS	X		
DE		X	
DEBUG-CONTENTS	X		
DEBUG-ITEM	X		
DEBUG-LINE	X		
DEBUG-NAME	X		
DEBUG-SUB-1	X		
DEBUG-SUB-2	X		
DEBUG-SUB-3	X		
DEBUGGING	X		
DECIMAL-POINT	X		
DECLARATIVES	X		
DEFAULT	X		
DELETE	X		
DELIMITED	X		
DELIMITER	X		
DEPENDING	X		
DESCENDING	X		
DESTINATION		X	
DETAIL		X	
DISABLE		X	
DISPLAY	X		
DISPLAY-1	X		
DIVIDE	X		
DIVISION	X		
DOWN	X		
DUPLICATES	X		
DYNAMIC	X		
EC			X

Table 87. **Reserved words** (continued)

Word	Reserved	Standard only	Potential reserved words
EGCS	X		
EGI		X	
EJECT	X		
ELSE	X		
EMI		X	
ENABLE		X	
END	X		
END-ACCEPT			X
END-ADD	X		
END-CALL	X		
END-COMPUTE	X		
END-DELETE	X		
END-DISPLAY			X
END-DIVIDE	X		
END-EVALUATE	X		
END-EXEC	X		
END-IF	X		
END-INVOKE	X		
END-JSON	X		
END-MULTIPLY	X		
END-OF-PAGE	X		
END-PERFORM	X		
END-READ	X		
END-RECEIVE		X	
END-RETURN	X		
END-REWRITE	X		
END-SEARCH	X		
END-START	X		
END-STRING	X		
END-SUBTRACT	X		
END-UNSTRING	X		
END-WRITE	X		
END-XML	X		
ENDING	X		

Table 87. **Reserved words** (continued)

Word	Reserved	Standard only	Potential reserved words
ENTER	X		
ENTRY	X		
ENVIRONMENT	X		
EO			X
EOP	X		
EQUAL	X		
ERROR	X		
ESI		X	
EVALUATE	X		
EVERY	X		
EXCEPTION	X		
EXCEPTION-OBJECT			X
EXEC	X		
EXECUTE	X		
EXIT	X		
EXTEND	X		
EXTERNAL	X		
FACTORY	X		
FALSE	X		
FD	X		
FILE	X		
FILE-CONTROL	X		
FILLER	X		
FINAL		X	
FIRST	X		
FLOAT-EXTENDED			X
FLOAT-LONG			X
FLOAT-SHORT			X
FOOTING	X		
FOR	X		
FORMAT			X
FREE	X		
FROM	X		
FUNCTION	X		

Table 87. **Reserved words** (continued)

Word	Reserved	Standard only	Potential reserved words
FUNCTION-ID			X
FUNCTION-POINTER	X		
GENERATE	X		
GET			X
GIVING	X		
GLOBAL	X		
GO	X		
GOBACK	X		
GREATER	X		
GROUP		X	
GROUP-USAGE	X		
HEADING		X	
HIGH-VALUE	X		
HIGH-VALUES	X		
I-O	X		
I-O-CONTROL	X		
ID	X		
IDENTIFICATION	X		
IF	X		
IN	X		
INDEX	X		
INDEXED	X		
INDICATE		X	
INHERITS	X		
INITIAL	X		
INITIALIZE	X		
INITIATE		X	
INPUT	X		
INPUT-OUTPUT	X		
INSERT	X		
INSPECT	X		
INSTALLATION	X		
INTERFACE			X
INTERFACE-ID			X

<i>Table 87. Reserved words (continued)</i>			
Word	Reserved	Standard only	Potential reserved words
INTO	X		
INVALID	X		
INVOKE	X		
IS	X		
JAVA	X		
JNIENVPTR	X		
JSON	X		
JSON-CODE	X		
JSON-STATUS	X		
JUST	X		
JUSTIFIED	X		
KANJI	X		
KEY	X		
LABEL	X		
LAST		X	
LEADING	X		
LEFT	X		
LENGTH	X		
LESS	X		
LIMIT	X		
LIMITS		X	
LINAGE	X		
LINAGE-COUNTER	X		
LINE	X		
LINE-COUNTER		X	
LINES	X		
LINKAGE	X		
LOCAL-STORAGE	X		
LOCALE			X
LOCK	X		
LOW-VALUE	X		
LOW-VALUES	X		
MEMORY	X		
MERGE	X		

Table 87. **Reserved words** (continued)

Word	Reserved	Standard only	Potential reserved words
MESSAGE		X	
METHOD	X		
METHOD-ID	X		
MINUS			X
MODE	X		
MODULES	X		
MORE-LABELS	X		
MOVE	X		
MULTIPLE	X		
MULTIPLY	X		
NATIONAL	X		
NATIONAL-EDITED	X		
NATIVE	X		
NEGATIVE	X		
NESTED			X
NEXT	X		
NO	X		
NOT	X		
NULL	X		
NULLS	X		
NUMBER		X	
NUMERIC	X		
NUMERIC-EDITED	X		
OBJECT	X		
OBJECT-COMPUTER	X		
OBJECT-REFERENCE			X
OCCURS	X		
OF	X		
OFF	X		
OMITTED	X		
ON	X		
OPEN	X		
OPTIONAL	X		
OPTIONS			X

Table 87. **Reserved words** (continued)

Word	Reserved	Standard only	Potential reserved words
OR	X		
ORDER	X		
ORGANIZATION	X		
OTHER	X		
OUTPUT	X		
OVERFLOW	X		
OVERRIDE	X		
PACKED-DECIMAL	X		
PADDING	X		
PAGE	X		
PAGE-COUNTER		X	
PASSWORD	X		
PERFORM	X		
PF		X	
PH		X	
PIC	X		
PICTURE	X		
PLUS		X	
POINTER	X		
POINTER-24			X
POINTER-31			X
POINTER-32	X		
POINTER-64			X
POSITION	X		
POSITIVE	X		
PRESENT			X
PRINTING		X	
PROCEDURE	X		
PROCEDURE-POINTER	X		
PROCEDURES	X		
PROCEED	X		
PROCESSING	X		
PROGRAM	X		
PROGRAM-ID	X		

Table 87. **Reserved words** (continued)

Word	Reserved	Standard only	Potential reserved words
PROGRAM-POINTER			X
PROPERTY			X
PROTOTYPE	X		
PURGE		X	
QUEUE		X	
QUOTE	X		
QUOTES	X		
RAISE			X
RAISING			X
RANDOM	X		
RD		X	
READ	X		
READY	X		
RECEIVE		X	
RECORD	X		
RECORDING	X		
RECORDS	X		
RECURSIVE	X		
REDEFINES	X		
REEL	X		
REFERENCE	X		
REFERENCES	X		
RELATIVE	X		
RELEASE	X		
RELOAD	X		
REMAINDER	X		
REMOVAL	X		
RENAMES	X		
REPLACE	X		
REPLACING	X		
REPORT		X	
REPORTING		X	
REPORTS		X	
REPOSITORY	X		

Table 87. **Reserved words** (continued)

Word	Reserved	Standard only	Potential reserved words
RERUN	X		
RESERVE	X		
RESET	X		
RESUME			X
RETRY			X
RETURN	X		
RETURN-CODE	X		
RETURNING	X		
REVERSED	X		
REWIND	X		
REWRITE	X		
RF		X	
RH		X	
RIGHT	X		
ROUNDED	X		
RUN	X		
SAME	X		
SCREEN			X
SD	X		
SEARCH	X		
SECTION	X		
SECURITY	X		
SEGMENT		X	
SEGMENT-LIMIT	X		
SELECT	X		
SELF	X		
SEND		X	
SENTENCE	X		
SEPARATE	X		
SEQUENCE	X		
SEQUENTIAL	X		
SERVICE	X		
SET	X		
SHARING			X

Table 87. **Reserved words** (continued)

Word	Reserved	Standard only	Potential reserved words
SHIFT-IN	X		
SHIFT-OUT	X		
SIGN	X		
SIZE	X		
SKIP1	X		
SKIP2	X		
SKIP3	X		
SORT	X		
SORT-CONTROL	X		
SORT-CORE-SIZE	X		
SORT-FILE-SIZE	X		
SORT-MERGE	X		
SORT-MESSAGE	X		
SORT-MODE-SIZE	X		
SORT-RETURN	X		
SOURCE		X	
SOURCE-COMPUTER	X		
SOURCES			X
SPACE	X		
SPACES	X		
SPECIAL-NAMES	X		
SQL	X		
SQLIMS	X		
STANDARD	X		
STANDARD-1	X		
STANDARD-2	X		
START	X		
STATUS	X		
STOP	X		
STRING	X		
SUB-QUEUE-1		X	
SUB-QUEUE-2		X	
SUB-QUEUE-3		X	
SUBTRACT	X		

Table 87. **Reserved words** (continued)

Word	Reserved	Standard only	Potential reserved words
SUM		X	
SUPER	X		
SUPPRESS	X		
SYMBOLIC	X		
SYNC	X		
SYNCHRONIZED	X		
SYSTEM-DEFAULT			X
TABLE		X	
TALLY	X		
TALLYING	X		
TAPE	X		
TERMINAL		X	
TERMINATE		X	
TEST	X		
TEXT		X	
THAN	X		
THEN	X		
THROUGH	X		
THRU	X		
TIME	X		
TIMES	X		
TITLE	X		
TO	X		
TOP	X		
TRACE	X		
TRAILING	X		
TRUE	X		
TYPE	X		
TYPDEF			X
UNIT	X		
UNIVERSAL			X
UNLOCK			X
UNSTRING	X		
UNTIL	X		

Table 87. **Reserved words** (continued)

Word	Reserved	Standard only	Potential reserved words
UP	X		
UPON	X		
USAGE	X		
USE	X		
USER-DEFAULT			X
USING	X		
UTF-8	X		
VAL-STATUS			X
VALID			X
VALIDATE			X
VALIDATE-STATUS			X
VALUE	X		
VALUES	X		
VARYING	X		
VOLATILE	X		
WHEN	X		
WHEN-COMPILED	X		
WITH	X		
WORDS	X		
WORKING-STORAGE	X		
WRITE	X		
WRITE-ONLY	X		
XML	X		
XML-CODE	X		
XML-EVENT	X		
XML-INFORMATION	X		
XML-NAMESPACE	X		
XML-NAMESPACE-PREFIX	X		
XML-NNAMESPACE	X		
XML-NNAMESPACE-PREFIX	X		
XML-NTEXT	X		
XML-SCHEMA	X		
XML-TEXT	X		
ZERO	X		

<i>Table 87. Reserved words (continued)</i>			
Word	Reserved	Standard only	Potential reserved words
ZEROES	X		
ZEROS	X		

Related references

[Appendix H, “Context-sensitive words,” on page 791](#)

Appendix H. Context-sensitive words

A context-sensitive word is a COBOL word that is reserved only in the general formats in which it is specified. If a context-sensitive word is used where the context-sensitive word is permitted in the general format, the word is treated as a keyword; otherwise it is treated as a user-defined word.

Table 88. Context-sensitive words	
Context-sensitive word	Language construct or context
BYTE-LENGTH	PICTURE clause
COMPAT	ENTRY-NAME phrase
CODEPAGE	ENCODING phrase
CYCLE	EXIT statement
DLL	ENTRY-INTERFACE phrase
ENTRY-INTERFACE	FUNCTION-ID paragraph
ENTRY-NAME	FUNCTION-ID paragraph
FIXED	CALL ... USING parameters
INITIALIZED	ALLOCATE statement
INTRINSIC	REPOSITORY paragraph
LOC	ALLOCATE statement
LONGMIXED	ENTRY-NAME phrase
LONGUPPER	ENTRY-NAME phrase
NAME	JSON GENERATE statement
	JSON PARSE statement
	XML GENERATE statement
PARAGRAPH	EXIT statement
RECURSIVE	PROGRAM-ID paragraph
YYYYDDD	ACCEPT statement
YYYYMMDD	ACCEPT statement

Related references

[“User-defined words” on page 12](#)

[Appendix G, “Reserved words,” on page 773](#)

Appendix I. ASCII considerations

The compiler supports the American National Standard Code for Information Interchange (ASCII) for magnetic tape files. Thus, the programmer can create and process tape files recorded in accordance with several standards.

The standards are:

- American National Standard Code for Information Interchange, X3.4-1977
- American National Standard Magnetic Tape Labels for Information Interchange, X3.27-1978
- American National Standard Recorded Magnetic Tape for Information Interchange (800 CPI, NRZI), X3.22-1967

Single-byte ASCII-encoded tape files, when read into the system, are automatically translated in the buffers into single-byte EBCDIC. Internal manipulation of data is performed exactly as if the ASCII files were single-byte EBCDIC-encoded files. For an output file, the system translates the EBCDIC characters into single-byte ASCII in the buffers before writing the file on tape. Therefore, there are special considerations concerning ASCII-encoded files when they are processed in COBOL.

This appendix also applies (with appropriate modifications) to the International Reference Version of the ISO 7-bit code defined in International Standard 646, *7-Bit Coded Character Set for Information Processing Interchange* (ISCII). The ISCII code set differs from ASCII only in the graphic representation of two code points:

- Ordinal number 37, which is a dollar sign in ASCII, but a lozenge in ISCII
- Ordinal number 127, which is a tilde (~) in ASCII, but an overline (or optionally a tilde) in ISCII.

The following paragraphs discuss the special considerations concerning ASCII-encoded (or ISCII-encoded) files. The information given for STANDARD-1 also applies to STANDARD-2 except where otherwise specified.

ENVIRONMENT DIVISION

In the ENVIRONMENT DIVISION, the OBJECT-COMPUTER, SPECIAL-NAMES, and FILE-CONTROL paragraphs are affected by the use of ASCII-encoded files.

OBJECT-COMPUTER and SPECIAL-NAMES paragraphs

When at least one file in the program is an ASCII-encoded file, the alphabet-name clause of the SPECIAL-NAMES paragraph must be specified; the alphabet-name must be associated with STANDARD-1 or STANDARD-2 (for ASCII or ISCII collating sequence or CODE SET, respectively).

When alphanumeric comparisons within the object program are to use the ASCII collating sequence, the PROGRAM COLLATING SEQUENCE clause of the OBJECT-COMPUTER paragraph must be specified; the alphabet-name used must also be specified as an alphabet-name in the SPECIAL-NAMES paragraph, and associated with STANDARD-1. For example:

```
Object-computer.  IBM-system
                  Program collating sequence is ASCII-sequence.
Special-names.    Alphabet ASCII-sequence is standard-1.
```

When both clauses are specified, the ASCII collating sequence is used in this program to determine the truth value of the following alphanumeric comparisons:

- Those explicitly specified in relation conditions
- Those explicitly specified in condition-name conditions
- Any alphanumeric sort or merge keys (unless the COLLATING SEQUENCE phrase is specified in the MERGE or SORT statement).

When the PROGRAM COLLATING SEQUENCE clause is omitted, the EBCDIC collating sequence is used for such comparisons.

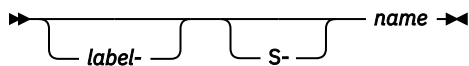
The PROGRAM COLLATING SEQUENCE clause, in conjunction with the alphabet-name clause, can be used to specify EBCDIC alphanumeric comparisons for an ASCII-encoded tape file or ASCII alphanumeric comparisons for an EBCDIC-encoded tape file.

The literal option of the alphabet-name clause can be used to process internal data in a collating sequence other than NATIVE or STANDARD-1.

FILE-CONTROL paragraph

For ASCII files, the ASSIGN clause assignment-name has the following format:

Format: assignment-name for QSAM files

►  name ►

The file must be a QSAM file assigned to a magnetic tape device.

label-

Documents the device and device class to which a file is assigned. If specified, it must end with a hyphen.

S-

The organization field. Optional for QSAM files, which always have sequential organization.

name

A required one-character to eight-character field that specifies the external name for this file.

I-O-CONTROL paragraph

The assignment-name in a RERUN clause must not specify an ASCII-encoded file.

ASCII-encoded files that contain checkpoint records cannot be processed.

DATA DIVISION

In the DATA DIVISION, there are special considerations for the FD entry and for data description entries.

For each logical file defined in the ENVIRONMENT DIVISION, there must be a corresponding FD entry and level-01 record description entry in the FILE SECTION of the DATA DIVISION.

FD Entry: CODE-SET clause

The FD Entry for an ASCII-encoded file must contain a CODE-SET clause; the alphabet-name must be associated with STANDARD-1 (for the ASCII code set) in the SPECIAL-NAMES paragraph.

For example:

```
Special-names. Alphabet ASCII-sequence is standard-1.  
FD  ASCII-file label records standard  
    Recording mode is f  
    Code-set is ASCII-sequence.
```

Data description entries

For ASCII files, the data description considerations listed in the topic apply.

- PICTURE clause specifications are valid for the following categories of data:

- Alphabetic
- Alphanumeric
- Alphanumeric-edited
- Numeric
- Numeric-edited
- For signed numeric items, the SIGN clause with the SEPARATE CHARACTER phrase must be specified.
- For the USAGE clause, only the DISPLAY phrase is valid.

PROCEDURE DIVISION

An ASCII-collated sort or merge operation can be specified in two ways as described in the topic.

- Through the PROGRAM COLLATING SEQUENCE clause in the OBJECT-COMPUTER paragraph. In this case, the ASCII collating sequence is used for alphanumeric comparisons explicitly specified in relation conditions and condition-name conditions.
- Through the COLLATING SEQUENCE phrase of the SORT or MERGE statement. In this case, only this sort or merge operation uses the ASCII collating sequence.

In either case, alphabet-name must be associated with STANDARD-1 (for ASCII collating sequence) in the SPECIAL-NAMES paragraph.

For this sort or merge operation, the COLLATING SEQUENCE phrase of the SORT or MERGE statement takes precedence over the PROGRAM COLLATING SEQUENCE clause in the OBJECT-COMPUTER paragraph.

If both the PROGRAM COLLATING SEQUENCE clause and the COLLATING SEQUENCE phrase are omitted (or if the one in effect specifies an EBCDIC collating sequence), the sort or merge is performed using the EBCDIC collating sequence.

Appendix J. Industry specifications

Enterprise COBOL supports various industry standards.

- ISO COBOL standards

- ISO 1989:1985, *Programming languages - COBOL*

ISO 1989:1985 is identical to ANSI INCITS 23-1985 (R2001), *Programming Languages - COBOL*

- ISO/IEC 1989/AMD1:1992, *Programming languages - COBOL: Intrinsic function module*

ISO/IEC 1989/AMD1:1992 is identical to ANSI INCITS 23a-1989 (R2001), *Programming Languages - Intrinsic Function Module for COBOL*

- ISO/IEC 1989/AMD2:1994, *Programming languages - Correction and clarification amendment for COBOL*

ISO/IEC 1989/AMD2:1994 is identical to ANSI INCITS 23b-1993 (R2001), *Programming Language - Correction Amendment for COBOL*

All required modules are supported at the highest level defined by the standard.

The following optional modules of the standard are supported:

- Intrinsic Functions (1 ITR 0,1)
- Debug (1 DEB 0,2)
- Segmentation (2 SEG 0,2)

The Report Writer optional module of the standard is supported with the optional IBM COBOL Report Writer Precompiler and Libraries (5798-DYR).

The following optional modules of the standard are not supported:

- Communications
- Debug (2 DEB 0,2)
- ISO/IEC 1989:2002, *Information technology - Programming languages - COBOL* (partial support)

ISO/IEC 1989:2002 is identical to ANSI INCITS 1989-2002 (R2013), *Information technology - Programming languages COBOL*

For a list of 2002 COBOL Standard features that are implemented since Enterprise COBOL Version 3, see [Appendix K, “2002/2014 COBOL Standard features implemented in Enterprise COBOL Version 3 or later versions,” on page 799.](#)

- ISO/IEC 1989:2014, *Information technology - Programming languages, their environments and system software interfaces - Programming language COBOL* (partial support)

ISO/IEC 1989:2014 is identical to ANSI INCITS 1989-2014, *Information technology - Programming languages, their environments and system software interfaces - Programming language COBOL*

For a list of 2014 COBOL Standard features that are implemented since Enterprise COBOL Version 3, see [Appendix K, “2002/2014 COBOL Standard features implemented in Enterprise COBOL Version 3 or later versions,” on page 799.](#)

- ANSI COBOL standards

- ANSI INCITS 23-1985 (R2001), *Programming Languages - COBOL*
- ANSI INCITS 23a-1989 (R2001), *Programming Languages - Intrinsic Function Module for COBOL*
- ANSI INCITS 23b-1993 (R2001), *Programming Language - Correction Amendment for COBOL*

All required modules are supported at the highest level defined by the standard.

The following optional modules of the standard are supported:

- Intrinsic Functions (1 ITR 0,1)
- Debug (1 DEB 0,2)
- Segmentation (2 SEG 0,2)

The following optional modules of the standard are not supported:

- Communications
- Debug (2 DEB 0,2)
- ANSI INCITS 1989-2002 (R2013), *Information technology - Programming languages COBOL* (partial support)
- ANSI INCITS 1989-2014, *Information technology - Programming languages, their environments and system software interfaces - Programming language COBOL* (partial support)
- International Reference Version of *ISO/IEC 646, 7-Bit Coded Character Set for Information Interchange*
- The 7-bit coded character set defined in *American National Standard X3.4-1977, Code for Information Interchange*

Enterprise COBOL has the following restriction related to COBOL standards:

- OPEN EXTEND is not supported for ASCII-encoded tapes (CODE-SET STANDARD-1 or STANDARD-2).

See *Option settings for 85 COBOL Standard conformance* in the *Enterprise COBOL Programming Guide* for specification of the compiler options and Language Environment runtime options that are required to support the aforementioned standards.

Acknowledgment

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Appendix K. 2002/2014 COBOL Standard features implemented in Enterprise COBOL Version 3 or later versions

Beginning with Enterprise COBOL Version 3, substantive changes are implemented according to the 2002 COBOL Standard and 2014 COBOL Standard. This topic lists those changes that will potentially affect existing COBOL programs and those changes that will not affect existing COBOL programs.

Table 89. 2002/2014 COBOL Standard features implemented in V3 or later versions that will potentially affect existing programs

Features	Notes
SPECIAL-NAMES paragraph, CURRENCY SIGN clause	The letters 'N', 'n', 'E' and 'e' are now used as picture symbols. The letter 'N' or 'E' can no longer be specified as a currency sign in the CURRENCY SIGN clause.
SAME clause	File-names referenced in a SAME clause shall be described in the FILE-CONTROL paragraph of the source element that contains the SAME clause.
Executable code production	The implementor is not required to produce an executable object program if a fatal exception condition for which checking is not enabled is detected by the compiler.
Exponentiation	If the value of an expression to be raised to a power is less than zero, the following condition shall be true for the exponent: (FUNCTION FRACTION-PART (exponent) = 0). Otherwise, the EC-SIZE-EXPONENTIATION exception exists and the size error condition is raised.
CORRESPONDING order	The order of execution of the implied statements created for corresponding operands for ADD, MOVE, and SUBTRACT with the CORRESPONDING phrase is defined to be the order of the specification of the operands in the group following the word CORRESPONDING. The previous COBOL standard did not specify an order. In addition, the evaluation of subscripts for the implied statements is done only once, at the start of the execution of the actual ADD, MOVE, or SUBTRACT statement. The previous COBOL standard implied this, but was not specific.
Sending and receiving operands	The terms sending operand and receiving operand have been defined.
Incompatible data clarification	The conditions that cause the incompatible data condition are specified explicitly. They are limited to boolean, numeric, and numeric-edited sending operands.

Table 89. 2002/2014 COBOL Standard features implemented in V3 or later versions that will potentially affect existing programs (continued)

Features	Notes
EVALUATE statement, sequence of execution	The sequence of evaluation of selection subjects and objects is now defined to be from left to right and selection objects are evaluated as each WHEN phrase is processed. When a WHEN phrase is selected, no more selection objects are evaluated.

Table 90. 2002/2014 COBOL Standard features implemented in V3 or later versions that will not affect existing programs

Features	Notes
ACCEPT statement four-digit year	The capability to access the four-digit year of the Gregorian calendar is added to the ACCEPT statement.
Apostrophe as quotation mark	The apostrophe character and the quotation mark can be used in the opening and closing delimiters of alphanumeric, boolean, and national literals. A given literal can use either the apostrophe or the quotation mark, but both the starting and ending characters are required to be the same. Whichever character is used, it is necessary to double that character to represent one occurrence of the character within the literal. Both formats can be used in a single source element.
Arithmetic operators	No space is required between a left parenthesis and unary operator or between a unary operator and a left parenthesis.
AT END phrase	The AT END phrase of the READ statement does not have to be specified if there is no applicable USE statement.
BINARY and floating point data	Two new representations of numeric data type are introduced, a binary representation that holds data in a machine-specific way and is not restricted to decimal ranges of values, and a floating-point representation. The floating-point type exists both in a numeric form, with a machine-specific representation, and in a numeric-edited form.
CALL argument level numbers (any)	CALL arguments can be elementary or groups with any level number. Formerly, they had to be elementary or have a level number of 1 or 77.
CALL BY CONTENT parameter difference	A parameter passed by content does not have to have the same description as the matching parameter in the called program.
CALL parameter length difference	The size of an argument in the USING phrase of the CALL statement can be greater than the size of the matching formal parameter if either the argument or the formal parameter is a group item. Formerly, the sizes were required to be the same.
CALL recursively	The capability of calling an active COBOL program has been added to COBOL.

Table 90. 2002/2014 COBOL Standard features implemented in V3 or later versions that will not affect existing programs (continued)

Features	Notes
COBOL words reserved in context	Certain words added to the COBOL standard are reserved only in the contexts in which they are specified and were not added to the reserved word list. See Appendix H, “Context-sensitive words,” on page 791 for details.
CODE clause (Report Writer)	The identifier phrase is provided in the CODE clause in the report description entry.
COLUMN clause	A relative form is provided using PLUS integer, by analogy with LINE; COLUMN RIGHT and COLUMN CENTER are provided, allowing alignment of a printable item at the right or center; and COL, COLS, and COLUMNS are allowed as synonyms for COLUMN.
COLUMN, LINE, SOURCE, and VALUE clauses	These clauses can have more than one operand in a report group description entry.
Comment lines anywhere in a compilation group	Comment lines can be written as any line in a compilation group, including before the identification division header.
Compiler directives	• The CALLINTERFACE compiler directive specifies the interface convention for CALL and SET statements. • The DEFINE directive defines or undefines a compilation variable. • The EVALUATE directive provides a multi-branch method of choosing the source lines to include in a compilation group. • The IF directive provides for a one-way or two-way conditional compilation.
Control data name	This is allowed to be omitted on TYPE CH/CF if only one control is defined.
Conversion from two-digit year to four-digit year	There are three functions for converting from a two-digit year to a four-digit year. DATE-TO-YYYYMMDD, DAY-TO-YYYYDDD, and YEAR-TO-YYYY convert from YYnnnn to YYYYnnnn, YYnnn to YYYYnnn, and YY to YYYY, respectively.

Table 90. 2002/2014 COBOL Standard features implemented in V3 or later versions that will not affect existing programs (continued)

Features	Notes
COPY statement	An alphanumeric literal can be specified in place of text-name-1 or library-name-1. When more than one COBOL library is referenced, text-name need not be qualified when the library text resides in the default library. The ability to nest COPY statements is provided. Library text incorporated as a result of processing a COPY statement without a REPLACING phrase can contain a COPY statement without a REPLACING phrase. A SUPPRESS PRINTING phrase is added to the COPY statement to suppress listing of library text incorporated as a result of COPY statement processing.
COPY and REPLACE statement partial word replacement	LEADING and TRAILING phrases of the REPLACE statement and the REPLACING phrase of the COPY statement allow replacement of partial words in source text and library text. This is useful for prefixing and postfixing names.
Currency sign extensions	The CURRENCY SIGN clause has been extended to allow for national characters and for multiple distinct currency signs, which can have any length.
DISPLAY statement	If the literal in a DISPLAY statement is numeric, it can be signed.
DIVIDE BY zero	DIVIDE BY zero conforms to the 2002 COBOL Standard. Nothing has changed in IBM COBOL, but the Standard is changed so that now Enterprise COBOL is conforming. (Enterprise COBOL was not conforming to the 85 COBOL Standard for DIVIDE BY zero.)
Dynamic-length elementary items	The addition of the DYNAMIC LENGTH clause provides the ability to describe a data item of varying size.
Dynamic storage allocation	ALLOCATE and FREE statements are provided for obtaining storage dynamically. This storage is addressed by pointer data items.
EXIT statement	The ability to immediately exit an inline PERFORM statement, a paragraph, or a section has been added.
EXIT PROGRAM allowed as other than last statement	EXIT PROGRAM is allowed to appear as other than the last statement in a consecutive sequence of imperative statements.
FILLER	FILLER is allowed in the report section.

Table 90. 2002/2014 COBOL Standard features implemented in V3 or later versions that will not affect existing programs (continued)

Features	Notes
Floating comment delimiters	• The floating comment indicator (*>) indicates a comment line if it is the first character string in the program-text area (Area A plus Area B), or indicates an inline comment if it is after one or more character strings in the program-text area. • The compiler directive indicator (>>) indicates a compiler directive line when followed by a compiler directive word - with or without an intervening space.
FUNCTION ALL INTRINSIC	The REPOSITORY paragraph supports a function specifier, and for some intrinsic functions, the keyword FUNCTION is optional when a function is included in the REPOSITORY paragraph.
GOBACK statement	A GOBACK statement has been added that always returns control, either to the operating system or to the calling runtime element.
Hexadecimal literals	The ability was added to specify alphanumeric, boolean, and national literals using hexadecimal (radix 16) notation.
Inline comment	A comment can be written on a line following any character-strings of the source text or library text that are written on that line. An inline comment is introduced by the two contiguous characters '*>'.
Index data item	The definition of an index data item can include the SYNCHRONIZED clause.
INITIALIZE statement, FILLER phrase	FILLER data items can be initialized with the INITIALIZE statement.
INITIALIZE statement, VALUE phrase	A VALUE phrase can be specified in the INITIALIZE statement to cause initialization of elementary data items to the literal specified in the VALUE clause of the associated data description entry.
Intrinsic function facility	Previously, the intrinsic function facility was a separate module and its implementation was optional. The intrinsic function facility is integrated into the specification and it shall be implemented by a conforming implementation

Table 90. 2002/2014 COBOL Standard features implemented in V3 or later versions that will not affect existing programs (continued)

Features	Notes
New intrinsic functions	New intrinsic functions are added: • ABS • BYTE-LENGTH • E • EXP • EXP10 • DATE-TO-YYYYMMDD • DAY-TO-YYYYDDD • DISPLAY-OF • NATIONAL-OF • NUMVAL-F • PI • SIGN • TEST-NUMVAL • TEST-NUMVAL-C • TEST-NUMVAL-F • TRIM • YEAR-TO-YYYY Support for industry standard date and time formats New date and time intrinsic functions are introduced that support encoding and decoding of date and time information to and from formats specified in ISO 8601, and that support encoding and decoding date and time information to and from integers that are suitable for arithmetic. These new intrinsic functions are: • COMBINED-DATETIME • FORMATTED-CURRENT-DATE • FORMATTED-DATE • FORMATTED-DATETIME • FORMATTED-TIME • INTEGER-OF-FORMATTED-DATE • SECONDS-FROM-FORMATTED-TIME • SECONDS-PAST-MIDNIGHT • TEST-DATE-YYYYMMDD • TEST-DAY-YYYYDDD • TEST-FORMATTED-DATETIME
INVALID KEY phrase	The INVALID KEY phrase does not have to be specified if there is no applicable USE statement.

Table 90. 2002/2014 COBOL Standard features implemented in V3 or later versions that will not affect existing programs (continued)

Features	Notes
LOCAL-STORAGE SECTION	A facility was added to define data that is set to its initial values each time a function, method, or program is activated. Each instance of this source element has its own copy of this data.
National character handling	The capability is added for using large character sets, such as ISO/IEC 10646-1, in source text and library text and in data at execution time. Class national and categories national and national-edited are specified by picture character-strings containing the symbol 'N'; national literals are identified by a separator N", N', NX", or NX'. Usage national specifies representation of data in a national character set. User-defined words can contain extended letters. Processing of data of class national is comparable to processing data of class alphanumeric, though there are some minor differences. Conversions between data of classes alphanumeric and national are provided by intrinsic functions.
Object orientation	Support for object-oriented programming has been added.
OCCURS clause	Repetition vertically or horizontally and a STEP phrase are added for Report Writer.
Optional word OF	Allowed after SUM.
Optional word FOR and ON	Allowed after TYPE CH or CF.
OR PAGE phrase of the CONTROL HEADING phrase	This enables the control heading group to be printed at the top of each page and after a control break.
PAGE FOOTING report group	Such a group is allowed to have all relative LINE clauses.
PAGE LIMIT clause	New COLUMNS phrase is provided to define maximum number of horizontal print positions in each report line and a LAST CONTROL HEADING phrase was added.
Paragraph-name	A paragraph-name is not required at the beginning of the procedure division or a section.
PERFORM statement	A common exit for multiple active PERFORM statements is allowed.

Table 90. 2002/2014 COBOL Standard features implemented in V3 or later versions that will not affect existing programs (continued)

Features	Notes
PICTURE clause	The maximum number of characters that can be specified in a picture character-string is increased from 30 to 50. The symbol 'E' can be used in a PICTURE character-string to specify a floating-point numeric-edited data item. The symbol 'N' can be used in a PICTURE character-string to specify a national or a national-edited data item. When the last symbol of a PICTURE character-string is a period or a comma, one or more spaces can precede the following separator period. It was unclear in the previous standard whether a space could precede the separator period in this context.
PLUS and MINUS	The symbol + or - is synonymous with PLUS or MINUS, respectively, in the COLUMN and LINE clauses.
Pointer data	A new class of data is introduced, a pointer type that holds data and program addresses in a machine-specific or system-specific way.
PRESENT WHEN clause	The PRESENT WHEN clauses allows lines and printable items, or groups of them, to be printed or not, depending on the truth value of a condition.
Program-names as literals	The option to specify a literal as the program-name to be externalized was added for names that are not valid COBOL words or need to be case-sensitive.
RECORD KEY and ALTERNATE RECORD KEY	Keys for indexed files can be made up from more than one component.
Report Writer	Previously, the Report Writer was a separate module and its implementation was optional. The Report Writer facility is integrated into the specification and it shall be implemented by a conforming implementation.
Report Writer national character support	The capability of printing national characters and alphanumeric characters in a report is provided.
Function specifier	The REPOSITORY paragraph supports function specifier, and for some intrinsic functions, the key word FUNCTION is optional when a function is included in the REPOSITORY paragraph.
SIGN clause in a report description entry	The SEPARATE phrase is no longer required in a report description entry and the SIGN clause can be specified at the group level.

Table 90. 2002/2014 COBOL Standard features implemented in V3 or later versions that will not affect existing programs (continued)

Features	Notes
SORT statement	A SORT statement can be used to sort a table. This sort can be done using the fields specified in the KEY phrase defining the table, by using the entire table element as the key, or by using specified key data items. GIVING files in a SORT statement can now be specified in the same SAME RECORD AREA clause.
SOURCE clause in a report description entry	The sending operand can be a function-identifier. An arithmetic-expression is allowed as an operand and a ROUNDED phrase was added.
SUM clause in a report description entry	The SUM clause was extended in the following ways: • Extension to total a repeating entry. • Now allowed in any TYPE of report group, not only control footing. • SUM of arithmetic-expression format. • Checks for overflow of a sum counter during totalling. • Any numeric report section item can be totalled, not just another sum counter. • ROUNDED phrase.
Underscore (<code>_</code>) character	The basic special characters of the COBOL character repertoire have been expanded to include the underscore (<code>_</code>) character, which can be used in the formation of COBOL words.
UNSTRING statement	The sending operand can be reference modified.
USE BEFORE REPORTING	The effect of GLOBAL in a report description and a USE declarative is further elucidated.
VALUE clause, WHEN SET TO FALSE phrase in data division	The WHEN SET TO FALSE phrase allows specification of a FALSE condition value. This value is moved to the associated conditional variable when the SET TO FALSE statement is executed for the associated condition-name.
VARYING clause	A VARYING clause is provided in the validate and Report Writer facilities to be used with an OCCURS clause.
WITH RESET phrase	This was added to the NEXT PAGE phrase of the NEXT GROUP clause to reset PAGE-COUNTER back to 1. Note: REPORT WRITER features such as CODE clause, COLUMN clause, control data name, CONTROL HEADING, PAGE FOOTING, PAGE LIMIT, SUM clause, and report description entries are supported via the optional Report Writer Precompiler.

Appendix L. Accessibility features for Enterprise COBOL for z/OS

Accessibility features assist users who have a disability, such as restricted mobility or limited vision, to use information technology content successfully. The accessibility features in z/OS provide accessibility for Enterprise COBOL for z/OS.

Accessibility features

z/OS includes the following major accessibility features:

- Interfaces that are commonly used by screen readers and screen-magnifier software
- Keyboard-only navigation
- Ability to customize display attributes such as color, contrast, and font size

z/OS uses the latest W3C Standard, [WAI-ARIA 1.0](http://www.w3.org/TR/wai-aria/) (<http://www.w3.org/TR/wai-aria/>), to ensure compliance to [US Section 508](https://www.access-board.gov/ict/) (<https://www.access-board.gov/ict/>) and [Web Content Accessibility Guidelines \(WCAG\) 2.0](http://www.w3.org/TR/WCAG20/) (<http://www.w3.org/TR/WCAG20/>). To take advantage of accessibility features, use the latest release of your screen reader in combination with the latest web browser that is supported by this product.

Keyboard navigation

Users can access z/OS user interfaces by using TSO/E or ISPF.

Users can also access z/OS services by using IBM Developer for z/OS.

For information about accessing these interfaces, see the following publications:

- *z/OS TSO/E Primer* (<http://publib.boulder.ibm.com/cgi-bin/bookmgr/BOOKS/ikj4p120>)
- *z/OS TSO/E User's Guide* (<http://publib.boulder.ibm.com/cgi-bin/bookmgr/BOOKS/ikj4c240/APPENDIX1.3>)
- *z/OS ISPF User's Guide Volume I* (<http://publib.boulder.ibm.com/cgi-bin/bookmgr/BOOKS/ispzug70>)

These guides describe how to use TSO/E and ISPF, including the use of keyboard shortcuts or function keys (PF keys). Each guide includes the default settings for the PF keys and explains how to modify their functions.

Interface information

The Enterprise COBOL for z/OS online product documentation is available in IBM Knowledge Center, which is viewable from a standard web browser.

PDF files have limited accessibility support. With PDF documentation, you can use optional font enlargement, high-contrast display settings, and can navigate by keyboard alone.

To enable your screen reader to accurately read syntax diagrams, source code examples, and text that contains period or comma PICTURE symbols, you must set the screen reader to speak all punctuation.

Assistive technology products work with the user interfaces that are found in z/OS. For specific guidance information, see the documentation for the assistive technology product that you use to access z/OS interfaces.

Related accessibility information

In addition to standard IBM help desk and support websites, IBM has established a TTY telephone service for use by deaf or hard of hearing customers to access sales and support services:

TTY service
800-IBM-3383 (800-426-3383)
(within North America)

IBM and accessibility

For more information about the commitment that IBM has to accessibility, see [IBM Accessibility](http://www.ibm.com/able) (www.ibm.com/able).

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Programming interface information

This Language Reference documents intended Programming Interfaces that allow the customer to write programs to obtain the services of Enterprise COBOL.

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List of resources

Enterprise COBOL for z/OS

COBOL for z/OS publications

You can find the following publications in the [Enterprise COBOL for z/OS library](#):

- *What's new*
- *Customization Guide*, SC27-8712-02
- *Language Reference*, SC27-8713-02
- *Programming Guide*, SC27-8714-02
- *Migration Guide*, GC27-8715-02
- *Performance Tuning Guide*, SC27-9202-01
- *Messages and Codes*, SC27-4648-01
- *Program Directory*, GI13-4526-02
- *Licensed Program Specifications*, GI13-4532-02

Softcopy publications

The following collection kits contain Enterprise COBOL and other product publications. You can find them at <http://www.ibm.com/e-business/linkweb/publications/servlet/pbi.wss>.

- *z/OS Software Products Collection*
- *z/OS and Software Products DVD Collection*

Support

If you have a problem using Enterprise COBOL for z/OS, see the following site that provides up-to-date support information: https://www.ibm.com/support/home/product/B984385H82239E03/Enterprise_COBOL_for_z/OS.

Related publications

z/OS library publications

You can find the following publications in the [z/OS library](#).

Run-Time Library Extensions

- *Common Debug Architecture Library Reference*
- *Common Debug Architecture User's Guide*
- *DWARF/ELF Extensions Library Reference*

z/Architecture

- *Principles of Operation*

z/OS DFSMS

- *Access Method Services for Catalogs*
- *Checkpoint/Restart*

- *Macro Instructions for Data Sets*
- *Using Data Sets*
- *Utilities*

z/OS DFSORT

- *Application Programming Guide*
- *Installation and Customization*

z/OS ISPF

- *Dialog Developer's Guide and Reference*
- *User's Guide Vol I*
- *User's Guide Vol II*

z/OS Language Environment

- *Concepts Guide*
- *Customization*
- *Debugging Guide*
- *Language Environment Vendor Interfaces*
- *Programming Guide*
- *Programming Reference*
- *Run-Time Messages*
- *Run-Time Application Migration Guide*
- *Writing Interlanguage Communication Applications*

z/OS MVS

- *JCL Reference*
- *JCL User's Guide*
- *Programming: Callable Services for High-Level Languages*
- *Program Management: User's Guide and Reference*
- *System Commands*
- *z/OS Unicode Services User's Guide and Reference*
- *z/OS XML System Services User's Guide and Reference*

z/OS TSO/E

- *Command Reference*
- *Primer*
- *User's Guide*

z/OS UNIX System Services

- *Command Reference*
- *Programming: Assembler Callable Services Reference*
- *User's Guide*

z/OS XL C/C++

- *Programming Guide*
- *Run-Time Library Reference*

CICS Transaction Server for z/OS

You can find the following publications in the [CICS library](#):

- *Developing CICS Applications*
- *API (EXEC CICS) Reference*
- *Developing CICS System Programs*
- *Global User Exit Reference*
- *XPI Reference*
- *Using EXCI with CICS*

COBOL Report Writer Precompiler

- *Programmer's Manual*, SC26-4301
- *Installation and Operation*, SC26-4302

Db2 for z/OS

You can find the following publications in the [Db2 library](#):

- *Application Programming and SQL Guide*
- *Command Reference*
- *SQL Reference*

IBM Debug for z/OS (formerly IBM Debug for z Systems and IBM Debug Tool for z/OS)

You can find information about IBM Debug for z/OS in the [IBM Debug for z/OS library](#).

IBM Developer for z/OS (formerly IBM Developer for z Systems)

You can find information about IBM Developer for z/OS in the [IBM Developer for z/OS library](#).

Note: IBM Developer for z/OS supersedes IBM Developer for z Systems and Rational Developer for z Systems.

You can find the following publications by searching their publication numbers in the [IBM Publications Center](#).

IMS

- *Application Programming API Reference*, SC18-9699
- *Application Programming Guide*, SC18-9698

WebSphere Application Server for z/OS

- *Applications*, SA22-7959

Softcopy publications for z/OS

The following collection kit contains z/OS and related product publications:

- *z/OS CD Collection Kit*, SK3T-4269

Java

- *IBM SDK for Java - Tools Documentation*, publib.boulder.ibm.com/infocenter/javasdk/tools/index.jsp
- *The Java 2 Enterprise Edition Developer's Guide*, download.oracle.com/javaee/1.2.1/devguide/html/DevGuideTOC.html
- *Java 2 on z/OS*, www.ibm.com/servers/eserver/zseries/software/java/

- *The Java EE 5 Tutorial*, download.oracle.com/javase/5/tutorial/doc/
- *The Java Language Specification, Third Edition*, by Gosling et al., java.sun.com/docs/books/jls/
- *The Java Native Interface*, download.oracle.com/javase/1.5.0/docs/guide/jni/
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JSON

- JavaScript Object Notation (JSON), www.json.org

Unicode and character representation

- *Unicode*, www.unicode.org/
- *Character Data Representation Architecture Reference and Registry*, SC09-2190

XML

- *Extensible Markup Language (XML)*, www.w3.org/XML/
- *Namespaces in XML 1.0*, www.w3.org/TR/xml-names/
- *Namespaces in XML 1.1*, www.w3.org/TR/xml-names11/
- *XML specification*, www.w3.org/TR/xml/

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Product Number: 5655-EC6

SC27-8713-03

